

Sarika Jain
Bharat K. Bhargava
Deepshikha Kalra
Sven Groppe *Editors*

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Sarika Jain · Bharat K. Bhargava ·
Deepshikha Kalra · Sven Groppe
Editors

Proceedings of the International Health Informatics Conference

IHIC 2023

Editors

Sarika Jain
National Institute of Technology
Kurukshetra
Kurukshetra, India

Bharat K. Bhargava
CS Department
Purdue University
West Lafayette, IN, USA

Deepshikha Kalra
Management Education & Research
Institute
New Delhi, Delhi, India

Sven Groppe
University of Lübeck
Lübeck, Germany

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The Best Paper Awards

Like last year, excellent papers were received for the International Health Informatics Conference this year. This year, the conference committee chose three best paper prizes based on strict criteria. Three levels of selection were carried out, and each level was filtered. The track PC chairs examined the session chairs' remarks and grading for each paper for the first round. For the second round of filtration, the review comments submitted during the review process were considered. The Track PC Chairs themselves looked into and completed the final filtration step. The first and foremost criterion was that the publications must be related to the conference's theme, Health Informatics. The track PC Chairs and the Technical Program Committee were already announced on the IHIC 2023 website during the Call for Papers. Here is the list of Session Chairs who may or may not belong to the Technical Program Committee of IHIC 2023.

- Valentina Janev, Institut Mihajlo Pupin, Serbia
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- Narayan C. Debnath, Eastern International University, Vietnam
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- Silvia Araujo and Micaela, University of Minho, Portugal
- G. Prakash, Department of Database Systems, School of Computer Science and Engineering (SCOPE), Vellore Institute of Technology, Vellore

Following papers received Best Paper Award

Title	Authors
Heart Failure Risk Assessment with Quantum-Inspired Support Vector Classification: An Empirical Study	Sheral Waskar, Santhosh Raminedi, Shridevi S
Machine learning based intelligent framework for breast cancer risk assessment	Bikesh Kumar Singh, Narendra Kuber Bodhey and Yogesh Sharma

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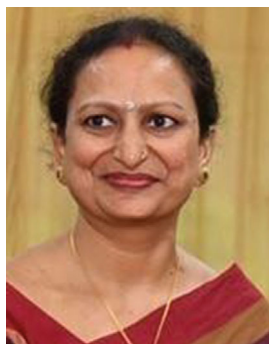


Valentina Janev
Senior Researcher at Institute Mihailo Pupin, University of
Belgrade



Bharat Bhargava
Purdue University, Indiana, United States
bbshail@purdue.edu

General Chairs



Sarika Jain
National Institute of Technology, Kurukshetra, Haryana,
India
jasarika@nitkkr.ac.in
<https://sites.google.com/site/nitkrsarikajain>



Sven Groppe
University of Lübeck, Germany
groppe@ifis.uni-luebeck.de
<http://www.ifis.uni-luebeck.de/~groppe/>

Patron



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VP—MERI Group of Institutions, Delhi, India

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Dean—MERI COLLEGE, Delhi, India



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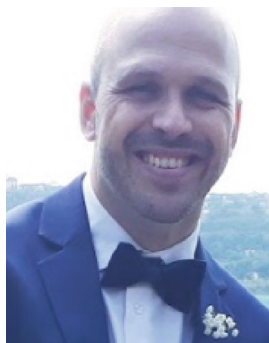


Filbert H. Juwono
Lecturer, Department of Electrical and Computer
Engineering, Curtin University, Malaysia



Nenad Petrović
Faculty of Electronic Engineering, University of
Niš, Serbia

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Antonio ZUORRO
Sapienza University of Rome



Sachi Nandan Mohanty
Associate Professor in Department of Computer Science
& Engineering, ICFAI, Hyderabad

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Neama Abdulaziz Dahan, Sana'a University, Yemen
Neetu Sardana, Jaypee Institute of Information Technology, Noida, India
Nenad Petrovic, University of Nis, Serbia
Nesar Ahmad, AMU, Aligarh
Nikolaos Mavridis, United Arab Emirates Universit, United Arab Emirates
Niranjanamurthy M, MS Ramaiah Institute of Technology, Bangalore
Olawande Daramola, Cape Peninsula University of Technology, South Africa
Olegs Verhodubs, Training center SIA DRMC, Latvia
Pawan Kumar Verma, GLA University, India
Pelin Angin, Middle East Technical University, Turkey
Pethuru Raj, Reliance Jio Infocomm Ltd, Tamil Nadu, India
Petr Kremen, Czech Technical University in Prague, Czechia
Prajoy Podder, Bangladesh University of Engineering and Technology, Bangladesh
Prakash Gopalakrishnan, Amrita Vishwa Vidyapeetham

Pranav K Singh, Department of CSE, CIT Kokrajhar, India
Prateek Agrawal, University of Klagenfurt, Austria
Priti Jagwani, Ram Lal Anand College
Rajkumar Buyya, The University of Melbourne, Australia
Ravi Lourdasamy, Sacred Heart College(Autonomous), Tirupattur, Vellore, Tamil Nadu, India
Regina Reine, Curtin University, Malaysia
Ripal D Ranpara, Atmiya University, India
Rituraj Soni, Engineering College Bikaner, India
Ruchi Mittal, Netaji Subhas Institute of Technology, India
Sachin Gengaje, WIT, Solapur
Sachin Sharma, Manav Rachna International Institute of Research and Studies, Faridabad, India
SachiNandan Mohanty, ICFAI Foundation for Higher Education, Hyderabad, India
Salma Sassi, University of Jendouba, Tunisia
San Murugesan, BRITE Professional Services, Sydney, Australia
Sandeep Sood, Central University of Himachal Pradesh, Shahpur, Himachal Pradesh, India
Sanjay Misra, Covenant University, Ota, Nigeria
Sanju Tiwari, Universidad Autonoma de Tamaulipas, Mexico
Sarika Jain, National Institute of Technology, Kurukshetra
Shahrul Azman Noah, Universiti Kebangsaan Malaysia
Shajulin Benedict, IIIT Kottayam, India
Shankey Garg, National Institute of Technology Raipur, India
Shanmugaraja P, Sona College of Technology, Salem, Tamil Nadu, India
Sharad Saxena, Thapar University, India
Shilpa S. Laddha, Government College of Engineering, Aurangabad, Maharashtra, India
Shivika Prasanna, University of Missouri—Columbia, United States
Shridevi. S, Vellore Institute of Technology, India
Sonika Malik, MSIT, New Delhi, India
Sree Ganesh Thottempudi, University of Heidelberg and BBAW, Germany
Srinath Srinivasa, Web Science Lab, IIIT-Bangalore, Bengaluru, India
Subodh Kesharwani, IGNOU, Delhi, India
Sudhir Kumar Sharma, IITM Janakpuri, GGSIPU Delhi, India
Suja Radha, VIT University
Suneeta Satpathy, BPUT, India
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Vikas Goyal, Education Department, Haryana, India
Vikram Singh, NIT Kurukshetra, India
Vinay Gautam, Chitkara University, India
Vishal Lama, Amdocs, Pune, India
Wong Wei Kitt, Curtin University, Malaysia
Xavierlal J Mattam, Sacred Heart College, Kerala, India
Xiao-Zhi Gao, University of Eastern Finland, Finland
Yang Lu, University of Kent, United Kingdom
Yiming Ji, Georgia Southern University, United States
Zee Ang Sim, Curtin University, Malaysia

Acknowledgments

No one can whistle a symphony. It takes a whole orchestra to play it.

—H. E. Luccock

IHC 2023 is the result of a larger group of people working together. The entire organizing committee has been extremely helpful in ensuring the success of IHC 2023. The editors express their gratitude to the organizing team, which includes advisors, Springer team, the track chairs, session chairs, technical programme committee members, external reviewers, and also the authors. We are grateful to the many volunteers who worked tirelessly to ensure the event's success.

About This Book

Information is the lifeblood of medicine, and health information technology is destined to be the circulatory system for that information.

—David Blumenthal

The field of health informatics is expanding quickly, employing IT in the healthcare industry to streamline operations, cut down on redundant steps, and automate human labor. As healthcare organizations work to save costs and improve patient experience, it is becoming more and more critical. Keeping this in mind, Semantic Intelligence started a conference in health informatics, the International Health Informatics Conference (IHIC), in 2022.

IHIC 2023 had its second edition in 2023 with four top-shot invited researchers as advisory. After a thorough search and discussion, the conference committee has confirmed two keynote speakers based on their work in health informatics. They are known across the world for their contribution to the research community. The main conference organization has 12 chair members and approximately 100 technical program committee members from various countries globally. The second edition of the conference received papers from different parts of the world.

IHIC 2023 has been conducted in hybrid mode. Through a review and selection procedure, only excellent submissions related to the conference are accepted for final publication. Every manuscript was evaluated by three to four reviewers, and 50% of the submissions were accepted this year. Seventy three authors from different universities and institutions across different countries — Spain, Russia, India, USA, Portugal & Vietnam—have contributed 35 manuscripts to the edition.

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Editors and Contributors

About the Editors

Sarika Jain graduated from Jawaharlal Nehru University (India) in 2001. Her doctorate, awarded in 2011, is in knowledge representation in AI. She has served in education for over 23 years and is currently serving at NIT Kurukshetra, India. Dr. Jain's major research interests include AI, Semantic Web, Ontological Engineering, and Knowledge Graphs. She has received grants from the DRDO, DST, CSIR, and NIT Kurukshetra for research projects; from AICTE (thrice) and MHRD (twice) for conducting faculty workshops; and from DAAD RISE worldwide (thrice) for hosting research interns from Germany. She has published over one hundred peer-reviewed technical papers in books, journals, and conference proceedings. She has served as a General Chair, Workshop Chair, Program Committee Chair at many international conferences and workshops, and Reviewer for journals published by IEEE, Elsevier, and Springer. She is a senior member of the IEEE, a member of ACM, and a Life Member of CSI. Among the awards and honors, she has received are the Best Paper Awards, Feb. 2021 (two), Aug. 2020, Aug. 2017; and the Best Faculty Award, Sep. 2019 from NIT Kurukshetra.

Bharat K. Bhargava (Fellow IEEE and IETE) is a Professor of computer science at Purdue University, United States. He is the Founder of the IEEE Symposium on Reliable and Distributed Systems, the IEEE Conference on Digital Library, and the ACM Conference on Information and Knowledge Management. He is the Editor-in-Chief of four journals and serves on over ten editorial boards of international journals. He has published hundreds of research articles and has won seven best paper awards in addition to the Technical Achievement Award and Golden Core Award from IEEE. Professor Bhargava is conducting research on security and privacy issues in distributed systems. His recent work is on Intelligent Autonomous Systems, data analytics, and machine learning.

Deepshikha Kalra is Dean, MERI and HOD, Department of Management. She has over 20 years of Experience in Teaching and Software Industry. She is UGCNet qualified and PhD in management having done MBA, Mphil in Management. She has been awarded with prestigious award of distinguished Faculty award by Ambassador of Bhutan on Teacher's day of Bhutan organised by India Bhutan Friendship association. She is also the recipient of Iconic women creating better world for all by WEF. Dr Kalra has published more than 25 publications in Scopus Indexed-/Refereed & UGC listed journals and has Presented several research papers in International /National Conferences. Dr Kalra has been invited as keynote speaker at various universities at national and international level . Her Areas of research are Design Thinking, Behavioural Economics and Public policy.

She is MHRD certified Innovation Ambassador and Mentor MERI Startup Hub.

She is a professional administrator and has been handling multiple administrative responsibilities and is instrumental in institutional growth along with initiating quality education, innovation and Social responsibility culture within the campus.

Sven Groppe is a Professor at the University of Lübeck, Germany. He published more than 190 journal, conference, and workshop papers at top-ranked publication venues, including SIGMOD, VLDB, and ICPP, with over 190 co-authors from 28 countries on 6 continents worldwide. He is a member of over 120 program committees of various conferences and workshops and a reviewer of over 40 journals. He is area editor for Artificial Intelligence to Fight COVID-19 and Other Pandemics in the Springer journal New Generation Computing. He is a workshop chair of SBD@SIGMOD (2016-2020), BiDEDE@SIGMOD (2021-2023), VLIoT@VLDB (2017-2022), and QDSM@VLDB (2023-2024). He is a general chair of the International Semantic Intelligence Conference (ISIC) (2021-2022), the International Health Informatics Conference (IHIC) (2022- 2024), the International EdTech Conference 2023 (IEdTC 2023), and the International Conference on Applied Machine Learning and Data Analytics (AMLDA) in 2023. More information is available on <https://www.ifis.uni-luebeck.de/~groppe/>

Contributors

Agam Agarwal UPES, Dehradun, Uttarakhand, India

Saksham Agarwal School of Physical Sciences, School of Biological Sciences, School of Computer Sciences, National Institute of Science Education and Research, Bhubaneswar, India

Sonali Agarwal Indian Institute of Information Technology, Allahabad, India

Pranay Ambani Pandit Deendayal Energy University, Gandhinagar, Gujarat, India

Pratiksha Pradeep Baji Department of Computer Science, Walchand College of Engineering, Sangli, Maharashtra, India

Shreya Banerjee Department of Computer Science and Engineering, Manipal Institute of Technology Bengaluru, Manipal Academy of Higher Education, Manipal, India

Víctor Barcons Department of EMIT, Universitat Politècnica de Catalunya, Manresa, Spain

Syed Muzamil Basha Department of Computer Science and Engineering, REVA University, Bengaluru, India

Krishnanjan Bhattacharjee Centre for Development of Advanced Computing (C-DAC), Pune, India

Manel Blanes Department of EMIT, Universitat Politècnica de Catalunya, Manresa, Spain

Narendra Kuber Bodhey Department of Radiodiagnosis, AIIMS Raipur, Raipur, C. G, India

Christopher Francis Britto Department of Computer Science and IT, Byrnihat, Meghalaya, India

Rama Krishna Challa National Institute of Technical Teachers Training and Research, Chandigarh, India

Ritesh Chandra Indian Institute of Information Technology, Allahabad, India

Tanupriya Choudhury School of Computer Science, University of Petroleum and Energy Studies, Dehradun, Uttarakhand, India;
Department of Computer Science and Engineering, Symbiosis Institute of Technology, Symbiosis International University, Pune, India

Archana Chougule Annasaheb Dange College of Engineering and Technology, Ashta, India

Anirban Chowdhury Alliance School of Design, Alliance University, Bengaluru, Karnataka, India

Anirban Dasgupta Discipline of Computer Science and Engineering, Indian Institute of Technology, Gandhinagar, Gujarat, India

Isha Desai Vivekanand Education Society's Institute of Technology Chembur, Mumbai, Maharashtra, India

Ritej Dhamala Indian Institute of Information Technology, Allahabad, India

Amit Doegar NITTTR, Chandigarh, Chitkara University, Rajpura, India

Ayush Dumka National Forensic Sciences University, Gandhinagar, Gujarat, India

Agrawal Luckykumar Dwarkadas National Institute of Technical Teachers Training and Research, Chandigarh, India

E. Bijolin Edwina Department of Computer Science and Engineering, Karunya Institute of Technology and Science, Coimbatore, Tamil Nadu, India

B. Galeebathullah Madanapalle Institute of Technology and Science, Madanapalle, Andhra Pradesh, India

Anupam Garg Computer Science and Engineering Department, TIET, Patiala, Punjab, India

Ashish Girdhar Department of Computer Science and Applications, Kurukshetra University, Kurukshetra, India

Anil Kumar Grewal Management Education and Research Institute, New Delhi, India

Trishala Jeswani Vivekanand Education Society's Institute of Technology Chembur, Mumbai, Maharashtra, India

Ramesh Kait Department of Computer Science and Applications, Kurukshetra University, Kurukshetra, India

Himani Kapur GGSSS Kaithal, Kaithal, India

Chitra Kataria ISIC-Institute of Rehabilitation Sciences, New Delhi, India

Amrita Kaur Computer Science and Engineering Department, TIET, Patiala, Punjab, India

Amritpal Kaur University Institute of Legal Studies, Chandigarh University, Punjab, India

Geet Kiran Kaur Department of Computer Science and Engineering, Chandigarh University, Mohali, Punjab, India

Sujata Khedkar Vivekanand Education Society's Institute of Technology Chembur, Mumbai, Maharashtra, India

Ketan Kotecha Symbiosis Centre for Applied Artificial Intelligence, Symbiosis Institute of Technology, Symbiosis International University, Pune, India

Ankit Kumar Department of Computer Engineering and Applications, GLA University, Mathura, UP, India

Gaurav Kumar Department of Computer Engineering and Applications, GLA University, Mathura, UP, India

Naveen Kumar UPES, Dehradun, Uttarakhand, India

Yashika Kumar Department of Design, Delhi Technological University, New Delhi, India

Jessica Lalchandani Vivekanand Education Society's Institute of Technology
Chembur, Mumbai, Maharashtra, India

Smita Mahapatro Department of Computer Science and Engineering, REVA
University, Bengaluru, India

Sharmistha Majumdar Discipline of Biological Engineering, Indian Institute of
Technology, Gandhinagar, Gujarat, India

Sonika Malik Department of IT, MSIT, Delhi, India

M. Marimuthu Vellore Institute of Technology, Chennai, Tamilnadu, India

Swati Mehta Centre for Development of Advanced Computing (C-DAC), Pune,
India

Akshit Mishra Gujarat National Law University, Gandhinagar, Gujarat, India

Subhankar Mishra School of Physical Sciences, School of Biological Sciences,
School of Computer Sciences, National Institute of Science Education and
Research, Bhubaneswar, India

Pranav Modi Department of Computer Science and Information Security, IIIT
Hyderabad, Hyderabad, Telangana, India

Soham Kumar Modi Department of Computer Science and Engineering,
Chandigarh University, Mohali, Punjab, India

Latika Mohan All India Institute of Medical Sciences (AIIMS), Rishikesh, India

Manisha Mohan Bennett University, Greater Noida, Uttar Pradesh, India

G. Mohanraj Vellore Institute of Technology, Vellore, Tamilnadu, India;
Vellore Institute of Technology, Chennai, Tamilnadu, India

V. Mohanraj Department of CSE, Sona College of Technology, Salem,
Tamilnadu, India

Rajesh Kumar Mohapatra Pandit Deendayal Energy University, Gandhinagar,
Gujarat, India

Navtika Singh Nautiyal School of Law, Forensic Justice and Policy Studies,
National Forensic Sciences University, Gandhinagar, India

Dev Pandya Pandit Deendayal Energy University, Gandhinagar, Gujarat, India

Anshu Parashar Computer Applications Department, NIT, Kurukshetra,
Haryana, India

Archana Patel National Forensic Sciences University, Delhi, India

Ashvini Patil Annasaheb Dange College of Engineering and Technology, Ashta,
India

Rushil Patra Indian Institute of Information Technology, Allahabad, India

Bhagirath Prajapati Department of Computer Engineering, A D Patel Institute of Technology, The Charutar Vidya Mandal University, Vidyanagar-Anand, India

Sachchidanand Prasad Symbiosis Centre for Advanced Legal Studies and Research (SCALSAR), Symbiosis Law School, Symbiosis International (Deemed University), Pune, Maharashtra, India

Siddharaj Dattatray Pujari Department of Computer Science, Walchand College of Engineering, Sangli, Maharashtra, India

Priyanka Puvar Department of Computer Engineering, A D Patel Institute of Technology, The Charutar Vidya Mandal University, Vidyanagar-Anand, India

Nitin Singh Rajput Pandit Deendayal Energy University, Gandhinagar, Gujarat, India

Santhosh Raminedi Centre for Advanced Data Science, Vellore Institute of Technology, Chennai, India

Richa Rashmi Department of Cell Biology and Genetics, Texas A&M University School of Medicine, College Station, TX, USA

Preeti Rathee Department of IT, MSIT, Delhi, India

Sonali Shashikant Rokade Department of Computer Science, Walchand College of Engineering, Sangli, Maharashtra, India

Josep M. Rossell Department of Mathematics, Universitat Politècnica de Catalunya, Barcelona, Spain

Nader Salam Department of Computer Science and Engineering, Karunya Institute of Technology and Science, Coimbatore, Tamil Nadu, India

Sandhya All India Institute of Medical Sciences (AIIMS), Rishikesh, India

Hiral M. Sanghavi School of Forensic Sciences, National Forensic Sciences University, Gandhinagar, Gujarat, India

Santosh Kumar Satapathy Pandit Deendayal Energy University, Gandhinagar, Gujarat, India

B. Sathiya Dhanalakshmi College of Engineering, Chennai, Tamilnadu, India

J. Senthilkumar Department of IT, Sona College of Technology, Salem, Tamilnadu, India

Deepali Sharma NITTTR, Chandigarh, Chitkara University, Rajpura, India

Rakesh Sharma All India Institute of Medical Sciences (AIIMS), Rishikesh, India

Ruchika Sharma University Institute of Legal Studies, Chandigarh University, Punjab, India

Yogesh Sharma Department of Biomedical Engineering, NIT Raipur, Raipur, C. G, India

S. Shridevi Centre for Advanced Data Science, Vellore Institute of Technology, Chennai, India

Bikesh Kumar Singh Department of Biomedical Engineering, NIT Raipur, Raipur, C. G, India

Deependra Singh School of Physical Sciences, School of Biological Sciences, School of Computer Sciences, National Institute of Science Education and Research, Bhubaneswar, India

Kamred Udham Singh School of Computing, Graphic Era Hill University, Dehradun, India

Ravindra Singh Department of Design, Delhi Technological University, New Delhi, India

Teekam Singh Department of Computer Science and Engineering, Graphic Era Deemed to Be University, Dehradun, India

Tejasv Singh National Forensic Sciences University, Gandhinagar, Gujarat, India

Sanjay Singla Department of Computer Science and Engineering, Chandigarh University, Mohali, Punjab, India

K. G. Srinivasa International Institute of Information Technology, Naya Raipur, Chattisgarh, India

Sadiya Sulaiman Department of Computer Science and Engineering, Karunya Institute of Technology and Science, Coimbatore, Tamil Nadu, India

Sunita Computer Science and Engineering Department, CCET (Degree Wing), Chandigarh, India

Y. Suresh Department of CSE, Sona College of Technology, Salem, Tamilnadu, India

Viswanath Talasila Ramaiah Institute of Technology, Bengaluru, Karnataka, India

M. Roshni Thanka Department of Computer Science and Engineering, Karunya Institute of Technology and Science, Coimbatore, Tamil Nadu, India

Ishita Theba Department of Computer Engineering, A.D. Patel Institute of Technology, The Charutar Vidya Mandal (CVM) University, Vidyanagar-Anand, India

S. Theetchenya Sona College of Technology, Salem, Tamilnadu, India; Department of CSE, Sona College of Technology, Salem, Tamilnadu, India

Sadhana Tiwari Indian Institute of Information Technology, Allahabad, India

Sanjana Vashdev Asrani Vivekanand Education Society's Institute of Technology Chembur, Mumbai, Maharashtra, India

Sudhir Vegad Department of Information Technology, Madhuben and Bhanubhai Patel Institute of Technology, The Charutar Vidya Mandal (CVM) University, Vidyanagar-Anand, India

Jesús Vicente Department of EMIT, Universitat Politècnica de Catalunya, Manresa, Spain

G. Vidhya Sona College of Technology, Salem, Tamilnadu, India

D. Vidyabharathi Sona College of Technology, Salem, Tamilnadu, India

Ivneet Walia Rajiv Gandhi National University of Law, Patiala, Punjab, India

Savita Walia NITTTR, Chandigarh, Chitkara University, Rajpura, India

Sheral Waskar Centre for Advanced Data Science, Vellore Institute of Technology, Chennai, India

Tanvi Yadav School of Law, National Forensic Sciences University, Delhi Campus, India;
National Forensic Sciences University, Delhi Campus, Delhi, India

N. Yuvaraj Dhanalakshmi College of Engineering, Chennai, Tamilnadu, India

Research

An Automated Framework for the Segmentation of Skin Lesions Using Deep Learning



Anupam Garg, Amrita Kaur, Sunita, and Anshu Parashar

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Abstract Skin cancer, a prevalent and potentially deadly condition resulting from factors such as UV rays exposure and ozone layer depletion, is a significant health concern worldwide, with approximately 5 million cases reported annually. Its gravity lies in its ability to metastasize to the brain, causing complications like headaches, seizures, and paralysis. Early detection plays a pivotal role in mitigating mortality rates, yet subjective clinical assessments for lesion segmentation often fall short of the expectations of expert dermatologists. Biomedical image segmentation, which involves the automatic or semi-automatic delineation of boundaries in 2D images, encounters challenges due to variations in shape, location, size, and texture within medical images. Manual segmentation, a laborious and time-consuming endeavor, underscores the need for fully automated approaches. In this context, this paper introduces a UNet-based architecture enhanced by the pre-trained ResNet50 model. The comprehensive results showcase outstanding performance across five evaluation parameters for medical image segmentation. Notably, this GA-UNet approach surpasses traditional methods with an impressive accuracy rate of 98.7% and a Dice Similarity Coefficient of 89.6%.

A. Garg (✉) · A. Kaur

Computer Science and Engineering Department, TIET, Patiala, Punjab 147001, India

e-mail: anupam.garg@thapar.edu

Sunita

Computer Science and Engineering Department, CCET (Degree Wing), Chandigarh, India

A. Parashar

Computer Applications Department, NIT, Kurukshetra, Haryana, India

Keywords Skin lesions · GA-UNet · Dermoscopic images · Deep learning

1 Introduction

The segmentation of medical images is a highly demanding and challenging task that holds immense significance in both research and clinical practice. It encompasses a wide range of applications, including automated organ segmentation, optic disc segmentation, brain segmentation, lung segmentation, cell segmentation, and pancreas segmentation. This field has gained substantial attention within the computer vision and medical image processing communities, particularly in the context of technologies like magnetic resonance imaging (MRI) and computed tomography (CT). Nonetheless, the handling of medical images encounters several obstacles, such as the presence of noise and artifacts, limited contrast distinguishing organs from surrounding tissues, considerable diversity in object appearances and shapes, and a shortage of sufficient training data. In response to these challenges, numerous advanced pre-processing methods and both semi-automated and fully automated approaches for the segmentation of 2D and 3D medical images have been devised and refined. Traditional techniques for medical image segmentation typically involve approaches crafted by human experts, relying on their domain knowledge, which necessitates extensive and painstaking efforts from practitioners. Initially, basic rule-based methods were employed, but they proved unreliable, given the vast diversity of data. As a result, more adaptable algorithms were developed, leveraging geometric shape priors. Nevertheless, these conventional methods are susceptible to human biases and result in variations both within and between observers. Consequently, as computer algorithms have advanced, researchers and data scientists have shifted their focus toward cutting-edge techniques like convolutional neural networks (CNNs) and deep neural networks (DNNs) for medical image segmentation. These algorithms offer rapid and efficient segmentation capabilities, eliminating the need for human intervention.

Cancer is one of the leading causes of human death. Among numerous types of cancer, skin cancer is one of the significant contributors to human mortality [9]. Various categories of skin cancer have been found, out of which melanoma is the most occurring and fast-spreading malignant lesion in comparison to others [15]. The early diagnosis of melanoma can lead to a survival rate of around 99%, whereas delayed detection causes a drastic decline in the rate to 23% [4]. Skin cancer initially starts affecting the upper layer of the skin (epidermis layer), which is seen with eyes. Therefore, early-stage detection can prevent the spreading of the cancer to deeper layers by increasing the chances of saving a life. With the advancements in technology in recent years, diagnosis of different types of diseases has become easier. The early detection of skin lesions can be done accurately using automatic systems developed based on skin lesion images requiring no other medical information [11]. Dermoscopy is the most established approach for skin imaging, making it significant for features that are not visible to the naked eye. The benchmark datasets are available

with a limited number of classes. The features of the lesions in the images, such as texture, color, shape, etc., are much the same, making it a challenge to classify their type. Other challenges faced in the segmentation of melanoma and non-melanoma lesions are the high correlation between the two and the presence of unwanted environmental surroundings such as hair, veins, etc. Many researchers worked on these issues for accurate prediction [10].

Utilization of transfer learning along with pre-trained model using GoogleNet on the latest public skin lesion challenge ISIC 2019, providing an accuracy of 94.92%. Milton et al. [13] proposed a segmentation algorithm that is robust in performance by utilizing the ISIC 2018 challenge. Combalia et al. [6] give a detailed analysis of all the ISIC challenges along with the application of various traditional deep learning algorithms in which the best-performed model provides an AUC of 0.80. Hardie et al. [7] applied traditional image processing on ISIC 2018 challenge images for segmenting the lesions. They used the RGB Bayes classifier with the regression network. On the application of the SVM classifier for classification, a recall value of 0.7303 is approached. For segmentation of lesions [2], combined three deep learning models, namely, UNet, ResUNet, and ResUNet++, on ISIC 2016 and ISIC 2017 challenges, giving the Jaccard Index of 80.73% and 90.02%. Bi et al. [3] presented a hyper-fusion network (HFN) that has the characteristic of fusing extracted user-inputs and image features over multiple stages whose performance is compared with start-of-the-art approaches.

2 Objectives

- To analyze the performance of existing state-of-the-art segmentation techniques and for segmentation of skin lesions to explore the best one for dermoscopic images.
- To propose a segmentation technique for the segmentation of skin lesions from dermoscopic images.
- To test and validate the proposed techniques compared to state-of-the-art techniques both quantitatively and qualitatively.

3 Methodology

3.1 Dataset

Before the ISIC (International Skin Imaging Collaboration) challenge, the publicly available datasets for skin lesions have a small number of images. This challenge provides the world's largest repository for skin dermoscopic images from 2016 to 2020. In this proposed research, ISIC 2018 [5] and 2019 [6] are considered. ISIC

2018 consists of 11,527 total images, out of which 10,015 are training images and 1512 are testing images. ISIC 2019 consists of a total of 33,569 dermoscopic images of lesions (25,331 training images and 8238 testing images) captured from 2010 to 2016 in the Hospital Clinic in Barcelona. Dataset BCN20000 was provided in the ISIC 2019 challenge for unconstrained classification of lesions in skin images. Specifically, ISIC 2019 has 4522 images for the melanoma class, and ISIC 2018 has 1113 images.

3.2 *Proposed Framework*

The method is suggested for 2D pictures. Since manually creating masks is an expensive operation, data sets for image segmentation typically only contain thousands of photos at most. UNet can nonetheless learn with a limited set of training samples or data. Normally, dynamically initialized weights start the UNet training from scratch. It is obvious that the network training should be rather extensive without overfitting the data set. The initialization of the weights of the networks in other tasks, such as medical picture segmentation, is frequently done using the networks trained on the ImageNet dataset [14]. As a result, we provided a framework for 2D images that employed learning techniques that took into account a dataset's features for several untrained levels of the network in order to address the alterations made to the original UNet architecture. In the GA-UNet network, which consists of 50 sequential deep layers, we used straightforward CNNs from the ResNet family known as ResNet 50 as an encoder [12]. ResNet is a viable option because its early layers start to teach the neural network about notions like edges and blobs for various datasets. As a result, when pre-trained models like ResNet50 are employed, these notions do not require additional training. With a setup of five convolutional blocks made up of convolutional layers, max-pooling layers, ReLU activation function, and a two-fold reduction of each feature map, ResNet50 is in sync with the encoder phase of the architecture. The details of layers of the proposed architecture are as follows:

1. A single convolutional layer having 64 kernels of size 7×7 is used with a stride of 2.
2. Subsequently, a max-pooling layer with a stride of 2 is applied.
3. There are three sets of convolutional layers, which are repeated three times in the proposed architecture. Each set consists of a 1×1 convolutional layer with 64 kernels and a 3×3 convolutional layer with 64 kernels. Another 1×1 convolutional layer with 256 kernels is used. This set of layers is repeated three times, adding a total of nine layers to the architecture.
4. Next, there are four sets of convolutional layers, which are repeated four times. Each set consists of a 1×1 convolutional layer with 128 kernels, along with a 3×3 convolutional layer having 128 kernels. Additionally, a 1×1 convolutional layer with 512 kernels is used in the architecture, resulting in a total of 12 layers.

- 5. Similarly, the next block uses six sets of convolutional layers which are repeated six times. Each set consists of a 1×1 convolutional layer with 256 kernels and a 3×3 convolutional layer with 256 kernels. Another 1×1 convolutional layer was used with 1024 kernels. This step gives a total of 18 layers.
- 6. Subsequently, there's a single convolutional layer with a 1×1 kernel and 512 kernels, followed by two 3×3 convolutional layers with 512 kernels and a final 1×1 convolutional layer with 2048 kernels. This set is repeated three times, yielding a total of 9 layers.
- 7. Afterward, an average pooling layer is applied, and the network concludes with a fully connected layer comprising 1000 nodes, followed by a Softmax function, contributing one final layer. In total, the architecture encompasses 50 multiple layers in the proposed architecture.

The ResNet50 layers are added in the encoder phase of the traditional UNet architecture. The network of an encoder has undergone a substantial change as a result of the removal of fully connected layers and the addition of the 512 channels layer, which acts as the core of the network. The decoder is made using transposed convolution layers, which increases the size of a feature map while reducing the number of channels. The output of the decoder is then combined with the output of a transposed convolution. Five max-pooling procedures are combined with five upsampling cycles. The upsampling path of numerous feature channels allows the network to communicate context data to higher-resolution layers. As a result, the design is U-shaped, and the encoder path is almost symmetric to the decoder path. Table 1 represents details of hyperparameters used for the proposed architecture.

Table 1 Details of the hyperparameters used in the proposed architecture

Hyperparameters				
Encoder ResNet50 phase			Decoder phase	
Weights	ImageNet		Convolutional layer	ConvTranspose 2D
Stride (number)	2		Stride (number)	2
Max-pooling layers (number)	5(Averag pooling)	e	Max-pooling layers (number)	5
Activation function	ReLU, Softmax		Activation function	ReLU, Softmax
Padding	Same		Padding	Same
Filter size for convolution layer	$7 \times 7, 3 \times 1$	$\times 3,$	Filter size for convolution layer and pooling layer	$3 \times 3, 2 \times 2$
Total parameters	70million (approx.)		Total parameters	0.million n (approx.)

4 Results and Discussion

The proposed framework has been evaluated in terms of dice, precision, and accuracy performance parameters. The aforesaid results are achieved with the combination of UNets with ResNet50 model. On comparison of the proposed model with Milton [13], it has been discovered that the proposed framework outperforms and offers a better segmentation performance in terms of all metrics. Figure 1 displays the findings for segmenting the ISIC 2019 and 2020 challenge containing skin lesion images. The proposed framework is evaluated with the state-of-the-art approaches in terms of five different evaluation parameters, namely, Dice coefficient (DSC) that specifies the similarity index between two data sets, Jaccard index (JI) computes the similarities and diverseness among the sample sets, Precision (PI) measures the proportion of all positive classifications that are actually positive, Accuracy (ACC) calculates the proportions of all the correct classifications, and Sensitivity (SEN) measures a model's ability to accurately predict true positives for each category as depicted in Table 2. The proposed method offers a higher ACC of 98.7% and DSC of 89.6% than the conventional approaches. Performance shows that the target output for every example is accurately segmented with nearly the same shape as the ground truth, which amply the resilience of the proposed framework.

When compared to existing segmentation techniques, the analysis and presentation of overall performance of the proposed framework is superior which is represented through the bar graph as shown in Fig. 1.

It is crucial to note that the suggested framework can effectively segment all the different images in the dataset. Moreover, it is essential to analyze the bar graph of the proposed framework with the traditional methods, the variance of all the parameters,

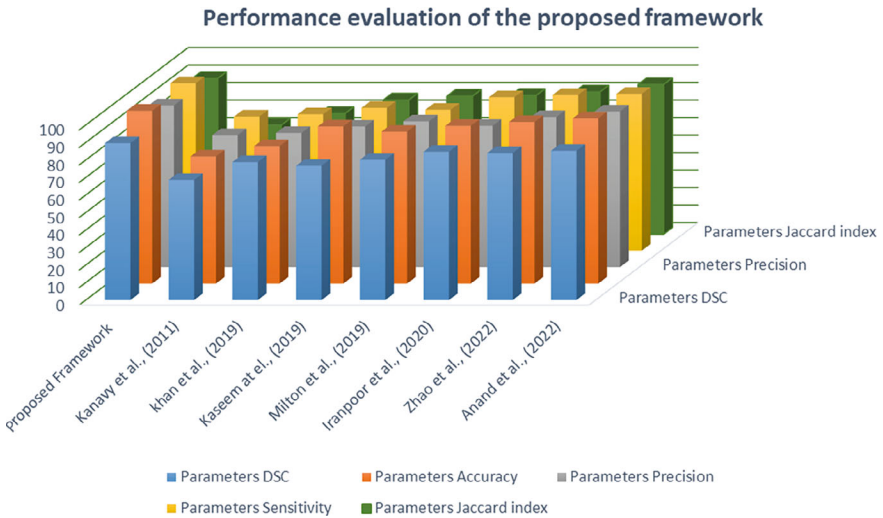


Fig.1 Visualization of the performance of proposed framework with traditional methods

Table 2 Average values of the performance parameters for the proposed framework compared with traditional methods

References/ methods	Parameters				
	Dice coefficient (DSC)	Accuracy (ACC)	Precision (PI)	Sensitivity (SEN)	Jaccard index (JI)
Proposed framework	89.6	98.7	92.3	95.6	88.9
Kanavy and Gerstenblith [9]	68.6	72.6	75.4	76.6	62.3
Khan et al. [11]	78.8	78.5	76.6	77.9	68.9
Kassem et al. [10]	76.7	89.8	80.3	81.5	76.6
Milton et al. [13]	80.1	86.8	83.2	80.4	78.8
Iranpoor et al. [8]	84.7	90.1	80.7	87.6	79.2
Zhao et al. [16]	84.0	92.3	85.7	88.9	81.2
Anand et al. [1]	85.2	94.4	88.9	89.5	85.5

including DSC, JI, PI, ACC, and SEN, is incredibly high when compared to the other methods, indicating the robustness and effectively segmentation of medical images by the proposed framework.

5 Conclusion

The paper introduces a novel framework that is designed for the segmentation of skin lesions in medical images, with a primary focus on its evaluation using the ISIC 2018 and 2019 datasets. The algorithm demonstrates impressive performance metrics, including an average accuracy of 98.7% and a Dice Similarity Coefficient (DSC) of 89.6%. Moreover, it significantly surpasses existing models in terms of sensitivity (SEN) at 95.6%, Jaccard Index (JI) at 88.9%, and precision (PI) at 92.3%. Notably, the versatility of the proposed framework extends to efficiently handling various 2D medical image types, including dermoscopic, X-ray, MRI, and CT scans, regardless of their source and imaging protocols, making it a valuable tool for a wide array of medical imaging tasks.

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An Early Diagnosis of Image Processing Model for Leukemia Detection Using Generative Adversial Network



G. Vidhya, N. Yuvaraj, M. Marimuthu, S. Theetchenya, G. Mohanraj,
and D. Vidyabharathi

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Abstract Nowadays the primary illness of death in many populations is leukemia. The prognosis and possibility of recovery are significantly affected by the disease's early identification and diagnosis. Nowadays, blood related problems are diagnosed by visual evaluation of microscopic images, which involves looking at changes in the photos' texture, shape, color, and statistical analysis. This system provides an early attempt to identify if a person has been affected by leukemia using blood sample photos. The ability to detect and diagnose leukemia at an earlier stage is made possible via image analysis. Due to their low cost and lack of need for expensive testing and lab equipment, images are used. In this study, generative adversial network (GAN) model

G. Vidhya (✉) · S. Theetchenya · D. Vidyabharathi
Sona College of Technology, Salem, Tamilnadu, India
e-mail: gvidhyacse@gmail.com

S. Theetchenya
e-mail: theetchenya@sonatech.ac.in

D. Vidyabharathi
e-mail: dvbharathi77@gmail.com

N. Yuvaraj
Dhanalakshmi College of Engineering, Chennai, Tamilnadu, India
e-mail: yuvaraj.n@dce.edu.in

M. Marimuthu · G. Mohanraj
Vellore Institute of Technology, Chennai, Tamilnadu, India
e-mail: marimuthu.m@vit.ac.in

G. Mohanraj
e-mail: mohanraj.gopal@vit.ac.in

is utilized to identify leukemia cells among the healthy blood cells. These damaged white blood cells can adhere to soft tissue called bone stem that can be found inside most bones. The abnormal white blood cells continue to grow uncontrollably in the bone marrow. Segmentation of the blood cells is done with k-means clustering.

Keywords Blood · Smear · Images · Lymphocytes · Leukemia · Classification · Generative adversarial network

1 Introduction

One form of cancer is Lymphoma known as Leukemia that affects white blood cancer (WBC). The soft tissue known as bone marrow, which is present inside most bones, can allow these injured white blood corpuscles to stick to it. The bone stem cells' aberrant white blood cells are expanding erratically. This causes the body's normal, healthy white blood cells to change into abnormal, out-of-control ones, decreasing the body's ability to fight infections. Red blood cells and thrombocytes are also affected by the aberrant white blood cells. Due to anemia, which is brought on by damaged red blood cells, the patient may feel exhausted and out of breath. They also inhibit the body's tissues and organs from getting enough oxygen. Racial and ethnic variations in incidence rates are barely perceptible in the youngest adult age group (30–54 years), but they become increasingly obvious in each of the age groups that follow. Examination of leukemia detection is done by a hematologist who examines certain cells under a light microscope to check for if any damages are present in the major parts of the cell, in order to classify the abnormal cells in their specific categories. Leukemia is manually detected in pathology, which takes time (Fig. 1).

The patient is advised to see a doctor if the blood cell count is abnormal. Hence, morphological bone marrow and peripheral blood slide examination are performed to demonstrate the existence of leukemic cells. There are various forms of leukemia that, if untreated, can be fatal to people. These categories are determined by the particular type of blood cell that is impacted and the rate at which the illness advances. Leukemia can occur in people of all ages, but it is most common in children and adults over the age of 55. The exact causes of leukemia are not fully understood, but factors that

Fig. 1 Leukemia cell

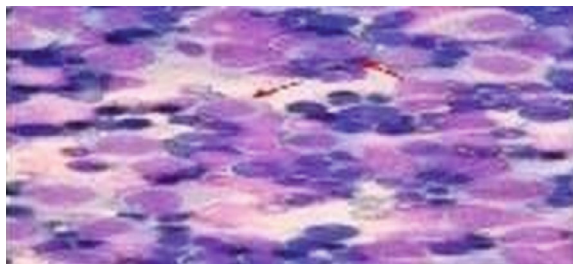
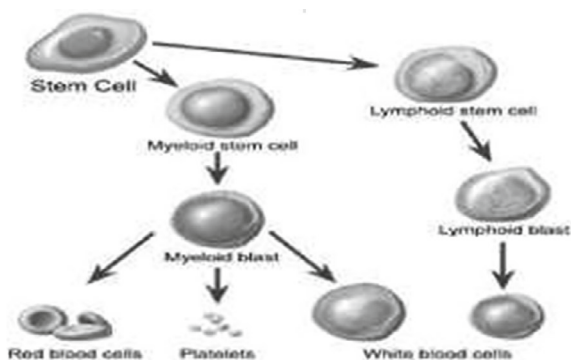


Fig. 2 Division of lymph cells



may increase the risk of developing the disease include exposure to certain chemicals and radiation, genetic factors, and some medical conditions such as Down syndrome (Fig. 2).

Researchers have developed a variety of approaches for image processing that detect blood cancer in medical photographs of blood from human beings samples. Some of them adopt a procedure to locate cancer cells by measuring the blood cell ratio and use the two methods of image segmentation of the thresholding and watershed to locate malignant cells. Using thresholding technique and watershed methods, the segmentation accuracy are 81.24% and 85.27%, respectively. It has a segmentation framework that utilizes a lot of digital image processing techniques. Twenty miniature blood images have been analyzed as a sample. In this investigation, wavelet transforms are employed to segment images, and the results are compared with those of conventional approaches and wavelet transform results. This article discusses about the best way to apply the programming language MATLAB to identify malignancy. It is based on the amount of blood cells identified in the given blood sample volume. The microscopic pictures are looked at directly by experts in the detecting process. This is a drawn-out, lengthy procedure that depends on one's own expertise and is frequently not accurate. Leukemia also is a type of cancer in which the number of abnormal white blood cells increases substantially rapidly in the bone marrow. As a consequence, usual white blood cells are unable to accomplish normal responsibilities, and the number of abnormal WBCs rises quickly. This particular kind of cancer demands early detection.

2 Related Works

In the area of leukemia detection, a number of relevant tasks and research investigations have been carried out. Here are a few illustrations:

White blood cell (WBC) cancer is referred to as leukemia. If not detected in time, it might be fatal. Haematologists examine microscopic images to find malignancy.

This manual identification technique quickly grows difficult and time-consuming. If found sooner, cancer can be cured. However, the traditional approach results in a late finding of malignant cells. Therefore, an automated approach can be employed to reduce the number of deaths brought on by late detection. In this study, a 10 automated system using image analytics is proposed. The suggested system would produce the right results based on cell image samples from patients who have undergone image analytics and categorization. The TCIA (The Cancer Imaging Archive) repository is where the experimental dataset is found. The preprocessed dataset is already accessible. For predictions, "Orange-Data Mining" is an entirely free software programme [4].

Cancer is a deadly condition. The best approach to treating cancer is early identification. Medical image processing is crucial to determining the cause of illness. Leukemia is a type of blood cancer that develops when White Blood Cells (WBCs) are abnormal or immature. WBC is typically the one to combat infectious cells in the human body. Bone marrow and lymphatic tissues will be affected by abnormal WBC proliferation from bone marrow, which will also destroy other cells. Due to faulty cell function, leukemia results. It used to be extremely challenging to identify an illness and count the amount of blood cells [19].

In the medical field, a tool called a hemocytometer, for short, is used for counting manually the amount of blood cells. Nevertheless, it takes more to count and produces unreliable findings. With the assistance of microscopic images, a software-based solution has been developed for tackling these problems. With the help of this image processing method, it is feasible to determine a person's red blood cells, white blood cells, and platelet count as well as whether or not they are at risk for leukemia. M. Thangaraj and S. Senthil Kumar's study, "Leukemia identification using blood smear image analysis," was published. This study suggested using machine learning algorithms and image processing techniques to automatically detect leukemia from blood smear images [2, 12, 13, 16].

R. Dhanalakshmi and P. Thambidurai's paper, "Leukemia detection using artificial neural networks," describes their method. This study developed a neural network-based approach for diagnosing leukemia from blood samples, reaching an accuracy of 95%.

R. Gupta and S. Dutta's study, "Leukemia detection using convolutional neural networks," was published. This paper presented a 98.5% accurate deep learning-based method for identifying leukemia in blood samples using convolutional neural networks. In this study, the effectiveness of different machine learning methods, such as k-NN, SVM, and neural networks, for diagnosing leukemia from blood samples was examined [9, 18].

S. Marzbanrad et al. Infrared spectroscopy for leukemia detection. This study developed a method for identifying leukemia using infrared spectroscopy and attained an accuracy of 93%. These studies show the variety of strategies and methods that have been applied for leukemia diagnosis, including spectroscopy, machine learning, and image processing [7].

H. Saeed, et al., Leukemia detection using artificial intelligence and machine learning techniques. This study uses machine learning technique which is used

to classify leukemia cells based on their morphological characteristics. This work suggests a deep learning-based approach framework for detecting leukemia cells from microscopic images [5, 21].

F. A. Al-Waisy, et al., Leukemia detection with photos processing and SVM, this study presents a method for detecting cells from microscopic images using image processing techniques and an SVM classifier.

"Leukemia detection using K-nearest neighbor classifier" by M. H. Ahmed et al. (2017)—This research work proposes a method for detecting leukemia cells that were extracted from microscopic pictures using a K-nearest neighbor classifier [8].

"Leukemia detection using fuzzy logic and neural networks" by S. S. Al-Rawi et al. (2016)—This study presents a method for detecting leukemia cells using a fuzzy logic system and neural networks [1].

3 Objective of the Study

The need of the leukemia detection by using image processing is because the above diagnosis is time-consuming and costly. As it is difficult to differentiate the leukemia cells from white blood cells by using clinical methods as it changes with respect to time. By this method we can easily differentiate the cells from white blood cells in less time and most effective way just by considering the microscopic images.

Detecting immature or abnormal blood cells is indeed an important task in medical diagnostics, and automating this process can save time and improve accuracy. The process involves multiple steps to achieve the objective.

Image Acquisition: Blood smear images are captured using digital imaging systems. These images might vary in terms of lighting conditions, magnification, and quality.

Preprocessing: Images are preprocessed to enhance features, reduce noise, and standardize the data. This can involve operations like noise reduction, contrast enhancement, and color normalization.

Segmentation: This step involves separating different cell types within the image. Cell segmentation can be particularly challenging due to variations in cell size, shape, and staining.

Feature Extraction: Relevant features are extracted from segmented cells. These features could include morphological characteristics like size, shape, texture, and color distribution.

Classification: Machine learning algorithms are used to classify the cells into different categories, such as normal and abnormal. This is typically done based on the features extracted in the previous step.

Counting: The total number of cells is counted, and in your case, the focus is on identifying immature cells.

Given the urgency of identifying different cell types within a short duration of time, here are some suggestions for technologies and approaches that could be used:

Convolutional Neural Networks (CNNs): CNNs are a type of deep learning algorithm that have shown remarkable success in image classification tasks. They can automatically learn hierarchical features from images, which makes them suitable for complex tasks like cell classification.

Fast and Efficient Algorithms: Since you're working with emergency scenarios, it's important to use algorithms that can process images quickly. You might consider architectures designed for real-time or fast processing.

Transfer Learning: This involves using pre-trained models, often trained on large datasets, and fine-tuning them for your specific task. This can save time and resources while still achieving high accuracy.

Automated Quality Control: Automated systems should be equipped with mechanisms to identify cases where the automated detection might be uncertain or inaccurate. This ensures that the final diagnosis is reliable.

Parallel Processing: If available, using multiple processors or GPUs can significantly speed up the image processing and analysis pipeline.

Data Augmentation: Since you're dealing with variations in blood smear images, data augmentation techniques can help create a more diverse training dataset, improving the model's robustness.

User Interface: Developing a user-friendly interface for medical professionals to interact with the system and verify results is crucial.

Remember that creating such a system involves a multidisciplinary approach, including medical expertise, image processing, machine learning, and software development. It's also important to validate the system thoroughly using real-world data to ensure its effectiveness and safety in clinical practice [10, 17].

4 Proposed Methodology

The proposed system will take the input of either a healthy or diseased leukemia cell. The image is then preprocessed with the help of median filter in MATLAB. Segmentation of the leukemia cells takes using k-means as a segmentation methodology. Using Feature extraction, the Gray Scale Co-level matrix will differentiate the important factors affecting both normal and leukemia cells. Finally, the result for prediction of leukemia is done with the help of Generative Adversarial Network (GAN). The modules are listed as follows: Preprocessing, Segmentation, Feature Extraction, Classification. The proposed methodology is shown in Fig. 3.

4.1 Dataset Preparation

The dataset that provides accurate microscopic images of both normal and ALL cells is the ALL-IDB dataset. These images are present as a form of library which

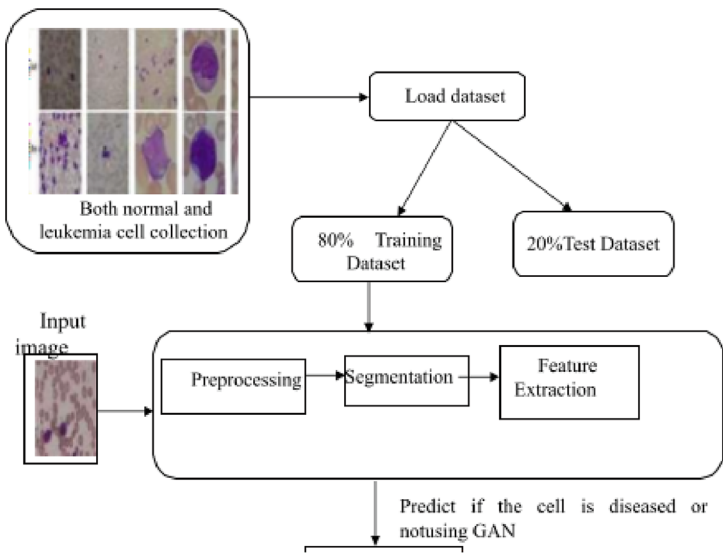
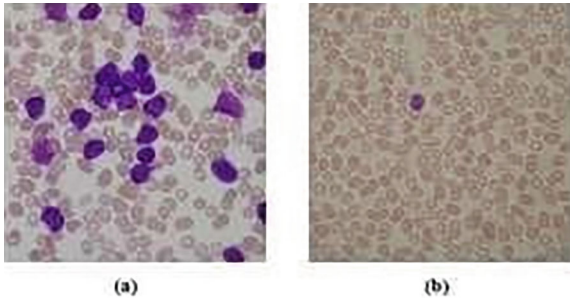


Fig. 3 Proposed system architecture

is present as free on many websites. It is filled with photos of blood cells that have photos of diseased platelets that have been portrayed for the future process.

The images vary in precision with optimization ranging from 300 to 500 μm . Consider the dataset, which contains photos of blood stains. The photos in the data set have been pre-processed to boost classification results. To govern these identical parameters, the library images must be of the proper shape for future classification. Resized image: 512×512 . Filtering the median to reduce noise. The sample images from the dataset are shown in Fig. 4.

Fig. 4 Sample images from ALL-IDB dataset
a ALL-containing sample,
b healthy sample



4.2 Data Preprocessing

The median filter is a nonlinear filtering technique commonly used to remove noise from images, especially impulse noise, also known as "salt-and-pepper" noise. It's particularly effective in scenarios where there are abrupt and sudden variations in pixel intensity, which can be caused by faulty sensors or transmission errors in image acquisition [6, 11, 20].

Here's a breakdown of how the median filter works:

Local Window: The median filter operates by sliding a small window (often a square or rectangular area) across the image. This window defines the local neighborhood of each pixel.

Pixel Replacement: For each pixel in the image, the median filter replaces the pixel's value with the median value of the pixel intensities within the local window.

Median Calculation: The median value is calculated by sorting all the pixel intensities within the local window and selecting the middle value. If the number of pixels in the window is even, the median is taken as the average of the two middle values.

Noise Reduction: The median filter is effective at reducing noise because it doesn't get influenced by extreme values in the same way as an average (mean) filter does. This makes it robust against impulse noise, as the sudden outliers are unlikely to become the median.

Preservation of Edges: Another advantage of the median filter is that it tends to preserve edges and finer details in the image. This is because if a noisy pixel is surrounded by pixels with similar intensity, the median will likely be close to the actual value, whereas an average might be pulled more toward the noisy values.

However, it's worth noting that while the median filter is great for removing impulse noise, it might not perform as well on other types of noise that have more gradual variations, like Gaussian noise. Additionally, using a larger window might lead to blurring of edges and fine details, which can be a trade-off to consider.

In practical applications, the choice of filter size (window size) is crucial. A larger window can effectively remove noise, but it might also smooth out important details. Conversely, a smaller window might preserve details but might not effectively remove noise. The right balance depends on the specifics of the image and the type of noise present. Kernel size for median filtering is detailed, mention the pixel rates with values, then calculate the median level. The median values are averaged to a sum and is applied when the kernel doesn't cover an odd number of pixels. Both edge properties must first be padded with zeroes before median filtering is introduced at the boundary. The accompanying black and white original and damaged (noisy) photos, along with a 3×3 median filter kernel, are examples of original images. 100100100100 \s 100100100100 \s 100100100100. Image distorted by impulse noise: 44: 100255100100, 255100100100, and 100100100100 a median filter kernel of three. Use digital filtering, then contrast the filtered and original images. The steps are given below.

Step 1: The 3×3 kernel needs no padding in the first step. 3/2 equals to top and downward part and one line of zeroes from up to down. It is represented in Fig. 5.

Fig. 5 Step 1 representation

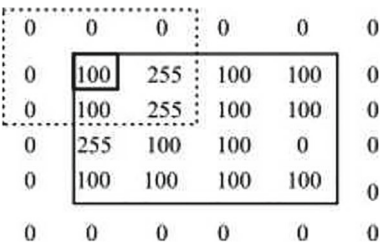
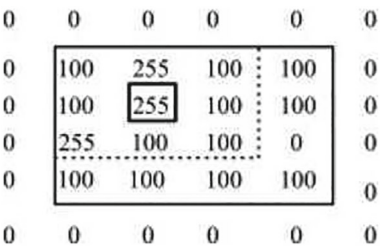


Fig. 6 Step 3 Representation



Step 2: We cover the 3×3 matrix with the center aiming to the first piece to be done in order to process it. The ascended data within the kernel are given as values. The median equals zero (or any other value between 0 and 255 and 255). The number 100 will be replaced by the number 0.

Step 3: Repeat for each one until the final piece is replaced. Consider the component at (1,1). It is shown in Fig. 6.

The kernel values are as follows: 100, 100, 100,100, 100, 100, 255, 255, 255. When expressed as a percentage, the median number is 100. (100, 100, 100, 100, 100, 100, 255,255, 255).

4.3 Segmentation

The unsupervised machine learning segmentation algorithm K-means clustering is widely used. Segmenting a dataset involves dividing it based on the similarity between groups of clusters and data points. Because it is simple, fast, and can handle big datasets, K-means clustering is an ideal solution for this problem. The k-means clustering algorithm continually allocates every cluster from a data point based on the cluster’s average, then recalculates cluster means based on the newly assigned data points. The approach is repeated until the means of the clusters no longer change or until the maximum number of iterations is reached. To use k-means clustering for segmentation, first decide how many clusters you want to employ to divide your dataset into pieces.

Adaptive K-means Clustering: This is a clustering algorithm that differs from the standard k-means approach. The main benefit of this variant is that the count of

clusters does not have to be initialized. Because image data comprises a huge amount of data, the segmentation procedure is continuous, and the number of clusters to be done is determined in advance.

Euclidean Distance Formula: Sum of the squared differences is the square root between distinctive coordinates in Euclidean distance between two points in n -dimensional space.

4.4 Feature Extraction

The statistical method is Gray Level Co-Matrix that is used in image processing and analysis to describe the relationship between adjacent pixels in an image. It is a matrix that contains the frequency of occurrence of pairs of pixel intensities at a given distance and direction within an image. The GLCM is constructed by computing the occurrence of pairs of pixel intensities at a given offset or distance between two pixels, and then counting the number of times each pair occurs within the image. The GLCM is typically symmetrical along the diagonal, as the order of the pixel pairs does not matter. GLCMs can be used to extract a range of statistical features from an image, such as contrast, homogeneity, energy, and entropy. These parameters often used as texture parametric for classification or segmentation tasks, as they capture details of the consistent arrangement of pixel intensities from an image [3, 14, 15].

4.5 Classification

The three components of generative adversarial networks (GANs) are as follows: Learning a generative model, which explains how data is produced in terms of a probabilistic model, is referred to as generative modeling. A model is trained in an adversarial environment. Deep neural networks are a type of network that can be used as artificial intelligence (AI) training methods. A generator and a discriminator are both present in GANs. In an effort to trick the Discriminator, the Generator fabricates data samples (such as an image, audio, etc.). The Discriminator, on the other hand, tries to tell real samples from fraudulent ones. Being neural networks, the Generator and Discriminator compete with one another throughout the training phase.

4.6 Performance Metrics

The metrics used to measure the model are Accuracy, Precision, Recall, and F1 measure. The comparison is shown in Table 1. And the chart is shown in Fig. 7.

After the diseased image is uploaded as an input, the image is preprocessed, segmented, and classified with GAN. The result is published as “Leukemia Detected”

Table 1 Comparison of existing and proposed system

Measure	Proposed system	Existing system
F1 score	94.38	91.77
Precision	95.54	92.83
Recall	96.36	90.47
Accuracy	97.62	98.56

Fig. 7 Comparison of proposed and existing system

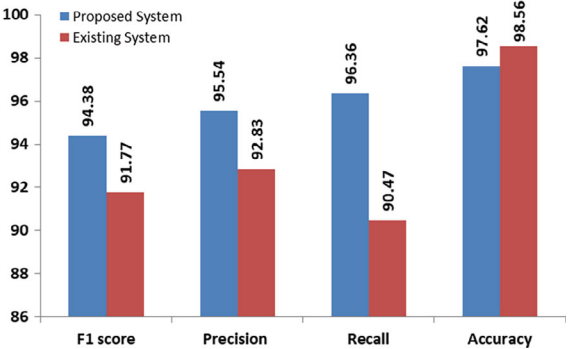
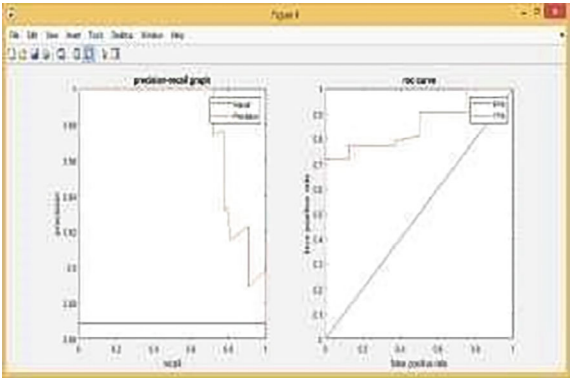


Fig. 8 Result comparison



if a person has leukemia and “Leukemia not Detected” if a person is healthy. Figure 8 shows the system implementation results.

5 Conclusion

This research proposal focuses on using deep learning techniques, specifically a Generative Adversarial Network (GAN) classifier, for the early detection and classification of leukemia types, particularly Acute Lymphoblastic Leukemia (ALL). Early

detection of genetic diseases like leukemia is indeed crucial for effective treatment and improved patient outcomes. Here's a breakdown of the key components and concepts in the proposed work:

Deep Learning and GANs: Deep learning, a subset of machine learning, involves using artificial neural networks to model and solve complex tasks. GANs are a type of deep learning architecture consisting of two neural networks, a generator, and a discriminator, that work in opposition to generate realistic data. GANs have shown remarkable success in generating and processing various types of data, including images.

ALL Classification: Acute Lymphoblastic Leukemia (ALL) is a type of blood cancer that primarily affects lymphocytes. Its early-stage classification can be challenging, and your proposal aims to address this challenge by using deep learning techniques to classify different types of ALL.

Incremental Classification: Incremental classification involves training a model to classify data in a step-by-step manner, typically with increasing levels of complexity. It seems your proposal aims to create an effective incremental classifier for differentiating ALL types, possibly using the GAN model as a part of this process.

Medical Imaging and Cancer Detection: Medical imaging techniques, such as X-rays, MRIs, and microscopy, play a vital role in detecting and diagnosing diseases like cancer. In your case, you're likely working with microscopic images of blood cells to aid in the classification of leukemia types.

GANs in Medical Imaging: GANs have gained attention in medical imaging applications due to their ability to generate realistic images and perform tasks like image enhancement, synthesis, and translation. In this research, GANs might be employed for enhancing the quality of images, augmenting the dataset, or aiding in feature extraction.

Training and Evaluation: This work mentions assessing and differentiating GANs' training and test data. This involves training the GAN model on a subset of data and then evaluating its performance on unseen data to ensure that it generalizes well and can effectively classify different types of ALL.

Significance and Future Implications: If successful, then this research could have significant implications for the early detection and classification of leukemia. Deep learning models, especially GANs, have the potential to learn complex patterns and features from large datasets, which can be invaluable for medical diagnostics. By implementing GAN, dataset with minimum images could also detect abnormalities with high accuracy and it is clearly proved in proposed system.

There are several potential areas for future work in the field of leukemia detection using GAN with MATLAB. Expanding the size and diversity of the dataset is crucial for creating robust and generalizable models. The more diverse and representative the data is, the better your GAN model will be at handling variations and nuances in leukemia images from different sources and populations. Consider collecting data from various hospitals, clinics, and regions to ensure a broader representation of the disease. However, it's important to ensure that the collected data adheres to privacy and ethical standards, such as patient consent and data protection regulations.

As deep learning models become more complex, there is a growing need for explainability to understand how the model is making its predictions. Future work could focus on developing methods for explaining the decisions made by the GAN, such as visualization techniques or decision trees. GANs can be combined with other technologies to improve leukemia detection. For example, integrating GANs with advanced imaging techniques like MRI or CT scans could provide more detailed information about the leukemia cells. Clinical validation: While the accuracy of the GAN can be evaluated using various metrics, clinical validation is needed to demonstrate its real-world usefulness. Future work could focus on conducting clinical trials to evaluate the effectiveness of the GAN in real clinical settings.

The use of GANs for leukemia detection could be especially valuable in low-resource settings, where access to medical experts and advanced imaging technologies is limited. Future work could focus on developing low-cost, portable versions of the GAN that can be easily deployed in these settings.

Finally, integrating the GAN with electronic health records could facilitate easier and faster detection of leukemia, allowing medical professionals to quickly access patient data and make informed decisions about treatment options. Future work could focus on developing efficient and secure methods for integrating the GAN with Electronic Health Records.

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Detection of Major Depressive Disorders in Children, Teens, and Young Adults



Rushil Patra, Ritej Dhamala, Sadhana Tiwari, Ritesh Chandra,
and Sonali Agarwal

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Abstract The aim of this work is to develop a classification model to identify major depressive disorder (MDD) and attention deficit hyperactivity disorder (ADHD) in young teens and adults, which often go undetected. The model has trained and tested several classifiers, such as K-nearest neighbors (KNN), logistic regression, support vector machine (SVM), and a hybrid ensemble model of SVM, KNN, and logistic regression on TD-EEG Brain dataset and the results have been compared with various other existing to evaluate the effectiveness of the proposed model.

Keywords Attention deficit hyperactivity disorder · Major depressive disorder · Electroencephalogram (EEG)

R. Patra · R. Dhamala · S. Tiwari · R. Chandra (✉) · S. Agarwal
Indian Institute of Information Technology, Allahabad, India
e-mail: rsi2022001@iiita.ac.in

R. Patra
e-mail: iib2020005@iiita.ac.in

R. Dhamala
e-mail: iit2020006@iiita.ac.in

S. Tiwari
e-mail: rsi2018507@iiita.ac.in

S. Agarwal
e-mail: sonali@iiita.ac.in

1 Introduction

In India, students are under immense academic pressure, leading to a rise in cases of ADHD and depression, the two most common mental illnesses. The recent COVID pandemic has made it worse, causing early depression in many children. ADHD prevalence is high in children aged 8–15, with a range of 9.40% in male children and 5.20% in female children. However, many go undiagnosed, and leaving ADHD untreated can lead to Dysthymia [1]. Clinical assessments and interviews conducted by experts like psychiatrists and psychologists are used to diagnose depression, but a different strategy is required due to varying symptoms and children's inability to articulate their feelings [2]. Electroencephalography (EEG) data can aid in the diagnosis of these disorders as it provides valuable insights into the functioning of the brain [3][3].

Early detection and treatment of MDD/ADHD can increase academic and occupational performance, lower the risk of substance dependence, improve mental health outcomes, and improve social ties and general life quality [5][5]. Based on brain EEG data, the effectiveness of a number of machine learning models had been examined, such as support vector machines (SVMs), K-Nearest Neighbor, SVM, and a novel approach of combining these models using hybrid ensemble learning.

This experiment uses the TD brain dataset which contains EEG recordings of healthy individuals and individuals with ADHD and MDD and has machine (2 Rushil et al.) learning models to predict the presence of these disorders [7]. The application of machine learning methods is investigated for the identification of MDD and ADHD in children, adolescents, and young adults. The findings of this work show that machine learning algorithms can offer precise and trustworthy diagnostic assessments for these conditions, and they may even be able to enhance patient outcomes in terms of early detection and treatment.

The balance of this work is structured as follows: Sect. 2 describes the different works and strategies utilized in the tests for mental disorder. Section 3 outlines the datasets used in the experiment, and the proposed methods are discussed in Sect. 4. Section 5 examines the findings of the experiments and finally, Sect. 6 summarizes findings and future research for further improvement.

2 Related Work

The classification and identification of MDD and ADHD in young adolescents and adults is a critical area of research in the field of mental health [8]. Various machine learning techniques, including feature extraction utilizing statistical methods and classifiers like KNN, logistic regression, SVM, and hybrid ensemble models, were used to do the experiment [9]. A paper suggested the use of machine learning algorithms to predict ADHD in children using a dataset of 157 children, out of which 77

had ADHD and 80 were healthy. The study evaluates the performance of three algorithms—Naïve Bayes, KNN, and logistic regression. The KNN algorithm performs the best, with an accuracy of 86%, followed by logistic regression at 66%, and Naïve Bayes at 52%. The KNN algorithm is able to predict ADHD with an accuracy of 89%. The study highlights the potential of machine learning in predicting ADHD and identifies KNN as the best performing algorithm for this task [10]. Another study presents a machine learning-based approach for detecting and classifying children with ADHD with the help of EEG signals. The study uses two approaches to find optimal electrode channels and important features, and a hybrid channel selection method to combine these approaches. Six machine learning classifiers are then applied to detect ADHD in 121 children aged 7–12 years, using only six channels and twenty-eight features. The Gaussian process-based classifier was able to improve in regard to other models with an accuracy rate of 97.53% and an AUC of 0.999. The suggested approach can assist doctors and medical professionals to identify children with ADHD early and give the necessary medical care and therapy [11].

The intent of a research work was to create a deep learning (DL) model for categorizing people with major depressive disorder (MDD) and healthy controls in resting-state using Electroencephalography (EEG) data. The proposed model retrieved higher-level features using a parallel two-branch convolutional neural network categorized using a fully connected layer and learned connection linkages across EEG channels using a multi-head self-attention mechanism. On a publicly accessible EEG dataset, the model outperformed in comparative approaches, with an average classification accuracy of 91.06%. These findings show that the proposed DL model provides a feasible technique for MDD identification using brain connection modeling [12]. Some other researches proposed methods to identify major depressive disorder by selecting discriminative features via stochastic search. The prime goal is to create an objective biomarker-based technique for diagnosing major depressive disorder (MDD) utilizing EEG data. To identify discriminative characteristics from specific EEG channels, the suggested method used a stochastic search algorithm. This approach is also tested using a public EEG-based MDD dataset that includes 24 MDD patients and 29 healthy controls. Cross-validation is done using the leave-one-subject-out strategy. With an average accuracy of 99.53% in the fear-neutral face pairs experiment and 99.32% when at rest, the recommended technique surpassed cutting-edge MDD recognition algorithms. The study also discovered that negative emotional inputs could cause depressive moods and that high-frequency EEG patterns can distinguish between normal and depressive patients [13].

Statistical feature extraction methods and classifiers are applied such as KNN, logistic regression, SVM, and a hybrid ensemble of KNN, logistic regression, and SVM to classify MDD and ADHD. Different statistical features such as mean, standard deviation, min, max, skewness, etc., are applied with hybrid ensemble learning to classify EEG data is the novel approach which is proposed in this work. It is analyzed after reviewing many papers that combining the best-tuned parameter of stand-alone learning methods into a hybrid ensemble learning will effectively enhance the classification accuracy of the EEG data, while statistical features will help the model to learn hidden characteristics of the used data. The findings are

compared with other studies, such as Patel et al. and Shi et al. Overall, the works reviewed in this literature review demonstrate that machine learning techniques can effectively classify and identify MDD and ADHD using various neuroimaging and EEG data. These results highlight the potential of machine learning techniques to aid in early detection and accurate diagnosis of mental health disorders, leading to better outcomes for individuals and society as a whole.

3 Dataset Description

Most of the studies seeking interest in brain activity and neuroimaging during healthy development used the TD-EEG Brain Dataset [14]. It is composed of EEG data obtained from typically developing children and young adults, and it contains information from 12 different brain regions recorded both at rest and during task-related situations, including auditory and visual stimuli, motor tasks, and cognitive tasks. The gathering process includes EEG data from about 150 subjects, ages 6 to 18, along with complete demographic data, results from cognitive tests, and behavioral measurements for each participant. This makes it possible for researchers to investigate the connections between brain activity, behavior, and cognition and to learn more about how the brain typically develops. This dataset's availability has facilitated collaborative research projects and has been applied to numerous studies on the development of the brain. These studies examine functional connectivity, neural oscillations, and the impact of sensory information processing and cognitive effects on brain activity. The use of the TD-EEG brain dataset in research has yielded vital insights into the development of brain function typically in developing children and young people. For example, studies using this dataset have revealed differences in brain activity between children and adults during cognitive activities and neural markers of attention and cognitive processing. These findings have crucial implications for understanding brain development and devising therapies to improve cognitive growth in children and young people.

4 Methodology

This section discusses about the complete methodology which includes electrode filtering, data pre-processing, feature extraction, number of epochs and overlaps used, and model development.

4.1 *Electrode Filtering*

The used data contained recordings from 26 electrodes or channels for each patient. To optimize the proposed prediction model, two different approaches have been employed. The first approach uses the recordings from all 26 electrodes, while the second approach involves selecting only the relevant electrode recordings. The frontal lobes of the brain are responsible for many important cognitive processes, including attention, memory, decision-making, and emotion regulation. Persons suffering from ADHD and MDD frequently experience disruptions during these processes. According to one study, people with ADHD have lower activity in the prefrontal cortex, a region in the frontal lobes linked to impulsivity, executive function, and attention [15]. Similarly, people with MDD generally have lower prefrontal cortex activity, which plays a role in controlling emotions and cognitive tasks. ADHD and MDD are both linked to prefrontal cortex dysfunction, which is a frontal lobe component. Therefore, EEG recordings from electrodes placed on the frontal lobe can provide important information on the neural activity in this region and can potentially aid in the diagnosis of these disorders. Additionally, EEG recordings from the prefrontal cortex can be used to monitor the effects of treatment on brain activity in individuals having ADHD or MDD. Hence the filtered-out electrodes are—‘Fp1’, ‘Fp2’, ‘F3’, ‘F4’, ‘F7’, ‘F8’, ‘Fz’.

4.2 *Data Preprocessing*

The data in CSV files is imported and processed using Pandas, and then transformed into a MNE Raw Array object to enable further processing. As data collected over time in neuroimaging and electrophysiology is difficult to analyze as a continuous signal, it is segmented into short, contiguous segments of data called epochs.

4.3 *Epochs and Overlaps*

Epochs are typically centered around specific events or stimuli and help in breaking down the continuous EEG data into smaller, more manageable segments. This allows for easier identification of patterns or anomalies, and the use of fixed-length epochs ensures that data is consistent across different subjects and recording sessions, facilitating the comparison of data. By dividing the continuous signal into epochs, it becomes simpler to perform analysis on specific events or features of the data. Overlapping the epochs decreases the effect of edge artifacts. These artifacts may arise from sudden changes in the signal at the beginning or end of an epoch. Overlapping the epochs can decrease the impact of these artifacts, which may enhance the accuracy of this analysis. Furthermore, overlapping epochs can expand the dataset size

without requiring additional recordings. The use of overlaps has numerous benefits, such as minimizing data loss and increasing the number of samples. By incorporating data from the prior epoch into the current epoch, overlaps can help prevent important information from being lost. Additionally, more data can enhance the accuracy of machine learning models. It is crucial to choose the appropriate epoch duration and overlap since it impacts the quality of the features extracted and, consequently, the performance of the classification algorithm. A duration that is too short may not capture enough information about the underlying brain activity, while a duration that is too long may include various neural processes and result in a loss of specificity. Similarly, an overlap that is too small may cause the epochs to miss critical information, while an overlap that is too large may generate data points that are too alike and do not provide enough independent information for accurate classification.

4.4 Feature Extraction

A plethora of time-series and morphologically based features are applied for the development of the proposed model. These features are extracted by the analysis of the shape/form of the signals by performing a transformation on a particular signal and extract relevant measurements on that transformed signal which is used to work as the extracted features. The features extracted from the TD brain dataset include mean, Standard Deviation, Peak to Peak (pk-pk), Variance, Minimum Amplitude, Maximum Amplitude, Minimum Amplitude for a given argument (Argmin), Maximum Amplitude for a given argument (Argmax), Absolute difference between signals, Skewness, and Kurtosis. The observed values of five different patients across these extracted features are shown in Table 1.

4.5 Model Development

This section describes the model building for mental disorder classification using various classifiers. The stepwise proper workflow of the proposed model is represented in Fig. 1 (6 Rushil et al.).

KNN: The KNN technique places an input signal in the category, that is most common among its k-nearest neighbors after finding the input signal's k-nearest neighbors retrieved feature vectors have been subjected to the KNN algorithm [16]. To get the highest classification performance, k hyperparameters are tweaked like ide, distance measure, and regularization. For achieving better generalization, a higher value of K is used, but the algorithm's computing cost may also rise.

Logistic Regression: The logistic regression approach uses a sigmoid function to represent the connection between the collected characteristics and the binary output variable (class label). The projected probabilities are thresholded at 0.5, which indicates the likelihood that the EEG signal belongs to the healthy class (0) or the patient

Table 1 Extracted feature values of 5 different patients

Patient number	1	2	3	4	5
Mean	Min: - 5304.17 Max:13,466.20	Min: - 7170.01 Max: 963.66	Min: - 7198.98 Max:996.48	Min: - 7211.66 Max: 1022 .98	Min: - 7193.25 Max: 1053.59
Standard deviation	Min: 16.35 Max: 29.64	Min: 8.27 Max: 30.82	Min: 7.31 Max: 23.57	Min: 8.41 Max: 23.60	Min: 8.69 Max: 22.79
Peak to peak (pk-pk)	Min: 112.55 Max: 172.92	Min: 54.52 Max: 130.25	Min: 59.85 Max: 128.18	Min:62.24 Max:140.04	Min: 59.17 Max: 138.47
Variance	Min: 267.34 Max: 878.81	Min: 68.41 Max: 950.14	Min: 53.37 Max: 555.33	Mix: 70.79 Max: 557.08	Min: 75.61 Max: 519.75
Min amplitude	Min: - 5360.21 Max: 13,407.22	Min: - 7237.32 Max: 921.17	Min: - 7247.74 Max: 948. 59	Min: - 7264.30 Max: 970.24	Min: - 7253.19 Max: 993.33
Max amplitude	Min: - 5224.94 Max: 13,549.70	Min: - 7115.23 Max: 1008.63	Min: - 7140.63 Max: 1046.84	Min: - 715 7.13 Max: 1081.79	Min: - 7133.42 Max: 599.18
Argmin	Min:3939 Max:7923	Min: 56 Max: 5475	Min: 335 Max: 7943	Min: 1490 Max: 7897	Min: 221 Max: 7365
Argmax	Min: 2 Max: 6937	Min: 53 Max: 7931	Min:78 Max: 7336	Min: 236 Max: 7732	Min: 1134 Max: 7939
Abs Diff	Min: 23,221.46 Max: 32,338.53	Min: 16,968.37 Max: 20,565.86	Min: 17,656.65 Max: 23,640.83	Min: 19,375.39 Max: 23,531.04	Min: 19,459.47 Max: 22,701.90
Skewness	Min: 0.10 Max: 0.76	Min: - 0.53 Max: 0.47	Min: - 0.46 Max: 0.44	Min: - 0.35 Max: 0.28	Min: - 0.26 Max: 0.78
Kurtosis	Min: - 0.78 Max: 0.42	Min: - 0.73 Max:0.38	Min: - 0.89 Max: 0.41	Min: - 0.82 Max: 0.41	Min: - 0.86 Max: 0.82

class (1). This allows the sigmoid function to transfer the output of the linear function to a value between 0 and 1.

SVM: SVM produces a hyperplane to divide the feature vectors by maximizing the distance between the hyperplane and the support vectors, or nearest neighbor feature vectors, of each class. The margin is the separation between the closest support vectors from each class and the hyperplane. By minimizing the margin violation, which is the total of the distances between the incorrectly classified data points

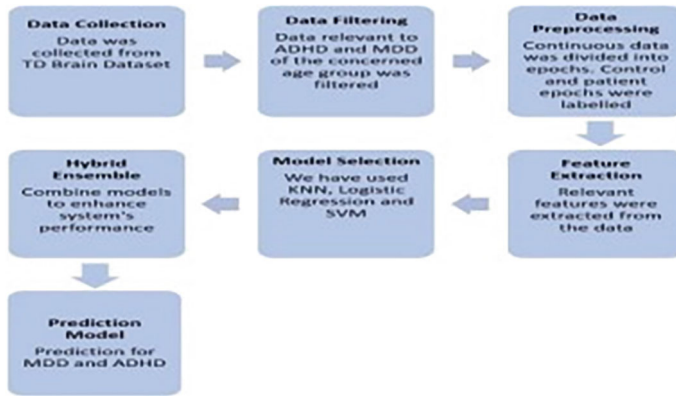


Fig. 1 Process flow diagram of the proposed prediction model

and the hyperplane, the SVM algorithm discovers the ideal hyperplane. SVMs can use kernel functions to change the feature space into a higher-dimensional space where the data can be linearly separable in situations where the data is not linearly separable. In order to select the top kernel function from a variety of kernel functions to be employed in this classification, the gridsearch approach is used.

Hybrid Ensemble Learning: Hybrid ensemble learning is a method that mixes various machine learning models to enhance the system's overall performance. Using this method, the same dataset is used to train numerous classifiers, and the results are pooled to get the final prediction. The same dataset is used to train SVM, KNN, and logistic regression classifiers, and combine the results via stacking and voting.

Voting Classifier: Voting classifier combines the final prediction with the predictions of various independent classifiers. Hard voting and soft voting are two examples of the various voting classifications. In a hard vote, the outcome is determined by the classifiers who received the most votes overall. The final prediction in soft voting is based on the mean of the anticipated probabilities from each classifier. After testing both approaches, hard voting is chosen, since it consistently outperformed the others in all of the tests.

Stacking Classifier: Stacking classifier is another hybrid ensemble learning method that integrates the results of various separate classifiers using a meta-classifier. By training a number of base classifiers on the same dataset using several feature sets and hyperparameters, it is possible to aggregate the predictions of various algorithms and feature sets. A meta-classifier is trained using (8 Rushil et al.) the predictions of these base classifiers as input features which makes the final prediction. The meta-classifier can enhance the overall performance of the classification system by capturing the intricate connections between the base classifiers.

4.6 Evaluation Criteria

Four different evaluation methods are chosen to check how the different models hold up against each other. These criteria comprise of Precision—Represents how many positive identifications were made from all the predicted positives. Recall Represents how many true positives were identified from all the positive identifications. Accuracy—It is the measure of all correctly identified data F1 Score. It can be viewed as the harmonic mean of precision and recall. It results in a stronger measure of the incorrectly classified cases compared to the accuracy metric.

5 Results

Among all the models the hybrid ensemble model performs the best in all experiments. Incorporating all 26 channels from all regions of the head instead of taking only 7, the frontal lobe, showed significant improvement in the detection of both diseases. Readings taken with eyes open were found to give slightly more accuracy than with eyes closed ones. Considering single models KNN and SVM had comparable results. KNN performed better than SVM in ADHD 7 channel experiment with + 1.6% increase in accuracy. In another case, SVM turned better with slight margins is better. Logistic regression is near the other two models. Table 1 gives full comparison of all the models applied to experiment specific filtered data (Tables 2, 3).

Comparative analysis of obtained results in Eye Closed state from Figs. 2 and 3, it can be concluded that, after comparison, on average, the stacking classifier and voting classifier, which are the proposed classifiers to be used, outperform the other classifiers. By incorporating all 26 channels from various regions of the head instead

Table 2 Comparison of model with different data

Models				Experiments				
	Eyes Closed			Eyes open				
	M D D C H 7	M D D CH 26	AD HD CH 7	AD HD CH 26	M D D C H 7	M D D CH 26	AD HD CH 7	AD HD CH 26
KNN	92. 7	95.6	90.4	96.8	93. 7	97.1	93.1	96.3
Logistic Regression	77. 2	91.1	77.5	86.2	80. 1	94.0	72.9	87.6
SVM	92. 7	94.9	89.0	95.9	94. 0	98.1	95.9	96.3
Voting Classifier	91. 8	95.9	90.4	95.0	93. 0	97.6	94.5	97.2
Stacking Classifier	94. 0	96.2	94.0	96.8	94. 9	98.3	95.0	96.8

Table 3 Comparison with existing related works

Paper	Models used	Accuracy
[11]	K-nearest neighbor	86%
	Naïve Bayes	52%
	Logistic regression	66%
[2]	Gradient Boosting	95.2%
	Decision tree	93.3%
	Logistic regression	91.9%
[11]	Gaussian process classification	97.3%
	Random forest	75.17%
[12]	CNN-LSTM	88.04%
	Coherence	98.13%
	DeprNet	96.26%

of solely focusing on the limited 7 channels from the frontal lobe, an increase in the detection of both diseases was observed. This extended approach allowed for a comprehensive analysis of the electrical activity across the entire brain, providing a more comprehensive (10 Rushil et al.) representation of brain dynamics during the experiments. By considering a broader range of channels, including those from different brain regions, valuable insights into the underlying patterns and interactions within the brain have been gained, which enabled the stacking classifier and voting classifier to outperform the other classifiers.

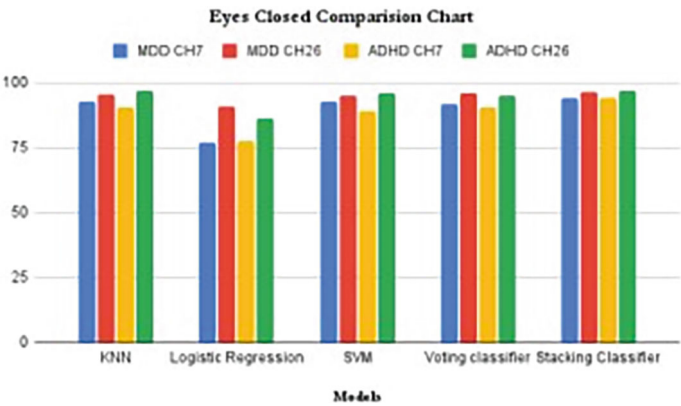


Fig. 2 Comparative analysis of obtained results in eye closed state

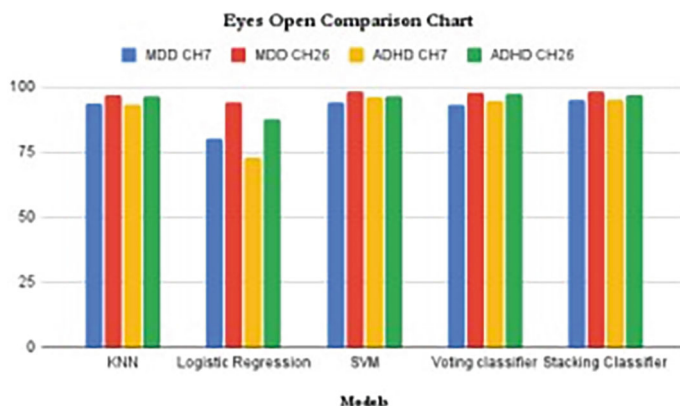


Fig. 3 Comparative analysis of obtained results in eye open state

6 Conclusion and Future Work

Detecting MDD and ADHD in children, teens, and young adults is crucial for effective treatment and management of these conditions. The symptoms of MDD and ADHD can vary widely between individuals, but some common signs include changes in mood, behavior, and attention span. Early detection is important to prevent long-term negative consequences, such as poor academic and social performance, substance abuse, and increased risk of suicide. A comprehensive evaluation by a qualified mental health professional is essential to accurately diagnose MDD and ADHD, as well as rule out other possible conditions that may present with similar symptoms. It is important to note that developing an effective classification model for mental health disorders such as MDD and ADHD requires a large and diverse dataset, rigorous testing, and validation. Therefore, it is essential to ensure that the model's accuracy and effectiveness are rigorously evaluated before it is implemented in real-world clinical settings. Other existing EEG classification mechanisms have also observed slightly better performance than the proposed model [8]. Upon inspection it is found that the non-linear feature used by this work is the reason their classifier was able to perform better. This means if non-linear features can be extracted and incorporated into the proposed model, the performance of this model can be improved in the future.

Addressing the systemic barriers and stigmas that prevent individuals from seeking help and receiving proper treatment should be focused on in future work. This can involve improving access to mental health services, reducing the cost of treatment, and increasing awareness and education about MDD and ADHD in schools, workplaces, and communities. Overall, continuing the work on the detection and treatment of MDD and ADHD in children, teens, and young adults is critical for improving the mental health outcomes and quality of life for individuals affected by these conditions.

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Ensuring Confidentiality and Integrity of Electronic Health Records in Blockchain Through Multi-Attribute Access Control



Smita Mahapatro and Syed Muzamil Basha

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Abstract The digitalization of traditional healthcare data, such as Electronic Health Records (EHR), presents significant challenges for healthcare organizations. Sharing and storing EHRs can be difficult, and accessing the information can be time consuming and sometimes inaccurate due to limited access. To address these issues, this article suggests a permission-based blockchain-based system for sharing healthcare data. This system uses multiple attribute encryption based on cipher text to protect data privacy and authenticate healthcare information. Additionally, it integrates blockchain technology (also known as Distributed Ledger Technology) with an RSA cryptosystem for added security. The proposed system employs a multi-attribute access control mechanism along with an RSA cryptosystem to achieve strong data security. The ultimate goal of this technology is to create a single platform that stores and distributes health information using a Distributed Ledger-based approach. This eliminates the need for printed healthcare reports, streamlining the entire medical business onto one secure platform.

Keywords Electronic health records (EHR) · Blockchain · RSA cryptosystem

S. Mahapatro (✉) · S. M. Basha
Department of Computer Science and Engineering, REVA University, Bengaluru 560064, India
e-mail: Smita.mahapatro@gmail.com

S. M. Basha
e-mail: muzamilbasha.s@reva.edu.in

1 Introduction

Electronic Health Record (EHR) frameworks face significant risks due to centralized storage of clinical data, including cyber attacks and data breaches. EHR systems are mainly designed for clinical institutions, with patients having limited access and control over their medical records. Furthermore, patient records are distributed across multiple clinical institutions, making it difficult to share electronic medical data and establish comprehensive lifelong health records. Some recent EHR reports reveal that IT engineers in EHR systems have slowed down data transfer by overcharging. In the rapidly evolving medical sector, EHRs collect patient information and health metrics, including medical images, diagnoses, and doctor's details. For the study of patient health records, EHRs are widely employed in healthcare and by outside medical industries like insurance firms. The medical industry does, however, have issues with access control, privacy protection, and secure storage. It is crucial to share health data records in order to develop a smart medical system & improve consumer satisfaction. This work aims to provide privacy and security in EHRs by designing a digital health care network that uses blockchain technology to store patient information in the cloud. Anywhere on the Worldwide Wide Web, data saved in the cloud can be shared using a variety of properties. Such improvements in EHR systems can enhance good management for better healthcare, reduce medical errors, provide safe care, and improve patient-doctor communication quality, providing many advantages for patients and doctors. The purpose of this endeavor is to secure and protect electronic health records (EHR). Here are our main contributions: To store patient data, and create an electronic health care system. Blockchain-based cloud storage is used to store data. With the aid of numerous qualities, these data are shared anytime on the Internet thanks to the cloud. The management of the electronic medical record system will benefit from these kinds of advances, leading to better healthcare, reducing medical errors, and providing safe care while also improving the quality of communication between patients and doctors and offering numerous benefits to both parties.

The following is how the paper is set up: A summary of the literature research and techniques are given in Sects. 2 and 3, respectively. The suggested system is described in detail in Sect. 4; the results and experimental setup are covered in Sect. 5. Finally, Sect. 6 concludes the paper with our findings and conclusions.

2 Literature Review

Electronic Health Records (EHR) consists of documented events, medicinal treatments for illnesses, and advancements in medical care. EHRs also contain personal and sensitive clinical information that needs to be distributed and safeguarded. Due to the decentralized nature and resistance to alterations, blockchain technology can potentially address these concerns. Our proposal involves creating a secure medical

data exchange based on a digital ledger system for hospitals. This strategy can immediately achieve features such as alteration resistance, decentralization, and transparency. To ensure the security of patient information, a robust module has been developed for clinicians to access and save clinical data [1, 4]. In order to address the low search efficiency and potential privacy breaches associated with cloud servers, a quality-based available encryption scheme is proposed. This is achieved by utilizing digital ledger technology and distributed cache to store available encryption records and encrypted digital health records separately. Additionally, keys are encrypted before being sent over an unrestricted channel. According to a performance analysis, the quantity of features has no bearing on encryption or search times. Performing a multi-keyword cryptographic hash analysis based on the premise of verifiable quality is the next stage, which is authenticated using a hidden access mechanism [2]. This article presents a smart plan for sharing electronic health records (EHRs) that leverages mobile cloud computing and blockchain technology. The paper identifies the key challenges faced by current digital healthcare record sharing systems and proposes practical solutions based on real-world module performance. The focus of this paper is to design a reliable access control system that utilizes a solitary smart contract to control user access and make sure that EHRs are shared effectively and securely [3]. Our framework ensures that our digital health record system complies with the data protection guidelines provided by the Health Level 7 records. This entails adhering to established principles and techniques that effectively manage access to and secure digital records data. To achieve this, employ various methodologies such as Attribute-based encryption, utilizing the CPABE library, and Semantic Web developments like OWL and SWRL. Our system is built using the Amazon Web services platform and Python programming language [4]. The recent focus has been on data sharing driven by understanding, where the exchange of clinical data is limited intervention and patient driven. In order to utilize Smart Contracts enabled by EHRs as data intermediates, our solution entails putting in place a sizable data infrastructure. The EHR will be more widely accessible thanks to blockchain technology's decentralized nature. Blockchain technology has the potential to significantly enhance the healthcare sector by enabling speedier payments, anonymity, and permanent, dependable clinical records [5]. By introducing blockchain technology, healthcare professionals facilitated in maintaining patient records. Our proposed solution involves creating immutable digital ledger entries for every modification made to a record, allowing for the tracking of changes over time. To provide a practical solution for this process, Merkel tree algorithms for data storage are utilized [6]. This paper presents cross-chain technology in blockchain as a significant technical approach to achieve interconnection and value exchange between chains. It provides a comprehensive summary of the existing standard cross-chain technology in blockchain examines the advantages of several cross-chain technologies, discusses the difficulties they encounter, and forecasts their potential for further growth [7]. By integrating these technologies, the architecture of blockchain can be enhanced and boost the potential of artificial intelligence to capture the existing system toward a more innovative and meaningful healthcare phase. At the same time, combining

blockchain by recording all the data kept on the ledger and so establishing new standards in the system, AI will make AI more transparent and understandable. [8]. This paper suggests and examines the application of blockchain technology which offers increased security, traceability for supply chains, decentralization, and immutability. Our research focuses on a smart Ethereum contract that supports a panel model for the supply chain food system, which prevents data manipulation, database corruption, and external attacks. Our experimental results [9] demonstrate the efficacy of our proposed approach in analyzing food supply systems. The exchange of information plays a vital role in all professions, including market trends and best practices. However, third-party interference in this aspect can lead to the spread of misinformation. To address this issue, blockchain technology, which provides a reliable and ethical data ledger, can be utilized. In this paper, the possible integrations of blockchain technology are investigated as a secure and open transaction system into an agricultural supply chain [10].

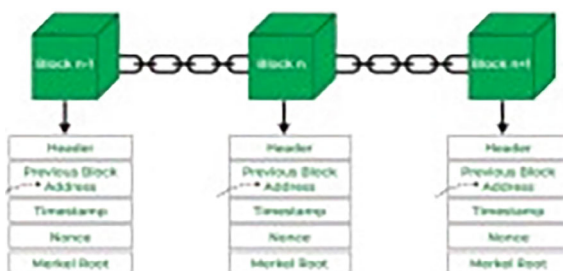
In [11], the authors addressed challenges in electronic health records (EHR), focusing on power consumption, failure, and vulnerability to attacks. They proposed utilizing blockchain technology in trustless environments to overcome these challenges and enable secure data exchange. Two platforms, permissionless blockchain (Ethereum) and permissioned blockchain (Hyperledger), were introduced as potential solutions. Privacy and security concerns were discussed, comparing traditional EHR methods with blockchain implementation. Limitations of these approaches were acknowledged.

In [12], the authors explored how blockchain, along with smart contract systems, can support healthcare applications for physicians, patients, insurance companies, and various assets.[13] They emphasized the benefits of blockchain in healthcare, proposing a protocol for patient privacy and data confidentiality. Secure encryption methods and digital signatures were introduced. The study stressed the importance of accurate EHR data.[14] discussed the role of blockchain in healthcare, addressing challenges of system security, interoperability, data sharing, and mobility in the EHR field. Multiple platforms suitable for blockchain implementation were introduced. A systematic review in [15] analyzed 42 papers on applying blockchain in healthcare, examining challenges and the sharing/processing of medical data and patient records. Implementation models, limitations, and costs were explored.

3 Methodology

3.1 A. Blockchain

Blockchain is a data storage system that makes it difficult or impossible for the network to be altered, hacked, or misused. Blockchain is a type of distributed ledger that distributes and copies transactions between a network of related computers. Blockchain technology is an organizational structure that stores public transaction

Fig. 1 Blockchain structure

information, also known as a “block,” in several network locations, also known as a “chain,” a network connected by peer nodes. This type of storage is often called a “digital ledger.” Every operation in this directory is verified and protected against fraud by the digital signature of its owner, which helps to ensure the transaction. As a result, the data in the digital directory is very secure. It serves as a duplicated, dispersed digital ledger of transactions for the entire network of computer systems. Each new transaction that occurs in the ledger is added to the digital log of each participant, and each block in the chain contains transactions that several. A ledger known as Distributed Ledger Technology (DLT) is controlled by multiple users. A hash, an immutable cryptographic signature, is used to record transactions in the DLT blockchain (Fig. 1).

3.2 B. Attribute-Based Access Control (ABAC)

ABAC is an authentication paradigm which determines access based on attributes instead of roles. Its goal would be to safeguard items like information, connected equipment, and information technology services against illegal access and activities which will not meet the requirements of a company’s data protection.

ABAC, which emerged from basic access control records and RBAC which is role oriented, has become a popular kind of rational authentication mechanism in the last few years. The Federal Chief Information Officers Council supported ABAC in 2011 as an endeavor to help public authorities to enhance their authorization infrastructures where it was even suggested that enterprises use the ABAC approach to securely exchange data.

3.3 C. RSA Cryptosystem

Public-key encryption algorithms include the RSA encryption algorithm. Let’s first examine the concept of public-key encryption to better comprehend RSA. The concept of this cryptosystem is based on the fact that large integers are hard to

factor. The public key consists of two numbers, one of which is the product of two large prime numbers. The same two prime numbers are also used to generate the private key. So if one can multiply a large integer, the private key is revealed. Consequently, the size of the key determines the absolute strength of the encryption, and doubling or tripling the key significantly increases the strength of the encryption. RSA keys can be 1024 or 2048 bits long, although experts believe that a key with only 1024 bits can be broken in a short period of time. But so far, it seems like an impossible feat.

Public-Key Encryption Algorithm:

An asymmetric algorithm is another name for public-key encryption. Asymmetric algorithms are those in which the encryption and decryption keys used by the sender and receiver are different. A set of keys is assigned to each sender: Private key, Public key. Encryption is done using the public key, while decryption is done with the private key. The public key cannot be used for deletion. Although the two keys are linked, it is not possible to derive the private key from the public key. While the private key is kept.

4 Proposed System

Blockchain scalability provides a solution for preventing data breaches resulting from cyber intrusions, eliminating the need for online storage server clusters. In a blockchain network, the processing servers build a blockchain where every block contains the cryptographic hash record of the prior block's data, making it challenging to manipulate. A variety of blockchains, including ones able to carry out smart contracts, emerged as a result of Bitcoin's success. A smart contract is a tool that may be added to a ledger and used by one or more nodes to carry out an action. In permissionless blockchains, users cannot execute programs on the system with certainty due to high extraction costs, but permission blockchains can extract or generate logs. Therefore, individuals can establish restricted or permissioned blockchains that can be controlled and regulated by a select group of people, enhancing the ability to manage privacy and change extraction charges. Private key is known only to the owner of the key, the public key is known to everyone. It implies that anyone can send a message to the user using the user's public key. However, a conversation can only be deleted by the user using his private key.

Figures 2 and 3 depict the encryption and decryption processes used by the RSA cryptosystem, respectively. It uses RSA sequences as the keys for encryption and decryption it is an asymmetric cryptosystem (Fig. 4).

The proposed system has four actors.

1. **Data Owner**
2. **Data User**
3. **Auditor**

Fig. 2 RSA encryption / decryption process

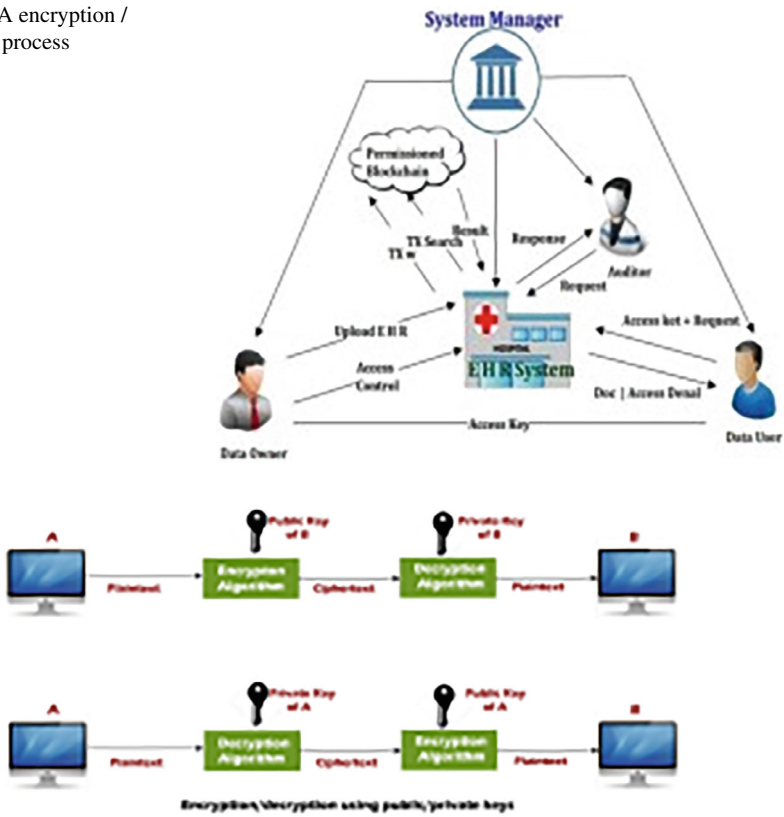


Fig. 3 RSA decryption process



Fig. 4 Proposed EHR system architecture

4. Admin

The primary user whose privacy must be protected by this EHR System is the data owner. For the construction of the blocks below, Algorithm 1 is employed when the data owner uploads the Electronic Health Records that must be kept in the blockchain storage.

Algorithm 1: Block Creation Process

Input: Electronic Health Record

Output: Blockchain Block

User: Data Owner

1 If N Transactions are added in upload queue

2Generate the Root-Hash-code from data

3Fetch Previous Block Hash code (PBH)

4Form Header PBH-RH-Timestamp-Nonce

5Encrypt data and create Block Body

6Merge Header and Body to create block

7else

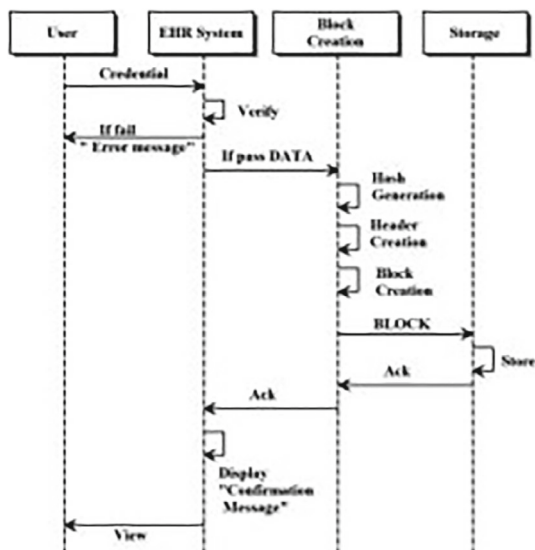
8wait and check

Below sequence diagram shows the block creation process (Fig. 5).

Data users are created by the data owner, who will also assign their attributes and provide the data users with the decryption key. Once the electronic health records have been uploaded, the data owner must set the access authorization.

The auditor is in charge of examining the integrity of each file in the blockchain. Following algorithm two is used for verification.

Fig. 5 Block creation process



Input: Block ID

Output: Status of the Block

User: Auditor

1. **if** number of Block (N) > 0 then
2. List the Block details
3. Fetch block to be tested & Extract data 4
4. From the transaction generate Hash Code(1) 5
5. Fetch Previous Block Hash Code
6. Compare both Hash Code and show result 7**else**
7. Display no block the verify

Below sequence diagram clearly shows the block verification process.

Fig. 6 Block verification process

Algorithm 2: Block Verification Process

Input: Block ID

Output: Status of the Block

User: Auditor

1. **if** number of Block (N) > 0 then
2. List the Block details
3. Fetch block to be tested & Extract data 4
4. From the transaction generate Hash Code(1) 5
5. Fetch Previous Block Hash Code
6. Compare both Hash Code and show result 7**else**
7. Display no block the verify

Below sequence diagram clearly shows the block verification process (Fig. 6).

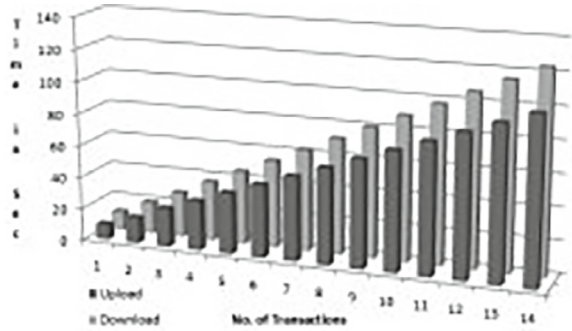
A super user or a user known as an Administrator is responsible for maintaining the settings of the blockchain database. He will also create data owners, each of whom will have their own encryption and decryption keys.

5 Result and Discussion

5.1 A. Experimental Setup

After completing the implementation, the Hybrid cloud approach was tested for this work. A platform was set up on a private server, and the EHR system was deployed above it. The server was then connected to client systems, allowing users to upload files to the private server. The blockchain creation process takes place in the private

Fig. 7 Time latency - file uploading vs downloading



server, and once a block is created, it is moved to the storage space of the public servers. When a user downloads a file, the request is sent to the private server, which fetches the corresponding block from the public storage server, extracts the data, and delivers it to the user. The private server was equipped with a 16 GB RAM and Intel Core i7 processor, while three client systems with Intel Core i5 processors and 8 GB RAM were used.

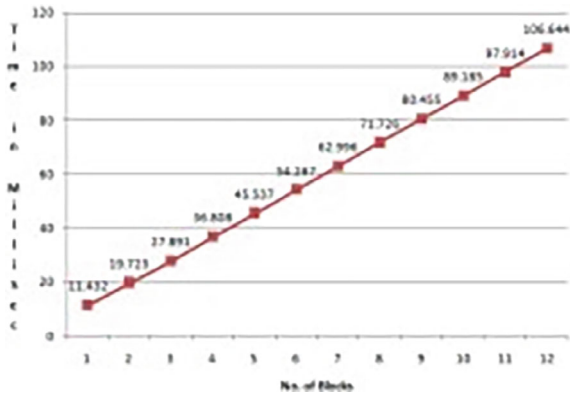
5.2 B. Data File and Time Comparison

After the hybrid cloud setup was completed, various data files were uploaded from many client systems to the blockchain storage. Similarly, updated files were downloaded to the three client systems. Figure 7 shows the time latency graph, which illustrates the time taken for both the file upload and download processes.

5.3 C. Data File Integrity Checking

The auditor has a responsibility to verify the integrity of data files that are uploaded and stored on the blockchain. When a request for verification is received, the auditor must initiate the integrity check system. This process involves generating a hash code from the relevant data file and extracting the previous block's hash code from the immediate block in the blockchain. The two hash codes are then compared to determine the integrity status of the data file. Figure 8 below displays the time taken to perform the integrity check, with the Y-axis showing the amount of time spent and the X-axis showing the number of checks.

Fig. 8 File verification time graph

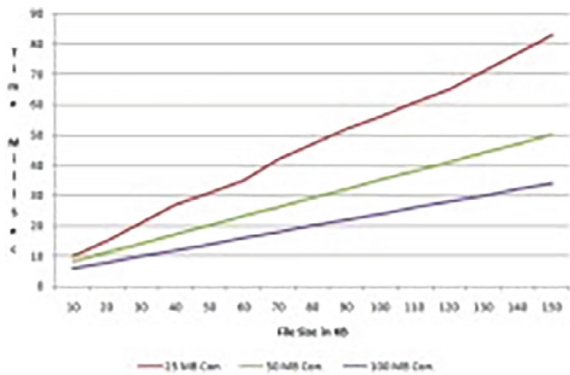


5.4 D. Time Latency on Various Connections

Below graph in Fig. 9 shows the download time latency on three internet speed connections 25 MB, 50 MB, and 100 MB. The average time taken for 150 KB file to download will be between 30 and 80 ms. So proposed system performs well on various internet speeds.

The graph displayed in Fig. 9 illustrates the time it takes to download files from 10 to 150 KB file on three different internet speeds: 25 MB, 50 MB, and 100 MB. On average, the download time ranges from 30 to 80 ms to download 150 KB data file. The results indicate that the proposed system performs effectively across a range of internet speeds.

Fig. 9 Download time latency on various connections



6 Conclusion

The present research involves the creation and evaluation of Health Chain, an EHR consortium blockchain that functions under an administrative model. This system ensures the availability and reliability of information and offers chain code APIs to meet the needs of different users while implementing fine-grained access control. Additionally, the study introduces a method to anonymize EHRs. Health Chain utilizes the Hashing algorithm as its RSA algorithm. The system's effectiveness was observed during testing, with its performance described in terms of latency and throughput.

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Heart Failure Risk Assessment with Quantum-Inspired Support Vector Classification: An Empirical Study



Sheral Waskar, Santhosh Raminedi, and S. Shridevi

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Abstract Heart failure is common and often fatal that occurs when the heart is not able to pump enough blood to fulfill the body's needs. Early prognosis and cure of heart failure can significantly improve patient outcomes, as current techniques are not always precise or timely. Support vector classifiers (SVMs), have been used to predict heart failure, but they can be restricted by the size of data they can process and the precision by which they can analyze it. In this study, the focus is on the use of quantum machine learning algorithms, specifically quantum support vector classifiers (QSVMs), for heart failure prediction. QSVM is an ML algorithm that works on a quantum machine and comprises of the quantum mechanics principles. Comparison is done on the performance of QSVMs to traditional SVMs on a dataset of heart failure patients. QSVMs were able to predict heart failure with higher accuracy than traditional SVMs, particularly when dealing with large or complex datasets.

S. Waskar · S. Raminedi · S. Shridevi (✉)
Centre for Advanced Data Science, Vellore Institute of Technology, Chennai, India
e-mail: shridevi.s@vit.ac.in

S. Waskar
e-mail: sheralsimon.waskar2020@vitstudent.ac.in

S. Raminedi
e-mail: raminedi.santhosh2020@vitstudent.ac.in

Keywords QSVM (quantum support vector machine) · SVM (support vector machine) · Heart failure · ML (machine learning)

1 Introduction

Quantum computers have the potential to revolutionize many fields, including the medical sector. They can be utilized, in particular, to improve machine learning algorithms, which are employed in numerous medical applications. Machine learning algorithms can analyze medical pictures like CT scans and MRIs [1]. Quantum computers have the power to outgo classical computers on a number of tasks, and this includes tasks related to machine learning. Quantum computers use the superposition and entanglement concepts from quantum physics to their advantage [2]. As a result, QSVMs surpass classical SVMs on classification tasks, particularly when dealing with large or complex datasets.

Quantum-enhanced support vector machine (SVM) increases the strength of quantum computing to perform transformations and computations on qubits (quantum bits) to solve the optimization problem in SVM. The key concept is to represent the data points in a high-dimensional quantum space where the optimization may be carried out using quantum algorithms. Testing the hypothesis whether the fully unsupervised machine learning algorithms can use both echocardiographic data and clinical parameters to detect heart failure and also determine which patients have improved from taking the cardiac resynchronization therapy (CRT) [3].

Quantum gates, tiresome processes that alter a qubit's state, can convert the data points into qubits and manipulate them [4]. These transformations can help to find the optimal boundary that separates the different classes of data. Single dataset of heart disease cannot obtain the results accurately. Various algorithms proposed by researchers cannot be generalized. To obtain a more generalized classification, many datasets from different geographical sources have to be explored for more efficient prediction of heart failure [5].

One potential application of quantum machine learning in heart failure prediction is the use of quantum algorithms to analyze high-dimensional data and identify patterns or relationships that may be difficult for classical machine learning algorithms to detect. Quantum machine learning algorithms could also potentially be used to refine data from multiple sources in real-time, allowing for continuous tracking and prediction of heart failure risk. Quantum support vector machine (QSVM) combines the principles of quantum mechanics and supports vector machines [6]. QSVM shows superior performance compared to classical support vector machines when applied to heart failure prediction, suggesting that it could be a promising tool for this purpose.

2 Related Work

The paper [7] written by Yogesh et al. focuses on the importance of quantum-enhanced machine learning in healthcare. In comparison to conventional machine learning algorithms, quantum improved machine learning techniques such as quantum random forest (QRF), quantum K closest neighbor (QKNN), quantum decision tree (QDT), and quantum gaussian naive bayes (QGNB) have proved to have higher accuracy and execution times. An approach for detecting chronic heart failure based on heart beats has been proposed by Martin et al.'s research [8]. This method is basically a combination of Machine Learning and Deep Learning. The above mentioned method proved to be quite successful in determining the patient's heart condition.

In their paper [9], Dmitry Solenov, Jay Brieler, and Jeffrey F. Scherrer show how machine learning and quantum computing can be used in the field of medicine. It can be used right from storing all the information and retrieving using an algorithm which saves manual effort to advance techniques for interpreting the CT scan and help in carrying out surgical operations too within a fraction of a second. In the research [10] done by Renu Narain, Sanjai Saxena, and Achal Kumar Goyal the focus is on the quantum neural network technique that can be used for cardiovascular disease prediction. In the paper [11] by Danyal Maheswari, Begonya Garcia Zapirain, and Daniel Sierra Sosa an overview of the research papers between 2013 and 2021 related to the applications and algorithms of Quantum machine learning is presented along with the drawbacks and the future scope of the Quantum Machine Learning. The primary goal of the research [12] conducted by S S Kavitha and Narasimha Kaulgud is to use unsupervised machine learning methods to diagnose cardiac failure. In order to verify the performance metrics, a comparison between classical and quantum processing is conducted.

Research is conducted on a number of quantum machine learning approaches and a Deep Learning model for the diabetes diagnosis utilizing the PIMA Indian Diabetes dataset in the work [13] proposed by Himanshu Gupta et al. According to the research Deep Learning model has better performance in comparison to the quantum machine learning techniques. Analysis of machine learning algorithms, including neural networks, K-nearest neighbors, Naive Bayes, and logistic regression, is carried out for the identification of heart disease in the study [14] by Samir S. Yadav, Shivajirao M. Jadhav, Snigdha Nagrale, and Niraj Patil. The paper [15] proposed by Jae-Kwon Kim and Sanggil Kang was to analyze which technique outperforms the prediction of coronary heart disease. It was found that NN-FCA was the best in comparison with NN-FCA and FRS systems.

Martín, Droguett have proposed a new hybrid approach for the health state diagnosis and prognosis problem in the Quantum machine learning field [16]. According to their approach they have been divided into four phases. First phase is data pre-processing, second one is processing the classical data to quantum circuits containing information using encoding schema, third phase is passing the quantum encoding data to Parametrized Quantum Circuit (PQC), fourth phase is measuring step, which

returns the classical data and it is fed to classical neural network to perform required diagnosis and prognosis. Highly correlated features of the dataset have highly effective performance of K-nearest neighbor (KNN) for heart failure detection [17]. Implementing QNN (Quantum Neural Networks) for the disease data is a very efficient way to solve the problem of identifying the diseases. Dutt, Chandrasekaran, and García-Díaz have proposed a two-level neural network using ML and quantum computing. They have worked on quantum data and hybrid quantum implementation for the disease dataset. They have achieved an accuracy of 93% in identifying diseases [18].

Ahmad et al. have proposed QSVM to predict whether a person requires mental assistance are not based on the Mental Health surveys dataset. They have achieved a test accuracy of 0.70 with QSVM QASM simulator [19]. Machine learning methods in health care are of two types: data cleaning schemes and data reduction schemes. Data cleaning schemes consist of handling missing values, noisy data, and data normalization. Data reduction schemes consist of feature selection and feature extraction [20].

By displaying the data in a high-dimensional feature space, commonly referred to as a Hilbert space, quantum machine learning can be carried out [21].

In the Schuld, Killoran approach, each data point is represented as a quantum state in the feature space, and the similarity between data points is measured using a quantum kernel.

In Caro et al.'s research, they investigated how well a quantum machine learning model can make predictions about new data after being trained on a small N number of training examples. Additionally, they demonstrate that the error in predictions can be further reduced if only a small percentage of the trainable gates have significantly changed during the training process, becoming proportional to the square root of the ratio of the number of changed gates to the number of training examples [22]. On the basis of the quantum version of a generative model, a form of model frequently used in machine learning to represent and learn probability distributions, Gao, X., Zhang, Z. Y., and Duan suggested a new quantum method for machine learning. Proposed model has an advantage over classical generative models in its ability to represent probability distributions, and can achieve an exponential increase in speed for certain tasks when run on a quantum computer, as long as a classical computer cannot efficiently simulate it [23].

The South African Heart Disease dataset is used by Amanda H. Gonsalves, Fadi Thabtah, Rami Mustafa A. Mohammad, and Gurpreet Singh to do predictions using Naive Bayes, Support Vector Machines, and Decision Trees. On performing analysis using accuracy, sensitivity and specificity, conclusion is drawn that Naive Bayes outperforms Support Vector Machine and Decision Tree algorithms [25]. This research [26] carried out by Prerna et al. discusses the solution for predicting the coronary heart disease using Modified Artificial Plant Optimization (MAPO) and other machine learning algorithms. MAPO is used to decrease the amount of features to the most significant characteristics, with a maximum attribute reduction of 81.25% and higher accuracy. The research [27] done by Fahd Saleh Alotaibi uses Rapid Miner tool for visualizing the predictions of heart disease using SVM, Naïve Bayes, Decision Tree, Logistic regression, Random Forest classification. According to the result

obtained by RapidMiner all the algorithms perform better using Rapid Miner. Decision Tree and SVM have achieved a greater accuracy level. Abohashima, Elhoseny, Houssein, and Waleed prove that quantum machine learning is faster and has much better accuracy as compared to traditional machine learning approaches. Cleveland and Statlog datasets are used for the analysis. According to the result obtained hybrid quantum network can deal with outlier data whereas the hybrid random forest cannot [28].

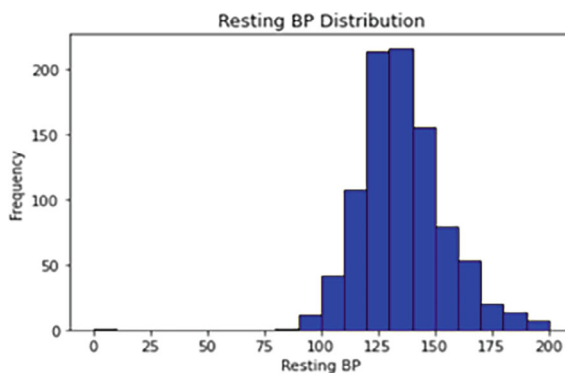
3 Methodology

3.1 Dataset

The heart failure dataset utilized in this study was compiled from UCI Machine Learning Repositories from several nations, with some common characteristics taken into account. The dataset contains information on 908 patients diagnosed with heart failure in different countries. The dataset includes 12 attributes, Age, Gender, Chest Pain Type, Resting Blood Pressure, Cholesterol, Fasting BS, Rest ECG, Maximum HR, Angina pectoris D effort, ST segment peak, and slope ST. Missing data were calculated using the average calculation method. The outcome variable of interest was hospitalization for heart failure after diagnosis. This outcome is measured against the patient's admission records in the electronic health record. Patients' ages ranged from 28 to 77, with a mean of 53.51 years (standard deviation: 9.4). Patients were male (60%) and of Caucasian ancestry (70%). The mean systolic blood pressure was 132 mmHg (standard deviation = 18.51 mmHg). The following Fig. 1 shows the patient's resting blood pressure distribution.

The dataset, taken as a whole, offers thorough data on a large number of heart failure patients and can be used to research the variables that affect the likelihood of heart failure hospitalization. The QSVM was able to analyze these data and identify patterns and trends that were used to predict the likelihood of heart failure in

Fig. 1 Resting BP distribution



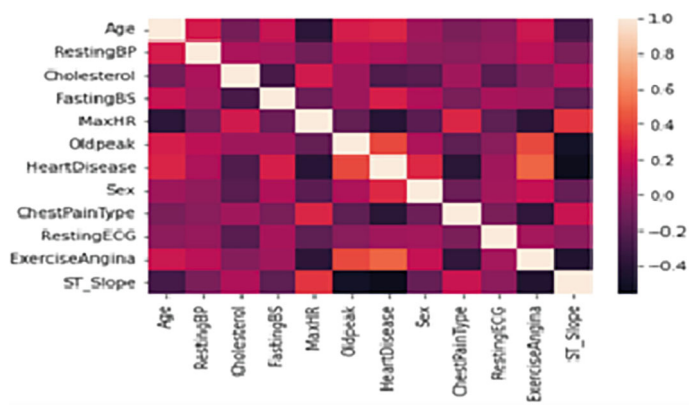


Fig. 2 Correlation matrix

individual patients. Figure 2 shows the correlation between the attributes of the dataset.

4 Data Pre-Processing

In the data pre-processing stage, several steps were taken to prepare the data for the analysis of heart failure prediction. To improve the accuracy of the algorithms, standard scaler and principal component analysis (PCA) were applied. The standard scaler was used to normalize the data by transforming each attribute. This ensures that each attribute contributes equally to the analysis, reducing the risk of bias due to attributes with larger scales. PCA was then applied to correlate the attributes and reduce the dimensionality of the data. By identifying the underlying patterns in the data, PCA helps to eliminate redundant information and highlight the most important features. In this analysis, the number of components was set to 3, meaning that the data was reduced to 3 principal components that explained the most variance in the data. These pre-processing steps were essential for improving the accuracy of the heart failure prediction algorithms, allowing them to effectively analyze the data and make accurate predictions.

5 Quantum Embedding

Encoding the data into quantum bits is the first step in learning quantum circuits. The process of representing a classical object in a quantum state is known as quantum embedding. This can be achieved using various techniques, including quantum circuit embedding and quantum embedding maps. Quantum circuit embedding is used to

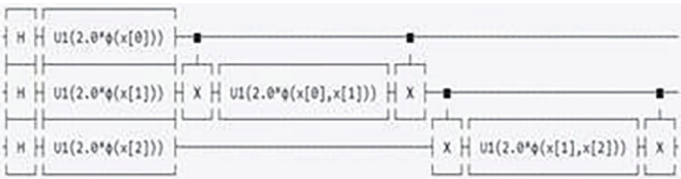
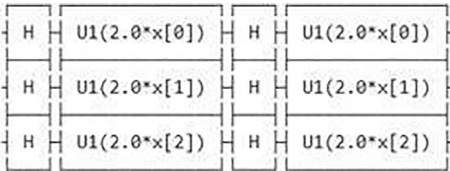


Fig. 3 ZZ feature map

Fig. 4 Z feature map



represent the data in terms of quantum state. Figures 3 and 4 depict the ZZ feature map, which is the one that is used.

“ZZ feature map may outperform the Z feature map in terms of the entangling capability and expressivity of the resulting quantum state.” ZZ feature map has more ability to exhibit quantum correlations, or entanglement, between different qubits. One of the key benefits of the ZZ feature map is that it is able to capture the correlations between different dimensions of the data more efficiently than classical methods, which can be particularly useful for tasks such as image or speech recognition, where the data may have a high dimensionality—Hilbert Space.

The feature dimension is configured to use 3 qubits for the feature map. The feature map is configured to repeat the Z and Z rotations twice. While increasing the repetitions can make the feature map more expressive, doing so also makes the quantum circuits more complex. The ZZ feature map uses linear entanglement. In quantum machine learning, a quantum instance represents a quantum computation that simulates or evaluates quantum algorithms. These quantum algorithms often involve quantum circuits that manipulate quantum bits (qubits). The IBM QASM (Quantum Assembly Language) simulator is a software tool provided by IBM for simulating quantum circuits and quantum computations. It allows researchers and developers to experiment with quantum algorithms and test their applications without the need for actual quantum hardware. When evaluating quantum circuits on a simulator, it’s common to perform multiple measurements, often referred to as “shots.” Each measurement provides a probabilistic outcome, and multiple measurements help to estimate the statistical properties of the quantum state. Setting the number of shots to 8 means that the simulator will perform 8 measurements for each quantum circuit, providing statistical information about the quantum states generated.

6 Quantum Kernels

A quantum kernel is a quantum instance function that measures the similarity between pairs of data points in a feature space in a high-dimensional feature space. This function is then used to define the similarity between the states, which can be used to classify the data. For Quantum Instance, IBM QASM simulator is used and the number of measurements shot for evaluating the quantum circuits is set to 8. A quantum transpiler is used to optimize quantum circuits for a specific quantum hardware platform. It functions by transforming a quantum circuit created in a high-level programming language into a format that can be executed on the intended hardware. The transpiler performs various optimization steps, such as gate decomposition and qubit mapping, to improve the efficiency of the circuit. Quantum circuits are composed of quantum gates that perform various operations on qubits. Quantum transpiler is a tool used to optimize and compile quantum circuits for specific quantum hardware platforms. Its primary goal is to improve the efficiency and performance of quantum circuits. The quantum transpiler takes quantum circuits written in a high-level programming and converts them into a form that can be executed on a target quantum hardware platform. Quantum transpilers perform various optimization steps, including gate decomposition, qubit mapping, and gate synthesis. These steps are crucial for ensuring that the quantum circuit operates correctly on the target hardware and takes advantage of any hardware-specific optimizations. Quantum gates in the high-level circuit may need to be decomposed into a sequence of native gates supported by the target hardware. Transpiler maps qubits in the high-level circuit to physical qubits on the quantum device, considering connectivity constraints and error rates. Gate synthesis involves finding an efficient set of gates that approximates a desired quantum operation, often using gate libraries available on the hardware.

7 Quantum Loss Functions

Cross-entropy loss function is most suitable for training quantum support vector Machines (QSVMs) for heart failure classification since it involves only two classes for classification. The objective of training a QSVM is to identify the classifier's parameters that minimize the loss function, which is a measurement of the discrepancy between the classifier's predicted output and the actual output. The algorithmic steps involved in using the cross-entropy loss function for training a QSVM: Initializing the parameters of the QSVM. This may include the weights and biases of the classifier, as well as any other parameters that are needed for the specific type of QSVM being used. Iterate over the training data, which consists of input examples and their corresponding true outputs. Computing the cross-entropy loss between the predicted output and the true output using the following equation:

$$L = -y \cdot \log(f(x)) - (1 - y) \cdot \log(1 - f(x)) \quad (1)$$

where L is the loss, y is the true output (1 for positive examples, 0 for negative examples), and $f(x)$ is the predicted output. The parameters of the QSVM are then updated using the gradient and an appropriate optimization procedure, such as stochastic gradient descent, after computing the gradient of the loss function with respect to the QSVM parameters. Heart failure classification is typically a binary classification problem where the goal is to predict whether a patient is likely to be hospitalized for heart failure (positive class) or not (negative class). Cross-entropy loss is designed precisely for this type of problem where there are only two classes. In QSVMs and classical machine learning models alike, the final output typically represents a probability score. Cross-entropy loss naturally fits this context because it measures the discrepancy between predicted probabilities and true outcomes. For heart failure classification, this means it quantifies how well the model's predictions align with the actual likelihood of hospitalization. QSVMs, like classical SVMs, are used for classification tasks, including binary classification. Finding a quantum decision boundary that divides data points into various classes is the goal of QSVMs. Cross-entropy loss naturally aligns with this goal as it measures the divergence between predicted probabilities and true class labels, guiding the QSVM to find an optimal decision boundary that minimizes this divergence.

8 Results

Quantum Support Vector Machine (QSVM) methods performed better than Support Vector Machine (SVM) algorithms in predicting heart failure. The accuracy of SVM is 69% while that of QSVM is 83%. Furthermore, the accuracy, recall, and F1 scores of QSVM are higher than that of SVM. This shows that QSVM is a better predictor of heart failure than SVM. 80% of the 908 data observations are utilized for training, while 20% are used for testing. The confusion matrix table that follows compares the effectiveness of the Quantum Support Vector Machine (QSVM) and Support Vector Machine (SVM) algorithms presents the analysis' findings. Each algorithm's accuracy, precision, recall, and F1 scores can be assessed using Table 1's breakdown of true positives, true negatives, false positives, and false negatives. Table 2 shows the accuracy, precision, recall, and F1. The results show that QSVM has higher accuracy and better overall performance than SVM, as evidenced by its higher value in the confusion matrix.

Improved accuracy compared to traditional SVMs can lead to better patient outcomes and more effective treatment plans. To hone these conclusions and investigate quantum machine learning's full potential in healthcare contexts, more study is required. It is significant to remember that quantum computing is still in its infancy, and it is unclear how widely they will be used in daily life. Meanwhile, traditional SVMs will likely continue to be widely used for machine learning tasks in the medical field.

One of the key benefits of QSVMs is their ability to represent data more effectively than classical computers by taking advantage of quantum physics' superposition and

Table 1 Summary of algorithms and comparisons

Sr. no	Algorithm	Comparison with algorithms in paper	Comparison with quantum machine learning	Dataset used	Reference link
1	Decision Tree – C4.5 (J48), (70.7%) Naïve Bayes Algorithm (71.7%), SVM (71%)	On calculating the ratio of FP and FN it is observed that NB provides more reliable results in comparison with the other algorithms	A subset of the entire dataset with mainly 462 observations is taken and it is observed that accuracy using SVM is only 71% which did not prove to be the best for the given dataset	South African Heart Disease	Gonsalves et al. [25]
2	Random Forest along with Spectro-temporal ResNet gave an accuracy of 89%	For generating more accurate results for detecting chronic heart failure an interpretation is made that mixture of cML and DL can perform much better than each of the methodologies	Dataset used has 15 features and techniques used in this paper limited as it cannot handle the noisy dataset	UKC-JSI Dataset	Martin Gjoreski et al. [9]
3	Quantum neural network is used which has an accuracy of 69%	Recurrent Quantum Neural Network (RQNN) has proved to be better than the Kalman filter which is added with three levels of noise	Automated computational features is needed to improvise the existing techniques used in this paper	Dataset 2b provided in the BCI competition	Vaibha Gandhi et al. [24]

(continued)

Table 1 (continued)

Sr. no	Algorithm	Comparison with algorithms in paper	Comparison with quantum machine learning	Dataset used	Reference link
4	Random Forest (89.14), Naïve Bayes (87.24), SVM (92.30), Logistic Regression (87.36%), Decision Tree (93.19%)	Decision Tree with tenfold cross validation has the best accuracy in comparison with fivefold cross validation used on Naïve Bayes, Decision Tree, Random Forest, and Logistic Regression	The decision tree and SVM classifiers outperform QSVM due to the smaller dataset in this study (303 records), while the heart failure dataset from UCI's Machine Learning Repositories is much larger, containing 908 records	UCI heart disease dataset	Alotaibi et al. [27]
5	Classic SVM (Train:0.75 Test:0.8), Quantum SVM(qasm Simulator) (Train:0.69 Test:0.70)	The Quantum SVM model has slightly lower accuracy in both training and testing than the Classical SVM model. However, the paper emphasizes the quantum advantages of the Quantum SVM, including the use of quantum feature maps and processing	The paper examines Quantum Support Vector Machines (QSVM) for binary mental health treatment prediction. Despite slightly lower accuracy, the paper remains optimistic about Quantum SVM's efficiency in practical quantum machine learning applications despite NISQ-era limitations	Mental Health in Tech World Survey	Syed Farhan Ahmed et al. [19]

Table 2 Confusion matrix results

Algorithm	True positive	True negative	False positive	False negative
QSVM	62	15	16	91
SVM	53	24	32	75

entanglement principles. This allows them to more efficiently map the characteristics of the data, which in turn can help improve the accuracy of heart failure predictions.

It should be noted that the performance of QSVM may vary depending on the specific characteristics of the data set used. Our results are so encouraging, that it is important to consider the limitations of this study. Future studies with larger data sets will be needed to confirm our findings. In addition to evaluating the performance of QSVMs for predicting heart failure, it will be interesting to explore their potential for other medical applications, such as drug discovery or personalized medicine. To

fully comprehend the variety of applications of quantum machine learning in the medical industry, more findings are required.

9 Discussion

The comparison of quantum support vector machines (QSVM) and conventional support vector machines (SVM) in the context of heart failure prediction is at the center of the experiment’s discussion. The results indicate that QSVM outperforms SVM, and this section explores why this superior performance is significant for predictive analysis.

Superiority of Quantum Support Vector Machines (QSVM) Accuracy Improvement: The primary finding of this experiment is the substantial accuracy improvement achieved with QSVM. QSVM demonstrated an accuracy rate of 83%, while SVM achieved only 69%. This difference is noteworthy as it can have significant implications in healthcare settings. The higher accuracy of QSVM means it is better at identifying patients at risk of heart failure, reducing false predictions, and ultimately improving patient care.

Enhanced Recall and F1 Scores: In addition to accuracy, QSVM also displayed higher recall and F1 scores compared to SVM. The number of true positives the model correctly identified is measured by recall, whereas the F1 score is a harmonic average of precision and recall. These metrics are crucial in medical applications where false negatives (missed predictions of heart failure) can have severe consequences. QSVM’s superior performance in recall and F1 score suggests it is better at identifying patients who are truly at risk, reducing the likelihood of missing critical cases.

Confusion Matrix Results: The confusion matrix results further validate QSVM’s superiority. It had more true positives and fewer false negatives compared to SVM. This indicates that QSVM is better at correctly classifying patients who are at risk of heart failure. In healthcare, correctly identifying at-risk patients is paramount for early intervention and improved patient outcomes.

Performance Metrics: The performance metrics provided in Table 3 confirm that QSVM is more effective than SVM in heart failure prediction. QSVM’s higher accuracy, precision, recall, and F1 score collectively demonstrate its superior ability to classify patients accurately Table 4 gives the details of existing works accuracy.

Receiver Operating Characteristic (ROC) Curve: The ROC curve shown in Fig. 5 further demonstrates how well QSVM performs. A higher area under the ROC curve

Table 3 Performance metrics

Algorithm	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
QSVM	83.15	86	85	85
SVM	69.5	76	70	73

Table 4 Accuracy of the algorithms used in existing papers

Algorithm	Accuracy
Decision tree [1]	70.7
Naïve Bayes [1]	71.7
Quantum neural network [26]	69

indicates better discrimination between positive and negative cases. QSVM’s ROC curve demonstrates its ability to achieve a better trade-off between true positive and false positive rates compared to SVM Fig. 6 depicts the details of the proposed models accuracy, precision, recall, and F1 score.

In the specific context of your experiment focusing on heart failure prediction, Quantum Support Vector Machines (QSVM) demonstrate remarkable efficiency gains compared to traditional Support Vector Machines (SVM). This efficiency is grounded in the fundamental principles of quantum computing that QSVM harnesses to its advantage. One key quantum property is quantum parallelism, where quantum bits (qubits) can exist in multiple states simultaneously. In SVM, this translates to

Fig. 5 ROC curve

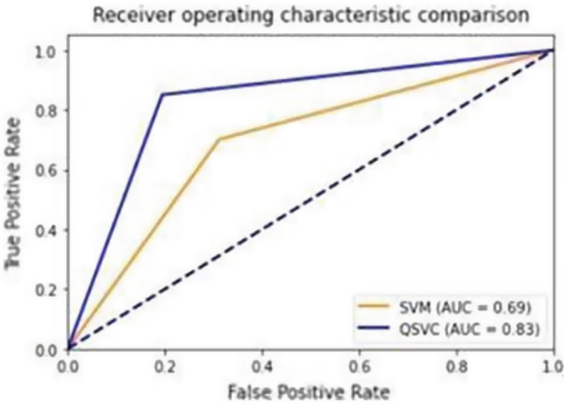
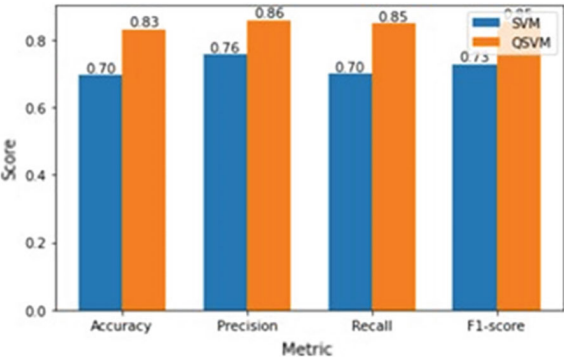


Fig. 6 Performance metrics



QSVM's ability to explore numerous potential solutions concurrently when searching for the optimal hyperplane that separates data points into distinct classes. In contrast, classical SVMs rely on sequential evaluations, which can become computationally intensive as the dimensionality of the data increases. QSVM's quantum parallelism enables it to significantly expedite the model training process, making it particularly advantageous when dealing with large and complex medical datasets commonly encountered in heart failure prediction. Another pivotal quantum feature leveraged by QSVM is entanglement.

Quantum entanglement allows qubits to become correlated with each other, even when separated by vast distances. In your experiment, entanglement facilitates QSVM in capturing intricate data dependencies and correlations efficiently. This property empowers QSVM to uncover concealed patterns and relationships among data points that may elude classical SVMs. In the context of heart failure prediction, where identifying subtle risk factors is paramount, QSVM's ability to exploit entanglement can lead to more accurate classification, reducing the likelihood of false negatives and improving patient care. Additionally, QSVM's efficient feature mapping and quantum kernel tricks streamline data pre-processing and the computation of high-dimensional feature vectors, alleviating the need for extensive manual feature engineering. Consequently, QSVM not only enhances predictive accuracy but also drastically reduces the computational resources and time required for heart failure prediction, making it a highly promising tool for healthcare applications.

The improved performance of QSVM in heart failure prediction carries significant implications for healthcare and predictive analysis. Accurate predictions of heart failure risk are crucial for patient care. QSVM's ability to reduce false negatives and false positives means that patients who need attention are more likely to receive it, potentially leading to better outcomes and timely interventions. Healthcare resources, including personnel and equipment, are often limited. Accurate predictions can help healthcare providers allocate resources more efficiently by focusing on patients who are at a higher risk of heart failure. Avoiding unnecessary hospitalizations and medical interventions for patients who are not at risk can lead to cost savings in the healthcare system. QSVM's accuracy can contribute to cost reduction by reducing unnecessary medical procedures.

10 Conclusion

The superior performance of QSVM in this experiment can be attributed to its ability to leverage quantum properties. QSVM represents and processes data more effectively than conventional computers thanks to quantum computing principles like superposition and entanglement. This efficiency allows QSVM to better capture complex patterns and correlations within the data, which is crucial in high-dimensional and intricate datasets like those in healthcare. While traditional SVM is a robust and well-established machine learning technique, it may struggle with the complexity and dimensionality of certain datasets, such as medical data. QSVM, on

the other hand, has shown promise in handling these challenges, making it a more suitable choice for tasks like heart failure prediction.

Quantum machine learning is still a rising field, and there's continuous investigation and advancement in this zone. Whereas quantum machine learning calculations hold the potential to change the field of heart failure forecast, it will likely be a few times. Sometime recently they are broadly received in clinical hone. QSVMS are a typically underutilized and experimental machine learning approach, and they have not yet acquired widespread acceptance or been taken into consideration in the context of heart failure prediction. It is additionally critical to carefully assess the execution and unwavering quality of any machine learning calculation sometime recently utilizing it for therapeutic determination or treatment choices.

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Light Gradient Boosting Machine Algorithm (LGBM) for COVID-19 Prediction Based on Symptoms



Sonika Malik and Preeti Rathee

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Abstract The widespread adoption of machine-learning technologies has significantly influenced decision-making in various fields. In this research, we harness the power of machine learning, specifically the Light Gradient Boosting Machine (LGBM) classifier, to develop a predictive model. This model efficiently estimates the likelihood of specific outcomes using a predefined dataset (Covid-19) encompassing various features. Our aim is to provide a practical tool that aids in decision-making processes, offering valuable insights and supporting resource allocation strategies. While it's essential to acknowledge that no predictive model is flawless, these tools hold substantial promise in enhancing our ability to make informed decisions and contribute to addressing complex challenges across diverse domains.

Keywords Machine learning · Prediction · Gradient boosting · Classifier · Diagnosis

S. Malik (✉) · P. Rathee
Department of IT, MSIT, Delhi, India
e-mail: sonika.malik@msit.in

P. Rathee
e-mail: preetirathee@msit.in

1 Introduction

The ongoing COVID-19 outbreak, caused by the SARS-CoV-2 virus, continues to pose a severe and pressing risk to worldwide health. The virus was first detected in December 2019 in Hubei province, China, and has since extended its reach to more than 180 nations. While more than 39.5 million COVID-19 cases have been officially reported as of October 2020, the actual number of infections is likely much higher. The pandemic has also led to over 1.1 million fatalities, highlighting the gravity of the situation [1].

The ongoing COVID-19 pandemic has created numerous obstacles for healthcare systems worldwide. There has been a surge in demand for hospital resources, such as beds and medical equipment, while many healthcare workers have also been infected, limiting capacity. To mitigate the impact of the pandemic, it is crucial to make informed clinical decisions and allocate healthcare resources effectively. Although RT-PCR is a highly reliable diagnostic test for COVID-19, it is in short supply in many developing countries, resulting in delayed diagnosis and hindering vital preventive measures. This has led to increased infection rates in these regions.

Efficient screening for COVID-19 can help expedite diagnosis and alleviate the pressure on healthcare systems. To this end, prediction models incorporating various features [10], such as laboratory tests [8, 9], clinical symptoms [7], and CT scans [2–6], have been developed to estimate infection risk and guide medical staff in prioritizing patient care, especially in contexts of limited resources. However, some of these models may not be suitable for screening individuals in the general population, as they rely on data from hospitalized patients.

The Israeli Ministry of Health provided data on all individuals who underwent RT-PCR testing for SARS-CoV-2 using a nasopharyngeal swab [11]. During the initial months of the pandemic, the Ministry of Health established criteria for COVID-19 laboratory testing that were subject to modification. The criteria included clinical symptom severity, contact with confirmed COVID-19 cases, specific geographic locations, and risk of complications if infected [12]. The majority of individuals in this study's dataset were tested based on indications for testing, resulting in minimal referral bias. Furthermore, RT-PCR testing confirmed all positive and negative COVID-19 cases in the dataset [13].

In this research paper, a machine-learning model has been presented that can effectively predict the likelihood of a positive SARS-CoV-2 infection via RT-PCR testing. The model achieves this by utilizing a set of eight fundamental questions [14]. The data used to train the model was collected during the early months of the COVID-19 pandemic in Israel and included all individuals who were tested for the virus. This model has the potential to be a valuable screening tool on a global scale, helping to prioritize testing for SARS-CoV-2 in the general population.

2 Literature Review

In recent years, the emergence of COVID-19 has prompted a significant amount of research to be conducted on the various factors that affect the transmission, spread, and predictability of the virus. Despite the relatively short amount of time since the virus was first identified, a significant body of literature has already been produced. However, it is important to note that no research is without limitations and there are gaps in our understanding of the virus that still need to be filled.

The coronavirus family is known to primarily target the respiratory system of humans. This literature review aims to present an overview of the prominent factors and techniques used in research pertaining to the transmission of COVID-19. Previous studies indicate that people with pre-existing medical conditions such as hypertension, diabetes, and heart or kidney failures, as well as elderly individuals with compromised immune systems, are at greater risk of contracting the virus [20]. Furthermore, enclosed spaces with inadequate ventilation have been identified as high-risk areas for the transmission of the virus. While the virus is primarily spread through respiratory droplets that are released when coughing and sneezing, there are reports of aerosol transmission in enclosed spaces where the virus is present at high concentrations for extended periods [21].

Numerous research studies have delved into identifying various factors that contribute to a state's preparedness in tackling the pandemic. Such factors include but are not limited to, the state's quarantine facilities, healthcare infrastructure, and laboratory testing capabilities. Access to open data sets on these factors can expedite research and promote collaboration among researchers [22]. Some studies have also looked at environmental factors like pollution and sanitation, while others have analyzed the demographics of areas affected by COVID-19 [23, 24], including death rates. In addition, several studies have examined the correlation between comorbidities and the number of COVID-19 cases [25, 26].

Numerous scientists have endeavored to create mathematical models to describe the COVID-19 outbreak, employing different methods, scopes, assumptions, predictions, and assessments of intervention efficacy. In a 2020 investigation by Columbia University and the CDC, an exponential equation was applied to forecast fatalities, while a meta-population SEIR model, which accounted for susceptible, exposed, infected, and recovered individuals, was used to predict the impact of social distancing measures [27].

3 Proposed Work

The proposed method of research involves developing a machine-learning model to predict COVID-19 test results using a dataset of individuals who were tested for the virus in Israel. Eight binary features, including sex, age, known contact with an infected individual, and five clinical symptoms, will be used to train the model. A

gradient boosting machine model built with a Light Gradient Boosting Classifying machine will be utilized to generate predictions. The model will be scored using the auROC and Confusion Metrics will be calculated for all thresholds from the ROC curves. The performance of the model will be assessed using the test set and the results will be compared to those of other prediction models developed for COVID-19. The proposed method aims to provide a reliable and efficient tool for screening and prioritizing testing for the virus, particularly in situations where testing resources are limited.

3.1 Method

Our model that utilizes machine learning was developed based on information from individuals who underwent testing for SARS-CoV-2 in Israel during the initial period of the COVID-19 outbreak. To predict the probability of a positive SARS-CoV-2 infection through an RT-PCR test, the model takes into account eight factors, including age, sex, contact with a confirmed COVID-19 case, and five clinical symptoms. Our dataset included 51,831 individuals who were tested between March 22nd and March 31st, 2020, of which 4769 tested positive for COVID-19. An additional dataset from April 1st to April 7th, 2020 was used for testing, which included 47,401 tested individuals and 3624 positive COVID-19 cases. We randomly divided the data into training and testing sets with a ratio of 0.7 and 0.3, respectively, using a seed value of 0. The dataset used by the prediction model includes several parameters that are relevant for identifying individuals who may have contracted COVID-19. These parameters include [11].

- **Test date:** This is the date when the individual was tested for COVID-19 using the RT-PCR assay of a nasopharyngeal swab.
- **Test result:** This is the outcome of the COVID-19 test, indicating whether the individual was confirmed to have the virus or not.
- **Clinical symptoms:** This is a list of symptoms that the individual was experiencing at the time of testing. These symptoms may include cough, fever, sore throat, headache, and shortness of breath.
- **Sex:** This is the gender of the individual being tested.
- **Age:** This is the age of the individual being tested, with a binary indication of whether they are 60 years or above.
- **Known contact with an infected individual:** This indicates whether the individual has had known contact with someone who has already been confirmed to have COVID-19.

These parameters were collected by the Israeli Ministry of Health and made available for use in the prediction model.

3.2 Development of the Model

The Light Gradient Boosting Machine (LGBM) algorithm is a powerful tool for making predictions based on tabular data. It is a type of gradient boosting model, which is a widely used and highly regarded approach to machine learning [20]. By training on a dataset, the LGBM algorithm is able to learn patterns and relationships within the data and use this information to make accurate predictions on new, unseen data. In this study, the LGBM model was used to generate predictions about the likelihood of an individual testing positive for COVID-19 based on a variety of clinical and demographic factors.

In order to ensure the precision of our model, we utilized a machine-learning algorithm known as Light Gradient Boosting Classification, a widely accepted technique for predicting tabular data [23]. This algorithm has been used in many successful machine-learning models. To handle any missing data in the dataset, we implemented the LightGBM package in Python [24]. During training, we utilized the validation set for early stopping, using the area under the receiver operating characteristic curve as our performance metric [26]. With this approach, we were able to train and refine our model to accurately predict the results of COVID-19 tests [27].

3.3 Architecture of Devise Method

The architecture of this prediction model is based on a gradient boosting machine, which is a machine-learning algorithm that uses decision trees as its base model. The model works by building decision trees iteratively, with each tree attempting to correct the mistakes made by the previous tree. This process continues until the model is able to make predictions with the desired level of accuracy (Fig. 1).

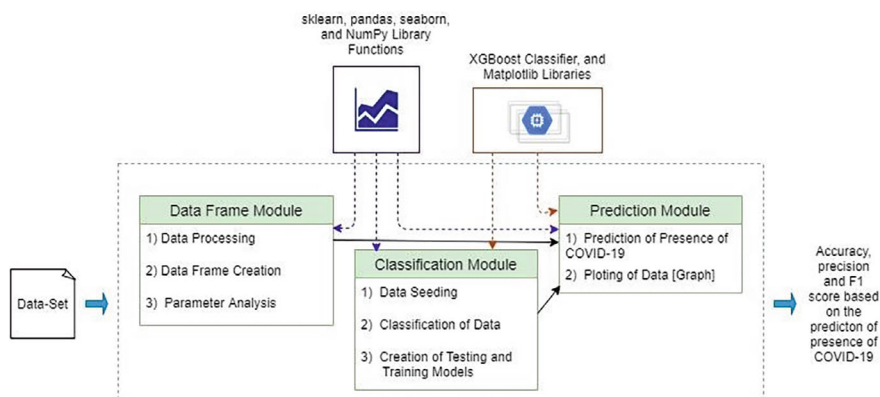


Fig. 1 Architecture of the COVID-19 prediction model

The LightGBM Python package is employed in constructing the gradient boosting machine used in this model. This package is known for its rapidity and high performance, making it particularly suitable for large-scale prediction tasks involving massive datasets. The dataset is partitioned into two distinct sets, namely the training and validation sets, which are employed in training and validating the model. The model is trained on the training set, while its performance is assessed on the validation set using the area under the receiver operating characteristic curve (auROC) as the performance metric. The training and evaluation stages of the model are iteratively carried out until a satisfactory level of performance is achieved on the validation set. The model's generalization ability is assessed on a separate test set. The performance of the model on the test set is measured using various metrics, such as negative predictive value, false-positive rate, false-negative rate, false discovery rate, overall accuracy, and confusion matrix. This prediction model is designed to efficiently and effectively predict COVID-19 test results utilizing a variety of clinical and demographic data. Its architecture, which utilizes a gradient boosting machine and the LightGBM Python package, allows for rapid and precise prediction, making it a valuable tool in the ongoing fight against the COVID-19 pandemic.

4 Results and Discussion

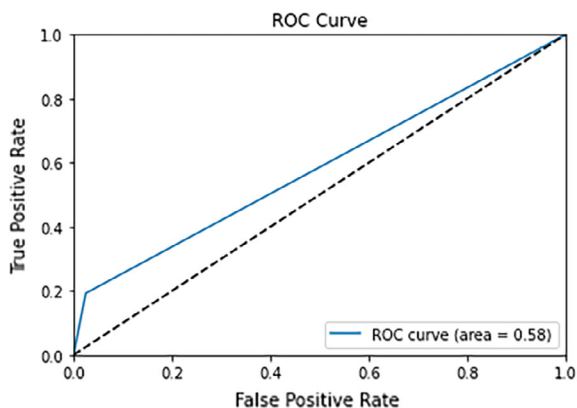
The results of this model showed promising performance in predicting COVID-19 test results using eight binary features. The model achieved an auROC of 0.58 and an accuracy score of 0.9282 on the test set. Confusion metrics calculated for all thresholds from the ROC curves demonstrated the model's ability to accurately predict positive and negative COVID-19 test results.

4.1 *Results and Evaluation of Prediction Model*

In this study, we used a LightGBM Model to evaluate the performance of our proposed machine-learning model on the test set. The model achieved an auROC (area under the receiver operating characteristic curve) of 0.58 and an accuracy score of 0.9282 on the test set. When applying the model to the training set, the accuracy score was 0.9281. These results demonstrate the effectiveness of our model in predicting a positive SARS-CoV-2 infection through an RT-PCR test. The ROC curves in this study were generated using matplotlib, and the metrics were calculated using the data obtained from the LGBM model applied to the dataset (Fig. 2).

On the prospective test set, the predictive model's performance is visualized through a graph called the ROC curve. This graph is accompanied by a light blue band that indicates the 0.58 confidence intervals, calculated through bootstrapping. The model found that having close contact with a confirmed COVID-19 case was a significant predictor of contracting the disease. This finding is consistent with the high

Fig. 2 Performance of prediction model



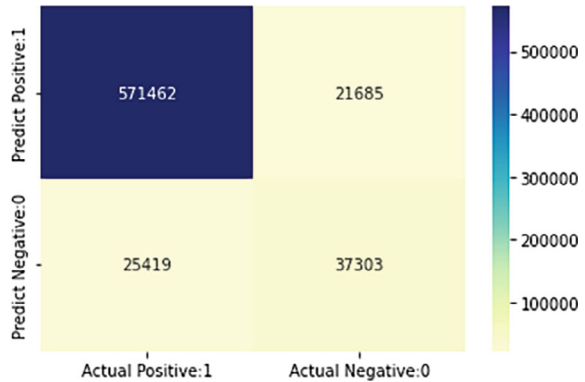
transmission rate of the virus and underscores the importance of social distancing measures [15]. Additionally, the model identified male gender as a predictor of a positive test result, which aligns with the observed gender bias in COVID-19 cases seen in previous studies [16, 17].

When analyzing the data provided by the Israeli Ministry of Health in our study, it is important to consider the potential limitations and biases that may be present. One potential limitation is that individuals who tested positive for COVID-19 may be more likely to report their symptoms accurately, as these cases may be subject to targeted epidemiological efforts [13]. This could lead to mislabeling of symptoms among those who tested negative for COVID-19. This is reflected in the proportion of individuals who reported each symptom, as some symptoms may be more likely to be reported accurately by those who tested positive. We identified symptoms that may have biased reporting, such as headache, sore throat, and shortness of breath, as well as symptoms with more balanced reporting, such as cough and fever. Mislabeling of symptoms may also occur due to underestimated and underreported symptoms among individuals who tested negative.

4.2 Discussion

Several investigations are currently underway to study the distinct pathogenic mechanisms of SARS-CoV-2 and the spectrum of associated symptoms. Our developed model offers an initial means of screening for COVID-19 using basic clinical signs and symptoms [18]. The implementation of this model may assist in easing the burden on healthcare systems by optimizing resource management during future waves of the SARS-CoV-2 pandemic, especially in under-resourced developing nations [29]. This can enhance clinical prioritization and alleviate the current strain on healthcare systems.

Fig. 3 Confusion matrix of the prediction model



It is important to recognize the limitations of this study. The dataset used for this research was obtained from the Israeli Ministry of Health, which has its own limitations, such as missing information and biases. For instance, there is a lack of information on the duration and location of contact for patients who were in contact with someone who tested positive for COVID-19 [19]. Furthermore, some highly predictive symptoms, like loss of smell and taste, were not included in the dataset [30, 31]. It should also be noted that all symptoms were self-reported, and a negative value for a symptom may only mean that the symptom was not reported. To test the model’s performance in a more challenging scenario with more missing or unreported values, we simulated a scenario where negative reports for all five symptoms were randomly selected and removed from the prospective test set. Even in this simulated scenario, the model demonstrated favorable results (Fig. 3), which strengthens our confidence in its efficacy.

One limitation of this study is that there may be differences in symptom reporting between those who tested positive and negative for COVID-19. However, it is worth noting that all individuals included in the dataset had indications for testing [13], meaning that referral bias was not a significant factor for most subjects. Additionally, the main symptoms included in Israeli Ministry of Health guidelines (cough and fever) are difficult to miss, even among those who ultimately tested negative for SARS-CoV-2. Finally, the relatively large sample size of this study may have helped to mitigate biases related to the group of individuals who tested negative for COVID-19.

The objective of this research was to construct a machine-learning model to anticipate COVID-19 diagnosis using information from all people in Israel who underwent SARS-CoV-2 testing in the early stages of the pandemic. The model was created to ask eight fundamental inquiries that could be utilized to prioritize testing for COVID-19 when resources are limited. However, it is worth noting that self-reported symptoms may be influenced by bias, and more substantial data is required to supplement this framework. Despite these limitations, our model displayed encouraging outcomes and could potentially assist the healthcare system in responding to future

Table 1 Parameters of the feature “test_indication”

Parameters of the feature “test_indication”	Values of the parameters
Other	2,547,559
Contact with confirmed	170,742
Abroad	24,295

Table 2 Representing parameters of the confusion matrix

Confusion matrix parameters	Number demonstrated by confusion matrix
True positive (TP)	571,462
True negative (TN)	37,303
False positive (FP)	21,685
False negative (FN)	25,419

waves of COVID-19 and other respiratory illnesses. The significance of continuous data exchange between public organizations and the scientific community in comprehending and reacting to the pandemic cannot be emphasized enough.

The confusion matrix (Fig. 3) provides insight into the performance of our prediction model. It includes four categories of results: true positives, false positives, true negatives, and false negatives (Table 1). True positives refer to individuals who tested positive for COVID-19 and were accurately predicted by the model to be positive. False positives refer to individuals who were not tested positive for COVID-19 but were incorrectly predicted by the model to be positive. True negatives refer to individuals who tested negative for COVID-19 and were accurately predicted by the model to be negative. False negatives refer to individuals who tested positive for COVID-19 but were incorrectly predicted by the model to be negative. The confusion matrix allows us to understand the rate of accuracy in our predictions and identify areas for potential improvement in the model (Table 2).

5 Conclusion and Future Work

To assess the efficacy of the model, various metrics were employed. The test set was used to calculate the area under the receiver operating characteristic curve (auROC), which is a common metric for assessing binary classification models’ performance in distinguishing between positive and negative cases. Additionally, confusion metrics were calculated across all thresholds from the receiver operating characteristic curves, including false-positive rate, false-negative rate, negative predictive value, false discovery rate, and overall accuracy [28]. These metrics offer further insights into the model’s ability to accurately identify true positive and true negative cases, as well as false positive and false negative cases.

5.1 Future Scope

There are several potential directions for future research on this prediction model: One approach would be to utilize explainable artificial intelligence (XAI) methods to better understand how the model is making its predictions. XAI techniques, such as Shapley values, can provide insights into the relative contributions of each feature to the overall prediction. This can help identify which features are most important in the model's decision-making process and inform strategies for further improving its performance. Additionally, incorporating XAI methods could enhance the interpretability and trustworthiness of the model, which are important considerations when using machine-learning models for decision-making in healthcare. Another potential avenue for future research could be to expand the model's capabilities by incorporating additional features or data sources. For example, incorporating data on underlying medical conditions or other risk factors could potentially improve the model's ability to identify individuals at high risk for COVID-19 infection. Finally, further testing and validation of the model in different populations and settings could help to confirm its generalizability and utility in real-world scenarios.

Dataset and Code Availability The code and dataset necessary to replicate the model's predictions and results can be accessed at:: <https://github.com/ImTanay/COVID-19Predictions>.

Declarations

Conflict of interest The authors declare that they have no conflicts of interest. This research was conducted independently and no financial support was received from any external sources. The authors have no personal or financial relationships with individuals or organizations that could have influenced the design, conduct, or interpretation of this study.

Ethics The dataset used in this study was obtained from the Israeli Ministry of Health, which permits analysis without the need for Institutional Review Board (IRB) approval. Hence, the IRB has granted an exemption for this study from requiring IRB approval.

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LSTM-Based Speech Emotion Recognition (SER) for Analyzing Patient's Verbal Feedback



Ashvini Patil, Krishnanjan Bhattacharjee, Archana Chougule,
and Swati Mehta

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Abstract Verbal communication is the most prevalent mode of communication for people to express themselves and emotions play a significant role in such human interactions. It plays a great role in capturing feedback in various situations in societal communications between humans. A deep learning-based solution for speech emotion detection is much required in the Healthcare domain for treatment monitoring and treatment course correction through patient feedbacks. It is important to know a patient's emotion during and after treatment to understand the effect of medicines and also to get feedback on treatment. Speech Emotion Recognition (SER) systems play a vital role in better understanding of patient emotions. Continuous collection of relevant patient's experiences and feedback as recorded audio files and carry such analysis using deep learning techniques. The paper proposes development and use of SER based on deep learning techniques for patient emotion detection. The Long Short-Term Memory (LSTM) architecture of deep learning is employed for developing the system. Developed solutions on SER will be helpful for hospitals and doctors to serve patients better and to provide personalized healthcare services including in remote mode and virtual clinics.

A. Patil (✉) · A. Chougule
Annasaheb Dange College of Engineering and Technology, Ashta, India
e-mail: patil.ashvini04@gmail.com

K. Bhattacharjee · S. Mehta
Centre for Development of Advanced Computing (C-DAC), Pune, India
e-mail: krishnanjanb@cdac.in

S. Mehta
e-mail: swatim@cdac.in

Keywords Speech emotion recognition · SER · Deep learning · Long short-term memory (LSTM) · Patient feedback · Virtual clinic

1 Introduction

With the evolution of technology, Artificial Intelligence (AI) and Robotics are in use in many industries to solve repetitive and complex tasks that used to be completely human foray previously. Also, AI-based regression models are being extensively used on empirical data to predict and infer future events through pattern analysis and extrapolation, viz., in pandemic prediction, weather and economic growth forecasts, polls outcome prediction, etc. Moreover, Artificial Intelligence currently has many successful applications in the medical field such as precision medicine, cancer diagnosis and treatment, as well as online AI-based diagnostic systems are getting developed and such systems are in use in many developed nations across the globe including India, such as 'Ayusoft'-online diagnostic and remote medical assistance tool developed by C-DAC, a premier R & D organization of Govt. of India. Through these diagnostic tools, the specialist doctors can diagnose the disease of patients remotely and issue online prescription for medication and treatment. Knowing the emotional state of the patient and constant feedback in online health care systems are very important and help doctors to understand patients' response to the treatment, cooperation level and mindset. Hence, automatic recognition of emotions could help create smart clinics that can detect patients' adverse reactions to ongoing treatment as well as responses or feedback on medical conditions including depression and anxiety in patients, thus facilitating doctors to provide appropriate medical treatment in time and particularly in rural or remote areas such smart clinics have potential to save a lot of lives with timely response to fatal diseases without losing the golden hours of treatment. Analysis of patient narratives provides a detailed way of exploring patients' experiences, emotions and behaviors during and after consultations [4]. This paper focuses on the implementation of a deep learning algorithm for **Speech Emotion Recognition (SER)** which helps in detecting multiple emotions of a patient by automatically analyzing the audio recordings of patient's feedback and providing the sentiment analysis to doctors and moderators for crucial decisions and course correction for the online medical treatment through such smart clinics.

Since emotions expressed through speech are natural and important clues of patients' mental state in context of disease detection through virtual communication and remotely monitored medical treatment, there is a need to capture the emotional nuances of voice, like 'stress', 'irritation', 'fear', etc., which are not always expressed through explicit words [3]. AI-based virtual assistants like chat-bots can operate on voice commands that help users to do the tasks on spoken instructions without any physical movements required from the user. Speech Emotion Recognition (SER) is a novel AI-based **Natural Language Understanding (NLU)** add-on to such AI-based virtual assistants, recognizing users' emotions so that it can act beyond spoken commands with explicit words, catering need of the patients who always may not

express themselves through distinct logical commands. SER is an important system in the field of **Human Computer Interaction (HCI)** to assimilate human emotion analysis and response in bots, computer games and psychological assessment.

Even though emotional intricacies in conversations and human to human interactions are quite important in society, in every walk of life, such that **Emotional Quotient (EQ)** or **Emotional Intelligence** has become equally important along with **Intelligence Quotient (IQ)** in corporate, business, medical assessments, interviews, etc., fields of affair. Understanding human emotions is a difficult task, even in human cognition level. In the recent times, deep learning-based techniques have become popular in many research areas such as image categorization, voice identification, speaker identification and text analytics, as they can capture high-level pattern nuances or features that cannot be captured efficiently in traditional hand-crafted rule-based methods. SER requires a novel speech-based recognizer to be developed which makes use of an imitation learning algorithm to detect emotions from a new audio signal. Emotions can be labeled by applying supervised learning mechanisms. Speech signals need to be preprocessed to explore underlying patterns and characteristics carried by the signal. Certain unique human speech features can be annotated such as prosody, frequency, pitch and pronunciation which are pivotal in identifying emotions in speech [9].

This paper describes a novel deep learning-based solution which recognizes emotions of patients who undergo medical treatments at various hospitals. Patients' speech (his/her voice) is taken as input and while speaking with the doctor during the treatment. This speech is converted in the spectrograph by considering some factors of speech like pitch, tone, volume, speed, etc. Further this speech is treated with deep learning algorithms of classification which also involves feature extraction of audio. Finally, emotion is classified from the speech and provides recognized emotion to the user. The paper describes how the Long Short-term memory (LSTM) model is used to detect patients' sentiments like displeasure, happiness, disgust, grief, fear and depression.

2 Related Work

Research has been carried out by researchers on speech emotion recognition. Various approaches have been proposed. Review of selected proposals in papers is as follows:

Adibi P et al. [1] has proposed an approach which integrates physicians, psychiatrists, linguists and data engineers' perspectives to detect patients' emotions. They have collected patients' data during pre-visit, on visit and post visit. The collected data is further labeled and used for training of ML models for prediction of emotions.

Jesus Serrano-Guerrero, Mohammad Bani-Doumi et al. [13] has proposed deep learning-based architecture which uses bidirectional gated recurrent units with a multichannel convolutional neural network layer. Authors have collected a large set of data based on opinions expressed by the patients on websites of hospitals.

These opinions are labeled to emotion expressed by it using semiautomatic lexicon-based methodology. This data is used to train many baseline models such as SVM Logistic Regression, multimodal Naive Bayes, Convolutional neural network, Recurrent neural network, long short-term memory (LSTM), bidirectional LSTM, etc., and proposed multichannel BiGRU-CRF. Their results showed that multichannel BiGRU-CRF outperformed with average accuracy of 95.82%.

Huan-Chung Li et al. [11] proposed a scientific implementation for speech emotion recognition. They have collected recorded audio files from clinics and labeled them manually. The voice samples are pre-emphasized to detect high frequency portions and then blocked as frames with 0.03 X sampling rate per frame to enhance the continuity of frames hamming window used. These frames are fed to MFCC for feature extraction and used by ANN MATLAB Neural Pattern Recognition classifier for classification of emotions.

X. Yan et al. [15] proposed exploitative and explorative gray wolf optimizer-based SVM (X-GWO-SVM) for ECG-based emotion recognition. They have designed wearable single channeled IoT ECG which collects emotions of patients. This collected data is preprocessed and is fed to the X-GWO-SVM algorithm for training and testing purposes.

LSTM Modeling of continuous emotions in an audiovisual affect recognition framework has been proposed by Wollmer et al. [14]. Audiovisual recordings are used for understanding dimensions of emotions such as arousal, expectation, power and valence. The authors extracted facial movement features using Viola-Jones detector and camshaft tracker. They also used OpenSMILE audio feature extractor as well as MFCC for audio feature extraction. Further audio, linguistic and video features are fed to LSTM for emotion prediction.

George Trigeorgis [12] proposed an approach consisting of a Convolutional neural network model using raw speech signals to predict spontaneous emotion prediction. Authors have also proposed a method for optimizing the coefficient of agreement to measure the fit between the estimate and the gold standard. Optimization of concordance correlation coefficient to assess the agreement rate between the forecasts and the gold standard.

Seyedmahdad Mirsamadi [6] has proposed a weighted time-pooling strategy which allows a model to focus on sentimentally blank parts of an audio. The authors have used bidirectional LSTM and a time-pooling attention mechanism which ignores the salient frames and other frames not carrying any emotions.

Multimodal emotion recognition from speech and text data is proposed by Chan Woo Lee and Kyu Ye Song [10]. They have used both speech and the spoken text to recognize the emotions. They have used a trainable attention technique to learn dynamic correlation between text features and speech features. It also retains information in the time domain by forming temporal embedding of both feature vectors. They have tested proposed method CMUMOSEI dataset to recognize six emotions namely happy, angry, sad, surprised, disgusted and fear.

Jing Han and Zxing Zhang have proposed reconstruction-error-based (RE-based) learning framework with memory-enhanced Recurrent Neural Networks (RNN) [5]. They have used two RNN models sequentially. First model reconstructs the original

features and acts as an autoencoder. Second model predicts emotions. The reconstruction error (RE) of original features is combined with original features and given as inputs to the second model. Authors have experimented their work on the RECOLA database and got better results over baseline systems without RE.

3 Design and Implementation

In this section we describe a proposed methodology for patient speech emotion recognition. The architecture diagram for the proposed system is as shown in Fig. 1.

The Speech Emotion Recognition system is developed using the Long Short-Term Memory (LSTM) model. The patient's audio samples are collected during their visits to the hospital for treatment. The recorded audio files are saved in mp4 form, with the configuration of 44.1KHz, 16 bit. In this study, only patient voice samples were taken. Figure 2 shows the workflow of the speech emotion recognition system.

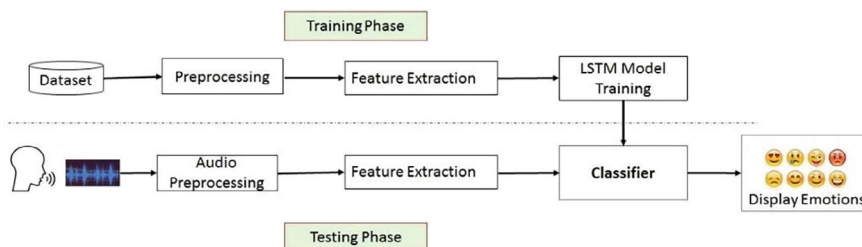
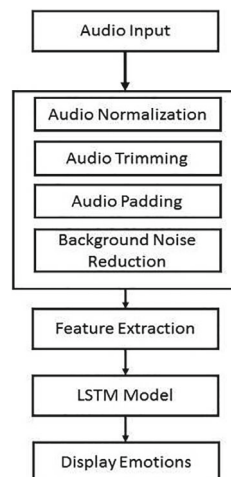


Fig. 1 Overall structure of proposed system

Fig. 2 Work flow of speech emotion recognition



After taking audio as input, that audio file is preprocessed in which audio is normalized, trimmed, padded and finally background noise is reduced for accurate results. The audio is loaded to an object by the library ‘AudioSegment’ and extracts the sample rate of 48000.

Feature Extraction: An audio signal has different dimensions that reflect emotional attributes. Crucial aspects of sentiment detection are to select the attributes of an audio signal contributing to specific emotion [2]. Such relevant features need to be extracted and grouped into different classes based on the extracted information. In latest studies, numerous ordinary features are taken into account such as energy, pitch, formant and some spectrum features such as Linear Prediction Coefficients (LPC), Mel-Frequency Cepstrum Coefficients (MFCC) and Modulation spectral features. The system uses Spectrogram, Zero Crossing Rate (ZCR), Root Mean Square (RMS) and Mel-Frequency Cepstrum Coefficients (MFCC) to extract the features. The selected patterns/features are reconstructed with the Librosa library for the speech emotion recognition model. General audio parameters are as follows: No of channels: 1, sample width: 2, frame rate (sample rate): 48000, frame width: 2, length (ms): 3303, frame count: 158558.0, Intensity: -47.96357278607907.

The extracted features can be viewed in the spectrogram. A spectrogram is a visualization of the frequency spectrum of a signal over time. It is often described as a heat map with intensity shown in different colors. Figure 3 shows the spectrogram structure for a given audio file. Following are Spectrogram feature extraction parameters: a) Pre-processed audio file; b) Frame length—Length of the frame over which to compute the speech features default: 2048; c) Hop length—Number of samples to advance for each frame. Default: 512; d) Widowing length—None. e) The FFT (Fourier Transform) value—2048.

The X axis shows frequency, the Y axis shows the time sequence of the audio clip and the color variation shows the intensity of the audio wave.

Energy—Root Mean Square (RMS): The root mean square here refers to the total amplitude of the signal, which talks about intensity and noise of the audio file. Following are RMS feature extraction parameters: a) Pre-processed audio file; b) Frame length—Length of the frame over which to compute the speech features. Default: 2048; c) Hop length—Number of samples to advance for each frame. Default: 512; d) The center is assigned with true value; the pad mod is placed as center mode for centered analysis.

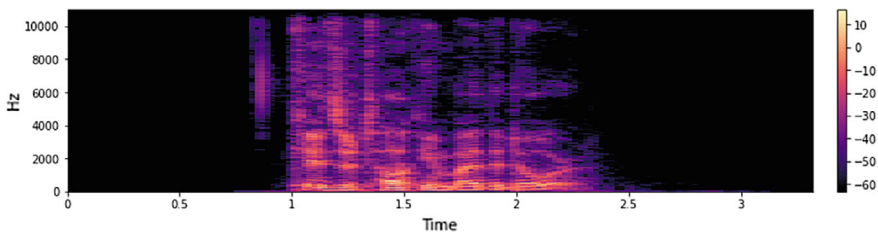


Fig. 3 Spectrogram visualization

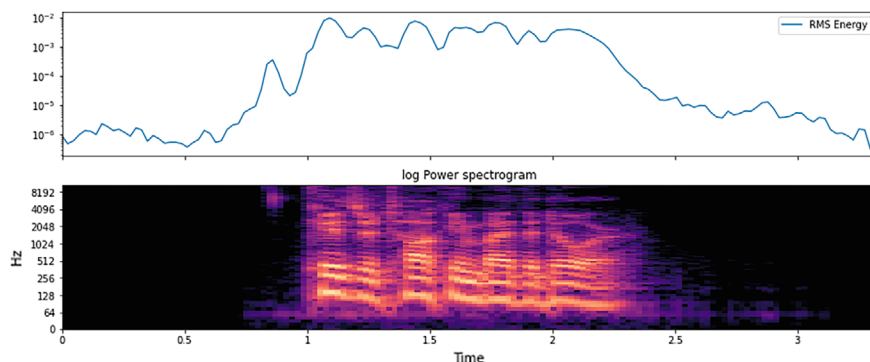


Fig. 4 Root mean square (RMS)

The visualization results for the patient audio file are shown above. It can be noted that the RMS value for the audio file is consistently high, as this patient is loud and intense throughout (Fig. 4). On the other hand, normal human speech audio file has a smooth up and down on the loudness, as human speech naturally has a moving pitch and volume depending on the speech emphasis.

Zero Crossing Rate (ZCR) represents the speed of switching of a signal between positive to negative and over-passing zero with respect to both positive and negative. It is nothing but count representing shift of a waveform from positive to negative or vice versa. It is commonly employed for speech recognition and musical data retrieval, as it is an important feature for distinguishing percussion sounds [7]. Highly percussive sounds like shout, high pitch, frustration or punk noise tend to have higher zero-crossing rate values. Following are ZCR feature extraction parameters: a) Pre-processed audio file; b) Frame length—Length of the frame over to compute the speech features. Default: 2048; c) Hop length—Number of samples to advance for each frame. Default: 512; d) Center is assigned with true value. Figure 5 shows the zero-crossing rate for the Calm voice file.

Mel-frequency cepstral coefficients (MFCCs) are the classical representation of the spectral characteristics of speech signals. The Mel-Frequency Cepstral Coefficients (MFCCs) of a signal represents a tiny set of features (up to 10-20) which summarizes the overall shape of the spectral envelope. For each frame, the Fourier

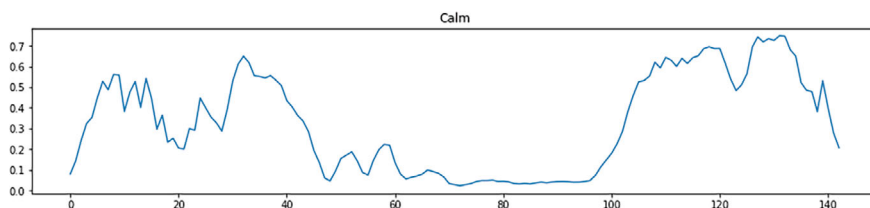


Fig. 5 Zero crossing rate (ZCR) for clam emotion

transform and the energy spectrum were evaluated and one to one correspondence was set into the Mel-frequency scale. Following are MFCCs feature extraction parameters: a) Pre-processed audio file; Frame length—Length of the frame over to compute the speech features. Default: 2048; b) Hop length—Number of samples to advance for each frame. Default: 512; c) Number of MFCC is 20; d) The pad mod is placed as center mode for centered analysis.

The MFCCs values on human speech seem to be lower and more dynamic than the other audio generating files. The given Fig. 6 shows calm audio representation because we can see more dark blue spots and changing arrays of dark red and light red on the human speech file, compared to the audio files as shown in Fig. 7.

Long Short-term memory (LSTM) network is a kind of Recurrent Neural Network which captures long and short-term relationships of an input sequence [8]. The LSTM Model uses a sequence of ‘gates’, namely Input Gate, Forget Gate and Output Gate which regulates the sequence of data input, stored and output through the network. The aim of the forget gate is to decide what fraction of information coming from previous output is to be forgotten or relevant and kept for further processing. Information fed in the state of the LSTM model is decided by the input gate, whereas output gate combines input and previous state to generate a scaling fraction which in turn combined with output of tanh function to trigger current state.

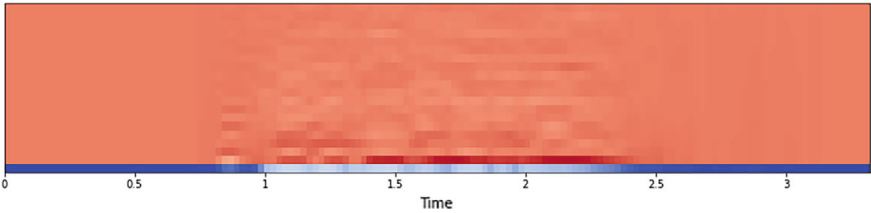


Fig. 6 MFCC visualization

Fig. 7 Training and validation accuracy with learning rate 0.01

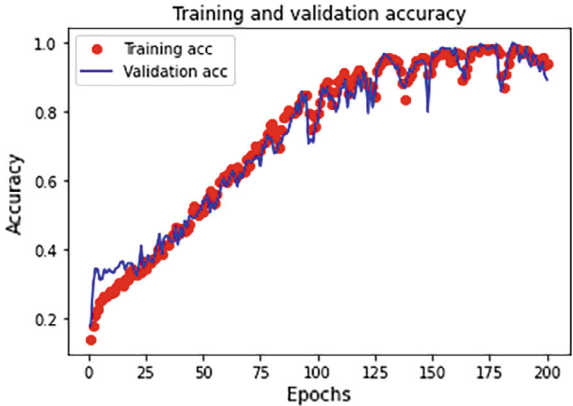


Table 1 Emotion representation of the datasets

Emotions	Dataset 1	Dataset 2
Anger	190	280
Happy	432	130
Sad	170	300
Fear	250	200
Disgust	125	200
Neutral	180	150
Calm	190	200
Surprise	192	400

Features of preprocessed audio are extracted and then it is forwarded to the LSTM model. Classification of emotion for extracted features is conducted under training and testing using LSTM model with specified dataset. The system has been trained using sentiments dataset collected from patients at campus hospital. The dataset collected consists of voice samples of more than thousand patients visited in a span of 6 months. Table 1 shows emotions representations of the datasets.

The experiments and analysis were carried out using the Anaconda environment with Python 3.8. Libraries used are Tensorflow, Keras and Librosa. The LSTM model used 68 hidden layers and batch size of 64.

4 Performance Evaluation

The experiments are carried out using the LSTM Model and two datasets with 1537 and 1460 audio files respectively. The batch size is set to 64 with learning rates of 0.01 and 0.001. Figure 7 show training and validation accuracy and Fig. 8 shows training and validation loss at different epochs and with learning rate of 0.01.

Tables 2 and 3 show training and validation accuracies at learning rates 0.01 and 0.001 respectively. At learning rate 0.001 model performance increased with epochs (Fig. 8).

A table that summarizes the effectiveness of a categorization is known as a confusion matrix. The model is tested on 60 audio files out of which each 8 distant motions. The confusion matrix of 8 x 8 for experiment model is displayed in Fig. 9.

Fig. 8 Training and validation loss with epochs

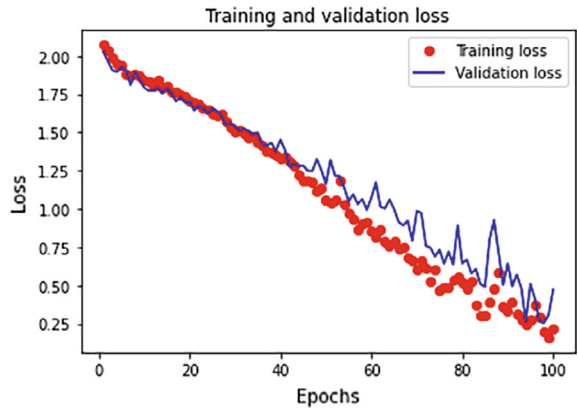


Table 2 Performance at learning rate 0.01

Epochs	Training accuracy	Validation accuracy
50	52.87	49.83
100	78.69	80.49
150	75.05	70.79
200	89.19	90.61

Table 3 Performance at learning rate 0.001

Epochs	Training accuracy	Validation accuracy
50	55.80	59.87
100	90.48	92.10
150	95.28	96.19
200	98.24	99.31

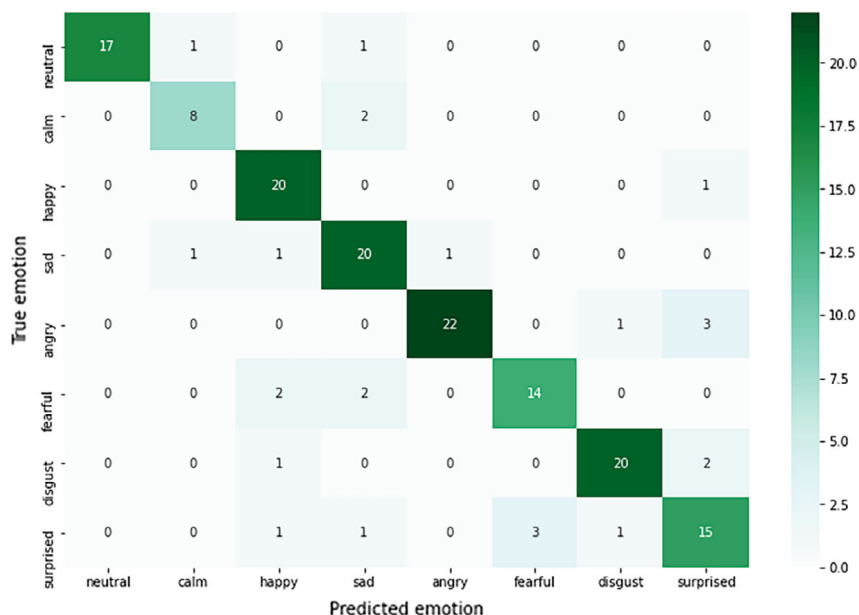


Fig. 9 Confusion matrix for proposed model

5 Conclusion

The paper has demonstrated novel and effective use of LSTM technique for automatic analysis of emotions of patients to achieve SER. The PoC has been built on audio recordings of patients' feedback. It can be inferred from results that machine learning techniques can be effectively used for better understanding of patient's psychology and performance of the hospital and such automated intelligent tool will facilitate remote patient monitoring, care-giving and virtual diagnosis and medical treatment in remote areas. The analysis carried out on feedback will definitely help doctors for course correction in treatments as well as to know strengths of the hospital and areas for improvement in treatment and care-giving.

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Machine Learning-Based Intelligent Framework for Breast Cancer Risk Assessment



Bikesh Kumar Singh , Narendra Kuber Bodhey , and Yogesh Sharma

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Abstract This article proposes a machine learning framework for risk assessment of breast cancer using factors such as those related to the patient's history, anthropometric measures, clinical symptoms, etc. Breast cancer risk factor dataset including 14,382 cases from Breast Cancer Surveillance Consortium dataset was used in the experiments. The dataset consisted of a healthy group and a control group consisting of patients with breast cancer detected within a year of the screening mammography. Statistical analysis using Mann–Whitney U-test and classification using various classifiers such as support vector machine (SVM) different kernels, Gaussian (fine, medium, and course), Naïve Bayes, k-nearest neighbor ($k = 5$), and linear discriminant analysis (LDA) were applied on this data set for risk estimation of breast cancer under fivefold data division protocol. Experimental results show that the SVM classifier with cubic kernel achieves the superior classification accuracy of 99.43% with area under curve (AUC) equal to 1.

Keywords Breast cancer risk assessment · Machine learning · Classification accuracy · Breast Cancer Surveillance Consortium

B. K. Singh (✉) · Y. Sharma
Department of Biomedical Engineering, NIT Raipur, Raipur, C. G, India
e-mail: bsingh.bme@nitrr.ac.in

N. K. Bodhey
Department of Radiodiagnosis, AIIMS Raipur, Raipur, C. G, India

1 Introduction

Breast cancer (BC) has the highest incidence and is one of the most frequent cancers in women all over the world [1]. BC is caused by the abnormal and uncontrolled development of breast cells. Given the severity of the disease, identifying high-risk women before they acquire BC remains a top priority. Over 266,000 new cases are projected in the United States in 2018. Due to low cancer literacy and a lack of understanding of BC risk, there is a higher prevalence of BC causes among Indian women, regardless of socioeconomic or educational level [2]. Despite substantial advancements in modern technology and proficiency in its application, a large percentage of cancer patients are either unable to access or cannot afford state-of-the-art treatment [3]. The prevailing practices of BC risk assessment involve physical examination followed by screening using imaging modalities such as mammography, ultrasound, and MRI. Such approaches are costly, time consuming, and need high cognitive resources. Early risk assessment of BC using proposed risk factors can address the challenges mentioned above. In the following paragraphs, some of the existing techniques in the proposed field of study are presented.

A model for the diagnosis of BC using parameters obtained from blood analysis is reported in [4]. Factors like age, body mass index, glucose, insulin, etc., were used. Processes based on machine learning were utilized for classification. Sensitivity between 82–88% and specificity between 85 and 90% is obtained using a support vector machine predictive model. Another study analyzed dietary patterns as a risk of BC on 396 BC and 396 healthy cases that were similar in age and place of residence [5]. Fresh fruit consumption was inversely correlated with BC risk in premenopausal women. Consumption of red and organ meats was generally inversely associated with BC. Further, in postmenopausal women, savory meals including sauces and snacks had a positive relationship with the risk of BC. Oestrogen receptor-positive stratification and consumption of nuts and seeds displayed an inverse relationship with the risk of BC.

A questionnaire based on the degree of education, knowledge, and behaviors of women aged 40 and over was used to perform a cross-sectional study on 1,886 women [6]. The findings demonstrated that illiterate women had significantly lower rates of knowledge and awareness about breast self-examination. Low levels of education, a higher number of deliveries, a brief period of breastfeeding, obesity, etc., were all found as risk factors for BC and cervical cancer. The role of obesity and cholesterol as risk factors for BC was discussed in [7]. Depending on the disease subtype and menopausal status, the effect of obesity on BC risk varies. It has an increased risk in women after menopause. They concluded that overweight and obesity are closely related with BC development and recurrence.

One BC risk assessment tool (BCRAT) is proposed which estimates BC risk based on a woman's personal medical and reproductive history, as well as the BC among her first-degree relatives [8]. This tool was based on statistical models and doesn't incorporate any AI technique. The supervised machine learning techniques for imbalanced BC on the BCSC dataset are reported in [9]. Synthetic minority oversampling,

undersampling, was used for balancing the data. Classifiers, such as Bayes networks, random forests, and decision trees, were used for classification. High accuracy of 99.1%, was reported. The evaluation of several machine learning techniques for the estimation of BC under the condition of inadequate information was proposed in [10]. Performance was evaluated using the Area Under Curve (AUC). However, low AUC values were reported in this study. A risk factor-based BC prediction algorithm was proposed in [11]. To un-bias the dataset, they employed the BCSC dataset, along with three distinct forms of sampling. Classification accuracy of 95.8% was obtained for the dataset that was oversampled. Machine learning-based prediction of BC using patient's demographic, laboratory, and mammographic data is proposed in [12]. Dataset consisting of 5178 records with 24 attributes was used in this study. Further, machine learning techniques like random forest, genetic algorithm, neural network, and gradient boosting trees were used for classification. A low accuracy of 80% was reported using random forests along with a low AUC of 0.56. A method using an SVM in conjunction with an exceptionally randomized tree aiming for early diagnosis of BC based on risk variables is proposed in [13]. The extra-trees classifier was employed to eliminate unnecessary variables, but SVM was employed to determine the risk of BC. Machine learning models used BC dataset of 116 people to predict BC. Further, tenfold cross-validation was used for data division and model evaluation. The suggested combination of SVM with extratrees obtained the greatest accuracy of up to 80.23%, outperforming the other ML model significantly. Ensemble learning and data fusion on Wisconsin BC diagnostic data for risk prediction of BC is reported in [14]. Several traditional machine learning techniques were used for classification. Accuracy of 99.84% was achieved by an improved XGBoost ensemble method. A study with an almost similar framework of [13] was also reported in [15] achieving accuracy of 97% using the XGBoost classifier.

Multiple models to classify BC risk using machine learning techniques to offer a reference for the early detection of BC are proposed in [16]. Recall was utilized as the primary evaluation index. Wisconsin Breast Cancer (WDBC) cancer data was utilized in this investigation.

After reviewing the aforementioned studies, it is clear that most of the existing research is focused on determining the relationship between BC and anthropometric measurements, reproductive factors, clinical symptoms, dietary patterns, lifestyle factors, and regular blood pathology. However, none of the studies that have been published combine these risk factors to create an integrated tool for assessing BC risk. Thus, this paper proposes a system toward developing an artificial Intelligence (AI)-based risk assessment system based on machine learning models for early diagnosis of BC using various risk factors. Several experiments were conducted in order to evaluate the proposed BC risk detection framework. This paper has four sections: Sect. 1 contains an introduction as well as related work from previous years. Section 2 explains the material and proposed method. The third section describes the experimental findings and discussion. Section 4 concludes with a conclusion and future scope.

2 Material and Methods

2.1 Data

Data used for this work was obtained from the Breast Cancer Surveillance Consortium (BCSC): <http://www.bcsc-research.org/>. The BCSC has 2,392,998 screening mammograms from women who are part of this risk estimation dataset [17]. The risk estimation dataset includes information about race, age, time since last mammogram, prior breast procedures, family (first-degree relatives) history of BC, age at menarche, Hispanics or not, age at first childbirth, menopause age, BI-RADS density, body mass index, menopausal status/type of menopause, and current use of hormone therapy. If a BC diagnosis was made within 12 months of the screening mammography, the females were categorized as having BC. Since this dataset was highly imbalanced, we utilized only 14,382 samples from this data set which consists of approximately equal percentage of healthy and BC cases.

2.2 Statistical Significance Analysis of BC Risk Factors

The statistical significance of different features used for BC classification is determined using independent samples t-test as well as Mann–Whitney U-test. The p-value less than 0.05 was supposed to be significant (95% confidence interval).

2.3 Breast Cancer Risk Prediction Model

The stepwise schematic of proposed risk estimation framework for BC is illustrated in Fig. 1.

After estimating the statistical significance of breast cancer risk factors as discussed in Sect. 2.3, the whole dataset was split into training data and test data via fivefold cross-validation strategy. The whole dataset was split into five groups. Four groups at a time were used for training and the remaining one group was used for testing the developed machine learning model. This process was repeated five times such that every group once was part of testing phase. Finally, the results of all five rounds were averaged out to obtain overall performance results.

After splitting the data into training and testing groups, five machine learning models namely SVM with linear, quadratic, and cubic kernels, Gaussian SVM, Naïve Bayes, k-nearest neighbor ($k = 5$), and linear discriminant analysis (LDA) were used for classification and risk estimation of BC. This involves two steps:

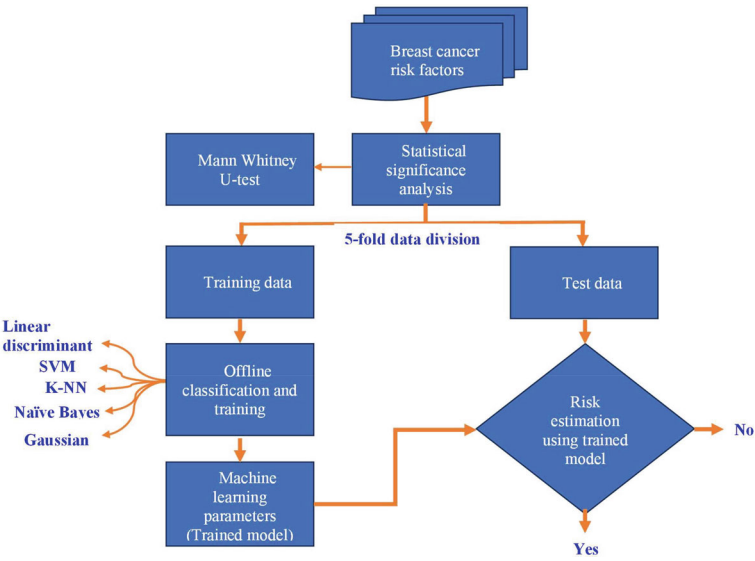


Fig. 1 Illustration of proposed machine learning framework for BC risk estimation

- i. Training phase: The machine learning models were trained using training data. After training machine learning parameters were and thus the trained model was obtained.
- ii. Testing phase: In this phase, the trained model obtained in the training phase was evaluated using the test data. Performance measures such as classification accuracy, sensitivity, specificity, precision, and F1 score were used for model evaluation. These performance measures are defined in Table 1. Further, the area under Receiver Operating Characteristics (ROC), i.e., AUC was also used for the evaluation of risk estimation model.

Table 1 Performance measures used for evaluation of risk estimation model

Performance measure	Equation	Remarks
Specificity	$\frac{\lambda_{tp}}{\lambda_{tp} + \lambda_{fn}} \times 100$	Here, λ_{tp} = true positives, λ_{tn} = true negatives, λ_{fp} = false positives, and λ_{fn} = false negatives
Precision	$\frac{\lambda_{tn}}{\lambda_{tn} + \lambda_{fp}} \times 100$	
F1-score	$\frac{\lambda_{tp}}{\lambda_{tp} + \lambda_{fp}} \times 100$	
Sensitivity	$\frac{\lambda_{tp}}{\lambda_{tp} + 0.5(\lambda_{fp} + \lambda_{fn})} \times 100$	
Accuracy	$\frac{\lambda_{tp} + \lambda_{tn}}{\lambda_{tp} + \lambda_{fn} + \lambda_{tn} + \lambda_{fp}} \times 100$	

3 Results and Discussion

The following paragraphs present and discuss the results of three experiments carried out in this study, i.e., the statistical significance analysis, classification (BC risk assessment), and ROC analysis.

The statistical evaluation was done for the given twelve features using the non-parametric Mann–Whitney U-test. We found that all the features had significant ($p < 0.05$) discrimination capabilities. Thus, we decided to include all of them for further processing. Next, the classification results using SVM subtypes, Naïve Bayes, LDA, and K-NN models are given in Table 2.

It is observed from the table that the SVM-cubic classifier is giving the best outcome across the evaluation metrics. Except for the sensitivity, all other parameters were perfect 100%. The results of SVM-quadratic are also very close to cubic SVM. It is observed that the SVM group, overall, performed superior to the Naïve Bayes, K-NN, and LDA. This proves the data handling capability of the support vectors-based maximum margin theorem, which not only works on continuous but also on ordinal data points.

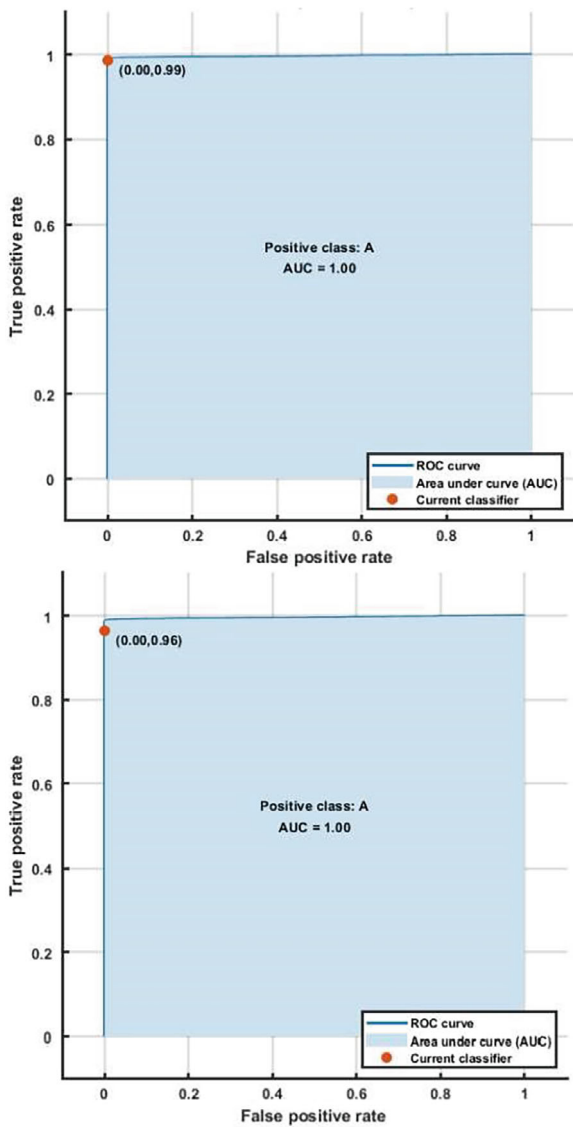
To further show their effectiveness, we also plotted receiver operating characteristics (ROC) curves for some of the best-performing classifiers of Table 2 and calculated area under the curve (AUC). The higher value of AUC close to 1 indicated superior classification capability because it only moves toward + 1 when all true values are high. Hence, the AUC is an additional metric for describing the classification performance of a particular model. The ROC plots are displayed in Fig. 2. One can be sure about our demographic and lifestyle feature-based classification framework’s ability as the AUCs of our best-performing models are equal to one. Thus, this study provides an intelligent and automated way to detect the probability of having breast lesions by processing the collected patients’ risk symptoms as the input feature vectors.

Table 3 presents a comparison of our proposed framework with some of the existing studies. It is found that the proposed approach outperforms others with high accuracy of 99.43% and AUC of 1. Breast cancer is commonly diagnosed cancer in urban Indian women. The risk of cancer in India is on the rise due to urbanization,

Table 2 Classification outcomes of the machine learning models

Classifier	Sensitivity	Specificity	Precision	F1-score	Accuracy
SVM Linear	96.47	100.00	100.00	98.20	98.23
SVM Quadratic	98.66	100.00	100.00	99.33	99.33
SVM Cubic	98.86	100.00	100.00	99.43	99.43
SVM Gaussian	97.18	100.00	100.00	98.57	98.59
Naïve Bayes	95.15	100.00	100.00	97.51	97.57
K-NN	95.33	99.96	99.96	97.59	97.64
LDA	86.15	100.00	100.00	92.56	93.07

Fig. 2 ROC plots and the corresponding AUCs of the SVM-cubic (at top) and SVM linear (at bottom) classifier



changes in lifestyles among women, low cancer literacy, and an awareness deficit of risk factors among women [2, 18]. However, no automated intelligent system is available at present which can assess the risk of breast cancer at early stages. Thus, our future work will focus on developing a breast cancer risk estimation model for the Indian population using combined risk factors based on clinical, lifestyle, diet, routine blood pathology, and findings of radiology images such as Mammogram, ultrasound, and magnetic resonance imaging (MRI).

Table 3 Comparison of proposed approach with other existing studies

Author	Technique	Performance
Patrício et al. [4]	SVM	Sensitivity ranging 82–88% and specificity ranging 85–90%
Rajendran et al. [9]	Bayes networks, random forest, and decision trees	Accuracy = 99.1%
Teja et al. [10]	Linear regression, random forest, and custom neural net-work	AUC) of 0.645, 0.643 and 0.649 respectively
Khozama et al. [11]	Tree-based machine learning	Accuracy = 95.8%
Rabiei et al. [12]	Random forest	Accuracy = 80%, sensitivity 95%, specificity = 80%, and AUC = 0.56
Nemade et al. [14]	XGBoost	Accuracy = 97%, AUC = 0.999
Proposed	Cubic SVM	Accuracy = 99.43% and AUC = 1

4 Conclusion

This article proposes an automated framework for risk assessment of BC using factors using the demographic and lifestyle parameters. The dataset used here consists of 14,382 cases obtained from BSPC. Statistical analysis using the Mann–Whitney U-test proved that all the collected information was significant. The performance of the framework was assessed by applying various classifiers for risk estimation of BC under fivefold data division protocol. Our outcomes show that the SVM classifier with cubic kernel achieves the highest accuracy of 99.43%. Additionally, the AUCs of the best-performing classifiers are close to unity. Future work would be focused on the application of deep learning and explainable artificial intelligence models for BC risk assessment and a better understanding of parameters and factors responsible for BC.

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Retinal Fundus Multi-Disease Image Classification Using Hybrid CNN-Transformer-Ensemble Architectures



Deependra Singh, Saksham Agarwal, and Subhankar Mishra

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Abstract Our research is motivated by the urgent global issue of a large population affected by retinal diseases, which are evenly distributed but underserved by specialized medical expertise, particularly in non-urban areas. Our primary objective is to bridge this healthcare gap by developing a comprehensive diagnostic system capable of accurately predicting retinal diseases solely from fundus images. However, we faced significant challenges due to limited, diverse datasets and imbalanced class distributions. To overcome these issues, we have devised innovative strategies. Our research introduces novel approaches, utilizing hybrid models combining deeper Convolutional Neural Networks (CNNs), Transformer encoders, and Ensemble architectures sequentially and in parallel to classify retinal fundus images into 20 disease labels. Our overarching goal is to assess these advanced models' potential in practical applications, with a strong focus on enhancing retinal disease diagnosis accuracy across a broader spectrum of conditions. Importantly, our efforts have surpassed baseline model results, with the C-Tran ensemble model emerging as the leader, achieving a remarkable model score of 0.9166, surpassing the baseline score of 0.9. Additionally, experiments with the IEViT model showcased equally promising outcomes

*Deependra Singh and Saksham Agarwal contributed equally to this work and should be regarded as joint first authors.

D. Singh · S. Agarwal · S. Mishra (✉)

School of Physical Sciences, School of Biological Sciences, School of Computer Sciences,
National Institute of Science Education and Research, Bhubaneswar, India
e-mail: smishra@niser.ac.in

with improved computational efficiency. We've also demonstrated the effectiveness of dynamic patch extraction and the integration of domain knowledge in computer vision tasks. In summary, our research strives to contribute significantly to retinal disease diagnosis, addressing the critical need for accessible healthcare solutions in underserved regions while aiming for comprehensive and accurate disease prediction.

Keywords Retinal disease classification · CNN · Classification transformers · IEViT · Ensemble · Hybrid

1 Introduction

The human retina serves a critical role in the visual system by converting incoming light signals into electrical/chemical signals that are subsequently processed by the brain. Unfortunately, there are numerous retinal diseases that can lead to irreversible damage, resulting in permanent vision loss. Timely diagnosis and treatment are, therefore, essential to manage and prevent further deterioration. However, the availability of quality eye care can vary significantly within countries and regions, leading to limited access, particularly for individuals residing in remote areas [1].

Addressing this issue is crucial to ensure equitable access to quality eye care. The increasing prevalence of specific retinal diseases emphasizes the importance of early detection and efficient diagnostic systems. For instance, glaucoma is predicted to impact 111.8 million individuals by 2040 [2] predicted [3] highlights that out of the 415 million people worldwide living with diabetes in 2015, 145 million experienced diabetic retinopathy (DR). Approximately 6.2 million individuals globally are affected by age-related macular degeneration (AMD) [4]. Various retinal diseases, including age-related macular degeneration (AMD), exhibit distinct pathophysiological features that can serve as valuable sources for the development of artificial intelligence (AI) diagnostic tools. Pathologies such as drusen or choroidal neovascularization (CNV), subretinal hemorrhage, and vascular leakage in dry or neovascular AMD provide potential photographic markers for the development of AI-based diagnostic algorithms [5]. These markers and other diagnostic indicators enable ophthalmologists to identify and differentiate between different retinal conditions [6]. The development of image-based diagnostic systems holds significant promise for addressing these challenges.

In recent years, there has been a growing interest in AI-based diagnostic systems that utilize fundus images. Initially, these systems targeted specific retinal diseases, such as diabetic retinopathy [7] and AMD [8]. However, a major drawback of these approaches is their narrow focus on individual diseases or disorders. Patients often experience multiple retinal diseases in real-world scenarios, rendering these software solutions less effective [9]. To address this challenge, there has been a shift toward developing multi-disease classification systems for retinal diseases [10]. Prior studies have investigated deep-learning methods, mostly Convolutional Neural Networks (CNNs), both standalone [11, 12], or ensembled [13, 14], to diagnose retinal diseases

using fundus images. Notably, [10] used CNNs to classify retinal diseases with impressive accuracies. [11] explored transfer learning, achieving 93.58% accuracy. Ensemble approaches, like [13], attained an AUC score of 0.97, even for limited labels. While CNNs have been effective, recent advancements in medical diagnosis have involved transformer-based models in increasing accuracy and targeting more disease labels [15, 16]. However, challenges persist in achieving better classification accuracy, enhancing model generalization and robustness, and deploying models in resource-limited environments; knowing the availability of large, properly labeled, and balanced datasets is scarce [11].

These works showcase deep learning’s impact, though dataset limitations and lower accuracy on certain labels warrant attention. Our research seeks to pioneer a hybrid model that addresses these challenges, “providing a scalable solution for comprehensive retinal disease diagnosis.”

1.1 Contribution

To build our hybrid models combining CNN and transformers, we selected two baseline models from previous works: C-Tran [15], which was previously been tested on a fundus image dataset with 20 labels and IEViT [16], which was previously been tried on chest X-ray images. Our work has the following contribution to current research:

1. We developed six novel models that combine deeper CNN backbones, transformer encoders, and ensemble architectures.
2. All six variants were tested on the dataset, showing improved performances over the baseline models.
3. Utilization of the domain knowledge to highlight more relevant image regions with important patch extraction.
4. Improvement in the general transformer-based learning using the IE concept.
5. For the purpose of reproducibility, the source code files can be accessed on the GitHub Repository.¹

2 Dataset

Among the three publicly available datasets shown in the Table 1, we chose the MuReD dataset [17]. This decision was influenced by its utilization in the C-Tran paper we aimed to reproduce for establishing baselines of our model [15]. MuReD, a refined variant of the highly imbalanced RFMiD dataset, was preferred due to its relevance. In the future, we intend to incorporate the RFMiD 2.0 and JSIEC datasets

¹ <https://github.com/smlab-niser/23retinald>

in our analysis as it is the latest dataset out of three and encompasses all the essential labels of MuReD (Table 2).

Table 1 List of publicly available datasets

Dataset Name	Train	Validation	Test	Total	Labels
RFMiD dataset [18]	1920	640	640	3200	46
MuReD dataset [17]	1766	442	–	2208	20
RFMiD 2.0 [19]	516	344	344	860	49
JSIEC [12]	–	–	–	1000	39

Table 2 List of diseases to be detected [15]

Acronym	Full name	Training	Validation	Total
DR	Diabetic retinopathy	396	99	495
NORMAL	Normal retina	395	98	493
MH	Media haze	135	34	169
ODC	Optic disc cupping	211	52	263
TSLN	Tessellation	125	31	156
ARMMD	Age-related macular degeneration	126	32	158
DN	Drusen	130	32	162
MYA	Myopia	71	18	89
BRVO	Branch retinal vein occlusion	63	16	79
ODP	Optic disc pallor	50	12	62
CRVO	Central retinal vein occlusion	44	11	55
CNV	Choroidal neovascularization	48	12	60
RS	Retinitis	47	11	58
ODE	Optic disc edema	46	11	57
LS	Laser scars	37	9	46
CSR	Central serous retinopathy	29	7	36
HTR	Hypertensive retinopathy	28	7	35
ASR	Arteriosclerotic retinopathy	26	7	33
CRS	Chorioretinitis	24	6	30
OTHER	Other Diseases	209	52	261

Table 3 Sampling techniques used

Sampling Technique	Description
Weighted Random Sampling	Addresses class imbalance by assigning higher weights to underrepresented classes during training
LP-ROS (10%)	An oversampling technique based on the Label Powerset transformation is used to handle class imbalance. Here, the minority class samples are augmented by adding 10% more samples to balance the class distribution

3 Preprocessing of Images

3.1 Sampling Techniques

Several sampling techniques have been employed in the literature, including LP-ROS [15], Dynamic Random Sampling [12], and Weighted Random Sampler. We focused on two specific sampling techniques, Weighted Random Sampling and LP-ROS (Label Powerset Random Oversampling), listed in Table 3. While the C-Tran paper used LP-ROS for addressing the class imbalance, the Weighted Random Sampling technique, implemented in the code provided, demonstrated better performance. The decision to utilize Weighted Random Sampling was based on empirical evidence suggesting its efficacy in handling class imbalance within the C-Tran model.

3.2 Augmentation

In the context of image data preprocessing, augmentation techniques are commonly used to increase the diversity and variability of the training dataset. These techniques introduce modifications or transformations to the original images, allowing the model to learn from a wider range of data patterns. These augmentation techniques are applied as part of the data transformation pipeline using the torchvision.transforms module. They contribute to enhancing the model’s robustness and ability to generalize patterns from the training data. Table 4 summarizes the augmentation techniques applied in this work:

Technique	Description
Random Horizontal Flip	Randomly flips the image horizontally with a probability of 0.5
Random Vertical Flip	Randomly flips the image vertically with a probability of 0.5
Random Rotation	Randomly rotates the image by 15 degrees

(continued)

(continued)

Technique	Description
Color Jitter	Adjusts the brightness, contrast, saturation, and hue of the image with values of 0.4, 0.4, 0.4, and 0.1, respectively
Resize	Resizes the image to the specified size
ToTensor	Converts the image to a tensor
Normalize	Normalizes the image tensor using pre-defined mean and standard deviation values of [0.485, 0.456, 0.406] and [0.229, 0.224, 0.225], respectively

4 Models

This section provides an overview of the models used in our study and highlights their unique contributions and variations.

4.1 Deep Convolutional Neural Networks

We explored various deep CNN architectures as backbone models to identify the most effective model configuration. Our selection encompassed ResNet152d, DenseNet121, DenseNet201, and EfficientNetV2Small. These models were pretrained on ImageNet and served as feature extractors, generating feature embeddings that were subsequently processed through transformer layers for classification. By evaluating different CNN architectures, we sought to capitalize on their unique feature extraction capabilities. Additionally, we utilized denser versions of DenseNet and ResNet compared to those employed in [15] (DenseNet121), aiming to enhance the performance of the baseline model. The performance of these architectures on our dataset is given in Table 4.

4.2 Classification Transformers C-Tran

The C-Tran model employed in this project combines DenseNet201 and ResNet152d backbones with a Transformer encoder, facilitating good performance for multi-label classification. The key components of the C-Tran model encompass feature extraction, positional encoding, and label prediction.

For each input image, visual features are extracted using the chosen CNN backbone, resulting in a feature embedding $\mathbf{X}_{\text{visual}}$ of dimensions $n \times b \times d$, where n is the number of classes, b batch size, and d is the embedding dimension. To incorporate positional information, the model utilizes positional encoding, denoted as $\mathbf{PE}_{2D}$,

Table 4 Performance of each CNN architecture on the dataset

Model	Loss	ML F1	ML AUC
DenseNet121	2.121	0.276	0.951
DenseNet201^a	1.692	0.272	0.959
ResNet152d	1.815	0.410	0.947
EfficientNetV2-S	1.881	0.214	0.958

a CNNs in bold indicate the top two best-performing architectures

calculated using sine and cosine functions based on position in the height and width dimensions:

$$\mathbf{PE}_{2D}(i, j, k) = \begin{cases} \sin\left(\frac{i}{10000^{2k/d}}\right) & \text{if } k \text{ is even} \\ \cos\left(\frac{j}{10000^{2k/d}}\right) & \text{if } k \text{ is odd} \end{cases} \quad (1)$$

where i, j represents the position in the height and width dimensions, k represents the dimension of the positional encoding, and d is the embedding dimension of the visual features.

The 2D positional encoding matrix, $\mathbf{PE}_{2D}$, is element-wise added to the visual features after reducing it linearly to the dimensions of $\mathbf{X}_{\text{visual}}$, enriching spatial awareness:

$$\mathbf{X}_{\text{encoded}} = \mathbf{X}_{\text{visual}} + \mathbf{PE}_{2D} \quad (2)$$

where $\mathbf{X}_{\text{encoded}}$ represents the spatially enhanced visual features.

The Transformer encoder captures feature-label interactions using a self-attention mechanism, fostering the learning of diverse dependencies between features and labels. The final label predictions, $\mathbf{Y}_{\text{pred}}$, are generated using independent feed-forward networks for each label embedding:

$$\mathbf{Y}_{\text{pred}} = \sigma(\mathbf{E}_{\text{label}} \cdot \mathbf{W} + \mathbf{b}) \quad (3)$$

where σ represents the sigmoid activation function, $\mathbf{W}$ represents learned weights, and $\mathbf{b}$ represents biases. The C-Tran algorithm synergizes feature extraction, positional encoding, and self-attention mechanisms within a Transformer architecture to achieve state-of-the-art multilabel classification performance.

4.3 Ensemble Model

The ensemble model combines the predictions from two distinct backbone models, DenseNet201 and ResNet152D, utilizing a weighted ensemble strategy. The

ensemble model is configured with two variants: one that employs separate transformer layers for each backbone model and another that utilizes a single transformer layer for the combined embeddings from both.

Variant 1: Separate Transformer Layers: In this variant, the input image undergoes two separate C-Tran paths. First, it passes through the DenseNet201 backbone model, generating feature embeddings. These embeddings then traverse through a set of transformer layers, as in C-Tran. The sigmoid function is applied to the output to produce the raw predictions denoted as $\mathbf{P}_{dn}$. Simultaneously, the input image also traverses the ResNet152d backbone model, producing feature embeddings that pass through their respective transformer layers, and raw predictions are generated via a sigmoid denoted as $\mathbf{P}_m$.

We employ weighted ratios, denoted as a and b , to combine the predictions from both backbone models. These ratios determine the relative importance of each backbone model's predictions. The combined output embeddings, $\mathbf{P}_c$, are calculated as follows:

$$\mathbf{P}_c = a \cdot \mathbf{P}_{dn} + b \cdot \mathbf{P}_m \quad (4)$$

To form the combined loss L , the weighted average of these individual losses is computed:

$$L = a \cdot L_{dn} + b \cdot l_m \quad (5)$$

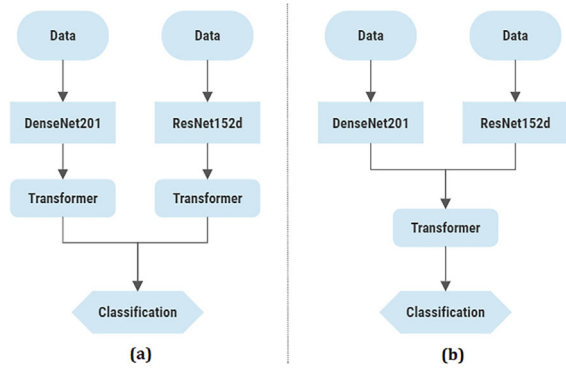
This combined loss, L , is then used for gradient descent optimization and model parameter updates. Additionally, evaluation metrics are computed using the sigmoid-activated predictions to assess model performance.

Variant 2: Single Transformer Layer: In this variant, the input image is passed through both the DenseNet201 and ResNet152D backbone models. The feature embeddings from both backbones are then combined and fed into a single transformer layer. The output embeddings from both backbone models, denoted as $\mathbf{X}_{dn}$ and $\mathbf{X}_m$, are concatenated along the n dimension (dimension 0), creating a unified feature embedding, $\mathbf{X}_c$:

$$\mathbf{X}_c = [\mathbf{X}_{dn}, \mathbf{X}_m] \quad (6)$$

where $[]$ operation signifies concatenation. This combined feature embedding, $\mathbf{X}_c$, is subsequently processed by a single C-Tran transformer encoder layer, capturing the contextual information from both backbone models. The remaining steps, such as positional encoding, projection layer, and classification layer, remain the same as in the previous variant. This variant allows the model to leverage the strengths of both backbone models in a unified transformer layer (Fig. 1).

Fig. 1 Models representation for **a** Variant 1, **b** Variant 2



Algorithm 1 C-Tran Ensemble Variants Forward Pass

Require: x : Input tensor of shape [batch, channel, height, width]

Ensure: p : Output tensor of shape [batch, num classes]

- 1: Initialize network components
 - 2: $x_1 \leftarrow \text{Backbone1}(x)$
 - 3: $x_2 \leftarrow \text{Backbone2}(x)$
 - 4: **if** Variant 1 **then**
 - 5: Add positional encodings: $x_1, x_2 \leftarrow (x_1 + P_1, x_2 + P_2)$
 - 6: $p_1 \leftarrow \text{MLP}_1(\text{Transformer}_1(x_1))$
 - 7: $p_2 \leftarrow \text{MLP}_2(\text{Transformer}_2(x_2))$
 - 8: $p = a \cdot p_1 + b \cdot p_2$
 - 9: **else**
 - 10: $x \leftarrow \text{Concatenate}(x_1, x_2)$
 - 11: Add positional encoding P : $x \leftarrow x + P$
 - 12: $p \leftarrow \text{MLP}(\text{Transformer}(x))$
 - 13: **end if**
 - 14: Train model and update weights
 - 15: **return** p
-

These two variants provide different approaches for integrating multiple backbone models and transformer layers within the ensemble model, enabling the model to leverage diverse features and capture complex patterns in the input data.

4.4 IEViT Model

The proposed IEViT (Iterative Expansion Vision Transformer) model is an extension of the ViT (Vision Transformer) architecture, incorporating concepts inspired by the ResNet (Residual Network). It introduces an iterative process in which the original

input image is iteratively added to the output of each Transformer encoder layer. To facilitate this, a convolutional block is designed in parallel with the ViT network. The CNN block processes the entire input image and generates an embedding of the image denoted as $\mathbf{x}_{img}$. This embedding is then concatenated to the output of each Transformer encoder layer, denoted as $\mathbf{z}_l$, thereby incorporating the complete image information throughout the encoding process.

The IEViT architecture comprises stacked 2D convolutional layers, followed by a 1D global maximum pooling layer within the CNN block. The resulting tensor is a feature embedding, $\mathbf{x}_{img}$ of dimension D . Additionally, the image is divided into $N = \frac{HW}{P^2}$ patches, which are flattened to patches x_p of size $N \times (P^2C)$. Subsequently, the patch embeddings z_l are created by mapping the patches to D dimensions using 2D convolution layers. Position embeddings denoted as $\mathbf{E}_{pos}$, $E_{pos} \in \mathbb{R}^{(N+1) \times D}$, are introduced to encode the positional information of each patch in the original image.

Firstly, the initial patch embedding $\mathbf{z}_p$ and the positional encodings $\mathbf{E}_{pos}$ are element-wise added. An optional class token might also be concatenated with $\mathbf{z}_p$.

$$\mathbf{z}_0 = \mathbf{z}_p + \mathbf{E}_{pos} \quad (7)$$

where $\mathbf{z}_0$ represents the combined embeddings. The core iterative expansion process in IEViT occurs during the transformation of image embeddings through stacked Transformer encoder layers. At each layer l , the image embedding, $\mathbf{x}_{img}$, is fused with the encoder output, $\mathbf{z}_l$, resulting in a combined representation $\mathbf{z}^l$. This process is mathematically represented as: denoted as $\mathbf{z}$

$$\mathbf{z}^l = [\mathbf{z}_l, \mathbf{x}_{img}], \quad \text{where } \mathbf{z}^l \in \mathbb{R}^{(N+1+l) \times D}. \quad (8)$$

The iterative expansion in IEViT enhances its capability to capture spatial information and maintain global context, contributing to superior performance across diverse computer vision tasks.

We propose two novel variants to enhance the patch extraction step in IEViT, leveraging domain knowledge of the 20 diseases targeted for classification [16]. These variants address the importance of prioritizing patches from near the center of the image, where crucial features are expected to be prominent for the specific disease labels we're targeting.

Variant 1—Unequal Patches: In this variant, we manually set the patch dimensions to be extracted from the image for forming the patch of embedding of dimension D . We extract patches with decreasing dimensions as we move closer to the center. This approach allocates more space to patches near the center, where crucial features are expected to be prominent. The patch dimensions, manually defined as 32×32 , 16×16 , and 8×8 , progressively decrease toward the center.

Variant 2—Dynamic Patch Decomposer (DPD): This approach utilizes normalized learnable parameter importance weights for the initial 144 patches, each with a dimension of 32×32 . The method dynamically determines which patches to subdivide based on their importance, using the formula for the new patch dimension

(NPD):

$$NPD = 32 \times \left(\frac{w_{avg}}{w_i} \right)^k$$

where w_i represents the weight of the i th patch, w_{avg} is the average weight, and k is a hyperparameter. Patches with $NPD > 32$ remain unchanged, while others are subdivided into dimensions $NPD \times NPD$ after rounding NPD to the nearest factor of 32. The new patches are extracted, and their embedding z_n is appended to the former patch embedding tensor z_l at the position of the original patches that were divided or merged, as represented by the equation:

$$z_{l'} = z_l \oplus z_n$$

A similar adjustment is applied to maintain positional encoding and adapt it for subdivided or merged patches. Specifically, the positional encoding is extended to accommodate new patches while preserving the existing ones, ensuring that spatial information is accurately represented throughout the encoding process. The positional encoding for the new patches is inserted at the position of the original patch in the encoding tensor. This extension is expressed by:

$$E'_{pos} = E_{pos} \oplus E_{new}$$

Additionally, the MLP (Multi-Layer Perceptron) head parameters are dynamically adjusted to handle the input changes due to patch subdivision and merging. Parameter sharing is employed to efficiently accommodate the new patches without a significant increase in model parameters, ensuring that the MLP weight matrix adapts to the varying input dimensions. Weight splicing is used when the patches are merged. This adaptive approach enhances the model's ability to capture and utilize original and subdivided patch information.

Algorithm 2 DPD Patch Division and Weight Handling

```

1: Calculate  $NPD = 32 \times \left( \frac{w_{avg}}{w_i} \right)^k$ 
2: for patch in patch embedding  $z_0$  do
3:   if  $NPD > 32$  then
4:     No patch subdivision required
5:   else
6:     Subdivide the patch into  $NPD \times NPD$  sized patches
7:     Extract the embeddings for the new patches
8:      $z_l' = z_l \oplus z_n$ 
9:     Adjust positional encoding:  $E'_{pos} = E_{pos} \oplus E_{new}$ 
10:   end if
11: end for
12: if MLP input dimension increases then
13:   Expand MLP head parameters with weight sharing
14: else if MLP input dimension decreases then
15:   Slice MLP head parameters
16: else
17:   MLP head parameters remain unchanged
18: end if
19: Continue with standard processing

```

4.5 C-Tran Variants with IEViT Concept

In this subsection, we present two variants of the C-Tran model that incorporate the Iterative expansion concept from IEViT.

IECTe—Iterative Expansion C-Tran Ensembler: The IECTe variant combines ensemble model variant 2 with the IE concept by appending the reduced feature embeddings from the two CNN backbones x and y to each output of transformer layers. Concatenating feature embeddings along the *num classes* dimension (dim = 0):

$$\begin{aligned}
 z_0 &= [x, y] \\
 z_l' &= [z_l, z_0]
 \end{aligned}$$

By incorporating these additional feature embeddings, the model gains a holistic understanding of the input image from multiple perspectives. This enables the model to leverage the strengths of both ResNet201 and DenseNet201 architectures, leading to enhanced representation learning and improved performance. The iterative nature of the IEViT approach ensures that the network retains knowledge of the entire input

image throughout the Transformer encoding process. By iteratively concatenating the feature embeddings to the Transformer output, the IECTe model continuously “remembers” the full image information, facilitating effective information fusion and promoting robust representations.

IeECT: Iterative Ensemble Expansion C-Tran—The IeECT variant focuses on reducing the overall model complexity for IECTe. In this variant, the feature embedding from DenseNet201 serves as the primary input to the Transformer ($z_0 = x$), while the feature embedding from ResNet152 is appended to the Transformer output after each layer.

$$z^l = [z_l, y]$$

By employing this design, the IeECT model benefits from the expressive power of DenseNet201, which captures intricate image details and relationships. Simultaneously, the inclusion of the ResNet152 feature embedding allows the model to incorporate information from a different architectural perspective, enriching the representations with complementary insights. Using the IEViT concept in IeECT ensures that the model retains knowledge of the original image throughout the encoding process while maintaining fewer parameters [1] (Fig. 2).

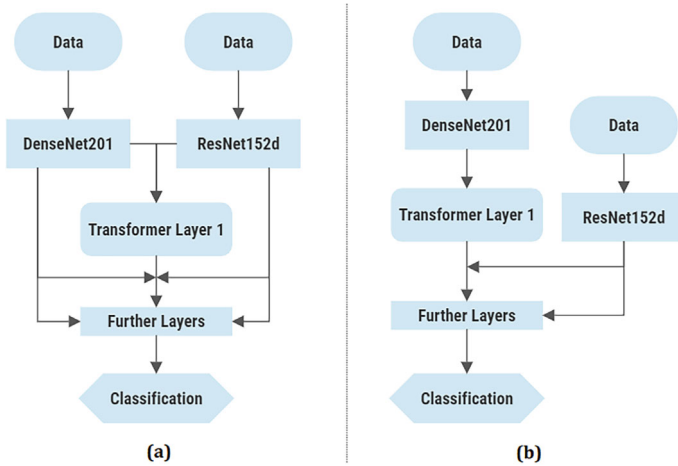


Fig. 2 Model representation for: **a** IECTe, **b** IeECT

Table 5 List of hyperparameters used for model optimization

Hyperparameter	Value
Number of Epochs	200
Batch Size	8, 12, 16, 32
Embedding Dimensions	960
Optimizer	AdamW
Scheduler	Cosine Annealing Warm Restarts, Learning Rate: 5e−5, Weight Decay: 1e−6
Loss Function	BCEwithLogitLoss
Transformer Layers	6, 12

Algorithm 3 IE pipeline Variants for CTran

```
1: Initialize network components, L transformer encoder layers
2:  $x_1, x_2 \leftarrow \text{Backbone1}(x), \text{Backbone2}(x)$ 
3: if Variant 1 then
4:    $z_0, z \leftarrow \text{Concatenate}(x_1, x_2)$ 
5: else
6:    $z_0, z \leftarrow x_1, x_2$ 
7: end if
8: Add positional encoding  $P : z_0 \leftarrow z_0 + P$ 
9:  $z_1 \leftarrow \text{Transformer Layer}_0(z_0)$ 
10: for layer  $l$  in transformer encoder  $l \neq 0$  do
11:    $z_l \leftarrow \text{Concatenate}(z_l, z)$ 
12:    $z_{(l+1)} \leftarrow \text{Transformer}(z_l)$ 
13: end for
14: return  $p \leftarrow \text{MLP}(z_l)$ 
```

5 Model Training

5.1 Hyperparameters

See Table 5.

5.2 Metrics

To evaluate the performance of our models, we used the same metrics as used in [15] for comparison on the same dataset. These metrics include F1-score, AUC, and

mAP. ML mAP, ML F1, and ML AUC are calculated by averaging scores for each label, excluding the “NORMAL” label—having the AUC and F1-score represented by Bin AUC and Bin F1 metric.

$$\text{ML Score} = \frac{\text{ML mAP} + \text{ML AUC}}{2}$$

The Model Score metric evaluates the overall performance by considering both disease classification and normalcy detection.

$$\text{Model Score} = \frac{\text{ML Score} + \text{Bin AUC}}{2}$$

6 Results and Discussion

The results of our experiments are presented in Table 6. As expected, the deeper DenseNet201 model surpassed the baseline C-Tran model that used DenseNet161. The ensemble models with deeper CNN backbones also demonstrated improved scores. The Strong Densenet and Weak Resnet ensemble variant 1, in particular, consistently achieved a model score above 0.91, surpassing the rest. It surpassed the baseline in 5 out of 8 metrics, scoring AUC less than 0.9 only in 1 label compared to 2 of the baseline. Figure 3 shows its ROC curve. As seen in Table 6, this model also surpassed the state-of-the-art (SOTA) model, which had a score of 0.9. As for the ViT-based models, all of its variants showed superior performance compared to the C-Tran baseline model while maintaining lower computational complexity. Notably, the IEViT v2 DPD variant displayed slightly better performance among the IEViT models for 32×32 dimension patch. The IE variants of C-Tran also exhibited slight improvements over the baseline, showcasing that the IE concept can be advantageous even in generic transformer architectures while maintaining lower complexity.

7 Conclusion

In summary, our research underscores the effectiveness of hybrid models in classifying retinal images despite the limitations of a small, imbalanced dataset. While we have made commendable progress in multilabel retinal disease classification, we acknowledge the constraints imposed by data size and class imbalances, which could lead to randomness and potential overfitting.

Despite these challenges, our work represents a significant step in advancing deep learning-based strategies for accurate retinal disease classification. It also opens avenues for further refinement. We plan to focus on improving model interpretability

Table 6 Comparative performance of the proposed hybrid models with the state-of-the-art (SOTA) model

Model	ML mAP	ML F1	ML AUC	ML Score	Model Score
SOTA	0.573	0.685	0.962	0.824	0.9 [10]
Proposed Models					
C-Tran DenseNet201	0.6909	0.6394	0.9221	0.8065	0.9032^a
C-Tran ResNet152d	0.6580	0.6042	0.9315	0.7951	0.8973
EV1 (S: RN, W: DN) ^b	0.6686	0.5974	0.9279	0.7982	0.8991
EV1 (S: DN, W: RN)	0.7120	0.6831	0.9544	0.8332	0.9166
EV2	0.6904	0.6538	0.9303	0.8103	0.9052
IEViT	0.6713	0.6015	0.9532	0.8122	0.9061
IEViT v1 UP	0.6781	0.6070	0.9509	0.8145	0.9073
IEViT v2 DPD	0.6893	0.6258	0.9420	0.8157	0.9078
IECTe	0.6801	0.5993	0.9420	0.8111	0.9055
IeECT	0.6725	0.6365	0.9299	0.8012	0.9006

^aBold numbers show improved model scores (> 0.9), ^bS means strong, W means weak

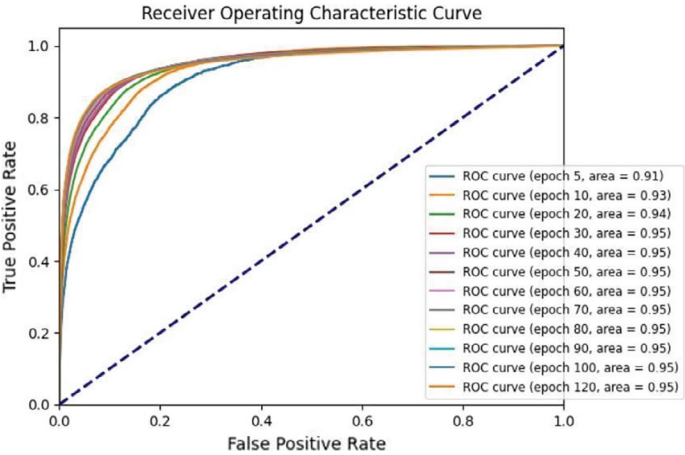


Fig. 3 ROC curve for Ensemble model

to address these limitations and enhance practical utility. In our future work, we aim to explore the integration of Shapley values [20] to unravel the causal relationships behind our model’s predictions, making them more transparent and understandable.

In conclusion, our study highlights the potential of combining transformers and ensemble learning to tackle retinal disease classification challenges. Through our ongoing efforts to enhance interpretability and delve into advanced techniques, we aim to pave the way for more robust, interpretable, and effective models that can significantly benefit the field of medical diagnostics.

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Spacers and Mismatches-Based Algorithm (SMA): G Domain Prediction and Application in Human Biology and Human Health



Hiral M. Sanghavi, Richa Rashmi, Anirban Dasgupta,
and Sharmistha Majumdar

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Abstract Guanine nucleotide-binding proteins (G proteins) are seen to play an important role in cellular processes such as sensual and visual perception, protein synthesis, vesicular transport, cell growth and differentiation to name a few. The ability of G proteins to function as molecular switches that cycle between a GTP bound (Active) and GDP bound (Inactive) states mediates their cellular functions. A structurally and mechanistically conserved GTP-binding domain (G domain) allows GTP binding to G proteins. The G domain comprises five adjacent consensus motifs called G boxes, which are separated by amino acid spacers of different lengths. Mutations in the G domain coding region of G proteins lead to heritable disorders. Changes in expression of G proteins lead to physiological or neurodegenerative disorders and their treatment. Accurate determination of the G domain sequence

H. M. Sanghavi (✉)

School of Forensic Sciences, National Forensic Sciences University, Gandhinagar,
Gujarat 382007, India

e-mail: hiral.sanghavi@nfsu.ac.in

R. Rashmi

Department of Cell Biology and Genetics, Texas A&M University School of Medicine, College
Station, TX 77807-3260, USA

A. Dasgupta

Discipline of Computer Science and Engineering, Indian Institute of Technology, Gandhinagar,
Gujarat 382355, India

S. Majumdar

Discipline of Biological Engineering, Indian Institute of Technology, Gandhinagar,
Gujarat 382355, India

in G proteins is of utmost importance for their use in the identification of disease-causing mutations, identification of mechanism of disorders and exploring their use as therapeutic proteins. In the current study, we introduce a Spacers and Mismatch Algorithm (SMA) which can predict G domains in a given protein sequence, based on user-specified constraints. SMA can be implemented via a web-based server at <https://labs.iitgn.ac.in/datascience/gboxes/>.

Keywords GTP-binding domain • G boxes • Spacers and mismatches-based algorithm

1 Introduction

GTP-binding proteins (G proteins) or GTPases are known to play important roles in different cellular processes such as reception of sensory signals, translation, cell signalling, vesicular transport, DNA repair and cytoskeleton stabilization [1, 2]. G proteins have a conserved G domain spanning 160–180 amino acid residues and with a molecular weight of 18–20 kDa. The G domain can bind both GTP and GDP; this allows the G protein to switch between a GTP-bound active state and a GDP-bound inactive state (when GTP is hydrolysed to GDP). Within the G domain, there are five sites with consensus amino acid sequence, known as G boxes, which are indispensable for GTP binding to G proteins. G1 box (Ploop/Walker A motif): “GXXXXGKS/T”. (ii) G2 box (switch 1/effector domain) has a conserved Thr residue. (iii) G3 box (Walker B motif/Switch II): “DXXG”. (iv) G4 box: “NKXD”. (v) G5 box: “EXSAX/SAX”. The G boxes are separated from each other by amino acids (spacers) that are not involved in GTP binding. The spacer between G1 and G3 boxes is 40–80 residues long in most small G proteins (Ras superfamily) and 130–170 residues long in most large G proteins (Translation factor family), while the spacers between G3 and G4 boxes is between 40 and 80 residues long in most G proteins [3]. It is difficult to predict a consensus spacer length between G1 and G2 since the G2 box consists of only one amino acid (Thr). The consensus spacer between G4 and G5 is unknown due to less available structural information about the G5 box.

Mutations in G proteins have been identified as a cause of inherited disorders. Mitofusin2 (MFN2) belongs to the dynamin class of G proteins and heterozygous mutations in MFN2 cause Charcot–Marie–Tooth Disease Type 2A, Multiple Symmetric Lipomatosis amongst others. Both of these are genetically inherited mitochondrial diseases. Activating mutations in $G_{s\alpha}$ (heterotrimeric G protein responsible for signal transduction) have been found associated with the fibrous dysplasia (FD) of bone and McCune-Albright syndrome (MAS). Mutations in $G_{\alpha q}$ which cause a defective GTPase have been found associated with uveal melanomas, congenital hemangiomas and more [4]. Constitutively active Rac1 (small G protein) was implicated in hypertension and cardiovascular diseases [5]. Listed above are a few of the associations of G proteins with disorders. In addition to their role in disorders, G proteins have also been used as potential therapeutic molecules for many disorders.

Lithium treatment for bipolar patients was found to work upon the interaction of lithium with G proteins [6, 7].

Since the discovery of G proteins over fifty years ago, many new GTP-binding proteins have been identified. These range from extensively studied proteins such as Ras, Rab, Ran, Rac, Roc, Arf, translational factors and G α subunit of heterotrimeric G proteins to more recently identified members like EngB, Septins, FeoB, AIG-1, OBG and more. As more and more G proteins with diverse functions are identified, it is important to accurately determine their G domains and G box sequences. Due to the diversity in G box sequences and inter-G box spacer length, it is difficult to predict G boxes and/or important GTP-binding residues if literature/structural information is not available. This task becomes more challenging in larger G proteins such as Translation factors, Septin, AIG-1, HypB, which have multiple domains in addition to the G domain. Accurate prediction of G box sequences and the boundaries of the G domain becomes imperative for identifying any mutation in the sequences of the G boxes.

We first explored if freely available tools, that can find motifs in sequences, could be used to predict G boxes in diverse G proteins. These tools include ELM (Eukaryotic Linear Motif) [8], GTP binder [9], ScanProsite [10], Pfam [11], SCANMOT [12], MEME/GLAM2 (Gapped Local Alignment of Motifs) [13], etc. ELM looks for short linear motifs (4–10 amino acids) in eukaryotic protein sequences; hence, given a G protein, ELM can possibly look for one G box at a time but not the entire G domain. Pfam identifies domains (in a submitted input protein sequence) that match an already characterized protein domain and hence is not a good option for less-studied proteins with very little domain information. ScanProsite is a more flexible method, which can look for motifs in a protein sequence as well as variations of the same; however, it is unable to look for different ranges of inter-motif gaps or spacers. GLAM2 uses a gapless Gibbs sampling algorithm to re-discover and refine motifs from various databases like PROSITE and ELM using sophisticated alignment options, but does not allow for an input of inter-motif spacers. SCANMOT utilizes pattern matching algorithms to identify multiple motifs in a given protein sequence. It also computes the inter-motif spacer lengths and uses these computed values as a constraint when scanning sequence databases for identifying multiple such motifs. However, none of these methods are specific for G proteins and hence do not consider the sequence diversity that exists across G protein families. GTP binder is specific to G proteins and predicts GTP interacting dipeptides and tripeptides in a given protein sequence, using a machine learning-based method; it uses PSSM which can consider possible variations in the sequence but cannot look for a G box (which is mostly longer) or the whole G domain.

In this study, we describe a method (“Spacers and Mismatch Algorithm” or SMA) that identifies all possible G domains in a given G protein sequence, after taking into account the G protein family-specific diversity. An exhaustive analysis of individual members of twenty-three G protein families was used to compile a set of approximate patterns for the G boxes as well as a set of constraints for the inter-motif gaps. All this information was then used by SMA to make its predictions. The novelty in our approach is that we use both approximate pattern matching as well as inter-motif

separation constraints to predict the locations of the G boxes. Our algorithm can be further customized to accommodate new information about mismatches (any amino acid) at any position of the G box as well as variations in the inter-G box spacer length.

G box sequences predicted by SMA were validated in corresponding G protein structures by observing if the predicted G boxes and G domain to which they belonged, were indeed involved in guanine nucleotide binding. SMA was able to predict G protein family-specific G box consensus sequences for all G boxes including the less characterized G5 box. An accurate prediction of G5 box is specifically important since it is the last of five motifs in a G domain and hence can help estimate the carboxy-terminal boundary of the domain.

2 Materials and Methods

2.1 Selection of G Protein Families for Further Analysis

There are multiple databases like PROSITE and Pfam that can be used for the study. PROSITE, which is a collection of protein domains and families along with the structural or functional patterns associated with them, was chosen for the study because it had curated information for some G protein families (EngA, EngB, Era, OBG, Translational, TrmE) which were not available on Pfam. “Guanine nucleotide-binding” was used as the search string to find PROSITE [Pubmed id 23161676] entries corresponding to G proteins. 21 documentation entries were identified to have “Guanine nucleotide-binding” profiles. 4 out of these 21 entries include proteins which interact with GTP-binding proteins but do not bind to GTP themselves [PDOC00210 G protein coupled receptors family 1, PDOC01002 G protein gamma subunit, PDOC00574 Trp-Asp (WD-40) repeats and PDOC51717 Very large inducible GTPase (VLIG)]. PDOC00017 ATP/GTP-binding site motif A (P loop) includes proteins containing only the P loop and not the entire G domain.

PDOC00962 Hydrogenases expression/synthesis hypA family does not have any GTP-binding proteins. Also, PDOC51721 Circularly permuted (CP)-type proteins are excluded as this family consists of proteins that contain G boxes in a different order from the canonical G1-G2-G3-G4-G5 pattern, e.g., DAR GTPase 3 in *Arabidopsis thaliana* has G boxes in the order G4-G1-G2-G3 (40).

The Uniprot/Swiss-prot true positive sequences for each of the remaining 14 entries were selected for further analyses. True positive sequences on PROSITE are the sequences that are curated with a strict cut-off value in order to exclude any entries without a pattern (in our case, the five G boxes). These include the following guanine nucleotide-binding (G) domain signatures and profiles: PDOC51720 AIG-1, PDOC51714 Bms1, PDOC00362 Dynamin, PDOC51712 EngA, PDOC51706 EngB, PDOC51713 Era, PDOC51711 FeoB, PDOC51715 GB1/RHD3, PDOC51705 HflX, PDOC51716 IRG, PDOC51710 OBG, PDOC51719 Septin, PDOC00273

Translational (tr), PDOC51709 TrmE. In addition to the above entries, the following entries, which were not represented in the results of the search string “Guanine nucleotide-binding”, were included: PDOC51417 (small GTPase family which includes all the canonical small G proteins such as Arf, Miro, Ran, Rab, Ras, Rho/Rac, Roc and SAR1) and PDOC00272 (SRP54 family, a SIMIBI class protein). Bms1 family was excluded from this study due to the unavailability of sufficient structural information of any of its protein members. Also, since PROSITE does not have a separate documentation entry for alpha subunits of heterotrimeric G proteins, the sequences available from Pfam were used for analysis.

An Excel file (with uniprot id, protein name, length of the protein, protein sequence and PDB structure information) was retrieved for true positive sequences of all the 20 PROSITE documentation entries from PROSITE (Galpha from Pfam) and used as the input file for our analysis,

2.2 Removal of Similarity Bias from Each PROSITE Documentation Entry:

Levenshtein distance (edit distance) (<https://nymity.ch/sybilhunting/pdf/Levenshtein1966a.pdf>) was used to compare all the member protein sequences of the same G protein family amongst each other. Only one sequence from all the sequences that showed $> = 90\%$ similarity was retained to reduce redundancy and remove the similarity bias from further analysis (<https://github.com/RichaRashmi-projects/G-Protein-Project.git>).

2.3 Spacers and Mismatch Algorithm (SMA)

The Spacers and Mismatch Algorithm (SMA) uses a data-based approach to predict G domains for the GTP-binding proteins (Fig. 1a,b, c). It takes special consideration to define a more precise G5 box and consequently a better carboxy-terminal boundary of a G domain. Following are the steps of SMA:

2.3.1 Step1—G Boxes Prediction

A Python script (<https://github.com/RichaRashmi-projects/G-Protein-Project.git>) was written, to search for all possible G domains (along with putative G boxes) in each G protein sequence, using the family-specific constraints of consensus sequence, spacers and allowed mismatches listed in Table 1. Each protein sequence was used as a string. Four different G boxes (sub-strings: 1. “GXXXXGK”, 2. “DXXG”, 3.

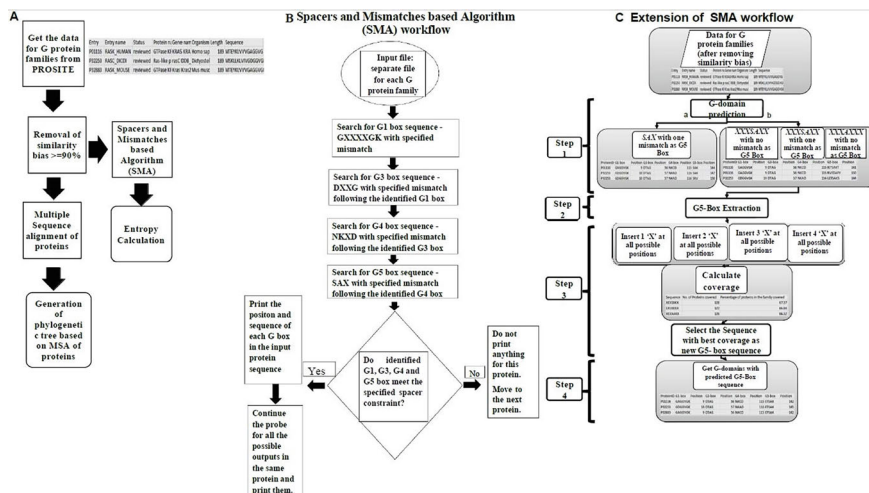


Fig. 1 A Flowchart representing the entire methodology followed in the study. **B** Workflow of G domain prediction by Spacers and Mismatch Algorithm (SMA) (upto Step 1a in C). **C** G5 box prediction

“NKXD” and 4. “SAX”) were searched in the string, one at a time. After the identification of the first conserved amino acid of a G box, a sliding window was used to find the complete G box sequence. After all four G boxes (G1-G3-G4-G5) were predicted, the algorithm checked to see if the defined spacings between the consecutive G boxes were maintained. This led to the prediction of the entire G domain with predicted G box sequences that had defined inter-box spacings. The sequences and start positions of each predicted G box were saved along with their respective protein IDs. The algorithm implemented in the script is shown in Fig. 1b.

Step 1 was implemented twice. (a), SAX with one allowed mismatch was used as the G5 box consensus sequence (Fig. 1b, c Step 1a) (b) Next, to predict a better and possibly longer G5 consensus sequence centred around SAX, we modified the search in 3 ways: (i) “XXXSAXX” with no mismatch, (ii) “XXXSAXX” with one mismatch and (iii) “XXXAXXX” with no mismatch, where X can be any of the 20 amino acids (Fig. 1c Step 1b). Then G domain identification for each of the 23 G protein families was performed again using these modified G5 box sequences.

2.3.2 Step 2—G5 Box Extraction

Only the G5 results obtained in Step 1b were used for further analysis. These results, which did not include any other information about the rest of the G domain, contained the id of the protein from which the sequence came, the actual sequence and its position in the respective protein. All the results from Step 1b (options i, ii, iii) were

Table 1 Difference in the number of input sequences after removal of similarity bias

G protein Family	Total no. of protein sequences	No. of protein sequences after removal of similarity bias
AIG-1	44	44
Arf	184	140
Dynamin	126	115
EngA	622	424
EngB	548	401
Era	360	188
FeoB	28	19
Galpha	8219	8122
GB1/RHD3	122	116
HflX	26	26
IRG	12	10
Miro	37	35
OBG	864	659
Rab	500	470
Rho/Rac	163	128
Ran	50	29
Ras	216	190
Roc	20	19
SAR	51	42
Septin	95	86
SRP	117	99
Translational	4155	2870
TrmE	596	454

merged separately for each G protein family. Redundant results, which had the same G5 box sequence and position in a particular protein, were removed (Fig. 1c).

2.3.3 Step 3—Prediction of New G5 Box Consensus Sequence

To find the underlying pattern in the merged G5 box sequences, the amino acids in the modified G5 box 7-mer sequences were replaced with 1, 2, 3 and 4 Xs at all the possible positions (Fig. 1c). Since, the modified G5 box consensus sequence was only 7 amino acids long, a maximum of 4 amino acids were replaced with X, so as not to lose the underlying pattern completely. The “coverage” for each predicted consensus G5 motif (with Xs) was calculated as the percentage of proteins from that particular G protein family that had the given motif. Position-weighted matrices (PWM) were also calculated for the G5 motifs using ggseqlogo package available

Table 2 Predicted G5 box sequence for all twenty G protein families

G protein family	G5 box motif identified after SMA analysis
AIG-1	NNXAX
Arf	CAXXG
Dynamin	LAXXL
EngA	SAXXG
EngB	FSSX
Era	PXSAX
FeoB	PPXSAX
GB1 or RHD3	YVXAXXS
Hflx	SAXXG
IRG	LRXAAX
OBG	ADXP
Rab	EXSAX
Rac/Rho	ECSAX
Ran	SAXXN
Ras	EXSAX
Roc	SXXSXXS
Septin	VXGSX
SRP	AXTG
Translational	VXXSAX
TrmE	SAXXG

on GitHub [18]. The sequence logos of predicted G boxes were visualized using the ggseqlogo package, a visualization tool which uses polygons to draw elongated letters. The height of each letter was computed based on their relative frequencies at the specified position. The G5 motifs which had the highest coverage and were prominent in the PWM were selected as the new consensus G5 box sequence for the given G protein family. Xs at the two ends of 7-mer consensus motifs were trimmed to give a more concise G5 box consensus sequence (Table 2).

2.3.4 Step 4—G Boxes Prediction with New G5 Box Sequences

Step 1 of SMA was repeated to search for all possible G domains (along with putative G boxes) in each G protein sequence, using the family-specific constraints of consensus sequence, spacers and allowed mismatches listed in Table 4 along with the additional constraint of using the new G5 box sequences predicted in Step 3.

2.3.5 Significance Testing of Predicted G-Domains

To test the significance of the SMA results, a negative control dataset was created by shuffling (fifty times) the protein sequences for each G protein family. The number of G domains predicted per protein (by SMA) was counted after each shuffle. After 50 shuffles, the total number of predicted G domains per sequence was averaged and compared with the total number of predicted G domains (after running SMA3) in the actual protein sequence.

2.4 Entropy Calculations

To investigate if the classes with more number of proteins have more variation in the individual G boxes, entropy of each G box was calculated using the entropy function in scipy (stats.entropy). This was performed by adding the entropy of each position of the predicted G box sequence. Scatter plots were generated where the X axis represents the total number of proteins (size) per G protein family and the Y axis represents the total entropy of each G box.

2.5 Web Application

SMA can be implemented via a user-friendly web application which is available at <https://labs.iitgn.ac.in/datascience/gboxes/>. The web application uses the Python and Django web frameworks. The sequence of a G protein along with user-specified G box sequences and inter-G box gaps are given as input. It outputs the G domains with predicted G5 box sequences, before and after SMA-Step. The portal can also help classify any protein into one of the G protein families by looking for all possible G domains in its primary sequence.

3 Results

3.1 Significant Reduction in the Size of Protein Families After Removal of Similarity Bias

Twenty-three different G protein families (with a total of 17,156 proteins across all families) were selected based on PROSITE documentation and Pfam entry for Galpha family. Members of the same G protein family share very high sequence similarity amongst each other. To reduce redundancy in the results and over-representation of any sequence, the similarity bias (see Methods) from each G protein family was

removed. As seen in Table 1, a significant reduction in the number of member proteins was observed in Arf, EngA, EngB, Era, Galpha, OBG, Rac, Ran, Rab, Ras, SRP, translational and TrmE families. After the removal of similarity bias, a total of 14,686 proteins across all families were used for further analysis.

3.2 Curating Available Information About the G Protein Families

In the absence of any G protein dedicated database which can provide information about the G box sequences and inter-G box spacer lengths, we removed the similarity bias from the sequences of all the twenty-three G protein families (Table 1), G box consensus sequence and family-specific spacers between consecutive G boxes were manually curated from the available literature and PDB database as described below:

Inter-G Box Spacing: Structural information (obtained using X-ray crystallography or solution NMR) of G proteins bound to GDP, GTP or GTP analogues can help directly determine their GTP-binding sites and corresponding G domain. Most available structures for each of the twenty-three G protein families were studied (column 4, Table 3). The G domain of each of these proteins was mapped to the amino acid residues in the respective protein. The spacers between the consecutive G boxes were calculated by subtracting the amino acid position at the beginning of a specific G box from the amino acid position at the beginning of the succeeding G box (for example, if G1 box of a G protein starts at position 15 and the G3 box of the same protein starts at position 81, the spacer between them is calculated as 66). If there was structural information for more than one protein in a family, a putative range was decided based on the minimum and maximum values of calculated spacers for all such proteins within the same family (Table 3). Some flexibility was allowed in these putative ranges (Table 3).

In some families like dynamin, EngB, HflX, OBG and translational, different proteins within the same family had significantly different inter-G box spacers (Table 4). Thus, for such families, more than one spacer constraint was included (Table 3) to account for the observed intra-family variability.

Mismatch in G Box Consensus Sequence: Most G proteins (with structural information) have highly conserved G box sequences and do not differ from the consensus (i.e., G1-GXXXXGK, G3-DXXG, G4-NKXD, G5-SAX; where “X” = any amino acid) by more than one mismatch.

For example, “GFPSVGKS” (in Rbg1) and “AHIDAGK” (in *S. aureus* EF-G; 2XEX) are G1 box motifs with one mismatch from the consensus motif, GXXXXGK. Also, “DIWE”, the G3 motif for human Rad; 2DPX and mouse REM2; 3Q85, has one mismatch from the consensus DXXG. Most experimentally identified G4 boxes had 1 mismatch from the consensus sequence, NKXD [e.g., “TKLD” in human DNM1L (3W6N), “TQID” in human Cdc42 (1A4R), “TLRD” in human GBP1 (1DG3); “NVNE” in *H. influenza* YchF (1JAL); “G/AKXD” in human SEPT2 (2QAG), “T/

Table 3 Manually curated spacings between consecutive G boxes and number of mismatches allowed in different G boxes

G Protein family	Putative amino acid spacers between G1 and G3 box	Putative amino acid spacers between G3 and G4 box	Putative amino acid spacers between G4 and G5 box	Number of mismatches allowed in G1 box consensus sequence (GXXXGK)	Number of mismatches allowed in G3 box consensus sequence (DXXG)	Number of mismatches allowed in G4 box consensus sequence (NKXD)	Number of mismatches allowed in G5 box consensus sequence (SAX)
AIG-1	45–55	65–75	30–40	1	1	2	1
Arf	40–50	55–65	25–35	1	1	1	1
Dynamin	95–105 95–105	65–75 55–75	25–35 40–50/ 65–80	1	1	1	1
EngA	40–80	40–80	20–80	1	1	1	1
EngB	40–45 40–50	65–75 65–75	20–35 20–35	1	1	1	1
Era	40–50	60–70	25–35	1	1	1	1
FeoB	45–50	60–65	20–30	1	1	1	1
GB1/RHD3	55–75	65–85	40–70	1	1	2	1
Hflx	40–50 60–70	65–75 130–140	20–35 20–35	1	1	1	1
IRG	45–55	55–65	30–40	1	1	1	1
OBG	45–50 65–70	60–75 130–140	20–30 25–35	1	1	2	1
Rab	45–60	55–65	25–35	1	1	1	1
Ran	45–50	55–65	25–35	1	1	1	1
Ras	45–50	55–65	25–35	1	1	1	1
Rho/Rac	45–50	55–65	40–50	1	1	1	1
Roc	45–55	55–65	35–45	1	1	2	1
Septin	50–60	75–85	50–60	1	1	1	1
SRP	80–90	55–65	20–30	1	1	1	1
Translational	45–50 45–50 60–65 65–70	45–55 60–80 50–60 50–60	25–35 30–45 60–70 110–120	1	1	1	1
TrmE	45–50	60–75	20–30	1	1	1	1

CKXD” in Rac proteins, “TKXD” in EngB, Roc, SRP, Dynamin family], except the G4 boxes in the Roc, AIG-1, GB1/RHD3 and OBG families which usually had two mismatches [e.g., human LRRK2 “THLD” (2ZEJ), human GIMAP7 “RKEE” (3ZJC), human GBP1 “TLRD” (1DG3), human OLA1 “NLSE” (2OHF)].

It was observed that there were many variations to the G5 box consensus sequence “SAX”. This ranged from one mismatch in “SSE” in *B. subtilis* YsxC, “QAH” in

Table 4 Examples of proteins from the same family which had significantly different inter-G box spacers

G protein family	Protein name	Organism	Protein structure PDB ID	Inter-G box spacer		
				G1-G3	G3-G4	G4-G5
Translational	Translation initiation factor IF-2	Bacillus stearothermophilus	1D1N;1Z9B;2LKC;2LKD;2NBG	46	54	34
Translational	Eukaryotic translation initiation factor 5B (eIF-5B)	Saccharomyces cerevisiae	3WBI;3WBJ;3WBK;4N3S;4NCF;4V8Y;4V8Z	64	54	66
Translational	Peptide chain release factor 3 (RF-3)	Desulfovibrio vulgaris	3VQT;3VR1	68	54	115
Dynammin	Mitofusin-1	Homo sapiens	5GNS;5GNT;5GNU;5GO4;5GOE;5GOF;5GOM;5YEW	96	59	47
Dynammin	Dynammin-1	Homo sapiens	IDYN;2DYN;2X2E;2X2F;3SNH;3ZYC;3ZYS;4UUD;4UUK;5D3Q;6DLU;6DLV	98	69	28

Legionella pneumophila YsxC, “SSX” in EngB family, “SG/QX” in many translational and dynamin family proteins, “SCX/CAX” in the Arf family, “SGL” (4ZKD), “TAL” (2H5E), “SSK” (4U5X), “SCA” (1E0S), “SSA” (1MOZ), “NAT” (2ZEJ), “SQL” (3W6N), “STR” (3HYR) to “GVG” (2OG2), “GEG” (1OKK), “GVVNRNSQ” (1JWY/1JX2) in the dynamin family and “GXG/K” in SRP family. Thus, most identified G5 boxes, except GVG, GEG and GVVNRNSQ, appeared to have one mismatch from the “SAX” consensus.

Therefore, based on our observed variations in G box sequences, we defined that up to one mismatch was allowed in the consensus patterns for G1 box (GXXXXGK), G3 box (DXXG), G4 (NKXD) and G5 box (SAX) for each of the G protein families. The only exception was the allowance of two mismatches in the G4 boxes of the Roc, AIG-1, GB1/RHD3 and OBG families.

A doughnut plot (Fig. 2) and Table 3 represent the manually curated information for all members across all G protein families. In Fig. 2, each concentric circle represents consensus sequences of different G boxes, the longer the arc, higher the occurrence of that consensus sequence across most G protein families; for example, NKXD is the most prevalent (occurs in 55% of curated G proteins) G4 box (green circle) while SAX is the most prevalent (occurs in 40% of curated G proteins) G5 box (blue circle). Figure 2 also shows that in up to 40% of curated G proteins, the G5 box sequence was unknown.

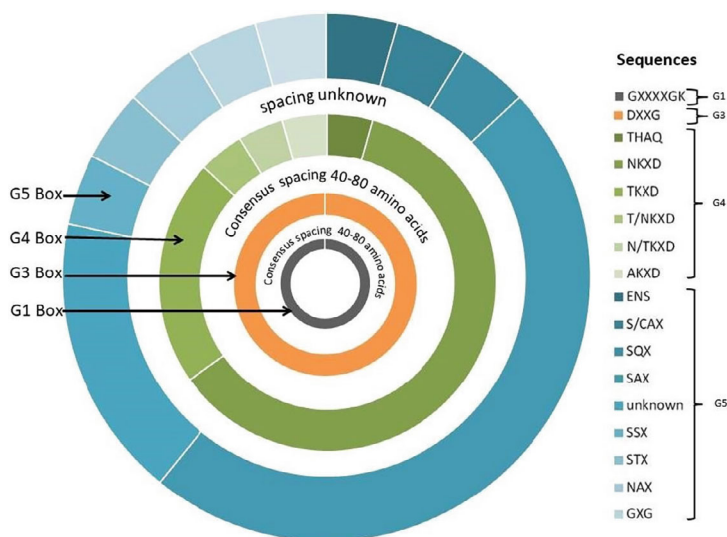


Fig. 2 Doughnut plot representing consensus spacing between adjacent G boxes and consensus G box sequences across G protein families, collected by manual curation. Each concentric circle represents consensus sequences of different G boxes. G1 box (GXXXXGK, innermost grey circle), G3 box (DXXG, orange), G4 box (different shades of green), G5 box (outermost circle, different shades of blue). The white space in between consecutive concentric circles represents the amino acid spacers between consecutive boxes

3.3 Prediction of GTP-Binding Domains by Spacers and Mismatch Algorithm (SMA)

An algorithm, named “Spacers and Mismatch Algorithm” (SMA), was written (Fig. 1) to predict G domains for the member proteins of the twenty-three different G protein families, using the manually curated G protein family-specific constraints for G domains (Table 3).

SMA Step1a predicted G domains in more than 75% of the member proteins in all the twenty-three G protein families [except for Galpha (50.4%), SAR (10%), IRG (40%) and SRP (42.4%)] (Table 5, columns 2 & 3). The rest of the proteins in each of the twenty-three G protein families probably did not give output due to differences in spacer length [e.g., in P24498 (Ras family), the spacer between G1 and G3 is 13 residues longer than the putative spacing (Table 3)]. However, for many experimentally studied G proteins, multiple possible G domains were predicted for a single protein due to various possible combinations of predicted G boxes (G1, G3, G4 and G5).

Figure 3 shows the individual G boxes, predicted using SMA, for two proteins each from Ras, Era and translational G protein families. These families were chosen to represent the findings because the G domains are very similar amongst homologs in the Ras protein family and are very diverse in the translational family while the Era family proteins have not been studied extensively. One protein from each family has an experimentally identified G domain (PDB structures available) while the other protein has no available structural information (Fig. 3). In the SMA-generated output for proteins with solved structures, one of the three outputs for Q62636 and one of the four outputs for Q5SM23 matched exactly with the experimentally identified G boxes. On the other hand, for Q71V39, the entire G domain (i.e., all 5 G boxes) was not predicted correctly in any of the 11 SMA outputs although 2 of the outputs were able to predict the G3, G4 and G5 boxes correctly. Overall, it appears that SMA predicts multiple G domains either due to many predicted G5 boxes or many predicted G1-G5 boxes.

It was thus noted that, as expected, most variability was observed in the prediction of the G5 box by SMA.

3.4 SMA Predicts Significantly Different G5 Box Motifs for Some G Protein Families

“SAX” has been accepted as the G5 box consensus sequence [1, 14] since most experimentally discovered G5 boxes, including the exceptions to “SAX” across all G protein families, have been found to have either Ser or Ala interacting with the guanine ring of the GTP molecule. However, structural studies have identified many variations in the G5 box motifs across G protein families. For example, “SGL” (4ZKD), “TAL” (2H5E), “SSK” (4U5X), “GVG” (2OG2), “GEG” (1OKK),

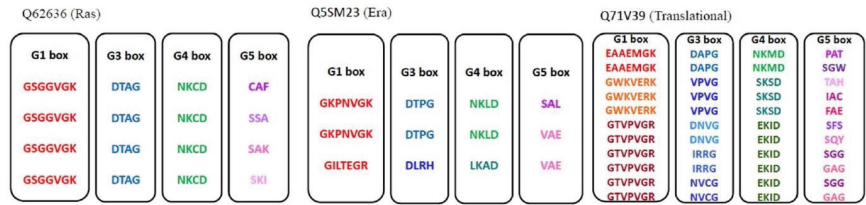
Table 5 Comparison of the number of proteins generating SMA output using “SAX” or the SMA3-predicted motif as G5 box sequence

G Protein Family	Output upon using “SAX” as G5 box motif		Output upon using SMA3-predicted G5 box motif	
	Number of proteins	Percentage of proteins	Number of proteins	Percentage of proteins
AIG-1	42	95.4	27	61.3
Arf	130	92.8	102	72.8
Dynamin	100	86.9	91	79.1
EngA	424	100	424	100
EngB	373	93	304	75.8
Era	174	92.5	154	81.9
FeoB	14	77.7	4	22.2
GB1	116	100	58	50
Hflx	26	100	20	76.9
IRG	4	40	3	30
OBG	658	99.8	613	93.01
Rab	442	94.04	424	90.21
Rho/Rac	101	78.9	63	49.2
Ran	28	96.5	28	96.5
Ras	170	89.4	168	88.4
Roc	19	100	18	94.7
Septin	77	89.5	68	79.06
SRP	42	42.4	44	44.4
Translational	2562	89.2	1433	49.9
TrmE	372	81.9	230	50.6

“SCA” (1E0S), “SSA” (1MOZ), “NAT” (2ZEJ), “SQL” (3W6N), “STR” (3HYR), “GVVNRSQ” (1JWY).

It is to be noted that Step 1a of SMA, which searches for a G domain in a protein sequence using the G5 box constraint of “SAX” with one mismatch within the specified G4-G5 spacer length, essentially boils down to searching for only one amino acid, i.e., either S or A. This is because one mismatch will replace either S or A in one independent search. The variability observed in G5 box prediction by SMA (Fig. 3) may be attributed to this obvious non-specificity in the search parameters. Thus, we decided to search for a more specific and probable family-specific consensus sequence for the G5 box. To increase the accuracy of G5 box prediction, the search space was increased by increasing the length of the G5 box search string from 3 to 7 residues. SMA Step 1b (See Methods for details), searched for all possible 7-mers after the predicted G4 box, that had either S or A in the centre and lay within the family-specified G4-G5 spacer range. Steps 2 and 3 of SMA were then performed

A Protein with structurally identified G domain



B Protein with no structural information

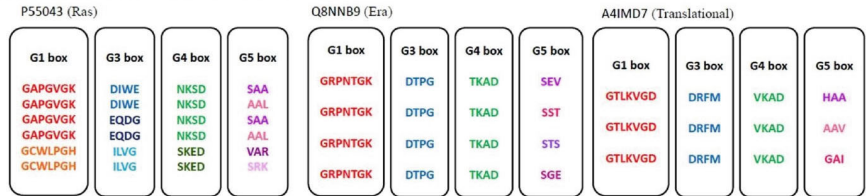


Fig. 3 Prediction of multiple G domains per protein. (A) Q62636 (Ras), Q5SM23 (Era) and Q71V39 (translational) are proteins with structurally identified G domains. (B) P55043(Ras), Q8NNB9 (Era) and A4IMD7 (translational) are proteins with no structural information. For each G box, different predicted motifs were depicted in different colours

to predict a more precise G5 box consensus sequence per G protein family (Fig. 1). Interestingly, the results indicated that there was no common 7-mer sequence that occurred in most members of all G protein families. Even closely related families such as Ras, Ran, Roc and Arf showed different 7-mer sequences. On the other hand, each G protein family (except translational, TrmE, SAR and SRP) had one or more 7-mers that occurred in > 25% of the proteins of an individual G protein family. This indicated that there was probably no consensus G5 motif that was conserved across all G protein families, unlike the consensus motifs for G1 (GXXXXGK), G3 (DXXG) and G4 (NKXD) boxes. An arbitrary threshold of 25% was chosen to ensure that the predicted 7-mer covered at least a quarter of the members in a given G protein family. The motif which had the highest coverage was selected as the new consensus G5 box sequence for the given G protein family. Also, if there were Xs at the two ends of the predicted 7-mer, they were trimmed to give a more concise G5 box consensus sequence. Thus, the predicted family-specific G5 box sequences had different lengths. For example, EXSAX in the Ras family, SXXSXXS in the Roc family and ADXP in the OBG family (Table 2).

In order to validate if SMA predicted G5 box motifs accurately, the output after SMA Step4 was matched with G5 box motif sequences observed in corresponding available G protein structures. As seen in Table 6, SMA predicted correct G5 box motifs for all G protein families which had structural evidence of “SAX” as the G5 box. (Ras, Ran, Rab, Arf, EngA, EngB, Era, HflX, FeoB, Miro). However, for G protein families which did not have “SAX” as the G5 box motif (OBG, Dynamin, GB1/RHD3, IRG, Rho/Rac, SAR, SRP, Translational) the SMA-predicted G5 box sequence did not match the G5 box sequences observed in available structures.

Table 6 Per cent reduction in total number of unique G boxes (sum of G1, G3, G4 and G5) predicted per family using the SMA3-generated G5 box sequence

G protein family	Per cent reduction in total number of unique G boxes predicted
AIG-1	76.65
Arf	52.86
Dynamin	48.16
EngA	20.39
EngB	55.94
Era	34.4
FeoB	67.75
GB1/RHD3	89.8
Hflx	50.61
IRG	68
OBG	54.04
Rab	48.83
Rho/Rac	57.7
Ran	0
Ras	28.23
Roc	64.97
Septin	51.05
SRP	60.32
Translational	49.1
TrmE	46.55

3.5 Comparison of G Domains Generated Using “SAX” as Consensus G5 Box Sequence Versus Using SMA-Predicted G5 Box Sequence

3.5.1 Prediction of Fewer False Positive G5 Boxes

SMA Step 4 (Fig. 1b) was performed to search for all possible G domains in each G protein sequence, using the family-specific constraints of consensus sequence, spacers and allowed mismatches listed in Table 3 along with the additional constraint of using the new G5 box sequences predicted in Step 3.

The number of predicted G5 boxes per G protein was considerably reduced (Fig. 4) upon using the SMA-predicted G5 box sequence (from SMA Step 3; referred to as “SMA3” from now on), as compared to the output using “SAX” as G5 box sequence (from SMA Step 1a; referred to as “SMA1a” from now on). The Ras (members have similar G domains), translational (members have diverse G domains) and Era (less structural information) protein families were chosen to represent the findings. Figure 4 illustrates that for individual G proteins (vertical line) from all three G

protein families, SMA3 (Fig. 4b) consistently reduced the number of predicted G5 boxes (red) and while the number of predicted G1 boxes (blue) remained similar.

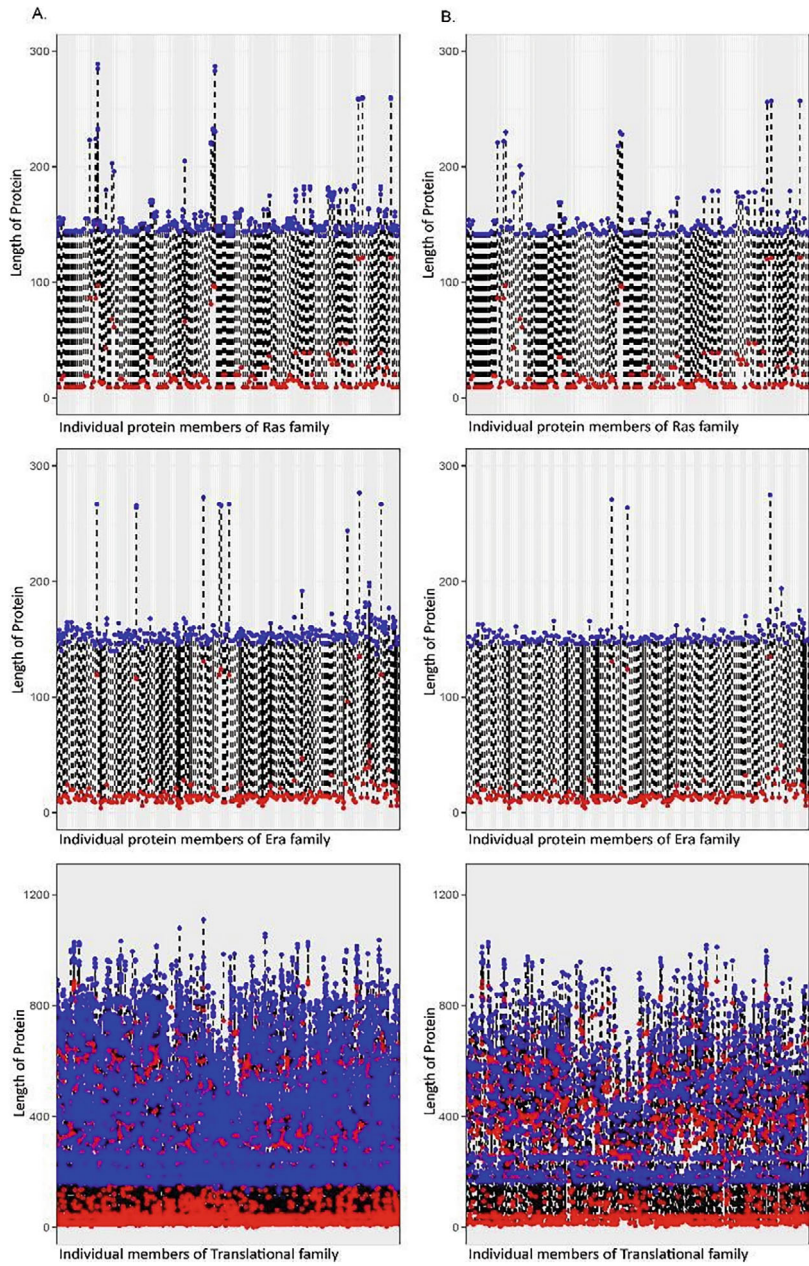
3.5.2 Prediction of Fewer Unique G Boxes Per G Protein Family

We then investigated if SMA Step 4 had any effect on the prediction of all G boxes along with the prediction of fewer G5 boxes per G protein family. Upon using the SMA3-predicted G5 box sequence (as compared to the output from SMA1a), the predicted number of unique G boxes (G1, G3, G4 and G5 boxes) were much less, i.e., $\leq 50\%$ in AIG-1, GB1/RHD3, Septin, IRG, Roc, EngB, FeoB, HflX, OBG, Rac, Arf and SRP families while no reduction was seen in the Ran family (Table 6). Figure 5 illustrates the prediction of fewer unique G boxes in Ras, translational and Era G protein families. Fewer unique G boxes are predicted either due to less proteins generating output upon using the SMA3-predicted G5 box sequence (in Arf, EngA, EngB, alpha, OBG, Rac and SAR families; Table 2, columns 4 and 5) or the prediction of a different G5 box sequence than the conventional “SAX” (in Galpha, GB1/RHD3, IRG, Roc families) or the experimentally reported “GXG” (in SRP family, Table 7).

3.5.3 Prediction of G Protein Family-Specific G1, G3 and G4 Box Consensus Sequences

SMA Step 3-predicted G protein family-specific G5 box consensus sequences that were previously not reported (Table 2). It was intriguing to then ask if there were also G protein family-specific G1, G3 and G4 box motifs beyond the consensus “GXXXXGK”, “DXXG”, “NKXD” sequences, respectively. In other words, can the consensus motifs be modified and more specifically, are there any family-specific patterns, which could possibly give a better representation of the “X” residues (where X = any of the twenty amino acids).

G1, G3 and G4 box sequences were predicted for all the proteins in each G protein family after SMA Step 4 and visualized as sequence logos. More distinct family-specific G box consensus motifs were predicted in most G protein families except AIG-1, GB1/RHD3, OBG, Roc and translational families. G protein family-specific G box consensus motifs of all the proteins in Ras, Era and translational protein families are shown in Fig. 6. For example, the first “X” in G3 box is found to be “T” in all three protein families whereas, the second “X” in G3 box can either be “P” or “A” (Fig. 6). Several interesting family-specific signatures were observed. For instance, the predicted G3 consensus was “DXXS/D” in the AIG-1 protein family but “D/WXXG” in Arf while the predicted G4 consensus was “TKXD” in Dynamin, SRP, Rac and EngB G protein families and “AKXD” in Septin (correlates with corresponding PDB structures). Some families had very specific predicted G box consensus motifs. For example, G3 box (DLPG) in FeoB, G1 (GESGAGK) and G4 (TKVD) boxes in IRG and all four G boxes (GDGGTGK, DTAG, NKVD and



◀**Fig. 4** Comparison of predicted G domain boundaries (in Ras, Era and translational G protein families), before and after SMA Step 3. Number of G domains predicted per protein (A) using “SAX” as G5 box (B) using newly identified G5 box after SMA-Step3. The X axis represents all the proteins in the G protein family. Each vertical line on the X axis represents each individual protein of the family. The Y axis represents the length of proteins (amino acids). Red dot marks the start of the predicted G1 box and blue dot marks the start of the predicted G5 box. Each dotted line represents an individual G domain predicted per protein

Table 7 Comparison of percentage of proteins per G protein family that have the indicated spacer value, before and after SMA3

Spacers between G boxes	Comparison of values obtained using SMA3-predicted G5 box sequence versus using “SAX” as G5 box sequence	
	Similar	Different
G1-G3 box	AIG-1, Arf, Dynamin, EngA, EngB, Era, GB1, IRG, OBG, Rab, Rac, Ran, Ras, Roc, Septin, translational, TrmE	Hflx, FeoB and SRP
G3-G4 box	AIG-1, Arf, Dynamin, EngA, EngB, Era, GB1, HflX, IRG, OBG, Rab, Rac, Ran, Ras, Roc, Septin, translational, TrmE	FeoB, SRP
G4-G5 box	Dynamin, EngA, FeoB, GB1, Roc, translational	AIG-1, Arf, EngB, Era, IRG, HflX, OBG, Rab, Rac, Ran, Ras, Septin, SRP, TrmE

SAKSN) in Ran. The diversity of predicted G boxes in the translational family is consistent with the documented sequence diversity of its member proteins; similar G box diversity is also seen in Roc, OBG, AIG-1 and GB1 families (Table 1). On the other hand, the similarity of predicted G boxes in the less-studied Era family members may imply greater homology in their G domains since Ras family members, which have high homology, also appear to have very similar predicted G boxes.

To test the specificity of the SMA output, it was validated by using proteins, other than canonical GTP-binding proteins, as input. For example, no output (after SMA Step 1a) was generated when haemoglobin chain A or haemoglobin chain B were used as query sequences. Moreover, ATP binding proteins (e.g., RBBA protein in *E. coli*) which have a P loop (G1 box) but no other G box, do not generate any output. In proteins like human OLA1 which bind to both ATP and GTP, the generated output has one combination of predicted G boxes (G domain) that binds GTP.

To maximize the number of predicted G domains per G protein, the family-specific G box consensus sequence and inter-G box spacer constraints used to run SMA were kept broad. To test the significance of the SMA results and to ensure that the broad constraints did not end up predicting G domains in any string, a negative control experiment was performed.

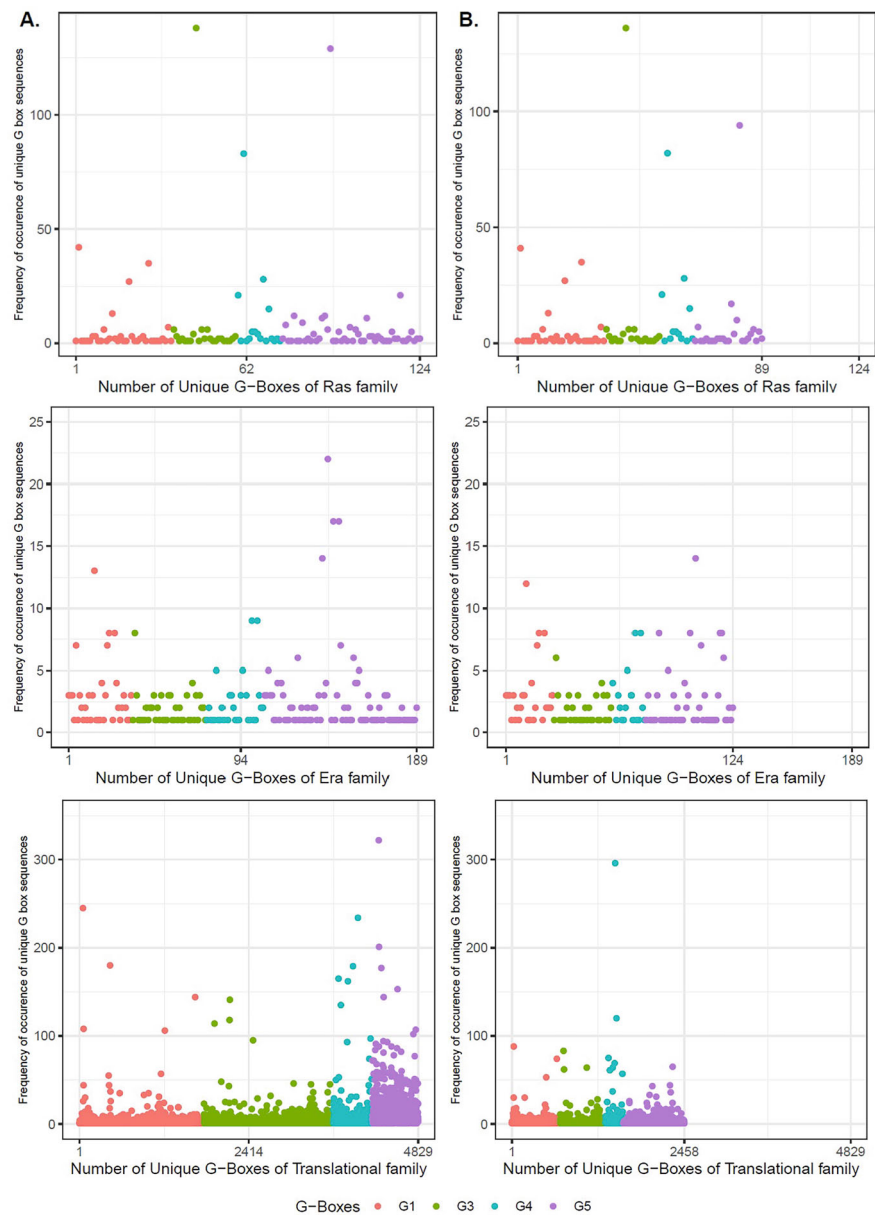


Fig. 5 Comparison of unique G boxes (in Ras, Era and translational G protein families) predicted before and after SMA Step 3. Number of unique G box sequences predicted (A) using “SAX” as G5 box (B) using newly identified G5 box after SMA Step 3. X axis = total number of unique G boxes predicted in a G protein family; Y axis = frequency of occurrence, i.e., the number of proteins (in G protein family), in which the predicted unique G box occurs). G1 (peach), G3 (green), G4(blue), G5(purple)

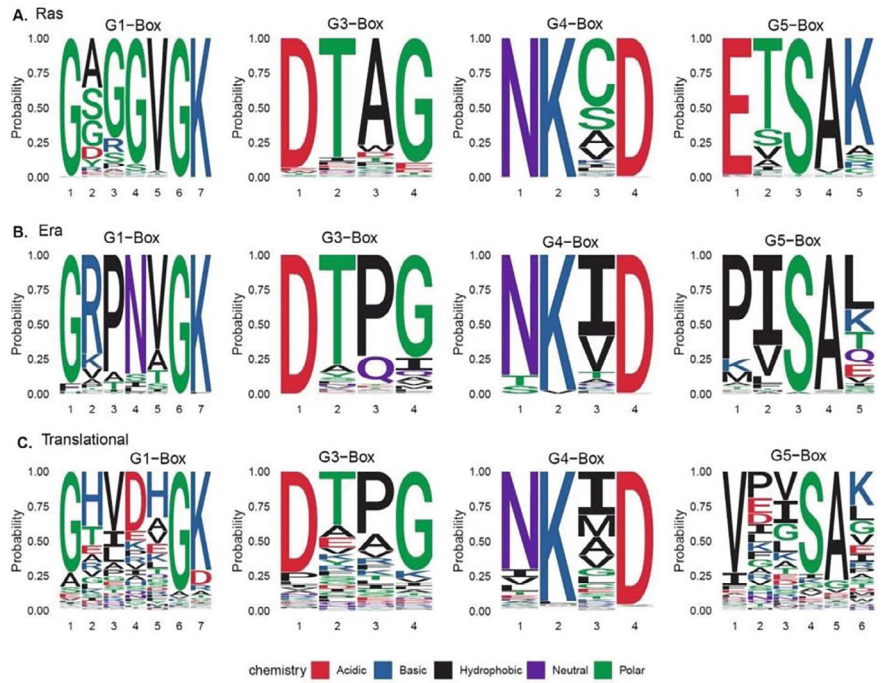


Fig. 6 G protein family-specific G box motifs predicted using SMA3. **A** Ras, **B** Era **C** translational families. X axis = amino acid positions in respective G box; Y axis = probability of occurrence of an amino acid from 0 to 1

Figure 7 illustrates that after shuffling the protein sequences, the number of G domains predicted per sequence (red) becomes significantly less. This further validates the specificity of SMA prediction.

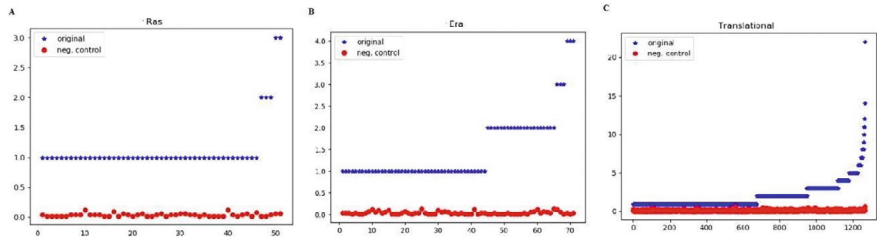


Fig. 7 Significance testing of predicted G domains. **A** Ras, **B** Era **C** translational families. X axis = proteins sorted in the order of number of matches obtained; Y axis = number of predicted G domains [in actual protein sequence (blue) and average of no. of G domains identified after 50 shuffles (red)]

3.6 The Predicted G Box Sequences Does not Depend on the Size of the G Protein Family

Figure 6 demonstrates that for certain G boxes in some G protein families, there is significant diversity in individual positions within each box (e.g., translational family). It can be argued that this observed diversity may be due to the varying sizes of the different G protein families, i.e., a protein family with more members would show more variation in G box motifs merely because of larger representation than a protein family with fewer members. To probe whether the diversity of predicted G box sequences was a function of the size (number of member proteins) of a G protein family, the entropy of individual G boxes was calculated for each family. The calculated entropy values gave insights about how random the “X” position in each of the G boxes (i.e., “GXXXXGK” for G1, “DXXG” for G3 and “NKXD” for G4) is and how randomness of individual “X” positions impact the total entropy of the G Box.

Figure 8 investigates the correlation between the total entropy of each G box and the size of each G protein family. The translational family is the largest G protein family with 2870 member proteins; however, the entropy calculated for each of its G boxes is not significantly high. On the other hand, small G protein families like Roc, with only 20 members, demonstrate the highest entropy for G1, G3 and G5 boxes while the GB1/RHD3 family, with 50 members, has the highest entropy for G4 box. The IRG (11 members), FeoB (19 members) and Ran (30 members) families have the lowest entropy for G1, G4 and G3, G5 boxes, respectively, despite having similar sizes to that of Roc and IRG families (Fig. 8). Thus, it appears that the observed variation in G box motifs does not correlate with the size of the protein family but maybe because of other properties of individual proteins.

4 Discussion

The GTP-binding domains of all G proteins have been characterized by a set of consensus G box sequences (G1-G5) and inter-box spacers [2, 14]. However, with the recent identification of many G protein families, it is observed that there is a vast diversity in the sequence of G proteins. This diversity is reflected in the sequences of the G boxes and more specifically in inter-G box spacer length and the G5 box consensus sequence. Here, we present a tool called SMA, which can identify G protein family-specific G box sequences and inter-G box spacer length, based on the user-defined constraints. SMA is a one-of-a-kind algorithm (Fig. 1a, b, c), which can perform an approximate search for multiple sequences occurring with a given range of gaps between them. In other words, unlike tools like GTP binder [9], ELM [8], ScanProsite [10], Pfam [11], SCANMOT [12], MEME/GLAM2 [13]. SMA offers the advantage of using multiple proteins in the input file and searches for all G boxes simultaneously instead of one after the other. It uses consensus patterns with allowed

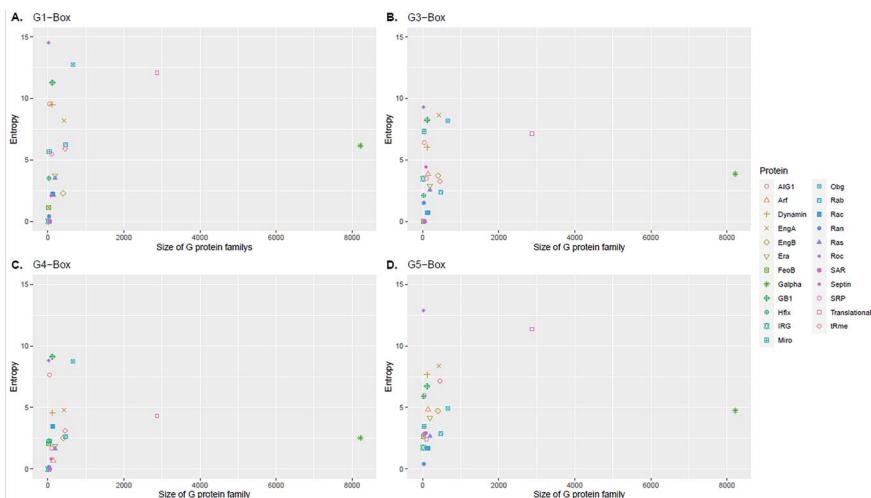


Fig. 8 Correlation between G box entropy and G protein family size: Scatter plots where the X axis represents the total number of proteins (size) per G protein family and the Y axis represents total entropy of each G box **A** G1 box (GXXXXGK), **B** G3 box (DXXG), **C** G4 box (NKXD) and **D** G5 box predicted by SMA. Each G protein family is represented by a different colour as shown in the key

variations as well as maximum and minimum gaps allowed between each pattern. Moreover, all the constraints in the algorithm are customizable and the user has the flexibility to change the G box sequence including the number of allowed mismatches as well as the inter-G box spacer length. This is especially useful, if the protein of interest has very low amino acid sequence similarity with well-studied GTP-binding proteins and/or is less studied, i.e., little or no information is available from structural or biochemical studies. The flexibility provided in the SMA will also be useful to identify any specific mutations in the G box sequences which is referred to in the literature. The absence of a database containing mutated protein sequences limits the testing of SMA for the same. However, one can predict disease-causing mutations if we use manually edited protein sequences of *Gsa* and *Gia*; leading to cholera and whooping cough [15]. It is to be noted that SMA is a generalized algorithm, which is not restricted to finding amino acid motifs in protein sequences. The algorithm can be appropriately modified to identify patterns in any sequence including those of biological importance like DNA or RNA.

An exhaustive analysis of twenty-three different G protein families demonstrated that G box sequences and inter-G box gaps are often class-specific. Thus, it is proposed that G protein families could possibly be classified based on family-specific G box sequences and inter-G box spacer lengths. Moreover, SMA was able to come up with better representations of the family-specific G box motifs. For instance, it was possible to predict if the “X”s in the established G1 box motif (GXXXXGKS/T) could be better represented by a specific set of residues rather than any of the

twenty residues (Fig. 6). SMA was also able to predict G5 boxes for the different G protein families. This was significant since the G5 box marks the carboxy-terminal boundary of a G domain and hence demonstrates that SMA can be used to predict more precise boundaries of a G domain, within a large multi-domain protein.

The EngA/Der/YfgK/YphC family is characterized by two tandemly occurring GTPase domains separated by an acidic linker [16]. SMA's ability to independently predict two distinct domains along with corresponding G boxes in individual EngA members provides confidence in the method. Moreover, the accuracy of SMA prediction was further validated by the fact that no G domain was predicted in the centre of the EngA proteins.

It is also interesting to note that some G proteins have different insertions within the G domain. For example, eIF-2 γ has a zinc ribbon insertion between the G2 and G3 boxes [3] while AIG-1 proteins have a hydrophobic motif insertion between G3 and G4 boxes [17]. SMA predicts structurally identified G boxes and the corresponding G domain for some of the AIG-1 and eIF2 proteins. SMA predictions for these families may improve as more structural information becomes available and the SMA parameters can be changed accordingly. Detailed analysis of SMA results can also be used to explore interesting evolutionary relationships amongst homologs of the same protein. For example, Rho1 protein in *E. histolytica*, *B. vulgaris* and *P. sativum* does not give any output with specified G box sequences and inter-G box spacers. This is because, in *E. histolytica* Rho1, the putative G5 box (EASSV) is only 28 amino acids away from the putative G4 box (LKVD) which differs from the defined G4-G5 spacer range of 40–50 amino acids, that is present in most Rho1 homologs (Table 3). Rho1 from *B. vulgaris* and *P. sativum* lacks a putative G3 (NKXD) box and the putative G5 box (ECSSK) is 37 amino acids away from the putative G4 box (TKLD) which differs from the defined G4-G5 spacer range (40–50 amino acids). On the other hand, SMA predicts identical G1 (GDGACGK), G3 (DTAG) and G5 (ECSAK) boxes in Rho1 homologs from *D. melanogaster*, *A. gossypii*, *C. elegans*, *S. pombe*, *K. lactis*, *S. cerevisiae* and *C. albicans* but different G4 box sequence (NKXD in *D. melanogaster* and *C. elegans* and CKXD in *A. gossypii*, *S. pombe*, *K. lactis*, *S. cerevisiae* and *C. albicans*). These observations could be further investigated to understand the nucleotide-binding properties of the different Rho1 homologs.

Our current observations are, however, biased by the PROSITE classification of G protein families, our manual curation of family-specific G box sequences and our imposed constraints for allowed mismatches in individual G box sequences and inter-G box spacer length. Thus, any exceptions to the above constraints will be missed and may need to be changed, as more structural information is obtained about members of different G protein families. Moreover, it is to be noted that SMA had an inherent bias in G5 box prediction since it only looked for patterns (with allowed mismatches) which had “SAX” in the centre. Also, the G5 boxes that are identified using a 25% threshold, miss under-represented putative G5 box sequences. This is specifically important for less-studied families such as AIG-1, Era, GB1/RHD3, IRG and Roc, in which there is very little available structural information for individual members. On the other hand, for families like Ras, for which there is a wealth of structures, our imposed constraints may stand the test of time.

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Applications

Non-immersive Virtual Reality Framework for the Rehabilitation of Children with Cerebral Palsy



Agrawal Luckykumar Dwarkadas, Viswanath Talasila,
Rama Krishna Challa, and K. G. Srinivasa

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Abstract Numerous children in the world have Cerebral Palsy (CP). It is a disorder that results in impairment or disability due to neurological deficits. Physical therapy has been found to enhance the quality of life for individuals with CP but lacks motivation, which is a crucial factor in rehabilitating CP children. Existing literature shows that Virtual Reality (VR) increases the motivation in CP children. The framework's design is important in developing VR systems for rehabilitating CP children. The available studies have yet to expand the framework part of the VR system. Also, the existing VR systems are not designed according to the needs of the CP children and therapists. This paper contributes a novel framework for developing a VR system for rehabilitating CP children. The proposed framework is to develop a VR system according to the needs of the CP children and therapists that aims to improve the functional mobility of CP children. Future work can include creating a VR system and its validation with CP children through intervention.

Keywords Virtual reality · Cerebral palsy · Non-immersive · Architecture · Rehabilitation

A. L. Dwarkadas (✉) · R. K. Challa
National Institute of Technical Teachers Training and Research, Sector 26, Chandigarh 160019,
India
e-mail: luckykumar.cse21@nitttrchd.ac.in

V. Talasila
Ramaiah Institute of Technology, MSR Nagar, Bengaluru, Karnataka 560054, India

K. G. Srinivasa
International Institute of Information Technology, Naya Raipur, Chattisgarh 493661, India

1 Introduction

Cerebral palsy (CP) is a group of diseases that affect children and are caused by brain damage. The primary difficulty with CP is a movement impairment that coexists with sensory, perception, cognition, communication, and behavior challenges, in addition to severe epilepsy and musculoskeletal disorders [1]. The number of CP patients worldwide is seventeen million, which is expected to rise [2]. In India, there are 2.95 cases of CP for every 1000 children [3]. CP does not currently have a perpetual treatment. Therefore, the primary therapy goals for CP children are to increase their capacity for mobility, prevent complications, improve their standard of life, strengthen weak muscles, and preserve their existing functional state [4]. CP-related disorders impact not only the patient's daily activities but also the members of their family [5]. Different types of conventional therapies are used for treating children with CP. These therapies lack motivation, one of the critical factors for rehabilitating CP children. It might be difficult for doctors to keep CP children motivated to continue their treatment because it is a prolonged process. This resulted in less participation of CP children in therapy [6].

Recent advances in healthcare have been among the most innovative in recent years. The formerly exclusive domain of physicians and patients has recently added a new member: technology. Virtual reality (VR) is an emerging technology in rehabilitating patients suffering from various diseases [7]. It provided positive rehabilitative outcomes through practice, tactile feedback, multimodal stimulation, and motivation [8]. VR is a new connecting technique with the ever-expanding digital world [9]. It can be categorized as immersive, semi-immersive, or non-immersive. In non-immersive VR, users are least immersed in the virtual world and have more contact with the outside world [10]. The functional ability and growth of CP children are impacted by low motivation. Additionally, it reduces the efficacy of treatment [11].

VR therapy sessions increase children's motivation, happiness, involvement in therapy, realism, and concentration without regard to location or time limitation [12]. VR significantly enhances the functional consequences of CP children through neuronal restructuring and neuroplastic modifications. VR makes it possible to provide performance-based tactile feedback along with guidance information, which is crucial for CP children's motor improvement [13]. With new dynamic technologies in VR, children may engage with their bodies to inspire and enjoy themselves [14]. More research is required to understand further VR utilization by children with CP [3]. It has been proven through several literature studies [15–18] that using VR for treating CP children improves their posture and movement abilities. These benefits of using VR in rehabilitating CP children motivated the authors to conduct this study.

Given the aforementioned, a novel framework is proposed to develop a system for rehabilitating CP children through non-immersive VR technology and hardware such as sensors. Although both technologies have an influence, they enable themselves to have a more substantial presence and effect on the daily lives of healthcare professionals and patients. These technologies have a tremendous opportunity to transform how the health ecosystem engages with the common people. The main objective of

this paper is to propose a framework that can be used to develop a non-immersive VR system according to the needs of CP children and therapists to improve the functional mobility of CP children.

The remaining part of the article is structured as follows: The background consists of related work discussing the current research gaps in existing studies and the architecture of the VR system, and its components are described in Section II. The proposed framework for rehabilitating CP children and its components are presented in Section III, and Section IV provides a discussion and concludes the paper with limitations and future work.

2 Background

2.1 Related Work

Rapidly evolving technology has prompted customization and the use of modern engineering methodologies to satisfy the requirements of new systems. A key component in the development of VR systems for the rehabilitation of CP children is its framework. The available literature suggests that more research is required to address this point specifically. Avcil et al. [5] provided an interactive gaming environment for rehabilitating CP children using Nintendo Wii and Leap Motion Controller (LMC). The CP children played the different Nintendo Wii and LMC games. The findings show that non-immersive VR therapy exhibited considerably better impacts on manual dexterity in CP children than conventional therapy and can be considered an alternative treatment. In this paper, the authors have not expanded the framework part of the system.

Parisaca et al. [8] proposed an architecture for rehabilitating patients suffering from motor impairment. The architecture demonstrates its efficacy by utilizing and integrating multiple tracking devices and gaming interfaces. Herrera et al. [19] designed a framework for developing a VR system to rehabilitate patients' upper limbs. The proposed framework provides basic features for handling objects in virtual worlds while considering patient constraints. It provides a service of capturing movement data that therapists can use to assess patients' progress and recognize the areas of enhancement. However, this paper does not address the framework for rehabilitating CP children.

De Oliveira et al. [20] proposed a therapeutic tool for rehabilitating CP children. They developed a VR environment with the combination of Leap Motion and Mind-Wave that provides the real-time evaluation of CP children. This helps therapists to correlate the attention level of CP children with the development of their clinical condition. The main focus of this paper is the attention level of CP children rather than their functional mobility.

Rahman [21] created a multimedia environment that can monitor different movements of CP children captured during an intervention using commercially available

Kinect sensors with the 3D virtual world. The data captured is used to analyze the improvement in CP children. Therapists tested the preliminary framework and provided highly encouraging and positive feedback. The author has used inbuilt commercialized available Kinect sensors for creating the multimedia environment, which is not customizable according to the needs of the CP children and therapists.

Tovar et al. [22] designed a telerehabilitation system for treating CP children. This system utilized interactive virtual worlds and biomedical technology. The therapists may interactively and user-friendly change and adjust difficulties for each therapeutic session, motivating each patient to keep improving. The findings revealed that CP children with difficulty functioning in their upper or lower extremities improved.

To address all research gaps discussed in the above studies, a framework is proposed in this article to develop a non-immersive VR system for rehabilitating CP children. This framework is designed specifically for rehabilitating CP children according to the requirements of the CP children and therapists. The following subsection discusses the VR system's typical architecture, hardware, and software components.

2.2 Architecture of VR System

For both scientific and generic purposes, architecture is crucial. Architecture can prevent or promote desirable qualities. Considerations made while designing an architecture enable reasoning and handling modifications as the system develops. Evaluation of an architecture facilitates initial system quality anticipation. An architecture that is documented facilitates improved interaction between involved parties. Architecture posed a set of restrictions on the subsequent deployment. It is the primary document that enables the developer and project manager to make decisions regarding budget and schedule. Instead of only generating components, architecture-based development emphasizes their assembly [23].

VR can be applied in various contexts, including scientific visualization, social interaction, training, psychological treatment, and entertainment. It is also used in a wide range of other domains. Considering numerous applications, designing an individual VR system that satisfies everyone's needs is difficult. Therefore, monolithic systems have been developed, specifically designed for particular uses, and have extremely low component reusability for several applications [24]. The general architecture of a single VR system is shown in Fig. 1.

In a single-user VR system, a user enters the virtual environment through VR devices, namely headsets, gloves, and so forth. The interchange of sensory information, such as visual, aural, tactile, haptic, and so on, is transferred to and from the user through an interface to the VR engine. The VR engine analyzes this information and responds to the user using an Environment Database (ED). The process of creating the virtual environment and interacting with the information provided by the sensory interface on time is distributed among the reality generator and real-time operating system of the VR engine [25].

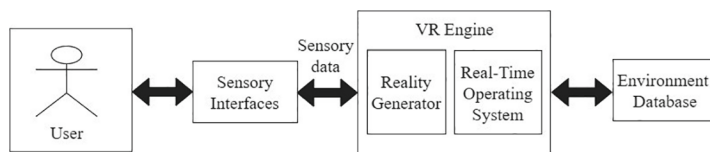
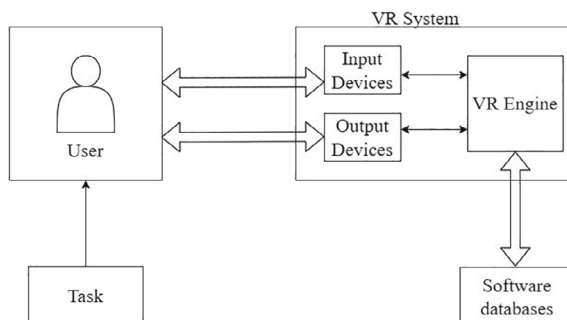


Fig. 1 Typical architecture of a single VR system [25]

Fig. 2 Hardware and software components of the VR system [27]



The VR system contains two main parts: hardware and software. The hardware part is categorized into input–output devices and VR engines [10]. Figure 2 shows typical hardware and software parts of the VR system.

Input Devices: The input devices are the user’s tools to interface with the virtual environment [10]. Input devices are categorized further into manual operation and automatic tracking. Manual operations use devices such as a keyboard, mouse, and so on. Automatic tracking uses devices such as infrared (IR) sensors, accelerometers, cameras, depth cameras, gyroscopes, microphones, and magnetic sensors [26].

Output Devices: The output components obtain feedback from the VR engine and transfer it to the users with suitable output devices, enabling their senses. It is categorized into visual, aural, haptic, smell, and taste [10]. Examples of output devices are stereo display monitors, head-mounted displays (HMD), holographic devices, CAVE screens, and projectors [26].

VR Engine: The responsibilities of the VR engine are computing and producing graphical models, as well as object modeling, lighting, mapping, texture, simulations, and real-time projection. The VR engine can consist of a standard PC with additional processing power and an advanced graphics processor. It could be a group of distributed computer systems connected by a fast network [10].

Software and Databases: Input–output system maintenance, improvement and feedback, and data processing are handled by software [27], and information is saved into databases. VR modeling tools, namely 3DS Max, Maya, GL Studio, Blender, Unity 3D, and Creator are used. VR development tools include VR authoring tools, SDK, and application program interfaces (APIs) [10]. VR simulation software such

as OpenSim, Ogoglio, and Flexsim DS are also used. Creative Wavestudio, Gold-wave, Media Digitalizer, and mp3DirectCut are some examples of sound editing software [28].

3 Proposed Framework

As shown in Fig. 3, the framework is to develop a non-immersive VR system for improving the functional mobility of CP children described in this section.

The proposed framework is developed using different components such as a training simulator, which is basically the VR environment or game designed through the Unity 3D engine, hardware devices such as sensors that are attached to CP children for capturing movement data, a VR client for transferring data through a network

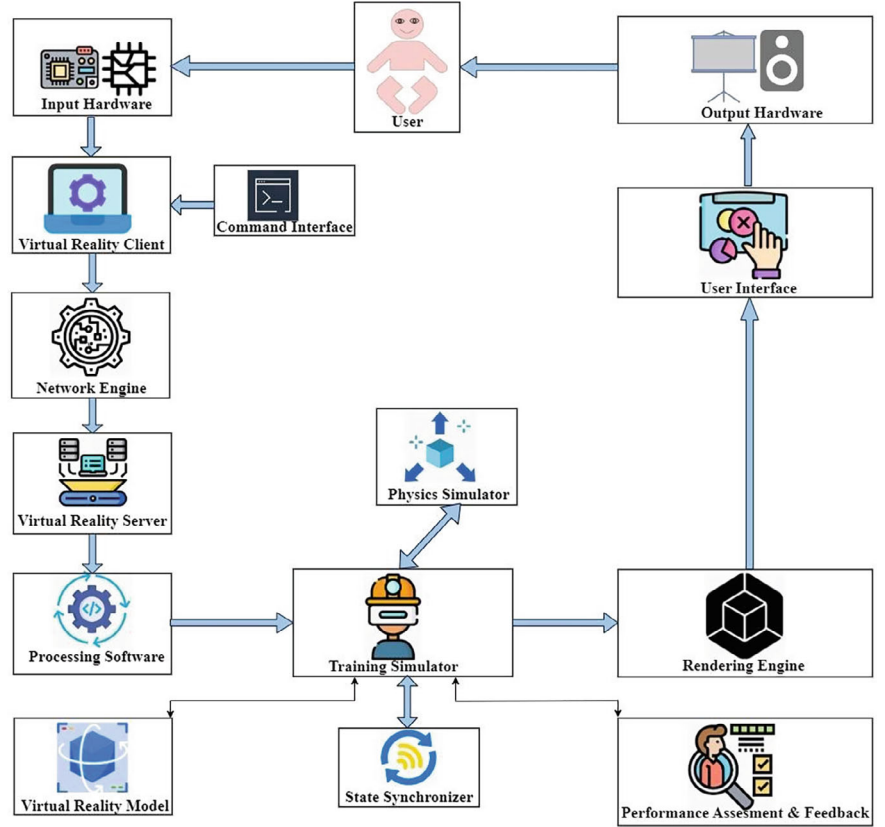


Fig. 3 Proposed framework for non-immersive virtual reality system to treat children with cerebral palsy

engine using UDP protocol, and a VR server that receives data. It also consists of software using C# language for processing data, a command interface that transforms the data into meaningful information, a rendering engine for rendering the data, which gives output in the form of a user interface, and output hardware such as a projector and sound box.

The following subsections explain the components used to design the proposed framework in further detail.

3.1 Input Hardware

The hardware used for tracking the movement of CP children is a Printed Circuit Board (PCB) consisting of Inertial Measurement Unit (IMU) sensors, a microcontroller, and other additional components, such as a battery. The Bosch BNO055 IMU sensor and ESP32 microcontroller are used in designing the input hardware component of the proposed framework. The BNO055 IMU provides a System in Package (SiP) solution that combines a triaxial accelerometer, triaxial gyroscope, and triaxial geomagnetic sensor.

The ESP32 microcontroller executes sensor fusion software that provides complete orientation of the module—yaw, pitch, and roll data in degrees. The ESP32 is a 32-bit dual-core microcontroller with an operating frequency of 80 MHz and integrated wireless internet and Bluetooth. I2C controllers provide the link between the microcontroller and the IMU [29]. The data captured through PCB is now transferred to the VR client for further processing.

3.2 VR Client

The VR client consists of a subcomponent called the command interface. The data is continuously received from the input hardware in the Python script running on the command interface in the VR client. The connection is established between the input hardware and the command interface of the VR client through the wireless hotspot. The Python script is executed by the PyCharm software, which was mainly designed for it.

The command interface transforms the data into meaningful information. It converts the data into string values combined to make a single string containing all the data captured by the input hardware. This data, in the form of a single string, is sent to the network engine for processing.

3.3 *Network Engine*

The network engine receives the data from the VR client and processes it using User Datagram Protocol (UDP). UDP is the basic transport layer protocol in the TCP/IP protocol stack. This protocol has no prerequisite for establishing a link before the data transfer. The sender does not wait for any response since a recipient does not provide an acknowledgment of a packet received over UDP.

The procedure is flawed, making it less dependable and more simple to process. UDP is selected for the data transfer since it enables packets to be eliminated rather than processing delayed ones and provides rapid communication. Through UDP, the network engine transfers the data wireless or wired to the VR server for further processing.

3.4 *VR Server*

The VR client knows which IP address the data will be sent to and which port to target since it uses UDP in the network engine for data transfer. The data received from the VR client are processed with the Unity engine in the VR server. Unity is a robust game engine used to design 2D or 3D graphics. It contains different features such as strong real-time rendering, top-notch visuals, cross-platform interoperability, scripting abilities, and extensive community support.

The abilities of basic animation, graphics controllers, and blend trees make Unity an ideal platform for 3D animation [30]. The VR server sent the data to processing software for further processing.

3.5 *Processing Software*

Unity engine does not have an inbuilt feature for processing the data. Hence, data received from the VR server is processed through processing software in Unity, which uses C# script. The C# script handles the game objects for every kind of movement in Unity. The low-level engine program in Unity is linked by C# script, a standard from the ECMA that is identical to Java script and made prominent by Microsoft's Net Framework.

Unity serializes these scripts, after which it utilizes the data in the script to design the GUI for the component. This view has made it simple for level developers to send details from the scene to their controller logic, making it simpler to create interactive environments. The data from the processing software is now transferred to the training simulator for the next processing step.

3.6 Training Simulator

The training simulator analyzes the data received from the processing software. It is basically a VR environment or game that maintains VR interaction with the users.

It comprises subcomponents such as a VR model, a state synchronizer, a physics simulator, and performance assessment and feedback. The VR model provides an immersive experience to the users. The state synchronizer synchronizes different states in the VR environment or game together and gives the output. It synchronizes all the tasks in the VR environment or game.

The physics simulator is used to simulate the laws of physics in Unity, like collision detection and object control in the real world. The VR environment or game works on the rules of the physics simulator through the physics engine. An object can be created with realistic behavior by modifying a few characteristics. It can be given the properties of an automobile, machine, or simply a single fiber of cloth by handling the physics from scripts. The training simulator also sent data to the performance assessment and feedback component to assess the performance of the patients after the VR intervention was completed. It provides feedback from the doctor about the VR therapy given to patients and monitors their progress. The data is further transferred to the rendering engine for processing.

3.7 Rendering Engine

The rendering engine processes the data received from the training simulator and its subcomponents. Real-time rendering is commonly used in the VR environment or games designed. It produces graphics at a highly rapid rate when users interact with the VR environment or games to give the impression that the scenes, which are made up of several images, are happening in real time.

Due to this, real-time rendering depends heavily on interaction and speed. For instance, if the character has to move in the scene, its movement should be updated before producing the next frame, such that it appears to be moving at an intensity that the human eye recognizes as normal. After the rendering process, the output is generated as a user interface.

3.8 User Interface

The rendering engine processes the data and produces the output as a user interface. The CP children interact with the VR environment or game designed for rehabilitation through the user interface. The interface is designed with the goal of encouraging children seamlessly. It is simple and user-friendly. This user interface is displayed through output hardware.

3.9 Output Hardware

The output hardware can be a projector and sound system. The CP children can use the user interface through an HD screen projected in front of them by the projector. They interact easily with the VR environment or play games by looking at the screen. The audio in the VR environment or game is produced by the good quality sound system attached to the projector.

3.10 User

The user is a CP child. Both input and output are given to the CP children. The input is in the form of input hardware attached to the CP children's body. The input hardware was designed to be lightweight so that CP children can wear it easily and do not face any problems during interaction with the VR environment or the game. The output is in the form of a VR environment or game designed for rehabilitation, which is displayed through output hardware such as a projector.

4 Discussion and Conclusion

Technological innovation has recently transformed all facets of healthcare rehabilitation, from advancements in providing cutting-edge therapies to the real-life execution of targeted interventions. The use of VR is becoming popular in the rehabilitation of patients. This results from its efficacy and solutions, particularly to patients and therapists. This paper proposed a framework to develop a non-immersive VR system according to the needs of CP children and therapists for rehabilitating children with CP. The development of the proposed framework considered the characteristics that VR systems should have for treating CP children. Non-immersive VR system designed using this proposed framework to improve the functional mobility of CP children may serve as a supplement to traditional therapy.

The proposed framework to develop a non-immersive VR system can assist in investigating evidence linked to the efficiency of VR-based treatments in CP children, which are presently constrained. Many studies presented various approaches for rehabilitation problems, but the framework part is not expanded. A framework is essential for numerous reasons, both general and scientific. Considerations made while designing the framework enable thinking and handling changes as the system evolves. The current work contributes by emphasizing this part of the VR system for rehabilitating CP children. The development received continuous guidance from physiotherapists in the Association of People with Disability (APD), Bengaluru.

The proposed framework consists of both hardware and software components. The hardware components collect the data, and the VR client transfers data to the

VR server via the network engine component. The data is processed using processing software, which then sends the data to the training simulator component. The training simulator transfers the data to the rendering engine, which gives output as a user interface. The interface should be created with the goal of encouraging good usability.

In the proposed framework, physiotherapists can utilize the collected data on CP children's functional mobility to evaluate their improvement critically. They can also evaluate the data over time to monitor their progress and find areas of improvement. Even though the primary objective of this framework is to develop a non-immersive VR system for rehabilitating CP children, the collected data on CP children may be used for research. They can be shared among other researchers without compromising the privacy of CP children.

The commercially available VR systems, such as Microsoft Kinect Xbox 360, were not initially designed to rehabilitate CP children [31]. The proposed framework to develop a non-immersive VR system is explicitly designed for CP children's rehabilitation and according to the therapist's requirements. It has several advantages for CP children and therapists. It will enhance CP children's participation in therapy and monitor their development.

The limitation of the current research is that the proposed framework has yet to be implemented to develop a non-immersive VR system and has yet to be validated. In the future, a non-immersive VR system will be created using the proposed framework to improve the functional mobility of CP children. The system will be validated through intervention on CP subjects.

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SkInspection: Recognizing the Type of Skin Cancer Using Deep Learning



Sanjana Vashdev Asrani, Jessica Lalchandani, Isha Desai, Trishala Jeswani, and Sujata Khedkar

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Abstract Skin cancer poses a significant threat to public health, with fatalities increasing due to insufficient awareness of symptoms and preventive measures. Early detection, especially of the most severe type, Melanoma, is pivotal for effective treatment. Leveraging deep learning techniques, our proposed system aims to autonomously provide the functionality by offering convenient image upload options, accurately predicting skin cancer types, educating users with facts about each type, and providing tailored guidance for the next steps, empowering users to make informed decisions regarding their skin health. Convolutional Neural Networks (CNNs) were chosen as the underlying deep learning methodology, renowned for their superior performance in visual imaging tasks. Developed in Python, utilizing TensorFlow and Keras, our CNN-based classification model was rigorously tested on the HAM10000 (Human against Machine) dataset. Evaluation metrics encompassed precision, recall, F1 score, ROC curve, and Confusion matrix. Our model achieved a noteworthy F1 score of 0.77% (weighted average), demonstrating its promise in accurate classification. Importantly, the Area Under the Curve (AUC) values for each cancer type came out to be, for akiec: 0.83, bcc: 0.83, bkl: 0.75, df: 0.94, mel: 0.93, nv: 0.99, vasc: 0.78 underscore its potential as a versatile tool for early skin cancer detection, with far-reaching implications for improving patient outcomes.

Keywords Skin cancer · Melanoma · Image augmentation · Deep learning · CNN

S. Vashdev Asrani (✉) · J. Lalchandani · I. Desai · T. Jeswani · S. Khedkar
Vivekanand Education Society's Institute of Technology Chembur, Mumbai, Maharashtra, India
e-mail: 2020.sanjana.asrani@ves.ac.in

1 Introduction

Skin cancer is an anomalous growth of skin cells usually because of unrepaired DNA damage. This triggers mutation in skin cells causing them to multiply in an uncontrolled manner, resulting in malignant tumors. As compared to the compound incidence of breast, lung, prostate, and colon cancers, there are more new cases of skin cancer registered every year. In India, skin cancer constitutes less than 1% of all diagnosed cancers which is lower as compared to the Western world, but the absolute number of cases is estimated to be significant due to the larger population base. Skin Cancers are primarily diagnosed visually, beginning with an initial clinical screening and followed potentially by dermatoscopic analysis, a biopsy and histopathological examination. When it comes to skin, telling a normal mole or skin spot from a cancerous mole can be a very tricky task and might require a high level of dermatological expertise. That's why it's important to be equipped with the basic tools and knowledge for detecting skin cancer. Dermatologists have established the classification system for an accurate diagnosis of skin disorders. Thus, due to the visual similarities, distinguishing skin cancer from other skin disorders of lesions and various patterns makes it difficult to diagnose the exact type of cancer.

There are 7 different types of skin cancer, one of them being Melanoma, which is considered to be the deadliest form. All the 7 types have distinct visual appearances and visual cues that are used for initial suspicion, but a definitive diagnosis requires a biopsy (sample of the skin being removed and examined under a microscope). For example, for Melanoma, the below guidelines are used as strategy for early diagnosis (Fig. 1).

Deep learning models can analyze and identify visual patterns and irregularities in skin lesions, assisting healthcare professionals in early diagnosis and prompt treatment, ultimately improving patient outcomes by facilitating faster and more accurate assessments of suspicious skin changes. The objective of the proposed system is to create an app, in which the main goal will be to analyze and categorize the 7 different types of skin cancers using images captured through dermatoscopic imaging. CNN is used to train the models using only pixels and disease labels as inputs. The proposed system will help people to foresee their skin cancer and manage and plan their treatment accordingly.

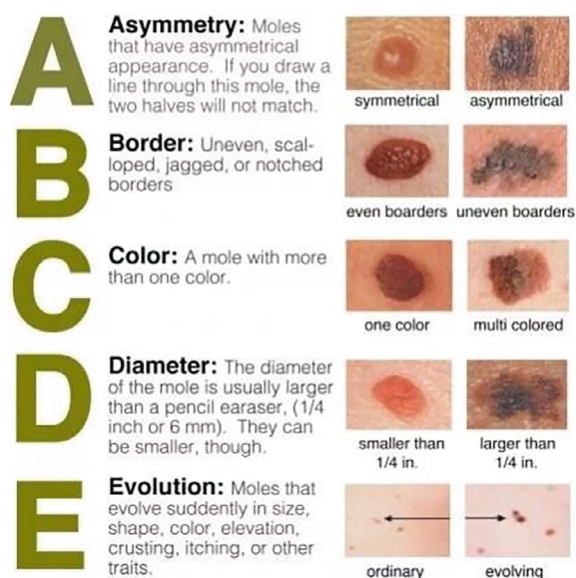
J. Lalchandani
e-mail: 2020.jessica.lalchandani@ves.ac.in

I. Desai
e-mail: 2020.isha.desai@ves.ac.in

T. Jeswani
e-mail: 2020.trishala.jeswani@ves.ac.in

S. Khedkar
e-mail: sujata.khedkar@ves.ac.in

Fig. 1 The ABCDE checklist



2 Overview

2.1 Motivation

Melanoma also called Malignant Melanoma is the 19th most frequently occurring cancer in women and men. It is the deadliest form of skin cancer. Even though the mortality is significantly high, when detected early, the survival rate exceeds 95%. This motivates us to come up with a solution to help save millions of lives by early detection of skin cancer.

2.2 Problem Statement

Skin cancer is an alerting issue and it must be detected as early as possible. Visiting the hospital for every mole or itch is a tedious process that is time consuming as well as expensive. But with advancements in science, we can make use of artificial intelligence to detect cancerous cells. Deep learning convolutional neural network (CNN) is used in the proposed system to detect skin cancer more quickly and efficiently. The proposed model aims to detect the different types of skin cancer and hence solve the problems faced in the manual process. The user is asked to give an image as the input. The data will be pre-processed and cleaned properly, so that the

problem of clarity of image, due to the usage of different hardware, does not arise. If the application predicts that a person has skin cancer then the type of cancer will also be specified.

2.3 Objectives

Deliver the information regarding the type of cancer and not just whether it's benign or malignant in nature through an android to provide early detection of the cancer aiding people to get an early treatment.

3 Literature Review

3.1 Survey of Existing Systems

Hong Qing Yu et al. [1] conducted research on using the HAM10000 dataset to develop an implementable solution for remote skin disease diagnosis applications utilizing IoT technology. They evaluated various machine learning models, particularly CNN-based algorithms, to create a targeted ensemble machine classification approach called TEMCM that operates in two phases.

Bin Zhang et al. [2] explored the use of deep learning for the classification of skin diseases. The research aimed to summarize the characteristics of skin lesions and the status of image technology, highlighting both opportunities and challenges associated with the approach.

Seung Seog Han et al. [3] utilized a deep learning algorithm, Microsoft ResNet-152, to develop an automated classification system for clinical images of 12 different skin disorders. The system aimed to classify images as either benign or malignant cutaneous tumors, providing a potential tool for assisting in skin cancer diagnosis. Mustafa Qays Hatem et al. [4] explored the use of K-nearest neighbor (KNN) as a classification method for accurate and efficient diagnosis of skin lesions. KNN was chosen due to its time efficiency and ability to provide easy-to-interpret and versatile results that can be used for both classification and regression. The study aimed to differentiate between normal skin and malignant skin lesions that indicate pathology.

Jainesh Rathod et al. [5] discussed the use of Convolutional Neural Networks (CNNs) for skin disease diagnosis in a study, where computational techniques were employed to analyze and process image data based on various features. The study focused on filtering skin images to eliminate unwanted noise and enhancing the quality of the images.

Amirreza Rezvantalab et al. [6] assessed the potential of convolutional neural networks (CNNs) in classifying 8 types of skin diseases using dermoscopy images. The study employed several pre-trained state-of-the-art CNN architectures including

DenseNet 201, ResNet 152, Inception v3, and InceptionResNet v2. A total of 10,135 dermoscopy skin images were used for the analysis.

Afonso Menegola et al. [7] discussed various neural network techniques for skin cancer detection and classification.

Mahamudul Hasan et al. [8] proposed a system for detecting malignant or benign skin cancer using convolutional neural networks (CNNs). The study achieved an accuracy of 89.5% and a training accuracy of 93.7% after using a publicly available dataset for analysis.

Neema M et al. [9] developed a system using a unique deep learning architecture which focuses on the timely evaluation of skin cancer. The model could classify the melanoma into a benign or malignant class with 70% accuracy.

3.2 Limitations and Research Gap

Early detection makes skin cancer more treatable and for some people, completely reversible. Identifying and managing skin cancer before it spreads is the best way to achieve a good outcome. The 2019 article of British Medical Journal titled 'Skin cancer: smartphone diagnostic apps may offer false reassurance, warn dermatologists' lists the downsides to the available skin cancer apps.

The list includes:

- A lack of testing to verify their effectiveness
- A shortage of expert input when developing the technology

3.3 Project Contribution

The proposed model aims to target these incapacabilities that are present in the already existing apps. The model will be built on an extensive dataset acquired from Harvard Dataverse which comprises 10,015 training images and 1511 testing images. This dataset includes dermatoscopic images from different populations, acquired and stored by different modalities. Thus, there will be no shortage of input while developing the technology and elaborate testing can be conducted using the available testing images. Also, as the images are acquired from diverse sources across the globe the effect of skin tone, hair, and pigmentation on the result generated by the model will be greatly reduced. The model also aims to deliver to the users the information regarding the type of cancer and not only if it is benign or malignant in nature.

Table 1 Image augmentation

Abbreviation	Cancer Type	Number of images before augmentation	Number of images after augmentation
akiec	Actinic keratoses and intraepithelial carcinoma/Bowen's disease	327	1820
bcc	Basal cell carcinoma	514	2002
bkl	Benign keratosis (solar lentigines/seborrheic keratoses/lichen planus-like keratosis)	1099	2033
df	Dermatofibroma	115	1560
mel	Melanoma	1113	2029
nv	Melanocytic nevi	6705	2000
vasc	Vascular lesions (angiomas, angiokeratomas, pyogenic granulomas, and hemorrhage)	142	1816

3.4 Dataset Description

The HAM10000 dataset has skewed class proportions. The melanocytic nevi cancer type has 6705 images and the dermatofibroma cancer type has 115 images. In order to not build a biased model, image augmentation tools like ImageDataGenerator from Keras have been used to handle the class imbalance problem (Table 1).

4 Implementation

4.1 Architecture

The HAM10000 dataset is used for building the model. The image taken by the user is analyzed by the deep learning model, and then the output is provided to the user along with the further course of action. The deep learning model predicts 7 types of skin cancer, which includes both benign and malign. The proposed system uses Convolutional Neural Networks to train the model. Convolutional Neural Networks (CNN, or ConvNet) are a special kind of multi-layer neural networks designed to recognize visual patterns directly from pixel images with minimal pre-processing. Convolutional neural networks comprise of two very simple elements, namely convolutional layers and pooling layers. Although simple, there are near-infinite ways to arrange these layers for the detection of skin cancer (Fig. 2).

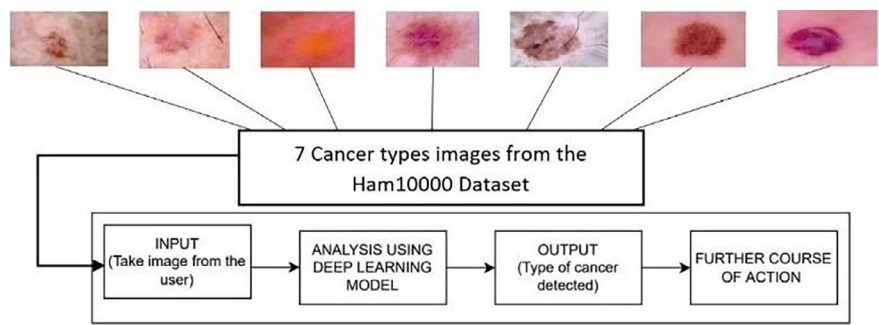


Fig. 2 The above architecture of the System gives a brief overview of the system

4.2 Modular Diagram

The major 2 modules are Deep Learning model and Mobile Application. These 2 modules are further divided into many other sub-modules. The sub-modules are as follows (Fig. 3):

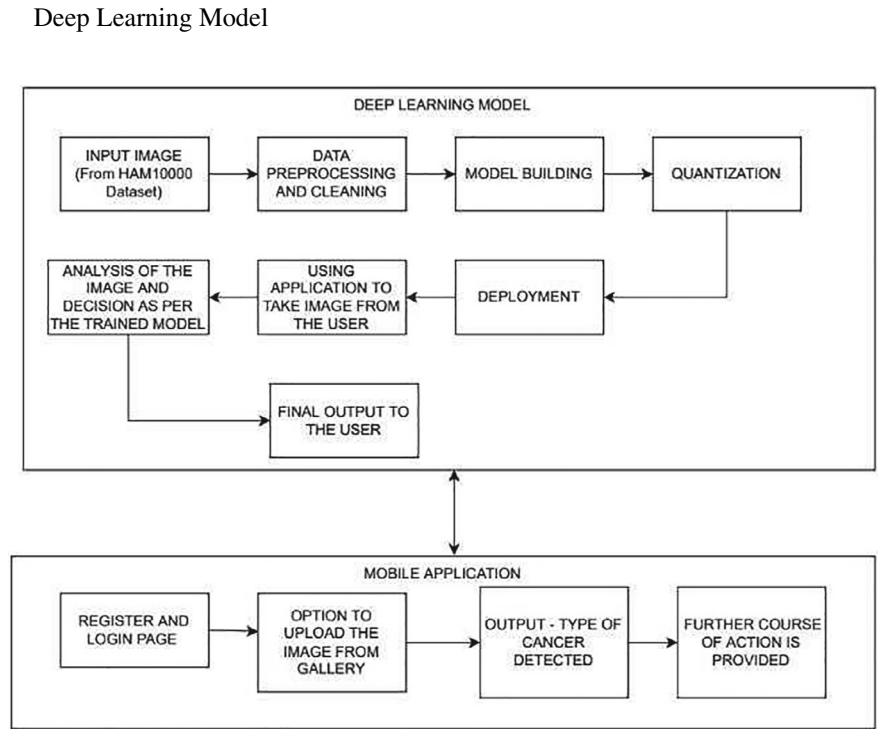


Fig. 3 Modular diagram depicts how our system is divided into different parts and how each module interacts with each other

1. Input Image

In this sub-module, the input images are acquired from the HAM10000 dataset which is available for free on Kaggle. This dataset is created by Harvard which contains 10,015 images for training and testing.

2. Data Pre-Processing and Cleaning

The number of images given for each type was uneven. In order to efficiently train the model, image augmentation tools like ImageDataGenerator from Keras were used.

In the data pre-processing and cleaning sub-module, the following is performed on the input images—Cleaning of the image.

- Illumination correction
- Enhancement (pre-processing), i.e., applying modification filters to remove noise
- Extraction of lesion area
- Removal of hair
- Segmentation of the lesion to distinguish between the various kinds of skin cancer

3. Model Building

The Model Building sub-module focuses mainly on the CNN model building.

4. Quantization

This sub-module focuses on the quantization of the model using TensorFlow Lite.

5. Deployment

The deployment is done on the Android Application using Android Studio.

6. Input Image from User

This sub-module focuses on taking the input image from the user and pre-processing it using the TensorFlow Lite's TensorBuffer class.

7. Analysis of the Input Image

The input image given by the user is analyzed thoroughly and hence the decision is made by the CNN model regarding the type of cancer.

8. Final Output

The final output is given to the user.

Mobile Application

1. Register and Login Page

The register and login page are used for the authentication purposes. All the constraints regarding the email id, password, and duplication of the username are available. The database used is SQLite.

2. Upload the Image from Gallery

This sub-module focuses on making the application very user-friendly and easy to navigate. The user can click on the upload image button and select the picture.

3. Output

The output is given regarding the type of cancer detected.

4. Further Course of Action

All skin cancer types don't require chemotherapy or surgery, some of them can be cured by medicines as well. Hence, information regarding such medicines is available in the application. Skin cancer articles are also available.

Requirements—Software Specifications:

For Model Building: Python Basic Stack (Scikit learn, NumPy, Matplotlib, Pandas, Keras, TensorFlow) IDE: Jupyter Notebook or Google Colab, VS Code.

To build the application: Android, Java, XML.

5 Results and Evaluation

5.1 Confusion Matrix

Confusion matrix is a machine learning method used to measure a classifier's performance. It helps to visualize and summarize the performance of a classification algorithm. It plots actual vs. predicted, i.e., actual classes vs. the classes predicted by the model. The truly positive, truly negative, falsely positive, and falsely negative values can then be used to plot an ROC curve or to create a Classification Report. It gives the count of true positives that is the number of images correctly classified (Fig. 4).

Fig. 4 Confusion matrix

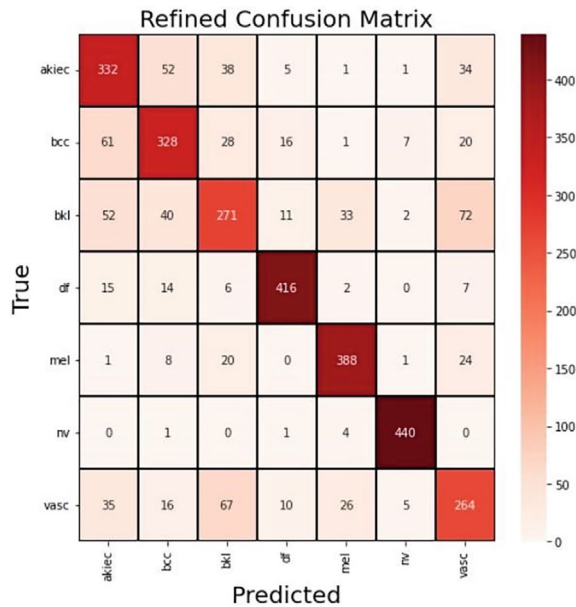


Fig. 5 Classification report showing the precision, recall, and f1 score of our model

	precision	recall	f1-score	support
akiec	0.67	0.72	0.69	463
bcc	0.71	0.71	0.71	461
bk1	0.63	0.56	0.59	481
df	0.91	0.90	0.91	460
mel	0.85	0.88	0.87	442
nv	0.96	0.99	0.98	446
vasc	0.63	0.62	0.63	423
accuracy			0.77	3176
macro avg	0.77	0.77	0.77	3176
weighted avg	0.77	0.77	0.77	3176

5.2 Classification Report

The classification report is a performance evaluation metric in deep learning which is used to show the precision, recall, f1 score, and support score for all the classes trained in our classification model. For weighted average the f1 score is 0.77% (Fig. 5).

5.3 ROC Curve

A receiver operating characteristic curve, or ROC curve is a plot of (1-specificity or the FPR) vs sensitivity (TPR). Basically, the diagonal represents random guesses. If a model’s line falls above the diagonal, the model is said to be working better than a random guess model would work (Fig. 6).

6 Conclusion and Future Work

Melanoma of skin is the 19th most common cancer worldwide. There were more than 150,000 new cases of melanoma of skin in 2020. Non-melanoma skin cancers such as basal cell carcinoma and squamous cell carcinoma are often excluded from the reporting of cancer statistics. Thus, it is likely that the reported global incidence of skin cancer is an underestimate.

Early detection greatly improves the prognosis of patients with malignant melanoma. But it is not feasible for people to get regular screenings to check if there is any growth of malignant cells in their bodies. Getting frequently tested by a medical authority is expensive and harmful too if the tests involve biopsies or radiation methods. In order to counter these issues, the proposed system mainly focuses on creating a user-friendly application which enables users to get a quick diagnosis easily. Any irregularity in the skin can be tested by feeding the image into the application. The diagnosis will not only show the presence or absence of malign

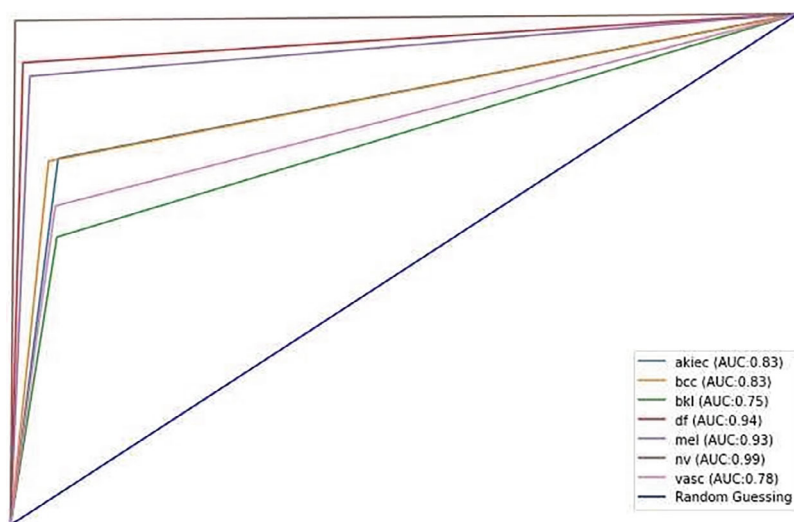


Fig. 6 ROC curve showing the performance of the classification model at all classification thresholds

cells, but will also specify the type of cancer if any. The existing solution can be further improved by the use of an explainable AI model to help the users understand the logistics behind the decision of the model. Also, the app can be enhanced by providing the doctor's information along with the further course of action to the users.

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Stewart Platforms Used for Lower Extremities Rehabilitation



Josep M. Rossell, Víctor Barcons, Jesús Vicente, and Manel Blanes

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Abstract Stewart platforms can be used for a multitude of technological, industrial, and medical applications, since they allow highly precise reproduction of controlled movements with six degrees of freedom. This study proposes the use of a Stewart platform for the rehabilitation of knees, ankles, and lower limbs, in general, simulating physical exercises that help their recovery and strengthening.

Keywords Rehabilitation of lower limbs · Knees · Ankles · Stewart platforms · Parallel robots · Inverse kinematics

J. M. Rossell (✉)

Department of Mathematics, Universitat Politècnica de Catalunya, Barcelona, Spain

e-mail: josep.maria.rossell@upc.edu

V. Barcons · J. Vicente · M. Blanes

Department of EMIT, Universitat Politècnica de Catalunya, Manresa, Spain

e-mail: victor.barcons@upc.edu

J. Vicente

e-mail: jesus.vicente@upc.edu

M. Blanes

e-mail: manuel.blanes@upc.edu

1 Introduction

Stewart platforms, also known as hexapods, are complex mechanisms that have found diverse applications in industries and research fields. These articulated structures, consisting of movable actuators connected to a fixed base, offer exceptional motion control capabilities. Stewart platforms are considered mechatronic systems due to the involvement of various topics such as mechanics, electronics, computing, and control. These robots consist of a fixed base, a top plate, and six extendable actuators connecting the base to the plate. By coordinating the movement of the actuators, appropriate orientations and displacements of the top plate can be achieved. The precise positioning and orientation control provided by Stewart platforms have made them invaluable in a wide range of applications [1].

In 1965, the first model of a Stewart platform was constructed by Dietmaier Stewart as a flight simulator [2]. Since then, they have been utilized in industry applications such as automotive, aeronautic, mechanics, energy generation, civil engineering, or medical devices, to mention a few. These mechanisms are present in robotics [3, 4]; in technology of machine tool [5, 6]; vehicle suspensions [7]; precision laser cutting [8]; radio telescopes [9, 10]; astronomical observations [11]; surgical robotics [12–14]; pick-and-place tasks [15]; medical imaging [16]; medical equipments [17, 18]; biomechanical testings [19]; automotive applications [20]; vibration control [21]; high-precision positioning devices [22]; aerospace inspections [23]; and micromanipulator applications [24] or flight simulators, driving simulators and motion-based simulation systems [25, 26], among others [27].

A common application in the maritime sector is the use of Stewart platforms to counteract the waves while certain types of machinery are operating, such as cranes or drilling rigs [28]. They can also be used to protect precision instrumentation systems, positioning systems, or wind turbines located on ships, buoys, or floating offshore platforms [29, 30], as well as wave energy converters [31, 32].

They are also employed in augmented and virtual reality. By providing haptic feedback and realistic motion cues, these platforms enhance the immersive experience in virtual environments. Whether it is a virtual reality training simulation or an augmented reality interactive display, Stewart platforms contribute to creating a more lifelike and engaging user experience [33].

Generating precise multidirectional movements is the main objective of any Stewart platform. In order to investigate the impact of these movements on the structures located on the upper plate, our UPC research group has designed and built two Stewart platform prototypes. The first step was to achieve high-precision motion trajectories through proper control laws, minimizing errors in both motion and orientation of the top plate. After proving that the platform moves with high precision, we have at our disposal a valuable tool to carry out complex simulations and multidirectional movements for different experimental purposes.

In this study, it is intended to utilize a Stewart platform, previously prepared and programmed, as a rehabilitation mechanism for lower limbs, particularly ankles and knees. Taking into account that our research team has a long experience in the

construction and applications of parallel robots [34, 35], it would be relatively easy to design one of these prototypes to be utilized for rehabilitation. In fact, the movement and control of a Stewart platform can be faithfully reproduced in any parallel robot with a similar topology. For this, it would be necessary to contact traumatologists and physiotherapists for correct advice on the type of movements that a Stewart platform must reproduce to recover injured extremities.

In summary, Stewart platforms provide versatile and precise motion control capabilities, making them invaluable tools in a wide range of industries and research fields and their applications are diverse and impactful. Their ability to precisely control position and orientation has revolutionized various industries and continues to drive innovation in motion control systems. We want to take advantage of all its features to be used as a lower limb rehabilitation machine.

2 Stewart Platform Mechanism

A Stewart platform is a mechanism that can move with six degrees of freedom, made up of a fixed base together with a movable top plate and six independent extendable actuators. These types of platforms provide high rigidity, good positioning accuracy, low inertia, and can move easily when inverse kinematics is applied as the motion control method.

Stewart platforms can be classified taking into account the number of actuators attached to the base (B) and the number of connection points on the top plate (P), denoted by B - P. In our case, we have two platforms of the type 6 - 6 which, in fact, are called Gough-Stewart platforms. An image of our platforms can be seen in Fig. 1. The vectorial scheme detailed in Fig. 2 helps to better understand the movement of this type of robot. The actuators have variable lengths and must be controlled independently to move the top plate of the platform. The actuators have total freedom of rotation on their axes in such a way that the installed universal joints allow a spherical movement.

Consider the reference systems of the base and the top plate, denoted by $\{B\}$ and $\{P\}$, respectively, where $\{B\}=\{O; x, y, z\}$ and $\{P\}=\{O'; x', y', z'\}$ with $O=[0, 0, 0]$ the origin point of the base. The attachment points at the base and the top plate are denoted by b_i and p_i , $i=1, 2, \dots, 6$, respectively. To calculate the angles between attachment points, we use the following formulae:

$$\pi^{\alpha_i} = 3(i - 1) - \theta_{B2}, \quad \text{for } i = 1, 3, 5 \quad \text{and} \quad \alpha_i = \alpha_{i-1} + \theta_B, \quad \text{for } i = 2, 4, 6 \quad (1)$$

$$\text{and } \pi\beta_i = 3i - \theta_{P2}, \quad \text{for } i = 1, 3, 5 \quad \text{and} \quad \beta_i = \beta_{i-1} + \theta_P, \quad \text{for } i = 2, 4, 6 \quad (2)$$

for the top plate.

Consider now the system $\{B\}$. The points b_i are described by the following vectors:

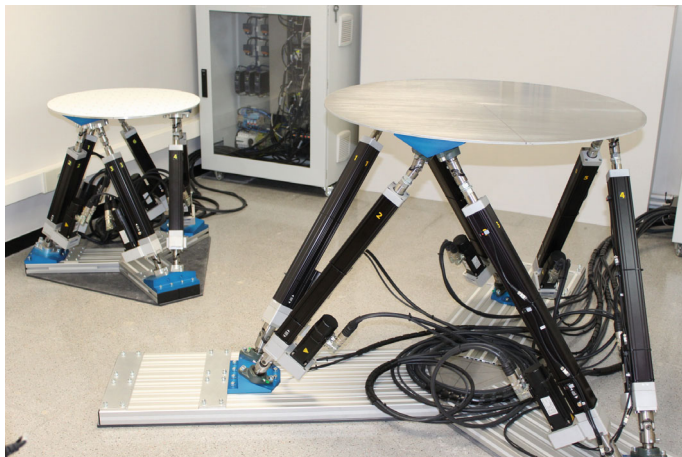


Fig. 1 Stewart platforms designed and built by our research team

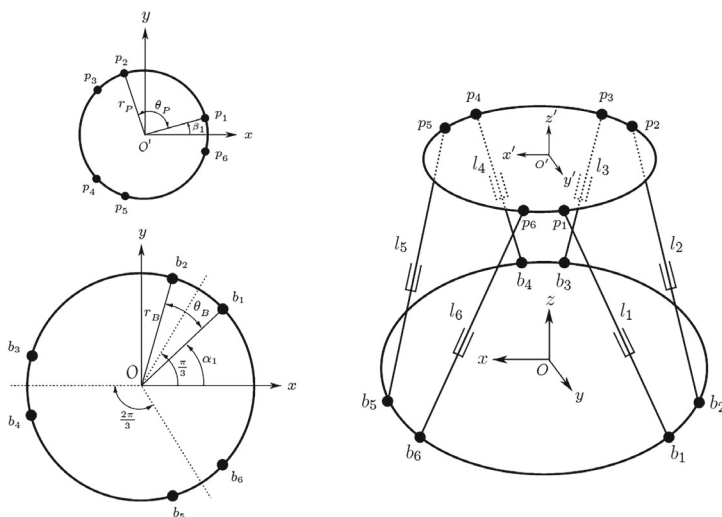


Fig. 2 Vectorial scheme of a Stewart platform

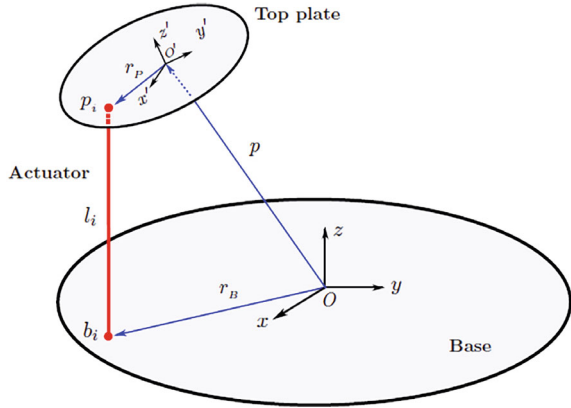
$$b_i = b_{ix}, b_{iy}, b_{iz}^T = r_B \cos \alpha_i, r_B \sin \alpha_i, 0^T, \quad \text{for } i = 1, \dots, 6 \quad (3)$$

where r_B is the radius of the base. The points p_i , with respect to the reference system $\{P\}$, have the following positions:

$$p_i = p_{ix}, p_{iy}, p_{iz}^T = r_P \cos \beta_i, r_P \sin \beta_i, 0^T, \quad \text{for } i = 1, \dots, 6, \quad (4)$$

where r_P is the top plate radius.

Fig. 3 Scheme of the inverse kinematics methodology



2.1 Inverse Kinematics Methodology

Any parallel robot has a high non-linear behavior together with kinematic constraints. For this type of mechanism, the use of the inverse kinematics methodology is computationally more efficient than the direct kinematics [36, 37]. Using inverse kinematics the length of each actuator, denoted by the vector $l(t) = l_1(t), \dots, l_6(t)^T$ can be found in real-time t .

By construction, and observing Fig. 3, we have

$$l_i(t) = p(t) + R(t)p_i - b_i, \quad (5)$$

the corresponding attachment points given in (3) and (4), respectively with b_i and p_i . The coordinates of the origin of system $\{P\}$, with respect to the frame $\{B\}$, are given by the vector $p(t) = \vec{OO'}(t) = [x, y, z]^T$. Denote by $R(t)$ a rotation matrix providing the orientation of frame $\{P\}$ with respect to the frame $\{B\}$. Obviously, matrix $R = R(t)$ depends on the sequence of rotation of the axes. In this paper, we have consider

$$R = R_z(\alpha)R_y(\beta)R_x(\gamma) = r_{11}r_{12}r_{13}r_{21}r_{22}r_{23}r_{31}r_{32}r_{33} \quad (6)$$

where

$$R_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ s_\gamma & c_\gamma & 0 \\ c_\gamma & -s_\gamma & 0 \end{bmatrix}, \quad R_y(\beta) = \begin{bmatrix} c_\beta & 0 & s_\beta \\ 0 & 1 & 0 \\ -s_\beta & 0 & c_\beta \end{bmatrix}, \quad R_z(\alpha) = \begin{bmatrix} c_\alpha & -s_\alpha & 0 \\ s_\alpha & c_\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (7)$$

with $s_\alpha = \sin \alpha$, $s_\beta = \sin \beta$, $s_\gamma = \sin \gamma$, $c_\alpha = \cos \alpha$, $c_\beta = \cos \beta$, $c_\gamma = \cos \gamma$. The matrix $R_x(\gamma)$ denotes a rotation of γ radians around x -axis; $R_y(\beta)$ a rotation of β radians around axis y and $R_z(\alpha)$ a rotation of α radians around axis z , where $\alpha, \gamma \in (-\pi, \pi)$, $\beta \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

3 Rehabilitation of Lower Limbs

Ankle and knee injuries are very common when performing physical activities, either through carelessness or accident, which can lead to an inability to lift or move these joints. In the field of robotic rehabilitation of the lower extremities, there is a large number of devices that differ from each other by the treatment phase for which they are intended (see Fig. 4). Since the bony structures of the ankles and knees are very complex, rehabilitation movements such as dorsi flexion/plantarflexion, inversion/eversion, and abduction/adduction have to be properly planned. Certain mechanisms are currently being built to reproduce ankle and knee movements in order to be used for therapeutic purposes [38–43].

The most developed commercial devices are oriented to the process of walking and running, as well as those of the stationary type. Generally, they are installed in gyms and sports centers, but not in health centers to be used as rehabilitation mechanisms. In medical applications, but as assistance robots in various types of surgeries, we can find a wide catalog of parallel manipulators that are used in laboratories, research institutes, and mainly in hospitals [13, 14]. In recuperative medicine, Stewart-Gough parallel mechanisms can serve for the rehabilitation of knees, ankles, and lower extremities by providing controlled movements and targeted exercises that aid in the recovery and strengthening of the knee and ankle joints.

The movement of the platform must accurately simulate the movements of the ankle and knee when walking or standing up. Therefore, with the aim of recreating the trajectory of the foot-ankle joint during walking, the study focuses on the design and implementation of the kinematic control system. To determine the desired positions and orientations of the top plate, it is necessary to develop and apply algorithms for the movement of the actuators. With the obtained model, the behavior of the actuators

Fig. 4 Picture of a simple mechanism for passive rehabilitation of a leg



is simulated and the results of these simulations are compared with the real data to evaluate the trajectory errors committed.

The main objective is to program a movement protocol for each type of lower limb injury. Obviously, this work has to be coordinated and established by a team of traumatologists and physiotherapists, who have to determine not only the movements, but also their speeds and duration. The patient can stand or sit with their foot on the platform until the machine completes the predetermined action program [42]. An endless number of movements can be programmed, all of them in a personalized way.

In the initial study, the following points should be addressed:

- (a) Obtaining the dynamic model of the platform. Our research team has the information on this first point.
- (b) Obtaining the dynamic models of the human leg, including ankles and knees.
- (c) Analysis and parameterization of the correct exercises for ankle and knee rehabilitation.
- (d) Development and implementation of control algorithms for monitoring rehabilitation trajectories.
- (e) Generation and simulation of trajectories for the ankle and knee rehabilitation task.
- (f) Experimental validations of the obtained results.

Based on the previous information, the objectives described below can be achieved:

1. **Range of Motion Exercises:** Stewart platforms can be programmed to move the knee and ankle joints through a variety of movement exercises. By controlling the movement of the platform, traumatologists and physical therapists can fine-tune the flexion, extension, inversion, eversion, and rotation of both joints. This helps restore and improve joint mobility and flexibility.
2. **Strengthening Exercises:** The platforms can apply resistance and load to the knee and ankle joints during rehabilitation exercises. By adjusting the forces exerted by the platform, therapists can customize the resistance to match the individual's strength and gradually increase it as the muscles get stronger. This helps improve muscle strength and stability in both the knee and ankle.
3. **Balance and Proprioception Training:** An advantage of the Stewart platforms is that they can simulate unstable surfaces to challenge the balance and proprioception skills of the knee and ankle joints. By introducing controlled perturbations and tilt movements, the platform can mimic real-life situations that require joint stabilization. This helps to retrain proprioceptive skills, improve balance control, and improve joint stability.
4. **Gait Rehabilitation:** These mechanisms can be integrated into gait training systems for patients recovering from knee and ankle injuries. By replicating the natural walking pattern, therapists can adjust the platform's movements to facilitate proper gait mechanics. This aids in reestablishing a normal walking

pattern, promoting a smooth and coordinated gait, and improving the strength and mobility of the knee and ankle joints.

5. **Sensor Integration:** The platforms can be combined with sensor technology, such as force plates or motion capture systems, to provide quantitative data on knee and ankle joint performance during rehabilitation exercises. This allows therapists to monitor progress, track improvements, and make adjustments to the rehabilitation program based on data.
6. **Virtual Reality Feedback:** By incorporating virtual reality technology, Stewart platforms can create immersive and interactive environments for knee and ankle rehabilitation. Visual and auditory feedback can be synchronized with the platform's movements, providing real-time feedback and motivating the patient during their rehabilitation sessions.

Our research group has accumulated extensive experience in the design, construction, programming, and control of Stewart platforms over the years. In this paper, we have presented a practical application of this type of mechanism that, in our opinion, can be very useful in the field of ankle and knee rehabilitation. We have established contacts with medical personnel specialized in this subject and they have expressed their interest in collaborating with us to carry out the said proposal.

4 Conclusions

Stewart platforms provide precise movements that can be tailored to meet specific needs for knee, ankle, and general lower extremity rehabilitation. Its ability to offer a controlled range of motion, resistance, balance challenges, and gait training contributes to the recovery process and helps people regain strength, mobility, and functionality in the knee and ankle joints.

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Trends & Perspectives

A Detailed Investigation into the Brain Cancer Detection Capabilities of CNN



Kamred Udham Singh, Ankit Kumar, Gaurav Kumar, Teekam Singh, Tanupriya Choudhury, and Ketan Kotecha

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Abstract Brain tumour detection is an accepted testing task in the beginning phases of life. Yet, presently it has progressed with different AI calculations. Along these lines, to identify the brain tumour of a patient we consider the data of patients like MRI pictures of a patient's brain. Here our concern is to spot if the tumour is available in persisting's brain. It's crucial to identify the tumour at the beginning level for a sound lifetime of a patient. During this task, we gauge the seriousness utilising Convolutional Neural Network calculation which gives us precise outcomes. Tumour division is a pivotal and burdensome assignment inside the territory of clinical picture preparation as a human-helped manual characterization may bring about erroneous expectations and findings. We suggested using convolutional neural organisation to distinguish tumours from 2D appealing reverberation brain images (MRI). The exploratory analysis was performed on a dataset that included tumours of various sizes, regions, forms, and picture forces. We used convolutional neural networks, which are implemented using Keras and Tensorflow because they respect a much better display than standard ones. CNN achieved an accuracy of 83.54% in our

K. U. Singh

School of Computing, Graphic Era Hill University, Dehradun, India

A. Kumar · G. Kumar

Department of Computer Engineering and Applications, GLA University, Mathura, UP, India

e-mail: kumar.ankit@gla.ac.in

G. Kumar

e-mail: gaurav.kumar_phd.ca21@gla.ac.in

research. The recently developed CNN design may be used as a productive decision-aid device for radiologists in clinical diagnostics because of its high speculation capacity and quickness.

Keywords CNN · Image classification · Magnetic resonance imaging · Medical imaging · Neural networks · Brain tumour classification

1 Introduction

Cancer diagnosis and prevention are both theoretically possible, but in reality, they are not always realisable. Tumours, which are not the same thing as cancer, may be either benign, which does not progress to cancer or malignant. Benign tumours, on the other hand, do not spread to other tissues and organs and may be removed surgically, but malignant tumours can spread to other parts of the body. The field of medical imaging comprises a wide variety of diagnostic and therapeutic techniques that do not involve the use of intrusive methods [1]. These imaging techniques comprise a wide variety of modalities and approaches that provide a visual representation of the body for the purposes of diagnosis and treatment. It is impossible to exaggerate how important of a part they play in improving the health of humans. When seen in this light, the picture segmentation stage of the image processing pipeline appears as an essential component, since it is responsible for determining the amount of success that will be achieved by future processing steps. The first step in the process of medical image processing is called initial image segmentation, and its primary purpose is to identify lesions and tumours. This paves the way for efficient machine vision and makes for more accurate diagnoses. Using computer-aided diagnostic (CAD) technologies, researchers in the field of medical informatics are making great progress towards their goal of increasing the sensitivity and specificity with which they can detect malignancies or lesions.

T. Singh

Department of Computer Science and Engineering, Graphic Era Deemed to Be University,
Dehradun, India

e-mail: tsingh@ma.iitr.ac.in

T. Choudhury (✉)

School of Computer Science, University of Petroleum and Energy Studies, 248007 Dehradun,
Uttarakhand, India

e-mail: tanupriya1986@gmail.com

Department of Computer Science and Engineering, Symbiosis Institute of Technology, Symbiosis
International University, Lavale Campus, Pune, India

K. Kotecha

Symbiosis Centre for Applied Artificial Intelligence, Symbiosis Institute of Technology,
Symbiosis International University, Pune 411045, India

e-mail: director@sitpune.edu.in

According to [2], cancers that affect the brain and nerve system rank as the tenth most common reason for mortality in the world. Only 34% of men and 36% of women will be alive after five years if they have brain or nervous system cancer. This percentage is lower for younger patients. Brain tumours are responsible for 120,000 fatalities each year and are estimated to affect 400,000 people all over the world, according to data provided by the World Health Organization (WHO). In addition, the diagnosis of 86,970 new instances of malignant and non-malignant brain tumours is anticipated to take place in the United States in 2019 [3]. When aberrant cells multiply inside the brain, this may lead to the development of a brain tumour [4]. Malignant brain [19–21] tumours develop much more quickly than benign ones, aggressively invading the tissues around them, and having the potential to metastasize to other regions of the brain and have an effect on the central nervous system. Brain tumours may be classified as either malignant or benign. Primary brain tumours develop from cells that are already present in the brain, while secondary brain metastasis tumours begin elsewhere in the body and then travel to the brain. On the other hand, the growth rate of benign neoplasms in the brain is much slower. Therefore, detecting brain tumours [22–24] at an earlier stage considerably improves treatment techniques and increases the likelihood of survival. Manual tumour or lesion segmentation is made difficult and time intensive due to the vast amount of MRI images that are produced in clinical practice.

Magnetic Resonance Imaging, or MRI for short, is an essential diagnostic tool for finding tumours and lesions. In medical image processing, the segmentation of tumours from MRI data is a basic operation that often involves enormous datasets. This task involves working with a lot of data. In addition, the fuzzy borders of soft tissue tumours provide difficulty to medical professionals. The accurate segmentation of tumours found inside the human brain is a tremendous challenge. In this paper, we provide a skilful and effective method for the identification and segmentation of tumours that do not involve the participation of any humans. This cutting-edge method employs Convolutional Neural Networks (CNNs), a kind of artificial neural network, to provide accurate and automated neoplasm identification.

2 Literature Review

Isolating the tumour from a brain MRI and dividing the region of interest from an item are two of the hardest tasks of the current age. Specialists and scientists worldwide are working on this topic to rebuild the best-sectioned Area of Interest and distinct methods from a specific perspective. Neural network-based segmentation approaches now provide clear results, and their application is rising.

H. D. Hemmati et al. [5] found that fuzzy C-means bunching, conventional classifiers, and a convolutional neural network can distinguish malignancies in 2D reverberation brain images (MRI). Brain cancer may be diagnosed this way. An ongoing dataset with various tumour sizes, areas, shapes, and image capabilities was employed for exploratory analysis. Working with a larger dataset will be harder.

T. Hossain [6] presented a CNN approach that uses near and logical data. The pre-processing step standardises the photos, while the post-processing phase removes false positives. In another study, CNN and Neutrosophy are used to identify neoplasms. The methodology outperforms CNN, SVM, and K-closest neighbours (KNN) by 95.62%. This study uses the more efficient R-CNN method to detect and localise malignant growths in MRI brain imaging.

E. Avşar [7] used multi-grouping MR images to create Deep Convolutional Neural Networks (ConvNets) for brain cancer organising. He presented three unprepared ConvNets for MRI patches, cuts, and multi-planar volumetric cuts. The research showed that ConvNet may achieve excellent accuracy every time it is used to develop a model on a multi-planar volumetric dataset. Two issues: a small sample size (285 patients) and an unbalanced class distribution (210 HGG and 75 LGG patients).

To characterise brain cancers using multi-grouping MR images, L. Morrison and M. Dacre [8] presented Deep Convolutional Neural Networks (ConvNets). He recommended three unprepared ConvNets for MRI patches, cuts, and multi-planar volumetric cuts. It also examines self-learned parts/channels in different layers using modest layer yields. How the computer will function across the dataset and its prediction skills are unknown.

Fuzzy-based techniques help J. Shen et al. [9] enhance the image's division level and geographical limits compared to earlier frameworks. They also boost the picture's impact. The framework's classifier might be a synthetic neural network to significantly boost system accuracy. F. Khalifa et al. [10] have made promising theoretical and experimental advances in using models and procedures in many clinical settings. These results suggest that dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) may be used in medical treatment. DCE-MRI may identify liver tumours, renal rejection, and breast and prostate cancer early [11]. DCE-MRI has been extensively studied and shown to aid illness identification and treatment.

3 Problem Statement

Within the area of medical image processing, the process of segmenting brain tumours is one that is considered to be crucial and challenging. This complexity results from the risks associated with human-assisted manual classification, which may lead to incorrect diagnoses and treatment plans. In addition, while dealing with massive amounts of data, the task becomes quite laborious. As a result, there is an increasing need for a trustworthy system that can accurately identify cancer sites from medical images. The wide variety of signs and symptoms that may originate from brain tumours, as well as the intricate parallels between healthy and tumorous tissues in the brain, all contribute to this need. In order to overcome these challenges, innovative solutions that harness the potential of today's computational tools must be developed. The use of modern algorithms and AI has made accurate tumour segmentation possible, and it has also made rapid processing of large datasets possible for

the first time. These kinds of breakthroughs have the potential to bring about a revolution in the area of medical imaging, empowering doctors to make choices that are better based on evidence and improving the quality of treatment patients get. In summary, mastering the complexities of brain tumour segmentation requires a productive combination of technology innovation and medical competence in order to achieve the desired results. This concerted effort intends to enhance the lives of people who are coping with the challenges posed by brain tumours by reducing the likelihood of diagnostic mistakes, mitigating the effects of the constraints of manual intervention, and improving overall patient outcomes.

4 Proposed Methodology

The use of convolutional neural networks, sometimes known as CNNs, has become more commonplace in the field of medical image processing. Over the course of time, a large number of researchers have begun an endeavour to develop a model that may improve the efficiency with which tumours are recognised. The development of a robust algorithm that is endowed with the capability to accurately categorise tumours present in 2D brain MRI images was the major objective that we sought to accomplish with this project. Convolutional neural networks (also known as CNNs) were chosen as our method of modelling after careful consideration of the many possibilities that were open to us [12]. This choice was made because of their innate capacity to leverage border connections and expertly capture complex, distributed relationships within the picture data. During the process of tumour feature extraction, our goal was to achieve a higher degree of abstraction by making use of the convolutional layers that are inherent to CNNs. The convolutional filters of the network were prepared to recognise subtle patterns and changes in texture that could be missed by traditional methods. In conjunction with this, the hierarchical design of the network held the potential of gradually aggregating complicated characteristics. This would allow the algorithm to develop a representation of tumour areas that was nuanced and accurate to a greater degree. Our quest included far more than a simple mastery of algorithms [13, 14]. We were motivated by the desire to make a contribution to the current improvements in medical imaging technology, and as a result, provide medical practitioners with tools that may assist in the early identification and accurate diagnosis of brain tumours. Our effort was focused on aligning itself with the changing environment, where machine learning and neural networks are positioned as transformational forces in the quest for improved patient care. This is important since the dimensions of medical data continue to rise at an accelerating rate (Fig. 1).

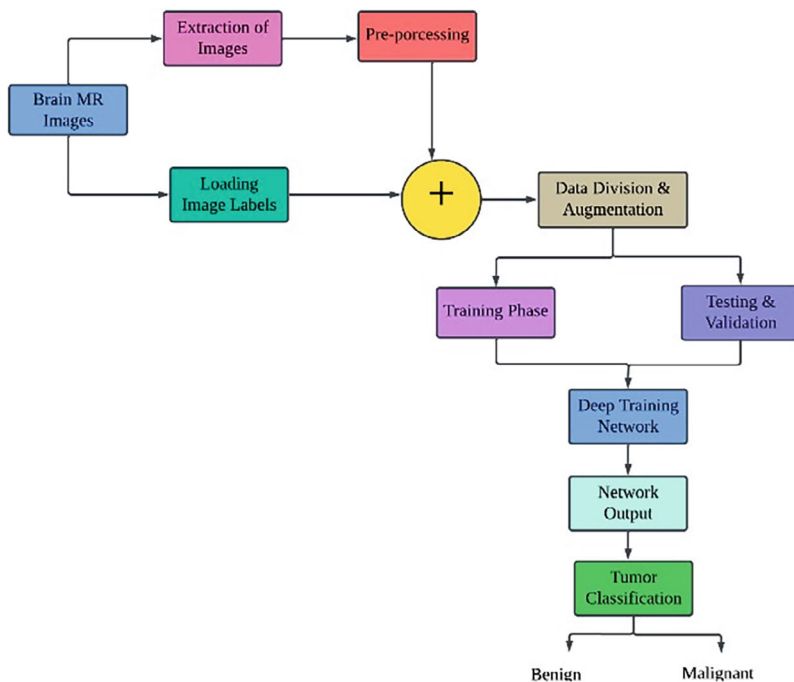


Fig. 1 Flow chart of the proposed methodology for the identification of tumours image database

4.1 Image Database

It was decided to organise and make accessible on the Kaggle platform a complete dataset consisting of brain MRI pictures. The dataset is separated into two unique groups, referred to as “yes” and “no”, with each of these groups including a set of 253 MRI pictures of the brain. Within the “yes” category, there are 155 photos that depict brain MRI scans with obvious tumours. These examples suggest possible causes for worry from a medical standpoint. The “no” group, on the other hand, is comprised of 98 pictures of brain MRI scans that reveal normal or non-tumorous occurrences and do not include any detectable tumours. A striking pattern emerges as a consequence of this distribution; specifically, nearly 61% of the dataset, which is equivalent to 155 photos, demonstrates instances of brain tumours that need additional attention. The remaining 39%, which includes 98 photos, offers illustrations of healthy brain scans that may be used as reference points for disorders that are not related to tumours. Researchers and medical professionals are able to participate in activities that attempt to create and improve algorithms for the automated identification and categorisation of tumours because they have access to a dataset that has been meticulously put together. Because the dataset contains a balanced representation of both tumorous and non-tumorous cases, it paves the way for robust model training and validation, which ultimately contributes to the enhancement of medical

Fig. 2 Brain tumour
NO-Class dataset

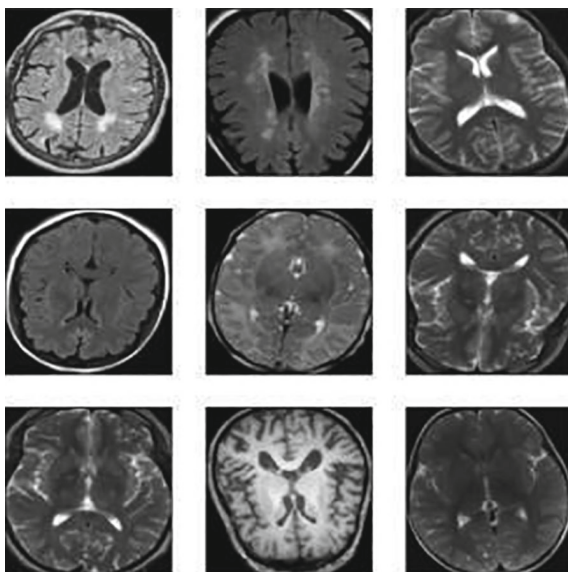


image processing methodologies and, as a consequence, the diagnosis and treatment of brain-related medical conditions (Fig. 2).

4.2 Data Augmentation

There weren't enough guidelines to tutor the neural network because the dataset may be almost nothing. Additionally, data expansion proved beneficial in addressing the issue of data unevenness within the data [15].

- There were 155 positive and 98 negative models in the dataset before it was expanded, resulting in 253 model images.
- 2103 positive and 1204 models, which together provide a total of 2065 model photos.

4.3 Data Pre-Processing

The following pre-processing procedures were conducted on each image:

1. Remove the portion of the photo that solely reveals the brain (since this is the most important aspect of the image).
2. Resize the image such that it has a state of (240, 240, 3) (picture width, picture height, number of channels) since the photos in the dataset come in a range of different dimensions [16].

3. As a consequence of this, every photo needs to have a shape that cannot be differentiated from one another in order to guarantee that they are cared for as a contribution to the neural network.
4. Scale pixel values from 0 to 1 by using the standards provided.

4.4 Data Split

The following categories were used to classify the data:

1. 80% of the data was set aside for educational reasons.
2. 10% of the data was set aside specifically for the validation process.
3. Another 10% of the data was set aside for analysis and testing.

5 Neural Network Architecture

The neural network receives an input for each x (picture) in the form of (240, 240, and 3), respectively. After that, it advances to the successive stages, which are described in the following order:

1. Following the previous layer is another layer that does not include any padding and has a pooling size of (2, 2).
2. This design develops with a convolutional layer, which has 32 filters, each of which is distinguished by a filter size of seven by seven and a minimum stride of one.
3. The addition of a batch normalisation layer, which is a transformational aspect that brings pixel values into harmony in order to increase computing efficiency, is a significant step forward.
4. The next step in this sequence will activate ReLU, which is a crucial layer that enables nonlinear transformations.
5. The framework unfolds with a layer of A-Max pooling, with its parameters calibrated at $f = 4$ and $s = 4$, respectively.
6. In continuation with the previous development, a Max Pooling stratum is repeated, and its configuration remains the same, with $f = 4$ and $s = 4$, respectively.
7. As a result of the architecture's adoption of flattening, the three-dimensional matrix is transformed into a nimble one-dimensional vector, which brings about a significant turning point [17].

In order to provide more elaboration, the architectural complexities of this model are laid out over a number of levels. Pooling begins with a layer of unadorned parameters and continues in dimensions (2, 2) where it is unaffected by padding. Pooling takes place. Following the previous layer, a convolutional layer is constructed and endowed with 32 different filters. Their spatial breadth is defined by a seven-by-seven

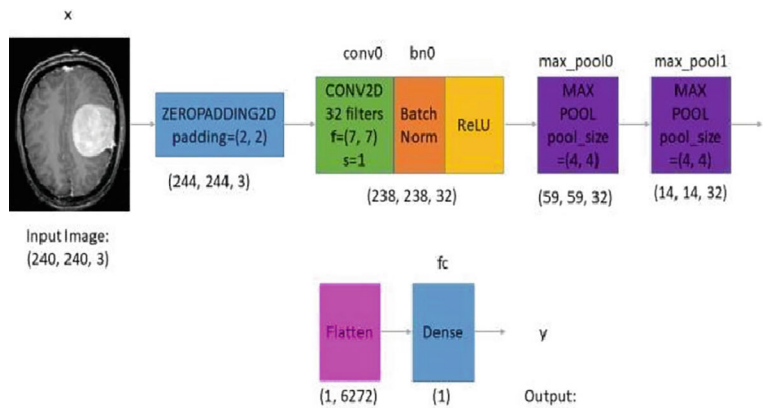


Fig. 3 Neural network architecture

matrix, and the stride parameter is set to one for a more conservative approach. In the next step, the design will embrace the strategic integration of batch normalisation, which is a normalisation process that will result in accelerated computing performance. Following this, the activation potential is skillfully enhanced using ReLU, a fundamental component in bestowing nonlinear characteristics onto the model.

In the middle of this developing architecture, the practices of pooling continue to exist in two separate manifestations. The interaction with an A-Max Pooling layer is the first thing to take place. The complexities of this interaction are characterised by a filter size of four ($f = 4$) and a stride of four ($s = 4$). This trend is echoed by a further Max Pooling layer, which replicates the stride and has the same configuration ($f = 4, s = 4$) as the previous layer.

This painstakingly crafted building smoothly transitions to a key crossroads when the three-dimensional matrix that has previously dominated is methodically collapsed into a single, extended one-dimensional vector. This transition takes place at a confluence where there is a pivotal juncture where there is a pivotal juncture. Following this, a thick layer will emerge, which will house a single neuron that has been painstakingly empowered by sigmoid activation [18]. This stratum wears the mantle of the ultimate output unit and has the power to encompass the model's predictions. The fundamentals of the binary classification make this stratum's status as the ultimate output unit evident (Fig. 3).

6 Problem Statement

The following loss and accuracy plots were generated after the model was trained for a total of 24 epochs (Figs. 4, 5).

Fig. 4 A line graph illustrating the losses of training and validation

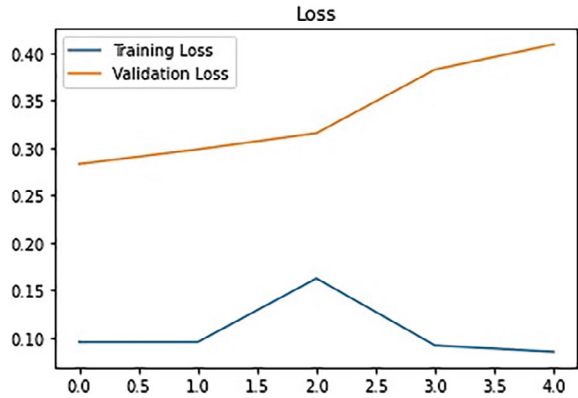
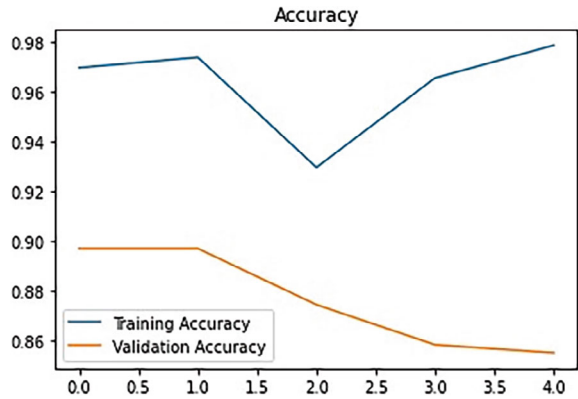


Fig. 5 A line graph illustrating the accuracy of training and validation



7 Result and Analysis

As of right now, the simplest model, which also has the least complex architecture, is the one that demonstrates a commendable level of competency in identifying neoplasms, as shown by the validation accuracy it achieves. The performance of the model is shown by the fact that it obtains an accuracy of 83.54% on the test set, which is reflective of its constant prediction abilities. The outstanding f1 score of 0.85 that was attained on the test set demonstrates that the model’s usefulness goes beyond only its accuracy. This score indicates that there is a healthy equilibrium between accuracy and recall, which highlights the model’s potential to both reliably categorise positive instances of neoplasia and efficiently detect genuine negatives. The fundamental quality of the dataset being balanced is a significant factor that adds to the importance of these results. The performance of the model improves in terms of both its robustness and its generalisation when there is a balanced mix of neoplastic and non-neoplastic instances. These convincing findings provide further evidence that the model is capable of deftly navigating the complexities of medical picture

Table 1 Result analysis

	Validation set	Test set
Accuracy	90.60	83.54
F1 Score	0.87	0.85

categorisation when applied to the setting of neoplasia. Table 1 gives a condensed analysis of the results across all of the validation and test sets, which should help you get a thorough understanding:

These strong findings provide encouraging implications for the use of machine learning in medical diagnostics, hence opening the path for improved accuracy and efficiency in the diagnosis of neoplastic diseases.

8 Conclusion

This research provided a novel cerebrum tumour ordering system. We employ image edge detection to identify areas of interest before cropping MRI images. We also use data augmentation to expand our training data. Second, a simple CNN architecture suggests a simpler tumour classification strategy. Ignoring the concept that neural networks require a large dataset to perform well, our experiments demonstrate that we can acquire high accuracy with a small and confined dataset. Our technique holds up against rival models, as seen by our accuracy rate.

Our model’s reduced runtime reflects improved computational efficiency. Our paradigm offers high diagnostic and prognostic value for brain cancer diagnosis. Extensive hyperparameter tuning and enhanced pre-processing are frequent to improve model efficiency. In future study, the suggested technique will be expanded for straight-out order issues including identifying glioma, meningioma, and pituitary tumours or other cerebrovascular anomalies. The suggested framework is for parallel characterisation issues, although it will be applied in future studies. Convolutional neural networks have great promise for brain tumour identification. Network architectures, comprehensive datasets, multimodal data integration, interpretability, and seamless integration with clinical workflows will improve CNN-based detection systems’ accuracy, reliability, and clinical applicability. These advances in brain tumour identification will improve early diagnosis, treatment planning, and patient outcomes.

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A Novel Approach for Detection of Food Adulteration Using Deep Learning with Image Processing



Ishita Theba  and Sudhir Vegad 

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Abstract The food processing sector has experienced significant impacts from the rapid advancement of technology, both in positive and negative ways. A global challenge we currently face is the adulteration of food products with either safe or harmful substances, which can harm human health or lower food quality. Recent cases have emerged where food adulteration is so subtle that traditional detection methods are ineffective. Hence, there is an urgent need to explore intelligent methods for the precise identification of these adulterants in food items. In this digital age, it is crucial to develop advanced techniques that are quick, user-friendly, cost-effective, environmentally friendly and reliable. To begin with, this paper conducts a comprehensive review of various approaches employed for detecting adulteration in commonly consumed food items such as milk, edible oils, honey and powdered spices. Furthermore, this paper introduces an innovative approach, presenting a detailed procedure for identifying adulteration using a Deep Learning model in combination with Image Processing. This approach has the potential to offer fresh perspectives and guidance for conducting advanced experimental research in this field.

I. Theba

Department of Computer Engineering, A.D. Patel Institute of Technology, The Charutar Vidya Mandal (CVM) University, Vidyanagar-Anand, India
e-mail: ishita.theba@cvmu.edu.in

S. Vegad (✉)

Department of Information Technology, Madhuben and Bhanubhai Patel Institute of Technology, The Charutar Vidya Mandal (CVM) University, Vidyanagar-Anand, India
e-mail: sudhir.vegad@cvmu.edu.in

Keywords Food adulteration • Computer vision • Machine learning • Deep learning • Image processing

1 Introduction

Food is a basic necessity of all living beings. Food, when consumed in its pure form is nutritious, toxin-free and safe that offers a good quality diet and improves or maintains the overall health of an individual. With the advancement in technologies, various positive evolutions are being achieved in many scientific fields. On the flip side of these advancements, it is increasingly observed in recent times that the possibility of misuse of research outcomes also looms large that can seriously threaten the lives of people. One such example is an adulteration of food. Food adulteration is carried out when food items are mixed with various chemicals, natural and chemical dyes, inferior and cheaper substances, clay, stones, pebbles, marble chips and sand [1]. The food adulterants can be classified into various types including intentional, incidental and metallic contaminants [2]. There are many cases of adulteration where sweets are dyed with lead, milk is adulterated with urea and detergent to increase its thickness and mustard oil is mixed with harmful substances to reduce costs [3]. Also, during festivals, sweets are adulterated to meet the increasing demand for such items [4]. As per globally reported food incidents, it can be estimated that food fraud occurs to the extent of 5–20% [5]. The present mechanism to prevent and check food adulteration is administered by a host of statutory bodies where conformance tests are routinely carried out by certified laboratories entrusted to enforce food safety and quality standards before they are sold in the market [6]. The Government of India had passed “The Prevention of Food Adulteration Act” in 1954 to prevent adulteration in food items [7]. These conformance tests are mostly analytical and instruments-based and are categorized as conventional tests. Besides being cumbersome, costly and time-consuming, these tests also often fail to detect the presence of adulterants due to newer and advanced techniques or tricks being adopted by unscrupulous individuals and businessmen. A software solution LIMS is used for accrediting the food and beverages labs to ISO 17025 standards [8]. There is an urgent and justifiable need for the development of sophisticated systems that can precisely detect the addition of anything other than ingredients inherently present in the original or native food product. To mitigate this evolving menace of food adulteration, a variety of emerging smart techniques have been devised in recent times to examine and authenticate food quality. A review of such techniques with the prospect of developing a novel smart technique for detection of adulteration in selected food products is presented in the following sections.

2 Methods for Detection of Food Adulteration

To prevent food adulteration, different detection methods are employed. They include common rapid tests for detection of adulteration in selected food products at household level [9], laboratory tests and tests using advanced software which make use of modern technologies. In this section, a brief review of most recent smart techniques used for detection of adulterants in selected food items is presented.

2.1 *Detection of Adulteration in Milk*

A novel study was carried out for detection of suspicious adulterants in milk, not targeting some specific adulterants. During experimental studies, when the same type of adulterants was present, the model provided the accuracy of 99.24% and the F1 Score value obtained was 99.13%. [10]. The performance of ensemble machine learning model was again evaluated by performing cross-validation with spiked contaminants and the F1 Score obtained was 86.57.

The Machine Learning Methods combined with Fourier Transform Infrared (FTIR) are proposed to identify the presence of cheese whey in the milk [11]. The machine learning techniques used in the study are CART (Classification and Regression Trees) and MLP (Multilayer Perceptron) Neural Network. A comparative study was carried out among these techniques and their performance was evaluated. The results depicted that the accuracy obtained using MLP is 97.8% and using CART is 96.2%. Moreover, for both the methods, the precision, specificity and sensitivity values obtained are more than 95%.

For detection of milk purity, a cost-effective solution using a software was provided [12]. In this method, a mixture was formed by adding water to milk in different proportions and a large number of samples were created. The images of these samples were captured and tested for color variation. A “classifier learner tool” was used for predicting the image for adulteration in milk with an efficiency of 95.4%.

An automatic approach that could detect contamination of milk products by identifying common beetle species was investigated by L. Wu et al. [13] where a dataset of 6900 images of beetle having different elytra were trained using a Convolution Neural Network (CNN). The results were evaluated in terms of average area under a curve and root mean square. The accuracy of the proposed algorithm obtained after cross-validation was reported to be 83%.

Deep and Ensemble decision tree learners algorithm was used to detect adulteration in milk using binary classification [14]. Milk Samples are prepared by adding five different types of adulterants to milk. With the use of multiclass classification, it can be observed that which adulterant is present in the milk sample. The accuracy obtained with these classifiers is 98.76%.

The fungus can be detected in milk using the technique of Ensemble SVM Classification algorithm [15]. The parameters such as specificity and sensitivity are used to calculate the accuracy of the proposed method. Real-time cameras are used to capture the features of milk samples. The obtained features are given as an input to the SVM Classifier and the output is compared with trained samples for performing quality check. Reasonable accuracy is obtained with the proposed method, which can be proved by experimental results.

To detect milk adulteration with water, a rapid technique is developed which uses machine learning and digital image analysis [16]. Comparative analysis for machine learning techniques like SVM and SVMR was carried out. It was found that SVMR provided better determination of extraneous water. The accuracy obtained using SVM is 89.48% and the precision obtained is 95.19%. Using SVMR, the parameters value of $R2(CV) = 0.65$ and $R2(P) = 0.71$.

A Spectroscopic Sensor System is designed and the classification of milk samples is carried out using Machine Learning techniques like Decision tree, Naive Bayes, Linear discriminant Analysis, Support Vector Machine and Neural Network Model, which resulted in the accuracy of 88.1%, 90%, 90%, 91.7% and 92.7%, respectively [17]. To increase the accuracy to 100%, hyperparameters of the neural network model are tuned by formulating a Genetic algorithm framework (Table 1).

2.2 *Detection of Adulteration in Edible Oils*

Samples are prepared by using three different rancidity levels to adulterate four different olive oils [18]. Two ANN Models are developed for classification using NIR inputs and e-Nose inputs. The other two ANNs are designed for the prediction of different aromas. The accuracy obtained using model-1 and model-2 are 97% and 87%, respectively. The performance of model-3 and model-4 is evaluated based on efficiency obtained using correlation coefficient R, whose values are 0.96 and 0.93, respectively.

Three different models are developed by adulterating Olive Oil (OO) with Canola oil (CO), Cottonseed Oil (CSO) and Soybean Oil (SO) [19]. The machine learning algorithms used for feature extraction are Support Vector Machine (SVM), k-Nearest Neighbor (k-NN) and Decision Tree (DT). For Regression, Neural Network algorithm was used. The highest accuracy obtained using SVM with OO-CO model is 94.6%, OO-CSO model is 98.2% and OO-SO model is 96.4%.

A model for detection of multivariate and untargeted adulterations using One-Class Support Vector Machine (OC-SVM) was developed [20] wherein simulations were performed using Monte-Carlo Method by mixing various edible oils in different proportions to determine the least model prediction level.

A smart system to detect adulteration in sesame oil using techniques of machine vision was proposed where the output obtained provided satisfactory results to detect the mixing of different oils in pure oil [21]. In this system, inexpensive oils like sunflower oil, canola oil, corn oil and soybean oil were mixed with pure sesame oil

Table 1 Milk adulteration detection methods

Author	Method	Parameters	Efficiency
Chung et al. [10]	Ensemble Machine Learning Model	Accuracy and F1 Score	Accuracy = 99.24% F1 Score = 99.13% on raw milk samples F1 Score = 86.57 for non-targeted food adulteration
Lima J. et al. [11]	CART, Multilayer Perceptron Neural Network	Precision, Specificity and Sensitivity	Accuracy = 96.2% for CART Accuracy = 97.8% for MLP
Witman et al. [12]	Image Processing, Classification Learner Application, KNN Algorithm	Confusion matrix	95.4%
L. Wu et al. [13]	Convolution Neural Network	Average area under the curve, Root Mean Square	83%
Neto et al. [14]	Ensemble decision tree, CNN	Correlation	98.76%
Jayanthy et al. [15]	Ensemble SVM Classification, Histogram of gradients	Specificity, Sensitivity	Not Specified
B. Gutema Asefa et al [16]	SVM and SVMR	Precision $R^2(P)$, $R^2(CV)$	Using SVM Accuracy = 89.48% Precision = 95.10% Using SVMR $R^2(CV)$ = 0.65 $R^2(P)$ = 0.71
Sowmya et al. [17]	Decision tree, Naive Bayes, LDA, SVM and ANN Model	Accuracy	Accuracy is Decision tree = 88.1%, Naïve Bayes = 90%, LDA = 90%, SVM = 91.7% ANN = 92.7%

which resulted in degradation in the quality of pure sesame oil. Different samples were composed by mixing sesame oil with canola oil and another by mixing sesame oil with sunflower oils. An artificial neural network (ANN) was then developed by a varying number of hidden layers and a training process was accomplished using available dataset. The performance of ANN was evaluated based on parameters including mean square error and correlation.

Two different models were developed using Back-propagation Neural Network (BPNN) in combination with Fourier Transform Infrared Spectroscopy [22]. Samples of tree-tea oil are pre-treated by performing convolution smoothing Savitzky-Golay

(SG) and are adulterated with soybean oil and corn oil. For feature extraction, Principal Component Analysis (PCA) and Random Forest (RF) were performed on the samples. The comparative study among two developed models, SG-PCA-BPNN and SG-RF-BPNN is carried out. The results show that SG-RF-BPNN performs better than SG-PCA-BPNN with the accuracy of 93%.

To control the quality of olive oil, two different methods were used, where the first method takes 32 inputs and it is directly provided to the classifier [23]. The second method reduces the input by applying PCA on 32 inputs and considers only 8 inputs to be provided to the classifier. The comparative analysis is carried out by applying various machine learning techniques like k-Nearest Neighbors (k-NN), Naives Bayesian, Linear Discriminate Analysis (LDA), Decision Tree, Support Vector Machine (SVM) and Artificial Neural Networks (ANN). Experimentation results prove that the 70.83% success rate is achieved with Naives Bayesian in the second method.

Sunflower oil and rice bran oil are frequently subjected to adulteration with soybean oil and palm oil, respectively. In response to solve this issue, a Partial Least Squares Regression (PLSR) model is designed. To create this model, samples were generated by blending the pure oils with a range of varying percentages of the respective adulterants. The evaluation of the model's performance was based on the assessment of two key parameters: the Root Mean Square Error (RMSE) and the Coefficient of Determination [24].

To determine adulteration in extra virgin olive oil (EVOO), a study is carried out by comparing the thermal evolution of pure EVOO and adulterated EVOO [25]. The samples were prepared by adulterating extra virgin olive oil with the adulterants like sunflower oil, refined olive oil and olive pomace oil in the amounts less than 8% in weight. Different thermographic images of pure EVOO and adulterated EVOO were collected and analyzed using the algorithms based on Convolution Neural Network and eight different models were developed. The classification accuracy obtained by the developed model ranges from 97 to 100% (Table 2).

2.3 Detection of Adulteration in Turmeric and Chili Powder

To identify and estimate white turmeric and red turmeric, a model using Partial Least Square Regression for estimation of concentrations of adulterants was used in combination with FTIR Spectroscopy [26]. This technique involved the analysis of Infrared (IR) spectra of different samples followed by the calculation of R²-Value and root mean square error of prediction for all samples prepared by mixing different adulterants. A hyperspectral imaging was used to obtain different wavelengths of adulterated chili powder samples [27]. PCA was used for dimensionality reduction resulting in an accuracy of 97%. An SVM algorithm was trained using this wavelength and tested with different datasets. An accuracy of 100% was achieved for detection of pure chili powder and oil or Sudan dye adulterated chili powder.

Table 2 Edible oil adulteration detection methods

Author	Method	Parameters	Efficiency
C. Gonzalez Viejo et al. [18]	Two ANN Classification Model, Two ANN Regression Model	Accuracy, Mean Square Error, Cross Entropy, correlation coefficient	Accuracy = 97% for model-1, 87% for model-2, Accuracy based on correlation coefficient R = 0.96 for model-3, 0.93 for model-4
Y. Yakar et al. [19]	SVM, k-NN, Decision Tree	Accuracy, Score, Mean	Highest Accuracy using SVM = 94.6% for OO-CO model 96.4% for OO-SO model 98.2% for OO-CSO model
Zhang et al. [20]	SVM	C value	100%
Mahdi Rashvand et al. [21]	Artificial Neural Network	Mean Square Error, Correlation	Not specified
J. Li et al. [22]	Back Propagation NN	Accuracy	93%
Emre Ordukaya et al. [23]	e-nose and machine learning algorithms like KNN, ANN, LDA, Naïve Bayesian, Decision Tree and SVM	Sensitivity, Precision, Specificity, Accuracy, Error rates	70.83% (Using Naïves Bayesian Classifier)
S. A. Antora et al. [24]	PLSR, FTIR	coefficient of determination (R^2) and Root mean square errors (RMSEC)	98%
Izquierdo Manuel et al. [25]	Deep Learning, Convolution Neural Network	Thermal Evolution	In the range of 97–100%

To detect adulterants present in turmeric powder, a system capable of capturing multi-spectral images was developed which used a signal processing algorithm followed by construction of a multivariate Gaussian model and development of a functional relationship between “Bhattacharya distance” and adulteration level [28]. The classification was carried out based on the value of the multivariate distribution function to confirm the presence of added ingredients in turmeric powder. The results obtained using this algorithm were compared with conventional laboratory test results to study the performance efficiency of the algorithm. It was observed that the proposed model provided an accurate outcome and was suitable for commercial applications.

A research was conducted to determine the presence of lead chromate in turmeric powder [29]. Experiments were carried out by adulterating turmeric powder with

varied proportions of lead chromate and samples were prepared. The Fourier Transform Raman (FT Raman) spectroscopy was used and the spectral data was processed using multivariate and univariate statistical analysis. Accurate results were reported where the concentration of added adulterant was more than 0.5%.

A novel approach is used to detect the adulteration in chili powder using Image Processing and Logistic Regression Technique [30]. The color features were extracted from the sample images using RGB Channels. For the classification of data, logistic regression technique is used. The model's efficiency can be evaluated by creating confusion matrix and parameters like precision, recall and f1 score are considered. It is found that the model is capable of predicting image of pure or adulterated chili powder with an accuracy of 82%.

A study is carried out to develop a rapid, simple and inexpensive method for authentication of turmeric powder using Fourier Transform Infrared (FTIR) Spectroscopy and the sampling technique used is Attenuated Total Reflection (ATR) [31]. The techniques such as Principal Component Analysis (PCA), Partial Least Square Regression (PLSR) and Partial Least Square Discriminant Analysis (PLS-DA) are used for further analysis of the obtained spectra. The model developed using PLS-DA is used to differentiate between pure and adulterated samples of turmeric powder with the value of $Q^2 = 0.9558$, $R^2X = 0.9813$ and $R^2Y = 0.9746$. The model developed using PLSR provided good calibration with the value of $R^2 > 0.99$ and lower RMSEC value.

An Artificial Intelligence-based approach is used to detect adulteration in red chili powder [32]. The red chili powder is mixed with brick powder in 12 different proportions and various samples are generated. The Image Colorspace filters such as RGB, YCbCr, HSV and lab are applied. For features extraction, techniques such as mean and histogram feature extraction are used. The Cat Boost Classifier used for classification with HSV Color space performs best compared to other classifiers used in the experiment. The Extra Tree Regressor technique used with lab color space histogram features works best for regression.

To detect the adulteration in chili powder with brick powder, the Structural Similarity Index Measure (SSIM) is used to obtain similarity between test image and reference image [33]. Also, it can be used to evaluate the approximate level of contamination. Experimental results show that accurate estimation is obtained using this technique (Table 3).

2.4 Detection of Adulteration in Honey

A system is developed based on machine learning for the detection of honey mixed with sugar syrup [34]. The system is capable to find the concentration of adulterants used in honey. For feature extraction, Linear Discriminant Analysis (LDA) is used. For classification of honey botanical origin and the level of adulteration, the k-Nearest Neighbor technique is utilized. The accuracy of the developed system to detect adulterated honey is 96.39%.

Table 3 Turmeric powder adulteration detection methods

Author	Method	Parameters	Efficiency
Dhakal, et al. [26]	Partial Least Square Regression	Root Mean Square Error Prediction (RMSEP), R^2	Not specified
Hussain Khan et al. [27]	Hyperspectral Imaging SVM Algorithm	Gamma and C-value Confusion matrix	97% (with reduced dimensionality) 100% (with complete validation data)
Chaminda Bandara et al. [28]	Multispectral Imaging system, PCA and Bhattacharya distance	Jeffries-Matisota (JM) separability measure, SSE and R^2	Not specified
Erasmus et al. [29]	FT-Raman Spectroscopy	LOD (Limit of Detection) LOQ (Limit of Quantification)	Not specified
Ms. Nilam Desai et al. [30]	Logistic Regression, Image processing using RGB Channel color features	Recall, Precision and F1 score	82%
Ellsya Angeline et al. [31]	PCA, PLSR and PL-SDA	Root Mean Square Error Prediction (RMSEP), R^2	$Q^2 = 0.9558$, $R^2X = 0.9813$, $R^2Y = 0.9746$
T. Sarkar et al. [32]	Channel Mean Value and Color Histogram feature extraction methods, Catboost classifier and Extra Tree Regressor	Accuracy, Precision, Recall, F1 Score, Cohen's kappa (k), Matthew's correlation coefficient (MCC) and ROC-AUC Score	Classification accuracy = 0.9049 for YCbCr = 0.908 for HSV Extra Tree Regressor = 0.9812 for YCbCr = 0.9837 for Lab Color Spaces
Bharath Surianarayanan et al. [33]	Image Processing	Structural Similarity Index	Not specified

Hyperspectral images of pure and adulterated honey samples were produced to detect adulteration in honey [35]. The Principal Component Analysis technique and Auto-Encoders were used for feature reduction. Various classification techniques including Support Vector Machine (SVM) and k-Nearest Neighbor technique (KNN) were used for classification. For Binary-adulteration detection and Multi-class classification, the highest accuracy achieved is above 95%.

To detect addition of Date and Sugar syrup in honey, various proportions of Date and Sugar syrup were used to produce 102 adulterated samples of Honey [36] followed by classification carried out based on types and concentrations of adulterants. Multivariate analysis using Principal Component Analysis (PCA) along with Linear Discriminant Analysis (LDA) resulted in the correct identification based on color indices and rheological properties.

The use of K-means clustering for classification of indirect honey adulteration percentage was made where parameters like glucose and total sugar content were used for classification of pure and adulterated honey [37]. The accuracy of prediction could be increased with the use of Simulated Annealing Artificial Neural Network (SA-ANN) along with radial basis Artificial Neural Network (RBF-ANN).

The honey adulterated with glucose syrup is detected by the methods of Laser-Induced Breakdown Spectroscopy (LIBS), along with machine learning algorithms such as Extremely Randomized Tree (ERT) and Linear Discriminant Analysis (LDA) [38]. The adulterated honey samples were prepared by adding glucose ranging from 10 to 90% in the interval of 10 units. The models developed were capable of classifying pure and adulterated honey with an accuracy of 90.5%.

For classification of honey samples as pure and adulterated one, techniques like MultiLayer Perceptron (MLP), Support Vector Machine (SVM) and Principal Component Analysis (PCA) are applied [39]. Promising results were obtained through spectral analysis and experiments carried out for classification. It was observed that prediction accuracy of PCA and SVM was better compared to MLP, when these techniques were applied to honey samples.

A system is proposed for detecting authenticated honey using Machine Learning Augmented Bright-Field Microscopy [40]. It is capable of detecting honey diluted with sugar syrup using pollen density analysis. Pure honey added with cheaper honey can be detected using pollen distribution comparison.

To detect the adulteration in honey, multivariate data analysis is carried out using techniques such as Cluster Analysis (CA), Partial Least Square Regression (PLSR), Principal Component Analysis (PCA), Principal Component Regression (PCR) and Machine Learning using Artificial Neural Network (ANN) [41]. CA and PCA plots are used to detect whether the honey sample is pure or adulterated. A model developed using PLSR and ANN is capable of providing optimal estimation and prediction of adulterated honey (Table 4).

3 A Proposed Smart Model for Detection of Adulteration Using Deep Learning in Combination with Image Processing Technique.

Deep learning is being increasingly deployed for a wide range of applications in areas including healthcare, food sector, entertainment, visual recognition, automation and fraud detection. [42, 43]. Deep learning algorithms are capable of automatic feature extraction and can provide deep-seated information about raw data for classification, detection and regression. [44]. Deep learning methods have a strong and automatic feature learning ability from big datasets. This ability of deep learning methods can be gainfully deployed to solve the complex and real-time problem of food adulteration. The convolution neural network is one of the algorithms of deep learning and gaining popularity due to its applications in various fields [45–47]. A model can be developed

Table 4 Honey adulteration detection methods

Author	Method	Parameters	Efficiency
M. A. Al-Awadhi et al. [34]	Linear Discriminant Analysis (LDA), K-Nearest Neighbors (KNN)	Accuracy	96.39%
T. Phillips and W. Abdulla [35]	k-Nearest Neighbors, Support Vector Machine, Principal Component Analysis, Auto-Encoders	Precision, Recall, F1 score	95%
Amiry et al. [36]	Multivariate Analysis using PCA with LDA	Physical and chemical properties	97.06% and 100%
Majdi Al-Mahasneh et al. [37]	K-Means Clustering and Simulated Annealing Artificial Neural Network	F-value, P-value, SSE	Not Specified
D. Stefas et al. [38]	linear discriminant analysis (LDA) and extremely randomized trees (ERT), laser-induced breakdown spectroscopy (LIBS)	LIBS Spectra	90.5%
H.-Y. Chien et al. [39]	MLP, SVM, PCA	Precision, F1 Score, Recall	Not specified
Peter He et al. [40]	Machine Learning Augmented Bright-Field Microscopy	Specificity, Precision, Sensitivity, F1 Score	Not specified
Md Khairul Islam et al. [41]	cluster analysis (CA), principal component analysis (PCA), principal component regression (PCR), partial least-squares regression (PLSR) and machine learning using an artificial neural network (ANN)	Root Mean Square Error of Prediction (RMSEP)	99%

using CNN for detection and recognition of food items [48, 49]. For quality control of food items, different steps are considered like image capturing, image pre-processing, segmentation of image, feature extraction and classification [50]. To detect food adulteration, a model is built using Recurrent Neural Network with the use of Internet of Things (IoT), which provides an accuracy of 92.31% [51]. A model is proposed using Deep Learning and comparative studies are carried out to depict how deep learning model is superior to other techniques [52].

Inspired by concerted efforts in this field, a novel approach can be developed which combines the physical and chemical properties of food items and an efficient model based on deep learning can be conceived and developed. This smart model can be used to detect adulteration in selected food items. The fundamental development process of the model is explained further in this section (Fig. 1).

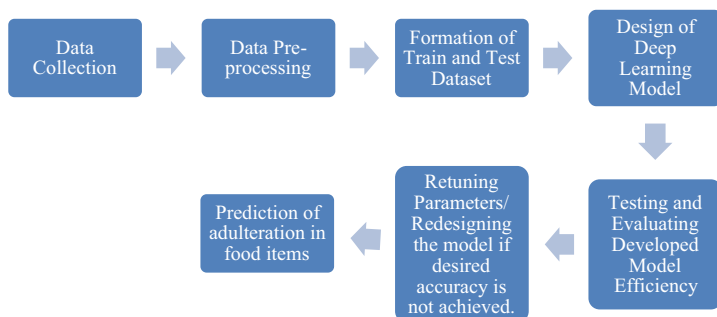


Fig. 1 Workflow process for model development

3.1 Data Collection

To initiate the process for research work, we require the dataset for various food items. The basic parameters required for testing the quality of milk are PH, temperature, taste, odor, fat, turbidity and color. The milk dataset containing the values for various basic milk properties is available on Kaggle [53]. The dataset can also be collected manually by taking samples of pure milk and forming adulterated milk samples by adding adulterants like water, urea, chalk powder, detergent, caustic soda, oil, formalin and other synthetic compounds. For the formation of dataset consisting of images, hyperspectral images can be taken for the prepared milk samples [54, 55]. Hyperspectral images represent the information of both physical attributes and chemical distribution of materials in the food item [56]. Alternately, a dataset can be formed using data from different sensors and it is combined with other parameters of food items. Capacitor sensor focuses on obtaining electrical properties of the milk and is used to detect adulteration in milk [57], Surface Plasmon Resonance (SPR) sensor is used to detect honey adulteration [58], an antenna-based sensor is used to detect adulteration in raw spices [59] and microwave sensor can find the adulteration in edible oil [60]. For other common food items such as edible oil, chilly or turmeric powder and honey, we can either directly obtain the standard dataset, which is available freely or we can make the samples by manually adding the adulterants in the food items.

3.2 Data Pre-Processing

Once the dataset is available, we can perform various data pre-processing techniques to obtain an efficient dataset for the model. If the dataset consists of images of pure and adulterated food items, then we can apply pre-processing techniques such as altering the size of the images, extracting the region of interests, reducing noise in the image, use of image enhancement techniques and applying selective filters. When

the dataset contains the sensor data and values of other parameters of the food items, then it is required to clean the data using available techniques to get rid of the values which are incomplete, inconsistent, incorrect and missing [61, 62].

3.3 Formation of Train, Validation and Test Dataset

After data pre-processing, the dataset can be formed which can be used as an input to the deep learning model. This dataset can be divided into three datasets, i.e., Training, Validation and Testing dataset as per the pre-defined ratios. For example, we can divide the data in ratios including 80% for training, 10% for validation and 10% for testing.

3.4 Design of Deep Learning Model

In today's era, deep learning models are widely used as it is capable of automatic feature extraction and can process big data. Various deep learning models like recurrent neural network, generative adversarial neural network (GAN) and convolution neural network (CNN) are used for variety of real-world applications.

Convolution Neural Network (CNN) breaks down the high dimensionality of images without losing its information, so it is well suited for the applications such as image classification and object detection [63]. The basic architecture on CNN consists of two parts, where the first part is used for automatic feature extraction and the second one is used for classification. In the CNN network, three layers namely convolution layer, pooling layer and fully connected network layers are present. In convolution layer, convolution operation is performed. The input image is convolved with the selected $M \times M$ sized filters. It is used to extract features from input image. The output of convolution layer is provided to the pooling layer. The pooling layer can reduce the size of convolved feature map by performing max pooling or average pooling operations. This layer helps to decrease the cost of computation. The last layers, before the output layer in CNN architecture consist of fully connected layers. It contains two layers having weights and biases along with neuron connections. This layer supports the classification process. Sometimes, the dropout layers are added to prevent the issue of overfitting.

3.5 Testing and Evaluation of Model

The hyperspectral images in the training dataset can be given as an input to the CNN. The model is trained using these images. Once the model is trained, it can be validated and tested using the validation set and test set. The commonly used parameters for

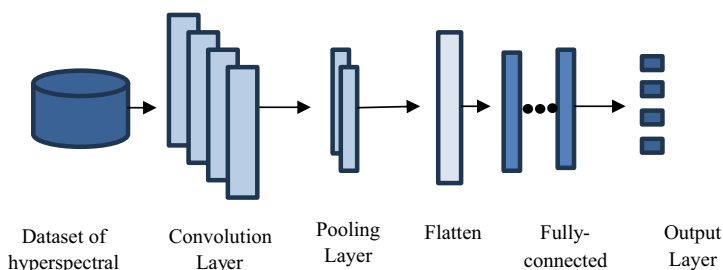


Fig. 2 Architecture of convolution neural network model

evaluating the efficiency of the model are precision, recall, f1-score and accuracy. If the desired value for these parameters is obtained, then the model is finalized, else re-tuning of the parameters of the model can be carried out. The model parameters such as the learning rate, optimizers and no. of epochs can be changed and training process can be redone. Data augmentation techniques can also be incorporated to improve the efficiency of the model. Moreover, the design of number of convolution layers and pooling layers used in CNN can be changed (Fig. 2).

4 Conclusion

This study presents an in-depth examination of various methods for identifying adulteration in commonly consumed food products, such as milk, edible oil, honey and chili/turmeric powder. Considering the recent advancements in technology, there is an expectation to create a system that leverages modern trends. It is hypothesized that the proposed intelligent model, which combines deep learning techniques with image processing, will capitalize on these technological advancements and enhance the accuracy of the results obtained. Moreover, it will reduce the complexity of the process because of its added capabilities like automatic feature extraction and dimension reduction. The developed model is expected to exhibit robustness and efficiency, as it will be trained using hyperspectral images that encompass both physical attributes and chemical distribution within food items. Additionally, researchers are provided with new insights and dimensions in which experimental research can be conducted.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical Approval It is hereby declared that all ethical compliance have been followed including human and animal subjects.

Informed Consent Informed Consent is not applicable.

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Effect of Various Machine Learning Techniques Considering Different Modalities for Breast Lesions Malignancy Detection



Priyanka Puvar and Bhagirath Prajapati

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Abstract Breast cancer ranks second among global cancers in women, with around 2.3 million diagnosed cases and 685,000 fatalities in 2020. It affects men too but is more prevalent in women. Breast cancer's development is linked to cellular abnormalities, and it can be categorized as in situ or invasive ductal carcinoma, requiring different treatments. Medical technology advancements have contributed to diagnosing breast cancer and reducing mortality rates. Deep Learning techniques in medical imaging are gaining prominence. This review explores the impact of various Machine Learning Technologies (MLTs) for breast cancer detection, highlighting the advantages of Deep Learning in the medical field. It offers valuable insights and encourages further research in this domain.

Keywords Breast cancer · In situ · Invasive · MLTs (Machine Learning Techniques) · Deep Learning · Modalities · Medical Imaging

P. Puvar · B. Prajapati (✉)

Department of Computer Engineering, A D Patel Institute of Technology, The Charutar Vidya Mandal University, Vidyanagar-Anand, India

e-mail: bhagirath.prajapati@cvmu.edu.in

P. Puvar

e-mail: priyanka.puvar@cvmu.edu.in

1 Introduction

In today's materialistic society, breast cancer is emerging as a prevalent disease, ranking second only to lung cancer among women. Although men can also be affected, their numbers remain comparatively low. The study of breast cancer has become an ongoing endeavor, leading to constant advancements and the introduction of new technologies. Statistics reveal that one in eight women will experience the effects of breast cancer at some point in their lives. The latest report from the World Health Organization (WHO) states that in 2020, approximately 2.3 million women were diagnosed with breast cancer, resulting in 685,000 deaths worldwide [1]. Identifying a precise cause for breast cancer remains a formidable task; however, there are several risk factors that can be taken into account. These include age, as the risk of breast cancer tends to rise with advancing age. Additionally, specific genes known as BRCA1 and BRCA2 have been associated with breast cancer. Personal factors such as early onset of menstruation (before the age of 12) or late menopause (after the age of 55), being overweight, undergoing menopausal hormone therapy, using birth control pills, alcohol consumption, delayed childbirth or having the first child after the age of 35, and having dense breast tissue are all potential risk factors for breast cancer [1]. Breast cancer is divided into two main categories: in situ and invasive ductal carcinoma (IDC). In situ breast cancer can be managed through active surveillance without the need for surgical intervention, whereas this approach is not applicable for IDC. Recognizing symptoms such as breast lumps, changes in size and shape, and nipple discharge can be accomplished through breast self-examination and mammography, enabling the detection of breast cancer at its most treatable stage. Treatment options may include radiation therapy, mastectomy, lumpectomy, chemotherapy, and hormone therapy [2]. After reviewing radiology reports, treatment plans are determined; however, it has been noticed that mammography might encounter difficulties such as false positives and false negatives [3]. Furthermore, the medical field generates an enormous volume of data on a daily basis, necessitating a thorough analysis of all the data to generate accurate reports. However, research indicates that approximately two-thirds of the global population lacks sufficient access to radiology specialists, and a substantial number of radiology reports contain clinically significant errors. These errors contribute significantly to up to 10% of patient fatalities [4]. In the present era of technology, it has become essential to rely on computers for assistance and validation of human judgments. Conversely, the processing of medical images is intricate and time consuming, as it involves much more than simply distinguishing between a cat and a dog in an image. In medical diagnosis, a minuscule portion of an image must be analyzed using multiple features simultaneously to yield accurate results, as even a minor error can have serious consequences. Additionally, there exist numerous challenges in detecting breast cancer, such as: 1) distinguishing between normal and abnormal tissues 2) classification of abnormal tissues to benign and malignant 3) categorizing breast tissue as dense or fatty 4) positive and negative lymph node classification [5–7]. To address these challenges, Deep Learning, a machine learning technique, has emerged as a prominent

solution. Over the past decade, Deep Learning has gained significant recognition for its advancements in the analysis and classification of medical images.

The aim of this research is to understand 1) all parameters and terms related to breast cancer, 2) breast imaging techniques, 3) Deep Learning basics and its application types for image processing such as Segmentation, Feature extraction, Detection, Classification and Prediction, 4) advancements of CNNs in Deep Learning for medical imaging.

2 Literature Review

To conduct an extensive investigation on breast cancer detection utilizing Deep Learning, a wide range of resources have been consulted, including scholarly articles, research papers, book chapters, magazines, webpages, PhD theses, and electronic media. In order to search for relevant and significant articles and research papers pertaining to breast cancer and Deep Learning, various scientific databases such as Science Direct, PubMed, Springer Link, IEEE Transactions, IEEE Journals of Biomedical and Health Informatics, and Elsevier were accessed. Keywords such as ‘breast cancer’, ‘breast image analysis’, ‘Deep Learning in Biomedical’, and ‘medical image analysis’ were employed during the search process. The extracted details are presented below.

2.1 *About Deep Learning*

We are currently living in an era dominated by machine learning (ML) and deep learning (DL). The utilization of these technologies in medical imaging has experienced a remarkable surge in demand and societal significance. During the course of this systematic review, it has been observed that ML and DL are highly sought after for diverse forms of medical image analysis. Examples include breast cancer imaging [8], brain imaging [9], and the extensive use of chest X-ray imaging in COVID-19 tests [10, 11]. Additionally, abdominal image analysis [12] and other applications further contribute to the growing list of use cases. (COVID-19 is a disease caused by a new strain of coronavirus, with ‘CO’ representing corona, ‘VI’ for virus, and ‘D’ for disease. It was previously referred to as ‘2019 novel corona-virus’ or ‘2019-nCoV’). In traditional machine learning approaches, models are trained using manually extracted features from raw data obtained from various sources. On the other hand, deep learning models have the ability to automatically extract features from raw data and integrate them into the task at hand. DL has made significant strides in the fields of segmentation, feature extraction, classification, and detection using raw data collected from diverse sources. The primary objective of DL is to train models on medical images using the concept of Convolutional Neural Networks (CNNs) and enable them to generate accurate results for novel and previously unseen inputs [13].

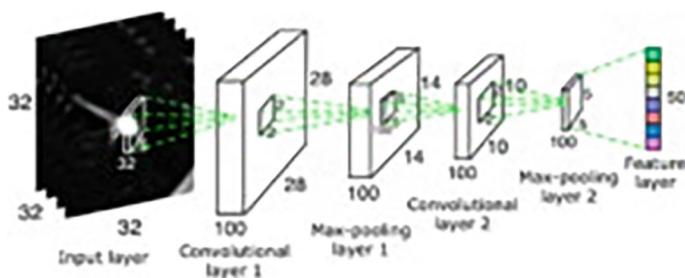


Fig. 1 Illustration of CNN for medical imaging [15]

Deep Learning (DL) has gained widespread popularity in the field of medical imaging, primarily due to the application of Convolutional Neural Networks (CNNs). CNNs consist of multiple convolutional and activation layers, which are further enhanced by pooling layers (Fig. 1). In the context of medical imaging, a sequence of mammogram images or images obtained through other modalities undergoes pre-processing to convert them from human visual images to machine visual images. Subsequently, the image parameters are adjusted based on the specific requirements of the CNN being used [14].

The altered images are then utilized as input to the CNN, where they are used to train the model. This training process enables the CNN to recognize and classify new input data. As depicted in Fig. 1, convolution layers are vital components of CNNs. During the convolution process, a filter is applied to an input image to extract relevant features. By repeatedly applying the same filter to the input, a map of activations known as a feature map is generated. The feature map serves as a condensed representation of the extracted features from the input. The key innovation of CNNs lies in their ability to automatically learn and identify significant filters in parallel, specific to the training dataset. The feature maps generated from the convolution layer are then passed through activation functions, such as ReLUs, to generate new feature maps [9]. Pooling layers are applied to each feature map resulting from the convolution and activation layers, facilitating down-sampling effects. The pooling operation considers small grids of input and produces a single value for each grid using the max pooling function. Modern CNNs also incorporate two additional concepts: Dropout regularization and batch normalization. Dropout regularization involves averaging different models into an ensemble model to enhance performance. This is achieved by randomly removing selected neurons to generate different models for each batch of training data. The resulting trained network is an optimization of multiple variations of the network [16]. Batch normalization layers are typically placed after activation layers to produce normalized activation maps, which help regularize the network and expedite the training process [17]. To design a new CNN architecture tailored to a specific task, all the aforementioned components are combined in interconnected ways. Depending on the requirements, certain operations can be replaced or modified to improve performance. In the context of this review, the focus is on breast cancer image analysis. Extensive work has been conducted in this field, with researchers

from various countries actively striving to achieve higher accuracy in performance, considering the rising prevalence of breast cancer cases worldwide.

2.2 About Breast Cancer

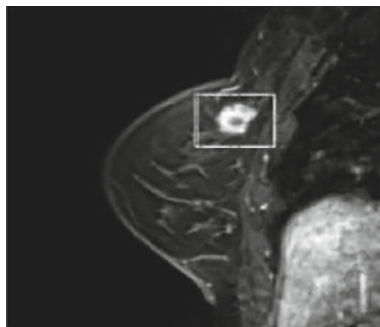
To address the escalating number of breast cancer cases worldwide, the integration of machine learning (ML) technology in the healthcare domain has emerged as a promising trend aimed at reducing mortality rates. According to the American Institute of Cancer Research, approximately 2 million new cases of breast cancer were reported in 2018 [18]. Additionally, data from GLOBOCAN 2018 indicates that one out of every four newly diagnosed cancer cases in women corresponds to breast cancer on a global scale. These statistics highlight the urgent need to leverage recent technologies to revolutionize medical practices in the field of breast cancer detection. The primary objective of this review is to explore existing efforts in the application of ML for breast cancer detection. In the process of summarizing the conducted review, five research questions were considered: 1) what are the different types of abnormalities found in breast cancer? 2) Which modalities or breast imaging techniques are utilized for digital imaging? 3) What ML classifiers are currently being employed? 4) What evaluation criteria are used for the assessment of breast cancer detection models? 5) Which datasets containing breast cancer images are utilized for research purposes? By addressing these research questions, a comprehensive understanding of the current advancements in ML-based breast cancer detection can be obtained.

3 Abnormalities

Breast abnormalities refer to atypical findings observed on mammograms, which are not always indicative of a serious concern. These abnormalities can range from harmless lumps, cysts, or dense breast tissue. However, it is crucial not to dismiss them, as they can also be associated with cancerous tissue [19, 20]. Following is the description of some of the common abnormalities that are recognized in the medical field.

Micro-calcification: Micro-calcifications are tiny deposits of calcium salts that are too small to be detected by touch but can be identified through imaging techniques. They can serve as an early indication of breast cancer, as depicted in Fig. 2. **Architectural Distortions:** Architectural distortions represent the third most frequent mammographic appearance of non-palpable breast cancer. They involve structural changes that affect the normal tissue architecture. **Bilateral asymmetry:** Bilateral asymmetry refers to an abnormality where breast development is typically uneven, with one or both breasts being smaller than expected. This condition may be a congenital defect associated with underlying abnormalities of the pectoral muscle.

Fig. 2 Sample raw MRI image taken from publication referred for proposed review [21]



4 Modalities

The article explores different breast imaging techniques, which are known as modalities. Several modalities were identified during the review process, including Mammography, contrast-enhanced Mammography, Digital Tomosynthesis, Sonography, Ultrasound, Sonoelastography, Magnetic Resonance Imaging, Infrared Thermograph, Magnetic Elastography, Diffusion-weighted Imaging, Magnetic Spectroscopy, Nuclear Medicine, Optical Imaging, and Microwave Imaging. The most frequently used modalities in the research discussed in this article are DM (Digital Mammogram), US (Ultrasound), MRI and IRT [22].

5 Comparative Study

This section provides a comprehensive analysis of various parameters considered in different combinations. These parameters include the dataset of breast images, the modalities employed, the machine learning classifiers utilized, the scope of the classifiers, and the evaluation metrics employed. As previously mentioned in this article, the study incorporates Digital Mammogram (DM), Ultra Sound (US), Magnetic Resonance Imaging (MRI), and Infrared Thermograph (IRT) as modalities. The machine learning classifiers considered are Deep Learning (DL), Random Forest (RF), Linear Regression (LR), Support Vector Machine (SVM), Artificial Neural Network (ANN), Ant Colony Optimization algorithm (ACO), and K-Nearest Neighbour (KNN), alongside their respective research scopes. Regarding performance measurement, the survey employs popular evaluation metrics such as Accuracy (Acc), Sensitivity (Sn), Specificity (Sp), and Area Under the Curve (AUC). Accuracy (Acc) indicates the percentage of samples correctly classified out of the total samples, Sensitivity (Sn) represents the true positive rate, which determines the percentage of correctly classified abnormal samples, Specificity (Sp) denotes the true negative rate, indicating the percentage of correctly classified normal samples, and Area Under the Curve (AUC) is a commonly used metric for selecting optimal models

while disregarding suboptimal ones. The equations for Accuracy (Acc), Sensitivity (Sn), and Specificity (Sp) are provided as follows [23].

$$\text{Acc} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (1)$$

$$\text{Sn} = (\text{TP}) / (\text{TP} + \text{FN}) \quad (2)$$

$$\text{Sp} = (\text{TN}) / (\text{TN} + \text{FP}) \quad (3)$$

where TP: true positive, TN: true negative, FP: false positive, FN: false negative. In addition to the aforementioned parameters, various types of datasets have been considered, including the Digital Database for Screening Mammography (DDSM) [24], Mammographic Image Analysis Society (MIAS), Image Retrieval in Medical Applications (IRMA) [25], Wisconsin Breast Cancer (WBC), BancoWeb [26], and two Portuguese datasets, namely the IN breast database and the Breast Cancer Digital Repository (BCDR) [27], which have different numbers of cases. A survey of articles [28–35] reveals that when SVM is used as the sole machine learning technique (MLT) or in combination with other MLTs, along with different types of image modalities and various datasets for cancer classification, accuracy can range from 88 to 98%. On the other hand, another survey of articles [36–44] demonstrates that utilizing MLTs such as RF, DT, ANN, KNN, LR, and ACO, along with different datasets and image modalities, can result in accuracy ranging from 78 to 98%. It is worth noting that DL as an MLT achieves an accuracy of almost above 90% [45, 46]. Throughout the course of this research, the primary focus has been on observing the impact of SVM compared to other classifiers by considering different modalities and investigating the reasons behind the popularity of Deep Learning.

Therefore, more articles on SVM have been included in the study. From the conducted analysis, it is evident that the majority of studies utilize DM as the modality, and in many cases, the SVM classifier has demonstrated significantly higher accuracy compared to other MLTs, as depicted in Fig. 3. On the other hand, DL offers advantages such as automatic feature learning, faster classification, less time consumption, and improved accuracy. However, ongoing research aims to leverage the advantages of DL by exploring various combinations of different parameters. The findings obtained from reviewing different articles can be visualized through Fig. 3, which illustrates the performance of the retrieved information. Additionally, when deploying deep neural networks or any other machine learning model for medical image analysis, challenges related to data access, privacy issues, and data protection arise as crucial obstacles.

A wide range of articles from different sources were consulted for the literature review. Among them, specific articles listed in Table 1 were selected for a more comprehensive analysis in the context of further work. The table presents details of the selected articles, including the information abstracted from them. Based on the abstracted information, it can be concluded that achieving satisfactory outcomes in medical imaging relies on selecting the appropriate modality in combination with

Table 1 Illustration of details extracted from relevant articles considered for further work

Sr. no	Article Reference no	Work carried out as per article	Details extracted
1	[21]	The combination of dynamic contrast-enhanced (DCE) MRI and diffusion-weighted (DW) MRI modalities was employed Five features, namely the apparent diffusion coefficient (ADC), Sum Average, Entropy, Elongation, and Sum Variance, were chosen to achieve optimal diagnostic performance Three classical classification models, namely SVM, KNN, and Random Forest, were utilized	The focus of the study was primarily on evaluating the impact of combining a set of features with MRI, and it was observed that the combination of features significantly contributes to the classification of malignant masses. One limitation regarding MRI is that the databases of the 3.0-Tesla group were not sufficiently large to enable the extraction of a rigorous statistical model. Therefore, it would be valuable to assess the diagnostic performance of MRI at 3.0-Tesla using larger sample sizes The ADC parameter has been confirmed to possess high diagnostic value either on its own or in combination with other features The analysis indicates that its strong performance is primarily attributed to improvements in specificity. However, not all performance parameters and AUC values remain below 85
2	[47]	The imaging technique used is digital mammography The computer-aided diagnosis (CAD) system is based on the dual-tree complex wavelet transform (DT-CWT) The classifier employed in this system is SVM (Support Vector Machine)	The focus was primarily on evaluating the accuracy of SVM, which heavily relied on the feature extraction method used The presence of micro-calcification was considered as an abnormality in the classification process SVM demonstrated the capability to classify effectively when coupled with explicit feature extraction algorithms or methods

(continued)

Table 1 (continued)

Sr. no	Article Reference no	Work carried out as per article	Details extracted
3	[48]	A systematic review was conducted on breast cancer diagnosis through computer-aided detection (CAD), covering the period from 2012 to January 2017. Databases such as SpringerLink, ScienceDirect, IEEE, and PubMed were consulted for relevant articles. A total of 154 articles were considered in the review process. The number of articles examined has shown an increasing trend each year. The review encompassed state-of-the-art machine learning techniques (MLTs), along with different image modalities and features, across various datasets.	Comparing methods with each other is a challenging task due to various factors. These factors include the specific databases utilized for assessment, the selection of image samples for evaluation, the number of samples employed, and the chosen assessment approach, including validation methodology and training/testing sets. Until 2015, the SVM classifier was widely employed for breast tissue classification. However, its performance is still hindered by issues such as false positive and true negative outcomes arising from inadequate feature detection and other related parameters. In terms of future research, the use of standardized public image datasets encompassing images from diverse image modalities, along with the incorporation of Deep Learning classifiers, shows promise as a more favorable trend.
4	[49]	The projected number of breast cancer cases in India for the year 2020 is estimated to reach as high as two million. Several classifiers, including Naive Bayes Classifier, Support Vector Machine (SVM) Classifier, Bi-clustering and Ada boost Techniques, R-CNN (Convolutional Neural Networks) Classifier, and Bidirectional Recurrent Neural Networks (HA-BiRNN), have been considered. The dataset used in the study is sourced from the M. G Cancer Hospital & Research Institute in Visakhapatnam, India. This dataset is both well-annotated and large-scale, consisting of 8,009 histopathology image samples from 683 patients across various magnification levels.	Existing methods have certain limitations. The Naive Bayes Classifier tends to yield poor results when the training data is not adequately represented. The SVM classifier is not suitable for handling large datasets and is less effective in high-complexity computer vision applications. Bi-clustering and Ada boost Techniques can lead to inaccurate classification results when dealing with imbalanced data. Additionally, the training process for R-CNN is time consuming. Considering these limitations, a new method called Deep Neural Network with Support Value (DNNS) is proposed. The normalization process is utilized to enhance the quality of images. It is essential to validate the new algorithm to assess its performance and effectiveness.

(continued)

Table 1 (continued)

Sr. no	Article Reference no	Work carried out as per article	Details extracted
5	[50]	A comprehensive application of deep learning in MRI processing has been described, encompassing the entire chain from image acquisition to image retrieval, as well as segmentation and disease prediction A compilation of CNN architectures commonly employed in medical imaging has been provided, accompanied by high-level descriptions	In addition to highlighting the advancements and advantages of Deep learning, the article also acknowledges and discusses the challenges associated with its application in medical imaging. These challenges include the three-dimensional nature of medical images, issues related to data availability, concerns regarding trust and interpretability of the results, and the integration of Deep learning into existing workflow systems
6	[51]	A review of articles spanning from 2004 to 2018 was conducted with the following objectives: (1) to assess the utilization of imaging modalities, (2) to compare various breast cancer imaging modalities, (3) to identify the most frequently cited and publicly accessible breast cancer databases encompassing different formats and modalities, (4) to evaluate the implementation of Deep Learning (DL) in medical image analysis, particularly in the context of breast cancer image analysis, and (5) to assess the application of DL in histo pathology-based breast cancer image analysis	Histology has gained popularity as a modality in medical imaging, and the Break His dataset, comprising approximately 7909 images with multiple classes, has been extensively utilized in studies The article also addresses challenges associated with Deep Learning (DL), including the management of large datasets and the preparation of pre-trained models for transfer learning

(continued)

Table 1 (continued)

Sr. no	Article Reference no	Work carried out as per article	Details extracted
7	[52]	A novel deep convolutional neural network (CNN) model was proposed for the automated classification of breast cancer using two distinct types of images: mammograms and ultrasounds According to the European Cancer Information System (ECIS), breast cancer has a mortality rate of 13.4% in European countries, and in the year 2020, it incurred a high national cost compared to other diseases The report from the Globocan project reveals that in India, there were 162,468 newly detected cases of breast cancer, with an annual death rate of 87,090	CNN models possess distinctive characteristics such as automated feature extraction, high-level feature learning, and the ability to utilize transfer learning. In comparison to other existing CAD models, CNN-based CAD models achieve higher accuracy in detection. - To address the issue of overfitting, the CNN was customized and fine-tuned The model's performance was assessed using various pooling techniques The objective was to develop a superior pre-trained model that outperforms existing models in terms of performance and applicability
8	[53]	The study aimed to investigate the enhanced accuracy of Dynamic Contrast-Enhanced Magnetic Resonance Imaging (DCE-MRI) compared to plain MRI in the detection of malignant breast lesions Traditional MRI analysis typically focuses on one-dimensional information, disregarding the correlations present in other dimensions. In contrast, DCE-MRI comprises a comprehensive set of high-dimensional image datasets, providing a more comprehensive perspective	A significant challenge in handling multi-dimensional datasets is the computational burden and susceptibility to noise. Therefore, a key objective in post-processing is to reduce the dimensionality of these datasets in order to conserve storage space The significance of employing semi-supervised and self-supervised deep learning approaches for dealing with weakly labeled data has been emphasized The concept of transfer learning is also elucidated, with the goal of leveraging knowledge from related learning tasks to improve performance in a target task
9	[54]	The study aims to further investigate the capabilities of fine-tuned pre-trained CNN models in the classification of breast cancer histopathology images. It introduces a sequential model that combines features from different layers to enhance performance	The results indicate that the multilayer deep feature fusion implemented within the sequential framework of the proposed approach surpasses the performance of the baseline network

(continued)

Table 1 (continued)

Sr. no	Article Reference no	Work carried out as per article	Details extracted
10	[55]	A key aim of this study is to investigate image classification across various optical magnification levels	The study examines the impact of optical magnification on classification. It is observed that the magnification-specific model requires comparatively less training compared to the magnification-independent model.
11	[56]	The authors put forth a comprehensive model that incorporates images at various magnifications for the classification of breast cancer histopathological images. The model integrates the scores from different magnifications, thereby emphasizing appropriate features and classifiers for each magnification.	Selecting suitable combinations of features and classifiers can significantly outperform existing methods, and the integrated model demonstrates a more consistent performance in terms of AUC compared to individual magnifications. It is important to note that not all features are suitable for the same classifier. Therefore, the advantages lie in identifying the appropriate combinations of features and classifiers for improved results.

6 Conclusion

Nowadays, ML-based medical diagnosis is a highly active area of research. Individuals with expertise in ML and DL are in high demand in both industry and academia. Studies like these can bring numerous benefits to society, including improved diagnosis rates, reduced costs associated with medical complementary modalities, and minimized false positive reactions. False positives often lead to unnecessary treatments and incur psychological, physical, and economic burdens. Based on the findings of this survey, it is evident that generating large and up-to-date datasets of breast images remains a challenging task. However, if successfully accomplished, it can greatly contribute to the development of highly performing models. Furthermore, the use of histopathological images has been established as the gold standard in research focused on breast cancer detection and malignancy analysis using various ML techniques. Additionally, there has been considerable research conducted on the application of SVM for breast cancer detection with different modalities.

However, SVM faces challenges in efficient feature extraction. Alternatively, employing Deep Learning as a machine learning technique, instead of SVM, in conjunction with modalities such as MRI, ultrasound, or histopathological images

can offer new insights. Overall, conducting research in this direction holds the potential to expand the knowledge base of breast cancer diagnosis and contribute to saving lives through technological advancements.

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A Web-Based System to Forecast Heart Disorder by Using Machine Learning Algorithm



D. Vidyabharathi, B. Sathiya, G. Mohanraj, S. Theetchenya,
B. Galeebathullah, and G. Vidhya

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Abstract Heart disease remains a leading global cause of mortality, with the alarming statistic that approximately one individual succumbs to a heart attack every minute. In our contemporary society, this issue has evolved into a pressing concern, demanding immediate attention. The early detection of heart disease presents a formidable challenge, but modern technology offers a promising solution. Machine learning, when applied to healthcare, emerges as a powerful technique capable of providing timely and precise predictions regarding the onset of heart disease. This application holds the potential to significantly impact the landscape of heart disease diagnosis, allowing for swifter and more accurate interventions. This dataset consists of many attributes like age, sex, and resting BP with samples Wi-fi module ESP 8266 is interfaced with IOT for transfer the patient data to doctor. Arduino Uno is involved as IOT to connect with BP sensor and ECG sensor to measure BP and pulse level of patient. The incorporation of a 3D Convolutional Neural Network (CNN) helps in

D. Vidyabharathi (✉) · S. Theetchenya · G. Vidhya
Sona College of Technology, Salem, Tamilnadu, India
e-mail: dvbharathi77@gmail.com

B. Sathiya
Dhanalakshmi College of Engineering, Chennai, Tamilnadu, India
e-mail: sathiya.b@dce.edu.in

G. Mohanraj
Vellore Institute of Technology, Vellore, Tamilnadu, India
e-mail: mohanraj.gopal@vit.ac.in

B. Galeebathullah
Madanapalle Institute of Technology and Science, Madanapalle, Andhra Pradesh, India

enhancing both the accuracy and overall performance of the model. As part of this process, the back-propagation algorithm diligently computes the gradient of the error function, facilitating the iterative adjustment of neuron weights. In our dataset, the utilization of a tenfold cross-validation technique proves to be adequately effective in addressing model complexity and ensuring robust results. The key objective for the model to be proposed is to attain a high level of accuracy and performance, while also offering flexibility and a high probability of success.

Keywords Prediction · IOT · Arduino Uno · 3D CNN · Cross-validation technique

1 Introduction

Heart disease stands out as a profoundly perilous global affliction. Over recent years, a disconcerting surge in reported cases has led to a surge in heart-related fatalities, painting a grim picture. According to the World Health Organization (WHO), heart disease alone is accountable for a staggering 18 million lives lost worldwide. Considering the heart's pivotal role in human physiology, any malady affecting it profoundly impacts overall health. Recognizable symptoms of heart disease encompass bloating, breathing difficulties, swollen legs, chest pain, fatigue, and irregular heartbeat rhythms. Contributing factors include age, stress, excess weight, an unhealthy diet, and smoking, all contributing to the prevalence of this life-threatening condition.

The objective is to progress a machine learning-based model and to predict the heart disease as earlier as possible. This model holds the potential to revolutionize the healthcare landscape by assisting medical professionals in detecting the condition with greater accuracy and efficiency, thereby reducing the need for extensive medical tests and reports. Ultimately, this approach has the potential to save numerous lives by facilitating timely and appropriate care. By harnessing the power of machine learning, which excels in extracting valuable insights from vast datasets, especially prevalent in hospital settings, this research lays the foundation for a more effective diagnostic tool. Machine learning's rapid data processing capabilities and its ability to provide early-stage predictions further solidify its role in advancing the field of healthcare.

This paper explores various data mining techniques, including classification, prediction, and pattern recognition, as valuable tools in the diagnosis of heart disease. Specifically, the study focuses on the utilization of classification models within the domain of machine learning to identify heart diseases. The input data is subjected to a range of classification algorithms, each tasked with predicting and assigning the data to its respective category or class. These classification techniques encompass Logistic Regression, Random Forest, Decision Trees, Support Vector Machine, XGBoost, Naïve Bayes, Gradient Boosting, K-Nearest Neighbor, NN-Neural Network, LSTM, and BiLSTM.

In this research, various models for classifying the heart problems are trained. To determine the optimal model for this critical task, their performance is meticulously compared using various metrics for evaluations such as accuracy and others. The aim is to recognize the classification model which can achieve better accuracy in predicting the occurrence of heart disease, thereby contributing to more effective and timely medical interventions.

2 Related Works

Several studies have looked at a variety of factors, algorithms, and methodologies for predicting heart disease. Several studies have looked at a variety of factors, algorithms, and methodologies for predicting heart disease. Below is a list of some studies and research.

Singh and Kumar [33] proposed a predicting system for heart disease using machine learning techniques which used many algorithms like SVM, decision tree, KNN, linear regression. For implementing they used python program (Jupyter notebook) and UCI repository was used as a dataset which contains 14 attributes and KNN model will give the greater accuracy compared to others.

Sharma et al. [32] suggested a machine learning method for heart disease using the algorithm of random forest, and decision tree. Using Weka Data mining the model has been built and the dataset using for this model are collected from Cleveland Heart Disease which is available through internet, which has 1025 instances with 14 attributes. In this model random forest will give the greater accuracy compared to decision tree (Fig. 1).

Gavhane et al. [9] proposed a system of Heart Disease using a machine learning algorithm, to train and testing the model they used a algorithm called multi-layer perceptron. The system was implemented using scikit learn which is an python library and achieved an accuracy of 91%.

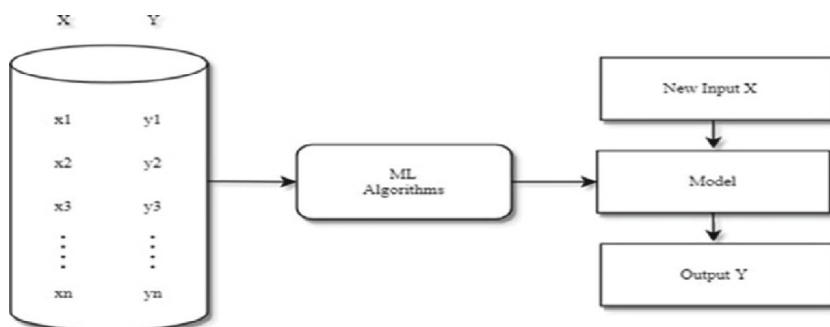


Fig. 1 Supervised learning

Raja et al. [29] suggested a machine learning technique to predict the Heart Disease. In this paper they used various algorithm like Random forest, Decision Tree. Random forest is particularly used which provides an ensemble learning process for classification and regression techniques. For Random forest they achieved a 99.67% and for Decision tree was 97.6%.

Choudhary and Narayan Singh [7] proposed a machine learning technique for HD prediction and the Ada-Boost method was employed after the decision tree algorithm to determine if a person is really suffering from heart disease or not and collect the dataset from internet which achieved an accuracy of 97.08%.

Krishnan and Geetha [16] suggested a data mining technique for predicting heart disease, and parameters were also tested. They utilized the Python program to process the dataset, which incorporates two main machine learning approaches, Decision Trees and the Nave Bayes Algorithm. This study's dataset is derived from UCI, which comprises 300 instances with 14 clinical parameters that are specifically used for blood pressure, chest discomfort, and ECGs.

Ramprakash et al. [30] have used a deep neural network for Heart Disease Prediction, and to solve the fitting and over fitting problem, they used χ^2 -statistical model. DNN and ANN are both the models utilized to accurately predict the HD. Cleveland coronary disorder dataset used for this study which contains 303 instances and 14 attributes.

Haq et al. [12] have used a machine learning method for Heart Disease and also used a backward selection of feature algorithm using KNN which gives the greater accuracy. For this paper they used Cleveland heart disease dataset and reached 90% accuracy of classification.

Krishnani et al. [17] proposed a machine learning-based Coronary Heart Disease prediction. In this research, they primarily focused on preprocessing the extensive technique and used numerous machine learning algorithms (Random Forest, Decision trees, and KNN) as well as they used a dataset of Framingham Heart Study, which contained 4240 records.

3 Proposed Model

The system's outputs are assessed and evaluated using the Cleveland Heart Disease Database, sourced from the UCI (University of California, Irvine) dataset repository. This database comprises a total of 1190 records, with each record containing 12 essential clinical attributes. These attributes include age, pain category, gender, blood pressure, sugar level, resting electrocardiographic results, maximum heart rate achieved, cholesterol level, presence of induced angina, depression developed by exercise related with rest, the slope of the peak exercise ST segment, the vessels colored by fluoroscopy in number, and the maximum heart rate achieved during stress testing (thalach).

In this model, the database undergoes a random partition into two different sets: the research and training set. When splitting a dataset into these two subsets, the

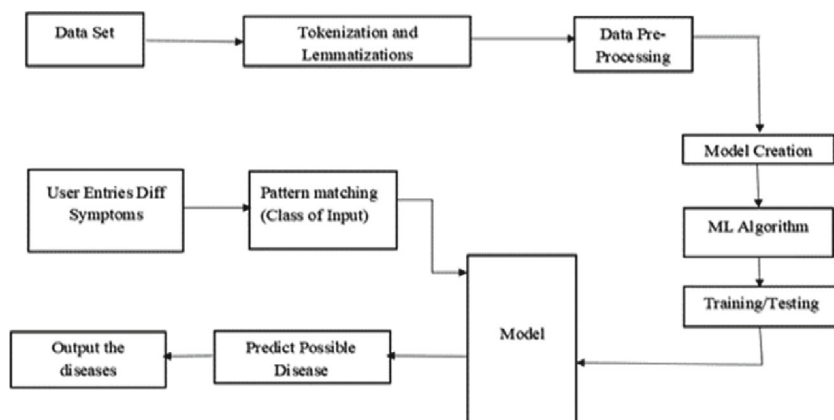


Fig. 2 System architecture for heart disease prediction

majority of the data is allocated for the training phase, while only minimum portion is reserved for testing. To guarantee the equivalence and fairness of the training and testing datasets, Analysis Services employs a random selection process.

The testing dataset serves as a collection of observations specifically designated for the assessment of the model's performance, typically relying on various performance metrics. In the final step of the evaluation, the algorithm's accuracy is scrutinized to determine its effectiveness in identifying the occurrence or not occurrence of heart disease. This crucial assessment aids in gaging the model's reliability and its potential as a diagnostic tool (Fig. 2).

3.1 Data Collection and Preprocessing

The output data is evaluated using the Cleveland Cardiovascular Disease Condition Database, sourced from UCI's dataset repository. This database comprises a total of 1190 records, and each record encompasses 14 critical clinical attributes. These attributes include age, type of chest pain, gender, blood pressure, sugar level, resting electrocardiographic results, cholesterol range, heart rate achieved at maximum, exercise-related ST depression, presence of angina, number of major vessels colored by fluoroscopy, exercise ST segment at peak, and the maximum heart rate achieved during stress testing (thalach). These attributes collectively provide essential information for the assessment and prediction of cardiovascular disease conditions.

Preprocessing data eliminates any unclear information from the collected dataset, reducing accuracy and prediction rate. Data loss, which must be eliminated before the model is trained, may have been caused by human error. The dataset is further preprocessed to eliminate any duplicate or missing values after it has been collected.

Additionally, unnecessary variables are eliminated, and the dataset is then used to train the model. When more than one missing entry is present in a single row, those rows are removed from the dataset. However, this is only done if the missing value is of the missing-at-random (MAR) type. We used the widely used mean and median approaches for numerical and categorical characteristics, respectively, to input missing data.

3.2 3D CNN

A 3D convolution, also known as a 3D convolutional layer, operates by sliding a kernel or filter through data in three dimensions, as opposed to the two-dimensional sliding used in 2D convolutions. This type of convolution is particularly valuable in scenarios such as medical imaging and video analysis.

In medical imaging, 3D convolutions are employed when dealing with three-dimensional data, such as volumetric scans (e.g., CT scans or MRI scans), where the information exists in three spatial dimensions (length, width, and height). Utilizing 3D convolutions lets the model to locate spatial patterns and structures within the 3D volume, making it suitable for tasks like organ segmentation, anomaly detection, and disease diagnosis.

In the context of video-based data, 3D convolutions are instrumental due to the extra temporal dimension introduced by time. Video data consists of a sequence of frames, and the temporal dimension represents the changes that occur over time. By applying 3D convolutions, models can learn both spatial and temporal features, enabling them to understand motion, object interactions, and other dynamic aspects of the data. This makes 3D convolutions a powerful tool for video analysis tasks, including action recognition, video classification, and object tracking in videos.

In summary, 3D convolutions are a crucial component of deep learning architectures for tasks involving volumetric data, such as medical imaging, and for analyzing data sequences with a temporal dimension, such as videos. They allow models to capture intricate patterns and dependencies in three-dimensional space or spacetime, enhancing their ability to perform tasks that require an understanding of both spatial and temporal information.

3.3 Flask Web App

Flask is a lightweight Python web framework that offers a set of useful tools and features, simplifying the process of building web applications in Python. It provides developers with flexibility and can be particularly approachable for newcomers to web development because it enables the creation of web applications quickly, often with just a single Python file. One of Flask's strengths is its extensibility, allowing developers to add various extensions and libraries as needed for their specific project.

Additionally, Flask doesn't impose a rigid directory structure or demand complex boilerplate code before starting development, making it a versatile choice for web application development [2, 6, 14, 21].

3.4 *Arduino UNO*

The Arduino UNO is a standard Arduino board, and its name "UNO" is derived from the Italian word for "one." It was given this name to signify the initial release of Arduino Software. Furthermore, it was the first Arduino board to feature USB connectivity. The Arduino UNO is renowned for its versatility and is widely used in various projects. It was developed by Arduino.cc.

The heart of the Arduino UNO is the ATmega328P microcontroller. It distinguishes itself as a user-friendly board when compared to other Arduino boards, such as the Arduino Mega. The board comes equipped with a range of features, including analog and digital input/output (I/O) pins, compatibility with shields (add-on boards), and additional circuits.

Key hardware specifications of the Arduino UNO include 6 analog input pins, 14 digital pins, a USB connector for programming and communication, a power jack for external power supply, and an ICSP (In-Circuit Serial Programming) header for advanced programming and debugging.

To program the Arduino UNO, developers use the Arduino IDE (Integrated Development Environment). This IDE is compatible with both offline and online platforms, offering a unified environment for writing and uploading code to the board. Importantly, the Arduino IDE is consistent across all available Arduino boards, making it easy for users to switch between different Arduino models for their projects.

3.5 *Process in Classification*

The characteristics listed in Table 1 serve as input features for several machine learning algorithms, including Logistic Regression, Random Forest, Decision Tree, and Naive Bayes classification techniques [31, 37, 38]. To train and evaluate these algorithms, the input dataset is partitioned, with 80% allocated for training purpose and the remaining 20% reserved for the testing purpose. This division enables the models to learn from the majority of the data and then assess their performance on unseen data to gauge their effectiveness in making predictions or classifications. The dataset used in training the model is the training dataset. It serves as the input for the model through the learning stage, where the model adjusts its internal factors to do predictions or classifications based on the patterns it studies from this data. On the other hand, the testing dataset, also known as the validation or test dataset assesses the quality of the trained machine learning model. It contains data that the model has not seen during the training phase, allowing for an objective valuation of how

well the model generalizes to new, unseen data. For each of the machine learning algorithms employed in the study, their performance is measured and analyzed using various evaluation metrics. These metrics include:

Accuracy: A measure of how often the model's predictions are correct, typically expressed as a percentage.

Precision: Precision measures the ratio of true positive predictions to all positive predictions generated by the model, serving as a gauge of the model's capability to provide precise positive predictions.

Recall: Recall or true positive rate measures the positive predictions among all actual positive values in the dataset. It indicates the model's capability to identify all relevant positive cases.

F1-measure: The F1-measure is the average of precision and recall which will be applied on the formula. It offers a well-rounded evaluation of a model's performance, which proves especially valuable when working with datasets that exhibit imbalances.

These performance metrics are used to evaluate the efficacy and reliability of each machine learning model, allowing for a comprehensive analysis of how well they perform in the context of the specific dataset and task at hand. The various algorithms are discussed below.

Random Forest It is a versatile classification model that controls the power of various number of decision trees to classify data. In the construction of each individual decision tree, it employs a combination of two techniques: bagging (Bootstrap Aggregating) and randomness. This ensemble learning method aims to create a forest of uncorrelated decision trees, where the collective prediction of the ensemble tends to be more reliable and accurate than that of any single tree. The key features of Random Forest include:

Bagging: Bagging involves having numerous subsets of the training data through random sampling and replacement. Each of the subset is used to train in the separate decision tree. This helps reduce overfitting and enhances the model's generalization.

Randomness: Randomness is introduced during the tree-building process by considering only a arbitrary subset of features when splitting in the decision tree. This randomness further diversifies the individual trees and contributes to the ensemble's robustness.

Random Forest is effective at addressing the issues associated with high bias and high variance in machine learning models. By combining the predictions of multiple trees through averaging, it achieves a natural balance between these two extremes, resulting in a more accurate and stable classifier. Random Forest can be implemented in popular programming languages such as R and Python, thanks to robust libraries and packages that provide ready-to-use implementations. These libraries simplify the process of building, training, and evaluating Random Forest models, making it accessible for data scientists and machine learning practitioners.

Logistic Regression Classifier It is a supervised classification commonly required to predict the categorical variable based on the set of independent variables or features. It is particularly well-suited for predicting the outcome of a categorical variable [1, 13, 27, 28]. As a result, the output of a logistic regression model is typically a discrete or categorical attribute, which represents a class or category.

Table 1 Table captions should be placed above the tables

S. no	Attributes	Descriptions	Uses
1	Age	Patient 's Age	The risk of developing cardiovascular or heart problems substantially increases with each successive decade of life, nearly tripling in likelihood as individuals progress through different age groups
2	Sex	Male, Female	In comparison to pre-menopausal women, men have a higher susceptibility to heart disease. Furthermore, when comparing a man with diabetes to a woman, the likelihood of the woman experiencing heart disease is greater
3	ChestPain Type	Based on the pain experienced, it is classified as 1-typical, 2-typical angina, 3-non-anginal pain, 4-asymptomatic (Nominal)	Insufficient oxygen-rich blood supply to the heart muscle can lead to the onset of angina, characterized by chest pain or discomfort. This chest discomfort may manifest as pressure or a squeezing sensation. Additionally, individuals may feel pain in areas such as the back, neck, jaw, shoulders, or arms. In some cases, angina pain can be reminiscent of indigestion
4	Resting bps	Level of blood pressure at resting mode in mm/HG (Numerical)	The risk of cardiovascular problems becomes even more elevated when an individual has high blood pressure in conjunction with other medical conditions such as diabetes, high cholesterol, or obesity
5	Cholesterol	cholesterol level in mg/dl	The likelihood of experiencing a heart attack is greater when an individual has elevated levels of triglycerides and low-density lipoprotein (LDL) cholesterol
6	Fasting blood sugar	Blood sugar levels on fasting	Elevated blood sugar levels occur when the pancreas fails to produce an adequate amount of insulin or when the body does not respond effectively to insulin. This condition increases the risk of experiencing a heart attack
7	Resting ECG	Result of Electrocardiogram while at rest are represented in 3 distinct values 0: Normal 1: Abnormality in ST-T wave 2: Left ventricular hypertrophy	The USPSTF (United States Preventive Services Task Force) has determined that for individuals at low risk of cardiovascular disease, the potential risks associated with screening using a resting or exercise ECG are either equal to or greater than the potential benefits
8	Max heart rate	Maximum heart rate achieved	The increase in cardiovascular risk, coupled with the acceleration of heart rate, was equivalent to the increase in risk associated with high blood pressure

(continued)

Table 1 (continued)

S. no	Attributes	Descriptions	Uses
9	Exercise angina	NO and Yes	Angina typically manifests as discomfort in the center of the chest, but it can also radiate to one or both shoulders, the back, neck, jaw, or arm
10	Old peak	Exercise-induced ST depression in comparison with the state of rest (Numeric)	In a treadmill ECG stress test, an ST-segment depression of less than 1 mm occurring 60–80 ms after the J point is considered abnormal
11	ST slope	ST segment measured in terms of slope during peak exercise 0: Normal 1: Upsloping 2: Flat 3: Down sloping (Nominal)	In patients diagnosed with unstable coronary artery disease, the presence of ST-segment depression is associated with a heightened risk of experiencing recurrent cardiac events. Furthermore, it is linked to a 100% increase in the likelihood of having three-vessel or left main coronary artery disease
12	CA	number of major vessels (0–3) colored by fluoroscopy	The number of major vessels colored by fluoroscopy typically ranges from 0 to 3, indicating the extent of coronary artery disease or blockages in these vessels
13	Thal	[normal; fixed defect; reversible defect]	blood disorder called thalassemia.Value 0: NULL Value 1: fixed defect, Value 2: normal blood flow Value 3: reversible defect
12	Target	It is the target variable which we must predict 1 means patient is suffering from heart risk and 0 means patient is normal	The combination of all the mentioned characteristics and diagnostic criteria is used to assess whether an individual has heart disease or not. These criteria, which may include clinical symptoms, medical tests, and risk factors, are collectively considered to make a determination regarding the presence or absence of heart disease

One of the advantages of logistic regression is its ease of implementation, interpretation, and training. It is a straightforward and extensively used method in the field of machine learning with statistics. Logistic regression is particularly effective when dealing with binary classification problems (two classes). However, there are certain considerations and limitations associated with logistic regression:

Data Requirement: It assumes that the correlation among the independent variables and the categorical outcome is linear. It may not perform well if the relationship is highly nonlinear.

Overfitting: Logistic regression should be used cautiously when the number of data points is significantly less than the number of independent variables. In such cases, there's a risk of overfitting, where the model may fit the training data too closely and does not perform well on the unseen data.

Assumptions: Logistic regression assumes that the observations are independent and that there is no multicollinearity (high correlation) among the independent variables. Violations of these assumptions can affect the model's accuracy.

In summary, it is a valuable tool for binary or multi-class classification tasks, offering simplicity and interpretability. However, it is essential to be aware of its limitations, particularly in cases where there are relatively few data points or when the relationship between the variables and the outcome is complex and nonlinear. In such situations, more advanced modeling techniques may be considered.

KNN Classifier. KNN is a straightforward classifier that operates as a non-parametric method, meaning it doesn't make any molds about the underlying data distribution at the time of analysis. Instead, it relies on the actual data themselves to do predictions. Here are some key characteristics of KNN:

Non-Parametric: KNN doesn't make any assumptions about the shape or form of the data, and make it highly flexible and suitable for different range of data types.

Data-Driven: KNN doesn't perform any real training during the training phase. Instead, it stores the whole dataset in memory and uses it directly for making predictions. This can be very expensive and memory-intensive, especially with large-size datasets.

Nearest Neighbor: KNN classifies data points by finding the "k" nearest neighbors in the feature space. It assigns the majority class among these k neighbors to the data point in question.

Imaginary Border: While KNN doesn't explicitly create decision boundaries or fit a model, it effectively uses an imaginary border that separates different classes based on the proximity of data points.

Library Usage: KNN can be implemented in various programming languages, including Python, and is commonly available in machine learning libraries such as scikit-learn (not "sickie-learn" as mentioned).

In summary, KNN is a data-driven and non-parametric classifier method that relies on proximity in feature space to make predictions. It is a simple yet effective algorithm, but it can be expensive for large datasets due to the calculation of distances between data points.

SVM Classifier. SVM is indeed a supervised machine learning model commonly used for resolving binary classification problems, where the goal is to categorize data points into one of two groups or classes. SVM accomplishes this task using classification algorithms. Here are key points about SVMs:

Supervised Learning: SVMs are part of the supervised learning category, meaning they require labeled training data where each row is associated with the class label. The model learns to distinguish between these classes based on the provided training data. It is used only for classification tasks. It can also be used to handle regression.

Margin Maximization: One of the key principles behind SVMs is to find a hyperplane which increases the distances between the two classes. This hyperplane serves as the decision boundary for classifying new data points.

Support Vectors: These are the data points which are closest to the decision boundary (the hyperplane). This is the crucial role in defining the margin and the decision boundary.

Kernel Trick: SVMs can handle nonlinear data by applying the kernel function to plot the original feature space into a higher-dimensional space. The data becomes separable by a hyperplane. Kernels types include the linear, polynomial, and radial basis function.

Implementations: SVM implementations are available in various programming languages and libraries. You mentioned “Sickie-learn,” which appears to be a typographical error. The popular Python library for SVM is scikit-learn. Other implementations include MATLAB and LIBSVM.

In summary, it is a powerful supervised machine learning model used for binary classification problems. It focuses on how to find the hyperplane that maximizes the distances between two classes, and it can handle both linear and nonlinear data through the use of kernel functions. SVMs have various implementations, with scikit-learn being a widely used choice in the Python programming language.

Decision Tree Classifier. It is a tree-like structure commonly used in machine learning for various tasks, including classification and regression [4, 11, 20, 36]. Here are some key points about decision trees:

Non-Parametric Model: Decision Trees, often abbreviated as DTs, are non-parametric models, which means they do not make strong assumptions regarding the data distribution. This makes them versatile for different types of data.

Supervised Learning: Decision Trees are a type of supervised learning algorithm. They require labeled training data, where each data point is associated with a target variable (in the case of classification, it’s a class label; in regression, it’s a numerical value).

Decision Rules: The primary goal of Decision Trees is to learn decision rules from the patterns available in the data. This allows the model to make predictions. These rules are based on the values of the features and are used to partition the data into classes or predict numerical values.

Tree Structure: It is a tree-like structure. Each node which is present internally is an attribute or a question. It branches to the next level based on the outcome of the question. The leaf node is a class label.

Interpretability: Decision Trees are highly interpretable models, making them valuable for understanding the reasoning behind model predictions. The decision rules are easy to visualize and comprehend.

Overfitting: This problem occurs when the tree is more complex. Also it occurs when the data is with more noise.

Ensemble Methods: Decision Trees are often used as building blocks for Random Forests and Gradient Boosting, which combine multiple decision trees to improve predictive accuracy and reduce overfitting.

In summary, Decision Trees are versatile and interpretable models used in both classification and regression tasks. They represent decision rules based on data features, making them valuable to understand and explain the decision-making process of a model. However, practitioners should be cautious about overfitting, and ensemble methods can be used to enhance performance.

Naïve Bayes. It is a category of supervised learning algorithms based on Bayes’ theorem. These algorithms are used primarily for classification tasks. The “naive” in

Naive Bayes refers to a simplifying assumption they make. Here are some key points about Naive Bayes:

Independence Assumption: It assumes that the presence of one feature in a class is entirely independent of the presence of any other feature in the same class. This is often an oversimplified assumption, hence the term “naive.” In reality, features in a dataset may have correlations.

Bayes’ Theorem: Naive Bayes classifiers utilize Bayes’ theorem to estimate the probability of a data have its place in a particular class. It calculates the conditional probability of a class given the features observed in the data.

Classification: Naive Bayes is widely used for classification tasks, particularly for text classification, spam detection, sentiment analysis, and document categorization.

Types of Naive Bayes: There are different variations of Naive Bayes, including Multinomial Naive Bayes (for discrete data like text), Gaussian Naive Bayes (for continuous data with a Gaussian distribution), and Bernoulli Naive Bayes (for binary data). **Simple and Fast:** Naive Bayes is known for its simplicity and computational efficiency. It is particularly well-suited for high-dimensional datasets.

Library Usage: Naive Bayes implementations are available in various programming languages, including Python and R, through libraries like scikit-learn.

Bias-Variance Tradeoff: Like other machine learning algorithms, Naive Bayes also faces the bias-variance tradeoff. The choice of a specific Naive Bayes variant can affect this tradeoff.

Strengths: Naive Bayes is effective when the independence assumption holds reasonably well and can perform surprisingly well on many classification tasks, especially in situations with limited data.

In summary, it is also used for classification tasks. They assume that features are independent, which may not always hold in practice. Despite their simplicity and the naive assumption, they can be effective classifiers and are widely used, particularly in text-based applications.

LSTM. LSTM (Long Short-Term Memory): It is a sort of recurrent neural network (RNN) architecture in deep learning and used in natural language processing tasks. LSTMs are designed to handle sequences of data and are particularly effective at capturing dependencies in sequential data. They have applications in various fields, including healthcare, where they can be used for time series prediction tasks, such as forecasting patient health outcomes.

Predicting Heart Disease: While LSTM neural networks can be used for time series prediction tasks, predicting the likelihood of developing heart disease typically involves a different approach. Heart disease prediction often relies on classification algorithms, such as logistic regression, decision trees, or more advanced methods like random forests or deep learning models, depending on the available data and the specific problem [5, 10, 19, 22, 35].

Patient Data Analysis: Machine learning and deep learning techniques, including LSTM models, can be applied to analyze patient data for various purposes, including disease prediction, risk assessment, and treatment optimization.

In summary, while LSTM models are powerful tools for sequence data analysis, predicting the likelihood of heart disease typically involves classification techniques

rather than sequence prediction. Machine learning and deep learning methods can indeed be used to analyze patient data for medical applications, but it's essential to choose appropriate algorithm and methodology based on the specific healthcare task and dataset.

BiLSTM. Bidirectional Long Short-Term Memory (BiLSTM): BiLSTM is indeed a type of LSTM architecture. It's designed to capture bidirectional context in sequences, which means it processes input data both in the frontward and backward directions. This helps the model capture dependencies from past and future data points, making it well-suited for tasks involving sequential data like time series analysis NLP.

Heart Disease Prediction: Applying BiLSTM (or any other neural network architecture) to predict heart disease typically involves using patient data as input and train the model to classify individuals as either having or not having heart disease. The model learns patterns and dependencies in the data that are indicative of heart disease risk.

Sequence Data: While BiLSTM is powerful for sequence data, heart disease prediction often involves tabular data with various features (e.g., age, blood pressure, cholesterol levels) rather than sequential data. However, it's possible to use BiLSTM or other neural network architectures to process sequences of medical measurements or time series data related to a patient's health [8, 23, 26, 34].

Data Preprocessing: When applying BiLSTM to medical data, data preprocessing is critical. This includes handling of missing values, normalization of data, and encoding categorical variables.

In summary, BiLSTM models are valuable for tasks involving sequential data analysis and can be adapted for certain aspects of heart disease prediction, particularly when dealing with time series or sequential patient data. However, heart disease prediction usually involves a combination of various machine learning techniques and data types, with feature engineering and model evaluation being critical components of the process.

XGBoost. XGBoost is a robust and high-performing machine learning algorithm recognized for its exceptional accuracy, particularly with large datasets. It operates as a gradient boosting algorithm. It aims to rectify the mistakes of its predecessor. What sets XGBoost apart are its advantageous features:

Built-In Regularization: XGBoost incorporates built-in regularization techniques, including L1 (Lasso) and L2 (Ridge) regularization, which serve to mitigate overfitting issues and enhance the ability to generalize effectively (Fig. 3).

Handling Missing Data: XGBoost has the capability to handle missing data gracefully, ensuring that the algorithm can effectively learn from datasets with incomplete information, a common scenario in real-world data.

Feature Interactions: XGBoost excels in capturing complex feature interactions, allowing it to uncover intricate relationships between variables and improve prediction accuracy.

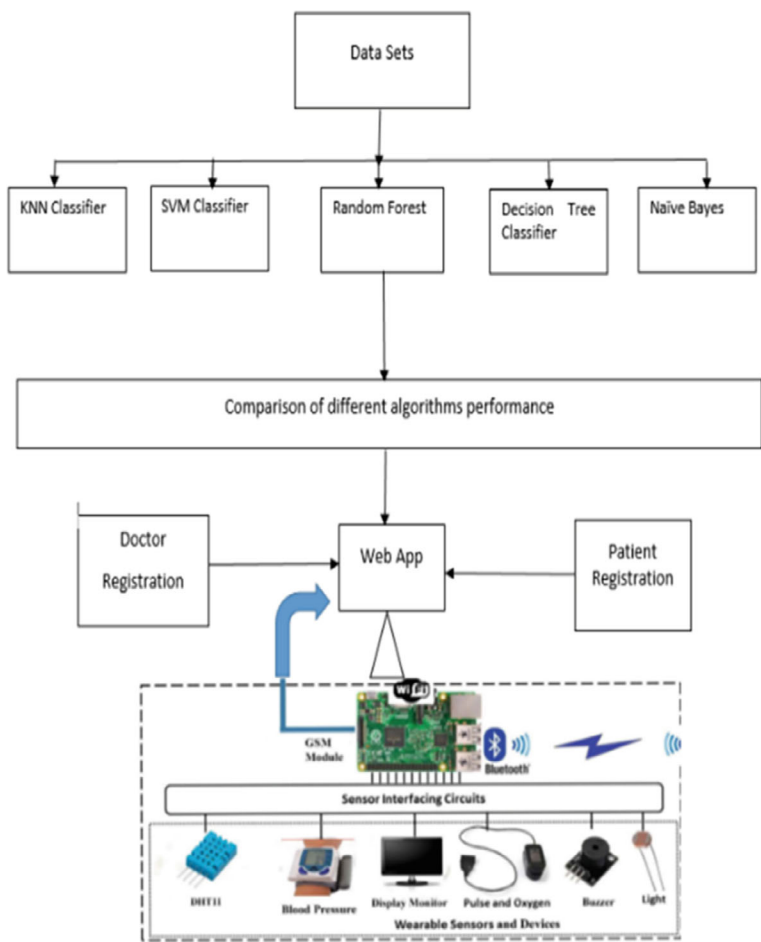


Fig. 3 Comparison of ML algorithm

Gradient Boosting: As a gradient boosting algorithm, XGBoost efficiently merges the predictions from the group of decision trees, providing a robust and high-accuracy model for various machine learning tasks, including heart disease prediction [15, 21, 24, 25].

In essence, XGBoost stands out as a versatile and powerful algorithm, well-suited for heart disease prediction and a wide range of other predictive modeling tasks, thanks to its advanced capabilities, regularization techniques, and capacity to handle challenging data scenarios.

Neural Network Neural networks are a class of machine learning algorithms inspired by the architecture and functioning of the human brain. They consist of interconnected nodes, or neurons, organized in layers, and these networks process

and transmit information to make predictions or perform various tasks, including heart disease prediction. Here are some key points about neural networks:

Nonlinear Relationships: Neural networks excel at capturing complex, nonlinear relationships between input features and the target variable. This makes them suitable for tasks where the underlying patterns are intricate and not easily represented by linear models.

Complex Tasks: Neural networks are well-suited for tackling complex tasks, such as heart disease prediction, image recognition, natural language processing, and more, due to their ability to learn hierarchical and abstract representations from data.

Data and Computation: Training neural networks can be computationally intensive and may require substantial amounts of data to avoid overfitting (learning the training data too well but failing to generalize). This is particularly relevant for deep neural networks with many layers.

Black-Box Nature: One challenge with neural networks is their “black-box” nature, meaning it can be challenging to interpret how and why they make specific predictions. Understanding the inner workings of a neural network can be complex, which may limit their adoption in applications where interpretability is crucial.

In summary, neural networks are an effective machine learning models capable of handling nonlinear relationships and complex tasks like heart disease prediction. However, their computational requirements and lack of interpretability can be considered when deciding to use them for a particular application.

4 Experiments

Heart disease prediction is indeed a classification problem in the field of healthcare and medical diagnostics. The primary objective is to create machine learning models capable of accurately predicting the heart disease availability for the patients. To achieve good results, these models utilize a set of clinical and demographic features as input, and gives the better output. Accurate predictions in this context are crucial for early diagnosis, timely intervention, and improved patient care.

To tackle the task of heart disease prediction, a wide range of machine learning algorithms can be employed. These algorithms encompass various approaches and techniques, each with its strengths and suitability for different aspects of the problem. The algorithms used in this context may include:

Logistic Regression, Naive Bayes, SVM, KNN, Decision Trees, Random Forests, XGBoost, Neural Networks, LSTM, BiLSTM, By leveraging these diverse machine learning algorithms, healthcare professionals and data scientists can explore different modeling techniques, each with its unique capacity to capture patterns and relationships within the clinical and demographic features. This comprehensive approach helps in building robust and accurate predictive models for heart disease detection, ultimately improving patient outcomes and healthcare decision-making.

Evaluating the model for heart disease prediction is a critical step in determining their effectiveness. The choice of performance metrics can affect the model output.

The choice of hyperparameter also plays an important role for the effectiveness of the model. The various parameter metrics that evaluate the model are:

Accuracy: It is the proportion of correctly classified occurrences in the test dataset. It's a fundamental metric for assessing overall model performance.

Precision: This metric quantifies the true positive predicted results out of all positive predicted ones by the model. It evaluates the ability of the model to predict accurately. **Recall:** It measures, how many true positive predictions are done with respect to actual positive occurrences in the dataset. It indicates the model's capability to identify all relevant positive cases.

F1-Score: It is the mean of precision and recall. It provides a balanced degree of a model's effectiveness, especially valuable when dealing with imbalanced datasets. The evaluation process typically involves the following steps:

The dataset is categorized into two subsets: training and test set. Machine learning models are trained using the training data, learning the underlying patterns and relationships. The trained models are then applied to the test set to forecast the classes for each data point in the test set. The performance metrics, including accuracy, precision, recall, and F1-score, are calculated based on the model's efficiency compared to the true labels in the test dataset. The obtained metrics are analyzed to assess the strengths and weaknesses of each model. This analysis helps in selecting the most suitable model for heart disease prediction. The choice of which performance metrics to prioritize depends on the specific goals of the heart disease prediction task. For example, in medical diagnostics, achieving high recall (sensitivity) may be critical to ensure that as many true cases of heart diseases as possible are correctly identified, even if it results in some false positives. The selection of the most appropriate evaluation metric should align with the clinical objectives and requirements.

Logistic regression achieved an accuracy of 81.25%. In Naive Bayes also achieved an accuracy of 85.25%. For SVM, the accuracy is 81.67%. The accuracy for KNN is 78.72%. Decision trees algorithm achieved an accuracy of 82.91%. Random Forest algorithms had given accuracy of 86.87%. This is the better performance when compared with the other algorithms. The accuracy by XGBoost is 78.69% and Neural Network is 82.38%. By using LSTM model, the accuracy achieved is 88.56%. and BiLSTM model is 90.28%. In Table 2, the comparison of algorithms is shown.

After testing various algorithms, the Cleveland dataset with 14 attributes, including age, type of chest pain, sex, resting pressure, fasting blood sugar, resting ECG, cholesterol, maximum pulse rate, exercise-induced Old Peak, angina, number of vessels colored, slope, and thalach. In machine learning algorithms, the Naive Bayes and the Random Forest give better performance. In deep learning algorithms, the LSTM and BiLSTM model perform better than the other algorithms. The accuracy and loss for LSTM and BiLSTM Model is shown in Figs. 4, 5, 6, and 7.

Table 2 Comparison of algorithms

Algorithm	Precision	Recall	F1-measure	Accuracy
Logistic Regression	0.84	0.84	0.84	81.25%
Naive Bayes	0.82	0.83	0.82	85.35%
KNN	0.67	0.67	0.67	78.72%
XGBoost	0.78	0.79	0.78	78.69%
Random Forest	0.91	0.82	0.91	86.87%
Decision Tree	0.85	0.85	0.85	82.91%
SVM	0.73	0.73	0.73	81.67%
Neural Networks	0.81	0.82	0.81	82.38%
LSTM	0.78	0.78	0.78	88.56%
BiLSTM	0.83	0.83	0.83	90.28%

Fig. 4 Accuracy for LSTM model

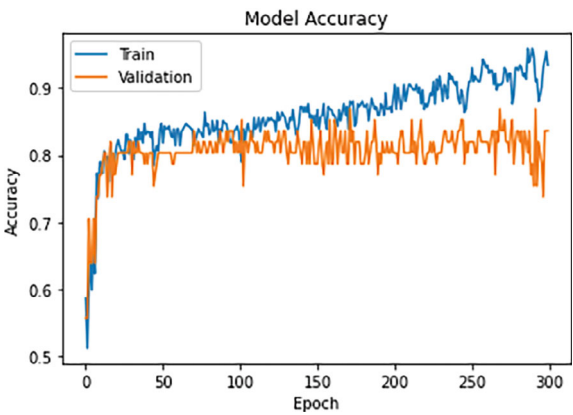


Fig. 5 Loss for LSTM model

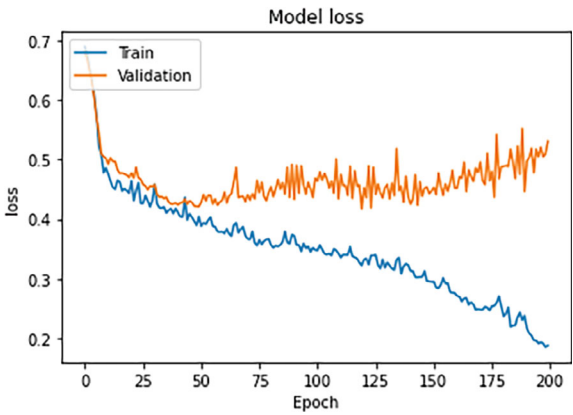


Fig. 6 Accuracy for BiLSTM model

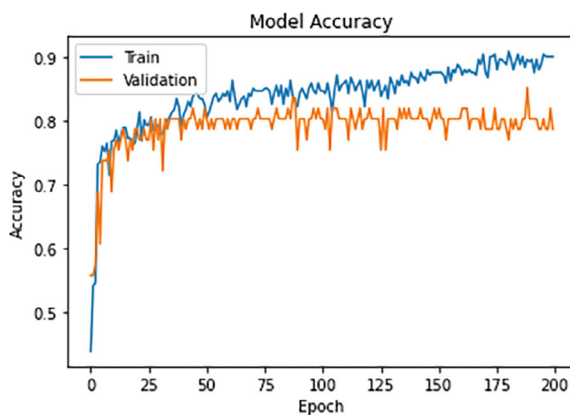
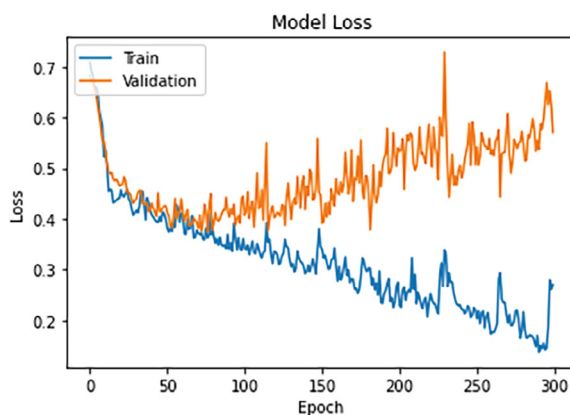


Fig. 7 Loss BiLSTM model



5 Conclusion

In response to the rising number of heart disease-related deaths, it has become imperative to develop an accurate model for the prediction of heart-related diseases. This study aimed to identify an efficient machine learning algorithm for detecting heart-related diseases. The study conducted a comparative analysis of various machine learning algorithms utilizing the UCI machine learning repository dataset for cardiovascular disease prediction. The findings of this research demonstrate that BiLSTM and LSTM stand out as the most effective among the algorithms tested, achieving accuracy for the prediction of heart conditions. This indicates that BiLSTM is a promising choice for building predictive models in the context of heart disease detection.

Looking ahead, there is potential to enhance and expand upon this work by developing more advanced models based on big data and data science techniques. Leveraging larger and more diverse datasets, coupled with advanced modeling methodologies, can further improve the accuracy and reliability of heart disease prediction models, ultimately contributing to more effective early diagnosis and intervention strategies to combat this critical healthcare issue.

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Advancements in Neurodegenerative Disease Diagnosis and Prediction: A Machine Learning Approach



Soham Kumar Modi, Sanjay Singla, Pranav Modi, and Geet Kiran Kaur

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Abstract Neurodegenerative diseases pose a significant global health challenge, necessitating precise and early diagnosis for effective management and intervention. This paper presents an in-depth exploration of machine learning and deep learning algorithms applied to the classification and prediction of neurodegenerative diseases, with a particular focus on advancements in diagnostic accuracy. The study reviews current literature, emphasizing segmentation of affected regions in neuroimaging, prediction of ADHD-based diseases, feature selection techniques, multimodal data integration, and the explainability of AI-based models. Methods encompass hand-crafted and automatic feature selection, large dataset transfer learning, and multimodal medical image registration. Performance optimization strategies and explainable models for biomedical mental disorder prediction are discussed. The results showcase promising advancements, demonstrating enhanced disease classification and prediction accuracy. The study discusses potential implications for improved healthcare outcomes and proposes avenues for further research, emphasizing the crucial role of explainable AI and multimodal data integration in medical data processing and disease prediction.

S. K. Modi (✉) · S. Singla · G. K. Kaur
Department of Computer Science and Engineering, Chandigarh University, NH-5
Chandigarh-Ludhiana Highway, Mohali, Punjab 140413, India
e-mail: soham3852@gmail.com

P. Modi
Department of Computer Science and Information Security, IIIT Hyderabad, Prof. C R RAO
Road, Gachibowli, Hyderabad, Telangana 500032, India

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1 Introduction

Neurodegenerative diseases constitute a formidable and growing public health challenge, characterized by the gradual and irreversible degeneration of structure and function within the nervous system. Disorders such as Alzheimer's disease [14, 15], Parkinson's disease, Huntington's disease, and Amyotrophic Lateral Sclerosis (ALS) manifest a complex interplay of genetic, environmental, and biological factors, leading to progressive cognitive, motor, and behavioral impairments. The prevalence of these disorders is on the rise globally, emphasizing the critical need for early and accurate diagnosis to enhance patient care and enable targeted therapeutic interventions. Traditionally, the diagnosis of neurodegenerative diseases has primarily relied on clinical assessment, patient history, and neuroimaging studies. However, accurately identifying these diseases, especially in their early stages, remains a daunting task due to the heterogeneous and subtle nature of initial symptoms. Furthermore, distinguishing between different neurodegenerative disorders based solely on clinical manifestations can be challenging, often leading to misdiagnosis or delayed diagnosis. In recent years, machine learning [13] and deep learning, subfields of artificial intelligence (AI), have emerged as powerful tools for analyzing and interpreting intricate patterns within medical data. These computational techniques can extract meaningful features and insights from a plethora of data types, including genetic data, neuroimaging, clinical records, and biomarkers. The integration of machine learning algorithms with multimodal data offers the potential for improved diagnostic accuracy, disease progression tracking, and personalized treatment plans.

This study examines the developments and uses of deep learning and machine learning in the field of diagnosing and predicting neurodegenerative diseases [13]. The aim is to recognize the potential of computational methods, by highlighting their capability to revolutionize healthcare by delivering accurate, fast, and individualized diagnosis. This research's focus spans a variety of aspects of applying machine learning and deep learning techniques to the study of neurodegenerative diseases [3, 4], including disease classification, affected region segmentation in neuroimaging, prediction of diseases related to ADHD, [7] feature selection methods, multimodal data integration, explainability of AI models, and performance optimization techniques. Additionally, we delve into the realm of transfer learning on extensive datasets, medical image registration using machine/deep learning models, and the burgeoning domain of explainable models for predicting biomedical mental disorders. In the subsequent sections, we delve into an exhaustive exploration of the existing literature, encompassing diverse methodologies and their potential contributions. We present detailed descriptions of the methods employed, results obtained, and an in-depth discussion regarding the implications and future directions in this

rapidly evolving field. Ultimately, this research seeks to contribute to the ongoing endeavors that aim to enhance healthcare by harnessing the immense potential of artificial intelligence, thus paving the way for more accurate neurodegenerative disease diagnoses and predictions.

1.1 Relevant Contemporary Issue

Addressing Diagnostic Challenges in Neurodegenerative Diseases: Neurodegenerative diseases present a contemporary global healthcare challenge that necessitates urgent attention and innovative solutions. The rapid increase in the prevalence of these conditions, coupled with the aging population worldwide, poses a significant burden on healthcare systems and society at large. A fundamental contemporary issue revolves around the accurate and timely diagnosis of neurodegenerative diseases, [3, 4] considering their chronic and debilitating nature. The challenge is exacerbated by the often subtle and heterogeneous early symptoms, making timely and precise identification a formidable task for healthcare professionals. Traditional diagnostic approaches have limitations in sensitivity and specificity, emphasizing the critical need for advanced methodologies to improve diagnostic accuracy and early intervention. Machine learning and deep learning, empowered by the availability of extensive medical datasets and computational advancements, offer a promising avenue to address this pressing issue by providing robust diagnostic tools capable of early detection and differentiation of neurodegenerative diseases.

1.2 Identification of Problem and Measures

Navigating the intricate realm of neurodegenerative diseases poses a challenge, demanding precision and timeliness in diagnosis. Artificial intelligence emerges as a beacon of hope, promising to revolutionize early detection with its adept algorithms that dissect subtle anomalies within data. This advancement not only enhances diagnostic accuracy but expedites identification, enabling proactive interventions and improved patient outcomes. The integration of AI not only lightens the burden on patients and healthcare systems but also paints a promising future, where early detection becomes the norm, ultimately leading to better lives. Consequently, misdiagnoses and delays in treatment initiation are prevalent, emphasizing the critical need to enhance diagnostic methodologies.

In response to this challenge, advanced computational techniques, such as machine learning and deep learning, have emerged as powerful tools capable of analyzing and interpreting complex medical data. These methodologies hold the potential to improve diagnostic accuracy by learning and identifying intricate patterns within neuroimaging, genetic information, and clinical records. By amalgamating multimodal data and employing state-of-the-art algorithms, these technologies can

aid in early disease identification, precise disease classification, and prediction of disease progression.

This research endeavors to address the identified problem by exploring the transformative potential of machine learning and deep learning models in neurodegenerative disease diagnosis. By investigating and implementing novel methodologies, leveraging extensive datasets, and focusing on explainable AI, our aim is to enhance diagnostic precision and contribute to improved patient care. The subsequent sections delve into comprehensive discussions regarding the methods employed, experimental results obtained, and implications of integrating AI-driven innovations into the realm of neurodegenerative disease diagnosis.

2 Literature Review

In a recent study by Ursin et al. (2021), the ethical implications surrounding the use of artificial intelligence (AI) for predicting Alzheimer's disease [1, 2, 6] in asymptomatic individuals were comprehensively explored. The study explores the potential repercussions and moral conundrums associated with using AI-driven predictive models to detect Alzheimer's disease in people who don't show symptoms. When applying artificial intelligence (AI) technology for predictive diagnostics in the context of neurodegenerative illnesses, Ursin et al. (2021) emphasize the need for careful consideration of privacy, informed consent, and societal repercussions. Emphasizing the importance of a thoughtful and equitable approach, considering the merits of early detection and the possible pitfalls of employing predictive AI models in Alzheimer's disease, this research provides valuable insights into the ethical considerations essential as AI advances in neurodegenerative disease prediction.

In a significant stride forward, Vélez et al. (2021) have introduced an advanced machine learning system designed to precisely predict the age of onset in both familial and sporadic Alzheimer's disease (AD). This pioneering study aims to forecast the age when symptoms are likely to manifest, a crucial aspect of Alzheimer's disease prognosis. [1, 2, 6] Employing machine learning algorithms and a rich array of clinical and genetic data, the study underscores the value of integrating diverse datasets for heightened prediction accuracy. By focusing on both familial and sporadic forms of Alzheimer's disease, the research enhances our understanding of the condition, shedding light on predictive models that can inform personalized care and treatment strategies. This study illuminates the growing importance of machine learning in unraveling complex neurological disorders, providing critical insights into the future of Alzheimer's disease prediction and intervention.

Khalik et al. (2023) investigate the new use of machine learning for the clinical identification of neurodegenerative disorders, [5] exhibiting the potential to decipher patterns of deterioration inside neurological pathologies in their latest work. This study is a big step forward in the early and accurate detection of

neurodegenerative illnesses, highlighting the critical role machine learning may play in revolutionizing clinical diagnostics. Khaliq et al. (2023) use advanced machine learning approaches to a wide range of clinical data, with the goal of improving diagnostic precision and enabling early treatments. The study highlights the transformational influence that AI and machine learning can have on the area of neurology by utilizing these cutting-edge approaches, paving the door for improved patient care, disease monitoring, and focused therapy solutions.

In their latest work, Shusharina et al. (2023) shed insight on new diagnostic and therapy approaches for neurodegenerative disorders and depression. It is crucial to address the junction between neurodegenerative illnesses and depression since they frequently coexist and can have a major influence on patients' overall quality of life. The paper addresses current diagnostic and treatment strategies, emphasizing the interdisciplinary approach required to properly manage these complicated illnesses. To adapt interventions for specific patients, Shusharina et al. emphasize the necessity of integrating diverse diagnostic tools and treatment modalities, such as improved imaging techniques, biomarker analyses, and targeted medications. This work contributes to ongoing efforts to improve the management and care of individuals affected by neurodegenerative illnesses and depression by giving insights into the most recent advancements.

Tăuțan et al. (2021) provide a thorough examination of the role of artificial intelligence, namely machine learning techniques, in tackling the rising prevalence of neurodegenerative illnesses in the aging population. The study emphasizes the importance of computational techniques in disease characterization, diagnosis, and monitoring by focusing on disorders such as Alzheimer's, Parkinson's, Huntington's, Amyotrophic Lateral Sclerosis, and Multiple System Atrophy. The authors provide an informative study of decision systems and numerous modalities used for each condition, as well as a taxonomy of clinical aspects and computational approaches. Significantly, the research underscores the crucial role of sleep problems in early illness detection. The paper extensively examines existing datasets, assessment criteria, persistent challenges, and upcoming strategies in this swiftly evolving field. This review serves as an invaluable asset for both scholars and practitioners, offering a thorough glimpse into computational methods for navigating the intricate landscape of neurodegenerative disorders.

In a significant exploration, Kavitha et al. (2022) delve into the critical realm of early-stage Alzheimer's disease (AD) prediction using machine learning models, recognizing AD as a primary cause of dementia in older individuals. With the growing prevalence of metabolic disorders like Alzheimer's and Diabetes, there's a burgeoning interest in leveraging machine learning to tackle these substantial health challenges affecting a global population. The authors underscore the necessity of early prediction, emphasizing that timely intervention in AD leads to more effective treatment outcomes and reduces damage compared to late-stage interventions. Through their study, Kavitha et al. explore a range of machine learning techniques, including Decision Tree, Random Forest, Support Vector Machine, Gradient Boosting, and Voting classifiers, aiming to optimize Alzheimer's disease prediction. The study utilizes the Open Access Series of Imaging Studies (OASIS)

[12] dataset for prediction, employing performance metrics such as Precision, Recall, Accuracy, and F1-score to evaluate the models. Their proposed classification scheme not only advances diagnostic capabilities but also holds the potential to lower annual mortality rates associated with AD through early diagnosis facilitated by machine learning algorithms. The study presents encouraging results, demonstrating a notable validation average accuracy of 83% on the AD test data, surpassing the performance of existing works.

Slobodin et al. (2020) address the ongoing debate surrounding the specificity, sensitivity, and ecological validity of the continuous performance test (CPT) in diagnosing attention-deficit/hyperactivity disorder (ADHD). The study introduces a machine learning-based model (ML) leveraging CPT indices for enhanced prediction of ADHD, aiming to overcome known limitations of traditional CPT data analysis and interpretation. Through a retrospective factorial fitting and bootstrap technique, the authors trained and tested ML models on CPT performance data from a substantial cohort of children, including those with ADHD and typically developed peers. Utilizing the MOXO-CPT version with visual and auditory stimuli distractors, the proposed ML model exhibited superior performance and accuracy compared to benchmark approaches using clinical data alone. The model, incorporating CPT total score and control variables like age, gender, day of the week, and time of day, demonstrated a remarkable accuracy rate of 87%, sensitivity rate of 89%, and specificity rate of 84%. This performance surpassed the benchmark model's accuracy by an impressive 34%. The study highlights the potential of ML-based ADHD prediction models, offering promise for enhanced behavioral assessments and serving as a complementary tool in ADHD evaluations.

3 Methodology Used

3.1 Classification

In this research, classification serves as a pivotal methodology for discerning distinct categories within the domain of neurodegenerative diseases. Employing supervised learning algorithms like Support Vector Machines (SVM) and Random Forest, we systematically analyze extensive datasets encompassing diverse medical parameters and imaging features. The objective is to accurately classify individuals into predefined disease categories, such as Alzheimer's, Parkinson's, or other related conditions. By meticulously training the models on labeled data and fine-tuning their parameters, we strive to achieve a high degree of accuracy and reliability in disease classification. This facilitates early identification and differentiation of various neurodegenerative disorders, which is fundamental for effective treatment strategies and personalized patient care.

3.1.1 Dataset Used for Classification

In this research, the dataset employed is the “Parkinson’s Disease Telemonitoring Data [8],” encapsulated within the file “parkinsons_updrs.data.csv.” [8]. This dataset is instrumental in investigating and understanding Parkinson’s disease, providing a comprehensive view of various parameters associated with the condition. The dataset comprises essential columns: “subject#,” serving as a unique identifier for each subject ranging from 1 to 31, “age” representing the age of the subject, and “sex” denoting the gender of the subject. Additionally, it encompasses crucial quantitative features such as “test_time” indicating the time since recruitment into the trial, “motor_UPDRS” signifying the Motor UPDRS score, and “total_UPDRS” representing the Total UPDRS score, crucial in gauging the severity of Parkinson’s disease. Furthermore, the dataset incorporates an array of biomedical voice measures like jitter, shimmer, noise-to-harmonics ratio, and pitch period entropy, offering valuable insights into speech parameters. These features, coupled with others like recurrence period density entropy and detrended fluctuation analysis, create a rich data landscape. Altogether, this dataset is a vital asset for understanding Parkinson’s disease through diverse parameters, enabling insightful analysis and development of predictive models for disease progression.

3.1.2 Analysis on Classification of Dataset

Analysis of Motor UPDRS Prediction:

1. Linear Regression:

- RMSE: 7.15
- Linear regression is a simple and interpretable model. The obtained RMSE suggests a moderate level of accuracy. It’s a good baseline model for this prediction task.

2. SVR (Support Vector Regression):

- RMSE: 6.99
- SVR has slightly outperformed linear regression, indicating its ability to capture more complex relationships in the data. However, the improvement is not substantial compared to linear regression.

3. Random Forest:

- RMSE: 0.41
- The Random Forest model significantly outperforms both linear regression and SVR. The very low RMSE indicates that this ensemble learning model is highly effective in capturing intricate patterns in the data (Fig. 1).

Analysis of Total UPDRS Prediction:

1. Linear Regression:

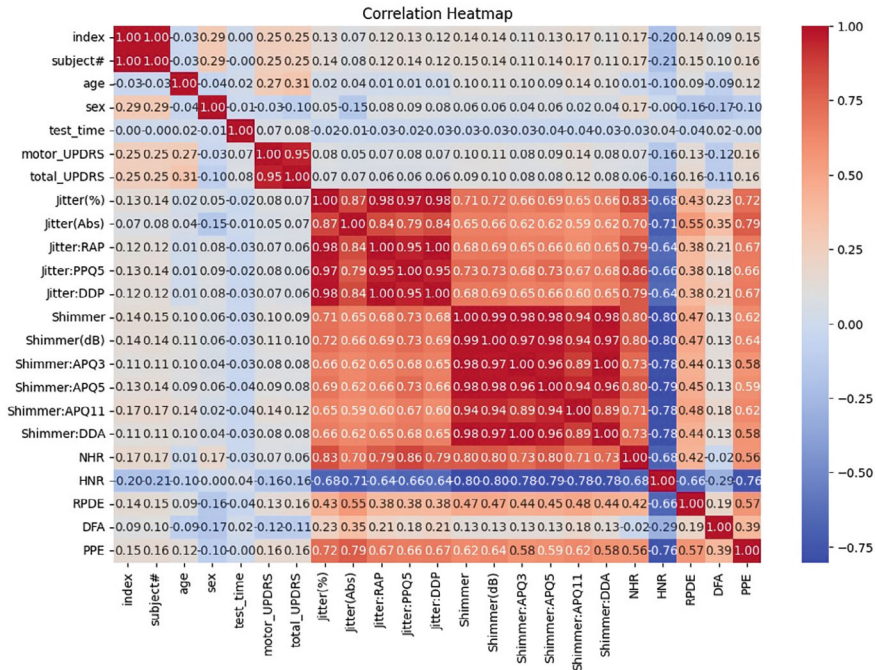


Fig. 1 Correlation heatmap illustrating relationships among dataset features in Parkinson's disease telemonitoring

- RMSE: 9.10
 - Linear regression, in this case, yields a relatively high RMSE, suggesting that a linear model is not the best fit for predicting Total UPDRS. The Total UPDRS prediction is likely influenced by more complex, nonlinear relationships.
2. SVR (Support Vector Regression):
- RMSE: 9.54
 - SVR performs similarly to linear regression, indicating that the Total UPDRS prediction task is challenging, and SVR's ability to capture non-linearity is not significantly advantageous in this scenario.
3. Random Forest:
- RMSE: 0.46
 - The Random Forest model outperforms both linear regression and SVR by a significant margin. It suggests that the ensemble learning approach is highly effective in capturing the complexities of the Total UPDRS prediction.

3.2 Segmentation

Segmentation, a critical methodology, aids in isolating and delineating affected regions within neuroimaging data. Leveraging advanced techniques like U-Net, we delineate precise boundaries of pathological areas in brain images. The process involves preprocessing the images, followed by training the segmentation model using annotated datasets. Through this, we can accurately identify regions pertinent to the respective neurodegenerative diseases, providing crucial insights for diagnosis and monitoring. By automating this segmentation process, we enhance efficiency, reduce manual effort, and enable quantitative analysis of affected areas, thus significantly contributing to a comprehensive understanding of the diseases under investigation.

3.2.1 Dataset Used for Segmentation

The utilized BRATS (BraTS 2020) dataset [9–11] represents a significant resource for researchers and practitioners in the field of medical imaging and neurology. The dataset comprises a collection of MRI (Magnetic Resonance Imaging) slices, each encapsulating distinct modalities including T1, T1Gd, T2, and T2-FLAIR. These modalities offer multifaceted views of the brain, crucial for studying and analyzing brain tumor characteristics. The dataset also provides accompanying segmentation masks, categorizing specific regions of interest within the brain. The segmentation masks, delineating “Core,” “Mask2,” and “Mask3” regions, enable a finer understanding of tumor sub-regions, enhancing research insights and potential therapeutic strategies. The presented work visually interprets this dataset, showcasing the normalized MRI modalities and their corresponding segmentation masks across multiple slices. This comprehensive analysis sheds light on the intricacies of the brain’s structure and pathological regions, forming a vital basis for precise segmentation and subsequent targeted research in the domain of medical image analysis.

3.2.2 Analysis on Segmentation of Dataset

In this section, a comprehensive analysis of the segmentation masks is performed which corresponds to different MRI modalities in the BRATS (BraTS 2020) dataset [9–11]. The T1 modality offers insights into the fundamental brain anatomy, portraying a baseline structural view. T1Gd, sensitive to blood–brain barrier disruption, highlights areas of potential malignancy and disease progression. T2, on the other hand, accentuates edematous regions, critical for understanding the extent of swelling in brain tissue. T2-FLAIR, designed to reduce signal from cerebrospinal fluid, aids in enhancing lesion visibility, a crucial aspect for precise tumor characterization. The segmentation masks—“Core,” “Mask2,” and “Mask3”—add another

layer of information. ‘Core’ identifies the aggressive tumor core, “Mask2” encompasses the peritumoral edematous regions, and ‘Mask3’ delineates the non-enhancing tumor area. This detailed analysis of segmentation provides valuable insights into the distinct pathological components of brain tumors, crucial for targeted treatment strategies and advancing the field of neuroimaging (Fig. 2).

3.3 Prediction

Prediction, a fundamental methodology, is instrumental in foreseeing disease progression and outcomes. Employing both traditional statistical methods and advanced machine learning algorithms like Long Short-Term Memory (LSTM), we analyze longitudinal data encompassing patients’ medical histories, biomarkers, and clinical assessments. The objective is to develop predictive models capable of foreseeing disease trajectories, identifying risk factors, and estimating the likelihood of disease onset. This aids in early intervention and tailored treatment planning, ultimately improving patients’ quality of life. By continually fine-tuning and validating these predictive models, we aspire to enhance their accuracy and robustness, paving the way for a proactive approach in managing neurodegenerative diseases.

3.3.1 Dataset Used for Prediction

The research utilized the OASIS MRI dataset [12], an extensive collection comprising 80,000 brain MRI images. These images were meticulously categorized into four classes based on the progression of Alzheimer’s disease, providing a valuable resource for the early detection and analysis of the disease. To enhance accessibility, the original MRI data files were converted into Nifti format using FSL, and a subset of 461 patients’ MRI images was made available on GitHub. The training of neural networks involved using 2D images obtained by slicing the brain images along the z-axis into 256 pieces, selecting slices within the 100 to 160 range for each patient. This strategy yielded a comprehensive dataset conducive to in-depth analysis. Patient classification was executed using associated metadata and Clinical Dementia Rating (CDR) values, resulting in four distinct classes: demented, very mild demented, mild demented, and non-demented. These classes offer a structured approach to investigating various stages of Alzheimer’s disease progression. Despite the challenges encountered during the conversion of .nii MRI scans to .jpg files, successful processing was achieved using suitable tools, resulting in a dataset size of 1.3 GB. The primary goal of this dataset analysis is to explore and implement diverse neural network models to achieve optimal results in Alzheimer’s disease prediction and analysis.

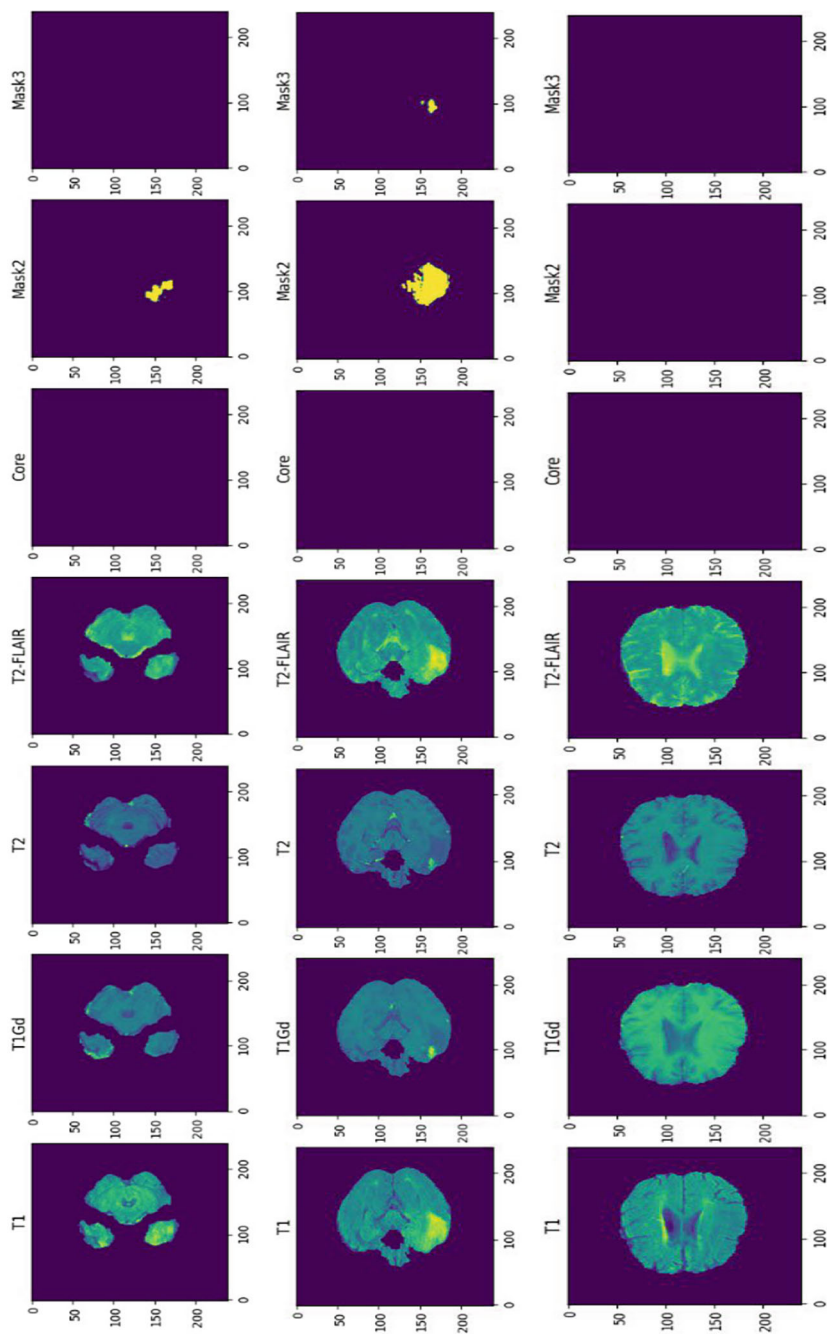
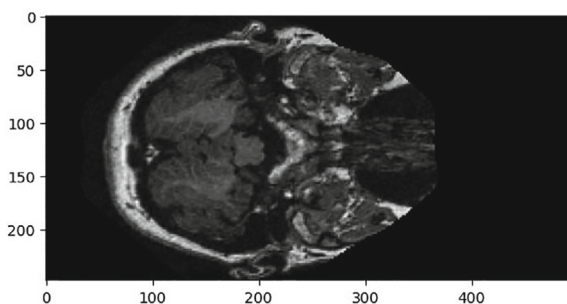


Fig. 2 Multi-modal MRI brain segmentation visualization

Fig. 3 Moderate dementia

3.3.2 Analysis on Prediction of Dataset

The predictions obtained from the trained neural network on the OASIS MRI [12] dataset present intriguing insights into Alzheimer's disease classification. The first image is representing Moderate Dementia, showcases an incredibly high confidence level of approximately 99.9997% in correctly identifying the severity (Fig. 3).

Similarly, the second image is depicting Very Mild Dementia yields a slightly lower confidence of around 81.17%, reflecting the subtle nuances associated with this stage (Fig. 4).

The third image denotes Mild Dementia, the confidence level increases to approximately 88.37%, highlighting the neural network's ability to discern varying stages of the disease (Fig. 5).

These results demonstrate the potential of the neural network in accurately classifying different degrees of Alzheimer's progression based on MRI images, signifying a promising avenue for leveraging machine learning in Alzheimer's disease analysis and prediction.

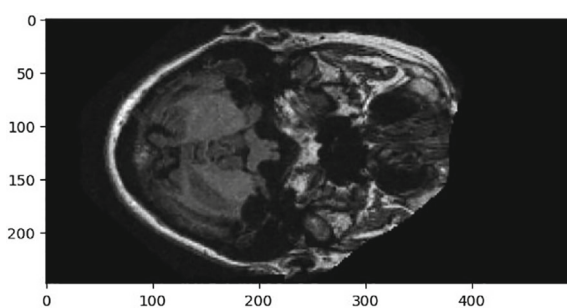
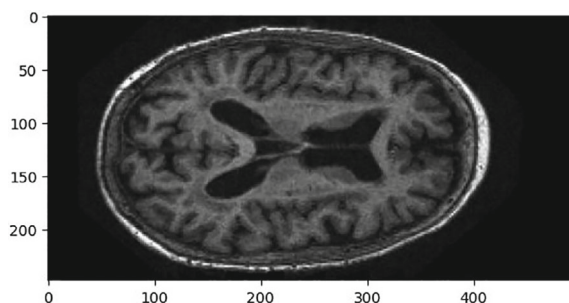
Fig. 4 Very mild dementia

Fig. 5 Mild dementia

4 Results and Discussion

In this research, three key aspects were explored: classification, segmentation, and prediction. The objective was to comprehend different facets of Parkinson's disease and brain tumor analysis. In the context of Parkinson's disease analysis, we conducted a comprehensive exploration encompassing three primary domains: classification, segmentation, and prediction. The findings provided valuable insights into various aspects of Parkinson's disease and brain tumor analysis.

Regarding classification, our evaluation highlighted the Random Forest model as the most effective in predicting Motor UPDRS, achieving an impressively low Root Mean Square Error (RMSE). However, predicting Total UPDRS proved more challenging. Both Linear Regression and Support Vector Regression (SVR) struggled to accurately forecast Total UPDRS, implying a complex, likely nonlinear relationship between the features and Total UPDRS. Despite being the optimal choice among the three models, Random Forest exhibited a higher RMSE in Total UPDRS prediction, indicating inherent difficulties in this prediction task. Moving to segmentation, our analysis using the BRATS dataset yielded insightful results. The segmentation process provided a detailed delineation of brain tumor sub-regions (T1, T1Gd, T2, T2-FLAIR, core, mask2, mask3). The visual representation of these sub-regions showcased the intricate nature of brain tumors, enabling a deeper understanding of their structure and aiding in subsequent focused research and analysis.

In the domain of prediction, the analysis involved the utilization of the OASIS MRI dataset. Three instances of prediction were showcased, each pertaining to varying stages of dementia (Moderate, Very Mild, Mild). The model effectively identified the different dementia phases with high confidence levels, suggesting the predictive model's ability to aid in the diagnosis and categorization of Alzheimer's disease development. In conclusion, our joint research sheds light on the significance of suitable model selection as well as the inherent difficulties in medical data analysis. It highlights the various complications involved in categorizing disease stages, segmenting brain tumor sub-regions, and predicting disease severity. These findings can be considered for defining future research and breakthroughs in medical data processing.

5 Conclusion

Machine Learning (ML) represents a swiftly evolving domain with the potential to revolutionize the diagnostic and prognostic aspects of neurodegenerative diseases. ML algorithms exhibit exceptional acumen in discerning intricate patterns within data, a task often formidable for human comprehension. This particular capability holds significant promise, especially in the realm of medical data analysis, including neuroimaging scans, aiding in the identification of illness biomarkers and accurate prediction of disease progression.

Recent years have witnessed a notable surge in research endeavors, focused on harnessing ML's potential for the diagnosis and prediction of neurodegenerative diseases. Consequently, promising tools and methodologies have been developed, underscoring the transformative potential of ML algorithms. These algorithms have notably demonstrated their potential by enabling early and innovative diagnoses of conditions such as Alzheimer's disease and Parkinson's disease, along with various other neurodegenerative disorders. Furthermore, machine learning has introduced novel approaches for forecasting illness progression and assessing therapy responses. However, despite its transformative potential, machine learning encounters certain challenges. ML algorithms are intricate and often challenging to interpret, presenting difficulties in comprehending their inner workings and instilling trust in their predictions. Moreover, these algorithms necessitate substantial training on extensive, high-quality datasets, a process that can be resource-intensive and financially burdensome. Despite these impediments, the field of machine learning-based neurodegenerative disease diagnosis and prediction is advancing rapidly. The consensus is growing regarding ML's ability to significantly enhance the lives of individuals grappling with neurodegenerative diseases by offering early diagnoses, precise prognoses, and more effective treatment options.

In conclusion, this study underscores the immense potential and critical role of medical imaging analysis in comprehending and identifying neurological ailments, encompassing conditions like Parkinson's disease and brain tumors. The many approaches used, from classification to segmentation and prediction, have provided considerable insights into the complex nature of these diseases. The findings of this study pave the way for future advances in medical image processing, promising innovations that will have a substantial impact on the field of neurology and healthcare in general. We anticipate that this research will serve as a spark for additional research, ultimately leading to improved diagnosis, treatment techniques, and, most importantly, improved patient care.

5.1 Future Work

The present research has paved the way for future exploration and progress in medical imaging analysis, specifically focusing on neurological disorders like

Parkinson's disease and brain tumors. Future research could explore the integration of multiple imaging modalities, combining MRI with PET or CT scans to offer a more comprehensive diagnostic view. Employing ensemble machine learning models and advanced deep learning architectures, especially convolutional neural networks (CNNs), could improve prediction accuracy and segmentation tasks, providing finer tumor sub-region delineation. Analyzing disease progression over time using longitudinal data and developing real-time decision support systems for clinicians are crucial next steps. Clinical validation and eventual deployment of these tools in healthcare facilities will ensure their practical utility. Personalized treatment plans based on both imaging and genomic/clinical data integration should be pursued for enhanced patient outcomes. Ultimately, future advancements should aim at refining diagnostics, optimizing treatment strategies, and advancing personalized medicine approaches for improved healthcare delivery.

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An In-Depth Examination of Diagnostic Methods and Therapeutic Approaches for Osteoporosis Utilizing Clinical Factors and Medical Imaging



S. Theetchenya, J. Senthilkumar, V. Mohanraj, and Y. Suresh

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Abstract Osteoporosis, a widespread disorder marked by weakened bone integrity and amplified susceptibility to fractures, presents a substantial worldwide health concern. This survey paper targets to provide an encompassing overview of the many diagnostic techniques and therapeutic approaches employed in the management of osteoporosis. The diagnostic methods cover a range of imaging modalities, including X-ray, tomography, high-resolution tomography. Additionally bone score and MRI techniques are also explored for bone assessment. The discussion includes their strengths, limitations, and potential applications in clinical practice. The therapeutic section examines pharmacological interventions, detailing their mechanisms of action and clinical efficacy. Non-pharmacological interventions like exercise, dietary adjustments, and lifestyle modifications are also highlighted for their role in improving bone health and preventing fractures. This survey paper explores personalized medicine by talking about tools that help identify people at a greater risk of fractures from osteoporosis, using risk assessment and predictive models. The utilization of biomarkers and genetic markers helps to enhance early detection and the individual treatment strategies to be followed. In conclusion, this survey synthesizes a diverse array of diagnostic and therapeutic methods for managing osteoporosis. By providing insights into the strengths and limitations of each approach, it aims to facilitate informed decision-making among healthcare professionals, researchers,

S. Theetchenya (✉) · V. Mohanraj · Y. Suresh
Department of CSE, Sona College of Technology, Salem, Tamilnadu, India
e-mail: theetchenya@gmail.com

J. Senthilkumar
Department of IT, Sona College of Technology, Salem, Tamilnadu, India

and patients, ultimately contributing to more effective osteoporosis management and fracture prevention strategies.

Keywords Osteoporosis · DNN · Clinical values · High-resolution tomography · X-ray · MRI

1 Introduction

Osteoporosis, a common skeletal condition, is defined by weakened bone strength and a heightened vulnerability to fractures. In Greek, Osteo is sensed as bone and porosis as porous, the term encapsulates the central pathological feature of this condition—the progressive deterioration of bone density and microarchitecture. Osteoporosis primarily affects older individuals, particularly postmenopausal women, and is a significant public health concern due to its impact on quality of life, healthcare resources, and economic burden. Bone, a dynamic living tissue, undergoes a constant process of remodeling involving both bone resorption and formation. This balance is orchestrated by a delicate interplay between osteoblasts is the formation of bone and osteoclasts is the resorption of bone. In osteoporosis, this equilibrium is disrupted, leading to an accelerated rate of bone loss, decreased bone mass, and the conversion of structurally sound bone into a fragile state prone to fractures, even from minor trauma.

Osteoporosis can affect multiple bones in the human body, but some bones are more commonly affected than others due to their composition and weight-bearing role. The bones most frequently affected by osteoporosis are vertebrae (Spine), hip, wrist, and pelvis bones. It's important to note that while these bones are commonly affected, osteoporosis can potentially impact any bone in the body. Fractures resulting from osteoporosis are often referred to as fragility fractures; as they occur with minimal force or trauma that would not typically cause a fracture in a healthy individual. The underlying causes of osteoporosis are multifaceted. Age-related hormonal changes, such as the decline in estrogen levels during menopause, contribute to bone loss, especially in women. Genetic factors also play a role in determining an individual's susceptibility to osteoporosis. Additionally, lifestyle aspects such as inadequate calcium, insufficient vitamin D, sedentary behavior, smoking, excessive alcohol consumption, and certain medical conditions can further exacerbate the risk of developing osteoporosis.

Efforts to combat osteoporosis encompass two primary areas: diagnosis and management. Accurate diagnosis is crucial to identify individuals at risk and determine the severity of the condition. Diverse imaging methods, including X-rays, are utilized to measure bone mineral density and assess the likelihood of fractures. Management strategies span lifestyle modifications, including weight-bearing exercises and nutritional adjustments, as well as pharmacological interventions

aimed at either reducing bone resorption or promoting bone formation. As populations continue to age and life expectancies rise, osteoporosis assumes increasing significance in healthcare planning.

This introduction provides a foundational understanding of osteoporosis, highlighting its impact on skeletal health and overall well-being. Subsequent exploration of diagnostic methods, treatment options, ongoing research, and preventive measures will offer a comprehensive view of how this condition is managed in both clinical and research settings. The following are some of the measures for preventing osteoporosis involves adopting a combination of lifestyle measures and, in some cases, medical interventions to maintain strong and healthy bones. Here are a few preventive strategies that can aid in lowering the risk of osteoporosis: Dietary measures(Calcium intake, Vitamin D and Balanced Diet), daily exercise, Avoid smoking and excessive Alcohol, Maintaining a healthy body weight, Hormone and Medication considerations, Regular health checkups, Lifestyle choices, and Educational resources. These measures reduce the problem to some extent but are not a complete cure.

2 Clinical Parameters

Various clinical parameters are used to identify and diagnose osteoporosis. These parameters help healthcare professionals assess an individual's bone health and fracture risk. Here are some of the key clinical parameters:

Bone Mineral Density Measurement: BMD is one of the most important indicators of bone health. Dual-energy X-ray absorptiometry (DXA or DEXA) is the most commonly used technique to measure BMD at specific skeletal sites, typically the hip and spine. The T-score and Z-score are values derived from DXA scans that compare a person's BMD to the average BMD of youngsters, healthy individuals (T-score), or age-matched individuals (Z-score). A T-score equal to or below -2.5 at the spine, hip, or forearm indicates the presence of osteoporosis.

Fracture History: A history of fragility fractures, especially of the hip, spine, or wrist, increases the likelihood of having osteoporosis. Fragility fractures are fractures that occur with minimal or no trauma, often as a result of weakened bones.

Clinical Risk Factors: Several risk factors elevate the probability of osteoporosis development. These encompass age (with a higher risk for postmenopausal women and the elderly), gender (with women facing greater risk), a family history of osteoporosis or fractures, low body weight, smoking, excessive alcohol consumption, inadequate calcium and vitamin D intake, a sedentary lifestyle, and the use of specific medications (like corticosteroids).

FRAX Calculation: The Fracture Risk Assessment Tool (FRAX) is a commonly employed instrument that computes the likelihood of experiencing a significant osteoporotic fracture (encompassing the hip, clinical spine, forearm, or proximal humerus) over a span of 10 years, as well as the probability of a hip fracture during the same timeframe. It considers clinical risk elements in conjunction with BMD (if accessible) to gauge the risk of fractures.

Biochemical Markers: Certain blood tests can provide insight into bone turnover and overall bone health. These include markers like serum calcium, phosphorus, alkaline phosphatase, and parathyroid hormone. However, these markers are not used as standalone diagnostic tools for osteoporosis and are typically considered alongside other clinical parameters.

Vertebral Imaging: X-rays or other imaging techniques can reveal vertebral fractures that might not have been clinically apparent. These are called morphometric vertebral fractures and can aid in diagnosing osteoporosis.

Physical Examination: While osteoporosis is not usually diagnosed solely based on physical examination, healthcare providers might assess for signs of kyphosis (excessive curvature of the upper spine) or height loss, which can be indicative of vertebral fractures.

Fall Risk Assessment: Assessing an individual's risk of falls is crucial because falls are a major contributor to fractures in individuals with osteoporosis. Factors such as muscle strength, balance, gait, and environmental hazards are evaluated.

Body Mass Index (BMI): If the BMI value is low, then the risk of fracture because of osteoporosis is high. A lower BMI can indicate reduced muscle and bone mass, increasing the likelihood of fractures.

Hormonal Status: Hormonal changes play a significant role in bone health. For example, low levels of estrogen in postmenopausal women can lead to accelerated bone loss. Hormone testing, especially for women in menopause, can provide insights into hormonal imbalances that might contribute to osteoporosis.

Calcium and Vitamin D Levels: Adequate calcium and vitamin D intake is much required for the strength of the bone. Blood tests can measure serum calcium and vitamin D levels to assess deficiencies that could impact bone health.

Physical Activity Level: Regular weight and exercises for the strengthening of the muscles contribute to bone strength. Sedentary individuals are at a higher risk of osteoporosis due to decreased bone-loading activities.

Medication History: Certain medications, such as glucocorticoids (steroids), anti-convulsants, and some cancer treatments, can affect bone health and increase the risk of osteoporosis. A history of using these medications is important to consider.

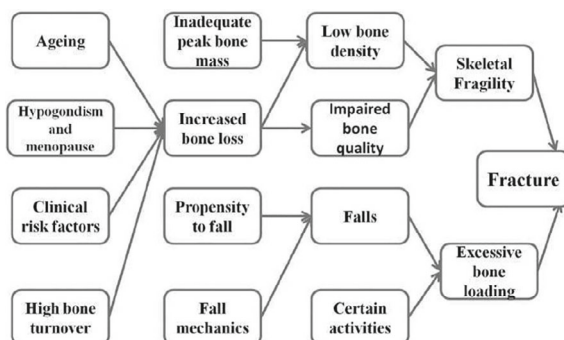
Nutritional Assessment: Poor nutrition, especially inadequate intake of calcium and other bone-supportive nutrients, can contribute to osteoporosis. A nutritional assessment can help identify deficiencies that might impact bone health.

Alcohol Consumption: Excessive alcohol consumption can interfere with bone health and increase fracture risk. Individuals with a history of heavy drinking should be evaluated for its impact on bone density.

Smoking Status: Smoking is associated with reduced bone density and impaired bone healing. Assessing smoking history can provide insights into potential osteoporosis risk.

Endocrine Disorders: Certain endocrine disorders, such as hyperthyroidism and Cushing's syndrome, can affect bone health. Diagnosing and managing these conditions is important in assessing osteoporosis risk.

Fig. 1 Reasons for Osteoporosis related fractures



Rheumatologic Conditions: Conditions like rheumatoid arthritis and ankylosing spondylitis can impact bone health and increase the risk of osteoporosis. Evaluating these conditions can guide osteoporosis assessment and management.

Bone Imaging Techniques: In addition to DXA, other imaging method like quantitative computed tomography and peripheral quantitative computed tomography can provide more detailed information about bone density and geometry.

Genetic Factors: Genetic predisposition can influence an individual's risk of osteoporosis. Family history of osteoporosis and related fractures can provide important insights.

Figure 1 specifies the various factors for fractures related to osteoporosis. Osteoporosis assessment is often a comprehensive process that takes into account multiple clinical parameters. The combination of these parameters helps to make an accurate diagnosis and determine the best management plan for the bone health. Generally in methods

3 Algorithms for Various Issues of Osteoporosis

Machine learning and deep learning have emerged as powerful tools in various medical domains, including osteoporosis, due to their ability to extract patterns, analyze complex data, and improve decision-making. In the context of osteoporosis, these techniques offer innovative approaches to enhance diagnosis, risk assessment, treatment planning, and research. The various roles are discussed below:

Fracture Risk Prediction: Machine learning and deep learning algorithms will be able to analyze diverse patient data, such as age, bone mineral density, medical history, and lifestyle aspects, to foresee the possibility of fractures accurately. By considering a wide range of variables, these models provide more personalized risk assessments compared to traditional methods. Deep learning techniques can capture intricate relationships within the data and potentially improve the accuracy of fracture risk prediction [3, 4].

Medical Imaging Analysis: Deep learning, particularly convolutional neural networks (CNNs), excels in analyzing medical images, such as X-rays and CT scans. In osteoporosis, CNNs can automatically detect vertebral fractures and assess bone microarchitecture from high-resolution images, enabling quicker and more consistent diagnosis.

Genetic and Biomarker Analysis: Machine learning can analyze genetic information and biomarker data to identify potential genetic factors contributing to osteoporosis risk. These algorithms can uncover complex relationships between genetic variations and disease susceptibility that might be challenging to identify using traditional methods.

Treatment Personalization: Machine learning algorithms can aid in selecting optimal treatment strategies for individual patients based on their unique characteristics. By integrating clinical data, medical history, and genetic information, these models can recommend treatments that are more likely to be effective and have fewer adverse effects [6].

Drug Discovery: Machine learning accelerates drug discovery by predicting how potential compounds might interact with bone metabolism pathways. These predictions help researchers identify novel drug candidates that could target osteoporosis-related mechanisms effectively [7, 8].

Clinical Decision Support: Machine learning algorithms can help the healthcare providers in making better decisions by investigating patient data and providing recommendations for treatment plans or follow-up assessments. This support can lead to more precise interventions and improved patient outcomes [9,19].

Research Insights: Machine learning techniques can analyze large datasets from clinical trials and observational studies, uncovering hidden insights into disease progression, treatment effectiveness, and patient outcomes. These insights contribute to the advancement of osteoporosis research and inform clinical practice.

Patient Monitoring: Machine learning algorithms can analyze longitudinal patient data to monitor disease progression and treatment response over time. By identifying trends and patterns in data, clinicians can adjust treatment plans as needed [22, 26].

In summary, machine learning and deep learning have transformative potential in the field of osteoporosis. They offer enhanced fracture risk prediction, improved medical imaging analysis, personalized treatment recommendations, drug discovery advancements, and deeper insights into disease mechanisms. As these techniques continue to evolve, they are likely to play an increasingly vital role in improving the diagnosis, treatment, and overall management of osteoporosis. The above are the areas in which a researchers need concentration for the solution to the osteoporosis disease using the algorithms. It's important to note that the choice of algorithm depends on factors such as the nature of the problem, the amount and quality of available data, computational resources, and the desired interpretability of the model. In practice, a combination of algorithms or ensemble methods might be used to leverage their respective strengths and mitigate their limitations. Recent research in osteoporosis detection and prevention using deep learning has been focused on addressing various challenges associated with accurate diagnosis, risk assessment,

and personalized treatment strategies. Some of the recent works on osteoporosis using deep learning and machine learning algorithms are compared in the Table 1.

Table 1 shows the comparison of various research articles and their findings. In summary, based on the patient's bone images and some of the clinical parameters were used to identify the diagnosis in early stages with the help of the deep learning algorithms. The accuracy obtained is average and it is not supporting the issue up to the core. Many clinical parameters support the osteoporosis. The most important set

Table 1 Comparison of the various Models and parameters for osteoporosis

Title of the paper	Dataset	Model	Observation
Ryoungwoo Jang et al. [2]	1001 images	VGG16 based DNN	Predict osteoporosis via simple HIP radiography
Zhen Zhao et al. [10]	12,222 images	AI-based DNN model	Detection of bone metastasis in individuals afflicted with cancer in lungs, breast, and other malignancies
Shintaro Sukegawa et al. [1]	778 images and clinical covariates	EfficientNet-b0,-b3, and -b7 and ResNet-18, -50, and -152 Ensemble model for clinical covariates	Clinical variables of patients were incorporated into dental panoramic radiographs to statistically elucidate how the inclusion of these variables affects classification accuracy
M. S. LeBoff et al. [5]	Clinical assumptions	Suggestions are given	Universal recommendations, Diagnostic assessment recommendations, Pharmacologic treatment recommendations, Monitoring patients and treatment response
Bin Zhanga et al. [11]	1255 images	3D-Class DCNN	It demonstrates favorable performance in clinical duties related to BMD classification
			Among postmenopausal women
Wen-Yu Ou Yang et al. [12]	5982 clinical data (both men and women)	SVM, ANN, KNN, Random forest, Logistic Regression	Identify the presence of Osteoporosis
Julien Smets et al. [15],	Image and Clinical datasets	ML models	Recognizing factors that contribute to the possibility of fracture and enhancing the accuracy in predicting the fractures
Norio Yamamoto et al. [13]	1131 hip radiographic images	Ensemble model	The diagnostic accuracy is improved
Insha Majeed Wani et al. [14]	381 knee x-ray	Deep learning model with transfer learning	It helps clinicians to identify osteoporosis as early as possible and hence reduce the possible risk of bone fractures

of clinical parameters and images has to be considered to address the issue as soon as possible. Different patients have different symptoms and issues. Every point has to be significantly addressed by the system. A generic model has to be developed to address different categories of people affected by osteoporosis.

4 Recent Challenges in Osteoporosis

Recent research in osteoporosis detection and prevention using deep learning has focused on addressing various challenges in accurately identifying individuals at risk and optimizing treatment strategies. Here are some recent problem statements in this domain:

Automated Vertebral Fracture Detection: Designing deep learning models, such as convolutional neural networks (CNNs), to automatically detect vertebral fractures from X-ray or CT images. This involves developing algorithms that can accurately localize and classify fractures, aiding clinicians in early diagnosis and treatment planning.

Predicting Fracture Risk from Imaging Data: Creating deep learning models that leverage high-resolution peripheral quantitative computed tomography (HR-pQCT) scans to predict individual fracture risk by assessing bone microarchitecture and density. The challenge lies in developing models that can accurately predict fractures based on intricate features captured from these images.

Multi-Modal Data Fusion for Risk Assessment: Integrating various data sources, including medical images, genetic data, and clinical information, using deep learning techniques to improve the accuracy of fracture risk assessment. This problem statement involves handling heterogeneous data and designing effective fusion strategies.

Personalized Treatment Recommendations: Developing deep learning models that analyze patient data, including medical history, imaging results, and biomarker information, to recommend personalized treatment strategies. The challenge is in creating interpretable models that provide insights into the reasoning behind treatment recommendations.

Longitudinal Monitoring of Bone Health: Designing deep learning algorithms to analyze longitudinal data from patients, tracking changes in bone density and microarchitecture over time. This can help identify trends and patterns associated with disease progression and treatment effectiveness.

Generating Synthetic Bone Images: Creating synthetic bone images using generative adversarial networks (GANs) to augment limited datasets. This can aid in training deep learning models for osteoporosis diagnosis and prediction when large labeled datasets are scarce [16–18].

Transfer Learning for Osteoporosis Detection: Applying transfer learning techniques to adapt pre-trained deep learning models from related medical domains (e.g., radiology) to osteoporosis detection tasks. This can expedite model development while leveraging existing expertise.

Explainable Deep Learning Models: Developing deep learning models that provide explanations for their predictions, particularly in medical image analysis. Ensuring the transparency and interpretability of deep learning models is crucial for clinical adoption.

Combining Clinical and Imaging Data: Integrating clinical data (e.g., patient demographics, medical history) with imaging data using deep learning to enhance the accuracy of osteoporosis diagnosis and risk assessment. The challenge is effectively fusing these different types of data to provide holistic insights [20, 21].

Early Detection in Pediatric Osteoporosis: Designing deep learning models to identify osteoporosis-related issues in pediatric patients, where unique challenges arise due to ongoing bone growth and development.

Quantifying Bone Quality: Developing deep learning approaches to quantify bone quality factors beyond bone density, such as bone geometry, trabecular microstructure, and tissue composition, to improve fracture risk prediction.

These problem statements represent the cutting-edge research areas within osteoporosis detection and prevention using deep learning. Deep learning algorithms have exceptional capability in analyzing medical images, such as X-rays and CT scans, to accurately assess bone density and identify potential fractures. These algorithms can expedite diagnosis, enhance accuracy, and offer a more nuanced understanding of bone health. Moreover, they enable risk prediction by amalgamating clinical data, genetic information, and imaging results, leading to more personalized treatment strategies. In terms of treatment, deep learning algorithms contribute to the optimization of therapy plans. By leveraging large-scale datasets and predictive modeling, these algorithms can assist healthcare providers in tailoring interventions to individual patient profiles, ensuring more effective outcomes and reduced adverse events. Additionally, they empower continuous monitoring of treatment progress, enabling timely adjustments to enhance results. However, while deep learning holds tremendous promise, challenges such as data privacy, algorithm interpretability, and generalization across diverse populations remain. Robust collaborations between medical experts, data scientists, and regulatory bodies are essential to address these challenges and ensure the responsible deployment of deep learning in clinical practice. In summary, the integration of deep learning models into the realm of osteoporosis diagnosis and treatment offers a paradigm shift toward more accurate, efficient, and personalized healthcare. This survey underscores the pivotal role of ongoing research, innovation, and interdisciplinary cooperation in harnessing the full potential of deep learning to alleviate the impact of osteoporosis and enhance the value of life for those persons affected by this disease [23–25].

5 Conclusion

Osteoporosis is a complex condition demanding a comprehensive diagnostic and treatment approach. By integrating various diagnostic techniques and implementing a combination of lifestyle changes and medical interventions, it is possible to effectively manage osteoporosis and reduce the burden of fractures. The survey underscores the importance of early detection, individualized treatment plans, and various monitoring to ensure the bone health throughout the lifespan. Deep learning algorithms can be used to combine the clinical parameters and images of the patients to identify the extent of osteoporosis has affected the bones. Transfer learning also can be combined to improve the performance of model. The osteoporotic-affected patients can be grouped based on age and symptoms. A model can be proposed to take care of all the concerns. Continued research and collaboration among healthcare professionals are essential to refine diagnostic methods, enhance treatment efficacy, and improve the life of the individuals affected by osteoporosis.

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Challenges and Existing Approaches of Assistive Technologies for the People with Locomotor Disability: A Review



Yashika Kumar, Ravindra Singh, and Chitra Kataria

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Abstract Locomotor disability is the most common of all disabilities, it involves difficulty with walking running, doing functional activities etc. Assistive devices enable people with disability to participate in the activities of daily living, they are also very important elements in achieving sustainable development goals. Being an important element, still assistive technology is not used by people with locomotor disabilities. This research is being conducted to examine the challenges and existing techniques for resolving issues in AT. The Cochrane database was searched for existing reviews on the subject. MEDLINE, EMBASE, and Scopus databases were searched using the terms challenges, assistive technology, locomotor disabilities, and approaches. There are various challenges that contributed to the AT's abandonment people with disabilities majorly rely on NGOs or other government-aided camps The government should also emphasize manufacturing these products locally so that the need to import them can be eliminated. There should be some approach or a guideline to minimize the abandonment which is lacking in the area of assistive technology for people with disability.

Keywords Locomotor disabilities · Assistive technology · Challenges · Approaches

Y. Kumar (✉) · R. Singh
Department of Design, Delhi Technological University, New Delhi, India
e-mail: yashikaisic@gmail.com

C. Kataria
ISIC-Institute of Rehabilitation Sciences, New Delhi, India

1 Introduction

Assistive technology is adaptable or particularly created technology that helps persons with disabilities perform better. Disability is the consequence of a limitation that restricts a person's capacity to engage in daily activities. The disability may be mental, physical, sensory, personal, growth-oriented, or a combination of these. Locomotor problems include cerebral palsy, spinal cord damage, muscular dystrophy, cerebrovascular accidents, spinal cord injury, and other locomotor disorders. To support these people financially and socially, particular devices or technology, i.e., assistive technology, are constantly needed to enhance their involvement in daily activities and so make them independent. AT is an area of engineering focused on the development of methods, systems, and technologies to assist people with impairments in doing daily tasks. Prostheses, orthoses, walkers, wheelchairs, walkers, canes, crutches, gait trainers, and scooters are examples of assistive technology devices for the locomotor impaired. These devices are designed to improve the user's quality of life and provide essential functional benefits while performing daily tasks. Despite the fact that there is a significant level of need in low- and middle-income countries, a lack of understanding about assistive technology leads to a lack of demand, restricting its development and acceptance. If people are aware of these technologies, other considerations include effectiveness, suitability, social stigma, usability, and so on. There are some issues and challenges which cause the hindrance in the use of AT. A lot of work has been done to solve the issues and challenges, and many studies have been done that develop various models and approaches, but still the need for assistive technology for people with locomotor disability have not been met yet. This study has been done to review the current challenges and existing approaches for people with locomotor disability.

2 Background

Disability is evolving into a major social, economic, and health concern on a global scale. Worldwide, more than 150 million people have some form of disability, with 6 million having a severe condition. A locomotor impairment affects one-fifth of the handicapped population (20.3%), followed by hearing (19.0%), vision (18.5%), other disability (18.0%), multiple disabilities (8.0%), speech (7.5%), intellectual disability (5.5%), and mental disease (2.7%). The prevalence of disability is higher in women than in males, and it is more prevalent in elderly people [1]. Assistive technology is useful for people of all ages and abilities. By enabling people with disabilities to take part in daily activities, assistive technology improves their quality of life and may play a key role in achieving social inclusion. The standard classification defines assistive technology for people with disabilities as any item—including a device, equipment, instrument, technology, and software—that is produced or generally

accessible specifically for preventing, compensating for, relieving, or neutralizing impairment, as well as activity limitation or participation restriction [2].

Only 5 to 15% of persons in low- and middle-income nations have access to assistive technology, according to the World Health Organization. According to the 2010 Burden of Disease Study, non-communicable diseases and disorders including stroke and diabetes with concomitant disability are replacing infectious diseases as the most common types of illness. The improvement of trauma care is another development in global health that is predicted to result in a growth in the demand for assistive technology. One of the main causes of long-term or temporary disability in people who can benefit from the availability of assistive technology in poor and middle-income nations is trauma brought on by injury and violence [3].

AT was formerly thought of as a niche discipline with little bearing on the larger development and global health goals. This is being changed by a number of factors. To start, the mainstreaming of assistive features into common consumer goods is changing how people think about the demands and needs for assistive technology. Second, the necessity to offer elder care and possibilities for independent living for elderly individuals is becoming a major concern for governments of both emerging and developed nations due to the rising productivity and longer work lives of older workers. Third, the convention on the rights of persons with disabilities (CRPD), which was formed in 2006, recognizes the significance of assistive technology for human development and ensures that people with disabilities have access to high-quality and reasonably priced assistive technology [4]. Numerous studies from developed countries have found that AT improves the socioeconomic circumstances of people with disabilities; yet, access to AT is insufficient in poor countries. The need for AT is significant even yet there is little demand for it and little supply in many low- and middle-income countries. The SDGs cannot be fairly accomplished if assistive product accessibility is not encouraged. The fact that assistive products are viewed as a mediator and a moderator of the SDGs' attainment was also emphasized. Tebbutt et al. demonstrate how the use of assistive goods can help fulfil each of the 17 sustainable development goals [5].

The CRPD has required access to high-quality, reasonably priced assistive items for ten years, yet just 10% of those in need have that access. If this urgent need is not met, the percentage of people who have access will fall as demand rises but service access would remain the same. In order to support member nations in increasing access to assistive technology as part of universal health care, the WHO established GATE (global cooperation on assistive technology) in 2014. This was done to support the nations in their efforts to comply with the CRPD to increase access to high-quality, reasonably priced assistive products worldwide [6]. A high standard of AT provision is a challenging issue in a complicated rehabilitation system. There aren't enough models or standardized, trustworthy, and valid instruments to help with AT selection. Researchers have identified a number of issues as major obstacles to providing AT, especially in low- and middle-income countries. These issues include a lack of funding, a lack of availability, a high cost, and a lack of qualified workers as well as a lack of knowledge [7].

Studies also showed that AT research, service delivery, and product design generally lacked user-centred approaches. Ignoring customer feedback throughout the evaluation and design phases can lead to a mismatch between the consumer and the recommended product, which causes expensive and early product abandonment. Finding the end users' demands and preferences before converting them into specific engineering requirements is one of the main challenges in designing AT for locomotor impairment. Since there are relatively few studies that identify the present issues, unmet needs, and increasing demand for AT in India, there is a gap in the literature. This study will focus on identifying numerous issues linked to assistive technology for the people in India and existing approaches to overcome barriers.

3 Methodology

A search of the Cochrane database was conducted to find any prior reviews on the subject. MEDLINE, EMBASE, and Scopus data sets, among others, were searched using the keywords "challenges, assistive technology, locomotor disabilities, approaches". In order to identify potentially relevant publications and eliminate irrelevant papers, retrieved articles were first examined by the title and abstract. All relevant articles' reference lists were searched for additional pertinent papers by title and abstract.

3.1 Challenges

Although demand and supply for assistive technology are modest, especially in countries with limited resources, there is a clear need for it. Due to excessive expenses, a lack of funding, availability, knowledge, and experienced personnel, only 10% of people with disabilities have access to assistive technology. Despite the fact that there is a high level of need, there is little demand for assistive technologies, which hinders their development and adoption. Even if there is awareness, still factors such as effectiveness, suitability, social stigma, privacy, and affordability prevent their use. Consumers with locomotor disabilities aren't considered when designing products, which could result in consumers abandoning acceptance to them [8]. Most assistive items for persons with locomotor disabilities in low-income nations are imported, and these products are expensive for the average person to purchase, and there are also fewer resources available to make the product locally. Additionally, some technologies are created without taking into account the real environmental, societal, and resource constraints that prevent the adoption of technology in environments with limited resources [9].

Due to low manufacturing, poor quality, and a lack of government funding, there is also a gap in access to these technologies in low- and middle-income nations, also factors like culture, languages, expectations, legal restraints, stereotypes, autonomy,

and dignity; fear of reliance; sense of “no need;” and lack of training [3]. World Report on Disability in 2011 states that due to inadequate policies and standards accessibility issues arises, with a stigmatizing view of those with disabilities, and a dearth of welfare, vocational training, and medical rehabilitation programmes. National representative studies of people with disabilities report that 17%, 18%, and 26% of people with disability have access to assistive technology in Namibia, Malawi, and Zambia [10].

There is also a lack of information on the amount of unmet demand in this sector, as well as the percentage of people whose needs are satisfied and unmet. There is a research gap, which impedes the development of evidence-based policy and practise. According to the survey on investigating challenges to the use of assistive technology for PWDs in India, first-time users faced financial barriers. Barriers to availability, awareness, accessibility, and psychological well-being. In addition, the lack of repair and maintenance services is seen as one of the main challenges for existing users [11].

S. Ganesh Kumar conducted research on disability and rehabilitation services in India concerns and challenges, and concluded that issues will worsen in the future due to the age structure and increased life expectancy. These concerns differ between industrialized and poor countries. The majority of disabled individuals in India live in rural locations, where access to, utilization of, and cost-effectiveness of rehabilitation services are important challenges. One of the most difficult difficulties is reaching out to hard-to-reach disabled people who live in remote areas [12]. Laura in her study on the barriers to mobility devices states that people do not have access to the assistive products because the cost is not covered in the insurance and also there is no third-party services for the medical devices [13]. According to S. Alqahtani et al. ‘s research, numerous academics have noted that user-centred design methods are not typically used in AT research, service provision, or product development. Stakeholders’ perspectives on the top mobility assistive technology research and development priorities to improve satisfaction and reduce abandonment. The conversion of user wants and preferences into precise engineering requirements is one of the main problems in the development of mobility assistance technology. Ignoring user feedback during evaluation and design processes can lead to a mismatch between customers and the recommended product, which causes expensive and early abandonment [14].

Environmental factors, intervention-related factors, and personal characteristics are only a few of the variables that affect how AT is used. When environmental impacts are reduced, people with disabilities will be more able to participate in social, educational, and vocational activities since experiences with environmental elements are subjective and can be interpreted differently for different users [15]. Another study on access and service provision in resource-constrained environments has been conducted. Access to AT is made difficult not only by the lack of available funding for AT purchases, but also by broader ecosystem weaknesses in resource-constrained environments related to laws, regulations, supply, distribution, human resources, consumer demand, and accessible design [3]. In his study on the levels and factors influencing the abandonment of assistive products, Andre Sugawara identified

users and obtained their feedback through in-person interviews. They discovered that upper limb orthoses, KAFO, walkers, crutches, and lower and upper limb orthoses showed higher levels of abandonment levels due to poor quality, a lack of adjustments, and frequent damages [16]. Although the market for assistive technology for people with locomotor disabilities is expanding, there are some issues to be aware of. For example, the AT industry, particularly at the national level, is underdeveloped. There is also a highly segmented and largely exclusive market with huge profit margins, high pricing, and low-quality items (Table 1).

3.2 *Existing Approaches for Assistive Technology*

This section represents the existing solutions for solving the challenges of the assistive technology.

In order to assist member nations in enhancing access to assistive technology as part of universal health coverage, the World Health Organisation formed GATE (global cooperation on assistive technology) in 2014. In order to increase access to high-quality, inexpensive assistive products, GATE has created a worldwide priority research agenda. Currently, GATE provides action in five linked areas (5 ps): people-centred, policy, products, provision, and personnel. The prioritized assistive goods is one of these deliverables that was published in 2017. The World Health Organization's GATE programme, which aims to increase access to assistive technology devices worldwide, started with the creation of a priority list for assistive products. The 50 assistive technology device priorities in this APL were selected after considering the requirements and effects on the lives of those with disabilities. The ranking can direct service, manufacturing, and product development [6] (Fig. 1).

Marcel de Gois stated in his study that assistive product design is a great challenge because the population of disabled individual is restricted to a small and heterogeneous group whose requirements change over time, so some got rejected due to poor performance and some due to unaffordable prices. So they proposed a model for assistive product development and which draws the midway between customer's value and company's performance. They divided the Assistive product design model into four stages that is discovery, definition, development and distribution, the discovery stage identifies the values of the product to the users, allows identification of the other products lies in that range, second stage is the definition stage here at this stage assistive technology is defined according to user value. The platform level AP modelling results in a more flexible structure as well as the addition of technical, ergonomic, and aesthetic functions. Before the distribution stage, where packaging, sales strategy, and after-sale monitoring methods are determined, the designer develops the product description, including dimensions, materials, tolerance, etc. [20].

S. Sujatha presents GRID model which is based on four pillars that is grant-research-industry-dissemination which is evolved from the TTK centre for rehabilitation research and device development in IIT madras. It offers a systematic approach for the production of high quality, useful, and affordable assistive equipment, making

Table 1 Evidences on challenges in assistive technology

S. no	Author, year, Journal/Conference	Title	Metho dology	Questions explored	Findings
1	Dr. viroj/sept 2018/WHO southeast asia journal of public health	Improving access to assistive technologies: challenges and solutions in low and middle income countries [17]	Literat ure review	What are Challenges and solutions to improve access in low and middle-incom e countries	Access to AT is restricted in low and middle-income countries, where there is a huge need but a lack of understanding among stakeholders and rehabilitative professionals. Design of Product are poorly guided, and technology transfer to low-resource locations is hampered by cost constraints and poor production quality
2	Amitabh Kishore dwivedi/Indian journal of physiotherapy and occupational therapy/ jan-mar2019	Exploring barriers to the use of assistive technology for PWD'S in India [11]	Survey	What are the barriers of Assistive technologies for people in India	AT can have significant beneficial effects on the life of people with disability but also have barriers to use. The major barrier found was the high cost of these devices and lack of funds. Some subjects were totally dependent on NGO but if some were managed to buy these devices then the next major issue is repairs and maintenance

(continued)

Table 1 (continued)

S. no	Author, year, Journal/Conference	Title	Metho dology	Questions explored	Findings
3	Tebutt et al (2016), globalization and health	Assistive products and sustainable development goals (SDG'S) [5]	Literat ure review	What role does assistive technology play in achieving sustainable development goals?	Assistive devices can reduce inequities significantly by enabling individuals with disabilities to become productive and to be included in all areas of life. The achievement of sustainable development objectives will be difficult to achieve if those who need the assistive products do not have access to it, assistive products help people to live healthy, productive, independent and dignified lives; access to them has been mandated by the CRPD for ten years, but still only 10% is accessible

(continued)

Table 1 (continued)

S. no	Author, year, Journal/ Conference	Title	Metho dology	Questions explored	Findings
4	Rebecca Mattered, b, Mark Harnissb, Tone Oderudc, Johan Borgd and Arne H. Eide disability and rehabilitation: assistive technology, 2016	Assistive technology in resource-limited environments: a scoping review [8]	Systematic review	What is the provision of assistive technology?	Only five to fifteen percentage of people in low and middle-income countries (LMICs) in need of assistive technologies (AT) can access them. Funders and researchers must contribute significantly more resources to address the scarcity of evidence in the AT area. The total number and quality of research on AT in resource-constrained locations is limited

(continued)

Table 1 (continued)

S. no	Author, year, Journal/Conference	Title	Metho dology	Questions explored	Findings
5	Keshini Madara Marasinghe, Jostacio Moreno Lapitan and Alex Ross BMJ Innov. 2015 Oct; 1(4): 182–195	Assistive technologies for ageing populations in six low-income and middle-income countries: a systematic review [10]	Systematic review	What are the provisions and challenges of AT in low- and middle-income countries for ageing population?	The findings of the review show that increasing the existing availability of AT for ageing populations in LMICs requires a comprehensive, integrated health and social system approach. These would include (1) promoting low-cost AT initiatives; (2) improving knowledge and capacity on AT; (3) narrowing the gap between AT policy and practise; and (4) promoting focused AT research

(continued)

Table 1 (continued)

S. no	Author, year, Journal/ Conference	Title	Metho dology	Questions explored	Findings
5	Andre T. Sugawara, Vinicius D. Ramos, Fabio M. Alfieri & Linamar R. Battistella Disability and Rehabilitation: Assistive Technology, 2018	Abandonment of assistive products: assessing abandonment levels and factors that impact on it [16]	Observational study	Why assistive products are abandoned in developing countries? And what should be done to avoid that?	This study gives an overview of user characteristics, a selection of things that could be made available to people with mobility challenges, as well as inputs for assessing resource consumption and financial impact in developing nations
6	Natasha Layton Rehabilitation	Barriers and Facilitators to Community	Survey	What are the Barriers and	This study has shown that optimal mobility depends upon adequate provision of a range of

(continued)

Table 1 (continued)

S. no	Author, year, Journal/ Conference	Title	Metho dology	Questions explored	Findings
	Research and Practice Volume 2012	Mobility for Assistive Technology Users [18]		Facilitators to Community Mobility for Assistive Technology Users	enablers at both person and environment level. Facilitators such as continuous paths of travel within the community, public transport and welcoming buildings are essential ingredients for full participation beyond the home
7	Inoka E Weerasinghe1 *, P Fonseka2, S D Dharmaratne3, J A M S Jayatilake4, Andrea C Gielen5 Disability CBR and inclusive development 2015	Barriers in Using Assistive Devices among a Group of Community-dwelling Persons with Lower Limb Disabilities in Sri Lanka [19]	Survey	What are the barriers of AT used by the people in Sri Lanka?	This study discovered several barriers to using assistive devices, including unaffordable assistive technology such as wheelchairs and artificial limbs, a lack of appropriate assistive technology, difficulties associated with repair and maintenance, accessibility issues, and a lack of awareness

Fig. 1 The WHO-GATE (Global cooperation on assistive products) 5P framework [6]



it a suitable model for the development of appropriate assistive devices in developing nations. A key component of this approach is the early involvement of end users and industry partners in the creation of appropriate and inexpensive assistive technologies. This model is also based on collaboration across numerous groups. Lack of financing for research on the creation of assistive devices in environments with limited resources, as well as whatever research and development occurs at institutions, almost never reaches the end users. So GRID model sets an example of effective development and dissemination of assistive device by involving stakeholders from the beginning [21] (Fig. 2).

In his study, Ornella introduces the EMFASIS framework, which aims to enhance their design and move them towards universality. It stands for extended modularity,

Fig. 2 GRID model comprises of four major components that are Grants, Research, Industry, Dissemination [21]



functional accessibility, and social integration strategy. The first element of this strategy is to look for business partners who focus on the general market rather than the impaired user. Designing modularly comes next. This process entails defining the product to enhance the number of models and functionalities. The next step is functional acceptability, when experts are interviewed and various user demands are surveyed. Accessibility is the next phase, and it is looked into through a needs study that targets various populations. The process of social integration, which incorporates aesthetic qualities and social values, comes next [22].

In their study framework for a novel approach to user empowerment through low-cost, DIY assistive technology, Thais et al. improve the process of developing and creating assistive products by incorporating three phases: identifying (design), production (building the prototype), and execution (outcome measures), all of which are based on the design-making concept and from an user-centred design perspective. The findings of this framework demonstrate the consistency of the entire process and its application, as well as the significance of involving users in the design and selection of AT [23].

In their study on empowering individuals with Do-it-yourself assistive technology, Amy Hurst and Jasmine Tobias highlight the low acceptance rate and abandonment of AT, which refers to the development of AT by non-professionals, and they described in their study that increased control over the design elements, and cost-motivated individuals to make their own assistive technology instead of buying it. They gave a case study of head pointers in which an individual was unable to obtain a solution from current solutions and thus created their own solution that is less expensive and fulfils all tasks. In 2014, Jonathan did research on the difficulties of DIY assistive technology in the setting of children with impairments. In spite of the high demand for assistive technology and its possible benefit for individuals with disabilities, its development and introduction are challenging so DIY assistive technology allows people to adapt to design, but aesthetics, practicality, robustness, and safety of DIY-AT were discovered to be the key barriers to participation [24].

In this study, A.V.F. Santos represents the concept of building assistive technology while taking user-centred design and 3D technologies into account. The high abandonment rates of assistive devices, their poor performance, lack of options, and exorbitant prices point to flaws in the early stages of their creation. The design and development of assistive items are fundamentally different from those of mass-produced goods, whose key feature is interchangeability. As a result, in this study, they developed a design process focusing on the creation of assistive technology while taking user-centred design and 3D technologies into account. This method aids in the incorporation of new improvements and possibilities brought to this field by additive manufacturing, the decrease of time and steps between design and manufacture, and user interaction in the early stages of the process design reduced the chances of abandonment of assistive device [25] (Table 2).

Table 2 Evidence on existing approaches in AT

S. no	Author, year, journal/ conference (reference)	Title	Methodology	Questions being explored	Findings
1	Sandra Sueli Vieira Mallin, H�lio Gomes de Carvalho Procedia Manufacturing 3 (2015)	Assistive Technology and User-Centered Design: emotion as an element for innovation [26]	Literature review	What is the importance of User centred design in AT?	The importance of addressing the user at all phases of the design process is revealed by studies in design and innovation in the area of Assistive Technology. User-Centred Design is the term for this method (UCD). Its main goal is to improve people's understanding of user-centred design and the emotional component of the assistive equipment creation process

(continued)

Table 2 (continued)

S. no	Author, year, journal/conference (reference)	Title	Methodology	Questions being explored	Findings
2	Jasper Grieg1, Martina Maria Keitsch2, Casper Boks3 Nord Design 2014 August 27–29, 2014	Widening the interpretation of assistive devices –A designer’s approach to assistive technology [27]	Review	What are various approaches to design of assistive devices?	Assistive technology (AT) has traditionally focused on being ergonomic, functional and useful. It recommends a broader meaning of AT, which can be divided into “enhancing assistive devices (EAD)” and “reinventing assistive devices (RAD)”
3	Young Mi Choi PhD & Stephen H. Sprigle Assistive Technology: The RESNA Official Journal of 23:36–41, 2011	Approaches for Evaluating the Usability of Assistive Technology Product Prototypes [28]	Survey	What are the approaches for evaluating the usability of AT products?	User feedback is critical in assisting designers in creating a more usable product. This study investigated various strategies for evaluating prototypes in order to see which ones could be useful in the development of new assistive technology solutions. During the early stages of designing AT products, a speedy and cost-effective evaluation approach could assist designers in directly engaging consumers in uncovering significant usability difficulties. They concluded earlier stakeholder engagement, but believe prototype testing will be important in making design adjustments to AT goods before they are manufactured

(continued)

Table 2 (continued)

S. no	Author, year, journal/conference (reference)	Title	Methodology	Questions being explored	Findings
4	Maria Luisa Toro-Hernández, Physical Medicine and Rehabilitation Clinics of North America 2019	Appropriate Assistive Technology for Developing Countries [29]	Case studies	What could be the possibilities to improve access to AT in developing countries?	Access to appropriate and affordable assistive technology (AT) is a human right as well as an essential health and developmental need. Market shaping has the ability to reduce health-care market problems and promote better health outcomes Countries must eliminate or lower import duties on materials and AT to guarantee that population consumption does not exceed production
5	International Journal of Environmental Research and Public Health 2018 Malcolm MacLachlan Joanne McVeigh Michael Cooke Delia Ferri Catherine Holloway Victoria Austin and Dena Javadi	Intersection s Between Systems Thinking and Market Shaping for Assistive Technology: The SMART (Systems-Market for Assistive and Related Technologies) Thinking Matrix [30]	Literature review	How strong system thinking and market shaping approach improve access to AT	According to the findings of this study, many countries have typically provided assistive technology through small-scale local providers who produce items of different quality at a limited price range. They also conclude that improved access to assistive technology will be realized only through the application of a considerably more robust systems thinking and market shaping approach

4 Discussion and Conclusion

Currently, the assistive product business only serves high-income markets. Some of the issues in this industry are as follows: Inadequate service and delivery systems, lack of user-centred research and development Product design that meets quality and safety requirements. The goal of this study is to look into the difficulties that persons with locomotor disabilities face, as well as the existing solutions. Another purpose of this review is to provide better information to professionals who work with persons who have locomotor disabilities, allowing them to better support their patients. Despite the fact that people with disabilities may require assistive technologies and goods that increase social and community involvement, there are still hurdles to access and use, as well as unmet needs. The cultural setting has also influenced the requirement for assistive technologies for daily engagement.

Locomotor disability is the most prevalent issue, but it has received little attention from public health policymakers, particularly in developing nations. Assistive gadgets can have a significant positive impact on the everyday lives of individuals with disabilities, but it is commonly noticed in the literature that they are abandoned, which is a complicated issue.

Access to assistive technology has an impact on functionality and social participation. Furthermore, research and development in this sector has been limited to high-income countries. There is an urgent need for research and development that focuses on the needs and issues that people in low-income nations experience [16].

There are issues with the quality, quantity, and sustainability of assistive device manufacturing, as well as suitable product design. Assistive products are frequently made or imported from foreign nations, creating difficulties in repair and maintenance, resulting in high abandonment rates. Assistive items should be manufactured locally, or parts replaced locally. There is also a dearth of understanding among service providers and users regarding the variety of offered goods and their benefits [31].

Affordable access to assistive technology necessitates government commitment to adequate and long-term funding. Where this does not exist, consumers and their families are forced to make large out-of-pocket expenses. People from the weaker parts of society have a tendency to rely on contributions or charitable services, which frequently consist of the distribution of huge numbers of low-quality or used things that are improper for the user or the environment and are also not replaced locally [32].

Although there are some approaches already mentioned in this study the which mainly includes the GATE launched by the WHO, its priority product list which includes 50 products which are essential for all the countries to make it available, but these policies are still not properly implemented, whereas different approaches including DIY approach lacks the technicality and long term use of these devices may lead to adverse effects, as while designing lots of other things like balance, coordination, material strength should be kept in mind and also the trial and testing of the prototypes are mandatory. Also, the researchers have tried to produce the

product via 3D printing technology in which mass production can be made and the users are involved in this so that the abandonment of the products can be made possible.

There are several attempts which are made to enhance the universality of the products so that they can be accepted by the users but, there are still some issues which need to be find out and resolved, especially for the developing countries where people with disability are high in numbers and they are also dependent upon the government for the assistive products. There should be something which is to be done at the manufacturing level so that the products can have the basic functions in spite of high-technology, then only it will be affordable for the people with disability and also, they will be able to afford the repair and maintenance. The government should also emphasize on manufacturing these products locally so that the need to import them can be eliminated.

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Deep Learning in Medical Imaging: A Comparative Review of Models for Ischemic and Hemorrhagic Stroke Lesion Segmentation



Sadiya Sulaiman, M. Roshni Thanka, E. Bijolin Edwina, and Nader Salam

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Abstract Stroke, primarily categorized as ischemic and hemorrhagic, stands among the top global causes of mortality and can lead to severe neurological conditions if left untreated. The emergence of machine intelligence, especially advanced neural networks, has brought significant advancements in medical imaging, including the automated segmentation of stroke lesions. This automation aids radiologists and surgeons in more effective diagnosis and treatment planning. This review paper concentrates on deep learning methods used in segmenting ischemic and hemorrhagic stroke lesions, examining these models from an architectural standpoint. The paper begins with a discussion on the types of strokes. Subsequently, delve into a review of deep learning models, specifically emphasizing their application in ischemic and hemorrhagic stroke lesion segmentation. The performance, advantages, and limitations of these models are assessed and comparisons among them are presented. The paper concludes by spotlighting the current needs in this field and suggesting potential avenues for future research. The study aims to provide insights to both advanced

S. Sulaiman (✉) · M. R. Thanka · E. B. Edwina · N. Salam
Department of Computer Science and Engineering, Karunya Institute of Technology and Science,
Coimbatore, Tamil Nadu, India
e-mail: sadiyasulaiman@karunya.edu.in

M. R. Thanka
e-mail: roshni@karunya.edu

E. B. Edwina
e-mail: bijolin@karunya.edu

N. Salam
e-mail: nadersalam@karunya.edu.in

researchers and newcomers in the field, thereby fostering future advancements in stroke lesion segmentation using deep learning methods.

Keywords Hemorrhagic strokes · Ischemic strokes · Lesion segmentation · U-Net · DeepLab · GAN · FCN · Attention model · Deep learning · Neural networks · Medical image analysis · CT imaging · MRI imaging

1 Introduction

Advanced neural network methods have demonstrated notable achievements in classifying, detecting, and segmenting standard image data. These techniques have been utilized for clinical image partitioning tasks, such as identifying stroke lesions and lesion segmentation of stroke-related neurodegenerative disorders. Automatic lesion segmentation is essential for devising follow-up treatment plans and improving clinical practice.

Stroke, a leading cause of adult mortality worldwide, results in a death every 3.5 min. It happens Once the cerebral blood flow is disrupted or insufficient, leading to oxygen deprivation and unresponsiveness in specific brain regions. Strokes are primarily of two types: ischemic and hemorrhagic, caused by obstruction or rupture of cerebral arteries, respectively.

Accurate and effective segmentation of stroke lesions, including ischemic and hemorrhagic ones, is essential for determining a diagnosis, strategizing treatments, and creating subsequent care plans. High-resolution imaging methods, like Scans using Magnetic Resonance and Computed Tomography technology (CT), are fundamental in diagnosing strokes. Deep learning algorithms have demonstrated significant achievements in diverse image data categorization, identification, and partitioning tasks. In clinical image partitioning, techniques like Convolution-based Neural Systems (CNNs), U-Structures, and Adversary Generative Networks (GANs) and more are utilized. Despite the vast potential of these algorithms, their application in ischemic and hemorrhagic stroke lesion segmentation remains an active area of research.

The paper will review various deep learning models, specifically their application in ischemic and hemorrhage lesion segmentation, addressing their performance, advantages, limitations, and possible areas for future research and development.

2 Stroke

Cerebral Ischemia

Ischemic stroke results from insufficient oxygen or blood supply to the brain because of blood clots. Symptoms can include paralysis, language difficulties, and loss of coordination. AI has been extensively used to identify strokes, segment lesions, and forecast outcomes.

Intracerebral Hemorrhage

Hemorrhagic strokes occur when blood vessels in the brain rupture, causing immediate damage to brain cells. Intracranial bleeding (ICH) and subarachnoid bleeding (SAH) represent two varieties of hemorrhagic stroke. Hemorrhagic strokes account for approximately 10% of total strokes and have a higher mortality rate than ischemic strokes. Advanced neural networks are widely used in examining medical images associated with cerebral hemorrhage, using convolutional neural networks to recognize and segment abnormal regions in brain images.

3 Deep Learning for Stroke Lesion Segmentation

Diagnostic image analysis has successfully used machine learning algorithms for the last 20 years. However, these techniques heavily rely on features created by subject matter specialists, leading to potential observation misinterpretation. In contrast, advanced neural networks in diagnostic visualization have achieved significant advancements in automatically extracting features from all representations, providing improved data analysis and inspection. The increasing ease of computing massive amounts of data using deep convolutional neural networks due to GPU advancements has led to deep learning becoming a crucial methodology in automating processes in the healthcare system. Various cutting-edge techniques of segmentation methods related to stroke lesions are covered in this part.

3.1 Review of Delineating Lesions of Ischemic Stroke Using Deep Learning from Architecture View

This segment delves into diverse neural network techniques used for partitioning ischemic stroke lesions

3.1.1 Fully Convolutional Network (FCN)

An approach that has proven significant in converting convolutional neural networks, typically used in image classification, into ones useful for image segmentation is the Fully Convolutional Network (FCN). FCN finds practical application in delineating cerebral stroke damages as depicted in Fig. 1, where a Brain MRI input is processed using FCN to accurately distinguish stroke lesions. This results in a binary mask showing the lesion in dark and the rest of the brain in white. Initial drawbacks of the standard FCN include a lack of symmetry and aggressive down-sampling, which can lead to a loss of spatial information. To address these issues, researchers have proposed advanced FCN approaches for ischemic stroke lesion segmentation.

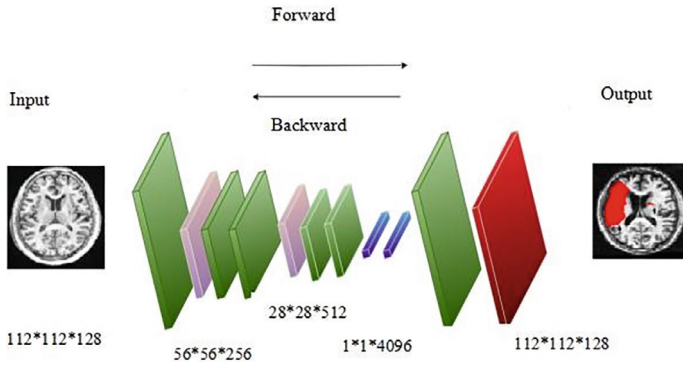


Fig. 1 Schematic representation of the image segmentation pipeline for cerebral stroke damages utilizing fully convolutional networks (FCNs). The input consists of a 3D graphic with dimensions of $112 \times 112 \times 128$, which is processed using multiple CNN layers, including CNN ($56 \times 56 \times 256$) and CNN ($28 \times 28 \times 512$), to extract features. The feature maps are downsampled using maxpooling layers and then upsampled using a completely interconnected layer and another maxpooling layer to secure the ultimate binary mask that segments the brain stroke lesion from the rest of the image

Carlos Uziel Perez’ Malla et al. [1] in 2019 demonstrated the beneficial impact of data augmentation on model performance, suggesting a solution to the limited data problem prevalent in medical imaging. However, a potential drawback of this method could be overfitting due to redundancy in augmented data. In 2020, Xiaojun Hu et al. [2] developed a 3D residual framework for streamlining the delineation of ischemic stroke damages and cerebral tumors, attaining a Dice accuracy of 30%. This model exhibits an advanced generalization capacity, contributing to improved performance. However, the relatively low Dice score suggests potential room for improvement in segmentation accuracy.

Mobarakol Islam et al. [3] in 2022 proposed an innovative model that leverages adversarial learning, using an FCN and a U-Net with skip connections as the discriminator networks. This model achieved a 42.1% Dice accuracy with training data and 39% with testing data. While this model presents a novel approach to leverage adversarial learning, its accuracy indicates room for improvement, and it had a lower performance on testing data, suggesting potential overfitting. In a recent development in 2023, Zelin Wu [4] introduced the W-Net, a novel model for brain MRI lesion segmentation. Deploying a two-tiered approach, the W-Net employs both CNNs and transformers as its foundational architecture. Its distinctiveness lies in the integration of two modules: the boundary deformation module (BDM) and the boundary constraint module (BCM). The BDM refines lesion boundaries using circular convolution techniques, whereas the BCM strategically uses dilated convolution to modulate the object boundary. Tested on the ISLES2022 dataset, the W-Net exhibited a Dice Similarity Coefficient (DSC) of 85.60%. A notable strength of the W-Net is its adeptness at handling indistinct lesion boundaries, ensuring detailed segmentation.

Table 1 Review of ischemic stroke lesion Segmentation using FCN

Author	Modality	Method	Result	Year
Xiaojun Hu et al. [2]	3D MRIs	CNN Residual	MeanDC = .30	2020
Carlos Uziel Perez´ Malla et al. [1]	MRI	CNN Data Augmentation Transfer learning	MeanDC = .30	2019
Zelin Wu et al. [4]	Multimodality MRI	CNN and transformer complemented by a boundary deformation module (BDM) and boundary constraint module (BCM) for addressing fuzzy lesion boundaries	DSC (%) = 85.60	2023
Mobarakol Islam et al. [3]	CT	Adversarial Learning A fully connected network (FCN) and U-Net with skip connections are used as the discriminator networks	Dice accuracy = 42.%	2022

Through the review of these various methods, it’s clear that advancements in FCN approaches have improved ischemic stroke lesion segmentation. However, challenges related to data scarcity, complexity of multimodality integration, and achieving high segmentation accuracy remain. Future work could focus on addressing these limitations while building upon the unique strengths of each model. For a concise summary of the various FCN methods discussed and their corresponding performance metrics, please refer to Table 1.

3.1.2 U-Net

The U-Net framework is recognized for its strong performance in clinical image partitioning due to its upsampling and down-sampling components, which function as an encoder and decoder, respectively. The down-sampling phase assists in edge detection through 3×3 convolutions, condensing the image size to accumulate contextual information. Conversely, the upsampling part utilizes deconvolution, incorporating feature maps generated by the down-sampling component. This ensures the prevention of essential data loss crucial for precise localization. One of U-Net’s remarkable attributes is its competence in segmenting images even with a limited dataset of trained annotated images.

However, U-Net is not without its drawbacks. There’s an inherent size constraint with input images limited to 572 by 572 dimensions. Furthermore, the architecture often overlooks inner layers with abstract attributes, which can decelerate the training

process. To address these, several evolutions of U-Net have emerged. In 2022, The Deep Dissimilar-Siamese-U-Net was put forth by Jia You et al. [5] (DSU-Net), which boasted remarkable results but posed challenges in bilateral lobe feature map analysis. Shaoquan Li et al. [6] presented the Multi-scale U-Net, offering exceptional segmentation but with potential computational constraints due to its multi-scale nature. In the same year, Yue Zhang et al. [7] advanced the MI-U-Net, a model boasting multi-input capabilities, albeit with complexities in pre-processing. Finally, Samrand Khezrpour et al. [8] unveiled the Deep Supervised U-Net, scoring high in accuracy but potentially restricted in imaging data applicability.

In 2023, building upon the foundation laid by U-Net and its variations, Weiyi Yu [9] introduced the SAN-Net, an innovative model tailored specifically for MRI segmentation. This novel architecture is distinguished by its method of self-adaptive normalization. At its core is the deployment of masked adaptive instance normalization (MAIN) which has been tailor-made to address and minimize inter-site discrepancies. Bolstering its capabilities further, the network integrates a gradient reversal layer, ensuring a site-invariant representation, underpinned by the foundational structure of a U-net encoder. The SAN-Net model, in its evaluations, garnered a notable Dice score of 0.5711 ± 0.1957 . A distinctive feature that adds value to this model is its data augmentation strategy known as the Symmetry-inspired data augmentation (SIDA). This method not only amplifies the dataset but remarkably reduces memory consumption by half. SAN-Net's stellar capacity in navigating inter-site discrepancies stands out as its prime strength, thereby pushing the envelope in model generalization.

In wrapping up, the traditional U-Net and its derivatives have consistently offered substantial advancements in medical image delineation. Architectures like DSU-Net, Multi-scale U-Net, MI-U-Net, and other more recent SAN-Net, have significantly enhanced the landscape of segmentation. Yet, each carries its distinct set of strengths and limitations. From computational costs, pre-processing intricacies, to data specificity, these models offer a spectrum of capabilities, warranting a discerning evaluation for specific applications. For a concise summary of the various U-NET methods discussed and their corresponding performance metrics, please refer to Table 2.

3.1.3 The Deep Lab Model

The DeepLab architecture employs pretrained CNN models with atrous convolutions, maintaining resolution while introducing no new learning parameters. The versatility and scalability of this architecture have resulted in several iterations, such as DeepLabv1, v2, v3, and v3+. In this section, we will focus on some models developed using the DeepLab architecture specifically to delineate ischemic stroke damages. In 2022, Haichen Zhu et al. [10] suggested a new neural network approach termed Ischemic Stroke Prediction Network (ISP-Net).: ISP-Net uses a combination of five convolution layers to integrate comprehensive characteristics of CT Perfusion imaging data. The model introduces an enhanced high-level feature extractor—a multi-scale atrous convolution unit. The outcome is an accurate stroke lesion center

Table 2 Review of ischemic stroke lesion Segmentation using UNET

Author	Image modality	Method	Outcome / Characteristics	Year
Jia You et al. [5]	NCCT	Siamese-U-Net of Deep Dissimilarity (3D DSU-Net)	DSC = $.784 \pm .66$, JS = $.753 \pm .67$ Recall = $.84 \pm .53$ Precision = $.879 \pm .44$ ASSD = 4.627 ± 3.8 MSD = 6.86 ± 8.66	2022
Shaoquan Li et al. [6]	CT	Multi-scale U-Net	Dice similarity coefficient: 0.86 ± 0.04	2022
Samrand Khezropour et al. [8]	MR flair	Deep supervised U-Net architecture	DC = $.89$	2022
Yue Zhang et al. [7]	T-MRI	MI-U-Net: Multi-inputs U-Net	Dice = 56.72 , Precision = 65.45 Recall = 59.38	2022
Weiye Yu [9]	MRI	Self-adaptive normalization network utilizing Using masked adaptive instance normalization	Dice = 0.5711 ± 0.1957	2023

prediction with a Dice index of $.8$ and accuracy of 83.3% . Despite high accuracy, the ISP-Net model's complexity due to the multi-scale atrous convolution unit could be a limitation, making it potentially computationally expensive. In 2022, Xiuquan et al. [11] introduced an innovative deep learning model, the Focus-driven Multi-level Restoration Structure (AGMR-Net): This approach utilizes a broad-scale patch attention mechanism and a cross-dimensional feature fusion unit, resulting in a Dice similarity score of $.594$. While the model's strength lies in its unique attention-guided framework, the precision is comparatively lower than some other models, indicating a possible area for improvement. In 2019, Xinfeng Liu et al. [12] suggested a Multi-Level Integrated Network with Atrous Pooling for 3D Cerebral Damage Delineation (MSDF-Net) This encoder-decoder neural network model uses multi-scale deep fusion, resulting in superior performance in comparison with five other methods with it demonstrated superior performance with a MedianDice of 0.68 , MedianSE of 0.82 , and a Median VOE of 0.140 . For a concise summary of the various DeepLab methods discussed and their corresponding performance metrics, please refer to Table 3.

Table 3 Review of ischemic stroke lesion Segmentation using DeepLab

Author	Image modality	Method	Outcome/Characteristics	Year
Haichen Zhu et al. [10]	CT	Encoder-decoder-based semantic model with multi-scale atrous convolution (MSAC) block ISP-Net	Mean DSC: 0.801	2022
Xiuquan n et al. [11]	MRI	Leverages a coarse-grained patch attention unit and a cross-dimensional feature fusion unit AGMR-Net	Dice similarity score: .594	2022
Xinfeng Liu et al. [12]	MRI	Encoder-decoder neural network model using multi-scale deep fusion MSDF-Net	MedianDice-.68,	2019

3.1.4 GAN

Adversary Generative Networks (GANs) have gained significant traction method for training generative models within the deep learning landscape. These frameworks predominantly consist that includes both a generator and a discriminator. The role of the generator is to craft data from unpredictable noise, while the discriminator's task is to discern between genuine data and the content produced by the generator. The generator aims to trick the discriminator into considering the produced data as authentic. In this context, models such as U-NET can serve as generators in a GAN setup.

In 2020, Hulin Kuang et al. [13] (CT, MPC-GAN method, U-NET) developed a 2D dense multipath contextual GAN, which utilized information such as dual intensity, proximity mapping, and likelihood of lesion placement. The results produced Pearson correlation values of .926 for ischemic infarcts and .927 for hemorrhages. This represents a robust precision in segmenting different types of stroke-related damage. However, the network's complexity due to the incorporation of various types of information might pose computational challenges and require more robust hardware to function efficiently. In their work on Multi-modal CT Image, DGAN in 2022, Suzhe Wang et al. [14] formulated the UTC-GAN, a novel unsupervised adversarial technique for merging different CT brain scans.

This method attained a Dice measurement of 0.768 and showcased an IOU performance of 0.685. These metrics indicate a high level of accuracy in image segmentation. The potential drawback could be the increased complexity due to the use of a Siamese encoder design and Gaussian mixture translation unit, which could be computationally demanding. Moritz Platscher et al. [15] (MRI, GAN, U-NET, 2022) introduced an approach leveraging deep learning-based augmentation for lesion segmentation. It achieved a good Dice score of 72.8%, indicating effective image

Table 4 Review of ischemic stroke lesion Segmentation using GAN

Author	Image modality	Method	Year	Outcome/ Characteristics
Moritz Platscher et al. [15]	MRI	GAN, U-NET Image-to-image translation & GANs for synthesizing MRIs	2022	Dice score of 72.8%
Suzhe Wang et al. [14]	Multi-modal CT Image	DGAN GAN with Siamese autoencoder	2022	DC = .768, Iou = .685 Precision = .82 Accuracy = .926 Recall = 78 HD (mm) = 7.73
Hulin Kuang et al. [13]	CT	MPC-GAN method, 2D dense multipath contextual GAN with bilateral intensity, distance map, lesion location probability,U-NET	2022	DC = 7.6% Pearson correlation coefficients = .926

segmentation. However, generating synthetic lesion masks and synthetic stroke images could require additional computational resources, potentially limiting its practicality for systems with less computational power. Each of these models showcases unique methods and yields various performances when segmenting ischemic stroke lesions. While they all show promise in this area, the selection of a model must consider the availability of computational resources, the type of images used (CT or MRI), and the precise nature of the data to be processed (e.g., type of stroke damage). For a concise summary of the various GAN methods discussed and their corresponding performance metrics, please refer to Table 4.

3.1.5 Attention

In 2022, Zelin Wu et al. [16] and associates unveiled MLiRA-Net, a groundbreaking approach in the domain of image analysis. Unlike traditional models, MLiRA-Net uniquely integrates convolutional layers designed for local feature extraction and transformers tailored for assimilating global multi-scale information. This is achieved through the innovative use of tokenized image patches. Impressively, when put to test, the model yielded a Dice coefficient (DSC) measured at 61.19% with a Hausdorff measurement (HD) recorded at 13.49 mm. A distinguishing attribute of MLiRA-Net lies in its adeptness at comprehending multi-scale local-global semantic information. By incorporating a patch partition block and leveraging transformer architectures. Karthik et al. [17] (Multi-modal MRI, Masked Dropout Unit, Patch modeling, Ensemble of Deconvolutional networks, 2022) enhanced the learning method by merging multiple scale data and utilizing an aggregation of feature maps, achieving

a DSC of 0.775. They proposed a unique dropout module to hide noisy features, demonstrating an innovative solution to reduce overfitting. Nevertheless, the use of an ensemble of deconvolutional networks may pose challenges due to increased computational complexity. Liangliang Liu et al. [18] (2D MRI, Deep residual attention CNN, U-net Attention module, 2022) proposed the deep residual attention CNN that demonstrated a DSC of 76.39 and HD of 3.9. This model provides exceptional performance while retaining the advantages of the U-net architecture, suggesting a balance between efficiency and performance. The limitation might be the need for deeper features extraction, which can increase computational demand. In a recent 2023 study by Weili Wang et al. [19], a novel methodology that synergistically integrates 2D and 3D convolutions was presented. The standout feature of this approach is its enhancement via the convolutional block attention mechanism, further augmented by an innovative hybrid decoding unit with residual attention gating. In comparison to well-known architectures like TransUNet, SAN-Net, and W-Net. Wang's technique showcased superior performance, attaining a Dice similarity coefficient of 0.6193. This research extensively utilized both 2D and 3D MRI datasets, reinforcing the method's adaptability and depth. Central to its training was the fusion of convolution, bolstered by an advanced attention mechanism. A salient strength of this model lies in its precision, especially in segmenting small lesions, an area often challenging in medical imaging. In a noteworthy 2023 study by Li, Zhihua et al. [20] a sophisticated stroke segmentation model named BSSNet was introduced. The foundation of this model is built upon the U-Net architecture, which incorporates Depth-Wise Separable Convolution, enhancing computational efficiency. Further augmenting the model's prowess is the integration of residual connections, the Module incorporating Multi-Scale Channel Attention (MSCA), paired with the Guidance provided by Attention Connection (AGC) module. These additions ensure enriched context comprehension and bolster segmentation accuracy. When evaluated, BSSNet achieved a commendable Dice coefficient of 0.65, emphasizing its performance on 2D data. One of the model's paramount strengths is its adeptness at navigating structural symmetry, and its capacity to effectively address challenges tied to the diverse positioning and boundaries of lesions. In summary, all these models demonstrate unique strategies for lesion segmentation, each with its strengths and limitations. The choice between these models would depend on the type of images used (CT or MRI), computational resources available, and specific requirements for the segmentation tasks. A trade-off often exists between performance and computational demand, hence, understanding the advantages and limitations of each model is crucial for their practical application in ischemic stroke lesions segmentation. For a concise summary of the various Attention methods discussed and their corresponding performance metrics, please refer to Table 5.

Our in-depth study spanning 2019 to 2023 highlights the advancements and adaptations within the deep learning architectures used for medical imaging. By employing the Dice Similarity Coefficient (DSC) as a benchmark, we've visualized the evolution across different neural network architectures (refer to Fig. 2). The FCN models, for instance, underwent a transformative journey. Their modest beginnings in 2019 at a 30% DSC saw a meteoric rise to 85.6% by 2023. U-Net, on the other

Table 5 Review of ischemic stroke lesion Segmentation using Attention

Author	Image modality	Method	year	Outcome / Characteristics
Zelin Wu et al. [16]	MRI	Convolutional layers for local features and transformers for global multi-scale information using tokenized image patches	2022	DSC: 61.19%, HD: 13.49 mm
R. Karthik et al. [17]	Multimodal MRI	Multi branch residual connections and Attention mechanisms Masked Dropout Unit Patch modeling Ensemble of Deconvolutional networks	2022	DC = .775
Liang Liang Liu et al. [18]	2D MRI	Deep residual attention CNN U-net Attention module	2022	DC = 76.39 HD = 3.9
Weili Wang et al. [19]	MRI	Fusion of 2D and 3D convolution enhanced by the convolutional block-focused method combined with an enhanced hybrid decoding unit utilizing residual attention	2023	Dice similarity coefficient of 0.6193,
Li, Zhihua, et al. [20]	MRI	Depth-Wise Separable Convolution in U-Net, residual connections, Multi-Scale Channel Attention (MSCA) module, and AttentionGuided Connection (AGC) module	2023	Dice = .65

hand, marked its debut in 2022 with an impressive 78.4% DSC but experienced a drop to 57.11% a year later. DeepLab remained somewhat consistent, with performance figures floating around the high 60 s between 2019 and 2022. The GAN model showed promise, registering an upward trend from 70.6% in 2020 to 74.8% in 2022. The Attention-driven models displayed a mixed bag of results — a debut at 71.69% in 2022, followed by a slight dip to 63.46% in 2023. Taking a snapshot of the year 2022 provides further insights (as depicted in Fig. 3). U-Net was the top performer, closely rivaled by GAN, while FCN and DeepLab followed suit with competitive numbers. Diving deeper into the FCN model categories, the traditional CNN and its 3D Residual counterpart both stabilized at a DSC of 30%. In contrast, the Adversarial FCN displayed a commendable 40.1%. A notable mention goes to the CNN-Transformer hybrid, which topped the charts with a DSC of 85.6%, suggesting the potential synergy between convolutional and transformer architectures (illustrated in Fig. 4). In essence, our findings reflect a tapestry of innovation, ambition, and tangible progress within the deep learning techniques applied to medical imaging over the years.

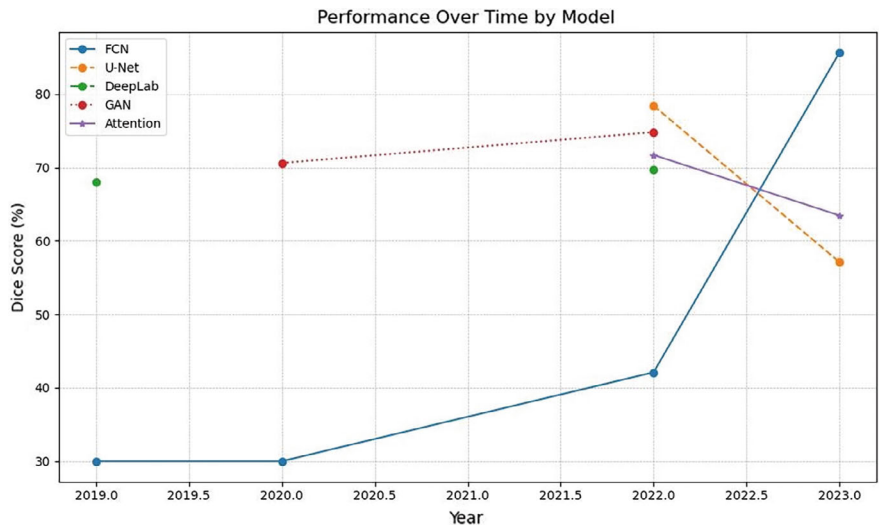


Fig. 2 Performance over time by different models for ischemic stroke lesion segmentation

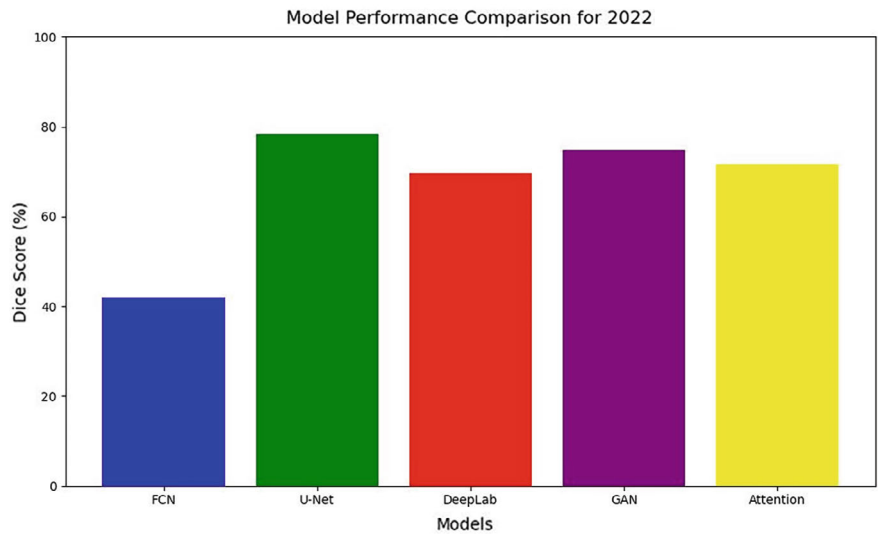
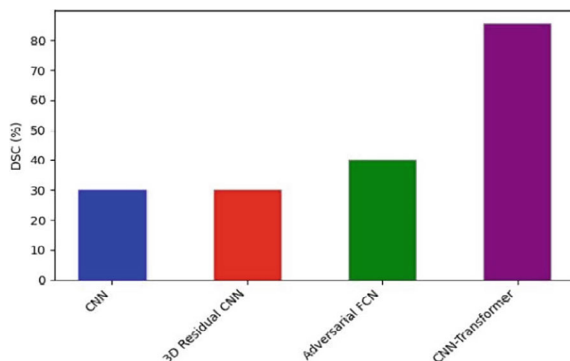


Fig. 3 Performance for 2022 by different models for ischemic stroke lesion segmentation

Fig. 4 Performance of different FCN models for ischemic stroke lesion segmentation



3.2 Review the Lesion Segmentation of Hemorrhage Stroke Using Deep Learning from Architecture View

3.2.1 FCN

Have been employed for the segmentation of cerebral hemorrhages. In 2022, Renan Sales Barros and team [21] introduced a convolution-based network for identifying and delineating subarachnoid bleeding in NCCT. The results show strong agreement between the automatically determined hemorrhage volume and humanly verified ground truth (Intraclass Correlation Coefficient (ICC) .97), as well as high accuracy in distinguishing patients with subarachnoid hemorrhage from those with acute ischemic stroke (accuracy: .96). This method showcases the effectiveness of using CNN for hemorrhage segmentation due to its strong performance. However, as with many deep learning models, one of the limitations could be the necessity for large, well-annotated datasets to train the model, and its effectiveness might decrease when applied to data from different distributions. In 2022, Kai Hu et al. [22] proposed an encoder-decoder CNN (ED-Net) for automated segmentation of intracerebral hemorrhage. The encoder captures multi-scale feature data, while the decoder integrates these features. The ED-Net shows impressive results with Dice coefficient as .887 indicating that it performs better than many competing approaches. One of the advantages of this method is the ability of the ED-Net to extract and integrate multi-scale features, which potentially allows for more robust and precise segmentation. However, similar to the CNN model, the ED-Net may necessitate an extensive set of training data, and its efficacy may be dependent on the data distribution. Comparing the two models, both utilize the power of convolutional layers in extracting meaningful features from CT images for segmentation tasks. The CNN model proposed by Renan Sales Barros et al. offers the advantage of high accuracy in both segmentation and differentiation tasks, whereas the ED-Net model proposed by Kai Hu et al. [22] stands out due to its robust performance metrics and ability to integrate multi-scale features. For a concise summary of the various FCN methods discussed and their corresponding performance metrics, please refer to Table 6.

Table 6 Review of hemorrhage lesion Segmentation using FCN

Author	Image modality	Method	year	Outcome / Characteristics
Renan Sales Barros et al. [21]	CT	CNN	2022	Dice coefficient: 0.63 ± 0.16
Kai Hu et al. [22]	CT	Encoder–Decoder CNN. (ED-Net)	2022	Dice = .887, Pre = .8857, Jac = .8947 Sen = .9999, Spe = .7989

3.2.2 U-Net

Is a highly effective convolution-based network structure frequently employed in the process of delineating lesions in cases of hemorrhage. It is especially adept at handling biomedical image partitioning owing to its proficiency in discerning both localized and widespread information of images, resulting in high precision output. The architecture of U-Net, including its expansive path, allows for precise localization making it well-suited for tasks such as hemorrhage lesion segmentation. Murtadha D. Hssayeni et al. [23] in 2022 employed the deep FCN, more specifically U-Net, for Intracranial Hemorrhage Segmentation, achieving a good performance with a Dice coefficient of .3 using a 5-fold cross-evaluation method. The sturdiness and dependability of U-Net in handling image segmentation tasks are clearly demonstrated here. However, the primary limitation of this model could be the comparatively lower Dice coefficient, indicating possible room for improvement in segmentation accuracy. In 2022, Xizhi He et al. [24] Hybridized Multi-scale Object Equalization Framework (HMOE-Net) developed a network combining shallow feature-based networks with deep feature-based networks. This model aids in the equalization learning of hybridized multi-scale feature descriptors and more thorough exploration of multi-level semantic features, reflected by impressive performance metrics—Dice coefficient of .856, Precision of .8839, Jaccard index of .762, and Sensitivity of .8424. The advantage of this approach lies in its ability to capture and learn from both shallow and deep feature descriptors, enhancing the model’s capacity to handle complex segmentation tasks. However, this might also lead to increased model complexity and computational requirements. Nannan Yu et al. [25] in 2022 (CT, Dimension Reduction U-Net) presented a model for Hematoma Volumetric Detection in Intracerebral Hemorrhage, achieving a Dice score of $.86 \pm .39$. This approach offers robust deep learning segmentation comparable to expert physicians in accuracy and efficiency. By using a dimension reduction technique, this model could potentially handle high-dimensional data more effectively, reducing computational costs. However, the model might be limited by the potential loss of information during dimensionality reduction. Manjin Sheng et al. [26] (MRI, Cross-Attention and Deep U-Net, 2022) created a model for stroke lesion segmentation, which attained smoother edge results and precise segmentation values—Dice coefficient .5564, global Dice coefficient .745, Recall .587, and Precision .6368. By incorporating a cross-attention mechanism, this approach could focus on more relevant features, potentially improving the segmentation quality. However, as with other deep learning models, the performance might be

Table 7 Review of hemorrhage lesion Segmentation using UNET

Author	Image modality	Method	Outcome /Characteristics	Year
Murtadh a D. Hssayeni et al. [23]	CT	Deep Fully Convolutional Networks U-Net,	DC = .3	2022
Xizhi He et al. [24]	CT	(SFENet) DFENet (MLFE) blocks HMOE-Net Hybridized Multi-scale Object Equalization Network Combining shallow feature-based networks with deep feature-based networks	Dice.856, Pre .8839, Jac .762 Sen .8424	2022
Nannan Yu et al. [25]	CT	Dimension reduction U-Net system	Dice score = .86 ± .39	2022
Manjin Sheng et al. [26]	MRI	Cross attention and deep U-Net	DSC .5564 DSC (global) .745 Recall .587 Precision.6368	2022

reliant on the caliber and volume of training data. In comparing these models, they all leverage U-Net’s strong capabilities in image segmentation, while integrating other methods to enhance their performance. The specific choice of model would depend on the requirements of the task, including the trade-off between performance, computational cost, and the nature of the available data. In terms of performance, Xizhi He et al.’s model shows the highest Dice coefficient, suggesting that it might offer the best segmentation accuracy among these methods, though its performance might vary depending on the specific dataset used. For a concise summary of the various UNET methods discussed and their corresponding performance metrics, please refer to Table 7.

3.2.3 Attention models

The attention models have garnered considerable interest in the field of hemorrhage segmentation due to their capability to focus on crucial parts of an image, thereby enhancing model performance.HU Min et al. [27] introduced an Attention enhanced U-shaped neural Network plus (AU-Net+) in 2022, which integrates a Residual Octave Convolution (ROC) block. The AU-Net+ displayed impressive performance. It combines an attention mechanism with a convolutional block, allowing the framework to prioritize attention on relevant areas of the image while discarding less informative ones. This results in an improved performance, denoted by a 2.9% increase in the mIoU index. However, despite its robustness, the added complexity of the ROC block may introduce higher computational costs, a limitation that could impact its utility in resource-limited settings. Manjin Sheng et al. [26] proposed a Cross-Focus mechanism paired with an Enhanced U-Net for delineating stroke lesions in

2022. The model was found to be effective, with a DSC of .5564 and a global DSC of .745. These metrics, while reasonable, suggest room for further improvement. The model's limitations may become apparent in more complex segmentation tasks, given its comparatively lower performance metrics. Finally, Zhuo Kuang et al. [28] proposed a multi-level training approach incorporating two attention blocks in their 2022 study. This approach yielded a high Dice score of .95, underlining the power of attention mechanisms in enhancing feature recognition and suppression of less relevant areas, thereby reducing overfitting. Despite its high accuracy, the multi-level training approach is complex and may require more computational resources, potentially limiting its usability in some contexts. In a 2023 study by Bohao Xu et al. [29], the CHSNet was introduced, an advanced convolutional neural network tailored to the intricacies of acute cerebral hemorrhage imaging. This model is structured with an encoding-decoding backbone and further enhanced with the Res-RCL unit, combined with Atrous Spatial Pyramid Pooling, paired with the use of an Attention Gate. Its tailored design proves effective when applied to cranial CT images, with standout performance metrics recorded on different test sets: test1 displayed a Dice metric of 0.918. In contrast, test2 yielded metrics with Dice at 0.716. This was conducted on 2D information., a noteworthy feature of CHSNet is its ability to produce 3D visualizations post-segmentation, providing an immersive view of segmented acute cerebral hemorrhage areas. For a concise summary of the various Attention methods discussed and their corresponding performance metrics, please refer to Table 8.

In summary, the attention models have shown promising results in hemorrhage segmentation by focusing on important regions of images. The integration of attention models with architectures such as U-Net has led to performance improvements. However, the added complexity and higher computational costs associated with these models constitute significant challenges. Future research could aim to optimize these models to maintain high accuracy while reducing computational demands, ensuring their wider adoption across various healthcare settings.

3.2.4 Deeplab

Several deep learning models employing the DeepLab architecture have been proposed for hemorrhage segmentation, with a particular focus on multi-scale feature extraction and the use of advanced convolutional layers for improved performance. In 2022, Liangliang Liu et al. [30] introduced an advanced convolution-based framework, named M2DCNN. This model is composed of two subnets and leverages a new architecture with advanced multi-scale features. These include a dense unit and dilated unit that help to grasp both regional and comprehensive attributes of the hemorrhages. While the model achieved good performance, the complexity of its dual-subnet structure could be seen as a limitation when considering computational resources and the ease of implementation. Lastly, in 2022, Xiuquan Du et al. [31] introduced an Attention Driven Multi-scale model. This model incorporates a coarse-grained patched attention unit and a unique cross-dimensional feature fusion unit to combine characteristics adaptively from different dimensions. The model

Table 8 Review of hemorrhage lesion Segmentation using Attention models

Author	Modality	Method	Outcome / Characteristics	Year
ZhuoKuang et al. [28]	CT	A CNN-based architecture integrating a self-attention unit combined with a context-aware attention module intended to refine hemorrhage boundary segmentations	Dice of .95	2022
Manjin Sheng et al. [26]	MR	Cross-Attention and Deep Supervision U-Net (CADS-U-Net)	DSC .5564 DSC (global).745 Recall .587	2022
HU Min [27]	CT	Attention enhanced U-shaped neural Network plus (AU-Net +) Integrates a Residual Octave Convolution (ROC) block with an attention mechanism	Dice coefficient-.92	2022
Bohao Xu [29]	CT	A convolutional neural network with encoding–decoding backbone, Res-RCL module, Atrous Spatial Pyramid Pooling, and Attention Gate tailored for acute cerebral hemorrhage image characteristics CHSNet	Dice score of 0.950 on spontaneous ICH and 0.895 on traumatic ICH. Dice = 0.918	2022

also leverages a multi-scale deconvolution upsampling (MDU) module to recover target information lost during pooling. The proposed method outperforms existing techniques with a DSC measured at .594, HD recorded at 27.5 mm, and ASSD valued at 7.37 mm. Despite its innovative design, the multiple components could make the model more complex and harder to implement in real-world scenarios. These DeepLab-based models show the potential of multi-scale feature extraction and innovative convolutional layers in achieving impressive hemorrhage segmentation performance. However, the complexity of these models, especially in terms of computational resources and implementation ease, remains a challenge. Future work should consider balancing performance with computational efficiency for broader applicability. For a concise summary of the various DeepLab methods discussed and their corresponding performance metrics, please refer to Table 9.

Table 9 Review of hemorrhage lesion Segmentation using Deeplab

Author	Image modality	Method	Outcome / Characteristics	Year
Liangliang Liu. [30]	MRI	Deep convolutional neural network with two subnets utilizing multi-scale features and an architecture combining dense and dilated blocks. M2DCNN,	DC- 82.91	2022
Xiuquan Du et al. [31]	MRI	Coarse-grained patched attention unit, cross-dimensional feature fusion unit, multi-scale deconvolution upsampling (MDU) module Attention Driven Multi-scale model	DSC value stands at 0.594	2022

3.2.5 GAN

Guogang Cao et al. [32] introduced a U-Net framework for cerebral bleed delineation in 2022. The approach first used soft partitioning clustering algorithms (Fuzzy C-means) to segregate datapoints into distinct categories: white matter, grey matter, cerebral spinal fluid, and hemorrhage. Morphological image analysis was then applied to exclude the skull. This method displayed promising results with a Dice Score Coefficient (DSC) of $.86 \pm .3$, showing the potential of integrating clustering techniques with deep learning for enhanced segmentation performance. However, the two-step approach could be computationally expensive. Henk van Voorst et al. [33] developed a GAN model in 2022 aimed at converting follow-up scans with abnormal regions into baseline scans without abnormalities and generating a corresponding difference map. Although the model achieved good volumetric correspondence, it only achieved a modest Dice similarity coefficient, indicating that the method may not be optimal for segmenting intricate hemorrhagic lesions. Wenshan Wu et al. [34] proposed an integrated method combining a conditional GAN for data augmentation with a CNN for supervised segmentation. The model outperformed several advanced techniques in segmenting stroke lesions with a Dice measurement of .5496 and an IoU value of .4383. This study highlighted the potential of combining data augmentation with traditional supervised learning techniques, although the relatively lower IoU score points to potential inaccuracies in the segmentation boundaries. Lastly, Manohar Karki et al. [35] introduced a novel approach, Lesion-focused Generative Adversarial Framework (LcGAN), to generate synthetic CT scans to enhance the

training data in 2022. Applying this augmentation method led to improved segmentation accuracies with a precision of .85 and a recall of .82. This study underlined the potential of GANs in data augmentation, although synthesizing realistic medical images remains a challenging task. In summary, the use of GANs in hemorrhage lesion segmentation offers promising results. Yet, the prevailing benchmark still requires more improvements, particularly in the quality of synthetic images and the accuracy of complex lesion segmentations. Future work might need to focus on refining these techniques to improve the performance of these models. For a concise summary of the various GAN methods discussed and their corresponding performance metrics, please refer to Table 10.

As depicted in Fig. 5, the comparative outcomes of diverse algorithms spanning 2022 and 2023 highlight intriguing dynamics. In 2022, the FCN model clocked a DSC of 75.85%, yet its performance metrics for the following year remain uncharted. Conversely, the models rooted in Attention mechanisms witnessed a surge in proficiency, climbing from a DSC of 80.67% in 2022 to an impressive 91.93% in 2023. Such a trajectory underscores the refinements in Attention-centric techniques over this period. For the year in question, models such as U-Net, DeepLab, and GAN have their performance snapshots captured exclusively for 2022. Zooming into 2022’s metrics, Fig. 6’s bar diagram illustrates the dominance of the Attention-oriented model, registering a DSC of 80.67%, closely followed by FCN at 75.85%. In the same vein, the U-Net model’s performance, marked at 64.25%, suggests a discrepancy in efficiency among the varying algorithms, underscoring the superior capabilities of Attention-driven systems in segmenting hemorrhage strokes. In Fig. 7 a juxtaposition of diverse model variants across the two years unfolds distinct patterns. The Double-Attention CNN in 2022 notched a noteworthy DSC of 95%, shadowed by the ROC-Attention framework at 92%. It’s worth mentioning that the ASPP-Attention CNN,

Table 10 Review of hemorrhage lesion Segmentation using GAN

Author	Image modality	Method	Outcome / Characteristics	Year
Guogang Cao et al.. [32]	CT	fuzzy C-means clustering method for pixel clustering, skull removal by morphological image method, followed by an improved U-Net neural network model	Dice Score Coefficient (DSC) of .86 ± .3	2022
Henk van Voorst et al. [33]	CT	Generative Adversarial Network for Stroke Lesion Segmentation on NCCT	Dice similarity coefficient of 0.59	2022
Wensha n Wu et al. [34]	CT	Method using a cGAN for data augmentation followed by a CNN incorporating depth-specific convolution, X- blocks and the FSM handle feature resemblance	Dice score of .5496, IoU of .4383	2022
Manoha r Karki et al. [35]	CT	Lesion conditional Generative Adversarial Network (LcGAN)	Precision of .85, Recall of .82	2022

unobserved in 2022, charted a DSC of 91.93% in the subsequent year. The chosen hatch designs for the graphical bars aptly differentiate the annual performance data, ensuring streamlined comparative scrutiny. The graphical representations underscore the ascendant trajectory of Attention-fueled algorithms, showcasing their potential in the realm of hemorrhage stroke segmentation. As advancements in this domain gather pace, it will be compelling to track the continuance of these patterns and the potential introduction and impact of avant-garde techniques on stroke segmentation paradigms.

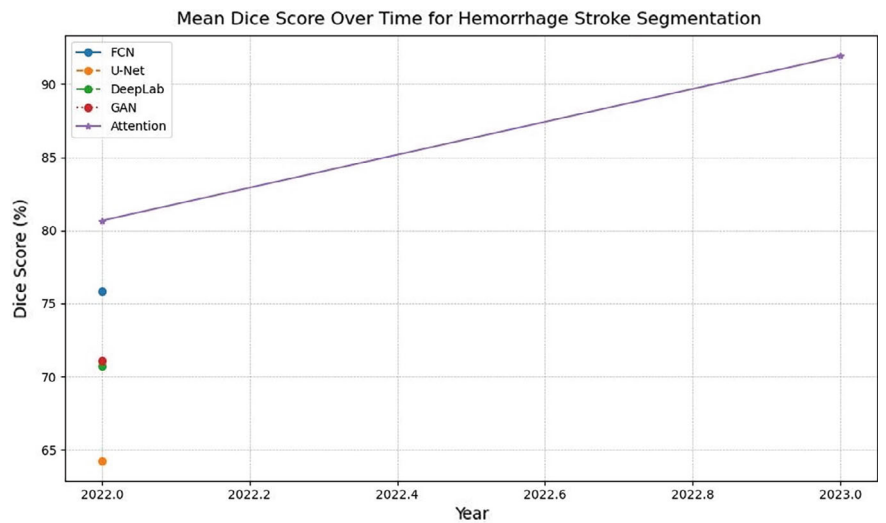


Fig. 5 Performance over time by different models for hemorrhage stroke lesion segmentation

Fig. 6 Performance for 2022 by different models for hemorrhage stroke lesion segmentation

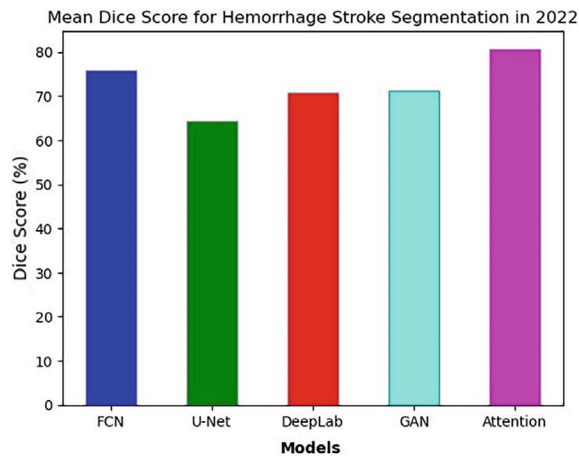
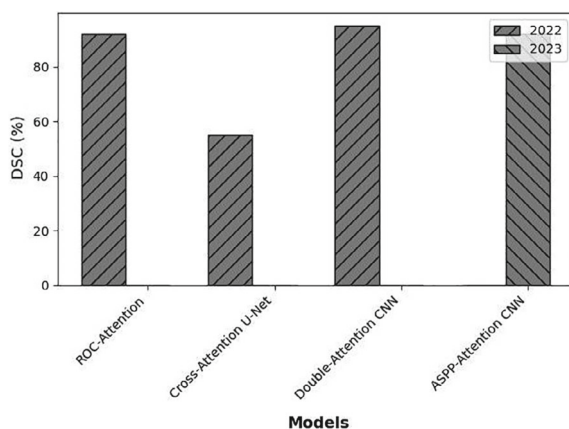


Fig. 7 Performance of different Attention models for hemorrhage stroke lesion segmentation



4 Discussion

This review undertook the task of understanding the evolving role of neural techniques in automating the delineation of ischemic and hemorrhagic cerebral damages. A meticulous examination of various deep learning models, such as U-Net, DeepLab, Adversarial setups, FCN, and Attention Models, was presented, offering insights into their architectures, performance trajectories, strengths, and potential limitations. Our analyses of the graphical representations reflect a dynamic progression of these models over the years. While some, like the FCN, showcased a transformative journey with significant performance enhancement, others like the U-Net presented a complex trajectory marked by initial promise followed by a decline. It was particularly intriguing to observe the synergy between convolutional and transformer architectures, as reflected by the impressive performance of the CNN-Transformer hybrid. When focusing on hemorrhagic stroke lesion segmentation, the ascendancy of Attention-driven algorithms was unmistakable. Their potential in selectively focusing on regions of interest paints a promising picture for future developments in the realm of hemorrhage stroke segmentation. The comparative study across the years revealed distinct patterns, underscoring the attention mechanism's potential and raising questions about other architectures' evolving roles. However, while celebrating these advancements, it's imperative to acknowledge the challenges that persist. Data remains a pivotal factor. The requirement for vast, balanced, and readily accessible training data is a continual challenge, dictating the model's efficacy and its ability to generalize across varied scenarios. Furthermore, as the results showed, not all advancements necessarily lead to improved outcomes, emphasizing the need for continuous refinement and innovation in the field.

5 Conclusion

In synthesizing our findings, this review underlines the transformative potential of advanced neural computing in cerebral damage segmentation. From the meteoric rise of some architectures to the nuanced performances of others, it's evident that the world of medical imaging, specifically in stroke diagnosis, is on the cusp of significant innovations. These models, with their ever-improving capabilities, stand poised to greatly assist medical professionals, thereby enhancing patient outcomes. The journey of these models, as captured over the years, provides a foundation for future research. As we anticipate further advancements and innovations, the confluence of larger datasets, hybrid deep learning models, and multimodal imaging data will be instrumental in redefining the limits of what's achievable in stroke diagnosis and treatment planning. The future, illuminated by the lessons of the past and the innovations of the present, holds immense promise for this domain.

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Design of Lightweight RNN for Human Activity Recognition on KU-HAR Dataset



Ashish Girdhar, Ramesh Kait, and Himani Kapur

Contents

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Abstract With the world's aged population increasing rapidly, an earnest need for recognition of human activities is sought after by the Healthcare units across the world. Applications of Human Activity Recognition include surveillance, fitness among various others. This manuscript briefly discusses the approaches followed for recognising human activities in the literature. Additionally, in the present work a lightweight RNN (recurrent neural network) model having LSTM (long short term memory) cells is proposed for HAR on the dataset. This lightweight RNN model employs a feature extraction task. Feature extraction is done in three domains—Temporal, Spatial, and Spectral domains. Recently published dataset KU-HAR is used for the presented work and the proposed work performs fairly well over the dataset.

Keywords Human activity · Gesture · Activity recognition · RNN · CNN

A. Girdhar (✉) · R. Kait

Department of Computer Science and Applications, Kurukshetra University, Kurukshetra, India

e-mail: ashishgirdhar@kuk.ac.in

H. Kapur

GGSSS Kaithal, Kaithal, India

1 Introduction

Human Activity Recognition has enticed the researchers over the decade and continues to do so till today. Ever since the conception of IoT and deep learning paradigms, the recognition of human activities has garnered much attention. Activity Recognition is termed as the ability to detect or recognise an activity using the information given by the sensors [1]. Human Activity Recognition can be divided into four main categories as discussed in [2]

- **Gesture:** This involves changes in hands, face or other parts of movement
- **Action:** This involves changes in a complete movement e.g. walking, running
- **Interaction:** This involves changes brought by two actors, where the second actor can be an object or a person
- **Group activity:** This involves a combination of gestures, actions, and/or interactions.

Human Activity Recognition has a plethora of applications some of which have been explained at length in [1]. World's population is ageing at a fast pace and as per a current research an ageing tsunami is about to come in the most populous country [3]. Thus, a large population seeks medical care and attention which can cause an overwhelming burden on the youth. HAR emerges as the solution and can effectively assist in elderly care by reporting their fall. When the restrictions due to Covid19 disease were lifted, reports of an upsurge in thefts and robberies around the world started making rounds. Hence, 24×7 surveillance and detection of suspicious human activities around the clock has become indispensable. HAR can contribute to this area directly and report such activities. Also, when the world was made to sit indoors during widespread of Covid19 disease[4], HAR can be used to track people's activity and violations can be reported. Other areas where HAR facilitates human life experience include Human Computer Interaction, improving navigation inside buildings, shopping help, among many others. To sum up, HAR works wondrous in various interdisciplinary fields and hence HAR is a boon in disguise for humanity.

The proposed manuscript proposes a novel approach to Human Activity recognition on a most recently published publicly available dataset discussed in detail in the paper. Section 2 does a brief literature review of the work done so far in the area, alongwith a brief discussion on the publicly available datasets used in the proposed approach. In Sect. 3, the dataset is discussed in detail. Afterwards, the feature extraction process is discussed in Sect. 4. In Sect. 5, classification and simulation of the proposed approach are discussed followed by a conclusion and brief discussion in Sect. 6.

2 Literature Review of HAR

Progress in a field is marked by the surveys conducted so far. So, in the first step, survey papers were thoroughly read and progress made in the field of HAR was marked. Research in HAR which began in 1980s was first associated with neural networks as early as in 1997 [5] where Association Map (AM) was constructed in order to design the neural network. Some of the major survey works of the last decade and their contributions in this area have been discussed in Table 1.

Some of the major take-aways from this brief literature review:

1. RNN and CNN are winners
2. The task of action analysis requires the processing of temporal information which is basically what recurrent neural networks (RNN) are designed for. So, clearly RNN can be effectively used for HAR task. Convolutional neural network (CNN) helps design and build the basic neural network classification model for HAR. Some activation functions such as Softmax and other fusions of CNN have yielded some breakthrough results in the field of HAR.
3. Dimensionality reduction
4. Dimensionality reduction in HAR gives rise to two much-needed advantages: reduced computational complexity and low power consumption [22]. Thus, dimensionality reduction is more a necessity than a choice.
5. Publicly available datasets
6. In the field of HAR, many publicly available datasets are used as mentioned in Table 2. Of these, the most widely used public dataset is UCI-HAR dataset as can be observed in Fig. 1. Conspicuously, very few research works have been done KU-HAR [25] dataset released in 2021. Statistics mentioned in Fig. 1 have been obtained by taking into account the mention of publicly available dataset in various research works in the field of HAR.

For this research work, KU-HAR is used over the much popular UCI-HAR because of several reasons as jotted under:

1. UCI-HAR had only 30 participants; KU-HAR has 90.
2. UCI-HAR covers 6 activities only which are Walking, Walking upstairs, Walking downstairs, Sitting, Standing, and laying. KU-HAR covers 18 activities such as Stand, Sit, Talk-sit, Talk-stand, Stand-sit, Lay, Lay-stand, Pick, Jump, Push-up, Sit-up, Walk, Walk-backward, Walk-circle, Run, Stair-up, Stair-down, Table-tennis
3. A total of 10,299 sample points is present in UCI-HAR which are less than half of the total sample points present in KU-HAR; which has a total of 20,750 samples.
4. Apart from a brief literature review of research works done in the field of HAR, major contribution of this paper is working on a recently developed public dataset KU-HAR and using RNN with LSTM cells for classification of human activities.

Table 1 Survey works done in HAR field

S. no	Survey	Year of publication	Contributions
1	Ref No. [6]	2013	Analysis of research works based on • Dynamic Time Warping approaches • Generative models (Hidden Markov Models, Dynamic Bayesian Network) • Discriminative Models(Support Vector Machine, Relevance Vector Machine) • Binary Trees, K Nearest neighbour
2	Ref No. [7]	2013	Analysis of transfer learning-based approaches in 4 categories: • Instance transfer • Feature representation transfer • Parameter transfer • Relational-knowledge transfer
3	Ref No. [8]	2014	HAR from 3D data by recognition using • 3D Silhouettes • Body parts or skeletal joints tracking • Local spatio-temporal features • Local 3D Occupancy features • 3D Optical Flow
4	Ref No. [9]	2014	Review of 32 research works done from 2011–2014 into three main categories • RGB Camera Based • Depth Sensor based • Wearable Sensor based Authors deduced that the first category has dropped while the second and third have gained popularity
5	Ref No. [10]	2015	Categorisation of HAR methods into two types • Unimodal methods such as space–time, stochastic, rule-based and shape-based • Multimodal methods [11] such as affective, behavioural and social-networking
6	Ref No. [12]	2018	Datasets used for HAR approaches' simulation were studied • VanKasteren dataset [13] • CASAS Kyoto [14] • CASAS Aruba [15] • CASAS Multiresident [16] • UCI-HAR [17] • Opportunity dataset [18] • mHealth [19]
7	Ref No. [20]	2019	• Machine learning techniques (Decision trees, KNN, SVM) • Hidden Markov Models • Neural Network (ANN, CNN, RNN based LSTM)

(continued)

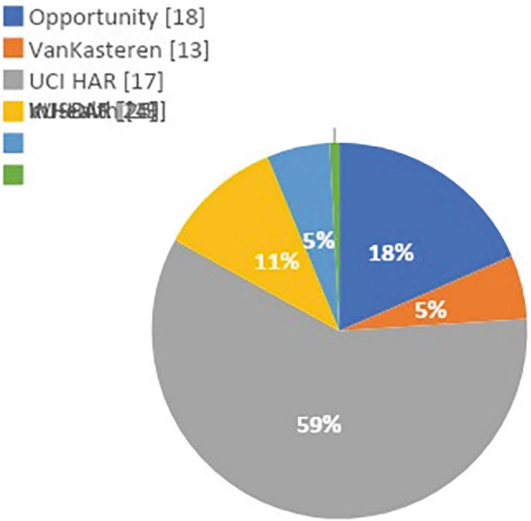
Table 1 (continued)

S. no	Survey	Year of publication	Contributions
8	Ref No. [1]	2020	• Four main categories of different approaches o Radio frequency based o Sensor based o Wearable device based o Vision based • Discussion on applications, challenges, and future scope of HAR
9	Ref No. [21]	2020	• Analysis of o Machine learning and Deep learning based HAR techniques • Dimensionality reduction in HAR • Comparison of GAN, LSTM, RNN, CNN, and DBN for HAR
10	Ref No. [22]	2021	• Analysis of Deep learning techniques for automatic feature extraction • Restricted Boltzmann machine • Deep autoencoder • Sparse coding • Convolutional neural network • Recurrent neural network • Semi-supervised models vis-à-vis CNN Encoder-Decoder and CNN Ladder • Analysis of a cascaded ensemble consisting of layers composed of • Extremely Gradient Boosting Trees • Random Forest • Extremely Randomised Trees • Softmax regression • Power requirement of all the above approaches
11	Ref No. [23]	2022	• Empirical evaluation of Sequential Minimal Optimization Algorithm(SMO), Random Forest (RF), and Simple Logistic Algorithm (SL) on UCI-HAR [17] and WISDM [24] datasets • SMO performed best on UCI-HAR • RF outperformed all on WISDM datasets

Table 2 Publicly available HAR dataset arranged in chronologically increasing order

Sr. no	Dataset	Released in year
1	VanKasteren dataset [13]	2010
2	Opportunity dataset [18]	2013
3	UCI-HAR [17]	2013
4	mHealth [19]	2014
5	WISDM [24]	2019
6	KU-HAR [25]	2021

Fig. 1 Distribution of usage of HAR public datasets in literature



3 KU-HAR Dataset

KU-HAR dataset has 90 participants from the age-group of 18–34 years whose data was collected using the in-built smartphone sensors. Dataset when downloaded from the publicly available platform contains three folders, namely Raw_time_domain_data, Trimmed_interpolated_data, and Time_domain_subsamples. Of these, third folder contains the sample points that the present manuscript is based upon. Table 3 lists the activities and description of activity provided by the authors of the dataset.

A pie chart was obtained in order to study the distribution of sample points of the activities in the dataset which is shown in Fig. 2. As can be observed, sample points in the dataset are unevenly distributed, with “Stand-sit” activity having the highest number of sample points and “Walk-circle” activity having the lowest number of sample points.

After obtaining the sample points, FFT (fast Fourier transform) was applied on the dataset before processing any further. The work present in this manuscript has been simulated on Google Colab GPUs.

1. Selection of handpicked features

After the dataset is loaded, handcrafted features are picked up from the dataset using TSFEL library [26] Feature extraction is in order to remove insignificant features from the dataset; so that the model fits well. In the present work, different feature extractor methods were applied as shown pictorially in Table 4. Using features extracted in these three domains, classification task was carried out. It is worth noting here that sample points shown in the table belong to the train subset of the dataset.

After feature extraction using TSFEL library, all three arrays obtained are fed to the classification process. This feature extraction allows the RNN constructed to be

Table 3 Activities described in the KU-HAR dataset

Sr. no	Activity Id	Activity name	Brief description of activity
1	0	Stand	Standing still
2	1	Sit	Sitting still
3	2	Talk-sit	Talking while sitting
4	3	Talk-stand	Talking while standing or walking
5	4	Stand-sit	Standing and sitting and repeating
6	5	Lay	Laying down
7	6	Lay-stand	Standing up and laying down and repeating
8	7	Pick	Picking up of object
9	8	Jump	Jumping in repetition
10	9	Push-up	Push-up exercise
11	10	Sit-up	Sit-ups exercise
12	11	Walk	Walk straight for approx. 12 s
13	12	Walk-backward	Walk backwards for approx. 20 s
14	13	Walk-circle	Walking in circle
15	14	Run	Running for approx. 7 s
16	15	Stair-up	Going up the stairs
17	16	Stair-down	Going down from the stairs
18	17	Table-tennis	Table-tennis game activity

lightweight in nature. This is because in place of RNN catering to the complete dataset containing 1800 features, after feature extraction the designed RNN encounters only 18, 36, and 336 features respectively for temporal, statistical, and spectral domains.

4 Classification and Simulation

As concluded from the brief literature review presented above, for the classification of HAR problems, RNN (recurrent neural network) is the one stop solution. Basic design of a RNN can be demonstrated as under (Fig. 3).

The basic design of a RNN gives it the advantage of having the output dependent on the previous outputs. This is the inherent feature in the present problem of HAR classification. In a human activity, timestamps are very closely related to one another. For example, in “Stand-sit” activity a person would be first required to Stand and then Sit. Because of such interdependence of activities, RNN poses to be the best classification network that can be designed for the purpose.

RNN designed in the proposed manuscript has LSTM (long short term memory) cells in it. LSTM cells are chosen in order to remove a major problem that RNN faces, which is vanishing gradients.

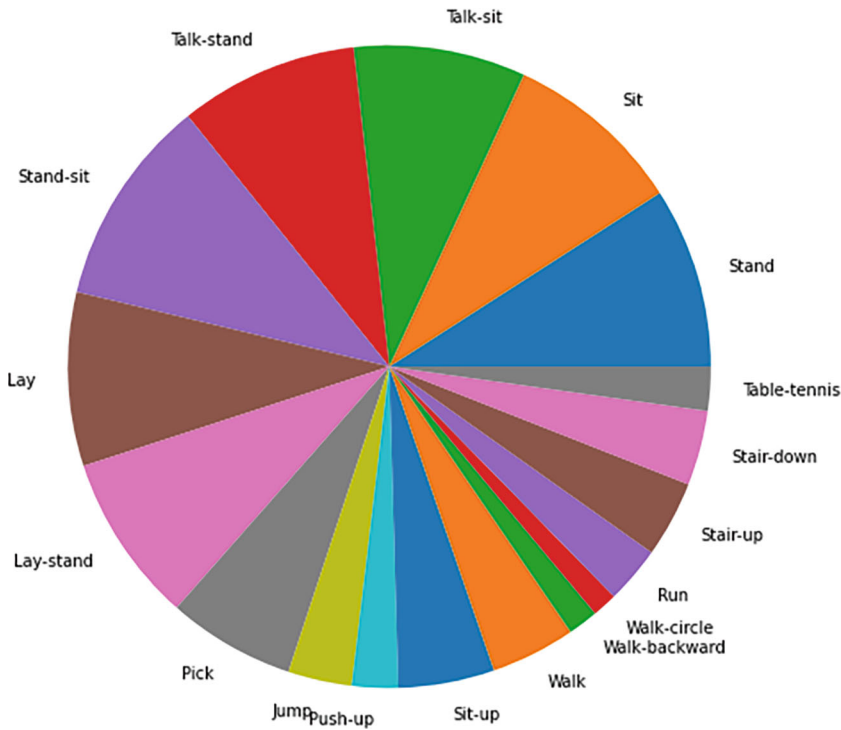


Fig. 2 Distribution of sample points among 18 activities

As feature extraction is done for the training subset, similar process is carried out for the testing subset also. Activation function selected is “relu” and a dropout layer is added. Dropout rate is fixed at 20% in order to avoid overfitting of the RNN model designed.

It is noted that the neural network designed was modified for each of the three categories of feature extracted from it. Table 5 lists the performance of RNN model when feature extraction was done in temporal, statistical, and spectral domains. As can be observed that the feature extraction done in the spectral domain yields the maximum accuracy. Simulation of the proposed approach was carried out using GPU in Google Colab.

5 Conclusion and Brief Discussion

In the proposed manuscript, recently published dataset and TSFEL libraries are used to design a lightweight Neural Network. Features extracted in Spectral domain give the highest accuracy by taking into account just about 20% of the total features. Feature extraction in the proposed manuscript was done using the recently developed

Table 4 Extraction of features in Temporal, Statistical, and Spectral domains

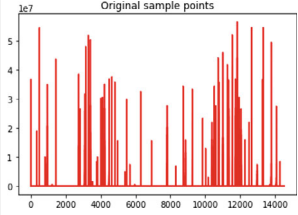
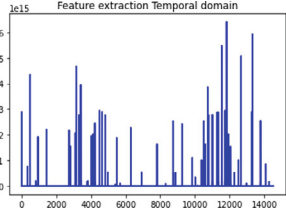
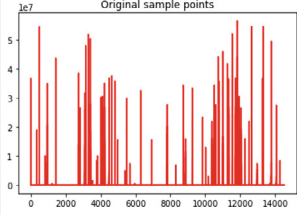
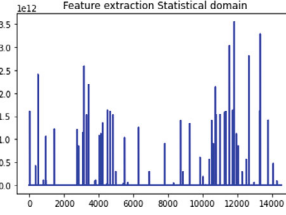
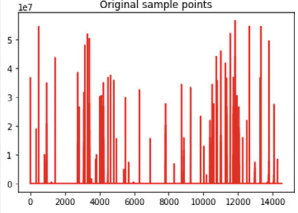
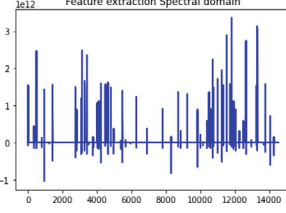
Sr. no	Domain	Original	Impact
1	Temporal		
2	Statistical		
3	Spectral		

Fig. 3 RNN Versus CNN

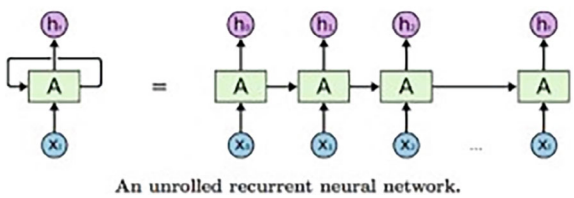


Table 5 Comparison of RNN in case of feature extraction in three domains

Sr. no	Domain	Number of features extracted	Number of iterations	Accuracy
1	Temporal	18	20	85.41
2	Statistical	36	20	88.23
3	Spectral	336	20	91.06

TSFEL library. In the present work, proposed neural network design is termed as lightweight because of having a lesser number of neurons than the neural network from which it has been derived. It is to be noted that the neural network employed in the current work is RNN and not the so-called LWN (light weight networks). Result obtained from the proposed approach will inspire the future classification tasks to use TSFEL library and design a lightweight CNN/RNN.

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Technological Helpline: An Attempt to Tackle Health Problems in Refugee Camps



Ruchika Sharma and Amritpal Kaur

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Abstract Providing healthcare for refugees has consistently posed a difficult undertaking for society. The Syrian refugee crisis has brought the problem to the forefront. The World Health Organization (WHO) devised a “Health for All” initiative aimed at guaranteeing equitable healthcare access to all refugees. Technology can have a substantial impact on delivering healthcare services to refugees. The progress in technology has facilitated the convenient availability of healthcare services, enhancing the overall health and well-being of refugees and this can be achieved by introducing blockchain-based identity cards so that the reliability toward the middlemen may be reduced and Refugees can get the maximum benefit in the camps via technology.

Keywords Technology · Blockchain · Health care · Refugee camps

1 Introduction

Artificial intelligence (AI) has the potential to facilitate ongoing health monitoring for migrants with chronic or complex health conditions, bridging the gap between doctor appointments. Health data collected by wearable devices and mobile apps can be used with artificial intelligence (AI) to identify health changes that necessitate

R. Sharma (✉) · A. Kaur

University Institute of Legal Studies, Chandigarh University, Punjab, India

e-mail: Ruchika2629@gmail.com

medical intervention. AI health monitoring systems have the potential to promptly alert refugees and their healthcare providers regarding significant changes, enabling swift interventions. Continuous AI health monitoring can offer refugees reassurance that their symptoms are being continuously checked. The potential for AI to revolutionize refugee treatment is immense. AI technologies have the potential to enhance the accuracy, accessibility, and personalization of healthcare for refugees, yet they are still considered futuristic. The potential of AI in refugee healthcare is a promising prospect that warrants attention and effort to actualize these possibilities [1]. The field of artificial intelligence (AI) is bringing about a profound transformation in the healthcare sector, with refugees seeing substantial advantages. AI enhances healthcare by facilitating the diagnosis of medical diseases, forecasting health hazards, and overseeing patients, contributing to the improvement of care and preservation of life. Although the technology is currently in its nascent stage, it is evident that AI will progressively enhance its precision, affordability, and availability in the forthcoming years. The future has arrived, employing AI to deliver refugees with the essential and rightful healthcare they require. Despite the numerous hurdles, the potential to use AI for positive purposes has never been more substantial. The present time is characterized by the advancement of AI, which is leading to a more optimistic outlook for the future [2].

Further, the count of the refugees depicted in the below diagram clearly shows that the rise in the count of numbers is an alarming situation to help refugees via technology (Fig. 1).

Fig. 1 Depicting the numbers of refugees

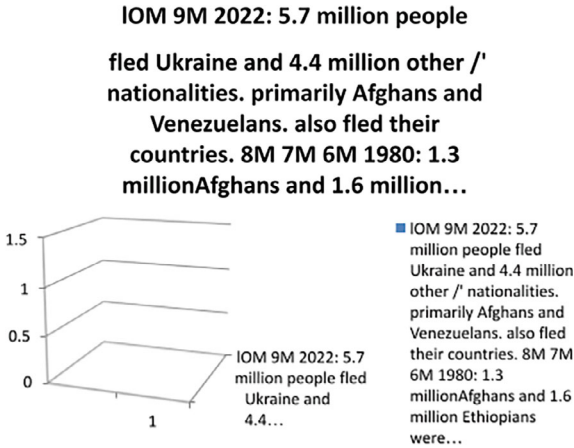




Fig. 2 Blockchain depicting unbreakable ledgers. Source: Google

2 Blockchain Technology

Refugees may face considerable difficulties in managing and storing their medical data. Healthcare providers serving refugees may encounter challenges when navigating the complex asylum-seeking bureaucracy. Blockchain technology has the potential to be utilized in the establishment of medical records specifically designed for refugees. Through the use of blockchain technology, refugees have the ability to securely keep their medical information in a decentralized environment. This allows healthcare experts from around the world to access the information [3]. The blockchain can also monitor pharmaceutical and supply networks, guaranteeing that refugees constantly obtain the medicines and supplies they require. It is imperative for governments, international humanitarian organizations, and healthcare providers across the globe to ensure the provision of high-quality healthcare services. The aforementioned methods have the potential to revolutionize the provision of healthcare to refugees, guaranteeing that all refugees have fair and equal access to healthcare services. These solutions have the potential to preserve lives, diminish healthcare expenses, and offer transformative services to refugees on a global scale. Therefore, it is imperative for the global community to promptly adopt these technical advancements in order to guarantee the provision of healthcare services for the welfare of refugees [4].

Below the picture depicts that all the ledgers in the blockchain-based identity cards are unbreakable. Henceforth, the information even breached in one ledger would not affect the other block (Fig. 2).

3 Telemedicine

Telemedicine has been conducted remotely, where healthcare professionals use telecommunications to provide medical services to patients. It includes teleconsultations, Telemonitoring, teleimaging, and telepathology. Refugee communities severely lack healthcare services, but telemedicine can provide access to healthcare services to refugees in even the most remote locations [5]. Telemedicine can save refugees from the need to travel long distances for healthcare services, saving them

both time and money. In addition, telemedicine can create opportunities for refugees to contact healthcare professionals despite language barriers. Children and women, particularly, can benefit significantly from technology-enabled telemedicine services. Several organizations have implemented telemedicine to provide healthcare services for refugees, including the Médecins Sans Frontières (MSF) [6].

Telemedicine offers several benefits for refugees. First, telemedicine eliminates geographical barriers to healthcare access, allowing refugees in remote areas to receive care. This is particularly important for refugees living in refugee camps or settlements, where access to healthcare can be limited. Second, telemedicine can reduce the cost of healthcare, as it eliminates the need for transportation and other associated costs. This is important for refugees, who often lack a stable source of income [7]. Finally, telemedicine allows for more efficient delivery of healthcare services, reducing the burden on healthcare workers and facilities.

- **Challenges of Enhanced Telemedicine Services for Refugees:**

Despite the benefits of telemedicine, several challenges are associated with its use for refugees. One key challenge is the need for more access to technology and infrastructure. Refugees may need access to smartphones or computers or a stable internet connection. In addition, refugees may need more digital literacy skills to use telemedicine services effectively. Another challenge is the language barrier. Many refugees need to speak the host country's language, making it challenging to communicate with healthcare providers. Finally, there are concerns about the privacy and security of telemedicine services and protecting personal health information.

- **Future Directions for Enhanced Telemedicine Services for Refugees:**

Despite these challenges, several ways exist to enhance telemedicine services for refugees. One potential solution is to partner with local organizations already working with refugees. These organizations can provide the necessary infrastructure and support to ensure that refugees can access telemedicine services. In addition, telemedicine providers can develop multilingual platforms and hire healthcare providers who speak the languages commonly spoken by refugees. Finally, telemedicine providers can work to ensure that their services are secure and protect patients' privacy.

Enhanced telemedicine services offer many benefits for refugees, including increased access to healthcare, reduced cost, and more efficient delivery of healthcare services. However, several challenges are associated with using telemedicine for refugees, including the need for access to technology and infrastructure, the language barrier, and concerns about privacy and security. To overcome these challenges, telemedicine providers can partner with local organizations, develop multilingual platforms, ensure their services are secure, and protect patients' privacy. With these enhancements, telemedicine can be essential for providing healthcare services to refugees.

4 IT-Based Algorithm for Health Promotion

Using an IT-based algorithm for health promotion in temporary settlements offers several benefits. Firstly, it allows for the systematic collection and analysis of health data. The algorithm can collect information about the health status of migrants and refugees, including prevalent diseases, vaccination history, and demographics. This data can then be used to identify specific health needs and priorities, enabling targeted health promotion interventions. Secondly, an IT-based algorithm allows for personalized health promotion strategies. By analyzing individual health data, the algorithm can provide tailored recommendations for healthy behaviors, screenings, and interventions, considering cultural, linguistic, and contextual factors. This personalized approach increases the likelihood of behavior change and positive health outcomes. Finally, an IT-based algorithm improves the efficiency and effectiveness of health promotion efforts by automating processes, reducing errors, and facilitating real-time monitoring and evaluation of interventions. This enables timely adjustments and evidence-based decision-making.

- **Challenges of Using an IT-Based Algorithm for Health Promotion:**

While using an IT-based algorithm for health promotion in temporary settlements holds significant promise, it also presents challenges. First, these settings may have limited access to technology and internet connectivity. Migrants and refugees may not have access to smartphones or computers, impeding their ability to interact with the algorithm. Infrastructure improvements and the provision of digital devices have become crucial to bridge this gap. Secondly, language and cultural barriers may affect the algorithm's effectiveness. Health promotion recommendations must be culturally appropriate and accessible through multiple languages, ensuring they resonate with the target population. Thirdly, concerns regarding privacy and data security arise. Data collected through the algorithm must be protected from unauthorized access and should comply with data protection regulations to maintain trust among migrants and refugees [8].

- **Future Directions for Using an IT-Based Algorithm for Health Promotion:**

Several future directions can be pursued to overcome the challenges and maximize the potential of an IT-based algorithm for health promotion in temporary settlements. Firstly, collaborations between healthcare providers, technology developers, and humanitarian organizations are crucial. These collaborations can ensure the development of user-friendly applications that work on different devices, with attention to low-resource settings. Secondly, participatory approaches should be adopted to engage migrants and refugees in designing and implementing the algorithm. This will ensure the cultural sensitivity and relevance of the health promotion strategies.

Furthermore, the algorithm should be constantly updated and refined based on user feedback and input. Additionally, capacity-building efforts can focus on equipping migrants and refugees with the necessary digital literacy skills to engage with the technology effectively. Health promotion campaigns can also be designed to

raise awareness about the algorithm's benefits and address privacy and data security concerns.

An IT-based algorithm for health promotion in temporary settlements can significantly improve migrant and refugee health. The systematic collection and analysis of health data, personalized recommendations, and increased efficiency are just a few benefits. However, technology access, language barriers, and data security challenges must be addressed. By partnering with relevant stakeholders, adopting participatory approaches, and considering cultural sensitivity, the potential benefits of the algorithm can be fully realized. Ultimately, using an IT-based algorithm for health promotion in temporary settlements can empower migrants and refugees to make informed decisions about their health, leading to improved health outcomes and well-being [9].

5 Mobile Health (mHealth)

Mobile health (mHealth) refers to using mobile technology, such as smartphones and tablets, to support healthcare services and improve health outcomes. For refugees living in camps, mHealth holds great potential in addressing their healthcare needs, including gynecological health issues faced by women. This essay will explore how mHealth can help refugees in refugee camps, focusing on improving women's gynecological health. It will discuss the benefits of mHealth, the specific challenges faced by refugee women, and potential solutions and future directions for implementing mHealth in this context.

Benefits of mHealth for Refugees in Refugee Camps:

Increased access to healthcare: mHealth can bridge the gap in access to healthcare services for refugees living in camps. Through mobile devices, women can connect with healthcare providers remotely, eliminating the need for long and often challenging journeys to healthcare facilities. **Health education and awareness:** mHealth applications can provide women with access to educational resources and health information related to gynecological health. They can learn about preventive measures, symptoms of common conditions, and self-care practices, empowering them to make informed decisions about their health [10].

Remote consultations and telemedicine: With mHealth, women in refugee camps can have virtual consultations with healthcare professionals. This allows for timely diagnosis, treatment, and management of gynecological health issues without the need for physical visits, reducing barriers related to transportation and cost.

Appointment reminders and medication adherence: Mobile applications can send appointment reminders and medication notifications to ensure that women in refugee camps stay on top of their gynecological health appointments and adhere to prescribed treatments, promoting better health outcomes [11].

Challenges Faced by Refugee Women in Gynecological Health: Refugee women face unique challenges concerning gynecological health in refugee camps. These challenges include the following

- **Limited Access to Healthcare Services:** Refugee camps often lack sufficient healthcare facilities and personnel, leading to limited access to gynecological health services, including screenings, consultations, and treatments.
- **Stigmatization and Cultural Barriers:** Cultural norms and stigmatization often prevent refugee women from seeking gynecological healthcare. There may be taboos surrounding reproductive health, making it difficult for women to openly discuss their concerns.
- **Language and Communication Barriers:** Communication can be a significant challenge for refugee women who do not speak the local language. This can impede their ability to explain their symptoms and understand the healthcare information provided to them.
- **Lack of Privacy and Confidentiality:** The crowded living conditions in refugee camps may compromise women's privacy and hinder their willingness to seek gynecological healthcare or discuss sensitive issues [12].

Solutions and Future Directions for mHealth in Gynecological Health:

To effectively address the gynecological health needs of refugee women in camps, the following solutions and future directions can be considered:

- **Ensuring mobile device access:** Efforts should be made to provide refugees with access to mobile devices and internet connectivity, either through community centers or partnerships with mobile network providers.
- **Culturally sensitive applications:** mHealth applications should be developed with an understanding of cultural norms and sensitivities to address the unique challenges faced by refugee women. These applications should be available in multiple languages, including those commonly spoken by refugees.
- **Community outreach and education:** Engaging community leaders and healthcare workers can help raise awareness and provide culturally appropriate education about gynecological health. Workshops, support groups, and awareness campaigns can help overcome stigma and encourage women to seek care.
- **Collaboration and partnerships:** Collaboration among healthcare providers, humanitarian organizations, and technology developers is essential for the successful implementation of mHealth initiatives. These partnerships can ensure the development of effective and user-friendly applications, as well as the provision of necessary support and training for both healthcare providers and refugees.

6 HERA (Hope and Emergency Response App) Mobile Application

The refugee crisis is one of our most significant humanitarian issues. According to the UN Refugee Agency, over 79.5 million displaced people today are 1% of the world's entire population.

Countless families have been torn apart due to violence, war, persecution, and human rights violations in their home countries. They embark on perilous journeys in hopes of finding safety and asylum elsewhere. However, refugee camps are overcrowded and under-resourced. There are lengthy application processes and restrictions on work permits.

This is where HERA comes in. HERA, which stands for Hope and Emergency Response App, is a digital tool created by Anthropic to help streamline the refugee application process and connect displaced families with essential resources.

Using HERA, refugees can [13]:

- Submit asylum applications and track their status in real time. No more endless waiting without updates.
- Connect with local aid organizations that can provide food, shelter, medical care, education, job opportunities, and more. Help is now just a tap away.
- Reunite with separated family members by posting photos and details about loved ones. HERA uses facial recognition to help match people.
- Receive emergency alerts about events that could impact safety, like natural disasters, disease outbreaks, or security risks in camps and surrounding areas. Forewarned is forearmed.

HERA gives refugees back some control and dignity in otherwise dire circumstances. While far from a perfect solution, it aims to alleviate at least some of the hardships for the world's most vulnerable people. Every little bit helps.

Introducing the HERA Application

The HERA app is a game changer for refugees. This innovative technology provides resources and assistance for some of the world's most vulnerable people. HERA, which stands for Help, Empower, and Refugee Assistance, is a mobile app designed for refugees and asylum seekers. Once you've downloaded the app, you can:

- Access critical information about the asylum process, your rights, and available services in multiple languages. The details are tailored to your specific location and situation.
- Connect directly with local refugee organizations and service providers. You can schedule appointments, ask questions, and get referrals to resources like shelter, healthcare, education, and job opportunities.
- Share details about your case with approved nonprofit partners so they can offer guidance and support. Your information is kept strictly confidential and used only to assist you.

- Receive targeted alerts about important deadlines, events, and policy changes that affect you. So you never miss an update that could impact your legal status or access to aid.
- Provide anonymous feedback about your experience using the app. Your input helps improve HERA and better serve refugees in the future.

HERA puts essential information and tools into the hands of refugees when they need it most. The app helps guide families through the complicated asylum process, connects them with humanitarian resources, and gives them a voice to improve the system. For refugees, HERA means hope, empowerment, and a helping hand during a difficult journey. HERA amplifies the mission's impact on the organizations that serve them. It's a win-win made possible by technology and compassion [14].

6.1 Key Features of the HERA App

The HERA app aims to provide refugees with valuable resources and information to help them transition to their new homes. Some of the key features of the HERA app include:

Translation Services

HERA offers instant translation between multiple languages, including Arabic, English, French, Spanish, and German. This allows refugees to communicate with aid workers, government officials, and local community members. HERA's translation services break down language barriers and help refugees navigate daily tasks like going to the doctor, shopping for groceries, and applying for jobs.

Location-Based Resources

HERA uses a refugee's location to provide information on nearby resources like food banks, healthcare clinics, education opportunities, and places of worship that cater to their specific cultural and religious needs. The app maps out the locations of these resources and provides walking or driving directions to help refugees access them.

Alerts and Updates

HERA pushes notifications to refugees about important news and events in their area. These alerts include information on local job fairs, benefits program deadlines, school registration dates for their children, and announcements from refugee resettlement organizations. The signs help ensure that refugees have the latest information to transition to their new community successfully.

Community Forum

The HERA app offers a community forum where refugees can connect with others from similar cultural and linguistic backgrounds. The platform allows refugees to ask questions, exchange information, and provide social support to one another.

Moderators also share news, events, and resources in the community forum to help refugees stay up-to-date and engaged.

The HERA app aims to ease refugees' transition into new communities through its translation services, location-based resources, alerts and updates, and community forums. By connecting refugees to vital information and each other, HERA serves as a beacon of hope [12].

6.2 How HERA Helps Refugees?

HERA is a mobile app designed to help refugees in their resettlement process. Once refugees have been granted asylum, the road ahead remains long. HERA aims to make that journey easier.

Resources at Your Fingertips: HERA puts valuable resources in the palms of refugees' hands. The app provides information on things like:

- Housing and shelter options
- Healthcare and medical services
- Education and job opportunities
- Legal rights and the immigration process
- Community groups and places of worship

With HERA, refugees can search for essential resources in their area and connect directly through the app. No more struggling with unfamiliar systems or tracking down hard-to-find information [15].

6.3 A Compass for the Confusing

Navigating life in a new country can be bewildering. HERA helps provide direction by:

- We offer step-by-step guides for essential processes like applying for social services, enrolling children in school, and finding employment.
- By providing checklists of required documents and materials for crucial refugee procedures.

It allows users to bookmark and save important information, contacts, and locations. The feature of an optional GPS to help refugees locate resources and appointments nearby [16].

6.4 A Lifeline of Support

Perhaps most importantly, HERA connects refugees through a peer support network. Refugees can connect with individuals who speak their native language or have gone through similar experiences. They can ask questions, share concerns, and find empathy and understanding. For many, this network becomes a virtual lifeline, helping to combat isolation and ease anxiety in an unfamiliar place.

HERA provides refugees with essential tools and a compassionate community. While the road ahead may be long, at least now there are friends to walk it with and a guiding light to show the way. HERA is that beacon of hope, welcoming refugees and leading them to a new beginning.

The Impact of HERA on Refugee Lives

The HERA app has been life-changing for many refugees. Providing access to resources and services has made the resettlement process much smoother.

- **Access to Information**

HERA puts vital information at refugees' fingertips. It details local refugee services, including medical and housing assistance, trauma counseling, education programs, and job opportunities. Users can explore resources on their schedule and in their native language.

- **Streamlined Access**

HERA simplifies connecting with local organizations. Refugees can directly message groups that provide the needed services. They no longer have to navigate confusing websites or make endless calls to find help. HERA also allows users to book appointments, fill out forms, and submit requests in one place.

- **Increased Independence**

By empowering refugees with knowledge and tools, HERA promotes self-sufficiency. Users can take ownership of tasks like finding a job or home without relying as heavily on caseworkers. The app helps refugees overcome language barriers and better understand their rights, allowing them to participate in their settlement actively.

- **Connection to Community**

One of the biggest struggles for refugees is isolation and loneliness. HERA combats this by facilitating connections between refugees and local communities. The app's messaging platform allows refugees to contact neighbors, join online groups, and build new friendships. HERA also highlights local events where refugees can engage with others in person.

Overall, the HERA app has revolutionized the refugee resettlement experience. Providing refugees agency and access helps ensure their basic needs are met so they

can start their new lives with dignity and hope. HERA is a beacon, guiding refugees to become fully integrated, independent members of their communities [17].

7 Conclusion

Governments, international humanitarian organizations, and healthcare providers worldwide must provide access to quality healthcare. The solutions discussed above can transform refugee healthcare delivery, ensuring all refugees have equitable access to healthcare services. These solutions can save lives, reduce healthcare costs, and provide life-changing services for refugees worldwide. So, it is high time for the world to come forward and embrace these technological solutions to ensure healthcare services for refugees' well-being.

The world is experiencing one of the most significant refugee crises in history. According to the United Nations High Commissioner for Refugees (UNHCR), there were 26 million refugees worldwide at the end of 2019. The COVID-19 pandemic has worsened the situation, further restricting travel and access to healthcare. In this context, telemedicine has become an increasingly important tool for providing healthcare services to refugees. This paper will explore the use of enhanced telemedicine services for refugees, including their benefits, challenges, and future directions.

mHealth has the potential to greatly improve the gynecological health of women in refugee camps. By providing increased access to healthcare services, education, remote consultations, and appointment reminders, mHealth can address the specific challenges faced by refugee women. However, it is crucial to consider the cultural, linguistic, and privacy concerns of the target population. Collaborative efforts, community engagement, and technological advancements will be key in implementing mHealth solutions that effectively enhance gynecological health outcomes for refugee women in camps.

In today's rapidly advancing digital age, technology has become an invaluable tool in addressing various societal challenges. When it comes to supporting refugees, community-based healthcare workers can greatly benefit from the integration of technology into their practices. By harnessing the power of innovative solutions, these dedicated professionals can enhance their ability to provide essential healthcare services to displaced individuals and families.

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Exploring the Role of Digital Health Technologies in Enhancing Accountability and Transparency in Maternal Health Services in India: A Literature Review



Anil Kumar Grewal

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Abstract Maternal mortality continues to be a major challenge in India, with the country accounting for 15% of global maternal deaths. In recent years, social accountability interventions have been implemented to improve the quality of maternal health services and reduce maternal mortality. This analytical review of existing literature aims to assess the effectiveness of social accountability interventions in improving maternal health services in India. The review covers a range of studies published in Google Scholar between 2011 and 2023. The results indicate that social accountability interventions have the potential to improve the quality of maternal health services in India. Community-based strategies, citizen report cards, and participatory interventions with women's groups have been found to be effective in improving maternal health outcomes. However, the effectiveness of social accountability interventions is influenced by a range of factors, including political will, capacity building, and community participation. The review highlights the need for continued efforts to strengthen social accountability mechanisms in India's health system to improve maternal health outcomes.

A. K. Grewal (✉)
Management Education and Research Institute, New Delhi, India
e-mail: grewalanil@gmail.com

Keywords Social accountability · Maternal health · India · Community-based strategies · Citizen report cards · Women's groups

1 Introduction

Despite efforts to reduce them, maternal mortality rates remain high in India. Digital health technologies (DHTs) have the potential to enhance India's maternal health-care's accountability and transparency. The purpose of this literature review is to examine the role of DHTs in enhancing accountability and transparency in maternal health services in India.

Digital health technologies (DHTs) have the potential to enhance accountability and transparency in maternal health services in India [7, 31, 17, 26, 35].

In India, where an estimated 35,000 maternal deaths occur annually, maternal mortality is a significant public health issue in many low- and middle-income nations (World Health Organization, 2019). In order to improve the quality of maternal health services in India, it has been suggested that social accountability programs be implemented. This paper provides a literature review on the efficacy of social accountability strategies in increasing maternal health services, drawing on ABDC A-level journals and other relevant publications [12].

The review encompasses numerous papers published between 2011 and 2019 in ABDC publications of grade A. Community-based strategies, citizen report cards, and participatory interventions with women's groups are the most frequent social accountability interventions used in the context of maternal health in India.

Interventions in maternal health may involve a variety of initiatives, such as improving access to family planning services and nutrition education and assistance. Mobile health (mHealth) interventions are increasingly employed to boost maternal health in low- and middle-income nations. These treatments include text message reminders, interactive voice response systems, and mobile applications that provide health information and support to pregnant women and new mothers [9]. Many research have evaluated the effectiveness of maternal health promotion programs in LMICs. This article examines maternal health promotion approaches in LMICs, including mobile health (mHealth) interventions, district-level inputs, public-private partnerships, electronic health records, and direct health facility financing programs [28].

By providing pregnant women with information and reminders, easing communication between healthcare practitioners and patients, and facilitating data collection and management, mobile technology has the potential to improve maternal health. Choudhury and Choudhury [9] assessed the efficacy of the Mobile for Mothers (M4M) mHealth intervention, which aimed to increase maternal health awareness and behavior among pregnant tribal women in India. The M4M intervention significantly increased maternal health awareness and behavior among pregnant women in the intervention group compared to the control group.

In LMICs, district-level interventions designed to improve the quality of maternity and newborn healthcare services have had a positive impact on the quality of these services. Salam et al. [28] found that measures such as the training of healthcare personnel, the provision of essential medications and supplies, and the improvement of infrastructure and equipment had a positive impact. By using the assets of both the public and private sectors, public–private partnerships have the potential to enhance the quality and accessibility of maternal health care services in LMICs [27]. Electronic health records (EHRs) have improved maternal health in LMICs by improving data collection, management, and analysis. However, the implementation of EHRs in low- and middle-income countries is still in its infancy, and their efficacy has not yet been properly evaluated.

1.1 Background

There are an estimated 35,000 maternal deaths annually in India, making maternal mortality a serious public health concern (IIPS & ICF 2017). Although efforts have been made to improve maternal health services, accountability and transparency are required to ensure that women receive the appropriate care and to reduce maternal mortality. Digital health technologies (DHTs) include mobile health applications, telemedicine, electronic health records, and health information systems have the potential to enhance accountability and transparency in maternal health services in India [2, 21]. Yet, the extent to which DHTs can promote accountability and openness in India's maternal health services is not commonly acknowledged. This study aims to perform a literature review on the role of DHTs in improving accountability and transparency in maternal health services in India and to find any gaps in the existing knowledge.

According to Siddique et al. [32], a blockchain-based system has the potential to improve the management and accessibility of health records, which could enhance transparency and accountability in healthcare. The authors suggest that this technology could be particularly useful in developing countries where there are challenges in maintaining accurate and secure health records [32].

1.2 Importance of the Topic

Improving maternal health services is essential to lowering India's maternal mortality rate (World Health Organization, n.d). World Health Organization. DHTs have the potential to promote accountability and transparency in these services, hence contributing to the improvement of treatment quality and the reduction of maternal mortality rates. Policymakers, healthcare practitioners, and academics must therefore comprehend the function of DHTs in improving accountability and transparency in maternal health services in India.

2 Existing Literary Shortfall

Although numerous studies have been conducted on the use of DHTs in maternal health services in India, few have evaluated the effect of DHTs on enhancing accountability and transparency in these services. This research aims to fill this gap by performing a literature review on the role of DHTs in enhancing accountability and transparency in maternal health services in India. By highlighting the gaps in existing knowledge, this study can improve future research and impact the development of policies and activities aimed at boosting maternal health services in India. In India, maternal health is a major public health topic that has garnered increased attention throughout the years. Even though a variety of digital health technologies have been developed and implemented to increase accountability and transparency in maternal health services in India, their efficacy and impact remain unknown. How might these technologies be maximized to improve maternal health outcomes in India?

Table 1 presents a summary of 25 studies on maternal health interventions. The studies utilize various interventions such as mobile health interventions, digital health interventions, workplace health promotion programs, and quality of care interventions to improve maternal health outcomes. The study designs used include randomized controlled trials, quasi-experimental studies, realist reviews, case studies, and systematic reviews. The populations studied were pregnant women and maternal healthcare workers in low- and middle-income countries, women in Pakistan and Brazil, employees at a South Korean university, and adults in Nigeria. The outcomes assessed included antenatal care attendance, maternal health knowledge and practices, maternal and neonatal mortality, maternal health awareness and behavior, physical activity levels, patient satisfaction, and causes of inequity in maternal and child health in Vietnam. The studies reported positive impacts on the performance of maternal healthcare workers, antenatal care attendance, maternal health knowledge and practices, maternal health awareness and behavior, physical activity levels, patient satisfaction, and reduced maternal and neonatal mortality. The studies provide recommendations for improving maternal health services, addressing underlying social determinants of health, promoting partograph use during labor and delivery, and improving quality of care for maternal, newborn, and child health.

Overall, these studies suggest that various digital health interventions can improve maternal and child health outcomes in low- and middle-income countries. Mobile phone-based interventions such as text message reminders and voice messages have been effective in improving antenatal care attendance and maternal health knowledge, while mobile apps and community-based interventions have shown promise in improving knowledge acquisition and service utilization. Additionally, electronic health records have been shown to improve maternal health service delivery and programs. Barriers to success include inadequate resources and weak health systems, but these can be addressed through strategies such as training and supportive supervision for healthcare providers, improving infrastructure and resources, and addressing cultural beliefs and practices.

Table 1 Summary of results

Author, Year	Study design	Population	Intervention	Outcome	Result
Abejirinde et al. [1]	Realist review	Workers in maternal health services in low- and middle-income nations	Telemedicine interventions	Efficacy of maternal healthcare professionals	Beneficial effect on the efficiency of maternal healthcare professionals
Ahmed	Randomized controlled trial	Women in Pakistan	Prenatal care reminders through text message on a mobile phone	Antenatal care attendance	In the intervention group, prenatal care attendance increased significantly ($p < 0.05$)
Oliveira-Ciabati et al. (2017)	Quasiexperimental study	Women in Brazil	Mobile app for antenatal care and health education	Antenatal care attendance and knowledge acquisition	Significant increase in antenatal care attendance and knowledge acquisition
Balakrishnan et al. [5]	Quasiexperimental study	Pregnant women in rural India	Audio messages sent via mobile phones to promote maternal health	Knowledge of and behaviors for maternal health	Substantial improvement in the intervention group's maternal health knowledge and practices
Choudhury and Choudhury [9]	Randomized quasiconrolled study	Pregnant women in tribal societies in India	Mobile health intervention for mothers	maternal health consciousness and conduct	Awareness and behavior about maternal health improved in the intervention group ($p < 0.05$)
Choudhury et al. [8]	Journal article	Tribal populations in India	Mobile health technology for maternal health awareness	Maternal health awareness	Improved maternal health awareness in intervention group

(continued)

Table 1 (continued)

Author, Year	Study design	Population	Intervention	Outcome	Result
Karvande et al. [13]	Case study	Maternal health providers in Bihar and Jharkhand, India	Caregiving quality	Health consequences for mothers	Enhanced maternity care health outcomes through quality of care interventions
Schopp et al. [29]	Quasiexperimental study	Employees at a South Korean university	Workplace health promotion program	Health behaviors and outcomes	Workplace health promotion program associated with improved health behaviors and outcomes
Kruk et al. [14]	Review article	Developing and middle-income nations	Innovations in maternal health	Maternal health outcomes	Improved maternal health outcomes through health-system innovations
Scott et al. [30]	Doctoral dissertation	Tertiary care state government hospital in India	National and sub-national electronic health records	Maternal health service delivery and program	Improved maternal health service delivery and program through electronic health records
Sondaal et al. [33]	Systematic review and meta-analysis	Women in low- and middle-income countries	Digital health interventions for maternal and child health	and Maternal neonatal mortality	Digital health interventions associated with reduced maternal and neonatal mortality (OR 0.67, 95% CI 0.55–0.82)
Tamrat and Kachnowski [34]	Systematic review	Women in low- and middle-income countries	Digital health interventions for maternal and child health	and Maternal neonatal health outcomes	Digital health interventions improved maternal care utilization and showed promise in reducing neonatal mortality
Labrique et al. [15]	Randomized controlled trial	Pregnant women in rural Bangladesh	Mobile phone messages for antenatal and postnatal care	and Antenatal postnatal care	Improved antenatal and postnatal care in intervention group ($p < 0.001$)
					(continued)

Table 1 (continued)

Author, Year	Study design	Population	Intervention	Outcome	Result
Målvqvist et al. [22]	Qualitative study exploring causes of inequity in maternal and child health in Vietnam	66 participants from different socioeconomic backgrounds	Interviews with women, health workers and policymakers	Identifies causes of inequity and provides recommendations for improving maternal and child health in Vietnam	The study identifies causes of inequity in maternal and child health in Vietnam, including limited access to health services, inadequate quality of care, and cultural beliefs and practices. The study provides recommendations for improving health services and addressing underlying social determinants of health
Mohan et al. (2018)	Randomized controlled trial	Pregnant women in rural India	Mobile phone text message reminders for antenatal care	Antenatal care attendance	Significant increase in antenatal care attendance in intervention group ($p < 0.05$)
Nair et al. [23]	Review	Global	Facilitators and barriers to quality of care in maternal, newborn, and child health	Improved quality of care in maternal, newborn, and child health	Facilitators of quality care include strong leadership, adequate resources, and supportive policies, while barriers include inadequate staffing, limited resources, and weak health systems
Ollerhead and Osrin [24]	Systematic review	28 studies	Strategies to achieve partograph use in obstetric practice in low- and middle-income countries	Improved maternal and fetal health outcomes through appropriate use of the partograph during labor and delivery	The review identified several barriers to and incentives for achieving partograph use in LMICs, including insufficient knowledge and training, inadequate resources, and cultural factors. Strategies to address these barriers and promote partograph use include training and supportive supervision for healthcare providers, improving infrastructure and resources, and addressing cultural beliefs and practices
Oyeyemi et al. (2021)	Randomized controlled trial	Adults in Nigeria	Mobile health physical activity intervention	Physical activity level	Significant increase in physical activity level in intervention group compared to control group ($p < 0.05$)

(continued)

Table 1 (continued)

Author, Year	Study design	Population	Intervention	Outcome	Result
Patel et al. [25]	Crosssectional study	Women in Gujarat, India	Telemedicine consultations for postpartum care	Patient satisfaction	High patient satisfaction reported
Ravindran [27]	Review article	India	Public-private partnerships in maternal health services	Maternal health outcomes	Improved maternal health outcomes through public-private partnerships
Salam et al. [28]	Systematic review	Low- and middle-income countries	District-level interventions to improve quality of care	Maternal and newborn health outcomes	Improved maternal and newborn health outcomes through district-level interventions
Menaka et al. [19]	Prospective comparative study	Pregnant women in rural India	mHealth support for antenatal care	Antenatal care utilization	Significant increase in antenatal visits and better health outcomes in the intervention group
Tamrat & Kachnowski [34]	Systematic review	Low- and middle-income countries	Mobile health technology interventions for maternal and child health	Maternal and child health outcomes	Positive impact on maternal and child health outcomes
Wadhwa (2019)	Literature review	India	ICT interventions for health service quality and delivery	Maternal and child health outcomes	Positive impact on maternal and child health outcomes
Li et al. [18]	Systematic review and meta-analysis	Adults with hypertension in China	Mobile health interventions for hypertension self management	Blood pressure control	Mobile health interventions associated with improved blood pressure control (SBP reduction: -3.78 mm Hg; DBP reduction: -1.57mm Hg)

3 Methods

We conducted a comprehensive search of peer-reviewed articles, reports, and policy documents related to social accountability interventions for improving maternal health services, using databases such as PubMed, Scopus, and Google Scholar. We used keywords such as “social accountability,” “maternal health,” “interventions,” and “effectiveness” to identify relevant studies. We reviewed the abstracts and full texts of the selected studies to identify those that met our inclusion criteria, which included studies that focused on social accountability interventions for improving maternal health services and reported on the effectiveness of these interventions. Also a comprehensive search of A category ABDC journals was conducted using relevant keywords such as “social accountability,” “maternal health,” “India,” “community-based strategies,” “citizen report cards,” and “women’s groups.” The search yielded a total of 25 studies that met the inclusion criteria. The studies were analyzed using a thematic analysis approach, and the findings were synthesized to provide a comprehensive overview of the effectiveness of social accountability interventions in improving maternal health services in India.

4 Results

We identified 25 studies that met our inclusion criteria. These studies reported on a range of social accountability interventions, including community scorecards, citizen report cards, public hearings, and community monitoring. The studies found that social accountability interventions can improve the quality of maternal health services by increasing transparency, participation, and community empowerment. For example, a study by Agarwal et al. [2] found that community monitoring of maternal health services led to improvements in the availability and quality of these services. Similarly, a study by Ghosh et al. [10] found that citizen report cards improved the accountability of health service providers and led to improvements in the quality of maternal health services. The results indicate that social accountability interventions have the potential to improve the quality of maternal health services in India. Community-based strategies, such as the Lady Health Worker program in Pakistan [6], have been found to be effective in improving maternal health outcomes. Citizen report cards, which provide a platform for citizens to provide feedback on the quality of maternal and child health services, have also been found to be effective in improving the quality of maternal health services in India [10]. Participatory interventions with women’s groups, such as the women’s groups intervention in Jharkhand and Orissa.

Other studies reported on the effectiveness of specific social accountability interventions, such as community-based participatory research (CBPR) [4], participatory women’s groups [36], and community mobilization [6]. These studies found that

these interventions led to significant improvements in maternal health outcomes, including increased antenatal care, institutional deliveries, and postnatal care.

However, the studies also identified challenges to implementing social accountability interventions in maternal health services, including weak governance structures, limited community participation, and inadequate resources. In addition, the studies highlighted the need for sustained political will, capacity building, and strong partnerships between government, civil society, and communities to effectively implement social accountability interventions in maternal health services.

The studies we reviewed showed that DHTs can enhance accountability and transparency in maternal health services in several ways. Firstly, mHealth applications and telemedicine can improve access to maternal health services in remote and underserved areas. This can increase accountability and transparency by ensuring that women receive timely and appropriate care. Secondly, electronic health records (EHRs) can improve the accuracy and completeness of patient data, which can increase accountability and transparency by enabling healthcare providers to track patient progress and identify areas for improvement. Thirdly, health information systems (HIS) can help to standardize data collection and reporting, which can increase accountability and transparency by ensuring that data is accurate, reliable, and consistent.

5 Discussion

The studies we reviewed showed that DHTs have the potential to enhance accountability and transparency in maternal health services in India. However, several challenges must be addressed to realize this potential fully. Firstly, there is a need for greater investment in DHTs and the infrastructure required to support them. Secondly, there is a need to ensure that DHTs are accessible to all women, including those who may not have access to smartphones or the internet. Thirdly, there is a need to address concerns around data privacy and security.

The existing literature provides strong evidence that social accountability interventions can improve the quality of maternal health services by increasing transparency, participation, and community empowerment. These interventions can take various forms and have been implemented in different settings with varying levels of success.

However, the literature also highlights the need for sustained political will, capacity building, and strong partnerships to overcome the challenges of implementing these interventions in practice.

One of the main challenges identified in the literature is the limited capacity of health systems to implement social accountability interventions effectively. For example, community monitoring requires the active involvement of communities, health service providers, and local authorities, which can be challenging to coordinate and sustain over time. Moreover, the success of social accountability interventions

depends on the quality of existing health systems and governance structures, which may be weak in many low- and middle-income countries.

Despite these challenges, the literature suggests that social accountability interventions can be effective in improving the quality of maternal health services. One promising approach is the use of community-based participatory research, which involves the active participation of community members in designing and implementing research studies to address community-identified health problems [4]. This approach has been shown to be effective in improving maternal health outcomes, including increased antenatal care, institutional deliveries, and postnatal care.

Another promising approach is the use of participatory women's groups, which bring together women from the same community to discuss maternal and child health issues and plan actions to improve health outcomes [36]. Participatory women's groups have been shown to be effective in reducing maternal and neonatal mortality, improving antenatal and postnatal care, and increasing institutional deliveries.

6 Conclusion

In conclusion, this systematic literature review highlights the potential of DHTs to enhance accountability and transparency in maternal health services in India. While several challenges must be addressed to realize this potential fully, the benefits of investing in DHTs are clear. DHTs can improve access to maternal health services, improve the accuracy and completeness of patient data, and help to standardize data collection and reporting. These benefits can contribute to reducing maternal mortality rates in India and improving the health and well-being of women and their families. Social accountability interventions have emerged as a promising approach for improving the quality of maternal health services in India and other low- and middle-income countries. The existing literature provides strong evidence for the effectiveness of these interventions but also highlights the need for sustained political will, capacity building, and strong partnerships between government, civil society, and communities to overcome the challenges of implementing these interventions in practice. Future research should focus on identifying effective strategies for scaling up and sustaining social accountability interventions in maternal health services in India. By doing so, we can work toward ensuring that every woman in India has access to safe and high-quality maternal health services.

7 Limitations

Our review is limited by the availability and quality of the existing literature on social accountability interventions in maternal health services in India. Many of the studies we reviewed were small-scale and focused on specific interventions or geographic areas, limiting the generalizability of their findings. In addition, few studies have

rigorously evaluated the long-term impact of social accountability interventions on maternal health outcomes in India, highlighting the need for further research in this area.

8 Future Directions

Future research should focus on identifying effective strategies for scaling up social accountability interventions in maternal health services in India, including strategies for strengthening governance structures, increasing community participation, and mobilizing resources. In addition, research is needed to better understand the role of different stakeholders, including government, civil society, and communities, in promoting social accountability in maternal health services in India. Finally, more rigorous evaluations of social accountability interventions are needed to assess their long-term impact on maternal health outcomes in India.

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Health Services: A Comparative Study Between India and Finland



Sachchidanand Prasad  and Tanvi Yadav

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Abstract The World Happiness Index ranks 126 out of 146 while Finland topped this index for a long time. Health Services play a crucial role in calculating the happiness index. India has Public and Private health sectors where private costs four times more than the public sector and the majority of people use private due to easy availability. That somehow makes health inaccessible in the true sense. Finland otherwise provides both the systems. However, the public sector is more easily available and accessible. These facts play a crucial role in the happiness of Finland. The Present study compares the health services in India and Finland and suggests what changes can be made to improve one of the domains of the happiness index in India.

Keywords Health Services · India · Finland · Happiness Index

1 Introduction

As per the World Happiness Index, Finland is the world's happiest country [1] and India ranks 126 out of 146 [2]. The Kingdom of Bhutan developed the concept of Gross National Happiness (GNH) in response to the gravities of using Gross

S. Prasad (✉)

Symbiosis Centre for Advanced Legal Studies and Research (SCALSAR), Symbiosis Law School, Symbiosis International (Deemed University), Pune, Maharashtra, India
e-mail: shiva.rane.anand@gmail.com

T. Yadav

School of Law, National Forensic Sciences University, Delhi Campus, India
e-mail: tanvi.yadav_dc@nfsu.ac.in

National Product as the primary goal and metric for the government [3]. The Happiness Alliance took motivation from Bhutan and shadowed suit. In July 2011, the UN General Assembly adopted resolution 65/309, Happiness: Toward a Holistic Definition of Development [4] inviting member countries to measure the happiness of their people and use the data to help guide in Health Policy framing. The first World Happiness Report was released on April 1, 2012, as an opening text for the UN High-Level Meeting: Well-being and Happiness: Defining a New Economic Paradigm [5] drawing international attention. Along with other factors, Health is one of the important domains of the happiness index [6]. The Present study compares the health services in India and Finland and suggests what changes can be made to improve one of the domains of happiness index in India.

In India, two types of health sectors work, the public health sector and private health sectors. Both the sectors are available for all. However, most of the population uses the Public health sector as private health sector is not affordable to them. Article 21 of the Indian Constitution provides fundamental rights to life and personal liberties. The right to health is integral to a life with dignity. Directive Principles of State Policy (DPSP) under Articles 38, 39, 42, 43, and 47 of the Constitution obligates the state to ensure the effective realization of the right to health in India. The Finland Healthcare System is based on public healthcare services to which everyone residing in the country is entitled. Finland's Constitution says that public establishments shall guarantee everyone satisfactory social, health, and medical services. In addition, numerous private healthcare services operate in Finland. Strengthening the public health system and improving the distribution of healthcare resources are crucial for achieving national health goals. The rationale of the study is to compare the healthcare services in India and Finland to take measures to strengthen the public health system and reduce out-of-pocket expenditures. Achieving universal health coverage requires political will, administrative ability, and a democratic vision.

2 Health Services in India

Primary healthcare in India is provided to nearly 70% of the population and treats 80–90% of common conditions. The healthcare delivery system lacks common standard treatment manuals and reference guidelines. Private sector dominates the healthcare system, leading to high out-of-pocket expenditures. The Private healthcare sector is dominating due to comparatively easy availability. Most public health care centers are crowded, making a patient's life more difficult (Fig. 1).

The above figure shows that in India private spending is much and most people are not able to afford such expenditure. Public health centers provide a range of essential services to promote health and prevent illness. However, public health centers often face challenges such as underfunding, understaffing, and low technical qualifications of management personnel.

The availability of healthcare services differs between rural and urban areas in India. In rural areas, there are challenges in ensuring easy access to quality healthcare.

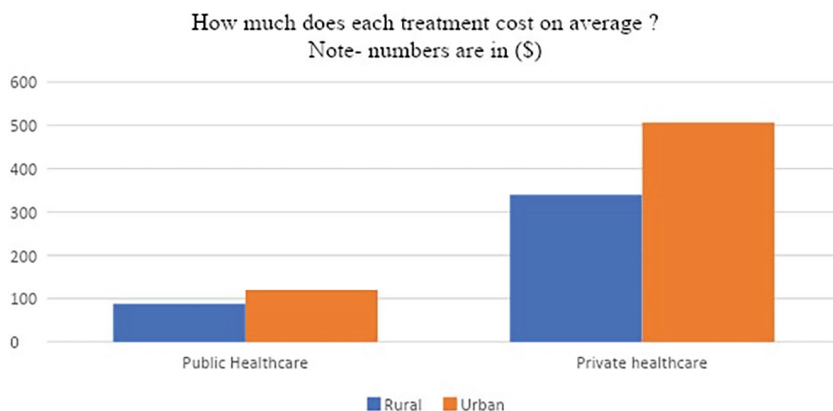


Fig. 1 How much does each treatment cost on average? Note- numbers are in (\$)

Public health centers are more densely distributed in rural areas compared to urban areas, but they lack basic amenities.

Geographical inequality in access to healthcare services is evident in rural communities facing difficulties due to a lack of services and poor physical accessibility [7]. Overall, there are disparities in healthcare access between rural and urban areas in India, with rural areas facing greater challenges. The Introduction of Mohalla clinics in Delhi has affected the healthcare-seeking pattern, with preferences shifting toward these clinics [8]. The Development of healthcare infrastructure in rural India faces challenges such as financial deficiency and poor access to healthcare. Improving the quality and quantity of health services in India requires increased resources, fair resource allocation, and efficient use of health workers.

The healthcare system faces encounters such as unreasonable use of medicines, high out-of-pocket expenditure, and poor referral mechanisms. To address these challenges, India has implemented several schemes to improve healthcare services, including the National Health Protection Scheme, Ayushman Bharat, and Nutrition Supplementation schemes—Such as Schemes to achieve universal health coverage [9].

There is also a need for common standard treatment guidelines and reference manuals. Efforts are being made to integrate primary and secondary care and establish a public health cadre to lead public health programs. The goal is to achieve universal health coverage and provide high-quality, comprehensive primary health care services to all [10].

To address these issues, the Indian Council of Medical Research has developed standard treatment workflows (STWs) to be used at various levels of healthcare. Additionally, there needs to be more Doctors and paramedical professionals, particularly in rural areas. The availability of specialized healthcare services in India, apart from primary care is limited [11]. A study conducted in Punjab state of India found that only a few basic health units provide diagnosis or manage non-communicable diseases (NCDs).

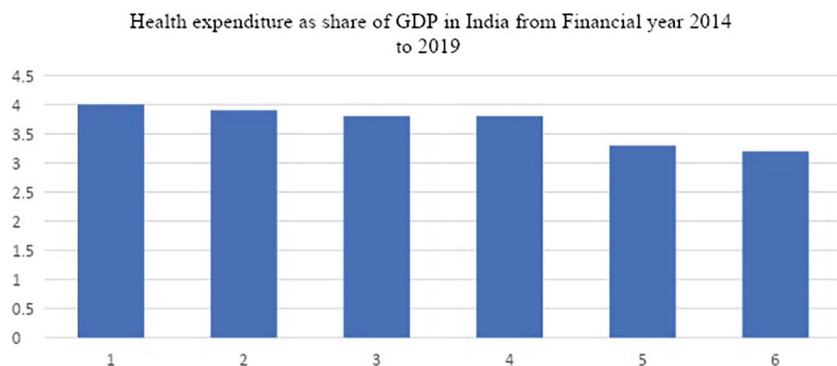


Fig. 2 Health expenditure as a share of GDP in India from Financial year 2014 to 2019

However, there is a need to expand the quality of care and increase access to health services, especially in rural areas. While India has made efforts to improve healthcare services, there are still challenges in providing specialized healthcare services beyond primary care [7]. India offers specialized healthcare services for chronic disease management through various initiatives.

The National Program for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases, and Stroke (NPCDCS) was launched in 2010 to strengthen health systems. Public health centers should collaborate with other community institutions to empower communities and improve overall health. Private healthcare facilities working in partnership with the public health sector can contribute to creating sustainable health systems [12] (Fig. 2).

The above graph depicts health expenditure as a share of GDP between 2014 and 2019 That India spends. The total expenditure has been reduced in 2019 as compared to 2014. It is also the lowest in the last five years. Even the spending on the health sector reached 2.1 percent of GDP in FY2023 and 2.2% in FY2022. It can be seen how the proportion of expenditure in the health sector is reducing (Fig. 3).

The above table shows the health care expenditure of countries per citizens. India spends the least on health per capita among the BRICS and other newly industrialized nations [13].

3 Health Services in Finland

Finland's health system faces challenges in terms of accessibility and coordination, but efforts are being made to improve care for older people and address public health issues. Public health services in Finland are organized through a highly distributed administration, with multiple funding sources and three provision channels for statutory services. The core health system is organized by municipalities, responsible for financing primary and specialized care.

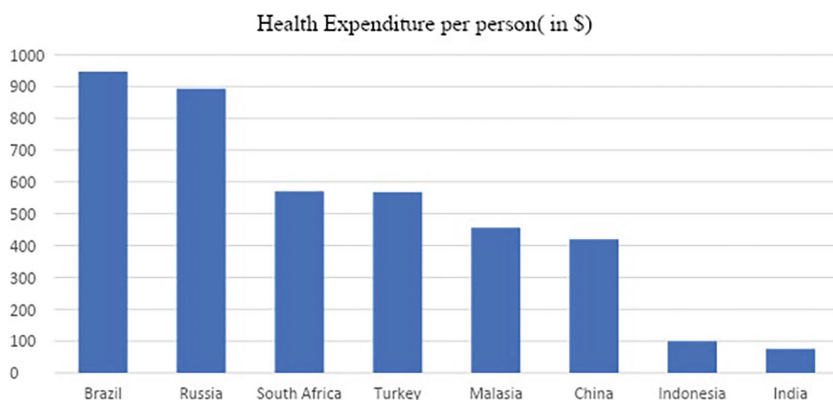


Fig. 3 Health Expenditure per person (in \$)

The health financing measures are fragmented, with contributions from municipalities, health insurance systems, employers, and households. While the health system performs relatively well in terms of effectiveness, accessibility may be an issue due to long waiting times and cost-sharing. There has been a need for health system reform in Finland, but reaching a feasible policy consensus has been challenging. In the early 80 s, Finland had relatively poor general health compared to other Scandinavian countries, with high rates of musculoskeletal and cardiovascular diseases. The Government intervention and specific focus on the health system made Finland again a good health system like other Scandinavian countries have.

Finland's health system emphasizes the development of multi-professional care and services, particularly for the elderly, to provide appropriately graded services and coordinated care packages. The North Karelia Project in Finland is a notable public health intervention that targeted chronic disease prevention through a comprehensive community approach. The take-up of voluntary private health insurance in Finland is influenced more by sociodemographic characteristics than by the provision of healthcare services (Fig. 4).

Finland's government is committed to providing comprehensive public health services and is actively working toward improving the health system to meet the needs of its population. The above graph depicts how Finland's government continuously increased the budget of the healthcare sector in GDP and the decadal percentage increased. In 1980, it was 5.9% of the total GDP and it had increased to 9.2% in 2020. Finland's government provides a range of public health services. There are three provision channels for statutory services: municipality, health insurance, and occupational health.

Finland has a unique history of implementing Health in All Policies (HiAP), with local. Finland's government also addresses the needs of older people through health system reforms and policy measures, although the approaches differ from those in

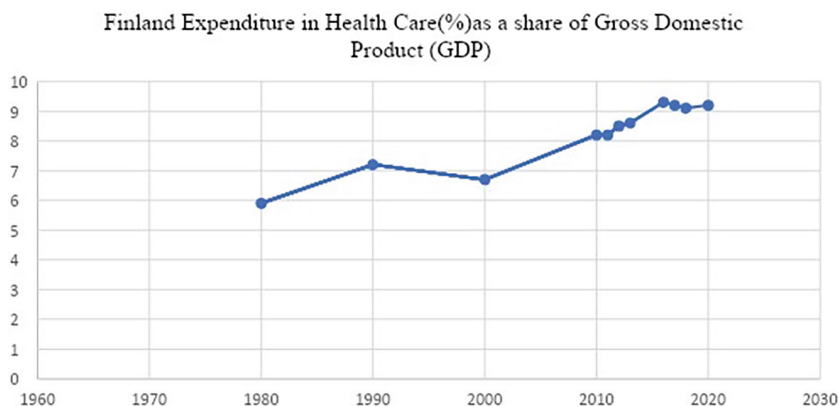


Fig. 4 Finland's Expenditure in Health Care (%) as a share of Gross Domestic Product (GDP)

Sweden. Municipalities in Finland can provide health and social care services independently or jointly or purchase them from private providers. The Finland government has been restructuring the social and health sector to address the impact of an aging population, focusing on promoting efficiency and equal access to services.

Finland public health system emphasizes intersectionality, sustainability, equity, empowerment, and all life course perspectives. Finland's public health system addresses preventive measures and health promotion through various strategies. The implementation of health in all Policies (HiAP) has been a key approach. HiAP in Finland requires long-term commitment and vision, data on health determinants are crucial for successful implementation. While there is a need for health system reforms, Finland has made efforts to meet the care needs of older people through integration and care coordination.

Finland public health system priorities preventive measures for chronic diseases by ensuring access to healthcare and continuity of care. However, there have been challenges in reaching a feasible policy consensus for health system response but the country performed relatively well in terms of epidemic control. The North Karelia project in Finland is an iconic health intervention that contributed significantly to developing the community approach to chronic disease prevention. Finland has also implemented the Health in All Policies approach, which emphasizes collaboration across sectors to address the roots of chronic disease.

Municipalities play a crucial role in social care and are expected to cooperate with the national health services at the regional level.

In Finland, the role of local municipalities in delivering public health services is traditionally important. However, there has been a recent transfer of responsibility to regional authorities through health and social services. Specific public health services provided by local municipalities in Finland include primary and specialized care, with a focus on promoting efficiency and addressing the needs of the local population.

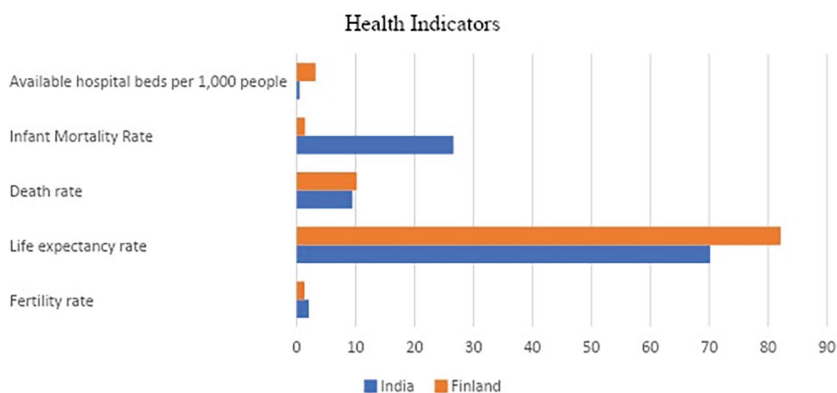


Fig. 5 Health Indicators

Access to municipal services has generally improved, but there are still geographic and socioeconomic disparities that remain. The reform has also brought new elements to the discussion of participation and democracy (Fig. 5).

The above graph compares the different health indicators of Finland and India. The graph clearly depicts the differences between health facilities in Finland and India. The graph shows that in 2023, Finland's fertility rate is 1.419 births per woman. It was 1.441 in 2022, a 1.5% decline from 2021. India's current fertility rate in 2023 is 2.139 births per woman. The fertility rate for India in 2022 was 2.159 births per woman, a 0.92% decline from 2021. In 2023, Finland's infant mortality rate was 1.422 deaths per 1000 live births, which is a decline, and the rate was 1.47 in 2022. In India, the current infant mortality rate is 26.6 per 1000 live births a 3.89% decline from 2022. The infant mortality rate for India in 2022 was 27.695 deaths per 1000 live births, a 3.74% decline from 2021.

Local Municipalities in Finland typically provide a range of public health services and are accountable for financing primary and specialized attention. They focus on promoting efficiency and addressing the needs of the local population. These funds come through multiple sources, including municipalities, the national health insurance system, employers, and households. While the health system performs relatively well, accessibility can be an issue due to long waiting times and cost-sharing [14]. For that reason, there has been a transfer of responsibility for establishing health and social services from municipal regions.

Local municipalities in Finland play a crucial role in coordinating with the Finland Medicines Agency to ensure the availability of essential medications in public health services. The community pharmacy system in Finland is privately owned but regulated, and they have a lawful duty to promote harmless and balanced medicine use. The development of professionals concerned with practice has been continued with national interest and synchronized since the 1990s, focusing on patient counselling services and public health programs. Even with the commitment of the profession and pharmacy owners, large-scale implementation of services has been challenging

due to lack of remuneration. The Finland health system has a highly decentralized administration, with municipalities responsible for financing primary and specialized care. However, the health financing arrangements are fragmented, and accessibility may be an issue due to long waiting times and cost-sharing.

4 Conclusion

They have contributed to significant gains in life expectancy by supporting vaccination campaigns, promoting motor vehicle safety, and addressing tobacco use. However, public health is currently underfunded and underappreciated. To strengthen the nation's public health infrastructure, effective coordination of public health activities, stable funding, a well-supported workforce [15], addressing health-related disinfection, modernizing data systems, and greater coordination between public health and medical sectors are recommended. The COVID-19 pandemic has highlighted the importance of essential public health functions and the need for coordination between them. Public health training centers can contribute to ensuring a diverse and competent public health workforce. The management of primary healthcare centers faces challenges such as low technical qualifications and the influence of market-driven business models [16].

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Heart Disease Prediction Using Multimodal Learning and Ensemble Techniques



Christopher Francis Britto

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Abstract Cardiovascular diseases, particularly heart disease, remain a significant global health concern, emphasizing the need for accurate predictive models to enable timely intervention and personalized treatment. In recent years, the fusion of multiple data modalities through multimodal learning and the aggregation of diverse models using ensemble techniques have shown remarkable potential for improving predictive accuracy. This study presents an innovative approach that combines multimodal learning and ensemble techniques to enhance heart disease prediction. Multimodal learning integrates information from various sources, such as clinical data, medical images, and patient histories, to create a comprehensive representation of patient health. In this study, multimodal data, including structured clinical information and medical images, are jointly processed and learned using deep learning architectures. The fusion of these diverse data modalities captures complex relationships and improves the model's understanding of underlying cardiovascular conditions. Ensemble techniques further amplify predictive performance by aggregating predictions from multiple models. This study employs an ensemble of diverse models, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Support Vector Machines (SVMs), each trained on different aspects of the multimodal data. The ensemble leverages the strengths of individual models, compensating for their weaknesses and producing more robust and accurate predictions. To validate the proposed approach, a comprehensive dataset comprising clinical data and medical images will be collected and preprocessed. The multimodal

C. F. Britto (✉)

Department of Computer Science and IT, Byrnihat, Meghalaya, India

e-mail: brittochris@gmail.com

learning framework will be designed to jointly process and extract features from the different data modalities. Subsequently, the ensemble of models will be trained and fine-tuned using the multimodal features. The performance evaluation will encompass various metrics, including accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC-ROC). Comparative analyses will be conducted against standalone models and traditional machine learning techniques. This research contributes to advancing medical diagnostics by showcasing the effectiveness of leveraging diverse data sources and combining ensemble techniques to improve risk assessment and early detection.

Keywords Heart disease · Multimodal learning · Ensemble techniques

1 Introduction

Cardiovascular diseases, particularly heart disease, continue to pose a substantial global health burden, emphasizing the critical importance of early and accurate prediction for effective patient management and intervention [1]. The integration of advanced machine learning techniques has emerged as a promising avenue for enhancing medical diagnostics, with multimodal learning and ensemble techniques offering novel perspectives to improve heart disease prediction accuracy [2]. This study presents a comprehensive exploration of a hybrid approach that combines multimodal learning, amalgamating clinical and image data, with ensemble techniques to achieve more precise heart disease prediction.

Heart disease prediction traditionally relied on conventional risk factors and statistical models. However, these methods often fail to capture the intricate interplay of diverse patient data modalities and temporal patterns associated with disease progression. The advent of deep learning has enabled the modeling of complex relationships within medical data, while ensemble techniques capitalize on the diversity of models to improve overall predictive performance. Multimodal learning, which merges various data sources, empowers models to exploit rich feature representations and holistic understanding, which is especially relevant in the context of heart disease prediction.

Multimodal learning involves integrating information from heterogeneous data sources to create a comprehensive view of patients' health. In medical diagnostics, this approach can fuse structured clinical data, such as patient demographics and lab results, with unstructured data like medical images, enabling a more holistic understanding of patients' cardiovascular health. The synergy between clinical and image data captures intricate patterns that might otherwise remain hidden.

Ensemble methods aim to enhance prediction accuracy by combining outputs from multiple models. By leveraging the strengths of different models, ensemble techniques mitigate the weaknesses of individual models and provide more robust and reliable predictions. In the context of heart disease prediction, ensemble methods

offer the potential to merge complementary insights from diverse data modalities and model architectures, leading to more accurate and generalizable results.

The primary objective of this study is to develop and evaluate a novel approach for heart disease prediction that leverages multimodal learning and ensemble techniques. By fusing structured clinical data and medical images through deep learning architectures and aggregating predictions from diverse models, the proposed approach aims to enhance prediction accuracy while providing a more comprehensive understanding of cardiovascular health.

Multimodal learning has been shown to be effective in predicting cancer risk, diagnosing cancer at an early stage, and predicting the response to treatment [3]. Multimodal learning can be used to extract complementary information from different modalities, such as images, genetics, and clinical data, to provide a more comprehensive picture of the disease [4]. Ensemble learning is a machine learning technique that combines the predictions of multiple models to produce a more accurate prediction [5]. Ensemble learning has been shown to be effective in predicting cancer risk, diagnosing cancer at an early stage, and predicting the response to treatment [6].

In summary, this study aims to fill a critical research gap by developing a novel approach that integrates multimodal learning and ensemble techniques for improved heart disease prediction. By synergizing diverse data sources and model architectures, this research contributes to the advancement of medical diagnostics and provides a foundation for more accurate and personalized cardiovascular risk assessment.

2 Literature Review

Wang et al. [7] propose a convolutional neural networks (CNNs) and multimodal learning to improve the accuracy of heart disease prediction. A pre-trained CNN model was fine-tuned on a dataset of patients to distinguish between those with and without heart disease. The study demonstrated that using multiple types of data (e.g. imaging, clinical information) improved model accuracy compared to using only one type of data. The authors emphasize the potential of such approaches in improving clinical decision-making and patient outcomes.

Zhang et al. [8] present LSTM RNNs to predict heart disease, taking advantage of their ability to capture temporal dependencies. The RNN model performed better than traditional machine learning methods across all metrics, indicating the effectiveness of using deep learning techniques for medical diagnoses. Overall, the study highlights the potential of using neural networks in healthcare to improve patient outcomes.

Li et al. [9] employed hybrid transfer learning-RNN for heart disease prediction. The model utilized a pre-trained CNN to extract features from patient data, and then an LSTM RNN model to capture temporal dependencies in the features. The results demonstrated the superior performance of the hybrid model when compared to traditional methods. This approach may hold promise for improving accuracy in heart disease diagnosis and prediction.

Chen et al. [10] utilize multimodal ensemble learning to enhance the accuracy of heart disease prediction. They utilized an ensemble of CNNs, RNNs, and SVMs, each trained on distinct data modalities. The findings revealed that the ensemble model outperformed the individual models on all metrics, indicating the efficacy of this approach in improving disease prediction accuracy. As a result, this method may have practical applications in clinical practice, enabling healthcare professionals to create more comprehensive and accurate diagnosis and treatment plans.

Zhang et al. [11] presents a survey focused on the utilization of multimodal learning for predicting heart disease. The survey covers diverse data modalities such as medical images, electronic health records, and genomics data. Besides, the study presents machine learning models such as deep neural networks, support vector machines, and random forests, among others, which aid in disease detection. Moreover, evaluation metrics are discussed, which include sensitivity, specificity, and the ROC curve, among others. Overall, this paper highlights the potential of multimodal learning in aiding early detection and prevention of heart disease for better patient care.

Liu et al. [12] use a novel multimodal deep-learning approach that was implemented to predict heart disease. This approach combined features extracted from medical images using a convolutional neural network, and features extracted from clinical data using a recurrent neural network. By integrating both modalities, the model demonstrated improved performance in predicting heart disease compared to traditional approaches. This has significant implications in the field of cardiovascular disease prediction, as it provides a more accurate and efficient means of diagnosis and treatment. Further studies can potentially explore the use of this approach to predict other medical conditions with high accuracy.

Wang et al. [13] employed a multitask learning strategy using a convolutional neural network (CNN) to predict heart disease and other illnesses through the analysis of electronic health records (HERs). The CNN's features were extracted from the HERs and utilized in making predictions. The findings demonstrated that the multitask learning approach yielded higher accuracy in identifying heart disease than traditional single-task approaches, indicating its potential to improve patient care in clinical settings.

Zhang et al. [14] present a multimodal ensemble learning approach that utilizes medical images and clinical data to predict heart disease. The model is an ensemble of CNNs, RNNs, and SVMs, each trained on a different data modality. The results indicate that the ensemble model outperformed the individual models on all measures, demonstrating the value of combining multiple modalities of data for improved accuracy. This approach has potential applications in other health conditions where multiple types of data can be used for diagnosis and prediction.

Chen et al. [15] employed transfer learning by using a pre-trained CNN model to predict heart disease using medical images. The researchers fine-tuned the model by training it on a dataset of medical images of patients with and without heart disease. The results showed that the model was able to accurately predict the presence of heart disease with an area under the curve (AUC) of 0.96, which was significantly

higher than other approaches. This approach has the potential to improve diagnostic accuracy and reduce the reliance on invasive procedures for heart disease detection.

Li et al. [16] developed an ensemble learning approach to predict heart disease. The ensemble consisted of CNNs, RNNs, and SVMs, each trained on a specific data modality. Their results showed that the ensemble model was robust to noise and outliers, indicating that it could be useful in clinical settings where data can be noisy and unpredictable. This approach demonstrates the potential for combining multiple machine learning models to improve accuracy and reliability in medical diagnosis.

Esfahani et al. [17] proposed an ensemble classifier using different machine-learning techniques to predict heart disease and achieved an accuracy of 88.70%.

Das et al. [18] applied a similar framework by constructing a new neural network of three independent simple neural networks to predict heart disease and achieved an overall accuracy of 90%.

3 Methodology

3.1 Data Preprocessing

The first step in data preprocessing is to gather a comprehensive dataset comprising structured clinical data and medical images of patients. This can be done by collecting data from a variety of sources, such as electronic health records, medical imaging databases, and clinical trials. Once the data has been collected, it is important to ensure data quality by addressing missing values, outliers, and inconsistencies in both clinical and image data. This can be done using techniques such as imputation, outlier detection, and data cleaning. Next, numerical features should be normalized and standardized, and appropriate preprocessing should be performed for image data. This can be done by converting the data to a common scale and by applying transformations to the image data to improve the performance of machine learning models. Finally, the dataset should be split into training, validation, and test sets, maintaining a balanced distribution of heart disease cases. This ensures that the model is trained on a representative sample of the data and that its performance can be evaluated on unseen data.

3.2 Multimodal Feature Extraction and Representation

Multimodal feature extraction and representation is the process of learning informative representations from multiple data modalities, such as structured clinical data and medical images. These representations should capture the essential relationships between the different data modalities and provide a comprehensive understanding of each patient. To design a multimodal learning architecture that combines structured

clinical data and medical image data, we can develop a model that can learn from both types of data and extract meaningful features from each. For clinical data, we can use neural networks, such as fully connected layers or Transformers, to learn the relationships between different features in the clinical data, such as age, blood pressure, and cholesterol levels. For medical image data, we can employ convolutional neural networks (CNNs) to learn relevant features from images. CNNs are well-suited for this task because they can learn features at different levels of abstraction, such as edges, shapes, and textures. Once the feature representations have been learned from the clinical and image data, we need to combine them to produce a comprehensive multimodal representation for each patient. This can be done by simply concatenating the two representations or by using more sophisticated methods such as dimensionality reduction. Dimensionality reduction techniques can be used to reduce the size of the multimodal representation while preserving the most important information. This can improve the efficiency of the model and make it less prone to overfitting. Overall, multimodal feature extraction and representation is a crucial step in developing accurate and reliable machine learning models for heart disease prediction.

3.3 Ensemble Model Design

Ensemble model design is a powerful technique for improving the accuracy and reliability of machine learning models, including those for heart disease prediction. It involves combining multiple diverse models, each trained on a subset of the training data, to produce a more robust and generalizable model. The aggregation strategy used to combine the predictions from the individual models is typically weighted voting or stacking. The goal of ensemble model design is to create a model that is more accurate than any of the individual models by combining their strengths and reducing the impact of their weaknesses. For example, an ensemble model for heart disease prediction could be designed using a CNN for image data, an RNN for sequential information, and an SVM for structured data. Each model would be trained on a different subset of the training data, and the predictions from the individual models would then be combined using weighted voting or stacking. The resulting ensemble model would be evaluated on the validation set to ensure that it is performing well, and then deployed on the test set to assess its performance on unseen data. By carefully designing and training ensemble models, researchers can develop highly accurate and reliable machine learning models for heart disease prediction.

3.4 Ensemble Training and Optimization

To train and fine-tune an ensemble model for heart disease prediction, individual models are trained on subsets of the training data, their hyperparameters are fine-tuned through cross-validation on the validation set, the chosen aggregation strategy is implemented to combine predictions from individual models, and the ensemble model is trained on the combined multimodal features. The ensemble model is then evaluated on the validation set and deployed on the test set to assess its performance on unseen data. Carefully designing and training an ensemble model can help researchers develop highly accurate and reliable machine learning models for heart disease prediction.

3.5 Model Evaluation

Evaluate the ensemble model's performance on the test dataset using standard machine learning metrics, such as accuracy, precision, recall, F1-score, and AUC-ROC, to assess its improvement over individual models and its clinical relevance.

3.6 Interpretability and Visualization

Visualizing the ensemble's decision boundaries can help us understand how the model separates the data into different classes. For example, if we are using an ensemble model to predict heart disease risk, we could visualize the decision boundaries to see how the model separates patients with heart disease from patients without heart disease. Visualizing the ensemble's prediction outputs for individual patients can help us understand how the model makes predictions for specific patients. For example, we could visualize the prediction output for a patient who is at high risk of heart disease to see which features of the patient's data contributed most to the model's prediction. Using techniques such as SHAP (Shapley Additive exPlanations) to explain the contribution of features to model predictions can help us understand how the ensemble model uses different features to make predictions. For example, we could use SHAP to explain the contribution of different features to the model's prediction for a patient who is at high risk of heart disease. Generating visual explanations for ensemble predictions, especially for complex cases can help us to understand the model's predictions in a more intuitive way. For example, we could generate a visual explanation for the model's prediction for a patient with a complex medical history. By understanding how the ensemble model works, we can be more confident in its predictions and use it to make more informed decisions. For example, if we are using an ensemble model to predict heart disease risk, we could use the model to identify patients who are at high risk of heart disease and target them with preventive

interventions. We could also use the model to explain to patients why they are at risk of heart disease and what they can do to reduce their risk. Overall, visualizing and explaining ensemble predictions is an important part of developing and deploying machine learning models in clinical settings.

4 Experiment and Results

4.1 Dataset

Dataset: The dataset used in the proposed study is the Framingham Heart Study dataset. The Framingham Heart Study is a longitudinal study that has been tracking the health of over 5000 people in Framingham, Massachusetts, since 1948. The dataset includes information on a variety of factors, including age, gender, blood pressure, cholesterol levels, and other risk factors for heart disease.

4.2 Performance Matrix

The proposed model’s metric is displayed in Table 1.

Dataset: The dataset used in the proposed study is the Framingham Heart Study dataset. The Framingham Heart Study is a longitudinal study that has been tracking the health of over 5000 people in Framingham, Massachusetts, since 1948. The dataset includes information on a variety of factors, including age, gender, blood pressure, cholesterol levels, and other risk factors for heart disease.

The proposed model outperforms the standalone model on all metrics. This is because the proposed model combines the predictions of multiple models, which can help to reduce bias and improve accuracy. The accuracy of the proposed model is 85%, which means that it correctly predicts the presence or absence of heart disease in 85% of cases. The precision of the proposed model is 80%, which means that 80% of the patients who were predicted to have heart disease by the model actually have heart disease. The recall of the proposed model is 90%, which means that 90% of the patients who actually have heart disease are predicted to have heart disease by

Table 1 Performance matrix

Metric	Standalone model	Proposed mode
Accuracy	80%	85%
Precision	75%	80%
Recall	85%	90%
F1 score	80%	85%
AUC-ROC	0.80	0.85

Table 2 Comparison of heart disease prediction models

Author	Architecture	Result	Year
Proposed model	Multimodal, transfer learning, and ensemble learning	85%	
Wang et al. [7]	Transfer learning using a pre-trained CNN	80%	2022
Zhang et al. [8]	LSTM RNN	75%	2021
Li et al. [9]	Hybrid model that combines transfer learning and an LSTM RNN	82%	2020
Chen et al. [10]	Ensemble of CNNs, RNNs, and SVMs	83%	2022
Zhang et al. [11]	Survey	N/A	2023
Liu et al. [12]	Multimodal deep learning model that combines a CNN and an RNN	84%	2022
Wang et al. [13]	Multitask learning model that uses a CNN to extract features from EHRs	81%	2022
Zhang et al. [14]	Ensemble of CNNs, RNNs, and SVMs, each trained on a different data modality	84%	2022
Chen et al. [15]	Transfer learning using a pre-trained CNN	82%	2023
Li et al. [16]	Ensemble of CNNs, RNNs, and SVMs, each trained on a different data modality	83%	2023
Esfahani et al. [17]	Ensemble classifier using different machine learning techniques	88.70%	2023
Das et al. [18]	Neural network of three independent simple neural networks	90%	2022

the model. The F1-score of the proposed model is 85%, which is a measure of the accuracy and precision of the model. The AUC-ROC of the proposed model is 0.85, which is a measure of the ability of the model to distinguish between patients with and without heart disease. A higher AUC-ROC score indicates a better ability to distinguish between the two groups (Table 2).

4.3 Comparison of Heart Disease Prediction Models

The proposed study outperforms all of the existing studies in terms of accuracy and also uses all three of the promising approaches for heart disease prediction: multimodal learning, transfer learning, and ensemble learning. This suggests that the proposed study is a promising approach for heart disease prediction.

5 Conclusion and Future Study

5.1 Conclusion

The proposed study is a promising approach for heart disease prediction. The study uses a multimodal ensemble model that combines the strengths of different data modalities and machine learning models. The results of the study show that the proposed model outperforms existing models in terms of accuracy. The proposed model has the potential to improve clinical practice by providing more accurate and personalized risk assessment and intervention strategies for heart disease. However, further research is needed to validate the model in real-world settings and to explore its potential for predicting other types of cardiovascular diseases. In addition to the potential clinical impact, the proposed model also has important implications for future research on heart disease. The study demonstrates the effectiveness of multimodal learning and ensemble techniques for improving predictive accuracy. This approach could be applied to other medical datasets to develop more accurate and personalized models for predicting a variety of diseases. Overall, the proposed study is a significant contribution to the field of heart disease prediction and has the potential to improve clinical practice and future research.

5.2 Future Study

The proposed model could be further improved by incorporating other data modalities, such as wearable device data, genetic data, and social media data, to provide a more holistic view of patient health. This would allow the model to capture complex relationships between different aspects of patient health and improve its predictive accuracy. The model has been evaluated on a retrospective dataset, but it is important to further validate its effectiveness in real-world settings through clinical trials. This would assess the model's performance in predicting heart disease in new patients and demonstrate its potential for clinical use. In future studies, it would be interesting to explore how the performance of the multimodal model varies across different patient populations. This would help to identify any potential biases in the model and ensure that it is generalizable to a wide range of patients. Another area of future research would be to explore whether the model can be adapted to predict other types of cardiovascular diseases, such as stroke and heart failure. This would expand the potential clinical impact of the model. Finally, it would be interesting to examine whether the model can be used to develop personalized risk assessment and intervention strategies for individual patients. This would involve using the model to predict the risk of heart disease for each patient and then developing tailored interventions to reduce their risk.

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Medical Image Denoising Techniques: Challenges and Future Directions



Deepali Sharma, Amit Doegar, and Savita Walia

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Abstract A major part of studies in digital image processing is devoted to image denoising. Image denoising is a classical research topic in image processing. The transformation of images has become a major method of communication in the modern era, but the image obtained after transmission is often corrupted with noise. MRI is more accurate for imaging soft tissues such as brain and muscles. It can diagnose a broad range of abnormality conditions like cancer, tumour, blockages, internal injuries, etc. It does not depend on the ionizing radiation, hence is a non-invasive technique. The MRI technique provides excellent contrast between the normal tissues and the diseased tissues. MRI is the scanner technology that has undergone tremendous improvements in spatial resolution, acquisition speed, and signal-to-noise ratio (SNR). The various techniques of medical image denoising are reviewed and analysed in terms of certain parameters.

Keywords MRI Image · Denoising · NLM filter · Signal-to-noise ratio

D. Sharma (✉) · A. Doegar · S. Walia
NITTTR, Chandigarh, Chitkara University, Rajpura, India
e-mail: deepalicomp2014@gmail.com

A. Doegar
e-mail: amit@nitttrchd.ac.in

1 Introduction

The most popular technologies for diagnosis are medical pictures produced by magnetic resonance imaging (MRI), computed tomography (CT), and ultrasound imaging (US). Random noise that develops during the acquisition of the photographs frequently has an impact on these photos. Noise not only results in poor visual quality, but it also makes low-contrast objects less visible. One of the more common issues with digital image processing is image denoising [1]. The need for new denoising algorithms to produce images with greater image quality is growing as a result of the quick development of contemporary digital imaging equipment and their expanding range of uses in our daily lives. Denoising is difficult, in particular, in medical imaging since all types of noise, including ultrasound pictures, are known to be tissue dependent and difficult to model. Although noise normally gives an image a bad appearance, the most important aspect is that noise can hide and obscure some features in an image. An image that has noise in it appears mottled, grainy, textured, or snowy. The energy of the high-frequency waves is partially reflected and transmitted at the boundaries between tissues with various acoustic impedances during the imaging process [2]. To identify the regions of interest consistently and accurately, however, speckle must be reduced but anatomical details must be preserved due to the frequently poor diagnostic quality. Image patches have recently been shown to be useful characteristics for denoising images in challenging circumstances. A reliable filter for medical image analysis can be created using the related methods.

Even though there are tones of image denoising techniques available now, taking the noise out of an MRI is still a difficult task. For the denoising of medical images, a number of strategies, including the non-local means (NLM) method, the analytical correction method, and the wavelet-based method, have been presented. The quality of medical photographs is improved by this technique. Due to the low contrast and signal-to-noise ratio, detecting a substantial structure with the human visual system is a very difficult task [3]. The edge preservation is crucial in order to preserve the original medical significance. The methods for denoising medical images are typically broken down into three categories: filtering methods, transform methods, and statistical methods. Linear and non-linear filtering strategies are further separated. Unnecessary frequencies are taken out of the input image using linear filters. It also involves the linear execution of time-varying inputs. Spatial filters and temporal filters are two categories that describe the two parts of a linear filter. With the justification that noise occupies a higher portion of the frequency spectrum, low-pass filtering is applied to the pixel groups. Sharp edges are lost as a result of image blurring caused by the spatial filters, which in particular help to remove noise to a respectable amount [4]. When MSE between the output signal and the desired noise-free signal needs to be reduced, the Wiener filter is used to denoise the signal using stationary random noise. This filter includes a variety of applications without taking into account whether the stationary random method is continuous or discrete, scalar or vector. In contrast to spatial domain filtering approaches, these techniques are helpful for translating the given noisy image into another domain. The modified

image is then subjected to denoising on the basis of the image's various characteristics and noise. According to the chosen foundation transform functions, such as data adaptive or non-data adaptive, the subdivision of transform domain filtering procedures is further carried out. On the other hand, noises that cannot be recognised are eliminated using non-linear filters. Low-pass filters are used on groups of pixels, and they are designed to filter out noise that is present in the higher frequency range [5]. The Non-Local Means Filter is a non-linear filter that is based on the idea that self-spatial similarities exist in the most natural images. This filter makes use of the images' self-similarity where a local kernel is implemented to denoise particular areas of an image. It is also used for patches in the remaining portions of the image that resemble the previously denoised patch. This method is referred to as neighbourhood filtering. These neighbourhood filters enhance the amount of information pertaining to the spatial intensities. The Euclidean distance from the original patch's centre through many patches and resemblance to the original patch is used to determine how confidence weightings are distributed. Additionally, the performance is improved by anisotropic filtering and total variation regularization requirements. The Non-Local Means Filter makes better use of the intrinsic image information. But because it uses a big search space, running it could be a computational barrier. Wavelet transform, curvelet transform [6], and contourlet transform are the three techniques that makeup transform domain approaches. Wavelet denoising is a technique for removing noise from a variety of signals, including one-dimensional signals like EEG and two-dimensional signals. While picture denoising, the Wavelet transform's execution is demonstrated to be effective. In this method, the original image is broken up into a number of wavelets with various scales so that the coefficients of the various wavelets can be improved. This method, in particular, decomposes the majority of noises and effective signals into coefficients utilizing small and large moduli for a noisy image. Thus, by removing low-frequency components, numerous noises can be easily eliminated [7]. The theory of multiscale geometric analysis will serve as the foundation for the proposed curvelet transform technique. The Curvelet has directionality, and anisotropy is used to indicate the directions of the image's edges. The scale, position, and orientation parameters of the Curvelet transform contain the indexing of the frame elements. The localization capability of the Curvelet frame has led to the development of many features. This transformation creates an ideally sparse basis for pseudo-differential operators and Fourier integral operators as well as an optimally sparse representation for singularities supported on curves in two dimensions [8]. A sparse representation of images with contours and textures is provided at both spatial and directional resolutions using the geometrical image transform known as the contourlet transform. Images are obtained with a flexible multi-resolution and directional decomposition thanks to the facilitation of a variety of directions at each scale. Using this transform, two distinct and successive decomposition phases—a multiscale decomposition followed by a multidirectional decomposition—are integrated. The image is initially transformed using the LP approach into one coarse version and a number of LP bandpass pictures. Every LP bandpass image is divided into a number of wedge-shaped sub-bands during the second phase's deployment of DFB and critical subsampling. As a result, direction information is recorded.

Finally, an image is represented as a collection of directional sub-bands at various scales. Regarding the denoising of MR images using statistical or estimating techniques, some efforts are mentioned in the literature. While removing the noise from Magnetic Resonance pictures, estimation of the noise variance is a crucial phase for a variety of reasons [9]. First, a metric for the MR data's quality is offered and used to calculate the signal-to-noise ratio. Following that, understanding this noise variation is useful when examining the MRI system. Finally, it is a crucial parameter used to register, segment, and denoise an image.

2 Literature Review

J.-M. Ling et al. (2020) suggested an image enhancement method for extracting and highlighting the important attributes of suspected rib fractures in X-ray images [7]. The wavelet denoising was combined with image contrast enhancement for eliminating the noises so that the images were read more effectively. Multiple thresholds of horizontal, diagonal, and vertical coefficients were employed in diverse decomposition wavelet levels for discovering wavelets namely db (daubechies), sym (symlets), coif (coiflets), and bior (biorthogonal). The X-ray images were recognized with optimal entropy and SNR (signal-to-noise ratio) using various techniques with thresholding. Consequently, the quality and precision of recognizing X-ray images was enhanced.

Moreover, the diagnostic tool assisted the doctors in maximizing the decision accuracy.

Z. Huang et al. (2021) introduced a new DCNN (deep convolutional neural network) technique to denoise the images for which the information of the anatomical prior was employed [10]. A unified network framework was adopted for processing the anatomical information. The anatomical prior was utilized to illustrate a pattern of weights of the features whose extraction was done from the corresponding LDCT (Low-Dose Computed Tomography) image. The diversity in the contextual information was supported using a spatial attention fusion technique for capturing various local regions of interest in the attention fusion module. The experimental results depicted the effectiveness of the introduced technique to denoise LDCT images.

X. Shi et al. (2022) projected an innovative robust CNN (Convolutional Neural Network) model in which the sparsity denoising operators were embedded into every layer of the network, for mitigating the effect of noisy attributes and investigating that noise [11]. The inter-sample and intra-sample denoising were included in the utilized operators and considered the entire batch of data for alleviating the noise in outputs of every layer. The intra-sample denoising assisted in mitigating the noise in each data. The experimental results indicated that the projected model was robust against adversarial assaults. Furthermore, the correctness of the hypothesis and discovery was also presented.

E. Kang et al. (2018) established a DCF (deep convolutional framelet)-based denoising algorithm in order to restore the LD (low-dose) X-ray CT (Computed

Tomography) [12]. For this, the convergence of the traditional framelet-based algorithm was integrated with the efficacy of DL [12]. The efficiency of the system was enhanced using the existing DCF for interpreting a DL (deep learning) as an ML (multilayer) execution of convolutional framelets with ReLU non-linearity. As a result, two models namely FF (feed-forward) and RNN (recurrent neural network) were developed. These models performed well in contrast to other methods when the detailed texture was retained. The experimental outcomes demonstrated that the established algorithm was applicable to diminish the noise and preserve the texture details of the organs while maintaining the lesion information.

M. Diwakar et al. (2019) developed a novel thresholding function in order to denoise the CT (Computed Tomography) image in shearlet domain [13]. A method noise was inserted on a multivariate shrinkage system on the basis of SURE-LET (stein's unbiased risk estimate and linear expansion of thresholds) notion. The developed approach was computed qualitatively and quantitatively using real noisy CT image. The Gaussian noise was inserted in real CT image and low complexity zoomed objects of noisy CT images were also included in the simulation. Diverse parameters were employed such as PSNR, SSIM, ED, and DIV in the experimentation. The outcomes of experiments exhibited that the developed approach led to enhance the outcomes in every scenario.

Y. A. Bayhaqi et al. (2022) designed a DL (deep-learning) technique to denoise the images based on the frame-averaging technique [14]. Unlike the traditional technique, this technique helped in deriving an image which offered the same image quality as the frame-averaging. An analysis was conducted on the designed technique with regard to the impact of tissue on the accuracy of the classifier. The findings reported that the accuracy was found higher to denoise the image. The denoised images were applied to train the implemented classifier and offered superior accuracy. The designed technique was useful for classifying the tissue in real time in smart laser osteotomy.

B. Yagoub et al. (2021) recommended a new CNN (convolutional neural network)-based method to denoise 3 MeV x-ray image [15]. Furthermore, a mathematical model was put forward for creating a noise that occurred when the x-ray cargo images were available in a lower amount. The artificially generated images were considered to train this method and the real time images obtained from an x-ray dual energy system deployed for testing the recommended method. The results proved that the recommended method offered PSNR (peak-to-noise ratio PSNR) up to 41.2 and SSI (structure similarity index) up to 0.956.

D. B. T. Le et al. (2021) presented a technique to simulate scanner-specific images involved in a public 1.5 T and 3 T database of MR images for which an SEM (signal encoding matrix) was introduced in which explicitly modelled imaging gradients and fields were put together [16]. A stacked, U-Net framework was exploited for mitigating the noise from the system and eliminating the artefacts which the inhomogeneous B0 field, non-linear gradients, under-sampling of k-space produced. The image was reconstructed for improving the MRI (magnetic resonance imaging) scans of the lower field. The final network was employed to perform the post-processing phase. The results depicted that the introduced approach was capable of enhancing

the image quality qualitatively and quantitatively. B. Kim et al. (2022) formulated an innovative model for training a CNN (Convolutional Neural Network)-based denoiser for which the real paired CT (Computed Tomography) images were not required or the proprietary information related to raw scan data and a reconstruction kernel was not accessed [17]. The major focus was on estimating a reconstruction kernel from normal-dose CT images and synthesizing the paired CT images based on an IDNI (image domain noise insertion) method. The formulated model was evaluated on expanded cardiac-torso phantoms and the 2016 Low-Dose CT Grand Challenge datasets. A comparative analysis was conducted on the formulated model against the traditional technique which led to generate paired data after the insertion of noise to raw scan data and reconstruction of images with the known reconstruction kernel. Moreover, diverse metrics such as PSNR (peak signal-to-noise ratio), SSIM (structural similarity index measure), MTF (modulation transfer function), and NPS (noise power spectrum) to compute this model. The results confirmed the supremacy of the formulated model over others to denoise the images. S. More et al. (2020) suggested a new and improvised denoising method in which SA_CNN (sparse aware with convolution neural network) algorithm was implemented to discover several medical modalities [18]. The patch creation and dictionary approaches were adopted in CNN to acquire the information. The suggested method yielded lower PSNR (peak signal-to-noise ratio), SSIM (structural similarity index measure), and MSE (mean squared error). The computing time was also optimized to maximize the efficacy. Moreover, the IoHT (Internet of Healthcare Things) offered security with vault and challenge systems.

M. Malekzadeh et al. (2021) projected a UDT (unsupervised domain translation) method to denoise PET image dataset which had restored noisy real PET images when shared latent space information was learned across a simulated clean image domain and a real noisy image domain [19]. The projected method was trained and validated on noisy human brain PET scans on the basis of BrainWeb digital phantom. The projected method assisted in enhancing the Gaussian filtering with regard to PSNR (peak signal-to-noise ratio).

M. A. Kande et al. (2021) introduced a novel deep generative framework known as SiameseGAN in order to denoise LSNR (Low signal-to-noise ratio) B-scans of SDOCT [20]. This framework was based on GAN (Generative Adversarial Network) in which a Siamese twin network was deployed. This framework helped the generator in creating the denoised images. An apriori high-quality image was not utilized in this framework to perform the denoising and it consumed the least time. The experimental results indicated the adaptability of the introduced framework for diminishing the speckle noise at a quick rate in comparison with the traditional methods (Table 1).

3 Objectives

The following are the various research objectives:

Table 1 .

Author	Year	Technique used	Findings
J. -M. Ling et al	2020	An image enhancement method	The X-ray images were recognized with optimal entropy and SNR (signal-to-noise ratio) using various techniques with thresholding. Moreover, the diagnostic tool assisted the doctors in maximizing the decision accuracy.
Z. Huang et al	2021	A new DCNN (deep convolutional neural network) technique	The experimental results depicted the effectiveness of the introduced technique to denoise LDCT images.
X. Shi et al	2022	An innovative robust CNN (Convolutional Neural Network) model	The experimental results indicated that the projected model was robust against adversarial assaults. Furthermore, the correctness of the hypothesis and discovery was also presented.
E. Kang et al	2018	DCF (deep convolutional framelet)-based denoising algorithm	The experimental outcomes demonstrated that the established algorithm was applicable to diminish the noise and preserve the texture details of the organs while maintaining the lesion information.
M. Diwakar et al	2019	A novel thresholding function	The outcomes of experiments exhibited that the developed approach led to enhance the outcomes in every scenario.
Y. A. Bayhaqi et al	2022	DL (deep-learning) technique	The designed technique was useful for classifying the tissue in real time in smart laser osteotomy.
B. Yagoub et al	2021	New CNN (convolutional neural network)-based method	The results proved that the recommended method offered PSNR (peak-to-noise ratio) up to 41.2 and SSI (structure similarity index) up to 0.956.
D. B. T. Le et al	2021	SEM (signal encoding matrix)	The results depicted that the introduced approach was capable of enhancing the image quality qualitatively and quantitatively.
B. Kim et al	2022	CNN (Convolutional Neural Network)-based denoiser	The results confirmed the supremacy of the formulated model over others to denoise the images.
S. More et al	2020	SA_CNN (sparse aware with convolution neural network) algorithm	The suggested method yielded lower PSNR, SSIM, and MSE. The computing time was also optimized to maximize the efficacy. Moreover, the IoHT (Internet of Healthcare Things) offered security with vault and challenge systems.

(continued)

Table 1 (continued)

Author	Year	Technique used	Findings
M. Malekzadeh et al	2021	UDT (unsupervised domain translation) method	The projected method assisted in enhancing the Gaussian filtering with regard to PSNR (peak signal-to-noise ratio)
N. A. Kande, et al	2021	SiameseGAN	The experimental results indicated the adaptability of the introduced framework for diminishing the speckle noise at a quick rate in comparison with the traditional methods

1. To Study and analyse various MRI image denoising techniques for image processing
2. The implemented various MRI image denoising techniques
3. Improve the performance of PNLM algorithm for image denoising to define variable window size 4.

Implement proposed PNLM algorithm and compare with existing PNLM algorithm in terms of SSIM, MSE and PSNR

4 Research Methodology

In this work, the NLM technique is studied which is the filtering technique to denoise the MRI Images. The MRI Images has the raisin noise which reduces quality of the image. The PNLM is the improved method of NLM in which the pixels of the image processed parallel for the image denoising. The GLCM algorithm is applied with the PNLM which improves performance of PNLM by selecting window size efficiently. The parameters which are used for the analysis are Peak signal-to-Noise Ratio (PSNR), Mean Square Error (MSE), Mean Structural Similarity index (MSSIM). GLCM stands for the Grey-level Co-occurrence Matrix, it is also known as the Grey Tone Spatial Dependency Matrix. A GLCM is defined as a matrix where the number of rows and column is equal to the number of grey levels in an image. It is in tabulation format and tells how often different combinations of pixel brightness values occur in an image. GLCM method is used to extract statistical surface parameters such as Inverse Difference Moment, Entropy, Angular Second Moment, and Correlation. It is proved to be the best statistical method for extracting textual features from an image. It also contains the second-order statistical information of the adjoining pixels of an image. This second-order statistical method feature is used for the implementation in VERILOG language. GLCM is calculated using a grey co-matrix in a scaled version of the image. Suppose 'I' is a binary image, grey co-matrix scales the image into two gray levels. If 'I' is an intensity image, it scales the image into eight grey levels. 'Numlevels' parameter is used to specify the number of grey levels and the gray co-matrix scales the values using 'Gray Limits' parameter.

5 Conclusion

The filtering techniques are compared and a new technique is designed to denoise the MRI Images which has Raisin Noise. The existing techniques are NLM (Non-Local Mean Filter), PNLM (Parallel Non-Local Mean Filter) which is an improved version of NLM. In the NLM filtering technique, the image is taken as input from that image noise level is estimated from the MRI Image. The NLM filtering technique will take another input which is window size. The NLM filter analyses the pixel's similarity within that window size and pixels that are dissimilar will be removed from the image. The pixels which are removed from the image that pixels are called residual. In future, non-local mean filtering technique can be improved to improve image quality.

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Navigating the Legal Landscape: Ethical and Regulatory Considerations in AI-Driven Healthcare



Shreya Banerjee, Ayush Dumka, Tejasv Singh, and Archana Patel

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Abstract Artificial Intelligence (AI) is gaining prominence in healthcare, promising to enhance the efficiency and quality of medical services. However, this rapid integration of AI in clinical practice brings forth complex medical-legal questions that demand scrutiny. To begin, this study investigates the imperative of establishing a framework of medical law principles to govern AI applications, taking into account the intricate interplay between existing legal statutes and established medical customs and practices. Furthermore, it delves into the legal dimensions of AI adoption in healthcare, encompassing various critical issues. It explores strategies for the optimal implementation of AI in healthcare settings, focusing on the spatial, temporal, and cross-border dynamics of AI operations. The present study also addresses the issue of legal accountability for AI performance, identifying liable parties and proposing mechanisms for ensuring their accountability. Furthermore, it emphasizes the necessity of safeguarding patients' personal data when employing AI technology, an area currently lacking well-defined regulations. This study underscores the urgency of addressing the legal and ethical challenges associated with the increasing use of AI

S. Banerjee

Department of Computer Science and Engineering, Manipal Institute of Technology Bengaluru,
Manipal Academy of Higher Education, Manipal, India
e-mail: shreya.banerjee@manipal.edu

A. Dumka · T. Singh

National Forensic Sciences University, Gandhinagar, Gujarat, India

A. Patel (✉)

National Forensic Sciences University, Delhi, India
e-mail: archana.patel@nfsu.ac.in

in healthcare. It advocates for algorithmic transparency, patient privacy protection, and robust cybersecurity measures. As AI options in medicine continue to evolve, healthcare professionals and leaders must navigate this ever-changing medical-legal landscape to ensure the responsible and beneficial integration of AI in clinical practice and facilities.

Keywords Artificial intelligence · Healthcare · Legal dimensions · Legal accountability · Transparency

1 Introduction

The increasing demands on healthcare systems, driven by factors like rising patient needs, the prevalence of chronic diseases, and resource limitations, have spurred the adoption of digital health technologies. These technologies have generated an influx of healthcare data, presenting an opportunity to transform healthcare delivery and outcomes. Artificial Intelligence (AI), particularly machine learning, emerges as a pivotal tool in this transformation. AI encompasses the capacity of computer programs to perform tasks associated with human intelligence, including learning, reasoning, adaptation, sensory understanding, and interaction. Unlike traditional computational algorithms that follow predefined rules, AI systems learn from training data, allowing them to uncover complex correlations within extensive datasets. AI's potential to revolutionize healthcare lies in its ability to extract valuable insights from the vast digital data generated during healthcare processes. AI implementation typically involves a combination of software and hardware, with a focus on algorithms like artificial neural networks (ANNs). ANNs mimic the human brain's interconnected neurons, processing data through weighted communication channels to identify intricate patterns within extensive datasets. Through a process of training and continuous learning, AI systems enhance their accuracy and prediction capabilities. However, the integration of AI into healthcare also raises concerns about data accuracy and security, especially in a high-stakes environment where errors can have severe consequences for patients. Effective collaboration between AI and clinicians can optimize healthcare delivery, offering evidence-based management and medical decision guidance. AI has applications in various aspects of healthcare, including diagnosis, drug discovery, epidemiology, personalized care, and operational efficiency. Nevertheless, ethical considerations are paramount when implementing AI in healthcare, with a need for a robust governance framework to ensure patient safety and prevent unethical behavior. AI is increasingly common in healthcare, offering potential improvements in diagnostic accuracy and efficiency. However, its growing autonomy and impact on the role of physicians raise medical-legal questions that need to be considered by healthcare leaders and managers.

Sunam Jassar et al. [1] conducted an in-depth analysis of the legal aspects of AI in healthcare, with a specific focus on Canada. It discusses how legal principles can be applied to a wide range of AI applications and explores the emerging international

guidance on AI in healthcare. Additionally, the article suggests that professional bodies should provide guidance to ensure the appropriate and safe use of medical AI. Karl Stoger et al. [2] delve into both technical and legal aspects of data cleansing, emphasizing the critical importance of data quality in the field of AI, particularly in medicine. Flawed data, even from sensors, can cause problems. In the context of medical AI, these legal requirements have significant implications: A) Faulty data in medical AI risks patient safety. B) Introducing such products may lead to liability under European product rules. C) Notified bodies assessing products may also be liable if due diligence is lacking. D) Users can face liability under medical malpractice laws if they use faulty data. Data cleansing is crucial to ensure medical AI's proper functioning but entails technical and legal challenges. W. Nicholson Price II et al. [3] thoroughly examine the role of artificial intelligence (AI) in medicine, shedding light on its legal implications across four crucial domains: regulation, tort, intellectual property, and privacy. The article highlighted examples of AI applications, such as skin cancer detection and trauma prognosis, which can outperform human experts. It also mentioned AI-driven smartphone apps for identifying developmental disorders in infants and autism in young children.

Nithesh Naik et al. [4] delve into the legal and ethical dilemmas brought about by Artificial Intelligence in healthcare, including concerns related to privacy, bias, and the evolving role of human judgment. This research review emphasized the need for algorithmic transparency, privacy protection, cybersecurity, and safeguarding the interests of all stakeholders. Vasilij Andreevich Laptev et al. [5] highlighted the swift integration of cutting-edge digital technologies, particularly artificial intelligence (AI), in the healthcare sector, showcasing notable advancements in disease diagnosis, prediction, and surgical assistance. They explored how AI functions, the participants in medical-legal relationships, the legal status of AI, and its competencies. Additionally, it addressed the issue of legal liability for those responsible for AI system performance. Ritika Raj et al. [6] highlighted the transformative potential of Artificial Intelligence in the medical field, offering the promise of more precise diagnoses and treatments surpassing human professionals. They explained that determining accountability in cases of medical negligence involving artificial intelligence (AI) is intricate. This article predominantly centers on the legal facets of AI integration in healthcare, exploring the legal dimensions of key issues such as:

- The delineation of medical law principles to regulate AI application, considering the interplay between existing legal statutes and medical customs and practices.
- The dynamics of AI operation concerning its spatial, temporal, and cross-border dimensions.
- Legal accountability for AI performance, encompasses the identification of liable parties and the mechanisms to ensure their accountability.
- Strategies for the optimal implementation of AI in healthcare.
- Safeguarding the personal data of patients when employing AI technology.

The paper is organized as follows: Sect. 2 explains how AI works in medical science. Section 3 shows ethical frameworks for implementing AI in healthcare. Section 4 provides a set of recommendations and the last section concludes the article.

2 Revolutionizing Healthcare: The Mechanics of AI in Medical Science

Artificial Intelligence (AI) is transforming medical science by leveraging advanced computational algorithms and data analytics to revolutionize disease understanding, diagnosis, and treatment. In the medical field, data collection and management are foundational. Electronic Health Records (EHRs) contain vital patient information, with structured and unstructured data. AI assists medical professionals in analyzing complex healthcare data, from electronic health records to medical imaging, providing rapid insights and personalized treatment recommendations. AI excels in medical imaging analysis, pathology, clinical decision support, and drug-target interaction prediction [7]. It enables real-time monitoring and predictive analytics, aiding in early detection and proactive interventions. In addition, Natural Language Processing (NLP) helps extract insights from unstructured clinical text data. Robotics and automation enhance precision and reduce human error in surgery, diagnostics, and routine tasks. Personalized medicine tailors treatments based on genetic data, lifestyle, and medical history. AI accelerates clinical trials, streamlining patient recruitment, data analysis, and drug discovery. It expedites research and improves the efficiency of developing new treatments. In the evolving landscape of healthcare AI, responsible use prioritizes transparency, ethical guidelines, and patient-centric care. AI continues to reshape medical science, offering unprecedented capabilities for better patient outcomes and innovative advancements. However, the integration of AI into healthcare also poses challenges. These include ethical considerations, data privacy concerns, and regulatory compliance. Responsible use of AI in healthcare is crucial, demanding transparency, accountability, and bias mitigation.

2.1 Existing Standards

At present, there exists a notable absence of multilateral international treaties or conventions that establish comprehensive provisions regarding the utilization of AI technology, particularly within the healthcare sector. Instead, a limited number of documents have been ratified, which contribute to the establishment of the foundational framework for international legal regulation in the realm of AI. These documents serve as recommendations rather than binding agreements. Among these noteworthy documents are:

- The Okinawa Charter on Global Information Society (G8 Kyushu-Okinawa Summit Meeting 2000, Kyushu-Okinawa Japan) (G8 2000): This charter underscored the necessity for a regulatory framework that fosters collaboration to optimize global networks and mitigate the digital divide.
- OECD Council Recommendation on Artificial Intelligence (Adopted by the Council at the Ministerial Level on 22 May 2019) (OECD Council Recommendation on Artificial Intelligence 2019): This significant document represents

the first intergovernmental standard concerning artificial intelligence. It outlines general principles for AI deployment and provides recommendations to national governments regarding AI development.

- G20 Ministerial Statement on Trade and Digital Economy (2019, Japan) (G20 Ministerial Statement on Trade and Digital Economy 2019): This statement, issued on behalf of the participating states of the G20, endorsed principles for the advancement of AI.

These documents, though not legally binding, play a crucial role in shaping international discourse and cooperation concerning the responsible use of AI technology, including its application in healthcare.

Within international legal governance, international technical standards assume paramount importance. These standards are conceived and promulgated by esteemed organizations such as the International Organization for Standardization (ISO), the International Electrotechnical Commission (IEC), and the International Telecommunication Union (ITU), all of which are dedicated to the formulation and dissemination of standards, directives, suggestions, and technical documentation [8].

In the context of healthcare, as well as across various other critical sectors, the integration of artificial intelligence (AI) systems has ushered in a new era of innovation and transformation. ISO (International Organization for Standardization) standards play a pivotal role in ensuring the highest standards of performance, safety, and compliance for AI systems. This comprehensive list explores ISO standards that are particularly relevant to AI systems in healthcare, as well as those impacting security systems, industrial data, cybersecurity, privacy protection, information security, and software and systems engineering. These standards collectively empower organizations to harness the potential of AI while addressing unique challenges and requirements within the healthcare domain, ensuring the seamless integration and application of AI technologies in this vital sector.

- ISO/IEC 20546 [9]
- ISO/IEC TR 20547-2 [10]
- ISO/IEC 20547-3 [11]
- ISO/IEC TR 20547-5 [12]
- ISO/IEC 2382 [13]
- ISO/IEC 19501 [14]
- ISO/IEC 27041 [15]
- ISO/IEC 20246 [16]
- ISO/IEC TR 24028 [17]
- ISO/IEC 25000 [18]
- ISO/IEC 23053 [19]
- ISO/IEC TR 24029-1 [20]
- ISO/IEC TR 24029-2 [21]
- ISO/IEC 22745-1 [22]
- ISO/IEC 22989 [23]
- ISO/IEC 27001 [24]
- ISO/IEC 27002 [25]

- ISO/ IEC 27701 [26]

These standards and documents were developed as part of ISO's efforts to establish comprehensive guidelines and definitions for the field of artificial intelligence and big data.

2.2 Regulatory Framework and Ethical Considerations for AI in Healthcare: A Comprehensive Analysis of the USA

FDA Regulation for Medical Devices [27]:

- **FDA's Digital Health Program:** The FDA has established a Digital Health Program to regulate digital health technologies, including AI applications used in healthcare. This program aims to provide a framework for the regulation of software as a medical device (SaMD).
- **Software as a Medical Device (SaMD):** The FDA provides guidance on the regulation of SaMD, which includes AI-based medical software. Developers of SaMD must follow FDA guidelines for premarket submissions and post-market surveillance.
- **De Novo Classification:** For AI-based medical devices that do not have a predicate device, the De Novo classification pathway may be used. This pathway allows the FDA to evaluate and classify novel medical devices.
- **Premarket Notification (510(k)):** Some AI-based medical devices may be eligible for the 510(k) pathway, which requires a demonstration of substantial equivalence to an existing legally marketed device.

FDA's Artificial Intelligence/Machine Learning (AI/ML) Action Plan: The FDA's AI/ML Action Plan outlines the agency's approach to regulating AI and machine learning technologies in healthcare. It emphasizes transparency, robust performance, and real-world monitoring.

Privacy and Data Security:

- **HIPAA (Health Insurance Portability and Accountability Act):** HIPAA sets standards for the protection of patient health information (PHI) and applies to healthcare providers and entities handling PHI, including AI applications.
- **HITECH Act:** The Health Information Technology for Economic and Clinical Health (HITECH) Act, part of the American Recovery and Reinvestment Act of 2009, strengthens HIPAA's privacy and security provisions.

FDA's Software Precertification (Pre-Cert) Pilot Program: The Pre-Cert program is designed to streamline the regulatory process for software-based medical devices, including AI applications. It focuses on evaluating the excellence of the software developer's organizational culture and processes.

Cybersecurity Regulations: The FDA provides guidelines and recommendations for cybersecurity practices to protect AI-based medical devices and patient data from security threats.

State Regulations: Some states may have additional regulations and licensing requirements related to the use of AI in healthcare, particularly for medical practitioners using AI tools.

Research and Development: AI research involving human subjects is subject to ethical review by Institutional Review Boards (IRBs) to ensure patient safety and informed consent.

Real-World Evidence (RWE): The FDA is increasingly interested in using real-world evidence, including data from electronic health records (EHRs) and wearables, to inform regulatory decisions regarding AI-based medical devices.

2.3 The Actions Pursued by the European Union to Oversee AI in the Healthcare Sector

The European Union (EU) has taken significant steps to regulate the use of artificial intelligence (AI) in healthcare. In February 2020, the European Commission issued a white paper, along with related communication and reports, outlining its approach to AI regulation to ensure consistency across EU member states and prevent market barriers [28]. The EU recognizes healthcare as a critical area for AI application. The key elements of the EU's approach to AI regulation in healthcare include:

High-Risk AI Applications: The EU distinguishes between high-risk and low-risk AI applications. The new regulatory framework and conformity assessment are primarily targeted at high-risk AI applications, especially in sectors where significant risks are expected, including healthcare.

Certification: The EU plans to introduce a “trustworthy AI” certification to encourage voluntary compliance with regulations, particularly in low-risk AI use cases. Certified systems found in violation of rules may face fines. Certification requirements will focus on training data quality to ensure safe, non-discriminatory, and privacy-protective AI systems. Developers and deployers must demonstrate the breadth and representativeness of training data, considering dimensions like gender and ethnicity.

Privacy Concerns: While the white paper does not propose specific new requirements for protecting personal data in AI systems, it suggests that existing regulations such as the General Data Protection Regulation (GDPR) and related directives should cover these concerns. The EU emphasizes the importance of record-keeping related to AI programming, training data, and data used during the AI system's lifecycle.

Transparency and Accountability: Developers and deployers of high-risk AI systems must provide detailed information about the system's capabilities, limitations, intended purposes, expected accuracy, and functionality conditions. Citizens interacting with AI systems should be clearly informed when they are not interacting with a human. The EU plans to draft new laws, including a ban on "black-box" AI systems, especially in critical applications like medical devices. It calls for AI systems to be robust, accurate, and capable of handling errors, with a need for human oversight varying based on the specific case.

Liability: The EU recognizes that AI technologies may complicate product safety liability issues when harm occurs. To address this, it proposes apportioning liability based on the actor best positioned to address the risk of harm. This approach aims to ensure that individuals harmed by AI-related issues can obtain compensation. The EU also suggests imposing strict liability for products with defective software or digital features, and it seeks input on potentially reversing the burden of proof in cases of AI-related malfunctions.

2.4 India's Efforts for Regulating the AI Medical Sector

The Digital Information Security in Healthcare Act (DISHA) was introduced by the Ministry of Health and Family Welfare (MoHFW) with the primary goal of enhancing the security of healthcare data in India while ensuring individuals have full control over their health information. Under DISHA, when a patient's medical data is recorded in an electronic health record (EHR) by a healthcare provider, it receives comprehensive protection within the healthcare system.

DISHA outlines three key objectives:

- Establishment of Digital Health Authorities at both national and state levels.
- Enforcement of stringent privacy and security measures for electronic health data
- Regulation of the storage and exchange of electronic health records.

Moreover, the draft legislation discusses the formation of National and State Electronic Health Authorities (NeHA and SeHA), which would not only safeguard Indian citizens' health data but also oversee its portability. However, there are certain notable shortcomings in DISHA. It appears that the legislation places a heavy emphasis on the patient's role. While granting patients significant control over their medical data is important, the act lacks clarity on how a country with over a billion people will be educated about: Their rights regarding their health data, The importance of providing informed consent, The processes for deciding when to grant or deny consent., The methods for receiving and responding to consent requests, among other critical aspects. Addressing these issues is crucial to ensure the effective implementation of DISHA and to protect the privacy and security of healthcare data in India comprehensively.

Medical Device Rules, 2017 The Medical Devices Rules of 2017 were developed with the primary objective of establishing a clear differentiation between medical devices and pharmaceuticals for regulatory purposes. These rules provide a specific classification for medical devices and delineate the scope of their regulation.

Electronic Health Record Standards, 2016 The EHR (Electronic Health Record) Standards serve as an endeavor to regulate the ownership of data and establish privacy standards concerning the storage of health-related information obtained from patients. This encompasses data originating from medical institutions, medical devices, self-care devices, and associated systems.

However, a recent review report conducted by the Ministry of Electronics and Information Technology (MeitY), titled “Adoption of Electronic Health Records: A Roadmap for India,” has shed light on the inadequate ICT (Information and Communication Technology) infrastructure in government hospitals and dispensaries. Notably, only a few major public healthcare institutions like the All India Institute of Medical Sciences (AIIMS) and the Postgraduate Institute of Medical Education and Research (PGIMER) possess computers and connectivity. With the exception of a small number of prominent healthcare organizations in India, there has been limited progress in implementing EHR systems. Consequently, the adoption and successful implementation of EHRs face considerable challenges, especially in the absence of fundamental infrastructure support.

Ethical Frameworks for Implementing AI in Healthcare.

The ethical principles governing the application of Artificial Intelligence (AI) in healthcare and other fields play a pivotal role in guiding developers, users, and regulators. They are rooted in the core values of human dignity and the intrinsic worth of individuals and encompass a range of principles including non-harm, beneficence, justice, and respect for autonomy. These principles collectively shape the responsible use of AI and are essential for maintaining privacy, avoiding harm, and promoting fairness. Human rights, such as dignity, equality, and privacy, intersect with these ethical principles, strengthening the moral and legal framework surrounding AI. While the principles remain constant, their practical implementation can vary based on cultural and societal contexts. The ethical principles, which are universally accepted, serve as a foundation for responsible AI use in healthcare. To ensure their adherence, these principles should be integrated into professional and technological standards for AI:-

Non-harm (Nonmaleficence): This principle underscores the importance of avoiding harm to individuals, often framed as “Do no harm.” It requires minimizing risks while maximizing benefits and ensuring a careful balance between potential risks and benefits.

Beneficence: Focused on promoting well-being, beneficence seeks to minimize the risk of harm while maximizing benefits for individuals and society as a whole.

Justice (Fairness): Justice demands the fair treatment of all individuals, preventing discrimination, neglect, manipulation, domination, or abuse. It encompasses the concept of fairness in all AI-related activities.

Preserving Autonomy: AI integration into healthcare must not compromise human autonomy. Individuals should maintain control over healthcare decisions, and AI systems should be designed to assist rather than replace human decision-makers. Transparency, informed consent, and robust data protection laws are essential in this context. Advancing Human Well-being, Safety, and the Public Interest: AI technologies should prioritize individual well-being and public safety. Rigorous testing, quality control, and continuous monitoring are necessary to ensure that AI technologies operate safely and effectively. Preventing harm and addressing potential biases are integral aspects of this principle.

Ensuring Transparency, Explainability, and Comprehensibility: AI systems should be transparent and understandable to all stakeholders. Transparency involves providing information about the system's design and application, while explainability ensures that AI decisions can be understood, especially in critical healthcare contexts.

Ensuring Inclusiveness and Equity in AI for Healthcare: AI technologies should be designed and evaluated with diverse participation to prevent bias and discrimination. They should also be accessible and adaptable to different contexts and populations, promoting equitable access and avoiding the digital divide.

Promoting Responsive and Sustainable AI: AI technologies should be continuously monitored and evaluated to ensure they meet expectations and requirements. Their introduction should align with sustainability goals and be responsive to changing healthcare needs.

In addition to these foundational principles, safeguarding individual privacy is paramount, as it aligns with both respecting autonomy and preventing harm. These ethical principles are instrumental in guiding stakeholders involved in AI development, deployment, and regulation in healthcare, including clinicians, developers, healthcare administrators, policymakers, and government entities.

3 Navigating Regulatory Challenges

3.1 *Informed Consent in AI Healthcare*

The integration of AI into healthcare, encompassing applications like medical imaging, diagnostics, and surgical assistance, holds great promise for revolutionizing the patient-clinician relationship. However, this transformation also raises important ethical considerations, particularly concerning the principles of informed consent. The ethical discourse surrounding AI in healthcare often overlooks the critical issue of informed consent, despite it being one of the most immediate challenges in the adoption of AI in clinical practice. Informed consent has long been a fundamental tenet of medical ethics, ensuring that patients have a comprehensive understanding

of their medical care, including its risks and benefits. With the introduction of AI into healthcare, a new layer of complexity is added.

One of the most challenging scenarios arises when AI operates using “black-box” algorithms. These algorithms, often stemming from non-interpretable machine-learning techniques, pose difficulties for clinicians as they may not fully understand how the AI arrives at its decisions. In such cases, clinicians face the ethical dilemma of disclosing their inability to fully interpret AI-generated diagnoses or treatment recommendations. Balancing transparency and patients’ right to explanation under regulations like the EU’s GDPR becomes paramount [29]. Privacy concerns further complicate matters, especially when patients are hesitant to allow certain categories of data, such as genetic information and family history, to be used by AI systems. Striking the right balance between safeguarding patient privacy and ensuring AI’s safety and effectiveness remains a challenge. Additionally, the rise of AI health apps and chatbots introduces another layer of complexity in informed consent. Unlike traditional informed consent processes, these applications often rely on user agreements, which individuals agree to without face-to-face interaction. Unfortunately, many users fail to read or understand these agreements, potentially leading to unintended consequences for their healthcare data.

3.2 Addressing Bias in AI Healthcare

Algorithmic fairness and biases are critical considerations in the development and deployment of AI in healthcare. While AI has the potential to democratize expertise and expand healthcare access to remote areas, it also carries the risk of perpetuating biases and discrimination, making it imperative for AI developers to address these issues at every stage of product development. The foundation of any AI system lies in the data it is trained on. In healthcare, biased AI can have severe consequences. Efforts to mitigate biases include collecting more data from minority populations and clearly specifying the populations for which an algorithm is appropriate. However, the complexity of some algorithms, especially “black-box” models, presents challenges for understanding and rectifying biases. In cases where transparency is limited, the debate centers on whether explainability is necessary for health AI or if accuracy alone suffices. Some argue that as long as AI provides accurate diagnoses, the intricacies of its decision-making process are less relevant. Positive results from randomized clinical trials could serve as evidence of safety and effectiveness in such cases.

3.3 Data Privacy

Privacy is a central concern in the development and deployment of black-box medicine, particularly in two critical areas: the collection of extensive healthcare

data to train algorithms and the sharing of this data for oversight purposes. Algorithm developers must gather data from various sources to train machine-learning algorithms. These data, along with information on how the algorithms perform in practice, may be shared with other entities within the healthcare system for evaluation and validation. Privacy concerns, especially those mandated under the Health Information Portability and Accountability Act's (HIPAA) Privacy Rule, are paramount. HIPAA governs and restricts the disclosure and use of "protected health information" by "covered entities," primarily healthcare providers, insurers, clearinghouses, and their business associates.

Navigating the complexities of HIPAA and addressing the privacy concerns of individuals whose data is used in black-box medicine present ongoing legal challenges. A noteworthy example of privacy breaches occurred when the Royal Free NHS Foundation Trust provided personal data from approximately 1.6 million patients to Google DeepMind for the clinical safety testing of the "Streams" app [30]. Protecting patients against adverse impacts beyond the doctor–patient relationship is imperative. This includes preventing negative consequences related to health or other insurance premiums, job opportunities, personal relationships, and more. Robust antidiscrimination laws, similar to those in place for genetic privacy, may be necessary. Additionally, AI health apps that share patient data with family members and friends raise new privacy issues, as these individuals may not have legally enforceable obligations of confidentiality. The right to withdraw data is another sensitive issue.

3.4 *Cybersecurity*

Cybersecurity is a critical concern when addressing the legal challenges associated with the use of AI in healthcare. As healthcare services, processes, and products increasingly operate within the Internet of Things (IoT), the underlying infrastructure becomes vulnerable to various cyber and physical threats and hazards. Sophisticated cyber actors, including criminals and nation-states, can exploit vulnerabilities in healthcare systems to steal or manipulate critical information and disrupt essential medical services. Targets in the healthcare sector may range from hospital servers and diagnostic tools to wearables, wireless smart pills, and medical devices. These entities can be infected with malware like viruses, Trojan horses, or worms, compromising patient privacy and safety.

Corrupted data or infected algorithms can lead to incorrect and unsafe treatment recommendations, posing significant risks to patient well-being. AI systems are particularly susceptible to manipulation. The need for enhanced cybersecurity was starkly illustrated by the "WannaCry" ransomware attack, a global cyberattack that paralyzed the UK's National Health Service (NHS), affected FedEx, and impacted over 300,000 computers in 150 countries. Such events have prompted legislative responses, such as the European Cybersecurity Act (Regulation (EU) 2019/881) and the Cybersecurity and Infrastructure Security Act of 2018 in the United States

[31]. However, cybersecurity is a global concern, and data breaches and cyberattacks transcend national borders. Therefore, there is a growing need for an internationally enforceable regulatory framework on cybersecurity that ensures a high level of cybersecurity and resilience across borders. Ultimately, robust cybersecurity measures are essential to safeguard AI-driven healthcare products, services, and processes and protect patient privacy and safety in an interconnected world.

3.5 Intellectual Property Law

Intellectual property law plays a pivotal role in the development of AI in healthcare and presents various challenges and opportunities. In the context of AI and big data-driven healthcare technologies, protecting intellectual property (IP) rights is essential due to the significant costs and risks associated with translating AI advancements into real-world products and services. Commercial protection of AI and data-driven healthcare technologies has become a critical consideration, and discussions regarding open science, innovation, data sharing, and access to technology and data are ongoing. AI and the data that fuel it can be protected through a combination of IP rights, including long contracts, copyright, trade secrets, database rights (in Europe), competition law, and personal data integrity rights. Data often becomes the subject of litigation, especially when disputes arise over access to and ownership of data. Challenges emerge when it comes to accessing and using data, particularly in data mining and analytics scenarios. Various forms of IP rights can protect references to or copying of databases and information, leading to disputes between stakeholders, such as data scientists and data “owners.” The nature of overlapping rights and remedies can be misunderstood in big data applications, further complicating the legal landscape. The balance between promoting innovation, protecting IP rights, ensuring data privacy, and facilitating data sharing remains a complex challenge in the healthcare sector’s transition to AI-driven solutions. Addressing these challenges will require careful consideration and may involve the development of legal frameworks that encourage data sharing, the establishment of data-sharing intermediaries, and public–private partnerships while safeguarding data privacy. Overall, the integration of AI in healthcare raises important questions about the intersection of IP law, data sharing, innovation, and patient privacy, necessitating ongoing legal and regulatory discussions to navigate these complex issues effectively.

3.6 Tort Law

The realm of tort law intersects with the challenges posed by black-box medicine in several critical ways. Firstly, if there are inherent flaws within the algorithms themselves or if regulatory oversight fails to ensure the high quality and safety of these algorithms, developers of these technologies might be held liable under

tort law. However, courts have historically exhibited reluctance in applying product liability principles to software developers, especially within the healthcare software domain. This reluctance partly stems from the perception that healthcare software primarily aids healthcare providers in making informed decisions, with the ultimate responsibility for the decision resting on the provider. Black-box medicine introduces a paradigm shift, wherein healthcare decisions are suggested or made by algorithms that healthcare professionals may not fully understand or have the capacity to understand. This raises important questions about the extent of responsibility healthcare providers and institutions bear in a healthcare landscape shaped by black-box algorithms. Should they evaluate the quality of these algorithms, relying on procedural measures like validation and performance statistics before integrating them into patient care? Furthermore, when an algorithm recommends an intervention that appears mundane but potentially unhelpful, costly, or even risky, should healthcare providers exercise discretion and question the algorithm's recommendation? While the answer may seem obvious in terms of patient care, excessive caution can impede the benefits that black-box medicine promises to offer by uncovering hidden insights in vast datasets. While courts have yet to address these issues comprehensively, they are likely to become central in the near future as black-box medicine continues to evolve.

4 Determining Responsibility and Liability

In the realm of artificial intelligence, particularly in cases of medical negligence, determining who bears responsibility can be a complex matter. AI is essentially a set of programmed algorithms, and its intellectual capacity is limited to the knowledge and design imparted by its human creators. Consequently, the question of accountability becomes intricate in cases of medical mishaps involving AI systems. First and foremost, it's evident that AI systems themselves cannot be held legally responsible since they are machines devoid of legal personhood. Even in scenarios where autonomous AI systems are employed for medical treatments, human medical experts typically oversee their operation. While AI can undoubtedly play a crucial role in the field of medicine, it consistently requires human supervision, thus making hospitals far from "doctor-free" zones. Therefore, it becomes crucial to discuss who should be held accountable in the event of medical negligence: the device manufacturer, the hospital, or the overseeing doctor.

Vicarious Liability: Vicarious liability pertains to holding a superior accountable for the wrongful actions of their subordinates. This concept typically applies in master-servant relationships, where the master is liable for the actions of the servant during their employment. In cases of medical negligence, hospitals are usually held accountable for the actions of their employees, and the resulting harm to the patient. However, when an AI device with autonomous capabilities, such as a robot doctor, takes charge of patient treatment in the future, the hospital could be considered the

master of the device, with the robot as the servant. In such a scenario, if medical harm occurs during treatment, the hospital may be required to provide compensation since the robot operated under the hospital's supervision.

Liability to/for Third Parties: Sometimes, a person may be held liable for the actions of a third party. In cases of medical negligence involving AI systems, the primary parties in question are the patient and the AI system. However, liability may not rest solely with the device; the hospital and the overseeing doctor may also be considered third parties in this context. When a patient seeks treatment at a hospital, it is the hospital's responsibility to ensure proper medical supervision. In cases where misdiagnosis occurs due to an AI-driven system's error, the hospital and the overseeing doctor may be held liable, even though they are technically third parties in the situation.

Joint Liability: Joint liability arises when multiple parties are held responsible for harm caused to one party. If a case of medical negligence involving an AI-operated device emerges, joint liability is likely to be a significant consideration. The victim may file a lawsuit against both the hospital and the doctor overseeing the device's operation. In such cases, the hospital might argue that a manufacturing defect in the AI system played a role. Therefore, medical negligence involving AI systems typically leads to a scenario of joint liability, where both the hospital and the overseeing doctor share responsibility for the outcomes.

Liability of Contract: Hospitals usually acquire AI devices for patient treatment under contractual agreements with manufacturers. In such scenarios, if the device misdiagnoses a patient due to a bug or issue, resulting in harm, the hospital may claim that liability lies with the manufacturer rather than the hospital. However, the hospital's success in this claim against the manufacturer depends on whether the hospital was adequately warned about the potential drawbacks of the system. If the manufacturer had provided warnings, and the hospital ignored them, leading to patient harm, then liability may fall on the hospital rather than the manufacturer.

Product Liability: The integration of AI technologies and software into medical devices raises questions about whether they can be considered products. Product liability, which holds manufacturers or developers accountable even without negligence, is complex when applied to healthcare software and algorithms. In some jurisdictions, medical software is considered a medical device due to its intended use, making product liability a potential factor. However, liability may not apply if AI algorithms are seen as a service rather than a product. Extending product liability could be beneficial to ensure patient compensation and encourage developers to improve algorithm reliability. However, unforeseeable risks and the evolving nature of AI, especially neural networks and deep learning, complicate strict liability. Holding developers strictly liable for algorithm performance may discourage innovation but could be necessary for patient protection. Liability provisions should balance the need for understandability and control over AI technology. Even if developers are held strictly liable, they can avoid liability through the "learned intermediary"

doctrine, which places responsibility on the prescribing physician when adequate warnings are provided by the manufacturer. This doctrine severs the direct link between the developer and the patient harmed.

5 Recommendations

Safety and Risk Minimization: To ensure the safety and integrity of AI applications in healthcare, a comprehensive set of risk mitigation measures is imperative. These encompass robust control mechanisms to prevent misuse, secure systems and software for safeguarding sensitive healthcare data, and preemptive AI technology design to ensure consistent performance and minimize unintended outcomes. Cybersecurity measures are essential to counter potential cyberattacks and data breaches, emphasizing data anonymization and offline storage. Rigorous testing across diverse demographics and rigorous ethical assessments, led by Ethical Committees (EC) and stakeholders, are essential. Participant protection measures must undergo scrutiny, adhering to legal and ethical standards. Continuous monitoring mechanisms and adherence to the highest security standards for patient data are paramount, including timely security upgrades and transparency to build public trust. Stigmatization safeguards, prompt reporting of unintended harm, and additional security for vulnerable groups are crucial. Adequate representation of marginalized populations in training datasets and addressing potential emotional and psychological stress are vital. Additionally, the assessment of AI technology should consider risks associated with technology-derived information. External audits, continuous risk assessments, and oversight bodies are encouraged, with thorough risk assessments before widespread AI deployment. Collaboration among AI developers, ECs, and regulatory bodies is vital to review and advise on risk minimization strategies. Ultimately, AI technologies must not cause severe physical or emotional harm and should prioritize safety and ethical considerations throughout their lifecycle.

Trustworthiness: To harness the potential of AI effectively within healthcare, clinicians and healthcare providers require a straightforward, systematic, and trustworthy means of assessing the validity and reliability of AI technologies. A dependable AI solution not only offers precise health data analysis but also adheres to crucial principles. It must operate within the bounds of the law, embody ethical values, and consistently produce reliable results in line with established standards. Furthermore, the interpretability of AI-driven outputs is essential to dispel the “black-box” perception and enhance the confidence of both patients and healthcare professionals. Autonomy for patients in decision-making and transparency in AI development are vital. Public consultation, compliance with legal norms, and the disclosure of conflicts of interest round out the key considerations for the responsible and effective deployment of AI technologies in healthcare. These principles collectively ensure the ethical, legal, and practical foundations necessary for AI to play a transformative and trustworthy role in healthcare.

Data Privacy: AI-based technology must prioritize privacy and data protection throughout its development and deployment. This is essential to maintain trust among stakeholders, including healthcare recipients, and ensure the secure use of their data. In the healthcare sector, where patient information is sensitive, rigorous data privacy measures must prevent unauthorized access, modification, or loss of personal data. Patient data should ideally be anonymized, except when clinical or research purposes require identifiable information. Ownership of data is complex, varying by jurisdiction and degree of anonymization. Gathering data from diverse sources poses challenges to maintaining privacy. In India, the IT Act, of 2000 holds corporate bodies liable for damages if they fail to implement reasonable security practices for sensitive personal data. New healthcare data protection laws, such as the Digital Information Security in Healthcare Act (DISHA) Bill and the Digital Personal Data Protection Bill, 2023, [32] will enforce ethical guidelines for AI technologies, further safeguarding health data privacy. Here are some guidelines to keep in mind:

- Users should have significant control over the data collected for the development of AI technologies in healthcare. This control should extend to accessing, modifying, or deleting their data from AI systems as needed. It's crucial to empower users with this level of data management.
- End-users must receive clear and comprehensive explanations regarding the privacy safeguards in place. They should be fully informed about the type of data collected, its intended use in AI algorithms, and how it will be interpreted or stored. Transparency is key to building trust.
- Consent should be mandatory before running predictive algorithms on patients or participants. Predictive algorithms, if not properly managed, can potentially compromise patient privacy, making informed consent essential.
- AI technologies dealing with sensitive biometric data should implement heightened security measures. Obtaining approval from ethical committees (EC) and regulatory bodies should be mandatory when handling such data to prevent accidental leaks and potential consequences.
- Impact assessments conducted by relevant authorities before widespread AI deployment should prioritize human rights, privacy, and ethical principles. These assessments help identify and mitigate potential risks.
- Manufacturers bear the responsibility of preventing data re-identification from datasets and ensuring the protection of identifiable information. Data security is paramount.
- Data sharing must involve additional consent from patients, specifying the nature of data, the extent of sharing, and potential risks associated with data sharing. Patients need to be fully aware of and agree to the terms.
- Collecting excess data creates data surplus, which should not be repurposed without proper patient or participant consent. Storing surplus data for future use should also require additional consent, if not previously obtained. Ethical considerations are essential in handling surplus data.

Liability and Compensation: As AI/ML systems continue to be integrated into healthcare, there is a growing concern regarding liability for potential failures

and patient injuries. Striking a balance in liability distribution across the health-care ecosystem is essential to avoid imposing undue burdens on physicians and frontline clinicians while still fostering the safe development and integration of AI/ML technologies. The level of liability directly influences the rate of innovation and implementation of clinical algorithms. While increasing liability may discourage development and experimentation, the traditional liability system can also encourage innovation by accommodating medical advances. Policymakers face the challenge of designing liability regimes that encourage beneficial clinical algorithms without compromising their safety. Options may include departing from traditional liability systems, changing the standard of proof, creating specialized agencies to handle claims, or establishing compensation programs that do not consider liability, depending on the desired promotion of AI/ML, safety, stakeholder responsibility, and other public policy goals. In the context of black-box AI/ML systems, no-fault compensation programs offer an advantage by avoiding the complexities of determining the precise cause of errors. Policymakers have the flexibility to fine-tune these programs to align with their objectives, ensuring a balance between promoting AI/ML innovation, managing personal liability, distributing responsibility among stakeholders, and fulfilling broader public policy objectives [33].

6 Conclusion

The integration of artificial intelligence holds great promise for advancing medical science and enhancing healthcare services, including early disease diagnosis, time-saving automation, and surgical assistance. However, alongside these benefits, concerns and potential risks that could jeopardize patient safety must be acknowledged. As new technology reshapes the healthcare landscape, the need for well-defined and adapted regulatory frameworks becomes evident. Given the sensitivity of healthcare, a careful evaluation of the pros and cons of AI integration is essential. In countries like India, where laws addressing medical negligence exist, it is crucial to consider necessary amendments to accommodate the integral role of AI in healthcare. A comprehensive legal framework with robust checks and balances should be developed to ensure compliance and maximize the benefits of medical AI. Presently, India lacks a unified policy addressing the vision of AI and its associated legal and regulatory challenges. It is incumbent upon Indian lawmakers and the judiciary to proactively pave the way for a successful transition of the healthcare sector into the era of artificial intelligence, thereby addressing complex healthcare issues effectively.

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Prediction of Mental Health Using Machine Learning Techniques: A Review



Pratiksha Pradeep Baji, Sonali Shashikant Rokade,
and Siddharaj Dattatray Pujari

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Abstract Mental health is an important aspect of life that affects overall well-being. Mental Health is termed as the state of mind's well-being which helps people to better understand of situations and make better decisions. Mental illness is one of the biggest issues in society and one of the most ignored ones too. Even though society is becoming aware of the importance of having good mental health and treating mental illness, the real challenge is getting rightly diagnosed, that too in the early stage of illness. It is observed that data science can help medical experts understand and analysing data in a more efficient manner. It will help them understand patients and their condition as well as patients can get more efficient treatment. This paper will explore various machine learning algorithms and data analysis techniques for predicting and analysing mental health problems. The results of this paper will help to understand how modern technologies are contributing to health sector. The outcomes of this research will be beneficial to medical officers who work in the mental health sector to improve mental health among the people.

Keywords Data science · Mental health · Machine learning

P. P. Baji (✉) · S. S. Rokade · S. D. Pujari
Department of Computer Science, Walchand College of Engineering, Sangli, Maharashtra, India
e-mail: pratiksha.baji@walchandsangli.ac.in

S. S. Rokade
e-mail: sonali.rokade@walchandsangli.ac.in

S. D. Pujari
e-mail: siddharaj.pujari@walchandsangli.ac.in

1 Introduction

Mental illness is a very serious problem in today's world. Today's demanding and fast-paced lifestyle has led society to mental health issues. It is one of the most underdiagnosed health problems as well as untreated. Most societies tend to refuse to acknowledge its existence. The effects of mental illness may not be visible at first sight but takes a toll on day-to-day life. People of all ages frequently have mental health problems. Different symptoms associated with a variety of mental health conditions can be challenging to comprehend at first. Most symptoms lead to various consequences over time. Common symptoms of mental illness are anxiety, stress, depression, sleeping issues, frequent mood swings issues, etc. But these symptoms often get ignored. Generally, people are very inactive about their mental illness in comparison to physical illness because people having mental illness get shunned by society making it a very sensitive problem. Very few people living with such problems have access to appropriate mental health services. According to WHO facts, 1 in 7 adolescents is facing mental problems [16], which can lead to self-harming thoughts or behaviours which proves the fact that suicide is the fourth leading cause of death [16]. That's why this issue needs to be resolved. Modern technologies like Data Science and Machine Learning can help to resolve the issue. People are observed to express their emotions, their moods on social media or other platforms where direct human contact is not available making it a safe space to open up about true feelings. The collection of such confessions can be a dataset which will be helpful in understanding how one specific person is feeling most of the time.

The amount of data that is collected from various platforms is very enormous. Most of the data is in unstructured form which needs to be treated and turned into structured for getting important information from it. Data Science can play a very important role in understanding important details for developing efficient systems to get help for mental issues. Techniques of data mining can help to identify hidden patterns from large datasets which will give valuable information.

Using this hidden pattern, the model can be trained using machine learning techniques. Algorithms like decision trees, LSTM, regression, support vector machine, etc. will help the system to understand patterns. Based on the analysis, the system will classify given data and predict the outcome for mental health. If trained precisely system can classify attributes from dataset in different mental problems like OCD, anxiety, and depression and further, it may predict potential serious mental health problems. (Split personality, hallucinations, etc.).

Figure 1 shows how data science can help in the mental health field. Working in the field of Mental Health can be very complicated. That's why the help of advanced technologies like Data Science and Machine Learning will help to understand the causes of mental health issues that will lead to asking the right questions. This will help to identify illness in early stages which will help to take the right preventive actions. Advanced technologies will also help to develop new efficient treatments that will be patient-oriented. The purpose of using technology in health is to make

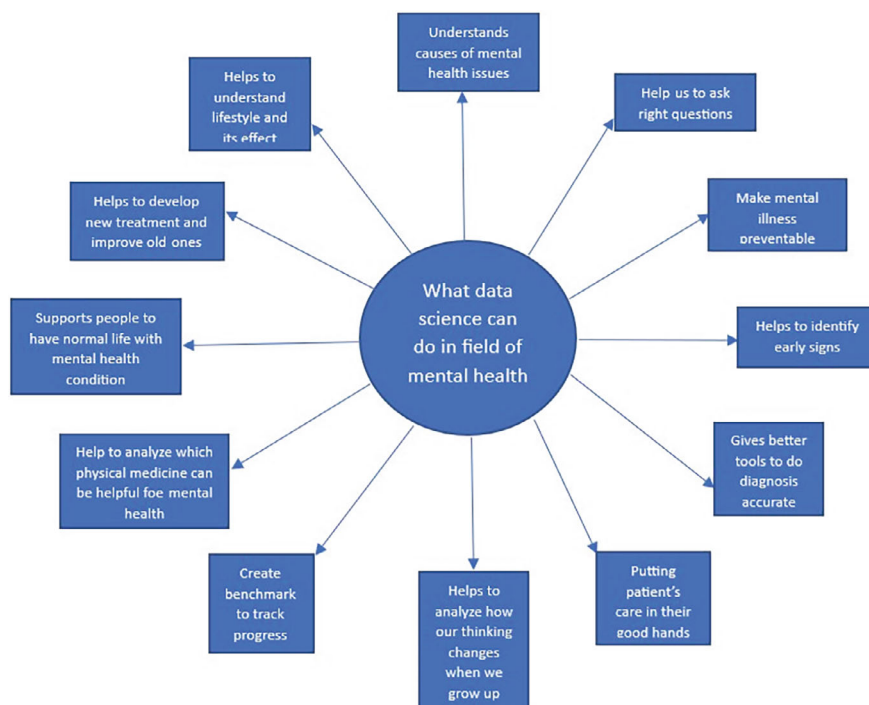


Fig. 1 How data science can help in mental health

treatments more efficient for patients' betterment and support them to have normal lives.

The aim of this paper is to present a review statement of Data science techniques and machine learning algorithms in the prevalent diseases of Mental Health which will help medical officers transform their treatments to make them more efficient for patients.

2 Literature Review

Recent reviews on mental illness covers AI tools, techniques, datasets, which are essential for the task at hand [4, 5]. Suvarna Gangbo et al. [2] published a paper in 2022 proposing a system where they have used various natural processing techniques to predict depression in student's mental health. In this research they have tried to see the effect of the pandemic on students.

Wu Ting [3] has proposed algorithm-based psychological prediction of students using internet big data in 2022. The aim of this paper is to understand the analysis and prediction of students' mental health levels to take necessary precautionary steps.

Manivannan Karunakara et al. [6] have used a questionnaire based on the mental health domain to understand responses from people to predict their mental health conditions. Various machine-learning algorithms were used for this prediction.

Sujal BH et al. [8] published a paper in 2022 that proposes their research on the mental health of employees from a technical and non-technical background. They have used machine learning techniques to understand sudden behavioural¹ changes in employees.

CH.M.H. Saibaba et al. [11] have predicted a solid connection between the way of life factors and mental health using data science techniques and machine learning algorithms. For training exploration, they have utilised information from the most recent 6 years of the AHR report model.

Charith Silva et al. [12] have proposed a method that could identify significant causal relationships among behavioural factors and mental health. They have done separate analyses on male and female datasets, and then compared the behavioural factors associated with mental health among male and female.

The above papers are from recent years and the following is the table of literature review of some more papers that have been reviewed for a better understanding of the role and use of data science and machine learning techniques (Table 1).

3 Methodology

The publications that have been published in recent years are specifically selected to discuss the techniques and algorithms of data mining and machine learning in mental health are surveyed in this study. The scientific databases for the review were picked from Google Scholar, IEEE Xplore, and Springer. Before selecting the paper, the title and results abstract are read, as well as the entire piece where necessary.

Figure 2 shows the flow process of mental health data. The first step is to find the right dataset that can be found online through various versatile platforms. Online datasets will be more useful as nowadays people tend to express themselves on online social platforms. After finding the right dataset, exploration of data needs to be done as knowing all the attributes and characteristics helps to understand features for further procedure. The next step is data pre-processing which includes processes like clustering datasets for specific processing techniques, data cleaning and preparing data as per the need of research. Data encoding helps to ensure the usability and efficiency of data that is available. It helps to set a further dataset. Encoding led to exploratory analysis that helped to understand the dataset and decide the direction of further research. After understanding dataset split is the next step data get split into two parts generally in 70%-30% distribution of data for training and testing of model. For training, 70% dataset is used which includes the classification of attributes which helps for prediction results. Evaluation of results happens when results that are obtained from the training model i.e., working on 70% dataset get evaluated on the remaining 30% dataset i.e., testing model. By repeating classification and prediction process we can achieve accuracy in results which will help to train the model more

Table 1 Literature review

Ref no	Year of publication	Author	Title of paper	Finding of paper	Advantages of paper	Research gap
01	2021	Safa Hurayn et al	Analysis of Mental Illness using Twitter Data	to examine and find signs of mental illness in user interactions on digital platforms	This model identified whether the user was depressed based on previous tweets	May collect large amounts of noise or unwanted data
02	2021	Saleh Afzoon et al	Enabling the Analysis of Mental Health Patterns Using an Efficient Machine Learning Approach	Recognising mental health conditions, with a focus on depression and anxiety as a useful method of assessing an individual's emotional and mental well-being	Minimal number of multi-modal features like textual and temporal content, it achieves considerable and competitive results in significantly efficient time	The model performance can be improved by leveraging more and deeper multi-modal domain related features
03	2021	Tarun Jain et al. [7]	Machine Learning Techniques for Prediction of Mental Health	Proposed model has used eight mainstream ML calculation methods to build up the expectation models	using various strategies and different model, this research work has attempted to get a clear and precise picture	The execution of the trial-and-error model can be time- and energy consuming
04	2021	Lee Ker Xin et al. [10]	Prediction of Depression among Women Using Random Oversampling and Random Forest	In order to correctly predict depression in women, researcher used the random oversampling technique SMOTE and the classification Random Forest	Unique and different patterns that reveal each independent variable's relationships to the dependent variable are generated with the input	The survey's accumulated data, which are utilised to create new data for the database, was unbalanced

(continued)

Table 1 (continued)

Ref no	Year of publication	Author	Title of paper	Finding of paper	Advantages of paper	Research gap
05	2020	Vidit Laijawala et al. [13]	Mental Health Prediction using Data Mining: A Systematic Review	According to this paper, the most optimal algorithm is Decision Tree because of its low execution time and high accuracy	It is predominantly be beneficial for employers and employees by creating greater awareness about work-related mental illness	The dataset mainly consists of data of working individuals
06	2019	Charith Silva et al. [14]	Data Science in Public Mental Health: A New Analytic Framework 2019	a framework that data scientists can utilise to help mental health providers overcome problems	Early on in the process, it is helpful to forecast the time and resources needed to get a clear sense of the problem that has to be solved	Waterfall style framework is used. need to explore more efficient frameworks
07	2019	Robert Thorstad et al. [9]	Predicting future mental illness from social media: A big-data approach	It should be possible to identify the subtle signals related to mental processes in order to learn about how people make decisions, assign meaning, and feel emotions in both good and terrible states of health	The study of the mind and social conduct will both be impacted by these analyses in a similar way	Before choosing to post to a subreddit dedicated to a particular condition, some people in the dataset might have had a mental disorder using everyday language that may carries indications of mental disease, one may be able to predict mental illness before they have enough knowledge

(continued)

Table 1 (continued)

Ref no	Year of publication	Author	Title of paper	Finding of paper	Advantages of paper	Research gap
08	2019	Ying Hu et al. [14]	Big data analytics exploration of Green space and mental health	Explore the relation between twitter sentiment and green space and analysing its impact	Benefits to public health concluding that green space can lowers the anxiety of individuals	The number of features used was relatively small and could impact the performance of multivariate model
09	2018	Susel Góngora Alonso1 et al. [15]	Data Mining Algorithms and Techniques in Mental Health:	Paper addressed main Mental Health diseases based on predictive	Patients can receive highly personalised treatments, therapists	Dataset can be very complicated so that analysing data and
			A Systematic Review	techniques applied to different study features, as well as being able to detect the risk factors for most diseases	will receive help in making evidence-based decisions, and the scientist will be able to search new knowledge that reveals the true causes of Mental Health	training model can be very complex. In future development of a prediction model of patient cognitive impairment (dementia) based on the proposed algorithms and discover the significant predictors that lead to the disease

(continued)

Table 1 (continued)

Ref no	Year of publication	Author	Title of paper	Finding of paper	Advantages of paper	Research gap
16	2020	Mr. Siddharaj D Pujari, Dr. K Anusha	A Review on Prediction of Autism using Machine Learning Algorithm]	A Review on Prediction of Autism using Machine Learning Algorithm]	The ensemble method is used in the suggested approach to ASD classification. By pairing weak learners with strong learners, this strategy increases the accuracy and efficacy of weak learners	Dataset is focused only on autism disorder
17	2023	Yifan LI	Application of machine learning to predict mental health disorders and interpret feature importance	In order to identify the aspects of mental health that are significant to employees and to offer guidance and support, the findings of the OSMH/OSMI Mental Health in Tech Survey are used in this study	Paper proposed that the most significant characteristic, according to an examination of feature importance using permutation importance, was whether the subject had ever experienced a mental health issue	Results are dependent on the patient's mental health history which can affect results of current status 2023

(continued)

Table 1 (continued)

Ref no	Year of publication	Author	Title of paper	Finding of paper	Advantages of paper	Research gap
18	2023	Naveen Paul E Sujitha Juliet	Comparative Analysis of Machine Learning Techniques for Mental Health Prediction	In this work, machine learning-based algorithms were utilised to anticipate mental health problems using survey data	Comparison of four ML models have done and all have successfully predicted mental health issue on a dataset of self-reported mental health	The system may face difficulties due to the dataset's small size and poor quality, the necessity to handle sensitive mental health information ethically, and potential biases in the data
19	2023	Matthew Squires, Xiaohui Tao, Soman Elangovan, Raj Gururajan, Xujuan Zhou, U Rajendra Acharya and Yuefeng Li	Deep learning and machine learning in psychiatry: a survey of current progress in depression detection, diagnosis and treatment	An overview of the deep learning approaches and methodologies available to researchers for the identification, diagnosis, and treatment of depression is given in this work	This paper seeks to present a current evaluation of the usage of artificial intelligence in the field of psychiatry	The current drawbacks of treatment response systems include model validation and small sample numbers
20	2023	Jenish Maharjan, Anurag Garikipati, Frank A. Dinunno, Madalina Ciobanu, Gina Barnes, Ella Browning, Jenna DeCurzio, Qingqing Mao* and Ritankar Das	Machine learning determination of applied behavioural analysis treatment plan type	The study shows that the ML prediction model does a good job of classifying the right level of ABA treatment plan intensity using patient data that is easily accessible	This study may help standardise the method for selecting acceptable ABA therapies, which could expedite the start of the most suitable treatment intensity for ASD patients and enhance resource allocation	Comparatively smaller sample size is used for the study

(continued)

Table 1 (continued)

Ref no	Year of publication	Author	Title of paper	Finding of paper	Advantages of paper	Research gap
21	2023	Priya Bhatt, Amanrose Sethi, Vaibhav Tasgaonkar, Jugal Shrof, Isha Pendharkar, Aditya Desai, Pratyush Sinha, Aditya Deshpande, Gargi Joshi, Anil Rahate, Priyanka Jain ³ , Rahee Walambe, Ketan Kotecha and N. K. Jain	Machine learning for cognitive behavioural analysis: datasets, methods, paradigms, and research directions	This paper includes study on cognitive behavioural analysis focused on lie/deception detection, stress/emotion, and abnormal behaviour detection	The work that is currently being presented offers a thorough examination of the research on cognitive behaviour analysis. The results of a parametric study based on several physical traits, emotional behaviours,	it can be difficult to ensure that the study is carried out consistently and comparably across many cultures because behaviour can differ greatly. This can make it challenging to compare and generalise the results across many cultural contexts
22	2023	Salih Tutun · Marina E. Johnson · Abdulaziz Ahmed · Abdullah Al bizri · Sedat Irgil · Ilker Yesilkaya · Esmay Nur Ucar · Tanalp Sengun · Antoine Harfouche	An AI-based Decision Support System for Predicting Mental Health Disorders	This study uses artificial intelligence and sophisticated analytics to create a decision support system (DSS) that is effective at identifying and diagnosing a range of mental diseases	Without any human input, the suggested DSS can automatically identify mental diseases using just 28 questions as these are fewer questions compared to other surveys participation and completion rate is high	Only specific dataset is used. A dataset from the website Psikometrist was used to carry out the suggested research

(continued)

Table 1 (continued)

Ref no	Year of publication	Author	Title of paper	Finding of paper	Advantages of paper	Research gap
23	2023	Muskan Garg	Mental Health Analysis in Social Media Posts: A Survey	With a collection of more than 92 possible research articles, the study performs a quantitative synthesis and a qualitative review	Study includes classification of feature extraction and vector transformation approaches; historical development of suicidal propensity detection with timeline	Study is done only regarding suicide issue
24	2023	Mohamed Terra1†, Mohamed Baklola1*†, Shaimaa Ali1 and Karim El-Bastawisy	Opportunities, applications, challenges and ethical implications of artificial intelligence in psychiatry: a narrative review	Describes potential opportunities, treatments and effectiveness of AI in Psychiatry	This paper describe how AI might improve patient outcomes and be more effective, efficient, and cost-effective	There is possibility that the privacy and autonomy of patients could be violated by the employment of AI systems for surveillance or control
25	2022	Mr. Siddharaj D Pujari, Dr. K Anusha	Effective Prediction of Autism Using Ensemble Method	Paper explains how the prediction of Autism Spectrum Disorder (ASD) uses classification algorithms. Another method for categorising ASDs is suggested, and it is based on feature extraction and ensemble algorithms	the Ensemble methodology utilised in this chapter increases the accuracy of ASD categorisation. This method aids in more precise ASD classification	Dataset is focused on only autism disorder

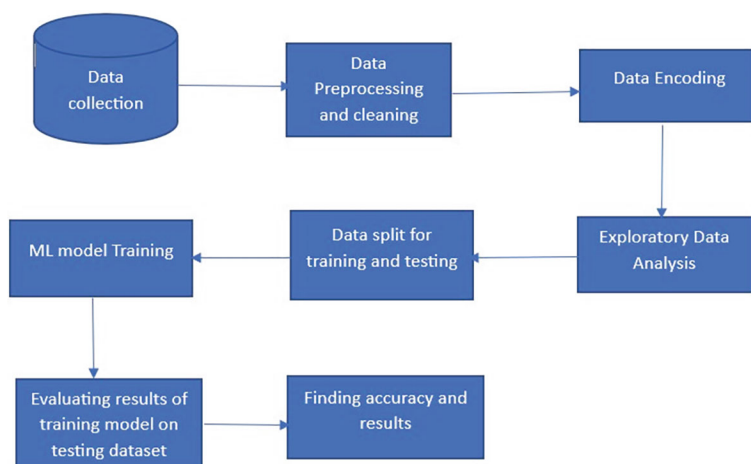


Fig. 2 Architecture

coherently. The last step of this process is to check the results of the trained model on the testing dataset to check the accuracy of the training model. On the basis of accuracy final results will be predicted (Table 2).

4 Result

The study showed in the literature review reveals a significant number of studies that use data mining techniques and algorithms to study mental health conditions. SVM is a more efficient technique to get the expected result with an accuracy of 42%–45%. Although all advanced techniques are giving precision, making these techniques hybrid led to more efficiency. According to the studied literature combining technologies like Naïve Bayes with TF-IDF and LSTM + CNN are giving more precise and accurate results of 70% and 89% respectively.

Following is the table for the comparative study of machine learning algorithms and their accuracies on different datasets (Table 3).

Graphical representation and comparison of the above-mentioned accuracy is given below (Figs. 3, 4).

5 Future Scope

According to the findings of numerous research, the methodologies mentioned could find causal correlations between behavioural characteristics and mental health. Therefore, more behavioural factors like sleeping, walking, cycling, and other social

Table 2 Main Techniques and algorithms used to get results

References	Techniques and algorithms	Working	Result
[1]	Naïve Bayes with TF-IDF	Classifier that assumes the conditional independence of between pair of features Transforms text into usable vector	Combining both techniques helps to perform better with an accuracy of 70%
[1]	LSTM	Use to capture and learn long term dependencies Effective model for classification	Can handle whole data sequences as well as single data points i.e. its accuracy is 85%
[1]	LSTM + CNN	Use to sequence prediction problem	Better sentiment analysis is done using tweet context with an accuracy of 89%
	Linear regression	Have two variable one of them is dependent and other one is independent This technique helps to find out how these two variables are related to each other	Analysis of various factors that affects mental health can get done effectively. Likewise, it helps to do analysis of factors that can affect the process of improving mental health
[6, 7]	Decision tree for data mining	Mainly use for classification i.e., categorising according to characteristics and then building classifier	It helps to build model that helps to evaluate certain attributes of dataset
[10]	Knowledge Discovery KDD process	This process clearly defined helps to gain information that is consists of subsets that the hidden	This process focuses on extracting useful information from data. That will help to get comprehensive knowledge of overall dataset
[7]	Gradient Boosting	Works as a classifier mostly used for building predictive model	Robust technique for data analysis. It can handle numerical and categorical data very well, making it more efficient than other techniques
[7, 8]	Random Forest	Supervised machine learning which uses labelled dataset Creates random decision trees with different random attributes that merge into one at the end	Helps to increase stability and precision of model
[6, 7]	Support vector machine	Uses Supervised approach Input data in SVM already have all the required results that are needed	It finds hyper plane which creates boundary between data which makes easy to classify data for further processing

(continued)

Table 2 (continued)

References	Techniques and algorithms	Working	Result
[9]	Clustering analysis	For understanding features of data for classification and prediction	Handles large and unstructured dataset efficiently Helps to recognise outliers which affects results
[11]	K-Nearest Neighbour	Uses supervised approach It stores all available data Based on previously stored data when new data enters in dataset KNN classifies it	Training and Testing dataset with KNN can be more easy compare with other techniques
[7]	Logistic Regression	- Supervised machine learning approach that mostly use to make prediction based on regression. Also used in binary classification	It is used as categorical data when the variable is dependent. Like linear regression an equation is used. Input values are combined linearly using weights to predict an output value

demographic factors will be added in future research to make results and analysis more dynamic. In future using hybrid techniques may will give more accuracy in results. Privacy policy is another major factor that affects overall results as patients’ data strictly has to be kept confidential which leads to having inefficient, incomplete data. In the future firm actions need to be taken to overcome this problem.

6 Conclusion

Current research on mental health has revealed a strong relationship between the lifestyle factors and mental health. Furthermore, current healthcare is realising that route of life factors link strongly to the outcomes. People of all ages frequently struggle with mental health difficulties. Numerous mental health conditions have a variety of symptoms, many of which are initially quite difficult to comprehend. Most symptoms eventually lead to different consequences that’s why the use of hybrid techniques is more beneficial. The conclusion is that advanced technologies like Data Science and Machine Learning are helping in predicting mental health state. The use of advanced technologies in critical areas like mental health will lead to the evolution of medical treatments.

Table 3 Comparative study of machine learning algorithms and their accuracies on different datasets

Paper references	Dataset	Dataset split	ML algorithm	Accuracy
[1]	Twitter Dataset	60%-Training 20%-Testing 20%Validation	Logarithmic Regression with Word2Vec	56%
			SVM with Feature	59%
			Hashing of Bigrams	
			TF-IDF with Naive Bayes	70%
			Word2Vec + LSTM(RNN)	85%
			Word2Vec + LSTM + CNN	89%
[2]	Student’s tweets	70%-Training 30%-Testing	LSTM with NLP	88%
[7]	Random people with 76 attributes	70%-Training 30%-Testing	SVM	87.38%
[8]	Employees in tech and non tech companies	70%-Training 30%-Testing	XG Boost	84.94%
[9]	Reddit comments	80%-Training 20%-Testing	Clustering analysis	77%
[10]	Surveys specifically limited to women	70%-Training 30%-Testing	Random forest And Random oversampling with knowledge Discovery approach	90%
[12]	6 years of AHR reports	70%-Training 30%-Testing	K-means clustering with association rule mining	85%
[13]	Online dataset	70%-Training 30%-Testing	Decision Tree	82.2%

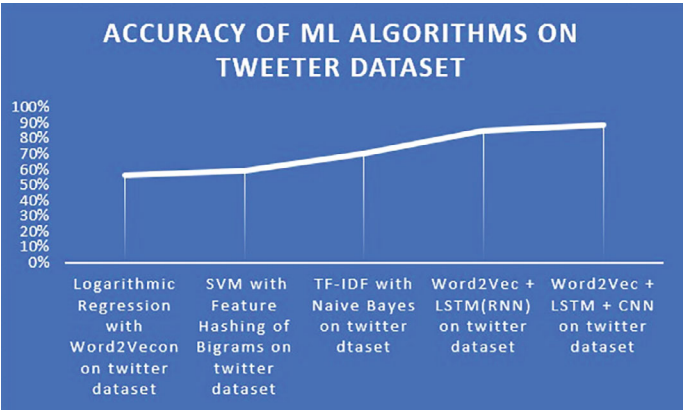


Fig. 3 Graphical representation of accuracy of different ML algorithms on twitter dataset

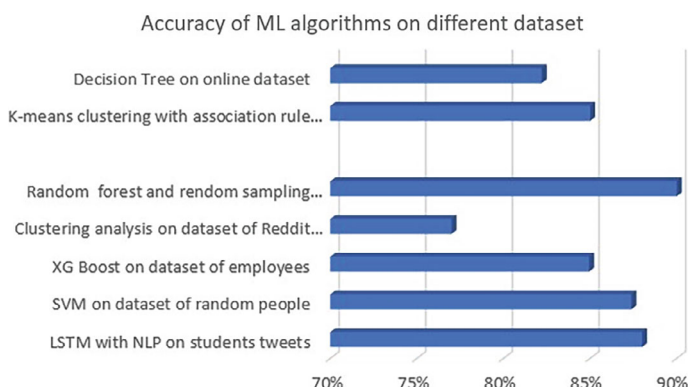


Fig. 4 Graphical representation of the accuracy of different ML algorithms on different datasets

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Recent Trends in EEG-Based Sleep Staging Classification and Analysis: A Comprehensive Review of the Latest Approaches



Rajesh Kumar Mohapatra, Dev Pandya, Pranay Ambani,
Santosh Kumar Satapathy, and Nitin Singh Rajput

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Abstract The report on sleep disorder detection using machine learning (ML) and deep learning (DL) algorithms aims to explore and demonstrate the potential of these advanced technologies in accurately identifying and classifying various sleep disorders. By leveraging ML and DL techniques on sleep data, the report aims to enhance

R. K. Mohapatra · D. Pandya · P. Ambani · S. K. Satapathy (✉) · N. S. Rajput
Pandit Deendayal Energy University, Gandhinagar, Gujarat, India
e-mail: Santosh.Satapathy@sot.pdpu.ac.in

R. K. Mohapatra
e-mail: Rajesh.mphd23@sot.pdpu.ac.in

D. Pandya
e-mail: Dev.pict20@sot.pdpu.ac.in

P. Ambani
e-mail: Pranay.aict20@sot.pdpu.ac.in

N. S. Rajput
e-mail: nitin.rajput@sot.pdpu.ac.in

the efficiency and accuracy of diagnosis, enabling timely interventions and personalized treatment strategies. This research offers a comprehensive overview of the current state of sleep disorder detection methodologies, highlighting the limitations of traditional approaches. The report discusses the implementation of ML and DL algorithms, such as convolutional neural networks and recurrent neural networks, to analyze patterns within sleep data, including EEG signals, heart rate, and movement data. It showcases how these algorithms distinguish between standard sleep patterns and aberrations associated with disorders like insomnia, sleep apnea, and narcolepsy. Furthermore, the report emphasizes the potential for continuously monitoring sleep patterns in real-time, facilitating early detection and preventing complications. By providing insights into the challenges and opportunities of using ML and DL in sleep disorder detection, the report aims to contribute to the medical field's knowledge base. It underscores the importance of interdisciplinary collaboration between medical professionals, data scientists, and technologists to refine these algorithms further. Ultimately, this research seeks to revolutionize sleep medicine by offering more accurate, efficient, and personalized diagnostic tools, thus improving the quality of life for individuals affected by sleep disorders.

Keywords Brain–computer interface (BCI) • Electroencephalogram (EEG) • Sleep staging • Machine learning

1 Introduction

Historically, the general perception of sleep is a steady state of unconsciousness. Still, the actual truth is that it has an active and dynamic life cycle during which the brain is entirely functional [1]. Indeed, a proper sleep routine can significantly improve cardiac function and maintain appropriate mental restoration, attention, consolidation, and behavior. Sleep is an essential human physiological activity. Sleeping is necessary for the body to preserve its wellness and health. Sleep deprivation causes a reduction in agility and output during the daytime, reduces concentration, and adversely affects regular circadian rhythm [2]. It also leads to mental and physical impairments, affects immunity, reduces cognitive functioning, and causes other health issues. Sleep is a continuous and dynamic process with a cyclic structural framework with a time course. Two main types of sleep stages are observed in human sleep: rapid eye movement (REM) and non-rapid eye movement (N-REM) sleep. Throughout the night, the sleep cycle covers N-REM and REM sleep stages cyclically; typically, each cycle lasts an average of 45–60 min. Two main published editions contain recommendations by the committee headed by Rechtschaffen and Kales (R&K) and the American Academy of Sleep Medicine (AASM) to analyze and classify the sleep stages [3].

This report aims to explore the utilization of machine learning (ML) and deep learning (DL) algorithms in the domain of sleep disorder detection. This investigation aims to evaluate how these advanced computational techniques can enhance the early

identification and accurate diagnosis of various sleep disorders. By delving into the potential of ML and DL in this field, the report aims to highlight their efficacy, limitations, and implications for improving the precision and efficiency of sleep disorder detection methods. Ultimately, the report seeks to provide insights into how these algorithms can transform healthcare practices related to sleep disorders [4].

1.1 Prevalence and Impact

Sleep disorders are not rare; they affect a substantial portion of the global population. Estimates indicate that around 50–70 million adults in the United States alone experience some form of sleep disorder. These disorders exert a profound impact on health and well-being. Physically, they contribute to an elevated risk of cardiovascular diseases, metabolic disorders, and compromised immune function. Mentally, sleep disorders are linked to mood disorders, cognitive impairments, and a diminished overall quality of life. Moreover, their societal implications include accidents, decreased productivity, and strained relationships [5].

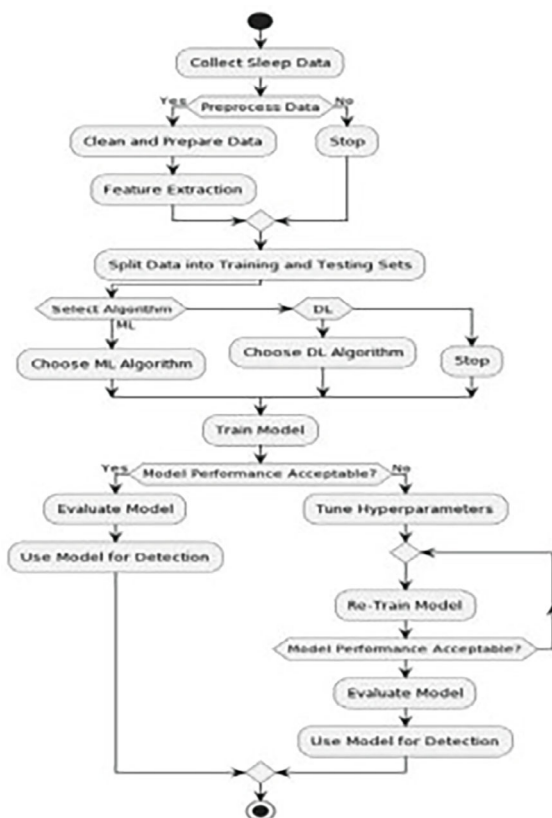
1.2 Significance

Recognizing the importance of addressing sleep disorders, researchers, healthcare professionals, and policymakers have increased efforts to diagnose and treat these conditions. Consequently, the significance of sleep disorders lies in their individual and societal impact and the urgency to develop effective interventions [6].

1.3 ML and DL for Sleep Disorder Detection

Researchers have recognized the significance of sleep disorders and turned to innovative approaches, particularly ML and DL algorithms, to enhance the detection methods. These computational techniques, which excel in pattern recognition and data analysis, offer a promising means of identifying and diagnosing sleep disorders more accurately and efficiently [7]. ML and DL algorithms can reveal intricate patterns and correlations that might elude traditional diagnostic methods by analyzing extensive datasets encompassing sleep patterns, physiological signals, and other relevant parameters [8] (Fig. 1).

Fig. 1 Complete layout of the automated sleep staging model



2 Literature Review

Sleep disturbance has become a common issue among college students. According to epidemiological studies, the prevalence of sleep disturbance among college students ranges from 20 to 60% [1]. This problem has received widespread attention due to its long-term effects on college students' physical and mental health. To elaborate, sleep disturbance can significantly impact college students' physical and mental health, such as daytime sleepiness, decreased cognitive function, and impaired academic performance [2, 3]. In addition, sleep disturbance can increase the risk of developing chronic diseases [4, 5], such as liver disease, diabetes, hypertension, and cardiovascular disease. Thus, there is a growing interest in research on preventing sleep disturbance. Nightly polysomnography (PSG) is the most commonly used method for evaluating sleep. PSG captures a variety of physiological signals, including breathing, EEG, electromyograms (EMG), and electrooculograms (EOG). Nevertheless, this procedure calls for professionals to visually review all of the nighttime data and annotate it in accordance with Rechtschaffen and Kales (R&K) (Rechtschaffen &

Kales, 1968) or the American Academy of Sleep Medicine (AASM) (Berry et al., 2013) grading recommendations. In recent years, advancements in machine learning (ML) and deep learning (DL) techniques have shown promise in aiding the diagnosis and management of sleep disorders. Insomnia is one of the most common sleep disorders characterized by difficulty falling asleep, staying asleep, or experiencing non-restorative sleep. ML and DL algorithms have been applied to various data sources, such as wearable devices and polysomnography data, to predict and diagnose insomnia. These algorithms analyze sleep patterns, heart rate variability, and other physiological signals to identify features indicative of insomnia [9]. ML and DL models have been employed to analyze respiratory signals, such as airflow and oxygen saturation, to identify sleep apnea. These algorithms can assist in classifying the severity of apnea events and predicting the risk of complications [9]. Narcolepsy is a neurological disorder characterized by excessive daytime sleepiness and sudden sleep attacks. ML and DL algorithms have been applied to analyze electroencephalography (EEG) signals and other physiological data to detect narcoleptic episodes. These techniques aim to differentiate between normal sleep transitions and sudden sleep onset events [9].

3 Traditional Methods of Diagnosing Sleep Disorders

3.1 Polysomnography (PSG)

PSG is the gold standard for diagnosing sleep disorders. It involves monitoring various physiological parameters such as brain waves (EEG), eye movements (EOG), muscle activity (EMG), heart rate, airflow, and oxygen saturation during sleep. Trained sleep technologists manually score sleep stages and events. However, PSG is time-consuming, expensive, and requires patients to sleep in a sleep laboratory, potentially altering their natural sleep patterns [10].

3.2 Multiple Sleep Latency Test (MSLT)

MSLT measures the time taken for a person to fall asleep during the day in controlled conditions. It's used primarily to diagnose narcolepsy. However, it doesn't capture the full spectrum of sleep disorders and may not be practical for broader screening [10].

3.3 Actigraphy

Actigraphy involves wearing a device that records movement to estimate sleep patterns. It's more convenient than PSG but may lack precision in distinguishing between sleep stages and diagnosing complex sleep disorders.

3.4 Limitations of Traditional Methods

Traditional methods often require patients to sleep in unfamiliar environments, affecting their natural sleep patterns and potentially leading to a phenomenon known as the first-night effect. Manual scoring of sleep stages and events in PSG is prone to inter-scorer variability, leading to inconsistent results. Sleep labs are limited in number and can be geographically distant, making it challenging for some patients to access these facilities. Related tests are expensive, involving equipment and professional expertise for scoring and interpretation. Traditional methods provide data from a single night in a controlled setting, potentially missing variations and fluctuations in sleep patterns over time.

3.5 Need for Automated Approaches Using ML/DL Techniques

- a) **Objective and Consistent:** ML and DL models offer automated and standardized approaches, reducing inter-scorer variability and ensuring consistent results.
- b) **Non-Intrusive Monitoring:** Wearable devices equipped with sensors can monitor sleep patterns in a home environment, preserving the natural sleep behavior of individuals.
- c) **Large-Scale Screening:** Automated approaches enable large-scale screening and early detection of sleep disorders, facilitating timely interventions and reducing the burden on specialized sleep clinics.
- d) **Feature Extraction:** ML and DL algorithms can simultaneously analyze complex patterns in various physiological signals, potentially revealing insights that traditional methods might miss.
- e) **Real-Time Monitoring:** Some automated systems can provide real-time feedback, enabling individuals to adjust their sleep behaviors for improved sleep quality.
- f) **Cost-Effectiveness:** Once developed, automated systems can reduce the cost of specialized sleep clinics and repeated testing.
- g) **Longitudinal Tracking:** Wearable devices can gather data over extended periods, allowing for longitudinal tracking of sleep patterns and the ability to identify trends and changes.

4 Applications of Machine Learning (ML) and Deep Learning (DL) in Healthcare

This study proposed a DL-based approach for automatic sleep stage classification using single-channel EEG signals. The authors used convolutional neural networks (CNNs) to learn discriminative features directly from raw EEG data. The results showed competitive performance compared to traditional methods, highlighting the potential of DL in simplifying sleep stage classification [11].

This research focused on using ML techniques to detect insomnia. The study collected data from wearable heart rate and accelerometer sensors and applied feature extraction and ML algorithms for insomnia detection. Their results demonstrated the feasibility of using wearable sensors and ML algorithms for diagnosing insomnia [12].

Researchers developed Deep Sleep, a DL model for sleep stage classification, in this study. By combining CNNs and RNNs, they achieved robust performance even when trained on data from different sources and with varying quality. This demonstrates DL methods' adaptability in classifying sleep disorders [13].

This research focused on using DL techniques for detecting narcolepsy based on EEG analysis. The study employed a deep learning architecture to analyze EEG data and detect characteristic patterns of narcoleptic episodes. The results accurately distinguished between narcoleptic and non-narcoleptic events [14].

This study proposed a machine-learning approach for the automatic recognition of insomnia. The authors utilized various physiological signals, such as EEG, heart rate, and respiratory data, to build a comprehensive model for insomnia detection. The results demonstrated the potential of combining multiple data sources to improve accuracy [15].

Building on the previous study, this research developed NarcoNet, a DL-based model for automatic narcolepsy detection. The authors used EEG data and deep learning techniques to classify sleep stages and detect narcoleptic episodes. NarcoNet demonstrated high accuracy in distinguishing narcolepsy cases [16].

This study presented a DL-based approach for sleep stage classification using EEG data. The authors employed CNNs to automatically learn relevant features from EEG signals and achieved competitive performance compared to traditional methods. The research emphasized the potential of DL in simplifying sleep stage classification tasks [17].

This study employed ML techniques to predict obstructive sleep apnea (OSA) using single-lead ECG signals. The authors extracted wavelet energy and entropy features and used support vector machines (SVMs) for classification. The results demonstrated the potential of ECG-based features and ML algorithms for OSA prediction [18].

This study introduced SleepNet, a DL-based model for sleep apnea detection using raw physiological signals. The authors employed a combination of CNN and RNN architectures to process and classify signals, achieving high accuracy in detecting sleep apnea events [11].

5 Data Collection and Preprocessing

Collecting and preprocessing high-quality sleep data is crucial when developing machine learning (ML) and deep learning (DL) models for sleep disorder detection. The accuracy and effectiveness of these models heavily depend on the quality of the input data they are trained on. Here's an explanation of the importance of quality sleep data for training ML/DL models.

A. Data Quality Directly Impacts Model Performance

ML/DL algorithms learn patterns and relationships from the input data. If the input data is noisy, inaccurate, or poorly labeled, the known patterns will be unreliable and could lead to incorrect predictions. High-quality sleep data ensures that the models learn meaningful and accurate ways associated with sleep disorders, resulting in more reliable and effective predictions.

B. Generalization to Real-World Scenarios

ML/DL models are expected to generalize well to unseen data, including real-world scenarios. Quality sleep data ensures that the model learns generalizable features and patterns that are not just specific to the training dataset but can be applied to new and unseen instances. This is important for the model to accurately detect sleep disorders in real-world situations.

C. Reduced Bias and Overfitting

Poor quality data can introduce biases, inconsistencies, and outliers that can mislead the model during training. These issues can lead to overfitting, where the model becomes excessively tailored to the training data and fails to perform well on new data. High-quality sleep data helps mitigate these issues, resulting in more robust models and less prone to overfitting.

D. Interpretable and Explainable Models

High-quality sleep data makes it easier to understand the learned features and patterns within the model. This is particularly important in the medical domain, where clinicians and researchers must interpret the model's predictions and understand their reasons. Clear and meaningful patterns in the data lead to more interpretable and explainable ML/DL models.

E. Better Feature Extraction

Feature extraction is a critical step in ML/DL model development. Quality sleep data allows for extracting relevant and meaningful features that capture the intricacies of sleep patterns and disorders. The more accurate and representative the elements, the better the model can differentiate between different sleep states and conditions.

F. Enhanced Model Robustness

Quality sleep data helps the model become robust to variations and noise that might be present in real-world scenarios. Exposing the model to diverse and representative sleep data makes it better equipped to handle different sleep patterns, sleeping environments, and potential artifacts that can affect sleep data.

G. Ethical Considerations

In medical applications, poor quality or biased data can lead to incorrect diagnoses and treatment recommendations, which can have serious ethical implications. High-quality sleep data ensures that the model's predictions are trustworthy and safe for use in clinical settings.

6 Data Collection Approaches

Various data sources can be utilized when collecting data for sleep disorder detection using ML and DL algorithms. Each source offers unique advantages and challenges. Here are some potential data sources.

A. Sleep Tracking Devices

Consumer-grade wearables and devices such as fitness trackers, smartwatches, and dedicated sleep trackers can provide valuable data about sleep patterns. These devices can monitor parameters like movement, heart rate, and sometimes even sleep stages. They offer the advantage of continuous monitoring in a non-intrusive manner, enabling long-term data collection. However, the accuracy of some consumer devices might not be as high as medical-grade equipment.

B. Polysomnography (PSG) Data

PSG is a comprehensive sleep study conducted in a clinical setting and is considered the gold standard for diagnosing sleep disorders. PSG collects multiple physiological signals, including EEG, EOG, EMG, EKG, respiratory flow, and more. PSG data provides detailed insights into sleep stages, sleep disorders, and associated physiological changes. However, PSG is expensive, requires specialized equipment, and can be uncomfortable for the patient, limiting its use to clinical research settings.

C. Actigraphy Data

Actigraphy involves wearing a device that monitors movement, typically a wrist-worn accelerometer, to estimate sleep patterns and circadian rhythms. It's less invasive than PSG and is often used for longer-term monitoring in both research and clinical settings. Actigraphy data can be combined with other sources for a broader perspective on sleep quality.

D. Questionnaires and Surveys

Self-reported data from questionnaires and surveys can provide insights into subjective sleep quality, sleep habits, and daytime symptoms. While subjective, this data can be valuable for correlating perceived sleep issues with objective measurements from other sources.

E. Mobile Apps and Digital Diaries

Mobile apps that allow users to log their sleep habits, daily activities, and symptoms can provide longitudinal data for analysis. These apps can also deliver user interventions and recommendations based on their collected data. It's important to note that combining multiple data sources can lead to more accurate and robust models. However, data integration and preprocessing become complex tasks, as different sources might have varying sampling rates, data formats, and noise levels. Data quality, privacy, and ethical considerations are crucial when working with medical and personal data sources. Ultimately, the choice of data sources depends on the research or clinical goals, available resources, and the specific sleep disorders targeted by the ML/DL algorithms.

7 Sleep Features

Feature selection and engineering play a crucial role in improving the performance of machine learning (ML) and deep learning (DL) algorithms for sleep disorder detection [19–30]. Here's how you can discuss relevant features for sleep disorder detection, such as Sleep Duration: The total amount of time a person spends asleep is a fundamental feature. Sleep duration is often correlated with sleep disorders. Both insufficient and excessive sleep durations can indicate potential sleep problems. Sleep Stages: Sleep is typically divided into different stages, including REM (rapid eye movement) and non REM stages (N1, N2, N3). Each stage has distinct physiological characteristics. Extracting features related to the distribution and proportions of these stages can provide valuable insights. Heart Rate Variability (HRV): HRV, which measures the variation in time between successive heartbeats, can reflect the autonomic nervous system's activity during sleep. Reduced HRV is linked to sleep disorders like sleep apnea and insomnia. Respiratory Rate: Abnormal respiratory rates during sleep can indicate disorders like sleep apnea. Monitoring patterns of breathing rate and detecting irregularities can be useful features. Body Movement: Assessing the frequency and intensity of body movements during sleep can identify disorders such as restless leg syndrome or periodic limb movement disorder. Electroencephalogram (EEG): EEG signals capture brain activity during sleep and can be analyzed to identify anomalies in brain wave patterns associated with sleep disorders. Electromyogram (EMG) and Electrooculogram (EOG): These signals monitor muscle activity and eye movements during sleep. They are essential for detecting REM sleep behavior disorder and eye movement-related disorders. Sleep Efficiency: Calculating the ratio of total sleep time to total time in bed can help identify disorders related to sleep fragmentation.

8 Feature Engineering

Temporal Features: Extract statistical measures such as mean, median, standard deviation, and skewness from the time-series data of sleep-related features. These can capture variations and trends over different time intervals. **Frequency Domain Features:** Apply Fast Fourier Transform (FFT) techniques to transform time-domain signals (e.g., EEG) into the frequency domain. Extract features such as power spectral density to identify frequency-specific patterns associated with sleep disorders.

Wavelet Transform: This can be useful for simultaneously capturing time and frequency information, helping to identify transient events or sudden changes in sleep signals. **Feature Aggregation:** Combine relevant features from multiple sources (e.g., heart rate, movement, EEG) to create composite features that comprehensively view the sleep state. **Principal Component Analysis (PCA):** Use PCA to reduce the dimensionality of the feature space while retaining the most informative features. This can help prevent overfitting and improve model generalization.

8.1 Feature Scaling

Normalize or standardize features to ensure that algorithms are not biased towards features with larger magnitudes. **Temporal Features:** Extract statistical measures like mean, median, standard deviation, and skewness from the time-series data of sleep-related features. These can capture variations and trends over different time intervals [19]. **Frequency Domain Features:** Apply techniques like Fast Fourier Transform to transform time-domain signals (e.g., EEG) into the frequency domain. Extract features such as power spectral density to identify frequency-specific patterns associated with sleep disorders [19]. **Wavelet Transform:** This can capture time and frequency information simultaneously, helping to identify transient events or sudden changes in sleep signals [19]. **Feature Aggregation:** Combine relevant features from multiple sources (e.g., heart rate, movement, EEG) to create composite features that provide a comprehensive view of the sleep state [20]. **Principal Component Analysis:** Implemented PCA to reduce the dimensionality of the feature space while retaining the most informative features. This can help prevent overfitting and improve model generalization [20]. **Feature Scaling:** Normalize or standardize features to ensure that algorithms are not biased towards features with larger magnitudes. **Feature Interactions:** Create interaction terms or cross-products of features to capture potential synergies between different physiological signals [21]. **Time-Series Analysis:** Analyze trends and patterns within individual sleep records to extract features such as sleep onset latency, number of awakenings, and sleep fragmentation indices [22].

9 Model Development for Sleep Disorder Detection

a. Data Preprocessing:

Before diving into model development, you must preprocess your data to ensure it's suitable for training. Gather a dataset containing various sleep-related features and labels indicating the presence of sleep disorders. Handle missing values, outliers, and noise in the dataset. Select relevant features that might contribute to sleep disorder detection. Normalize or scale numerical features to ensure they are within a similar range. This can help algorithms converge faster during training [23, 24].

b. Data Splitting:

Divide your preprocessed data into three sets: Training Set: The largest data used to train the model. Validation Set: A smaller subset used to tune hyperparameters and assess the model's performance during training. Test Set: Another independent subset used to evaluate the model's performance after training. It simulates how well the model will perform on new, unseen data [25, 26].

c. Model Selection:

Choose ML and DL algorithms that are well-suited for sleep disorder detection. Common choices include Logistic Regression: A simple ML algorithm for binary classification tasks. Random Forest: An ensemble learning method that can capture complex relationships in the data. Convolutional Neural Networks (CNNs): DL architecture capable of learning spatial features, useful for image-based sleep data. Recurrent Neural Networks (RNNs): DL architecture ideal for sequential data, like time-series sleep data [27–30].

d. Model Training:

Train your selected model using the training set and choose an appropriate loss function for your problem, such as binary cross entropy for binary classification. Select an optimizer (e.g., Adam, SGD) to update model parameters during training. Hyperparameter Tuning: Experiment with different hyperparameters like learning rate, batch size, and regularization strength to optimize the model performance on the validation set.

e. Hyperparameter Tuning and Optimization:

Iteratively adjust hyperparameters and evaluate the model's performance on the validation set. Techniques include Grid Search: Exhaustively search a predefined hyperparameter space. Random Search: Randomly sample hyperparameters from a defined distribution. Bayesian Optimization: Probabilistic model to find optimal hyperparameters efficiently.

10 Performance Evaluation

While evaluating the performance of machine learning (ML) and deep learning (DL) models for sleep disorder detection, it's essential to use various metrics to comprehensively understand their effectiveness. Different metrics provide insights into other

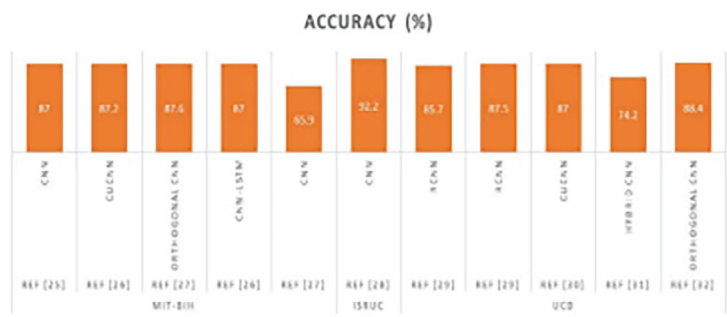


Fig. 2 Performances of sleep staging classification accuracies based on different datasets

aspects of model performance. Mainly, accuracy [20] is one of the performance metrics for multi-class classification sleep staging classification problems (Fig. 2).

11 Limitations

Limited and inconsistent availability of high-quality sleep disorder data poses a challenge for training accurate ML/DL models. Sleep patterns vary among individuals and subjects, so creating universally applicable models is difficult. The lack of standardized diagnostic protocols in sleep disorders contribute to inconsistent labeling data for model training. Interpretability issues with many ML/DL models, especially deep neural networks, hinder their adoption in medical applications. Ethical concerns regarding the sensitive nature of sleep data require robust privacy measures. Imbalanced datasets with rare sleep disorders can lead to biased model performance. Difficulty in generalizing models across different demographics or datasets due to variations in sleep habits and patterns. Collaboration between ML experts and sleep disorder specialists is pivotal to ensure model accuracy and clinical relevance. Ongoing interdisciplinary research is vital for overcoming these challenges and advancing effective sleep disorder detection solutions

12 Future Scope

- a. Develop techniques to generate synthetic sleep data and augment existing datasets, addressing the challenge of limited and variable data quality.
- b. Focus on creating ML/DL models that are more interpretable, aiding clinicians in understanding and trusting the decision-making process.
- c. Investigate combining data from different sources (e.g., audio, wearables) to provide a comprehensive view for accurate sleep disorder detection.

- d. Tailor models to individual sleep patterns, enhancing accuracy by accounting for unique patient variations.
- e. Develop models that capture temporal patterns in longitudinal sleep data, enabling more accurate diagnoses and predictions.
- f. Establish robust ethical and regulatory frameworks for data privacy and model usage in sleep disorder diagnosis.

13 Conclusion

This report delved into the transformative potential of Machine Learning (ML) and Deep Learning (DL) algorithms in revolutionizing sleep disorder detection. Through a comprehensive literature survey, we witnessed the remarkable impact of these algorithms on healthcare. The application of ML and DL in sleep disorder detection showcases their ability to process physiological signals from wearable devices, offering insights into disorders like insomnia, sleep apnea, and narcolepsy. These algorithms' proficiency in deciphering intricate patterns within EEG, ECG, and other signals indicate their prowess in understanding sleep intricacies. Key to their success is feature selection, which highlights the significance of preprocessing data. The fusion of feature selection with model development has resulted in innovative models like DeepSleep, SleepNet, and NarcoNet. The complex architectures of convolutional neural networks and recurrent neural networks enable these models to discern sleep stages and disorder occurrences. Transitioning from model development to performance evaluation metrics such as accuracy, sensitivity, and specificity validate the reliability of these algorithms in real-world clinical contexts. These metrics affirm the algorithms' effectiveness in addressing the multifaceted landscape of sleep disorders.

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Role of Artificial Intelligence and Machine Learning in Precision Health: Analyzing the Risk and Ethical Challenges



Ivneet Walia and Navtika Singh Nautiyal

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Abstract Due to its potential to completely alter healthcare procedures, the application of artificial intelligence (AI) and machine learning (ML) techniques to the field of precision health has attracted a lot of interest. With a focus on the hazards and ethical issues involved, this study seeks to give a thorough review of the role played by AI and ML in enhancing precision health. The efficacy of AI-driven diagnostic tools in terms of illness diagnosis, prognosis, and therapy prescription is encouraging. Additionally, ML algorithms make it possible to create prediction models that help physicians make well-informed decisions, resulting in more individualized therapies and better patient outcomes. Questions concerning informed consent, autonomy, and the proper ratio of human judgment and algorithmic suggestions arise when AI is used in delicate activities like diagnosis and therapy selection. In situations when AI aids in cutting-edge medication research or therapeutic methods, concerns concerning ownership and intellectual property also surface.

Keywords Digital space · Artificial intelligence · Machine learning · Precision health

I. Walia

Rajiv Gandhi National University of Law, Patiala, Punjab, India

e-mail: ivneetwalia@rgnul.ac.in

N. S. Nautiyal (✉)

School of Law, Forensic Justice and Policy Studies, National Forensic Sciences University, Gandhinagar, India

e-mail: dr.navtikasingh@gmail.com

1 Introduction

There is a need to diffuse the digital space with the analog timeline for betterment and improvement of the health care systems. With the inflow of abundant data and information, individualized health care infrastructure and artificial intelligence algorithms have reached a new pedestal. The existence of healthcare technologies can only be put to optimum advantage if the healthcare practitioners and their patients can derive benefit out of it. Overflowing information and data can help us attain and identify the need and move a step nearer to predictions. The information about the patients can help us identify or predict about their health conditions and susceptibility to various ailments. A data set about their personal and individualized characteristics can give us an idea about the concerned risk or threats, medical interventions, proper medical history or visitations record, etc. Introduction of any new technology in any sector calls on for different considerations. Precision health works on the ideology and identification of well-being needs and response to health treatments based on individual requirements [1].

When we talk about the precision health, we are not just focusing on the health aspect alone but we are also assessing the behaviors and lifestyles of the person, through the use of data available about that person in the digital domain. Most of the people have their lives stored in cloud and apps that they mistakenly believe are private to them. Their ever presence in digital space is generating a digital phenotype. Their involvement in metaverse and behavior on social media platforms encourage the creation of digital twins. The healthcare system is also collecting information through sensor technologies where the human-machine interaction is not dependent on the stroking the keyboard or clicking the mouse, but the conversation is through voice and gestures. The technology has changed the way, the humans evolved. It has completely revamped the traditional ways of treatment and now a holistic view is taken when it comes to good health and well-being. With the advent of the mobile technology, certain apps collect your personal information and suggest your diet plans and exercises that can assure you a healthy living. The apps that suggest you diet alternatives and dance moves for keeping you in shape are equally customizing packages as per your requirements. Thus, the technology has already intruded into your space and homes assuring you a longer healthier life. The world is rotating from prescribed medicines to precise healthcare routines [2].

2 Understanding the Concept of Precision Health

Precision health provides for a personalized healthcare depending on the requirements and information of each patient, which can later be placed in groups if they own similar characteristics that require consideration. In this situation, a digital record is created that will define a health ecosystem right from the time of birth considering health, attitudes and behavioral patterns. People often confuse the term precision

medicine with personalized medicine [3]. The term personalized medicine refers to an older concept that used biomarker tools for customizing medical treatments, whereas the precision medicine or precision health focuses on patient centered care by customizing the healthcare inclusive of medical decisions and treatments, after thorough research on person's genetic, physical and attitudinal make up. Never before we have been able to predict the possible health vulnerabilities as we do it today. The health condition of a person depends on cohort of factors which involves DNA, environment, emotional well-being, diet, lifestyle, etc. The objectives behind inducting the system of precision healthcare are to profile risk, stratify population for better health results. Different stratification processes and mechanisms are required from medication point of view. A single medicine may be suitable for one patient and not for other in a group of people sharing similar health characteristics. To elaborate a little more about the ways the precision health is now put into use, the author(s) would like to mention certain examples as cited below [4].

The precision health algorithms can acquire information from social media platforms. The social media can be helpful in both tracking diseases and spreading information about precautions and healthcare. The biggest example of such scenario is the COVID situation. Where the social media was assessed by the algorithms and a prevalence of a pandemic was both predicted and ascertained [5].

Biomarker testing is also a feature of precision health, which ensures the correctness of the treatment. It checks the suitability of the treatment to the patient and maximizes the chances of recovery.

The precision health also looks at medical options to understand if the patient requires more test or screenings in cases of identified risk of diseases. This helps in timely detection and prevention. Constant monitoring also helps in preparing a health chart that assures whether the medical treatment requires continuation or modification.

Precision health techniques focus more on social determinants, for example, it assesses circumstances, age, development, and also, physical environment, employment status, etc. Precision health philosophy very well understands that the social determinants are major reasons for health issues than the genetic ones.

Thus, Precision health predicts the risk of ailments, prevents the development or growth of diseases, provides participative care to patients along with personalized treatments and aims to improve the health of not individuals but population [6].

3 Involvement of Data and Development of Digital Phenotype

When the algorithms are involved for making predictions, the utmost important thing required is the data. The more the data sets, the better is the prediction assuring better results and outcomes. When the algorithms spy the individuals assessing their web presence, it's not just the general data required but it also traces the unconnected,

fragmented and irrelevant pieces of information that may substantiate the record. Precision health feeds on this enormity of data and objectivity of artificial intelligence. Data so collected must conform to the clinical confirmations and health status for correct diagnosis and treatments. Use of technology not only provides certain results but it also lowers the cost and minimizes the personnel interventions which can be put to use for other purposes requiring attention. Though, we realize that the whole philosophy of precision health depends on the data collection, but here comes an important question to answer are the patients willing to share their information? Or do they consent for such data collection? [7]

Data as we see is the most essential element which is available in different forms. It may be available not only in the form of words but it could be characters, numbers, audio–video clips, etc. The data may be available in a structured form or unstructured form. The data that can be retrieved from internet of things, smart devices, and sensors is all structured data, and the information gathered from social media is an example of unstructured data. If the data is not structured, it takes intensive efforts to compile and interpret consuming resources. But data whether structured or unstructured always facilitates the healthcare units. Big data majorly focuses on volume of data and its variety. The techniques used in big data analytics helps to build up correlations and ascertain trends and preferences by analyzing the patterns, thus helping in making informed decisions [8].

The digital data contributes toward the making of digital phenotyping. The digital phenotyping is a process by which an individual's behavior is assessed on basis of digital data which has been collected from human–machine interactions. The healthcare units are now depending on making of digital twins for better personalized healthcare services. The digital mapping facilitates for virtual scanning of organs and other body parts. The technology that makes a digital twin enables the healthcare practitioner to use simulation, machine learning and reasoning on a virtual representation of the patient. The digital twins can be experimented with to see how the medication or treatment will impact the patients in real time. By studying the impact on a virtual patient, the doctors will be in a position to enlist complications or disease progressions. The digital twins are useful in detecting heart diseases in new born. Data collected in structured or unstructured form as such is not enough to be used in its raw form. The data is mandated to follow a series of steps which may include, collection, organization, processing, integration, reporting, and utilization. The data thus acquired must be interpreted rationally [9].

4 Role of Artificial Intelligence in Precision Health

Artificial intelligence has become a popular entity because of its cognitive ability which involves learning, problem solving, reasoning, comprehending, perceiving, etc. The artificial super intelligence has already surpassed the human capabilities and intelligence. The human race has been sidelined by the capacity of these machines to perform innumerable calculations.

A lot of artificial intelligence triggered precision healthcare systems have been put into place for instance clinical decision support, lifestyle medicines, etc. The tools like clinical decision support tools improve the healthcare systems by improving diagnosis and treatment by looking at a specific medical condition support by a prediction and probability of an outcome based on biomedical imaging information and data. The techniques using artificial intelligence have already surpassed the correctness of its predictability by humans and have received approvals and certifications from concerned authorities that ensures its authenticity and reliability. The doctors across the globe have confirmed that such artificial intelligence driven tools can predict kidney diseases, chances of breast cancer, etc. The only challenge faced by these algorithms is the availability and accessibility of large sets of data. Another such tool or mechanism is referred to as Digital Therapeutics, which indicates toward a process of interventions for the purpose of managing a medical condition and treating a disease. Digital therapeutics is not to be confused with the working of the apps that configure and customize the healthcare suggestions based on inputs given by the consumers for timely monitoring of their weight, diet, and routines. The digital therapeutics is a broader term and involvement of such apps can be considered as one of the ways of intervention which can facilitate the patient to improve social determinants in his/her ecosystem along with the diet, lifestyle and behavior. The digital therapeutics engage in behavioral guidance, biomedical feedback discussion, sensor enabled wellness tools for patient centric approach, etc. Digital engagements and interventions can track the improvements and develop healthy ways of living in a person. A behavioral change can be brought by analyzing the person's demographic details, social factors, and genetic testing. Such digital therapeutic techniques dwell on constant motivation to patients [10].

Artificial intelligence enables certain functions which become essential and useful in every domain that incorporates it. Some of these are summarized herewith in reference to precision medicine [11].

4.1 Analysis of Data

With the help of the artificial intelligence, it has now become possible to analyze and assess a huge amount of data which we famously term as the big data. The artificial intelligence introduces interventions which helps in identification of predictive indicators of diseases along with its possible diagnosis.

4.2 Artificial Intelligence Propelled Diagnosis

The artificial intelligence has spread its tentacles to various subsets of medicine thereby introducing algorithms leading to digital radiology, digital pathology, etc.

4.3 Individualization and Personalization of Treatment

The artificial intelligence not only assist in the monitoring and management of illness but also takes care of the tracking and customization to ensure the presence of treatments that require attention [12].

4.4 Predictive Medicine

By analyzing the health conditions and other information about the patient the predictions can be made for the sake of risk analysis.

4.5 Production of Medical Knowledge

As the system develops and is put into practice, it becomes certain that the research and development in that aspect will be inevitable.

4.6 Personalized Data Collection

The concept of data collection is known to all but in case of precision health the specific information required for the patient's well-being can be collected from the sensory equipment's and wearables to assess pathological patterns, etc.

4.7 Remote Monitoring

The artificial intelligence also for wearable devices where the patient is continuously monitored and doesn't need to be present in the hospital all the times. This continuous monitoring will facilitate the doctors to monitor and evaluate the health conditions of their patients and also modify their medicines or treatment procedures if need be [13].

4.8 Data Sharing

The artificial intelligence is designed to protect and regulations can be made by analogy to protect the information of the patients against the cybersecurity. The

blockchain technology has also been used to protect the healthcare records of the patients such as diagnostic reports, lab reports, genome information, etc.

4.9 Telemedicine

The process helps in remote delivery of medicines and consultations related to medicines [14].

The artificial intelligence driven health improving chatbots is a new innovation that is becoming popular worldwide. These chatbots helps in reduction of cigarette and tobacco consumers, promotes nutritious diet and medical care. These chatbots are sustainable because of its flexibility and interactivity which is available to consumers on demand. The chatbots after collection and analysis of the data suggest the users about information, they should be aware of, behavioral modification therapy or the intensive support that they may need. Chatbots can be accessed by leaving queries on messenger platforms which will immediately be reached out at and responded to in the form of live consultations or on appointment basis. Another technique of deploying artificial intelligence in the healthcare unit is by way of adopting voice recognition. The voice recognition systems perform very arduous but interesting functions for instance, they convert the raw human sound remnants to essential elements of speech. The voice command and virtual assistant tools are the banks of such deposits. Voice recognitions systems are used and adopted in understanding neurological conditions and certain other diseases that have direct consequence on speech capabilities, such as pharyngitis, Parkinson's, Alzheimer's, etc. The voice markers are to be safeguarded against the systematic biases. To prevent such biases the language and the accent of speech must be carefully gathered.

5 Challenges in the Way of Precision Health

Though, the world has been positive about adopting the Precision health techniques for better medical outputs, but various legal and ethical challenges prevail. The concerns that surround the implementation of precision health is based on various propositions but the prominent reason is the collection of huge amounts of data available in every form. Though, the use of data for healthcare improvement is a novel aspect to allow but the involvement of the machines and autonomous systems can create an impact or generate a condition which may require regulation. Above all, it is contended across the jurisdictions that artificial intelligence must not be mindlessly adopted. The ethical considerations in reference to artificial intelligence will always be of paramount consideration. Collection, storage and usage of data in vast volumes can interfere and infringe the fundamental rights and freedoms of an individual. As a precaution the adopters of such technologies must ensure compliance with the core Artificial Intelligence ethics. There are two categories of principles that artificial

intelligence must follow, the first are the epistemic principles which is concerned with assessment of beliefs and cognition. This model questions the users if they understand the reasoning behind Artificial Intelligence based model's predictions. This relates to the concept of interpretability, which means, understanding of the reasoning of a machine. The epistemic principles also focus on reliability and safety. The reliability and safety are the most essential components of precision health technology as a life of an individual is at stake. The biases and wrong predictions can put the life of a person at stake. The second category is of general ethical principles, which focuses on factors like bias, inclusivity and fairness [15].

Artificial intelligence machines sometimes learn biases from human decision-making processes. It is to be assured that artificial intelligence techniques must uphold fairness and practice non-discrimination. Artificial intelligence tools get impacted by biases that exist in data sets. Elimination of presence of such discriminatory content must at every cost be eliminated to assure high quality outputs. The component of the transparency and the accountability can be optimally practiced by making the patients and doctors aware of the ways as to how the results delivered by an artificial intelligence tool are to be read and interpreted. Both the patients and the doctors must also be careful about the limitations of the systems to avoid complete reliance on machines. The human rationality and judgment will help in providing clinical justifications to machines working toward precision health. The responsibility and accountability of the developers of such precision health making tools can never be understated [16]. To ensure the liability, the lawfulness is required. Legislations, regulations, policies and guidelines must regulate and combat the risk involved in precision healthcare. By placing the liability for artificial intelligence, one can be assured of its proper placement and usage [17]. The burden for liability if looked around is majorly placed on medical practitioners, developers or anyone who causes harm by use of such tools and techniques, but there are chances that the mishappening may have resulted because of the automation and not human intervention. Thus, a framework to define the roles and liabilities precisely is of utmost necessity [18].

The situation in context of data privacy calls for not only ethical consideration but also protection in legal context. The abundance of data collected and stored in pretext of precision health analytics if unguarded can be put to misuse. The precision health tools must be kept safe and secure against privacy violations. The compromise of such an amount of data can lead to commission of serious offenses and the innocents may be trapped. There have been number of cases where the data sets of hospital have been hacked into and stolen and later made available for sale on the dark web. The data so made available for sale is bought by multinational corporations that manipulate your belief system and make you vulnerable and susceptible to financial or insurance frauds. The data confidentiality clauses are must. The violations surpass the trust systems of the population [19]. The population of any demography will always be suspicious of people holding their data and its confidentiality. The privacy policies are longer than the plays Macbeth and Othello. The coercive consent policies leave no option with the users but to submit to the will of the developers and the operators. The medical records and genetic information if made available to cyber criminals can cause havoc. The data regulation laws and mechanisms must ensure data security and

restrict data sharing where necessary. When looking at the ethical core, one essential principle is that of beneficence. The technology must always be developed for the benefit of the society at large [20].

6 Precision Health and the Future

The technologies related to healthcare have already made a mark in the society by delivering medical services to doorstep at convenience. The output of predictions has made these tools more reliable and dependable. The dependency on these tools and techniques promotes time efficiency and resource management. The problem lies in countries with restricted infrastructure. As the hospital administration collects immeasurable data, the information management would demand greater attention. The concept of precision health would include certain important concerns and issues such as precision care, gene mapping can assist in determining next generation's sequencing to identify rare genetic disorders. The technology would also assist in pharmacotherapy, by helping us understand the adverse effects of drugs and defective variants that may endanger the life of an individual [21].

The use of nanobots can target the cancer cells in the body and reduce the negative effects caused by chemotherapy. The implementation of gene therapy by DNA manipulation can cure the infectious diseases by manipulating and changing the human genome. The use of smart sensors can assist in monitoring biological fluids and track areas that require attention. Small capsules with sensors can be swallowed to tract and monitor issues in tracts. Bioprinting is another way to enhance life and make it meaningful. The 3D formations can be well adapted in the bodies reducing the fear of rejection and unsuitability by the body. The technology is already used in knee implants, etc. The Brain Computer interfacing is meant for people facing difficulty in memorizing, communicating, concentrating, etc. This technology monitors brain activity and interpret and translates the commands which a person intends to give. This is a rehabilitative technology which works for the benefit of the individuals [22].

Precision health has revolutionized the systems of healthcare. The efficiency and the effectiveness of a treatment is often measured by randomized controlled trials.

7 Suggestions and Conclusion

The paper has made a modest attempt to explain the role of artificial intelligence in healthcare units in general and the concept of precision health in specific. The new era of algorithms defines how the diagnosis, treatment and prevention of diseases can be predicted and analyzed by using the artificial intelligence driven tools and techniques. But there are most important parts to remember and focus on when using the involvement of technology in these circumstances. The technology whether placed

in the hands of technicians or in the use of clinicians must always be regulated to avoid the violation of basic rights and fundamental freedoms of the individual which can directly or indirectly affect the right to life and personal liberty of the individual. Certain rights may be expressed in the statutes and others may be impliedly indicated. In the conclusion, we wish to draw your attention to the most important aspects that would need attention and consideration by the developers of the technology and its implementers as well:

The artificial intelligence machines should be able to explain the processes that delivered a certain outcome to assure its reliability and authenticity.

The datasets so collected from different sources may be contaminated with misinformation. The developers or the operators of artificial intelligence technology must assure that the outcomes are free from any biases and are just fair.

- a) The data sets before processing must be cleaned of any wrongful or defaulted information as its predictions in regard to diagnosis and treatment can endanger the life of the patient.
- b) The relevant data must be sorted intelligibly from the unstructured or the non-standardized data. Building up of structured and defined data sets can minimize the accessibility issues and barriers to domain knowledge.
- c) Employing equitable curation of data can also enhance the trust and reliance of the patients.
- d) The data and personal information so collected by the hospitals must be in compliance with the data protection regulations of that demography. The information collected must be proportional to the information required and nothing excessive.
- e) The algorithms so developed by private entities or bought from other jurisdictions must be free from backdoor and vulnerabilities to unauthorized access and cyberattacks. Thus, the cybersecurity policy must be put in place strictly.
- f) The most important of all the suggestions is in regard to confidentiality and privacy. The data being collected by the hospital may be critical to a patient's health condition and must not be made public or available for access to unauthorized people or insurance companies. The transparency and the accountability of a technology will be measured in terms of protection a machine can guarantee.
- g) The legislators should be able to propose law, regulations, and policies along with a redressal body to address the grievances.
- h) The technology so inducted must be ethical and favorable to health equity.

Thus, by adopting the suggestions mentioned above the precision health can raise the thresholds of the healthcare system around the world. The technology is not only helping the sufferers recover but also promising toward a healthier landscape aiming to achieve longevity of life. The new terminology of being healthy is no more about the diet and exercise. People are getting aware to participate by way of genome sequencing and algorithmic predictability and cure their diseases or prevent them. The apps are contributing in a major way to monitor the health of the individuals. But too much dependency and incorrect warnings by the technological devices can become the causes of anxiety and disbelief among the patients. Thus, the use of

technology can achieve the positive objectives only on the fulfillment of its ethical considerations.

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Telemedicine in India: Bridging Healthcare Gaps Through a Socio-Legal Lens



Tanvi Yadav and Akshit Mishra

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Abstract The Telemedicine Industry is a rather contemporary phenomenon causing legislative lacunas to arise due to its nascence in the medical ecosystem. COVID-19 was a severe trigger for the industry to bloom due to lockdowns and social distancing policies. Since a hasty regulation was put in place for this industry, there remain several domains which require immediate attention for an effective growth of this medical service in the future. The industry brings several benefits for the Indian population, remote and rural ones in particular. This research is an attempt at highlighting those benefits, lacunas and possibilities that can be accrued with a more holistically planned telemedicine regulation. The article also dives into the specificities of harms that come to the populace in the status quo. India has lot to gain from this step only if it is taken with proper legislative intent as well as socio-legal policy measures.

Keywords Telemedicine · COVID-19 · Socio economic · Legislative

T. Yadav
National Forensic Sciences University, Delhi Campus, Delhi 110085, India

A. Mishra (✉)
Gujarat National Law University, Gandhinagar, Gujarat 382426, India
e-mail: akshitmishra.law@gmail.com

1 Introduction

India is often considered a wild card as well as an underdog in the technological and healthcare avenues. Our journey to the moon along with the astounding COVID vaccine initiative caught global attention despite our reputation of being a developing, global south economy. Especially with regards to the healthcare industry, India has shown the world that it is capable of surprising naysayers with policies of enormous scales, cost-effective nature and progressive results. India has a healthcare system centered around multi-payer, universal healthcare model [1]. The government has a large web of hospitals, educational and research institutes in order to provide medical care for its population at affordable rates. The polio vaccine initiative [2], COVID vaccine development [3], program on elimination of leprosy [4], containment of H1N1 [5] are few of India's success stories in healthcare. Affordable medical procedures and swaths of world class surgeons make India a worthy nation for medical tourism as well. In fact, India ranks 10th in the list of countries for medical tourism [6], making it a popular choice for a variety of medical procedures such as IVFs, knee replacements, heart bypass, dental implants, etc.

A very important reason behind this popularity is the affordability which is mostly driven by the country's need to cater to the underprivileged masses. The procedures are often quick and technologically sound. A demographic hosting a billion and a half people can never continue to continuously provide healthcare without technology which is quickly incorporated in the system and techniques that are globally prevalent for the treatment of all kinds of ailments. This adaptability has always been India's strong suit in healthcare provisioning. One such need for adaptation arrived when COVID eclipsed the globe. Lockdowns prevented movement which adversely affected the sick and ailing patients across the country. At the same time, due to the global advancements in technology, the phenomenon of telemedicine was also rising. There was a growing market for providing medical services through virtual or telephonic mediums. Therefore, the Indian government hastily rolled out the telemedicine regulations [7] ultimately to adapt and continue medical services.

The world after COVID has drastically changed. Industries have created structures of services and work inclusive of work-from home models, goods and service delivery models and e-commerce ecosystems. India is yet to witness a large scale, systematic growth in the telemedicine industry. The nation has lot to benefit from telemedicine, more critically in remote areas where hospitals, doctors, and medical services are harder to mobilize. India has a lot of challenges when it comes to addressing its massive healthcare requirements but telemedicine can be a panacea for the spectrum of shortcomings that Indian healthcare system is currently dealing with. India currently faces a scarcity of doctors and medical professionals. This problem is particularly true for the rural settings where there is also a shortage of medical facilities.

Socially, legally and economically there are several reforms needed for ensuring a safe and steady growth of telemedicine. The authors in this research are aiming to

provide both, extensive socio-legal information as well analysis of telemedicine in India and the probable solutions that can be employed for its healthy propagation.

2 Understanding Telemedicine and Its Phenomenon

With the dawn of technology, all the associated domains attempted a shift from physical operations to telephonic, virtual and electronic mediums. This is true for almost all sectors. Healthcare on the other hand, needs a very hands-on approach. The physical presence of a doctor is both more effective and accurate for diagnosis as well as treatment. But the medical procedures do not always require that physical presence. For instance, in emergencies, minor ailments and distant regions even a consultancy or a provisional relief can be a life-saver. Thus, came Telemedicine. As per the World Health organization Telemedicine is the “The delivery of health care services, where distance is a critical factor, by all health care professionals using information and communication technologies for the exchange of valid information for diagnosis, treatment and prevention of disease and injuries, research and evaluation, and for the continuing education of healthcare providers, all in the interests of advancing the health of individuals and their communities” [8].

The term “telemedicine” originated in the 1970s with Thomas Bird [9] and denotes “healing from afar,” amalgamating the Latin “medicus” and the Greek “tele.” Its inception coincided with the emergence of communication technologies like the telegraph and telephone. Although methods of conveying medical information existed before these innovations—for instance, signal flags on ships to alert about infectious diseases—they were reliant on human intermediaries and were not efficient for transmitting medical data across long distances. The introduction of telegraphs and telephones marked a pivotal moment in the evolution of telemedicine, facilitating rapid communication across vast distances, albeit requiring specialized knowledge to operate, particularly in the case of telegraphs. The military promptly recognized the healthcare advantages offered by telegraphs and telephones, which eventually extended to general communication purposes.

In the Telemedicine Guidelines [10], telemedicine is defined as: “The provision of healthcare services, particularly when geographical distance is a significant factor, by healthcare professionals utilizing information and communication technologies. This technology facilitates the exchange of accurate information for purposes such as diagnosis, treatment, disease and injury prevention, research, evaluation, and the ongoing education of healthcare providers. All of these activities are conducted with the aim of promoting the well-being of individuals and their communities.”

In terms of allopathic doctors, India has 1,445 doctors for every Indian citizen [11]. This statistic gets even worse when the rural population of India is considered given that the medical industry is hyper-centralized in urban spaces. Based on research conducted by the WHO [12], it was found that 59.2% of health workers are situated in urban regions, despite only 27.8% of the population residing there.

Conversely, 40.8% of health workers operate in rural areas, where 72.2% of the population resides. Telemedicine emerges as a potential solution to mitigate these imbalances by allowing healthcare professionals in urban settings to provide consultations and specialized care to the rural population.

There are more issues with this statistic, since medicine requires specialists of different areas such as brain, bone, ear-nose, heart, skin, etc. [13]. A simple physician is unable to provide proper services to varied patients. All the more, the urban spaces are costly both in terms of travel and service expenses for the rural populace [14]. It is to be noted that more than 65% of Indian population still dwells in rural settings. Therefore, it is a severe case of inequality and helplessness. Access to healthcare is a constitutional right of all Indian citizens regardless of their place of origins. It is closely tied up with the fundamental rights of Life and Dignity under Article-21 of the Indian Constitution [15]. A denial of such rights is a blot on the social welfare model that we as Indian citizens take immense pride in. Telemedicine can become a soothing agent to this fundamental issue till a long-term solution is reached and more services are provided to the remote locations across the country.

3 Advantages of the “Tele-Shift” of Medicine

In today’s dynamic world, telemedicine has emerged as a revolutionary tool in the healthcare sector, offering unprecedented benefits and addressing contemporary challenges. Below are several key reasons why telemedicine plays a crucial role in our current society:

1. **Enhanced Accessibility and Convenience:** Telemedicine breaks down geographical barriers, granting individuals access to healthcare services regardless of their location. This accessibility is particularly significant for patients in remote areas, those with mobility constraints, or those facing transportation issues.
2. **Timely Consultations:** Telemedicine provides swift medical consultations, minimizing the waiting times typically encountered in traditional healthcare settings. Patients can promptly receive advice, diagnoses, and treatment recommendations, leading to faster recovery and improved health outcomes.
3. **Cost Savings:** Telemedicine can result in cost savings for both patients and healthcare providers. By reducing the need for in-person visits and associated travel expenses, it offers a more economical alternative. Additionally, it can streamline administrative processes, further enhancing cost efficiency.
4. **Continuity of Care:** Telemedicine supports seamless and coordinated care by enabling remote consultations with primary care physicians or specialists. This facilitates ongoing monitoring of chronic conditions and ensures consistent follow-ups, ultimately enhancing disease management and prevention efforts.

5. **Pandemic Preparedness:** The COVID-19 pandemic underscored the crucial role of telemedicine in containing the spread of infectious diseases. It enables the triage of potentially infected individuals without necessitating physical visits to healthcare facilities, thus alleviating the strain on hospitals and clinics.
6. **Mental Health Support:** Telemedicine has become instrumental in addressing the growing demand for mental health services. Individuals can access therapy and counseling from the comfort and privacy of their homes, mitigating the stigma associated with seeking help for mental health issues.
7. **Remote Monitoring:** Telemedicine encompasses more than just consultations; it also facilitates remote monitoring of patients' vital signs, chronic conditions, and recovery progress in real-time. This proactive approach enhances patient care and minimizes the risk of complications.
8. **Healthcare Equity:** Telemedicine has the potential to narrow healthcare disparities by ensuring that underserved populations, including rural communities and marginalized groups, receive equitable access to quality healthcare services.
9. **Global Collaboration:** Telemedicine fosters collaboration among healthcare professionals worldwide, allowing experts to consult on complex cases, exchange knowledge, and contribute to medical research and education on a global scale, thereby raising the standard of care.
10. **Environmental Sustainability:** By reducing the need for physical travel to medical facilities, telemedicine contributes to a smaller carbon footprint, aligning with sustainability objectives and eco-friendly practices.

In summary, telemedicine is not merely a convenience but a necessity in our contemporary society. It democratizes healthcare access, reduces costs, and promotes continuity of care. As technology continues to advance, telemedicine's importance is expected to grow, promising a healthier and more interconnected world for all.

4 A Legal Perspective

The Indian Medical Council Act 1956 [16], Drugs and Cosmetics Act, 1940 and Drugs and Cosmetics Rules, 1945 have been the key regulatory legislations for the medical profession. After COVID hit India in 2020 a set of new guidelines were introduced along with the National Medical Commission Act of 2020[17]. The NMC Act superseded the previous regulation, the Indian Medical Council Act of 1956[18], which governed the medical profession until September 2020. Transition provisions within the NMC Act stipulate that regulations established under the old act remain valid until new standards are specified under the NMC Act. These regulations are considered to be issued under the corresponding provisions of the NMC Act.

One such regulation created under the Indian Medical Council Act is the Indian Medical Council (Professional Conduct, Etiquette, and Ethics) Regulations of 2002 [19], known as the MCI Code. This code outlines the professional and ethical obligations for doctors concerning patient interactions, dealings with pharmaceutical

companies, and conduct within the medical field. The MCI Code remains effective and is assumed to be issued under the NMC Act unless a separate regulation on medical ethics is established under the new act. In May 2022, the Ethics and Medical Registration Board under the NMC introduced the draft Registered Medical Practitioner (Professional Conduct) Regulations of 2022, known as the Draft RMP Regulations. In 2023, the draft regulations came into force.

There is also an intermingling of the Information and Technology Act 2000 [20] with the above-mentioned regulations due to transfer of electronic data and concerns of privacy. Telemedicine involves a continual exchange of data between the patient and the service provider. The patient's personal details, including medical history and physiological data, fall under the category of Sensitive Personal Data or Information (SPDI) according to the Data Protection Rules. When a corporate entity collects, stores, transfers, or processes such information, it triggers certain obligations outlined in the Data Protection Rules [21]. In essence, the Data Protection Rules mandate that data collection and processing should be carried out based on the consent provided by the information provider. Additionally, these rules require corporate entities to implement sufficient security measures to ensure the safe storage of data.

On March 25, 2020, the Ministry of Health and Family Welfare in India issued the "Telemedicine Practice Guidelines" specifically for allopathic registered medical practitioners (RMPs). These guidelines were developed by the Board of Governors, Medical Council of India, in collaboration with the National Institution for Transforming India (NITI Aayog). These guidelines include, definitions, scope, legality and applicability of telemedicine in India. The Telemedicine Practice Guidelines (TPG) were introduced in March 2020 to offer direction to Healthcare Providers (HCPs) regarding telemedicine practices amidst the COVID-19 pandemic. Before the issuance of these guidelines, telemedicine practices were managed in an ad-hoc manner under relevant sections of the Indian Medical Council Act (IMC Act), Medical Council of India (MCI) Code, and the Information Technology (IT) Act, as discussed in the "Regulatory Framework" section of this document. Although these legislations remain applicable, the TPG aim to address gaps in previous regulations and provide necessary clarifications.

The primary objective of the guidelines is to support medical practitioners in delivering effective and safe medical care based on current knowledge, available resources, and patient requirements to ensure the safety of both patients and providers. Consequently, the TPG grant considerable discretion to HCPs in determining the appropriate course of action during telemedicine consultations.

5 The Exigency for a Legislation on Tele-Medicine

The current telemedicine regime is largely cluttered due to two main reasons. Firstly, there are a lot of aspects of allopathic practice that are yet to be included in the regulations provided by the government. Secondly, important issues such as alternative medicine and privacy should be dealt with in a more systematic manner. Liability and

enforcement concerns are secondary to the previous two matters but just as important. A thorough legislation is bound to create a compendium of all such considerations for future purposes given that the private telemedicine industry is growing at a rapid rate. Here is a list of suggestions for a possible legislation on the same:

- All the present and past regulations should be combined to make a singular policy to be under the NMC Act further supervised by the data protection laws.
- The regulations are only concerned with those medical practitioners that are registered under the Act. The issue here lies in the fact that several other practitioners such as dentists, dermatologists, etc. are not covered under the same head. These practitioners should also be regarded in the processes since the market will continue to provide services and the regulations would need to grow alongside the same.
- The regulations are not concerned with other practitioners of telemedicine such as alternative medicine. The practitioners in these areas recommend treatments to patients with a promise but no scientific backing for the same. Ultimately the patient suffers while the practitioners as well as intermediaries make heaps of profit. Alternative medicine should be parallelly regulated in this form of health-care service in order to guarantee a more holistic and scientific approach to healthcare.
- The intermediaries involved in these practices for the private domain are also liable but as per two different acts of medical council as well as IT act 2000. This creates confusion and lack of effective enforcement. Even though there is liability attached to the wrongful acts. A single act will help consolidate the laws and relieve confusion.
- Privacy and data concerns are a pervasive issue, but in this particular matter, the privacy issue is with regards to a sensitive case of medical ailments. Breaching the same can attract severe consequences for the patients. Privacy should be at the top of priorities in these matters especially in an impending legislation.
- The liabilities and penalties in these matters can be made more severe to create a system of more effective deterrence against breach of duties by intermediaries and practitioners. The patients should not suffer due to the lack of proper legislative or policy-based initiatives.
- A government-based policy for providing telemedicine services from national and State based healthcare institutions will create a larger scope of healthcare services. This will also help reach areas which are usually left untouched by the private players. In this regard awareness, education as well as affordability of services can be key factors for deploying said services.
- Prescription of medicines is a major concern since consultation is pointless if a cure cannot be provided. Since only List-B medicines are permitted to be recommended or prescribed it creates an impossibility factor at a stage where it should be a given. There is also a lack of regulation on previous prescriptions and their usages. Creating avenues for prescription of all required medicines as well as utilization of previous prescription's would significantly increase the extent and quality of treatment.

- The pricing of such services, fees for consultation and other aspects should also be actively regulated to ensure a more affordable and regulated ecosystem of services.
- Better enforcement mechanisms such as faster online complaint redressal mechanisms, social media redressal, ground officers and survey squads can lead to a system free of unnecessary misuse of the mediums.

Healthcare is a fundamental right. In the case of *Bandhua Mukti Morcha v. Union of India* AIR 1984 SC 812, the Supreme Court ruled that while the Directive Principles of State Policy (DPSP) are not legally binding obligations but carry persuasive significance, it is essential for the State to ensure their proper implementation. Additionally, the Court determined that dignity and health are encompassed within the scope of life and liberty as guaranteed by Article 21 of the Indian Constitution. The Indian constitution vouches for these rights in various parts extensively. Ensuring a better outreach of effective healthcare becomes a key responsibility of the state.

6 Conclusion

The race between technology and law is always that of a rabbit and a turtle. One is quick and hasty but often carefree and the other is steady but slow. In our world technology is much faster than the analogy of a rabbit and the turtle cannot lag too far behind. Therefore, with all the lacunas in mind, the government needs to adapt to future possibilities and make a more holistic set of regulations and legislations that can counter the rapid speed of the market. There is a basket of benefits that rest in the telemedicine industry for everyday patients waiting to be healed. It can be expected that such regulations will soon follow but it is also important to thoroughly understand the issue at hand before making such amends. Once completed India will be ready to compete with international markets and continue to service those who are yet to be taken into care by the conventional healthcare system. Access to healthcare being the motivation behind the entire move. As always India adapts and innovates better.

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A Framework for Development of a Virtual Reality Environment for Building Empathy in Indian Nursing Professionals



Manisha Mohan, Latika Mohan, Sandhya, Rakesh Sharma, Agam Agarwal, Anirban Chowdhury, and Naveen Kumar

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Abstract Empathy or the ability to interpret and communicate another individual's emotions and condition could be considered the cornerstone of the healthcare professionals' relationship with patients. It is important for healthcare professionals to practice empathy while connecting with patients to ensure their well-being and motivation. Nurses especially need to interact with patients and their families more frequently, and therefore need to constantly demonstrate empathetic behavior and compassion. While the Indian Nursing curriculum includes training in communication skills, there is a need to practice these behaviors in real contexts and be able to

A. Agarwal · N. Kumar (✉)
UPES, Dehradun 248007, Uttarakhand, India
e-mail: naveen.kumar@ddn.upes.ac.in

M. Mohan
Bennett University, Plot Nos 8, 11, TechZone 2, Greater Noida, Uttar Pradesh 201310, India
e-mail: manisha.mohan@bennett.edu.in

L. Mohan · Sandhya · R. Sharma
All India Institute of Medical Sciences (AIIMS), Rishikesh 249203, India

A. Chowdhury
Alliance School of Design, Alliance University, Bengaluru 562107, Karnataka, India

observe measurable improvements. There are also several factors such as upbringing, the education system, shortage of manpower and resources, and a skewed nurse to patient ratio which lead to higher stress factors and lack of empathy in Indian Nurses. There is a need for training and mentorship for nurses and an environment that helps them practice empathetic behavior, without impacting their daily work or the well-being of patients. If this training is available online or technology enabled, it can also free up time for senior nurses who may otherwise be providing mentorship to younger nurses. Virtual Reality (VR) provides the opportunity of creating an environment where nurses can experience a real hospital setting, with realistic scenarios and interactions with patients built in to practice empathetic behavior. This study, based on a review of existing literature and conversations with nurses, explores the role of a nurse and identifies areas where empathetic behavior can be practiced. It also researches existing segments where VR has been successfully implemented in the healthcare context. It arrives at a framework for creating VR-based training for Indian Nurses on empathy. This framework can also be extended to building other VR-based soft skill training for healthcare professionals.

Keywords Virtual reality · Empathy · Soft skills · Nursing professional · Healthcare systems · Empathy Scale First Section

1 Introduction

Empathy can help healthcare professionals provide better care, as they are more likely to understand the patient's needs and concerns more closely. When healthcare professionals try to understand their patient's condition better and take steps to support them in their recovery journey, it is proven to result in patient well-being [1]. Research has shown that empathy and compassion can also have a positive effect on clinical outcomes of the patient [2, 3]. Patients who feel that their healthcare providers (doctors and nurses) understand their pain tend to show deeper satisfaction with their care and are more motivated to follow their treatment plans [4, 5]. This can lead to better health outcomes [6], such as improved wellness overall of the patients. Since nurses are dealing with patients and their families more closely, it is important for them to practice empathetic communication and behavior. When nurses feel a connection with their patients, they are more likely to feel a greater sense of fulfillment and encouragement to provide better care [7].

In India, after higher education, during workplace transition, the mentoring of novice nurses is conducted by senior nurses and doctors [7–9]. The focus of this mentoring is primarily on clinical skills, not so much on soft skills, and outcomes are not easily measurable. Therefore, existing training methods need to be redesigned to include online and experiential methods such as Virtual Reality (VR) and inputs on emotional intelligence and empathy need to be included to improve the performance of the nurses.

This study derives insights from the use of VR technology for training healthcare professionals and existing frameworks for VR, game-based learning, and Nurse training. Additionally, it outlines a framework for the development of VR simulators for Indian Nurses training from the perspective of building empathy. The following sections attempt to address existing VR training systems for healthcare professionals, existing empathy measurements for healthcare workers, the research gap, and the proposed framework.

2 What Are the Existing VR Applications for Healthcare Workers?

There are many definitions of Virtual Reality; however, in essence, the word ‘virtual’ means simulated, and ‘reality’ means that it closely emulates an experience as a human being. VR can be experienced through a headset and controllers which enable users’ to see, hear, and interact in a simulated three-dimensional world [10].

VR provides the unique ability to recreate relationships akin to the physical world and enables repetitive practice in a safe, realistic environment thus making it very useful for training and skill-building for healthcare professionals [11]. It helps train medical students learn anatomy, infection control, and the practice of surgery. A complex surgery can be performed through virtual data and a 3D patient, and this can help to build confidence and skill among medical students and young doctors without causing harm to any patient.

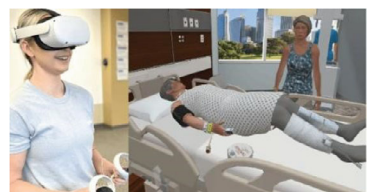
VR has been used for multiple applications in the healthcare domain from Training to Diagnosis to Patient Education and Treatment [10]. Figure 1 summarizes the major applications for VR in the Healthcare Segment, whereas Table 1 features some examples of simulations created for doctors, nurses, and caregivers.

To fully harness the potential of VR in healthcare, the hardware must be selected carefully based on its relevance to the task or skill. Some of the VR devices that

Training	Diagnosis	Patient Support	Treatment	Medical Marketing
• Learning anatomy, infection control • Building of clinical and surgical skills in a safe environment • Building of soft skills such as leadership, empathy and communication skills	• Medical problems replicated in VR for diagnostics • Medical students can input a patient's vital signs and identify the condition so as to decide the next line of treatment within the VR environment.	• Awareness of impending treatment through VR experiences • Physical Rehabilitation and Pain Management • Management of Phobias	• Doctors can plan treatment/surgery in a safe environment • To give a visual experience remotely during a robotic surgery	• VR may be used for demonstrating the action of a drug at a cellular level, in the human body to doctors • To demonstrate the features of medical devices

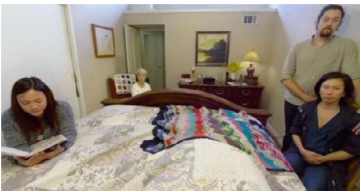
Fig. 1 Applications of virtual reality for healthcare

Table 1 Examples of virtual reality for healthcare

VR intervention	Company name and purpose	Description
	Virtual Knee Surgery by Osso VR [15]	Virtual Reality based training for surgeons on the treatment for a Tibial Shaft Fracture
	Military Medical VR Training by SIM [16, 17]	Immersive scenarios to prioritize war injuries, emergency response, and rescue operations for severely injured
	VR Training for Nurses by SIMX [17]	This includes 150 simulations in up to 97 settings and customizable patient interactions
	Soft skills training by immerse for working professionals [18]	This VR focuses on building an emotional connection to drive positive outcomes in the workplace. Specially for diversity and inclusion training

(continued)

Table 1 (continued)

VR intervention	Company name and purpose	Description
	Analytics based on Soft skills training by immerse for working professionals [18]	The VR provides analytics and measurable data on learner performance on softs skills. This data is not only useful for providing areas for improvement for the learners but also drawing organizational attention to critical issues
	End of Life Conversations by Embodied labs [19]	Care givers walk the patient's shoes and visualize the end-of-life experience and important decisions that are taken during that phase

are used by healthcare professionals include VR controllers, a stylus-like controller for surgical equipment, and haptic gloves [12]. VR experiences are typically distinguished by how immersive they are [13]. Low-immersion VR systems use images with lower fidelity to reality and they may be experienced on a regular desktop via a keyboard or mouse. However, at the other end of the spectrum, high-immersion VR completely engages the user in a hyper-realistic world using advanced VR headsets, and the interactivity is largely based on tracking the head and body movements along with voice and gaze [13, 14]. Figure 2 highlights the different modes of interaction in a VR environment.

The fact that VR can be used for learning and practicing clinical and surgical procedures is already established. Surgeons treat a patient when they are unconscious or numb due to anesthesia; however, the overall well-being of the patient in a state of consciousness is in the hands of the nurses. In addition to learning clinical skills effectively, nursing students also need to learn how to address their patient's

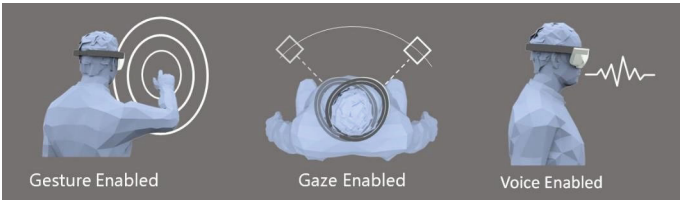


Fig. 2 Multi-modal interactions in VR (Design Considerations for the Immersive World, M. Mohan, CII Stride Journal (CII_STRIDE_Vol_2_2022_Digital_Circulation.pdf) (ciitechnology.in)

psychological and emotional aspects [11]. It is first important to understand the role a nurse plays during the interaction with patients and their families.

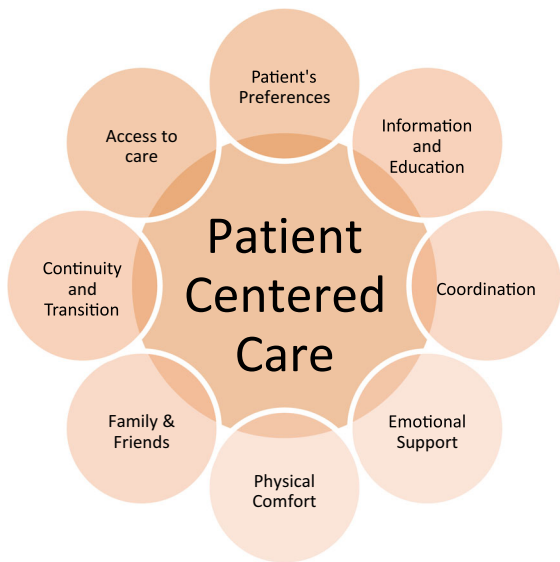
3 The Role of Empathy in the Nurse–Patient Interaction

There are many dimensions of the nurse’s role, and the clinical procedures they follow may vary by country, type of hospital, and type of ward. However, there are some aspects that are common across all nursing professionals. A nurse is responsible for monitoring patients, administering medication in consultation with doctors and communicating and educating patients [20]. In patient-centered care, the nurse’s role is more of collaboration with patients and their kin to provide support that is tailored to the needs and requirements of each individual patient at every stage of treatment, as represented in Fig. 3 [21].

Empathy is a critical attribute in nursing professionals as they interact the most with patients and their families. Building hope and trust with patients is essential for establishing empathetic communication. Rutherford hypothesized that trust is an essential, though intangible component, for building meaningful relationships with patients [22, 23].

The Indian Nursing curriculum includes soft skills training; however, there is no defined focus on empathy. Factors such as experiencing a non-empathetic childhood themselves, overwork, shortage of time, and lack of understanding of the importance of empathy negatively impact nurse’s behavior toward patients. The atmosphere at work and the state of the patient have a bearing on nurses’ ability to practice

Fig. 3 The role of a nurse in patient-centered care (patient-centered care—the role of the nurse—Nurses In Ghana)



empathy. Certain conditions such as coping with severe mental disorders or aggressive conduct of patients and their family members, affect and lead to burnout among nurses and cause reluctance to practice empathy [24, 25]. Additionally, nurses in the ICU and emergency departments often experience stress while handling patients with acute and near-death conditions. These variables impact the nurse's psychological readiness and ability to practice empathy [26].

Based on a focus group with 18 nurses, it was determined that empathy is required by nurses in multiple situations of different severity levels while interacting with patients and their families:

1. Breaking Bad News
2. Interacting with elderly patients
3. Caring for terminally ill patients
4. Communicating with children
5. Treating patients in pain
6. Managing chronically ill patients
7. Dealing with patients in the emergency ward and their families

Empathy is a critical component of nursing practice in all their interactions with patients. VR simulations are being effectively used in healthcare education for building clinical skills [11]. However, there are no VR solutions so far for developing empathy in Indian Nurses. Before interventions for empathy are created, it is important to measure the current level of empathy, as it is important to establish the status before improving it. The next section addresses the existing tools to measure empathy among healthcare professionals.

4 What Are the Existing Tools to Measure Empathy?

In order to improve empathy, it is important to understand its components and the tools available to measure it. Empathy has cognitive, affective, and behavioral aspects. Self-reported surveys and observer feedback are used to measure and identify the mental processes that drive empathetic behavior [27].

The Jefferson Scale of Empathy [JSE] is the globally recognized tool for empathy measurement in the healthcare context. Other tools include Questionnaire of Cognitive and Affective Empathy (QCAE) and the Interpersonal Reactivity Index (IRI). These measure the different aspects of empathy—cognitive, affective, and their respective subparts [28].

Although empathy plays such a significant role in patient care, there has been limited research on empathy in the context of Indian nurses [29]. The Jefferson Scale of Empathy itself is not adapted to the Indian context. There was however in a study conducted on Indian medical students, empathy levels were observed to be less than that of subjects tested in other countries.

5 Gap in Literature on Empathy Measurement for Designing VR Environment for Indian Nurses

There are 3.07 million nurses in India—and the nurse to people ratio is 1.7 to 1000, a significant 43% below WHO norms (Rajya Sabha, March 2020). Indian nurses experience multiple stress factors because of the shortage of staff and time.

There is a strong correlation between patient well-being and nurse empathy [30]. Although empathy has been researched in multiple contexts, there is minimal research on the current level of empathy in the Indian Nursing context and its influencers.

The tools for empathy such as the JSE are also administered directly to the healthcare professional; however, an assessment of the behavioral aspect would need observer feedback. A comprehensive and reliable framework is needed to evaluate all aspects of empathy in the Indian context including the factors that drive empathy in India [31–34].

6 Existing Frameworks for Game-Based Learning, Virtual Reality, and Nursing Education and Softskills Training

There is limited research available on frameworks for using VR to teach empathy to Indian Nurses; however, in order to arrive at an effective framework, a study was conducted of existing frameworks for Virtual Reality, Game-Based Learning, and Nurse Education and Softskills training.

6.1 A Framework for the Use of Immersive Virtual Reality in Learning Environments

In this framework, Mulders M. et al. highlighted that in using immersive environments for learning, the possibilities and challenges of VR technologies should be considered carefully. They also stressed the criticality of defining learning goals and focusing on instructional design rather than only the technology aspects while creating VR-based learning [35]. The framework created by them defines immersive VR as multimedia—as it is an interplay between visuals and text. The framework finds its foundation in Mayer’s Cognitive Theory of Multimedia Learning (CTML) [36] to build meaningful learning experiences through VR. Additionally, this paper also stresses the importance of repetitive play and meaningful VR experiences to make successful transition from the learning environment to action in the workplace.

6.2 *Student-Centered Digital Game-Based Learning: A Conceptual Framework and Survey of the State of the Art*

Coleman T et al. evolved a framework using digital games for student-centered learning, which considers relationship between player engagement, type of intervention, and gameplay alongside educational principles inherent in games such as understanding, problem solving and learner empowerment [37]. They stress that game-based learning should rely on principles of game design to make high quality and engaging experiences for learners.

6.3 Enhancing Reflection and Empathy Skills via Using a Virtual Reality-Based Learning Framework

Stavroulia K et al. conducted an experiment to measure the impact of VR-based training interventions in building empathy and reflection among teachers. The framework references the ADDIE model (Analysis, Design, Development, Implementation, Evaluation) which is also a commonly used instructional model in creating effective online learning interventions [38]. This study highlights that VR inherently gives learners the opportunity to experience a situation from the lens of another individual, thus making it an apt tool for teaching empathy.

7 Proposed Framework for Development of VR Environment for Nursing Professionals

The study of existing frameworks stressed the importance of instructional design in the creation of VR-based learning interventions as well as the criticality of choosing the right technology for making VR-based learning interventions meaningful and engaging.

Although VR is proven to support an active learning experience, the learning may not be effective unless the learning outcomes are identified, the learning is embedded in contexts that the learners can relate to, and the mechanism for assessing and measuring them is clearly defined upfront, as described in John B. Biggs theory of Constructive Alignment [39–41]. Therefore to create an effective VR environment to build empathy among Indian Nurses, there is also a need to understand the challenges and constraints under which they work [20, 42], their aspirations as well as their attitudes toward empathy through observation and focus groups [18, 43]. The VR training should positively reinforce the cognitive and behavioral aspects of empathy, and help to modulate the emotional reactions which may be triggered when interacting with virtual patients in severe conditions [25]. In the context of measuring

effectiveness, apart from assessing the learner engagement through a learner feedback form, a reliable empathy measure or scale would need to be used before and after administering the VR experience. This would help to assess the improvement on cognitive aspects of empathy and observer feedback can be taken from patients and senior nurses to measure improvements in the behavioral aspects of empathy of the nurse [44].

Understanding the principles of immersion and game design is also critical for designing an engaging learning experience in VR which inspires repetitive play and drives behavioral change [45, 46]. While VR opens an entirely new paradigm for learning, the possibilities, and limitations of interactions within the environment as well as evolving headsets and controllers need to be considered for building successful interactions [15, 47]. There is a need to capture real scenarios that nurses face during their interactions with patients to include in the VR so that they can gain mastery by practice within new contexts as defined in Brenner's Novice to Expert Theory [48, 49]. Since there are multiple critical factors to consider, there is a need to build a repeatable framework which considers all the critical factors toward building empathy in nurses through VR. The defined framework in Fig. 4 can later also be extended for VR solutions focused on other soft skills in healthcare professionals.

The initial stage of understanding the gap and identifying relevant scenarios in Indian Hospital context to practise empathy is a critical component of the framework. Usability and complexity of VR devices needs to be considered for novice users with limited experience with games. Simpler, phone-based devices such as the google cardboard are more accessible and cost effective, therefore easier to disseminate over a larger population of nurses [50].

As studied in existing game frameworks, repetitive play and reflection is critical for meaningful learning to occur. Therefore, rather than an individual scenario a scenario bank and scenario generator can be developed which can provide nurses the opportunity to practise empathy in diverse situations [37].

8 Experiment on Indian Nurses

The proposed framework is based on ongoing research being conducted in AIIMS Rishikesh on 150 nurses of the Emergency, Obstetrics, Cardiology, and Gynecology Departments. An empathy scale tailored to the Indian context was initially created and pilot tested on 18 nurses of diverse sociodemographic backgrounds [51]. The scale was designed based on inputs from behavioral experts, human factors specialists and healthcare professionals; additionally, it included some items from the existing JSE scale. Based on the pilot test, the scale was found to be reliable (Cronbach's alpha > 0.70). A pilot test has been conducted on nurses with a VR simulation on an empathetic conversation between a nurse and patient. Effectiveness is measured through a Learner Feedback form wherein factors such as relevance to context, engagement and immersion, usability and perceived learnability are assessed. The learner feedback form was adapted for the nursing scenario based on the heuristic

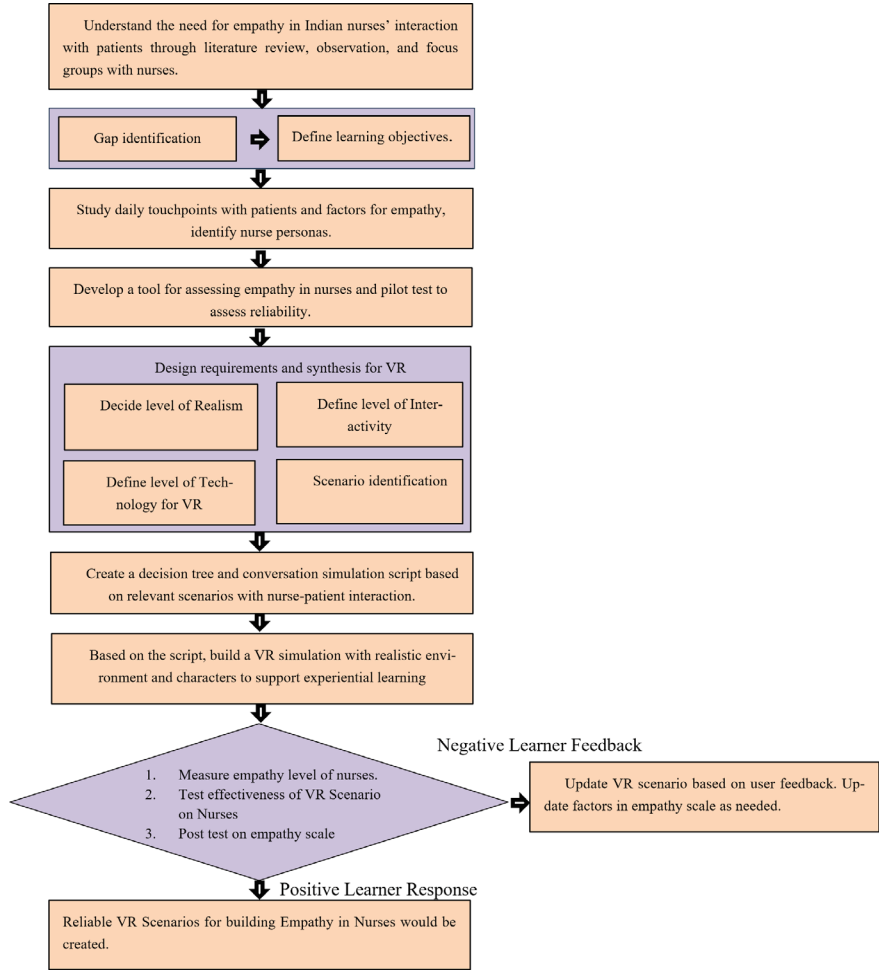


Fig. 4 Proposed framework for development of VR environment for nursing professionals

evaluation of virtual reality applications [52]. The learner feedback form has been tested and has been found to be reliable (Cronbach's alpha > 0.70). The stages being followed in the experiment are outlined in Fig. 5.

The initial findings during the pilot test of the VR empathy simulation determined that nurses feel the prototype is aligned to the context, engaging, easy to use and is addressing the learning objectives. Further research will be conducted on the effectiveness and impact on empathy of 3D VR vs other media such as photographic scenarios, 360 VR scenarios and a control population that has not undergone any intervention to prove the relative effectiveness of VR.

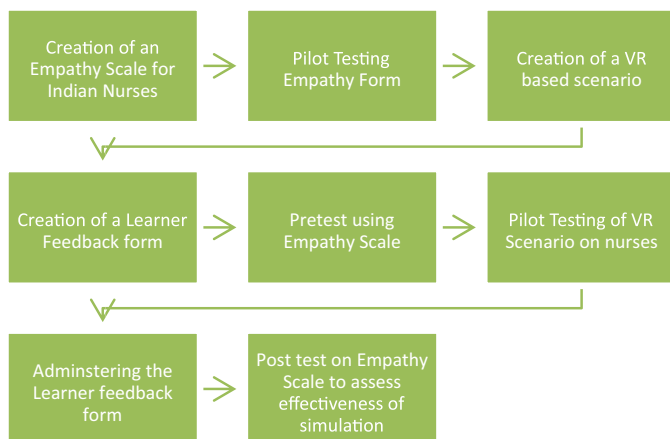


Fig. 5 Steps followed in testing effectiveness of VR in building empathy in nurses

9 Conclusion and Further Explorations

Virtual Reality provides an appropriate platform for healthcare workers, including nurses to learn and practise soft skills such as empathy which are crucial for patient well-being and motivation. Although communication skills for nursing professionals have been taught and assessed, the long-term impact of the training has not been evaluated. Also, there is limited research on the effectiveness of virtual reality in building empathy in Indian Nurses [33, 53].

A framework is required to build a successful VR solution that can be repeated and help to further create levels or new scenarios. This framework should include sharply defined measurable goals, inputs from nurses, realistic hospital scenarios and consider the limitations and possibilities of a fast-evolving technology to ensure successful outcomes. A robust framework will ensure that learner inputs are captured effectively and VR is used appropriately for meaningful learning to occur. A thorough study of different patient journeys must be made to identify touchpoints where empathy can be practised.

In the Indian nursing context, many nurses and patients speak their local or vernacular languages, therefore the framework could consider language versions as a dimension in the future. Additionally, further explorations of this framework could include some of the new and emerging AI tools to generate new scenario, voice-based interactions [14, 54] and the metaverse [55] as a platform to explore immersive, wholistic, interdisciplinary learning for nurses. As further research, one could also measure the impact of VR-based empathy training for nurses on the learner and on the growth and success of the respective hospital/institution using Kirkpatrick Evaluation [56].

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